PFL-AI
Body of Knowledge

The reference for the PCI AI Project Finance Leader
Project economics · structuring · financial mathematics · the governed use of AI

FIRST EDITION — DRAFT FOR EDITORIAL AND TECHNICAL REVIEW
NOT FOR RELEASE OR DISTRIBUTION
PROJECT CONTROLS INSTITUTE GLOBAL

Finance intelligently. Control predictively. Deliver successfully.

AI proposes; the professional verifies, decides and remains accountable.

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PART ONE

Foundations

Domains 1–4 — the profession, accounting foundations, financial mathematics and investment appraisal: the financial grammar of project finance leadership.

01

DOMAIN 1

Foundations of Project Finance Leadership

Group: Foundations (Domain 1 of 4 in Part One). **Target:** ~66 pages. **Binds to:** the PCI Book Pattern Specification and the shared registries (`docs/books/registries/`). This domain fixes the book's core vocabulary — recourse, SPV, sponsor, bankability, CFADS (introduced by name, built fully in Domain 10) — and the stakeholder map every later domain assumes. British English; USD (+SAR where useful, indicative `USD 1 ≈ SAR 3.75`).

Why this domain exists

Project finance is a distinctive answer to a distinctive problem: how to fund a large, long-lived, single-purpose asset whose only security is itself — its contracts and the cash they will produce. This domain establishes what makes that answer work, and what the leader at the centre of it actually does. It maps the role across the project lifecycle (KA 1.1), builds the discipline's three-cornered logic — value, cash and risk (KA 1.2) — and grounds the profession's obligations: fiduciary awareness, independence, and the governed use of AI (KA 1.3). Everything later in the book is a specialisation of this domain: the mathematics (Domains 3–4), the structures (Domains 5, 9, 12), the lender's machinery (Domains 10, 13–15) and the AI curriculum (Domain 16) all stand on the concepts fixed here. A reader who finishes only this domain should already reason like the profession: follow the cash, price the risk, know who bears it, and stay accountable.

Learning objectives. After this domain a candidate can: describe the project finance leader's role at each lifecycle stage; place a financing on the recourse spectrum and explain what limited recourse buys and costs; explain the SPV's purpose and the interests of each party around it; describe the infrastructure-finance market's asset classes and investors; explain why cash, not profit, is the binding constraint and demonstrate the difference; explain leverage's amplification of equity returns and of risk with a computed example; state the risk-return-bankability logic; and apply the profession's ethical and responsible-AI obligations to realistic situations.

The master thread. Kestrel Water SPC — whose loan, availability stream and investment case Domains 3 and 4 priced — began here: a sponsor group weighing how to finance a desalination plant at all. This domain tells that part of the story; Domain 5 takes the project through development to bankability.

KNOWLEDGE AREA 1.1

The project finance leader and the financing landscape

Topics: 1.1.1 the role across the lifecycle · 1.1.2 corporate versus project finance — the recourse spectrum · 1.1.3 the SPV and its stakeholders · 1.1.4 the infrastructure-finance market.

1.1.1 The role across the lifecycle

The project finance leader is the person accountable for a project's **financial integrity end to end** — not the deal-closer alone, and not the accountant alone. The role changes costume by stage while keeping one spine:

Stage	What the finance leader owns
Development	Screening economics (Domain 4); funding development spend at risk; shaping a financeable concept (Domain 5)
Structuring	Capital structure and funding sources (Domain 9); risk allocation into contracts (Domains 11–12)
Execution (financial close)	Due diligence, model audit, documentation, conditions precedent (Domain 13)
Construction	Drawdowns, cost-to-complete, lender reporting (Domain 14)
Operations	Covenant compliance, waterfall management, distributions, refinancing (Domain 15)
Maturity/exit	Handback, sale, restructuring where needed (Domain 15)

The spine is a single question asked at every stage: **will the cash arrive, and who is exposed if it does not?** The leader's authority rests on being the person in the room who can answer it with evidence.

1.1.2 Corporate versus project finance — the recourse spectrum

Definitions. In **corporate (balance-sheet) finance**, lenders lend to a company and are repaid from its whole cash flow; every asset stands behind every debt. In **project finance**, lenders lend to a ring-fenced project and are repaid **only from that project's cash flows**, with security over its assets and contracts — **non-recourse** to the sponsors, or **limited recourse** where sponsors give bounded support (a completion guarantee, a cost-overrun facility). Real deals sit on a spectrum between the poles.

What limited recourse buys sponsors: risk containment (a failed project cannot sink the parent), balance-sheet capacity, the ability to share a mega-project among partners, and discipline — lenders' due diligence becomes a second pair of eyes on every assumption. What it costs: higher margins and fees (lenders carry risk they cannot chase a parent for), heavy transaction and diligence costs, long documentation timelines, and covenant control over the project's cash (Domain 10). The break-even is scale and risk-shape: single-asset, contract-backed, capital-intensive projects with long lives are where the machinery pays.

Fig 1.1.1 — The recourse spectrum

1.1.3 The SPV and its stakeholders

Definition. The **special-purpose vehicle (SPV)** is a company created to do exactly one thing — own, build, finance and operate the project — and legally incapable of doing anything else. Ring-fencing is what makes non-recourse lending possible: the SPV's contracts are its assets, and every major relationship is written down (Domain 12 builds the contract matrix in full).

The parties and what each optimises:

Party	Wants	Watches
Sponsors (equity)	Return on equity; contained risk; distributions	Equity IRR, distribution tests
Lenders	Repayment with margin; downside protection	DSCR/LLCR, covenants, security (Domain 10)
Offtaker / grantor	Reliable service at agreed price	Availability, tariffs, handback condition
EPC contractor	Construction margin	Variations, delay LDs (Domain 12)
Operator (O&M)	Fee; performance regime it can meet	Availability/output guarantees
Government / regulator	Delivery of policy outcomes; compliance	Permits, obligations, public interest
Community & environment	Benefit without harm	E&S performance (Domain 11)

The finance leader's daily craft is reconciling these optimisations inside one cash flow — which is why the cash waterfall (Domain 15) reads like a peace treaty.

Fig 1.1.2 — The SPV at the centre of its contracts

1.1.4 The infrastructure-finance market

The asset classes and their financing habits, in one professional sweep: **transport** (toll roads, rail, airports — patronage or availability models); **power and renewables** (PPAs, capacity markets; the energy transition's build-out is the market's largest engine); **water** (desalination and treatment concessions — Kestrel's world); **digital infrastructure** (data centres, towers, fibre — shorter refresh cycles, credit-tenant leases); **social infrastructure** (hospitals, schools, housing under availability PPPs); and **natural resources** (commodity-linked, price-hedged). The capital comes from commercial banks (construction-phase specialists), institutional investors and infrastructure funds (long-dated operations-phase capital), export credit agencies and development banks (Domain 9), and bond markets (refinancing bankable operating assets). The leader's market literacy is matching **asset shape to capital shape**: construction risk to banks and ECAs, stabilised cash flows to institutions — mismatches are expensive at best and fatal at close.

AI in this KA

Market screening and precedent research are natural AI accelerants — summarising comparable transactions, extracting terms from public disclosures, drafting stakeholder maps. The governed habits start in Domain 1 because the failure modes do: a fluent but invented "precedent transaction" is this profession's textbook hallucination case. Sources are verified against the registry discipline (every claimed deal traced to a public record), and the stakeholder analysis an AI drafts is walked, party by party, by someone who has sat across from those parties. **AI proposes; the professional verifies, decides and remains accountable.**

KEY TERMS — KA 1.1

Term	Meaning
Recourse / non-recourse / limited recourse	Whom the lender can pursue: the sponsor's balance sheet; the project only; the project plus bounded sponsor support.
SPV	Ring-fenced single-purpose project company; the borrower and contract hub.
Sponsor	Equity investor promoting the project (this book's project-finance sense).
Offtaker	The buyer of the project's output or service under contract.
Ring-fencing	Legal isolation of the project's assets, contracts and cash.
Asset-capital matching	Pairing project risk phases with the capital suited to hold them.

SAMPLE MCQS — KA 1.1

MCQ 1.1-A `[1.1.2 · Application]` A sponsor gives lenders a guarantee that covers cost overruns until the plant passes its completion test, after which lenders may look only to project cash flows. This financing is best described as: - A. full recourse - B. non-recourse - C. limited recourse - D. unsecured corporate lending

Rationale: Bounded sponsor support (here, to completion) between the poles is the defining shape of limited recourse. A would expose the sponsor for the loan's life; B would mean no sponsor support at all; D abandons both the security package and the ring-fence.

MCQ 1.1-B `[1.1.3 · Analysis]` Which single feature of the SPV makes non-recourse lending possible? - A. its tax registration - B. legal ring-fencing: the SPV can conduct only the project, so its contracts and cash are isolated and chargeable - C. its sponsors' credit ratings - D. the size of its share capital

Rationale: Lenders can accept project-only recourse because the ring-fence guarantees no other business can dilute, encumber or divert the cash they are lending against. C reverses the concept (sponsor credit is what non-recourse lending does without); A and D are administrative facts, not the mechanism.

MCQ 1.1-C `[1.1.4 · Application]` A fund holding long-dated pension liabilities wants infrastructure exposure. The asset-capital matching principle points it toward: - A. construction-phase risk in a greenfield project - B. stabilised operating assets with contracted cash flows - C. development-stage equity at risk - D. short-term bridge lending

Rationale: Long-dated stable liabilities match long-dated stable cash flows. A and C sit where construction and development specialists (banks, ECAs, developers) hold the risk; D matches a treasury desk, not a pension profile.

SELF-CHECK — KA 1.1

1. State the finance leader's one recurring question. — Will the cash arrive, and who is exposed if it does not?
2. Name two things limited recourse buys a sponsor and two things it costs. — Buys: risk containment, balance-sheet capacity (also partnering, lender discipline). Costs: pricing and fees, transaction complexity (also covenant control, time).
3. Why does every party around the SPV get a contract? — Non-recourse credit is built from contracts: each relationship must be enforceable because the cash flow is the only security.

KNOWLEDGE AREA 1.2

Value, cash and risk: the discipline's logic

Topics: 1.2.1 value creation in projects · 1.2.2 cash as the binding constraint · 1.2.3 leverage, risk and the bankability triangle.

1.2.1 Value creation in projects

A project creates value when the present value of what it will produce exceeds what it costs to build and run — Domain 4's NPV, stated in words. The foundation point is where value can be created or destroyed by financing: structure does not conjure value from a bad project, but it can (1) allocate each risk to the party who bears it cheapest — lowering the priced-in premiums; (2) match capital to risk phase — lowering the blended cost of funds; and (3) impose diligence and covenant discipline that keeps forecast value from leaking in execution. The corollary the profession lives by: **financing engineering amplifies project quality; it never substitutes for it.**

1.2.2 Cash, not profit, is the binding constraint

The principle. Profit is an opinion about periods (Domain 2's accrual model); **cash pays debt service**. Projects die of cash exhaustion, usually while reporting profits.

WORKED EXAMPLE

Worked example 1.2.2 — profitable and out of cash.

1. **Setup.** In its first operating quarter a project company recognises revenue of USD 10,000,000 against costs of USD 8,000,000. But customers have paid only 7,000,000 of the revenue (receivables +3,000,000); spare-parts inventory was built up by 1,000,000; and suppliers extended 500,000 of additional credit (payables +500,000).
2. **Formula.** Operating cash flow = profit − Δreceivables − Δinventory + Δpayables.
3. **Substitution.** `2,000,000 − 3,000,000 − 1,000,000 + 500,000`.
4. **Result.** Profit **+USD 2,000,000**; operating cash flow **−USD 1,500,000**.
5. **Interpretation.** The same quarter is a success in the income statement and a crisis in the bank account: a 2.0m "profitable" company is 1.5m short of the cash its debt service assumed. This is why lenders size and test debt against **CFADS** — cash flow available for debt service (defined fully in Domain 10) — and why every model in Domain 6 is a cash model first. Domain 2 builds the full accrual-to-cash bridge.

1.2.3 Leverage, risk and the bankability triangle

Leverage amplifies. Debt is cheaper than equity, and fixed: whatever the project earns, debt service is owed. That fixity cuts both ways.

WORKED EXAMPLE

Worked example 1.2.3 — the two faces of leverage.

1. **Setup.** A project costs USD 100,000,000 and produces steady operating cash of USD 12,000,000 per year. Compare an all-equity structure with 70 % debt (USD 70,000,000, interest-only at 6.0 % = 4,200,000 per year), in the base case and with cash down 25 % and 50 %.
2. **Formula.** Unlevered return = cash / 100,000,000. Equity cash = cash − 4,200,000; levered return = equity cash / 30,000,000.
3. **Substitution.** Base: `12.0 − 4.2 = 7.8` on 30. Down 25 %: `9.0 − 4.2 = 4.8`. Down 50 %: `6.0 − 4.2 = 1.8`.
4. **Result.**

Scenario	Project cash	Unlevered return	Equity cash	Levered return
Base	12,000,000	12.0 %	7,800,000	**26.0 %**
−25 %	9,000,000	9.0 %	4,800,000	**16.0 %**
−50 %	6,000,000	6.0 %	1,800,000	**6.0 %**
−65 %	4,200,000	4.2 %	0	**0.0 %**

1. **Interpretation.** Leverage more than doubles the base-case equity return (26 % vs 12 %) — and makes the downside three times steeper (26 → 6 % as cash halves, versus 12 → 6 % unlevered). At a 65 % cash decline the equity earns nothing and the lender is next in line. Gearing is chosen, not maximised: Domain 9 structures it, Domain 10 shows how lenders cap it with coverage ratios sized precisely against scenarios like this table.

The bankability triangle. Three tests every financing must pass simultaneously: **value** (the project is worth doing — Domain 4), **cash** (the flows arrive in the periods that need them — Domains 6–8), and **risk allocation** (each hazard sits with a party able and bound to bear it — Domains 11–12). A project strong on two corners and weak on one is not two-thirds bankable; it is unbankable until the corner is fixed. **Bankability** (built fully in Domain 5) is precisely the state of passing all three tests in the eyes of the capital being asked to commit.

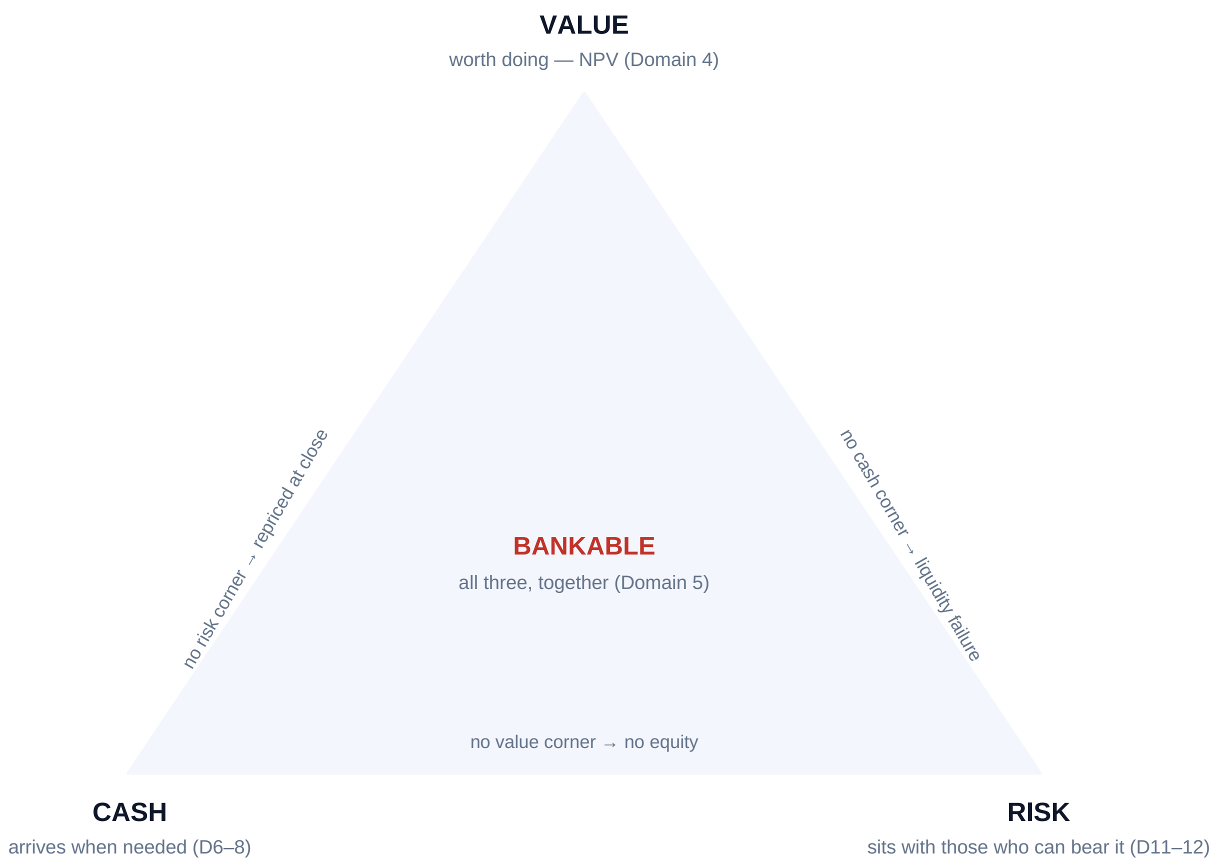

Fig 1.2.1 — The bankability triangle

AI in this KA

Scenario tables like 1.2.3's are ideal machine work — and the place where a subtly wrong fixed-charge assumption (interest-only vs amortising; Domain 3's shapes) silently reshapes every row. The governed pattern: the machine drafts the grid; the analyst re-derives one row by hand and checks the boundary case (where equity cash crosses zero) analytically; the leader reads the downside rows first, because that is what the structure is actually being designed against.

KEY TERMS — KA 1.2

Term	Meaning
CFADS	Cash flow available for debt service (name fixed here; machinery in Domain 10).
Working capital drag	Cash absorbed by receivables and inventory ahead of profit.
Leverage / gearing	Debt's share of funding; amplifier of equity return and risk.
Fixed charge	Debt service owed regardless of performance.
Bankability triangle	Value + cash + risk allocation, passed together.

SAMPLE MCQS — KA 1.2

MCQ 1.2-A `[1.2.2 · Application]` A company reports quarterly profit of 2,000,000 while receivables rose 3,000,000, inventory rose 1,000,000 and payables rose 500,000. Its operating cash flow is: - A. +2,000,000 - B. −1,500,000 - C. +500,000 - D. −2,500,000

Rationale: `2.0 − 3.0 − 1.0 + 0.5 = −1.5m`. A stops at profit; C nets only the payables against profit and forgets the asset build; D subtracts the payables increase instead of adding it — supplier credit is a cash source.

MCQ 1.2-B `[1.2.3 · Application]` In the leverage example (70m debt, interest-only 6 %; equity 30m), project cash of 9,000,000 produces a levered equity return of: - A. 9.0 % - B. 16.0 % - C. 26.0 % - D. 30.0 %

Rationale: `(9.0 − 4.2)/30 = 16.0 %`. A is the unlevered return; C is the base-case levered return; D divides project cash by equity without paying the lender first.

MCQ 1.2-C `[1.2.3 · Analysis]` A project shows strong NPV and well-allocated risks, but its revenue arrives seasonally while debt service is quarterly and level. The bankability verdict is: - A. bankable — two of three corners suffice - B. unbankable as structured: the cash corner fails; reshape the debt profile or add liquidity support - C. unbankable permanently — reject the project - D. bankable if the sponsors accept a higher equity IRR

Rationale: The triangle is conjunctive: mistimed cash defeats value and allocation. The cure is structural (sculpted or seasonal debt service, reserve accounts — Domains 9–10), not rejection (C) and not a return adjustment that changes nothing about timing (D).

SELF-CHECK — KA 1.2

1. Why do lenders test CFADS rather than profit? — Debt service is paid in cash; accrual profit can coexist with cash exhaustion (WE 1.2.2).
2. State leverage's two faces in one sentence each. — It multiplies the equity return on the same project cash; it makes every downside steeper because debt service is fixed.
3. Why is the triangle conjunctive? — Each corner defeats the financing alone: no value → no equity; no cash timing → default risk regardless of value; no allocation → risk premia or collapse at close.

KNOWLEDGE AREA 1.3

Ethics, fiduciary awareness and responsible AI

Topics: 1.3.1 obligations and duties · 1.3.2 conflicts and independence · 1.3.3 the responsible-AI principle in finance.

1.3.1 Obligations and duties

The project finance leader acts inside a lattice of duties: **fiduciary-type duties** to the employer or client (loyalty, care, confidentiality); **contractual duties** under mandates and finance documents; **statutory duties** (companies law, anti-bribery and corruption, sanctions, market conduct); and **professional duties** — competence, candour, and records that let others check the work. Two standing disciplines follow. Candour about numbers: forecasts are presented with their assumptions and sensitivities, never as certainties (Domain 4, KA 4.3.3); an optimistic case knowingly presented as a base case is a misrepresentation, whatever the spreadsheet says. Candour about limits: this book's own rule — educational reference, not individualized advice; jurisdiction-specific matters go to qualified counsel and advisers — is the same professional humility applied to oneself.

1.3.2 Conflicts and independence

Project finance concentrates conflicts because the same institutions recur in many roles: the adviser who would earn a success fee on close; the sponsor-affiliated contractor pricing the EPC; the bank advising the government while lending to bidders. The professional machinery is disclosure and separation: conflicts are declared before engagement, managed with information barriers or declined; advice and self-interest are never silently blended. The leader's test for any arrangement is the daylight test — would every party, seeing the full fee and relationship map, still regard the advice as independent? Where the answer wavers, independence has already failed. (Case study B applies this to a live tender; Domain 13's diligence streams exist partly to give lenders advice that passes the test.)

1.3.3 The responsible-AI principle in finance

The suite principle — **AI proposes; the professional verifies, decides and remains accountable** — lands hardest in finance, where machine output looks like the work product itself (a model, a memo, a covenant summary). Domain 16 builds the full governance architecture; the foundations fixed here:

- **Verification is not optional and not delegable to the tool.** Golden-answer checks for calculations (the discipline this book applies to itself); source-tracing for claims; document-against-summary checks for AI-read contracts.
- **Accountability cannot be transferred.** "The model said so" is never a defence — the signing professional owns the output as if hand-made.
- **Confidentiality travels with the data.** Deal information entering an AI tool is a disclosure; it happens only within approved, contracted environments.
- **Material AI use is disclosed** within the team and, where it touches deliverables, to the client — the daylight test again, applied to method.

KEY TERMS — KA 1.3

Term	Meaning
Fiduciary awareness	Acting in the principal's interest with loyalty, care and confidentiality.
Conflict of interest	An interest that could bias judgment; declared, managed or declined.
Daylight test	Would full disclosure of interests leave the advice trusted?
Responsible-AI principle	AI proposes; the professional verifies, decides, remains accountable.
Verification duty	The named human's obligation to check machine output before reliance.

SAMPLE MCQS — KA 1.3

MCQ 1.3-A `[1.3.1 · Analysis]` An analyst is asked to present the upside case as the base case "because the committee needs confidence". The professional response is: - A. comply — labels are a presentation choice - B. decline: presenting a knowingly optimistic case as the base misrepresents the forecast; offer the honest base with sensitivities instead - C. comply, but keep a private note of the true base - D. resign immediately without discussion

Rationale: Candour about numbers is a duty, not a style (A); a private note documents the misrepresentation without preventing it (C); B both refuses the breach and offers the legitimate route to confidence — evidence. D skips the professional obligation to fix the problem before escalating personal exits.

MCQ 1.3-B `[1.3.2 · Application]` A bank advising a grantor on a tender also wishes to lend to one of the bidders. The minimum acceptable handling is: - A. proceed — different departments are involved - B. disclose the dual role to the grantor, and either obtain informed consent with effective information barriers or decline one role - C. keep the lending discussion confidential until after award - D. advise the grantor to select that bidder

Rationale: Disclosure plus genuine separation (or declination) is the standing machinery; department labels alone are not barriers (A); concealment converts a conflict into misconduct (C); D is the conflict operating in the open.

MCQ 1.3-C `[1.3.3 · Recall]` Under the PCI responsible-AI principle, responsibility for an AI-drafted covenant summary used in a credit paper rests with: - A. the AI vendor - B. the model itself - C. the professional who verified, signed and used it - D. nobody, if the tool was approved

Rationale: Accountability cannot be delegated to software or its supplier; tool approval governs which tools may be used, never who answers for the output.

SELF-CHECK — KA 1.3

1. State the daylight test. — Would every party, seeing the full relationship and fee map, still trust the advice as independent?
2. What three checks make AI-assisted work professionally usable? — Recomputed numbers (golden checks), traced sources, document-against-summary verification — by a named human.
3. Why does confidentiality bind AI use? — Data entering a tool is a disclosure; it must stay within approved, contracted environments.

Industry variations — Domain 1

The foundations flex by sector mainly in who the counterparties are: in **power and water**, the offtaker is often a state utility — sovereign credit and political risk enter the stakeholder map (Domain 11); in **transport**, the "offtaker" may be the travelling public — demand risk reshapes the whole triangle; in **social PPPs**, the grantor's availability payment makes government the cash engine and handback condition a first-order obligation; in **digital infrastructure**, corporate credit-tenants replace state offtakers and refresh cycles shorten every horizon; in **natural resources**, the market itself is the offtaker and hedging policy joins the foundations. The leader's first map in any new sector: who pays, under what compulsion, and what can stop them.

Case study — Domain 1: how Kestrel chose project finance (water)

Situation. Kestrel's two sponsors — an international water operator and a regional infrastructure fund — faced a USD 100,000,000 plant with a 25-year availability offtake. Corporate borrowing was available to the operator at attractive rates; the fund could not guarantee anything beyond its equity.

Analysis. Corporate route: cheapest debt, fastest close — but the operator alone carries 100 % of construction and performance risk on its balance sheet, the fund cannot participate on equal terms, and a project failure would impair the operator's whole credit. Project route: an SPV with limited-recourse debt (completion support only), 70/30 gearing — pricier debt and eighteen months of structuring, but risk contained at the ring-fence, the partners aligned through one shareholders' agreement, and lender diligence pressure-testing every assumption (the discipline dividend of 1.2.1). The leverage table of WE 1.2.3 gave the equity story; the triangle gave the test — value (Domain 4's +16.18m NPV), cash (the availability stream priced in Domain 3), risk (allocated through the contract matrix built later in Domain 12).

The decision. Project finance — not because it was cheaper (it was not), but because it made the partnership possible, contained the downside, and converted lender scrutiny into project quality. The minute records the recourse position, the support obligations and their sunset at completion — the exact vocabulary of KA 1.1.2.

What the domain teaches here. The financing route is a risk and partnership decision before it is a cost decision; the cheapest debt attached to the wrong recourse shape is the expensive option.

Case study B — Domain 1: the adviser with two hats (transport tender)

Situation. A grantor's financial adviser on a toll-road tender was discovered — after preferred-bidder selection — to hold an advisory mandate for the winning consortium's lead sponsor on an unrelated deal, undisclosed. The losing bidders challenged; the award was suspended pending review.

What happened. The review found no evidence the evaluation was slanted — and it did not matter. The undisclosed relationship alone failed the daylight test: the tender was re-run with a new adviser at a cost of fourteen months and the grantor's credibility, and the adviser's firm lost its public-sector practice in the jurisdiction. A one-line disclosure at engagement, with barriers or a declined mandate, would have cost nothing.

What the domain teaches here. Conflicts are priced at discovery, not at occurrence — and the price is paid in time, trust and franchise, not fees. Independence is an asset that only disclosure can insure.

Executive perspective — Domain 1

What a project finance director cannot delegate in this domain:

- **The recourse position.** Exactly what the sponsor stands behind, until when, capped at what — the director signs this sentence personally and re-reads it before every support call.
- **The stakeholder map's honesty.** Every party's real incentive, including the uncomfortable ones, on one page the board has seen.
- **The cash question.** Asked in every meeting until it embarrasses no one: will the cash arrive, and who is exposed if it does not?
- **The conflicts register.** Kept current, disclosed early, tested against daylight — because the director's own relationships are usually the largest entries.
- **The AI accountability line.** Named humans own machine output; the director owns the culture that makes that real (Domain 16 gives the machinery; Domain 1 gives the law).

Calculation exercises — Domain 1

Exercise 1.1 Profit 3,500,000; receivables +2,200,000; inventory +900,000; payables +600,000. Operating cash flow? Solution. `3.5 − 2.2 − 0.9 + 0.6 =` **+USD 1,000,000**. Common error: sign on payables (supplier credit is a source: +0.6, not −0.6; the wrong sign gives −0.2m and a false alarm).

Exercise 1.2 The WE 1.2.3 project refinances to 80 % debt (80,000,000 interest-only at 6.5 %). Rebuild the base-case levered return and the cash decline at which equity income reaches zero. Solution. Debt service `80 × 0.065 = 5,200,000`; equity 20,000,000; base equity cash `12.0 − 5.2 = 6.8` → **34.0 %**. Zero at project cash = 5,200,000 — a **56.7 % decline** (from 12.0). Versus the 70/30 case: two points more base return (34 vs 26 %... on a third less equity) bought a materially nearer cliff (−56.7 % vs −65 %). Common error: comparing levered percentages without comparing the cliffs.

Exercise 1.3 Classify: (a) parent guarantees debt until completion test, then released; (b) parent comfort letter, non-binding; (c) no support, reserves funded from cash flow. Solution. (a) limited recourse; (b) effectively non-recourse in law — comfort letters are generally not enforceable guarantees (jurisdiction-specific; counsel confirms); (c) non-recourse with structural mitigation. Common error: treating a comfort letter as recourse — lenders price it as goodwill, not security.

Practitioner's toolkit — Domain 1

Adoption-ready artefacts; adapt headings to your organisation, then keep them stable.

Toolkit 1.T.1 — Stakeholder map (one page per project)

Per party: name and role · what they optimise · contract binding them (Domain 12 reference) · cash flows to/from the SPV · veto points and consents · relationship owner. Rule: the map is board-visible and includes the uncomfortable incentives.

Toolkit 1.T.2 — Financing-route decision record

Options considered (corporate / limited recourse / non-recourse / hybrid) · recourse sentence per option (who stands behind what, until when, capped at what) · pricing and cost deltas · partnership and balance-sheet effects · risk containment analysis (triangle test per option) · decision, rationale, decision-maker, date.

Toolkit 1.T.3 — Conflicts and AI-use register

Conflicts: relationship · parties affected · disclosure date · handling (barriers/consent/ declined) · review date. AI use: tool and environment · data classification cleared · verification steps and named verifier · disclosure status. One register, one owner, standing agenda item.

Exam preparation — Domain 1

The traps. Recourse classifications (comfort letter ≠ guarantee — Exercise 1.3) · payables sign in the cash bridge (Exercise 1.1) · levered-return arithmetic that skips debt service (MCQ 1.2-B distractor D) · reading the triangle as two-out-of-three · "sponsor" meaning equity investor in this book's project-finance chapters (terminology registry) · assigning AI accountability anywhere but the signing professional.

Reflection questions. 1. Take a project you know: write its recourse sentence in under 25 words. Who stands behind what, until when, capped at what? 2. Which corner of the bankability triangle does your current project stress most — and what structural (not cosmetic) fix would close it? 3. What in your team's current AI usage would fail the daylight test if disclosed in full — and what changes tomorrow because you asked?

Domain 1 summary

Project finance funds single-purpose assets against their own cash, made possible by the ring-fenced SPV and priced along the recourse spectrum; the leader's role is the financial integrity of that machine across the whole lifecycle, under one recurring question — will the cash arrive, and who is exposed if it does not? The discipline's logic triangulates value, cash and risk: financing amplifies project quality but never substitutes for it; cash, not profit, binds (a profitable quarter can be a cash crisis); leverage multiplies returns and steepens every downside, which is why lenders will later cap it with coverage machinery. Around the technique stands the profession: fiduciary-grade candour about numbers and limits, conflicts managed in daylight, and machine assistance governed by the suite principle — AI proposes; the professional verifies, decides and remains accountable. Domain 2 builds the accounting the cash bridge assumed; Domain 5 takes Kestrel from concept to bankability.

02

DOMAIN 2

Accounting and Financial-Statement Foundations

Group: Foundations (Domain 2 of 4 in Part One). **Target:** ~70 pages. **Binds to:** the PCI Book Pattern Specification and the shared registries (`docs/books/registries/`). This domain builds the accrual-to-cash bridge Domain 1 promised and supplies the statement vocabulary every later domain assumes — including the `CFADS` that Domain 10 turns into coverage ratios. Standards are named and described in this book's own words; no standard's text is reproduced, and nothing here is accounting advice for a specific entity or jurisdiction. British English; USD (+SAR where useful, indicative `USD 1 ≈ SAR 3.75`).

Why this domain exists

Domain 1 established that **cash, not profit, is the binding constraint** — and then left an obligation outstanding: if profit is not the thing that pays debt service, why does anyone compute it, and how do you get from one to the other? This domain discharges that obligation. It builds the accrual model and what recognition actually means (KA 2.1); assembles the three statements and the articulation that binds them into one system (KA 2.2); works the treatments that decide a project's reported numbers — working capital, revenue and cost recognition, capital versus operating expenditure, provisions (KA 2.3); and closes with ratio interpretation and the interfaces between project reporting and corporate accounts (KA 2.4). A project finance leader is not an accountant, and this domain does not try to make one. It makes something more specific: a leader who can read a set of statements, find the cash inside them, and tell when an accounting choice has changed the picture without changing the economics.

Learning objectives. After this domain a candidate can: explain accrual accounting and the recognition tests that drive it; describe each of the three statements and what question it answers; **articulate** a statement set — derive the cash-flow statement from profit and the balance-sheet movements, and prove it reconciles; explain how working capital consumes cash and compute its effect on `CFADS` and therefore on a coverage ratio; distinguish capital from operating expenditure and quantify the profit effect of the choice while showing cash is unaffected; explain provisions and contingent liabilities in principle; interpret the core ratios including interest cover; describe the interfaces between project cost systems and corporate reporting; and govern AI-assisted analysis of financial statements.

The master statements. Kestrel Water SPC — whose loan, appraisal and financing decision Domains 1, 3 and 4 built — now reports its **first full operating year**. The plant cost **USD 60,000,000** (Domain 4's I_0), depreciated straight-line over **25 years**. The senior loan is Domain 3's **USD 42,000,000 at 6.0 % over 12 years**, so year-one interest is **USD 2,520,000** and the annual instalment is **USD 5,009,635**. Revenue is **USD 12,000,000**, cash operating costs **USD 4,500,000**, and tax is charged at **20 %**. Every figure in KA 2.2–2.4 derives from these.

KNOWLEDGE AREA 2.1

The accrual model

Topics: 2.1.1 accrual versus cash accounting · 2.1.2 recognition · 2.1.3 the statements as one system.

2.1.1 Accrual versus cash accounting

Definitions. Cash accounting records a transaction when money moves. **Accrual accounting** records the effects of transactions when they occur, regardless of when money moves — revenue when it is earned, expenses when they are incurred. Accrual is the basis of general-purpose financial reporting because cash timing is a poor guide to performance: a project that invoices in December and collects in February did the work in December, and a cash-basis account would report a loss followed by a windfall.

The two bases answer different questions, and a finance leader needs both:

Basis	Answers	Where it governs
Accrual	Did we perform profitably in this period?	Statutory accounts, covenants defined on profit, tax
Cash	Can we pay what falls due?	Debt service, drawdowns, liquidity, `CFADS` (Domain 10)

The professional consequence. Accrual and cash diverge, and the divergence is information, not noise. A project reporting healthy profit while operating cash falls is telling you something precise — usually that working capital is absorbing the difference (KA 2.3.1). Domain 1's worked example was exactly this case; this domain now shows the machinery that produces it.

2.1.2 Recognition

Definition. Recognition is recording an item in the statements as an asset, liability, income or expense. It is governed by tests, not by preference, and the tests are the substance of what accounting standards do. Described in principle (IFRS is the framework referenced throughout this book by name):

- An **asset** is a present economic resource controlled by the entity as a result of past events — recognised when it exists and can be measured reliably. Control, not legal title, is the operative idea.
- A **liability** is a present obligation to transfer an economic resource as a result of past events. A future intention is not a liability, however certain it feels.
- **Income and expenses** are recognised as the underlying changes in assets and liabilities occur, which is why revenue recognition follows performance (KA 2.3.2) rather than invoicing.

Measurement then asks at what amount — historical cost, amortised cost, fair value — and the choice materially changes reported figures without changing any cash flow. That sentence is the whole reason a finance leader reads statements sceptically: **recognition and measurement policies are part of the answer, so they must be part of the question.**

2.1.3 The statements as one system

The three primary statements are not three reports; they are three views of one set of facts, locked together by identities:

```
Assets = Liabilities + Equity                              (the balance sheet identity)
Closing equity = Opening equity + Profit − Distributions   (+ capital contributed)
Closing cash   = Opening cash + Operating + Investing + Financing cash flows
```

Because they share one underlying record, a change in one propagates to all three — the property called **articulation**, demonstrated arithmetically in KA 2.2.4. Articulation is the reader's single most powerful tool: a statement set that does not articulate contains an error or an omission, and finding where it fails localises the problem immediately.

AI in this KA

Statement analysis is a genuine AI strength — extracting line items, restating across periods, flagging unusual movements — and it has a specific, dangerous blind spot: a model reads the numbers fluently and the policies not at all. Two projects with identical economics and different capitalisation or revenue-recognition policies produce different statements, and an assistant comparing them will report a difference in performance that does not exist. The governed habit: ask what policy produced each figure before comparing any two entities or periods, and verify extracted line items against the source statement. **AI proposes; the professional verifies, decides and remains accountable.**

KEY TERMS — KA 2.1

Term	Meaning
Accrual basis	Recognising the effects of transactions when they occur, not when cash moves.
Recognition	Recording an item in the statements, subject to definition and measurement tests.
Measurement	The amount at which a recognised item is carried (cost, amortised cost, fair value).
Articulation	The locking of the three statements to one underlying record via identities.
Accounting policy	The permitted choice that shapes reported figures without changing cash.

SAMPLE MCQS — KA 2.1

MCQ 2.1-A `[2.1.1 · Analysis]` A project reports rising profit and falling operating cash flow over three quarters, with no change in accounting policy. The most likely explanation is: - A. the profit figures must be erroneous - B. working capital is absorbing cash — receivables and/or inventory are growing faster than payables - C. depreciation has increased - D. the two measures are unrelated, so no explanation is needed

Rationale: Accrual profit and operating cash diverge principally through working-capital movements (KA 2.3.1). C would raise operating cash relative to profit (a non-cash charge added back), and D denies the articulation the statements are built on.

MCQ 2.1-B `[2.1.2 · Application]` A sponsor board has firmly resolved to fund a plant upgrade next year. At this year end this is: - A. a liability, because the decision is certain - B. not a liability — there is no present obligation from a past event; an intention is not an obligation - C. a provision, because the amount is estimable - D. a contingent asset

Rationale: Liability recognition requires a present obligation arising from a past event (2.1.2). Certainty of intent is irrelevant; a provision (C) still requires an obligation to exist, and D inverts the direction entirely.

MCQ 2.1-C [2.1.3 · Recall] A statement set where the cash-flow statement's closing cash does not equal the balance sheet's cash indicates: - A. a normal difference in presentation - B. an error or omission — the statements articulate to one record, so they must reconcile - C. the use of accrual rather than cash accounting - D. a foreign-currency effect that requires no action

Rationale: Articulation is an identity (2.1.3). FX and presentation differences are disclosed and reconciled, not left unbalanced — an unreconciled set is a defect, and locating the break is the reader's fastest diagnostic.

SELF-CHECK — KA 2.1

1. Which basis governs debt service, and which governs covenants defined on profit? — Cash for debt service; accrual for profit-defined covenants. A leader needs both.
2. Why is an intention never a liability? — Recognition requires a present obligation arising from a past event.
3. What does a failure to articulate tell you? — That there is an error or omission, and where the break occurs localises it.

KNOWLEDGE AREA 2.2

The three statements and their articulation

Topics: 2.2.1 the income statement · 2.2.2 the balance sheet · 2.2.3 the cash-flow statement · 2.2.4 articulation demonstrated.

2.2.1 The income statement

What it answers: did the entity perform profitably in the period? Its project-finance-relevant structure runs down from revenue through the layers a lender reads:

WORKED EXAMPLE

Worked example 2.2.1 — Kestrel's first operating year.

1. **Setup.** Revenue USD 12,000,000; cash operating costs USD 4,500,000; plant USD 60,000,000 depreciated straight-line over 25 years; senior interest USD 2,520,000 (Domain 3's year-one figure); tax 20 %.
2. **Formula.** `EBITDA` = revenue − cash operating costs; `EBIT` = `EBITDA` − depreciation; profit before tax = `EBIT` − interest; net income = PBT − tax. Depreciation = (cost − residual)/ useful life.
3. **Substitution.** `EBITDA = 12,000,000 − 4,500,000`; depreciation `= 60,000,000/25`; `EBIT = 7,500,000 − 2,400,000`; `PBT = 5,100,000 − 2,520,000`; tax `= 2,580,000 × 0.20`.
4. **Result.**

Line	USD
Revenue	12,000,000
Cash operating costs	(4,500,000)
EBITDA	**7,500,000**
Depreciation	(2,400,000)
EBIT	**5,100,000**
Interest	(2,520,000)
Profit before tax	**2,580,000**
Tax at 20 %	(516,000)
Net income	**2,064,000**

1. **Interpretation.** Four layers, four different questions. `EBITDA` approximates the cash the operation generates before financing, tax and the accounting for past capital spend — which is why lenders start there (Domain 10). `EBIT` charges the asset's consumption, so it measures operating performance including capital intensity. **PBT** is after the cost of the capital structure. **Net income** is the shareholders' accounting return — and note it is USD 2,064,000 while `EBITDA` is USD 7,500,000: on a capital-intensive, levered project the gap between the two is mostly depreciation and interest, and confusing them is the commonest error in project financial conversation.

2.2.2 The balance sheet

What it answers: what does the entity own and owe at a point in time? For an SPV the structure is unusually clean — one asset of consequence and one financing structure:

	USD
Plant, net (60,000,000 − 2,400,000 depreciation)	57,600,000
Receivables	900,000
Cash	balancing
Total assets	
Payables	300,000
Senior debt (42,000,000 − 2,489,635 year-one principal)	39,510,365
Equity (contributed + retained)	balancing

Two project-specific points. **The asset is consumed on paper while it produces cash** — depreciation reduces the carrying amount without any payment. And **the debt balance falls by principal only**: of Kestrel's USD 5,009,635 instalment, USD 2,520,000 is interest (an expense) and **USD 2,489,635 is principal** (a balance-sheet movement, not an expense) — exactly Domain 3's schedule. That split is why debt service never appears as a single line in the income statement, and why a reader who looks only at profit cannot see whether the loan is being repaid.

2.2.3 The cash-flow statement

What it answers: where did cash come from and go? Three sections: **operating** (the trading cycle), **investing** (asset purchases and disposals), **financing** (debt drawn and repaid, equity contributed, distributions paid). The indirect method — the one a project leader will usually meet — starts from profit and undoes the accruals:

```
Net income
  + non-cash charges (depreciation, amortisation, provisions)
  − increases in receivables and inventory      (cash absorbed)
  + increases in payables                       (cash released)
  = operating cash flow
```

2.2.4 Articulation demonstrated

WORKED EXAMPLE

Worked example 2.2.4 — from profit to cash, and proving it ties.

1. **Setup.** Kestrel's income statement above, plus the balance-sheet movements: receivables rose **USD 900,000**, payables rose **USD 300,000**. Derive operating cash flow.
2. **Formula.** Operating cash = net income + depreciation − Δreceivables + Δpayables.
3. **Substitution.** `2,064,000 + 2,400,000 − 900,000 + 300,000`.
4. **Result. Operating cash flow USD 3,864,000** — against net income of USD 2,064,000. The reconciliation: depreciation adds back **+2,400,000** (a charge that moved no cash), working capital absorbs **−600,000** net.
5. **Interpretation.** Read the bridge in both directions. Profit understates the period's cash by the depreciation of an asset paid for years ago; working capital takes back 600,000 of it to fund growth in the trading cycle. Both facts are invisible in the income statement and decisive for a lender. Note also what operating cash flow is **not**: it is before debt principal (a financing flow) and before capex — so a project can show positive operating cash and still fail to cover its obligations. Domain 10's coverage ratios exist precisely to test the thing this statement does not.

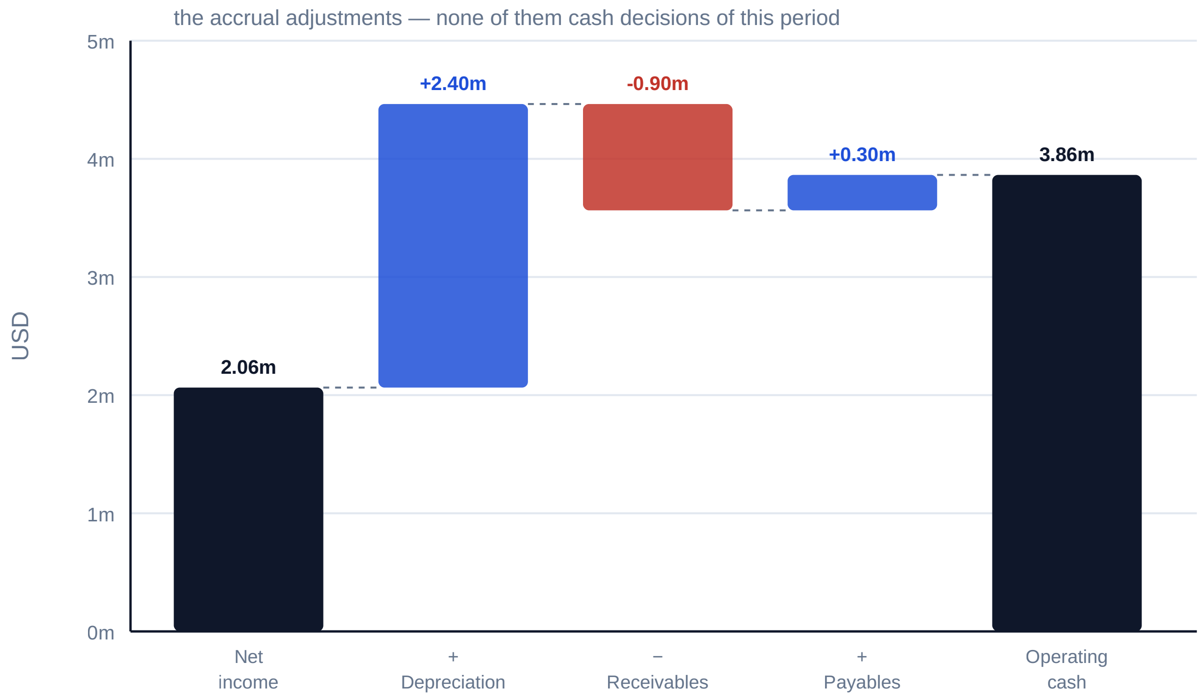

Fig 2.2.1 — Kestrel's accrual-to-cash bridge

KEY TERMS — KA 2.2

Term	Meaning
`EBITDA`	Earnings before interest, tax, depreciation and amortisation.
`EBIT`	Operating profit after depreciation; performance including capital consumption.
Depreciation	(cost − residual)/useful life; a non-cash charge for asset consumption.
Indirect method	Deriving operating cash flow from profit by undoing accruals.
Working-capital movement	Change in receivables, inventory and payables; absorbs or releases cash.
Principal versus interest	Principal is a balance-sheet movement; only interest is an expense.

SAMPLE MCQS — KA 2.2

MCQ 2.2-A `[2.2.1 · Application]` Revenue 12,000,000; cash operating costs 4,500,000; depreciation 2,400,000; interest 2,520,000; tax 20 %. Net income is: - A. USD 2,580,000 - B. USD 2,064,000 - C. USD 4,080,000 - D. USD 7,500,000

Rationale: `PBT 2,580,000 × (1 − 0.20) = 2,064,000`. A is PBT before tax; C taxes `EBIT` instead of PBT; D is `EBITDA`.

MCQ 2.2-B `[2.2.4 · Application]` Net income 2,064,000; depreciation 2,400,000; receivables +900,000; payables +300,000. Operating cash flow is: - A. USD 4,464,000 - B. USD 3,864,000 - C. USD 3,264,000 - D. USD 2,064,000

Rationale: `2,064,000 + 2,400,000 − 900,000 + 300,000 = 3,864,000`. A omits the working-capital movements; C subtracts the payables increase instead of adding it (supplier credit is a cash source); D stops at profit.

MCQ 2.2-C [2.2.2 · Analysis] Kestrel's annual instalment is 5,009,635, of which 2,520,000 is interest. The income statement expense and the balance-sheet effect are: - A. expense 5,009,635; debt falls by 5,009,635 - B. expense 2,520,000; debt falls by 2,489,635 - C. expense 2,489,635; debt falls by 2,520,000 - D. expense 5,009,635; debt unchanged

Rationale: Only interest is an expense; principal is a balance-sheet movement. A expenses the whole instalment (the classic error), C reverses the two components, D ignores repayment.

MCQ 2.2-D [2.2.3 · Recall] Repayment of debt principal appears in the cash-flow statement under: - A. operating activities - B. investing activities - C. financing activities - D. it does not appear, being a balance-sheet movement

Rationale: Principal flows are financing. D confuses the income statement's silence on principal with absence from the cash-flow statement, where every cash movement appears.

SELF-CHECK — KA 2.2

1. Why is *EBITDA* USD 7,500,000 while net income is USD 2,064,000? — Depreciation, interest and tax lie between them; on a capital-intensive levered project that gap is structural.
2. State the accrual-to-cash bridge for Kestrel. — 2,064,000 + 2,400,000 depreciation − 900,000 receivables + 300,000 payables = 3,864,000.
3. What obligations does operating cash flow not cover? — Debt principal and capex; hence Domain 10's coverage ratios.

KNOWLEDGE AREA 2.3

Project-relevant treatments

Topics: 2.3.1 working capital · 2.3.2 revenue and cost recognition · 2.3.3 capital versus operating expenditure · 2.3.4 provisions and contingencies.

2.3.1 Working capital and why lenders care

Definition. Working capital is the cash tied up in the trading cycle — receivables plus inventory less payables. Growth in it **consumes** cash; reduction **releases** it. For a project, the dominant driver is the gap between doing work and being paid for it, which is why payment terms are a financing decision (Domain 1's cash discipline; Domain 14's drawdowns).

WORKED EXAMPLE

Worked example 2.3.1 — what working capital does to a coverage ratio.

1. **Setup.** Kestrel's `EBITDA` USD 7,500,000; tax USD 516,000; the net working-capital increase of USD 600,000 from KA 2.2.4. Annual debt service USD 5,009,635 (Domain 3). Compute the `DSCR` — Domain 10's central ratio — first ignoring working capital, then including it.
2. **Formula.** `CFADS` = `EBITDA` − tax − Δworking capital; `DSCR` = `CFADS` ÷ debt service.
3. **Substitution.** Ignoring WC: `7,500,000 − 516,000 = 6,984,000`; `÷ 5,009,635`. Including WC: `6,984,000 − 600,000 = 6,384,000`; `÷ 5,009,635`.
4. **Result.** `DSCR` **1.39** ignoring working capital; **1.27** including it — a fall of **0.12** from one balance-sheet movement.
5. **Interpretation.** Twelve basis points of coverage may decide whether a covenant holds (typical senior covenants sit at 1.20–1.30, so this project is comfortable at 1.39 and much closer to the line at 1.27). The lesson is definitional rather than arithmetical: `CFADS` **is a defined term in the finance documents, and whether it is struck before or after working-capital movements changes the ratio it produces.** A leader who quotes a `DSCR` without knowing its `CFADS` definition is quoting an opinion. Domain 10 builds the full machinery; Domain 13's model audit checks that the model implements the documented definition.

2.3.2 Revenue and cost recognition

The principle. Revenue is recognised as performance occurs — when control of the promised goods or services transfers to the customer — not when the invoice is raised or the cash arrives. For long-duration work this means recognising **over time** as the obligation is satisfied, which requires a defensible measure of progress. IFRS 15 is the reference framework, described here in principle only.

Measuring progress uses either **input** methods (costs incurred relative to total expected costs — the familiar percentage-of-completion approach) or **output** methods (units delivered, milestones surveyed). The choice matters and is not free: an input measure can be distorted by inefficiency (spending more "earns" more revenue) and an output measure by lumpy deliverables. This is the same measurement-integrity problem the delivery book handles as earned value (PML-AI, Domain 7, KA 7.3.1) — and the two disciplines should use the same progress evidence, because a project claiming 60 % complete for revenue and 48 % for earned value has a story to explain.

Costs follow performance symmetrically: costs of fulfilling the obligation are expensed as incurred, certain fulfilment costs may be capitalised, and **expected losses are recognised immediately and in full** the moment a contract is expected to be loss-making — a deliberate asymmetry that stops a known loss being spread quietly across future periods.

2.3.3 Capital versus operating expenditure

The distinction. Capital expenditure (capex) creates or enhances an asset with benefits beyond the current period; it is capitalised to the balance sheet and depreciated over its useful life. **Operating expenditure (opex)** is consumed in the period and expensed immediately. The same cash outflow, classified differently, produces very different profit.

WORKED EXAMPLE

Worked example 2.3.3 — the same USD 1,200,000, two treatments.

1. **Setup.** Kestrel spends USD 1,200,000 on a control-system overhaul. If it extends the asset's capability it is capex, depreciated over 10 years; if it restores existing performance it is maintenance opex.
2. **Formula.** Capex year-one charge = spend ÷ useful life. Opex year-one charge = full spend.
3. **Substitution.** Capex: `1,200,000/10 = 120,000`. Opex: `1,200,000`.
4. **Result.** Year-one profit is **USD 1,080,000 lower** under the opex treatment. **Cash is identical in both cases** — USD 1,200,000 leaves the account either way.
5. **Interpretation.** This is the clearest demonstration in the domain that accounting choice moves reported performance without moving economics, and it cuts both ways. Capitalising flatters current profit and burdens future periods with depreciation; it also inflates the asset base against which returns are measured. Because the classification is judgment applied to facts — does this enhance or merely restore? — it is a standing area of audit attention, and a leader should expect to justify it rather than assert it. Note the covenant consequence: profit-based covenants respond to the choice while cash-based ones (`DSCR`) do not, which is part of why lenders prefer cash-based tests (Domain 10).

2.3.4 Provisions and contingencies

Definitions. A **provision** is a liability of uncertain timing or amount, recognised when a present obligation exists from a past event, settlement is probable, and a reliable estimate can be made (IAS 37 is the reference framework, in principle). A **contingent liability** fails one of those tests — typically because settlement is only possible rather than probable — and is **disclosed, not recognised**. A **contingent asset** is treated still more cautiously: disclosed when probable, recognised only when virtually certain, an asymmetry that deliberately resists optimism.

Project-relevant instances: **decommissioning and site-restoration obligations** (recognised at the present value of the expected cost when the obligation arises — often at construction, decades before the spend, and a major balance-sheet item for energy and resources projects); **warranty and defect obligations**; **onerous contracts** (the loss recognised in full, per 2.3.2); and **disputed claims**, where the provision/disclosure boundary is genuinely contested and legal advice governs. The leader's discipline here is narrow and important: **a provision is not a reserve to be created in good years and released in bad ones** — that is earnings management, and the recognition tests exist to prevent it.

AI in this KA

These four treatments are exactly where AI assistance is least reliable, because each turns on judgment applied to specific facts: is this capex or maintenance, is settlement probable, does control transfer over time. A model will produce a confident answer that reads like a conclusion and is actually a summary of how similar questions are usually answered. The governed position: use AI to marshal the facts, locate the relevant framework and draft the argument on both sides; never to reach the conclusion. Conclusions on recognition and measurement belong to the entity's finance function and auditors, and jurisdiction-specific treatment to qualified advisers — the boundary this book applies to itself (Domain 1, KA 1.3.1).

KEY TERMS — KA 2.3

Term	Meaning
Working capital	Receivables + inventory − payables; growth consumes cash.
CFADS	Cash flow available for debt service — a defined term whose definition changes the ratio.
Input / output progress measures	Cost-based vs delivery-based measures of performance to date.
Onerous contract	An expected-loss contract; the loss is recognised immediately and in full.
Capex / opex	Capitalised and depreciated vs expensed in the period; same cash, different profit.
Provision / contingent liability	Recognised (present obligation, probable, estimable) vs disclosed.

SAMPLE MCQS — KA 2.3

MCQ 2.3-A [2.3.1 · Application] EBITDA 7,500,000; tax 516,000; working-capital increase 600,000; debt service 5,009,635. DSCR including the working-capital movement is: - A. 1.39 - B. 1.27 - C. 1.50 - D. 1.20

Rationale: CFADS = 7,500,000 − 516,000 − 600,000 = 6,384,000; ÷ 5,009,635 = 1.27. A excludes the working-capital movement (the definitional point of the example); C ignores tax as well; D is a typical covenant threshold, not this calculation.

MCQ 2.3-B [2.3.3 · Application] USD 1,200,000 is spent on an overhaul. Capitalised over 10 years versus expensed immediately, the year-one differences are: - A. profit lower by 1,200,000 if capitalised; cash differs - B. profit lower by 1,080,000 if expensed; cash identical - C. profit identical; cash lower by 1,080,000 if expensed - D. profit lower by 120,000 if expensed; cash identical

Rationale: Expensing charges 1,200,000 against capitalising's 120,000 — a 1,080,000 difference — and the cash outflow is the same either way. A inverts which treatment charges more; C confuses which statement is affected; D states the capitalised charge as the difference.

MCQ 2.3-C [2.3.4 · Analysis] A contractor faces a claim it considers possible but not probable, with a reliably estimable amount. The treatment is: - A. recognise a provision, since the amount is estimable - B. disclose as a contingent liability; recognition requires probable settlement - C. no disclosure, since settlement is not probable - D. recognise a contingent asset

Rationale: Estimability alone is insufficient — probability of settlement is the failed test, so disclosure rather than recognition (2.3.4). C hides information users need; D reverses the direction.

MCQ 2.3-D [2.3.2 · Analysis] A contractor recognises revenue on 60 % completion using a cost-input measure while its earned-value system reports 48 % complete. The soundest reading is: - A. no issue — the two systems serve different purposes - B. the divergence needs explaining: an input measure inflated by inefficiency can recognise revenue ahead of performance - C. the earned-value figure must be wrong - D. revenue should be restated to 48 % automatically

Rationale: Cost-input measures reward spending, so inefficiency can raise apparent progress (2.3.2) — the gap is a signal to investigate, not to dismiss (A) or resolve by assumption (C, D).

SELF-CHECK — KA 2.3

1. Why can two projects with identical economics report different *DSCR*s? — Because CFADS is a defined term; its definition (e.g. before or after working capital) changes the ratio.

2. What does the capex/opex choice change, and what does it not? — Reported profit and the asset base; not cash.
3. When is a provision recognised rather than disclosed? — Present obligation from a past event, probable settlement, reliable estimate.

KNOWLEDGE AREA 2.4

Ratio interpretation and project interfaces

Topics: 2.4.1 the ratio families · 2.4.2 interest cover and the leverage view · 2.4.3 project systems and corporate reporting.

2.4.1 The ratio families

Ratios are comparisons, and their only value is in what they are compared against — the same entity over time, or a genuine peer on the same policies (KA 2.1.2's warning). Four families:

Family	Asks	Examples
Liquidity	Can it pay what falls due soon?	Current ratio, quick ratio
Leverage	How much of the funding is debt?	Debt/equity, debt/EBITDA
Coverage	Can earnings or cash service the debt?	Interest cover, DSCR (Domain 10)
Return	What is earned on the capital used?	Return on equity, return on capital employed

For project finance the **coverage** family dominates, because a ring-fenced SPV's whole credit case is whether its cash flow services its debt (Domain 1, KA 1.1.2).

2.4.2 Interest cover and the leverage view

WORKED EXAMPLE

Worked example 2.4.2 — Kestrel's cover and gearing.

1. **Setup.** EBIT USD 5,100,000; interest USD 2,520,000; senior debt at year end USD 39,510,365; EBITDA USD 7,500,000.
2. **Formula.** Interest cover = EBIT ÷ interest. Debt/EBITDA = debt ÷ EBITDA.
3. **Substitution.** `5,100,000 / 2,520,000`; `39,510,365 / 7,500,000`.
4. **Result.** Interest cover **2.02×**; debt/EBITDA **5.27×**.
5. **Interpretation.** Both are true and they point in different directions, which is the point of reading a set. Cover of 2.02× says earnings comfortably absorb the interest charge; debt of 5.27× EBITDA says the balance sheet is heavily geared — normal for contracted infrastructure and alarming for a merchant business, because what makes high leverage tolerable is **revenue certainty**, not the ratio itself (Domain 7's revenue models; Domain 11's risk allocation). Note too that interest cover is an accrual measure and ignores principal entirely: Kestrel must also find USD 2,489,635 of principal, which is why DSCR (1.27 including working capital, 2.3.1) is the ratio a lender actually covenants.

2.4.3 Project systems and corporate reporting

A project's cost system and the corporate accounts describe the same money in different languages, and the interfaces are where reconciliation errors and disputes live. The four that matter:

- **Commitments versus actuals.** A purchase order commits funds; accounting recognises cost on delivery or performance. Project systems track both; the ledger recognises one (PML-AI, Domain 7, KA 7.2.1).
- **Accruals at period end.** Work received but not invoiced must reach the ledger, or cost performance flatters and then lurches — the same defect PML-AI diagnoses as a stepped `CPI`.
- **Capitalisation boundaries.** Which project costs form part of the asset (and from what date capitalisation begins and ends) determines both the depreciation base and reported profit (2.3.3). Borrowing costs during construction are a specific instance — capitalised into the asset while it is being built, expensed thereafter.
- **Cut-off and period discipline.** A cost is in one period only. Projects run continuously and ledgers close monthly; every reconciliation dispute is ultimately a cut-off question.

The leader's obligation is not to run these reconciliations but to insist they exist and to know which number is being quoted. "The project has spent USD 40 million" can mean committed, incurred, invoiced, paid or capitalised — five different figures, all defensible, differing by material amounts.

KEY TERMS — KA 2.4

Term	Meaning
Interest cover	`EBIT` ÷ interest; an accrual coverage measure that ignores principal.
Debt/`EBITDA`	A leverage measure; tolerable level depends on revenue certainty.
Capitalisation boundary	Which costs and which dates form part of the asset.
Cut-off	Assigning a transaction to exactly one period.
Commitment vs actual vs paid	Distinct measures of "spend" that must never be conflated.

SAMPLE MCQS — KA 2.4

MCQ 2.4-A `[2.4.2 · Application]` `EBIT` 5,100,000 and interest 2,520,000. Interest cover is: - A. 1.27× - B. 2.02× - C. 2.98× - D. 0.49×

Rationale: `5,100,000/2,520,000 = 2.02`. A is the `DSCR` from 2.3.1; C uses `EBITDA` in the numerator; D inverts the ratio.

MCQ 2.4-B `[2.4.2 · Analysis]` A project shows interest cover 2.02× and debt/`EBITDA` 5.27×. The soundest interpretation is: - A. the ratios contradict each other, so one must be wrong - B. earnings service the interest comfortably while leverage is high — tolerable given contracted revenue, but dependent on that certainty - C. the project is over-leveraged regardless of revenue structure - D. interest cover is the only relevant measure for a lender

Rationale: The two measure different things and both are true (2.4.2); what makes high leverage acceptable is revenue certainty. D ignores principal, which is why `DSCR` is covenanted.

MCQ 2.4-C `[2.4.3 · Analysis]` A sponsor asks "how much has the project spent?". The professional response is: - A. quote the paid figure, as it is the most conservative - B. ask which measure is meant — committed, incurred, invoiced, paid or capitalised — since they differ materially and all are defensible - C. quote the committed figure, as it is the most complete - D. quote the capitalised figure, since it appears in the accounts

Rationale: Five defensible figures exist (2.4.3); answering without establishing which one is meant guarantees a later dispute. Each of A, C and D picks one arbitrarily.

SELF-CHECK — KA 2.4

1. Why does project finance emphasise coverage over return ratios? — A ring-fenced SPV's credit case is whether its cash services its debt.
2. What does interest cover ignore? — Principal repayment; hence `DSCR`.
3. Name the five meanings of "spend". — Committed, incurred, invoiced, paid, capitalised.

Advanced topics — Domain 2

2.A.1 Deferred tax, in principle

Accounting profit and taxable profit differ — most commonly because tax depreciation (capital allowances) runs on a different profile from accounting depreciation. Where the difference is **temporary**, deferred tax recognises the future consequence: accelerated tax depreciation reduces tax now and creates a **deferred tax liability** that unwinds as the difference reverses. For project models the practical significance is that **cash tax and accounting tax are different lines with different timing**, and it is cash tax that enters `CFADS` (Domain 6's model, Domain 10's ratios). Modelling accounting tax as if it were cash tax is a standard model-audit finding.

2.A.2 Leases and off-balance-sheet intuitions

Under current thinking a lessee generally recognises a right-of-use asset and a lease liability, so the older intuition that operating leases keep obligations off the balance sheet no longer holds (IFRS 16 is the reference framework, in principle). Two consequences for a finance leader: leverage ratios computed across periods spanning the change are not comparable, and covenant definitions written before it may capture or exclude lease liabilities in ways nobody intended — a live reason to read the definitions in finance documents rather than assume them.

2.A.3 The reviewer's statement eye

Invariants to test on any statement set before relying on it: the balance sheet balances; closing cash on the cash-flow statement equals balance-sheet cash; closing equity equals opening plus profit less distributions plus contributions; depreciation in the cash-flow statement equals the income-statement charge; the movement in debt equals drawings less principal repaid; principal appears in financing and interest in operating (or as disclosed); working-capital movements reconcile to balance-sheet deltas; and any ratio quoted can be recomputed from the face of the statements. A set that fails any of these has an error, an omission, or a policy that needs explaining — and the failure point localises it.

Industry variations — Domain 2

- **Energy and resources.** Decommissioning provisions are first-order balance-sheet items, recognised decades before the cash flows; commodity revenue makes recognition timing and hedge accounting material.
- **Transport concessions and PPPs.** The central question is what the operator actually holds — a financial asset (an unconditional right to cash from the grantor), an intangible right to charge users, or a mixture — and the answer changes the whole statement profile for identical economics. Framework treatment is specialised and advice-led.
- **Water and regulated utilities.** Regulatory asset bases and their depreciation profiles may diverge from statutory accounting; two "asset values" coexist and must never be conflated.
- **Digital infrastructure.** Shorter useful lives and heavy refresh capex make the capex/opex boundary (2.3.3) a recurring judgment with large profit consequences.
- **Construction and EPC.** Over-time revenue recognition and onerous-contract provisions are the operative treatments, and the input-versus-output progress choice (2.3.2) is where accounting and project controls must agree or explain.

Case study — Domain 2: the profitable year that nearly broke a covenant (water)

Situation. Kestrel Water SPC's first operating year closes on the figures above: net income USD 2,064,000, `EBITDA` USD 7,500,000 — a good year by any account. The finance director's draft lender report quotes `DSCR` **1.39** and describes headroom as comfortable against the facility's **1.25** covenant.

The problem. The facility's definition of `CFADS` is struck **after** movements in working capital. Kestrel's receivables had grown USD 900,000 as the offtaker's payment process settled into a slower rhythm than modelled, against a USD 300,000 rise in payables — a net USD 600,000 absorbed. On the documented definition, `CFADS` is USD 6,384,000 and `DSCR` is **1.27**, not 1.39. Headroom against the 1.25 covenant is USD 0.02 of ratio — roughly **USD 100,000 of cash** — not the comfortable margin reported.

What was done. The report was corrected before issue. More consequentially, the near-miss changed three things: collections became a monitored operational metric with a named owner (not a finance afterthought); the model's working-capital assumptions were re-based on nine months of actual collection behaviour rather than the financial-close assumption; and the treasury team sized a working-capital facility as a liquidity buffer. The following year's `DSCR` came in at 1.41 on the documented definition.

What the domain teaches here. The arithmetic here is trivial; the professional content is definitional. A ratio is only as good as the defined term inside it, and the definition lives in the finance documents, not in convention. It also shows the accrual/cash divergence of KA 2.1.1 doing real damage — a genuinely profitable year came within USD 100,000 of a covenant breach because of a balance-sheet movement no one was watching.

Case study B — Domain 2: capitalised into a better-looking year (digital infrastructure)

Situation. A data-centre operator overhauled cooling across three sites for USD 9,000,000, capitalised as an enhancement and depreciated over 12 years — a year-one charge of USD 750,000 rather than USD 9,000,000, lifting reported profit by USD 8,250,000 relative to expensing it.

What happened. The auditors challenged the classification: on inspection, roughly two-thirds of the spend restored the original design capacity rather than extending it, which is maintenance. The restatement moved USD 6,000,000 into operating expense, turned a reported profit into a loss, and breached a profit-based covenant in a holding-company facility. No cash flow changed at any point, and the project's `DSCR` — a cash-based test — was unaffected throughout.

What the domain teaches here. Accounting classification is judgment applied to facts, and it is auditable. The wider lesson is why lenders to projects prefer cash-based covenants: the `DSCR` was indifferent to a classification argument that flipped the profit-based test from comfortable to breached. A leader should know which of their covenants are exposed to accounting judgment and which are not.

Executive perspective — Domain 2

What a project finance director cannot delegate in this domain:

- **The definitions.** `CFADS`, `EBITDA`, "net debt", "spend" — the director asks what the defined term is in the documents before quoting any ratio built on it (Case study A).
- **The accrual/cash bridge.** Being able to explain, unprompted, why profit and cash differ this period — and what is absorbing the difference.
- **The policy exposures.** Which covenants can be moved by an accounting judgment (capex/opex, revenue timing, provisions) and which cannot.
- **Working capital as an operational matter.** Collections and payment terms are cash decisions with covenant consequences, owned by named people, not finance hygiene.
- **The interface discipline.** Insisting project and ledger reconcile, and never letting the five meanings of "spend" circulate interchangeably.

Calculation exercises — Domain 2

> **Exercise 2.1** Revenue 15,000,000; cash operating costs 5,800,000; asset 75,000,000 depreciated over 30 years; interest 2,100,000; tax 20 %. Build the income statement to net income. Solution. `EBITDA` **9,200,000**; depreciation `75,000,000/30 =` **2,500,000**; `EBIT` **6,700,000**; PBT **4,600,000**; tax **920,000**; net income **3,680,000**. Common error: taxing `EBIT` rather than PBT (giving 1,340,000 and overstating tax by 420,000).

> **Exercise 2.2** From Exercise 2.1, receivables rose 1,100,000, inventory rose 200,000 and payables rose 450,000. Compute operating cash flow. Solution. `3,680,000 + 2,500,000 − 1,100,000 − 200,000 + 450,000 =` **USD 5,330,000**. Common error: signing payables negative — supplier credit is a cash source.

> **Exercise 2.3** Using Exercise 2.1–2.2 and annual debt service of 4,400,000, compute `DSCR` both before and after working-capital movements (`CFADS` = `EBITDA` − tax [− ΔWC]). Solution. Before: `(9,200,000 − 920,000)/4,400,000 = 8,280,000/4,400,000 =` **1.88**. After: ΔWC `1,100,000 + 200,000 − 450,000 = 850,000`; `7,430,000/4,400,000 =` **1.69**. Common error: using net income instead of `EBITDA` in `CFADS` (double-counting interest, which debt service already includes).

> **Exercise 2.4** USD 2,400,000 is spent on plant modification, capitalised over 8 years versus expensed. State the year-one profit difference and the cash difference. Solution. Capitalised charge `2,400,000/8 =` **300,000**; expensed **2,400,000**; profit difference **USD 2,100,000**; cash difference **nil**. Common error: assuming a cash difference because the profit effect is large.

Practitioner's toolkit — Domain 2

Adoption-ready artefacts; adapt headings to your organisation, then keep them stable.

Toolkit 2.T.1 — Defined-terms sheet (one per financing)

For each term used in a covenant or report — `CFADS`, `EBITDA`, net debt, distributable cash, project costs — record: the **document and clause** defining it, the definition in plain words, what is included and excluded (working capital? cash tax or accrued? maintenance capex?), the model line implementing it, and the person who confirmed the match. A ratio quoted without a row here is not reportable (Case study A).

Toolkit 2.T.2 — Accrual-to-cash bridge (standing monthly schedule)

Net income · + depreciation and amortisation · + other non-cash charges (provisions) · − Δreceivables · − Δinventory · + Δpayables · = operating cash flow · − capex · − principal · − distributions · = movement in cash, **tied to the cash balance**. Rule: it is published even in good months, because its purpose is to make divergence visible before it is material.

Toolkit 2.T.3 — Statement-integrity checklist

- [ ] Balance sheet balances; closing cash ties to the cash-flow statement.
- [ ] Closing equity = opening + profit − distributions + contributions.
- [ ] Depreciation in the cash-flow statement equals the income-statement charge.
- [ ] Debt movement = drawings − principal repaid; only interest is expensed.
- [ ] Working-capital movements reconcile to balance-sheet deltas.
- [ ] Every quoted ratio recomputable from the face of the statements, on its defined terms.
- [ ] Capitalisation boundary and any policy change disclosed and explained.

Exam preparation — Domain 2

The traps. Expensing the whole debt instalment instead of interest only (MCQ 2.2-C) · taxing `EBIT` rather than PBT (Exercise 2.1) · signing payables the wrong way in the cash bridge (Exercise 2.2) · using net income in `CFADS` and double-counting interest (Exercise 2.3) · assuming a cash difference from a capex/opex choice (Exercise 2.4) · quoting a `DSCR` without its `CFADS` definition (2.3.1) · recognising a provision on estimability alone without probability (2.3.4) · treating accounting tax as cash tax (2.A.1) · comparing leverage across a period spanning a policy change (2.A.2).

Reflection questions. 1. For your current financing: what exactly does CFADS include, and where is that written down? 2. Which of your covenants could be moved by an accounting judgment rather than a change in economics — and who reviews those judgments? 3. When someone last told you what a project had "spent", which of the five measures was it — and did you ask?

Domain 2 summary

Accrual accounting records effects when they occur and cash accounting records money when it moves; both matter, because covenants are written on each and they diverge in ways that are information rather than noise. Recognition and measurement are governed by tests, and the policies chosen shape reported figures without touching economics — which is why statements are read sceptically and why the three of them, locked together by articulation, are stronger evidence than any one alone. Kestrel's first year demonstrates the machinery end to end: EBITDA 7,500,000 descending through depreciation, interest and tax to net income 2,064,000, then bridged back to operating cash flow of 3,864,000 by adding non-cash depreciation and deducting the 600,000 that working capital absorbed — a bridge that also explains why only interest is an expense while the 2,489,635 of principal is a balance-sheet movement. The project-relevant treatments each carry a professional edge: working capital moved Kestrel's DSCR from 1.39 to 1.27 on the documented CFADS definition, within USD 100,000 of a covenant; the capex/opex choice moved a year's profit by 1,080,000 on 1,200,000 of spend while cash was unchanged; revenue recognition follows performance and must agree with the progress evidence delivery uses; and provisions are recognised on tests, never created as reserves. Ratios are comparisons whose value lies in what they are compared against, coverage dominates in project finance, and the five meanings of "spend" must never circulate interchangeably. Domain 3 supplies the discounting these statements are valued with; Domain 6 turns them into a model; Domain 10 turns CFADS into the covenants a lender actually enforces.

03

DOMAIN 3

Time Value of Money and Financial Mathematics (quantitative flagship)

Group: Foundations (Domain 3 of 4 in Part One). **Target:** ~72 pages. **Binds to:** the PCI Book Pattern Specification and the shared registries (`docs/books/registries/`). This domain is the definitive home of the discounting symbols — `PV(x)`, `FV(x)`, `DF(t)`, `r`, `n`, `i_nom`, `i_real`, `π`, annuity payment `A` — every later domain restates them from here. Because this book is discounting-heavy, `PV` written bare is reserved for Earned Value contexts only; present value is always written `PV(x)` or in words. British English; USD (+SAR where useful, indicative `USD 1 ≈ SAR 3.75`).

Why this domain exists

Every judgment a project finance leader makes — whether an investment is worth making (Domain 4), how much debt a project can carry (Domain 10), what a concession's payment stream is worth (Domain 7), whether a delay destroys value (Domain 8) — reduces at some point to one question: what is money at one date worth at another date? Time value of money (TVM) is the machinery that answers it. This domain builds that machinery from first principles: interest and growth (KA 3.1), the annuity family and loan schedules that debt structures are made of (KA 3.2), and the adjustments — inflation, escalation and currency — that separate a defensible model from a spreadsheet accident (KA 3.3). Nothing in this domain is optional equipment: the financial models of Domain 6 are, in the end, disciplined arrangements of exactly these calculations, and the coverage ratios lenders live by (Domain 10) are discounted cash flows wearing covenants.

Learning objectives. After this domain a candidate can: distinguish simple and compound interest and compute either; move any cash amount across time with `FV(x)` and `PV(x)` and read a discount-factor table; value level streams with annuity and perpetuity formulae; build and check a loan schedule (annuity, level-principal and bullet); convert between nominal, periodic and effective rates; keep nominal and real quantities consistent using the Fisher relation; escalate costs and revenues defensibly; state how currency and forward rates enter project cash flows; and apply the governed-AI rule to any machine-produced calculation of these kinds.

The master financing. One fictional project runs through this domain and returns in Domains 4, 6 and 10. **Kestrel Water SPC** is a special-purpose company developing a seawater desalination plant. Its senior lenders offer a **USD 42,000,000** loan repayable over **12 years at 6.0 %** annual interest; the offtaker will pay an **availability payment of USD 5,600,000 per year for 25 years**; Kestrel's board evaluates offers at a discount rate of **8.0 %**. All three numbers are used repeatedly below — by the end of the domain the reader can price every side of Kestrel's deal.

KNOWLEDGE AREA 3.1

Interest and value

Topics: 3.1.1 simple and compound interest · 3.1.2 present and future value · 3.1.3 discount factors.

3.1.1 Simple and compound interest

Definitions. Interest is the price of money over time. Under **simple interest** the charge accrues on the original principal only:

```
FV(x) = x × (1 + r × n)        — simple interest
```

Under **compound interest** each period's interest joins the principal and itself earns interest:

```
FV(x) = x × (1 + r)^n          — compound interest
```

where `x` is the amount today (currency), `r` the interest rate per period (ratio) and `n` the number of periods (count). Project finance is a compound-interest world: loans, discounting, escalation and returns all compound. Simple interest survives only in narrow corners — short-duration instruments, some penalty-interest clauses, and certain Islamic-finance structures that avoid interest altogether and price time differently (Domain 9, KA 9.3).

The principle. Compounding is growth on growth. Its effect looks negligible over one or two periods and decisive over ten: the gap between the two formulae widens geometrically, which is why long-life infrastructure — concessions of 25 to 40 years — is acutely sensitive to the rate used.

WORKED EXAMPLE

Worked example 3.1.1 — the same deposit, two rules.

1. **Setup.** USD 100,000 is placed for 3 years at 8.0 % per year. Compare simple and compound growth.
2. **Formula.** Simple: `FV(x) = x(1 + rn)`. Compound: `FV(x) = x(1 + r)^n`, with `x` = 100,000, `r` = 0.08, `n` = 3.
3. **Substitution.** Simple: `100,000 × (1 + 0.08 × 3) = 100,000 × 1.24`. Compound: $100{,}000 \times 1.08^3 = 100{,}000 \times 1.259712$.
4. **Result.** Simple: **USD 124,000**. Compound: **USD 125,971** (125,971.20 at full precision).
5. **Interpretation.** Three years of compounding adds USD 1,971 over simple interest — under 2 %. Over a 25-year concession life the same 8 % compounds to $1.08^{25} \approx 6.85$ times the principal, while simple interest reaches only 3.0 times. The professional habit this example teaches: never extrapolate a short-horizon intuition to a long-horizon contract.

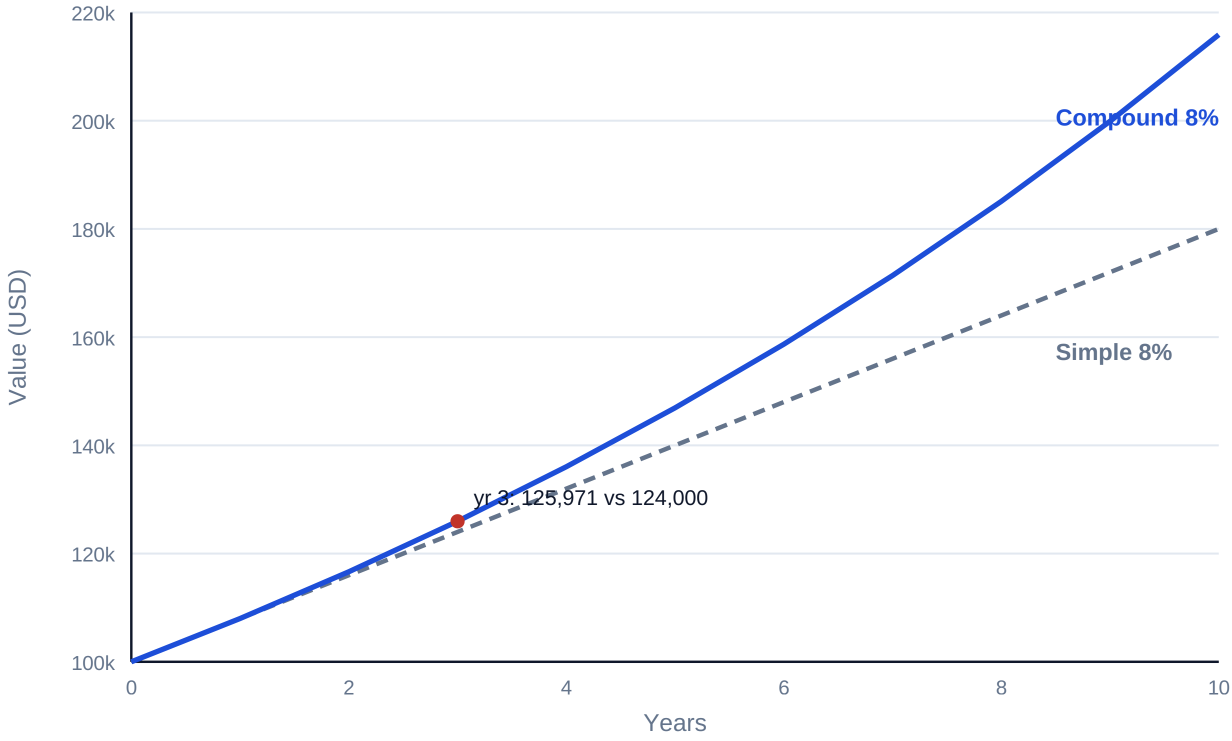

Fig 3.1.1 — Simple versus compound growth of USD 100,000 at 8 %

3.1.2 Present and future value

Definitions. Compounding runs both ways. **Future value** carries an amount forward; **present value** brings a future amount back:

```
FV(x) = x × (1 + r)^n          PV(x) = x / (1 + r)^n
```

`PV(x)` answers the only question a bid evaluation, a settlement offer or a buy-out negotiation really asks: what is that future money worth today, at the return I could otherwise earn? The rate `r` used in this role is called the **discount rate**; choosing it is a Domain 4 judgment (and, for a levered project, a Domain 9/10 conversation) — this KA takes it as given.

WORKED EXAMPLE

Worked example 3.1.2 — pricing a deferred receipt.

1. **Setup.** Kestrel will receive a USD 500,000 connection rebate exactly 5 years from now. The board discounts at 7.0 % for this risk. What is the rebate worth today?
2. **Formula.** `PV(x) = x / (1 + r)^n`, with `x` = 500,000, `r` = 0.07, `n` = 5.
3. **Substitution.** `PV(x) = 500,000 /` 1.07^5 `= 500,000 / 1.402552`.
4. **Result. USD 356,493** (356,493.09 at full precision; indicatively ≈ SAR 1,336,849 at `USD 1 ≈ SAR 3.75`).
5. **Interpretation.** A five-year wait at 7 % costs the rebate just under 29 % of its face value. If a counterparty offered USD 380,000 cash today to extinguish the obligation, Kestrel should accept — and if offered USD 330,000, decline. `PV(x)` turns "later" into a number that can be compared, which is the whole trade of this profession.

Common pitfall. Discounting with simple interest — `500,000 / (1 + 0.07 × 5) = 370,370` — overstates the value by USD 13,877 here. The error grows with horizon and rate; audit any model whose discount factors were "simplified".

3.1.3 Discount factors

Definition. The **discount factor** for period `t` is the present value of one currency unit received at `t`:

```
DF(t) = 1 / (1 + r)^t          so that   PV(x at t) = x × DF(t)
```

Discount factors are how models industrialise discounting: compute the factor row once, multiply every cash-flow row by it (Domain 6, KA 6.1). They also make review fast — an experienced reviewer reads a factor table the way a scheduler reads float.

At `r` = 10 %:

t (years)	1	2	3	4	5
`DF(t)`	0.9091	0.8264	0.7513	0.6830	0.6209

Two sanity rules a reviewer applies on sight: factors must **decline monotonically**, and each factor must equal the previous factor divided by `(1 + r)`. A factor table that violates either has a hard-coded cell or a broken formula — a Domain 6 model-check classic (KA 6.4).

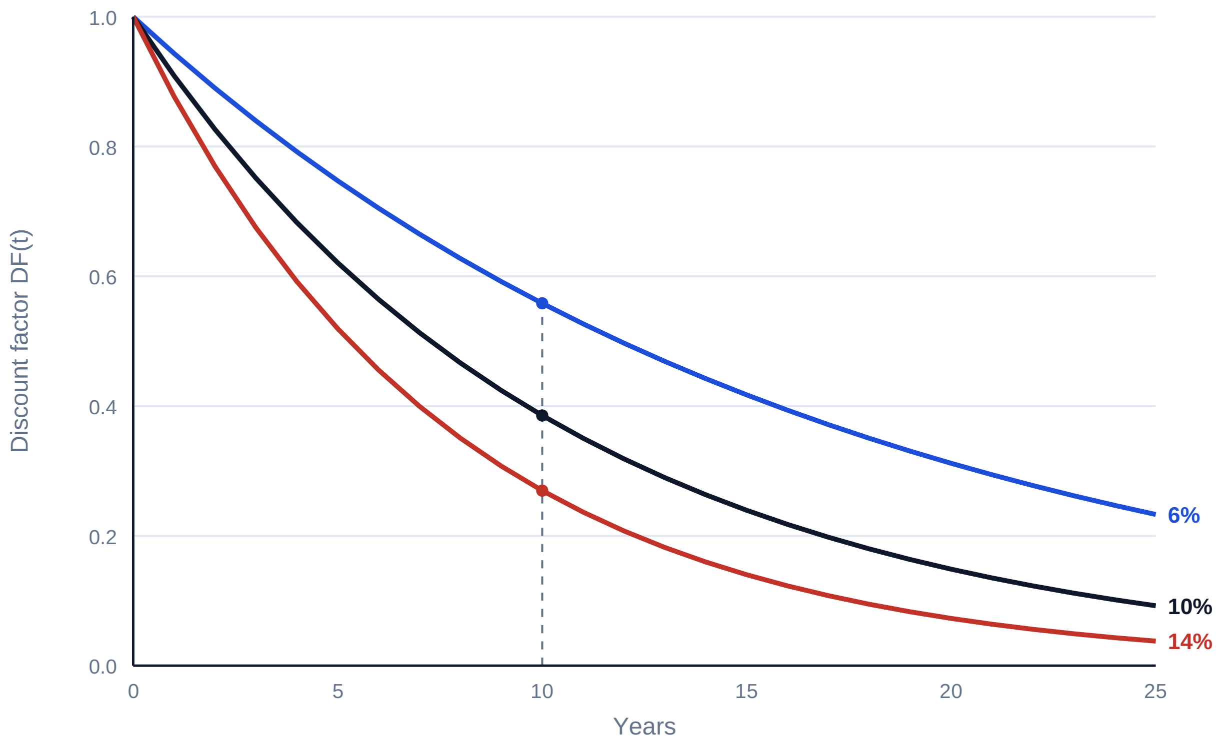

Fig 3.1.2 — Discount-factor curves at 6 %, 10 % and 14 %

Rounding discipline. Factors are displayed to four decimals but **calculations use full precision**. Multiplying a USD 900,000,000 programme cash flow by a four-decimal factor can move results by tens of thousands; the display is for the reader, never for the arithmetic (see the registry's decimal-arithmetic rule).

AI in this KA

Spreadsheet copilots will happily draft TVM formulae, and large-language-model assistants will "explain" a discount factor with perfect fluency and an inverted exponent. The governed position (the PCI principle: **AI proposes; the professional verifies, decides and remains accountable**) is mechanical here because verification is cheap: recompute one factor by hand, check monotonicity, check `DF(1) × (1+r) = 1`. An AI-drafted factor row that no human recomputed is not a calculation; it is a claim.

KEY TERMS — KA 3.1

Term	Meaning
Simple interest	Interest accruing on original principal only: `FV(x) = x(1 + rn)`.
Compound interest	Interest accruing on principal and accumulated interest: `FV(x) = x(1 + r)^n`.
Present value `PV(x)`	Today's worth of a future amount at discount rate `r`.
Future value `FV(x)`	The compounded worth of an amount at a future date.
Discount rate	The rate `r` used to move value across time; the required return for the risk.
Discount factor `DF(t)`	`1/(1+r)^t`; the PV of one unit received at `t`.

SAMPLE MCQS — KA 3.1

MCQ 3.1-A `[3.1.2 · Application]` A payment of USD 500,000 is due in 5 years; the discount rate is 7 %. Its present value is closest to: - A. USD 370,370 - B. USD 356,493 - C. USD 381,448 - D. USD 701,276

Rationale: $500{,}000 / 1.07^5 = 356{,}493$. A discounts with simple interest `(1 + 0.35)`; C uses four years instead of five; D compounds forward instead of discounting back.

MCQ 3.1-B `[3.1.3 · Application]` At a 10 % discount rate, the discount factor for year 3 is: - A. 0.7000 - B. 0.8264 - C. 0.7513 - D. 0.6830

Rationale: $1/1.10^3 = 0.7513$. A subtracts 10 % three times (simple); B is the year-2 factor; D is the year-4 factor — the two most common off-by-one-period errors.

MCQ 3.1-C `[3.1.1 · Analysis]` A lender quotes 25-year money and a borrower's analyst tests affordability using simple interest "as a conservative shortcut". The analyst's error is that: - A. simple interest overstates the debt cost, so the test is merely too strict - B. compound growth exceeds simple growth over long horizons, so the shortcut materially understates the true accumulation - C. the two methods converge over long horizons, so the shortcut is harmless - D. simple interest cannot be computed for horizons beyond ten years

Rationale: Compounding produces growth on growth: at 8 % over 25 years the compound multiple is ≈ 6.85 versus 3.0 simple — the shortcut is anti-conservative, the opposite of the analyst's claim (so A is wrong); the methods diverge rather than converge (C); D is arithmetic nonsense.

MCQ 3.1-F `[3.1.2 · Application]` USD 250,000 is invested at 7 % compound for 6 years. Its future value is closest to: - A. USD 355,000 - B. USD 350,638 - C. USD 375,183 - D. USD 166,586

Rationale: $250{,}000 \times 1.07^6 = 375{,}183$. A is simple interest (`1 + 0.07 × 6`); B compounds only 5 years; D divides instead of multiplying — discounting when the question asks for growth.

MCQ 3.1-D `[3.1.3 · Recall]` Which statement about discount factors is an invariant a reviewer can test without knowing the project? - A. `DF(t)` rises when cash flows are contracted - B. `DF(t+1) = DF(t) / (1 + r)` for every `t` - C. `DF(t)` equals `1 − r × t` - D. factors below 0.5 indicate an error

Rationale: Each factor is the prior factor discounted one more period — B holds for any `r`. A confuses risk of flows with the factor row; C is the simple-interest approximation; D is false — any long horizon at a normal rate passes 0.5 (see Fig 3.1.2).

MCQ 3.1-E `[3.1.2 · Application]` At 9 %, which is worth more today: USD 400,000 in 2 years, or USD 470,000 in 4 years? - A. the 470,000 — larger amounts always win - B. the 400,000: PV 336,672 vs 332,960 - C. the 470,000: PV 395,590 vs 336,672 - D. they are equal at 9 %

Rationale: $400{,}000/1.09^2 = 336{,}672$; $470{,}000/1.09^4 = 332{,}960$ — the earlier, smaller amount wins by USD 3,712. A ignores discounting entirely; C discounts the 470,000 across only two periods instead of four; D asserts an equality the arithmetic denies.

SELF-CHECK — KA 3.1

1. Why must discount factors decline monotonically? — Because `(1+r)^t` grows with `t` for any positive `r`; a rising factor implies a negative rate or a broken formula.
2. Your model shows *`DF(4) = 0.6830`* and *`DF(5) = 0.6209`* at 10 %. One cell check confirms both. Which? — `0.6830 / 1.10 = 0.6209`: each factor is the prior factor divided by `(1+r)`.
3. A colleague says "8 % for three years is 24 %". — Only under simple interest; compounded it is $1.08^3 - 1 = 25.97\ \%$, and the gap widens every further year.

KNOWLEDGE AREA 3.2

Annuities and loan schedules

Topics: 3.2.1 annuities and perpetuities · 3.2.2 loan schedules · 3.2.3 compounding frequency and effective rates.

3.2.1 Annuities and perpetuities

Definitions. An **annuity** is a level stream — the same amount each period for `n` periods. Project finance is built from annuities and near-annuities: availability payments, capacity charges, lease rentals, O&M fees, debt service. The present value of an ordinary annuity (payments at period-ends) of amount `A`:

```
PV(annuity) = A × AF(r, n)      where   AF(r, n) = (1 − (1 + r)^−n) / r
```

`AF(r, n)` is the **annuity factor** — the sum of the discount factors `DF(1) … DF(n)`. A **perpetuity** is the limiting case as `n → ∞`:

```
PV(perpetuity) = A / r
```

WORKED EXAMPLE

Worked example 3.2.1 — valuing Kestrel's availability stream.

1. **Setup.** Kestrel's offtaker will pay USD 3,000,000 per year for 20 years under an early tariff proposal, paid annually in arrears. The board discounts at 9.0 %. Value the stream.
2. **Formula.** `PV = A × AF(r, n)`, `AF(r, n) = (1 − (1+r)^−n)/r`, with `A` = 3,000,000, `r` = 0.09, `n` = 20.
3. **Substitution.** $AF(0.09, 20) = (1 - 1.09^{-20})/0.09 = 9.128546$; `PV = 3,000,000 × 9.128546`.
4. **Result. USD 27,385,637** (using the full-precision factor).
5. **Interpretation.** Twenty years of payments are worth barely nine years' worth of face value — discounting at 9 % has priced away more than half the nominal USD 60,000,000 total. Sensitivity: at a 10 % rate the factor falls to 8.513564 and the value to USD 25,540,691 — a 6.7 % value loss for one point of rate, the shape of sensitivity every concession bidder lives with (Domain 7, KA 7.4).

Common pitfall — the rounded factor. Using a four-decimal factor (9.1285) gives USD 27,385,500 — USD 137 adrift here, but scale the stream by 100× and the drift is five figures. Factors are display; arithmetic is full precision (KA 3.1.3).

Timing variants. An **annuity-due** (payments in advance, common in leases and some availability regimes) is worth `(1 + r)` times the ordinary annuity; a **deferred annuity** starts after a gap and is discounted twice — as a stream, then back across the deferral. Misreading the payment convention in a concession agreement misprices the whole stream by one period's discounting — check the contract, not the habit (Domain 12, KA 12.2).

WORKED EXAMPLE

Worked example 3.2.1b — the lease quoted in advance.

1. **Setup.** Kestrel's operations base is leased at USD 500,000 per year for 5 years, payable **in advance**. Discount rate 8.0 %. Value the obligation.
2. **Formula.** `PV(due) = A × AF(r, n) × (1 + r)`, with `A` = 500,000, `r` = 0.08, `n` = 5.
3. **Substitution.** `AF(0.08, 5) = 3.992710`; ordinary value `500,000 × 3.992710 = 1,996,355`; due value `1,996,355 × 1.08`.
4. **Result. USD 2,156,063** (2,156,063.42). The in-advance convention adds USD 159,708 — exactly one period's discounting on the whole stream.
5. **Interpretation.** Eight per cent of the stream's value hangs on one word in the lease. Reading "payable annually in advance" as an ordinary annuity is the commonest annuity mistake in diligence reviews (Domain 13, KA 13.1) — and it always flatters the tenant's model.

WORKED EXAMPLE

Worked example 3.2.1c — the deferred availability stream.

1. **Setup.** A grantor variant offers Kestrel the same USD 5,600,000 × 25-year availability supplement, but with the **first payment at the end of year 4** (a three-year deferral while the plant ramps up). Discount rate 8.0 %.
2. **Formula.** `PV = A × AF(r, n) / (1 + r)^d`, `d` = 3 (the stream's own valuation point sits at the end of year 3, one period before its first payment).
3. **Substitution.** `5,600,000 × 10.674776 = 59,778,747` (the stream valued at end-year 3); $59{,}778{,}747 / 1.08^3 = 59{,}778{,}747 / 1.259712$.
4. **Result. USD 47,454,296**.
5. **Interpretation.** The three-year wait costs USD 12.3 million — a fifth of the stream's value — which is precisely what the grantor saves by deferring. Any negotiation over payment timing is a negotiation over value at compound rates; Case study A prices the whole family of such trades.

3.2.2 Loan schedules

The three canonical shapes. A loan of principal `P` over `n` periods at rate `r` can be scheduled three ways, and a project finance leader must read all three on sight:

Shape	Rule	Debt-service profile	Where seen
Annuity (equal instalment)	Payment `A = P × r / (1 − (1+r)^−n)` each period	Level total; interest share falls, principal share rises	Most project term loans; mortgages
Level principal	Principal `P/n` each period + interest on balance	Front-loaded total, declining	Some ECA and development-bank facilities (Domain 9)
Bullet	Interest only; principal entire at maturity	Low until the final balloon	Bonds; mini-perm structures; refinancing plays (Domain 15)

WORKED EXAMPLE

Worked example 3.2.2 — Kestrel's senior loan.

1. **Setup.** USD 42,000,000, 12 annual instalments, 6.0 % — annuity shape. Find the instalment and build the first three schedule rows.
2. **Formula.** `A = P × r / (1 − (1+r)^−n)`, with `P` = 42,000,000, `r` = 0.06, `n` = 12. Each row: interest = opening balance × `r`; principal = `A` − interest; closing = opening − principal.
3. **Substitution.** $A = 42{,}000{,}000 \times 0.06 / (1 - 1.06^{-12}) = 42{,}000{,}000 / 8.383844$.
4. **Result.** `A` = **USD 5,009,635** per year (5,009,635.23; indicatively ≈ SAR 18,786,132). Schedule:

Year	Opening balance	Interest (6 %)	Principal	Closing balance
1	42,000,000	2,520,000	2,489,635	39,510,365
2	39,510,365	2,370,622	2,639,013	36,871,351
3	36,871,351	2,212,281	2,797,354	34,073,997

1. **Interpretation.** The instalment never changes; its composition does — year 1 is 50 % interest, and by the final year almost all principal. Two checks a reviewer runs instantly: the closing balance after year 12 must be exactly zero (a residual means a formula error), and total principal across all rows must equal `P`. This schedule is the direct input to Kestrel's debt service line — and therefore to its DSCR — in Domains 6 and 10.

WORKED EXAMPLE

Worked example 3.2.2b — the same loan, level-principal.

1. **Setup.** Kestrel's ECA co-lender offers the same USD 42,000,000 over 12 years at 6.0 % but on a **level-principal** schedule. Build the first two rows and the final row, and compare lifetime interest across all three shapes.
2. **Formula.** Principal each year `P/n = 42,000,000/12 = 3,500,000`; interest = opening balance × 6 %; total = principal + interest.
3. **Substitution.** Year 1: `3,500,000 + 42,000,000 × 0.06`. Year 2: `3,500,000 + 38,500,000 × 0.06`. Year 12: `3,500,000 + 3,500,000 × 0.06`.
4. **Result.** Year 1 **USD 6,020,000** · year 2 **USD 5,810,000** · year 12 **USD 3,710,000** — a declining profile. Lifetime interest: level-principal **USD 16,380,000** · annuity **USD 18,115,623** · bullet **USD 30,240,000**.
5. **Interpretation.** Level-principal is the cheapest shape in total interest because principal retires fastest — but its year-1 debt service is USD 1,010,365 heavier than the annuity's, exactly when a ramping project is cash-poorest. Shape selection is a cash-timing decision, not a cost minimisation: Domain 10 (KA 10.1) sizes debt against the early-year coverage this choice determines.

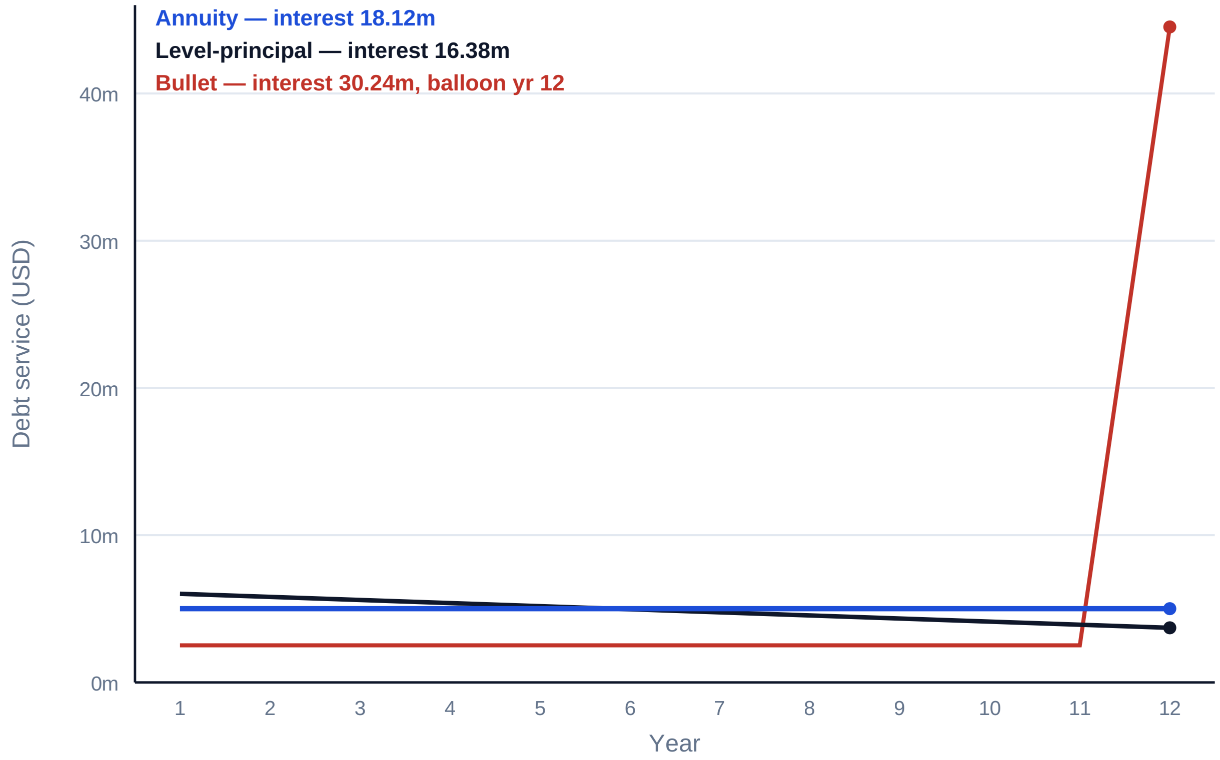

Fig 3.2.2 — Three shapes, one loan: annual debt service under annuity, level-principal and

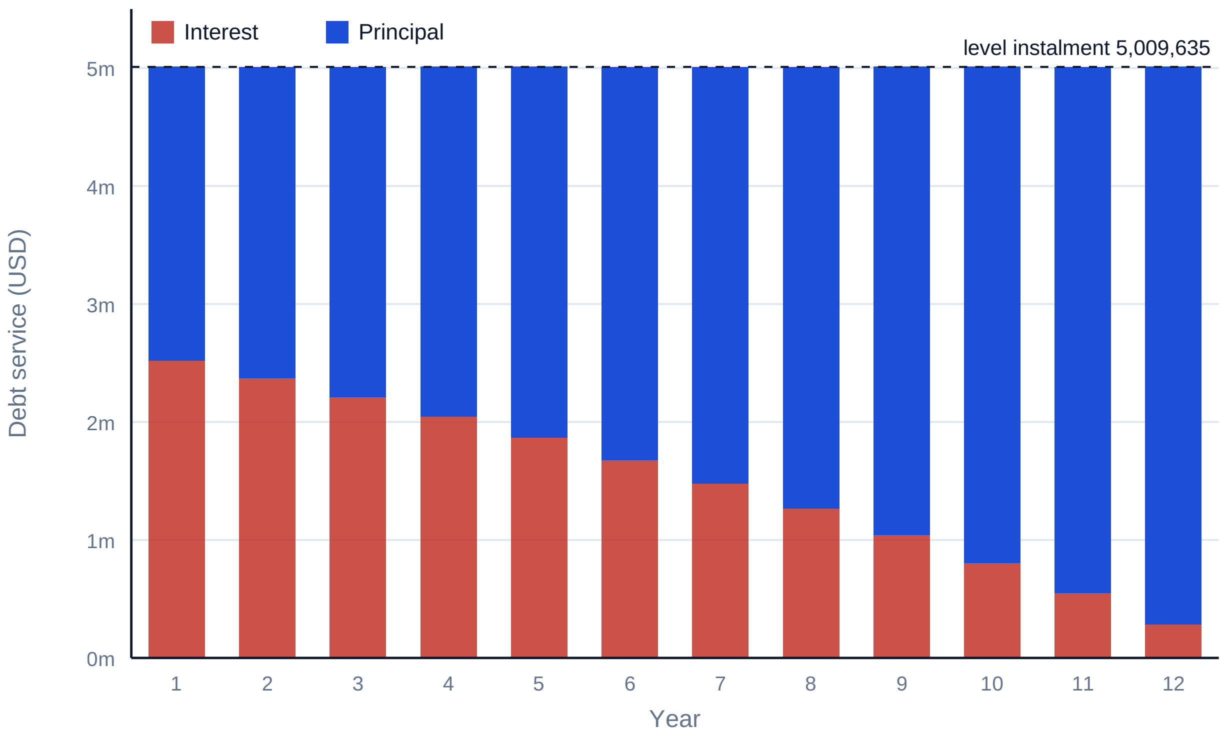

Fig 3.2.1 — Anatomy of an annuity loan: Kestrel's USD 42,000,000, 12 years at 6 %

Common pitfall — the bullet that looks cheap. A bullet's early debt service is a fraction of an annuity's, and a cash-strapped sponsor will feel the temptation. The full price appears at maturity as refinancing risk — the balloon must be refinanced at whatever the market then charges (Domain 15, KA 15.3). Case study B prices this trade.

3.2.3 Compounding frequency and effective rates

Definitions. A quoted **nominal annual rate** `i_nom` compounded `m` times per year applies `i_nom/m` per period. The **effective annual rate (EAR)** is the once-a-year rate that produces the same growth:

```
EAR = (1 + i_nom / m)^m − 1
```

WORKED EXAMPLE

Worked example 3.2.3 — the same 6 %, four ways.

1. **Setup.** A facility is quoted at "6 % nominal". Find the effective annual rate if compounding is annual, semi-annual, quarterly and monthly.
2. **Formula.** `EAR = (1 + 0.06/m)^m − 1` for `m` = 1, 2, 4, 12.
3. **Substitution.** `m=4`: $(1 + 0.015)^4 - 1 = 1.061364 - 1$.
4. **Result.** Annual **6.000 %** · semi-annual **6.090 %** · quarterly **6.136 %** · monthly **6.168 %**.
5. **Interpretation.** "Six per cent" is not one price; it is four prices in this example alone, and the differences compound over a 12-year loan. Term sheets are compared on effective rates, never on quoted nominals — and loan models must use the **periodic** rate `i_nom/m` with `n × m` periods, not the EAR with `n` years, or debt service will be misstated (Domain 6, KA 6.3).

AI in this KA

Amortisation is where AI-assisted modelling earns its keep — and where it must be caged. Generating a 360-row monthly schedule is exactly the mechanical work a copilot does well; whether the schedule uses the periodic rate correctly, honours the day-count convention in the actual facility agreement, and zeroes the final balance is exactly what it gets wrong silently. The governed workflow (Domain 6, KA 6.4; Domain 16): AI drafts the schedule; the professional runs the three deterministic checks (final balance zero; Σ principal = `P`; payment recomputed independently) before any output leaves the model.

KEY TERMS — KA 3.2

Term	Meaning
Annuity / annuity-due	Level periodic stream, in arrears / in advance.
Annuity factor `AF(r,n)`	`(1 − (1+r)^−n)/r`; PV of 1 per period for `n` periods.
Perpetuity	Level stream forever; `PV = A/r`.
Annuity, level-principal, bullet	The three canonical loan shapes.
Balloon / refinancing risk	The bullet's maturity principal and the risk of the rate then prevailing.
Nominal rate / EAR	Quoted annual rate with compounding frequency / its once-a-year equivalent.

SAMPLE MCQS — KA 3.2

MCQ 3.2-A `[3.2.2 · Application]` A USD 42,000,000 loan is repaid by 12 equal annual instalments at 6 %. The instalment is closest to: - A. USD 3,500,000 - B. USD 2,520,000 - C. USD 5,009,635 - D. USD 5,706,454

Rationale: $42{,}000{,}000 \times 0.06/(1 - 1.06^{-12}) = 5{,}009{,}635$. A divides principal by 12 and ignores interest; B is interest-only on the full balance; D uses a 10-year annuity factor — the wrong-tenor error.

MCQ 3.2-B `[3.2.3 · Application]` A 6 % nominal rate compounded quarterly has an effective annual rate of: - A. 6.00 % - B. 6.09 % - C. 6.14 % - D. 6.17 %

Rationale: $(1 + 0.015)^4 - 1 = 6.136\ \% \approx 6.14\ \%$. A is the nominal itself; B is semi-annual compounding; D is monthly — each a frequency misread.

MCQ 3.2-C `[3.2.1 · Analysis]` Two otherwise identical concession bids value the same 20-year availability stream. Bid X used the full-precision annuity factor; Bid Y rounded the factor to four decimals. The most defensible statement is: - A. the bids differ by an amount that grows with the size of the stream, and only X's practice passes model audit - B. rounding the factor is conservative, so Y understates value and is safer - C. the difference is always immaterial because four decimals is industry standard - D. Y's approach is wrong only if the discount rate exceeds 10 %

Rationale: Factor rounding error scales linearly with the cash flows and has no consistent conservative direction (B, D wrong); materiality depends on scale, not convention (C wrong); model audit standards require full-precision arithmetic with display-only rounding.

MCQ 3.2-F `[3.2.1 · Analysis]` A wayleave pays USD 800,000 per year; the discount rate is 8 %. Modelling it as a 30-year annuity instead of a perpetuity understates its value by: - A. USD 993,773 — the present value of the post-year-30 tail - B. nothing material — thirty years is effectively forever - C. USD 4,000,000 — one-third of the perpetuity value - D. it overstates value, because perpetuities are riskier

Rationale: Perpetuity $800{,}000/0.08 = 10{,}000{,}000$; 30-year annuity $800{,}000 \times 11.257783 = 9{,}006{,}227$; the gap — 9.9 % of value — is the discounted tail. B waves away a million dollars; C invents a fraction; D confuses valuation arithmetic with a risk adjustment that belongs in the rate, not the formula choice.

MCQ 3.2-D `[3.2.1 · Application]` A 5-year, USD 500,000-per-year lease payable **in advance** is valued at 8 %. Its present value is closest to: - A. USD 1,996,355 - B. USD 2,156,063 - C. USD 2,500,000 - D. USD 1,848,477

Rationale: Annuity-due = ordinary annuity × $(1+r)$: $500{,}000 \times 3.992710 \times 1.08 = 2{,}156{,}063$. A prices it as an ordinary annuity (the classic misread); C is the undiscounted total; D divides the ordinary value by 1.08 — the adjustment applied backwards.

MCQ 3.2-E `[3.2.2 · Analysis]` For the same principal, tenor and rate, which repayment shape has the lowest lifetime interest, and why? - A. the annuity — its instalments are level, so interest is averaged down - B. the bullet — deferral shrinks the money's time exposure - C. level-principal — the balance falls fastest, so less principal is outstanding for less time - D. all three are equal because rate and tenor are equal

Rationale: Interest is rate × outstanding balance × time; level-principal retires balance fastest (Kestrel: 16.38m vs 18.12m annuity vs 30.24m bullet). A's "averaging" is not a mechanism; B reverses the truth — the bullet maximises exposure; D ignores the balance path entirely.

SELF-CHECK — KA 3.2

1. What two instant checks validate any amortising schedule? — Final closing balance exactly zero; total principal equals the original loan.
2. Why is an annuity-due worth $(1+r)$ × the ordinary annuity? — Every payment arrives one period earlier, so each escapes one period of discounting.
3. A term sheet quotes "5.9 % monthly"; another "6.0 % annual". Which is dearer? — The first: $(1 + 0.059/12)^{12} - 1 = 6.06\ \%$ effective, above 6.00 %.

KNOWLEDGE AREA 3.3

Real-world adjustments: inflation, escalation and currency

Topics: 3.3.1 nominal and real rates · 3.3.2 inflation and escalation · 3.3.3 currency effects · 3.3.4 day-count conventions.

3.3.1 Nominal and real rates — the Fisher relation

Definitions. A **nominal** rate i_{nom} is quoted in money terms; a **real** rate i_{real} is measured in purchasing power, after inflation π. They are linked by the **Fisher relation**:

$$(1 + i_{nom}) = (1 + i_{real}) \times (1 + \pi) \quad \text{so} \quad i_{real} = (1 + i_{nom})/(1 + \pi) - 1$$

The subtraction shortcut $i_{real} \approx i_{nom} - \pi$ is a first-order approximation only — usable for intuition, never for models.

WORKED EXAMPLE

Worked example 3.3.1 — the real cost of Kestrel's equity hurdle.

1. **Setup.** Kestrel's shareholders require a 9.0 % nominal return; inflation is expected at 3.0 %. What real return are they actually demanding?
2. **Formula.** $i_{real} = (1 + i_{nom})/(1 + \pi) - 1$, with i_{nom} = 0.09, π = 0.03.
3. **Substitution.** $i_{real} = 1.09 / 1.03 - 1 = 1.058252 - 1$.
4. **Result. 5.83 %** real (5.8252 % at full precision). The subtraction shortcut says 6.00 % — 17 basis points high.
5. **Interpretation.** Seventeen basis points sounds academic until it is applied to a 25-year stream: mispricing the hurdle by that much moves a large concession's value by roughly one and a half years' worth of payments. Models must live entirely in one world — nominal cash flows with nominal rates, or real with real. **Mixing them is the single most common TVM defect found in model audits** (Domain 6, KA 6.4; Domain 13, KA 13.2).

3.3.2 Inflation and escalation

Definitions. Inflation is the general price level's drift; **escalation** is the contractual or forecast growth of a specific price — a tariff indexed to CPI, labour rates under a collective agreement, steel under a commodity formula. An amount x escalating at rate e for n periods becomes $x \times (1 + e)^n$. Escalation compounds, exactly like interest — and like interest, it is routinely underestimated over long horizons.

WORKED EXAMPLE

Worked example 3.3.2 — an O&M contract's third-year price.

1. **Setup.** Kestrel's O&M contract prices year-0 services at USD 10,000,000, escalating 4.0 % annually. What is the year-3 price?
2. **Formula.** $x_n = x_0 \times (1 + e)^n$, with x_0 = 10,000,000, e = 0.04, n = 3.
3. **Substitution.** $10{,}000{,}000 \times 1.04^3 = 10{,}000{,}000 \times 1.124864$.
4. **Result. USD 11,248,640**.
5. **Interpretation.** Simple-escalation thinking says "+12 %" (USD 11,200,000); compounding adds USD 48,640 by year 3 and the gap widens every year after. In a 25-year operating model the compound-vs-simple escalation choice changes lifecycle cost by whole percentage points — Domain 8 (KA 8.2) builds the full escalated cash-flow machinery on this rule.

Indexation discipline. Real contracts escalate on published indices with definitions, lags, caps and floors (Domain 7, KA 7.3; Domain 12). A model's escalation assumptions must cite the contractual index and its mechanics, not a bare percentage — the toolkit's escalation register (3.T.3) exists for exactly this.

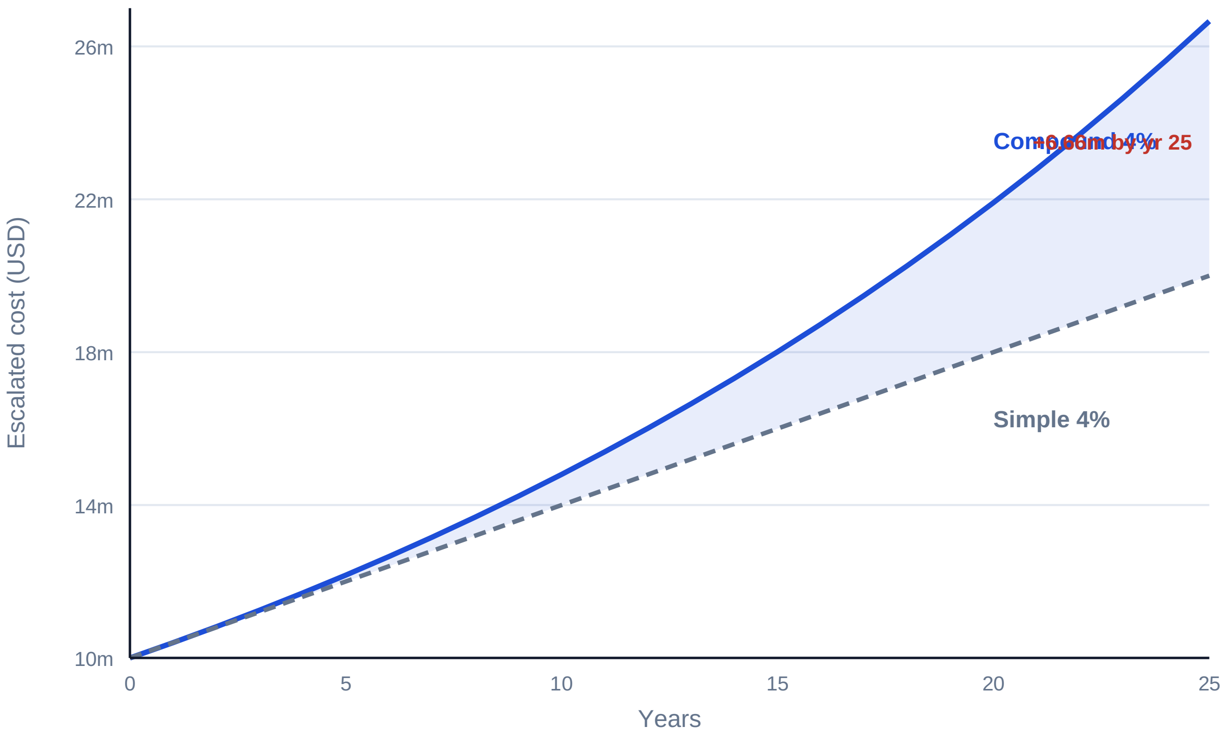

Fig 3.3.2 — Compound versus simple escalation of a USD 10,000,000 cost at 4 %

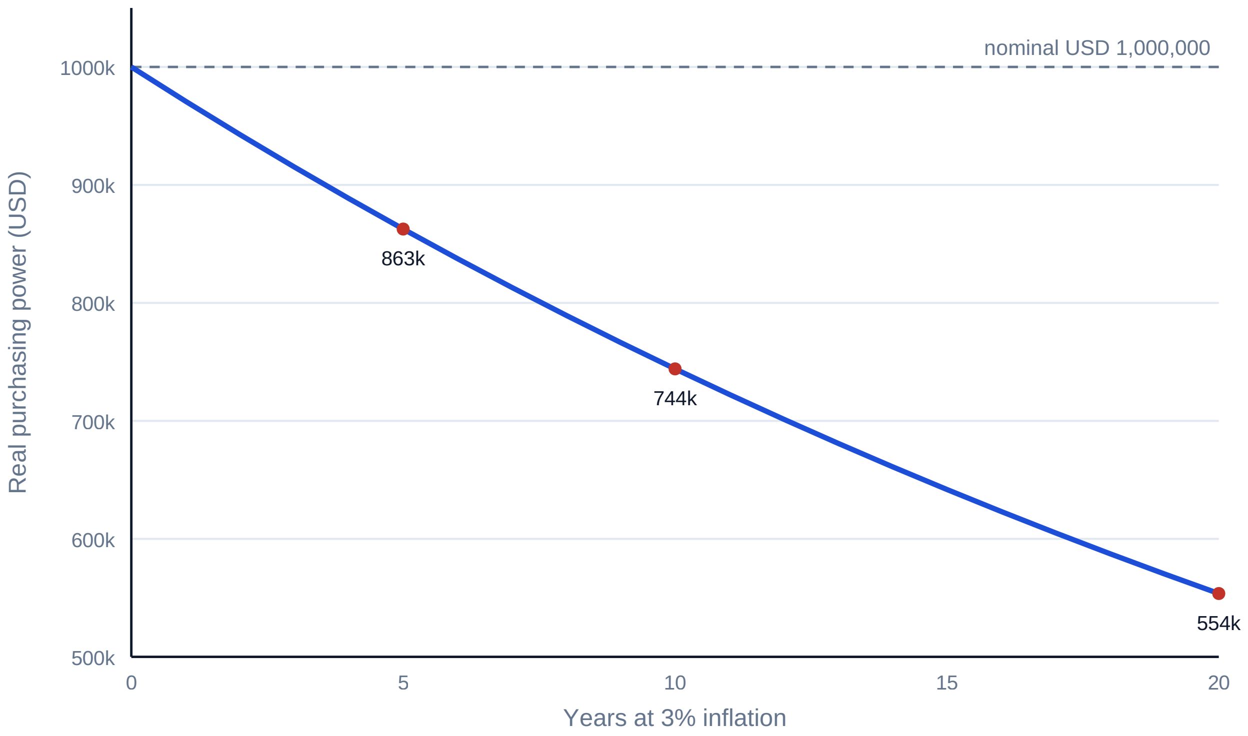

Fig 3.3.1 — What 3 % inflation does to USD 1,000,000 of purchasing power

3.3.3 Currency effects

The problem. Project cash flows often arrive in one currency while debt service, equipment or returns are owed in another. Two rates matter: the **spot rate** now, and the **forward rate** at which future exchange can be contracted. Forwards are priced off interest differentials (covered interest parity, stated here in its textbook form):

```
F ≈ S × (1 + i_quote) / (1 + i_base)
```

WORKED EXAMPLE

Worked example 3.3.3 — a one-year SAR/USD forward.

1. **Setup.** Spot `USD 1 = SAR 3.7500` (the indicative peg rate used throughout this book). One-year money costs 5.0 % in USD and 5.5 % in SAR. Estimate the one-year forward.
2. **Formula.** `F = S × (1 + i_SAR)/(1 + i_USD)`, with `S` = 3.7500.
3. **Substitution.** `F = 3.7500 × 1.055 / 1.05`.
4. **Result. SAR 3.7679 per USD**.
5. **Interpretation.** The forward is not a forecast; it is arithmetic — the rate at which borrowing in one currency and lending in the other breaks even. A project that leaves a currency mismatch unhedged is not "expecting the spot to hold"; it is running an open position whose size Domain 11 (KA 11.3) teaches you to measure and whose contractual mitigations Domain 12 allocates. **Assumption honesty:** actual pegs, spreads and convertibility restrictions are jurisdiction- and time-specific; every currency figure in this book is illustrative.

3.3.4 Day-count conventions

The problem. "Three months' interest" is not one number. Facility agreements specify a **day-count convention** — how days are counted and over what year they are divided — and the convention moves real money on large balances:

Convention	Rule	Typical home
30/360	Every month 30 days, year 360	Bonds, some term loans
actual/360	Actual days elapsed, year 360	Money markets, most floating-rate loans
actual/365	Actual days elapsed, year 365	Sterling markets, some jurisdictions

WORKED EXAMPLE

Worked example 3.3.4 — one quarter, three conventions.

1. **Setup.** Interest accrues on Kestrel's USD 42,000,000 drawn balance at 6.0 % for one calendar quarter that contains **92 actual days**. Compute the interest under each convention.
2. **Formula.** Interest = balance × rate × (days counted / year basis).
3. **Substitution.** 30/360: `42,000,000 × 0.06 × 90/360`. actual/360: `× 92/360`. actual/365: `× 92/365`.
4. **Result.** 30/360 **USD 630,000** · actual/365 **USD 635,178** · actual/360 **USD 644,000**.
5. **Interpretation.** The spread between conventions is USD 14,000 on one quarter of one drawdown — and actual/360 is systematically the most expensive because it counts every real day over a short year. Models must implement the convention named in the facility agreement (Domain 12), and the assumption register (3.T.1) records it per instrument; "6 % divided by 4" is not a convention, it is a guess.

AI in this domain — the systematic view

TVM is deterministic, which makes it the safest place in finance to use machine assistance and the most dangerous place to trust it unchecked: a wrong exponent produces a plausible number, not an absurd one. The domain's governed workflow, applied wherever Domains 4–16 discount anything:

1. **AI proposes** — drafts the factor table, schedule or escalated series.
2. **Deterministic checks** — monotone factors; `DF(1)(1+r) = 1`; schedule zeroes; Σ principal = `P`; one hand-recomputed cell per block.
3. **Assumption register** — rate, basis (nominal/real), compounding frequency, index, and currency of every stream written down (toolkit 3.T.1).
4. **The professional decides and remains accountable** — no AI-produced number reaches a board paper, bid or lender report without a named human owner (Domain 16, KA 16.4).

KEY TERMS — KA 3.3

Term	Meaning
Nominal / real rate	Money-terms rate / purchasing-power rate; linked by Fisher.
Fisher relation	`(1+i_nom) = (1+i_real)(1+π)`.
Inflation π **/ escalation** e	General price drift / specific contractual price growth.
Indexation	Contractual escalation by a published index, with lags, caps, floors.
Spot / forward rate	Exchange rate now / contracted for a future date.
Covered interest parity	Forward ≈ spot × interest-ratio; the no-arbitrage forward.

SAMPLE MCQS — KA 3.3

MCQ 3.3-A `[3.3.1 · Application]` Nominal return 9 %, inflation 3 %. The real return is closest to: - A. 6.00 % - B. 5.83 % - C. 12.27 % - D. 3.00 %

Rationale: `1.09/1.03 − 1 = 5.83 %`. A is the subtraction approximation; C multiplies `1.09 × 1.03 − 1` (compounding inflation on instead of off); D confuses the inflation rate itself with the real return.

MCQ 3.3-B `[3.3.2 · Application]` A USD 10,000,000 cost escalating at 4 % per year is, in year 3: - A. USD 11,200,000 - B. USD 10,400,000 - C. USD 12,486,400 - D. USD 11,248,640

Rationale: $10{,}000{,}000 \times 1.04^3 = 11{,}248{,}640$. A escalates simply (3 × 4 %); B stops at one year; C applies four periods' escalation with a decimal slip.

MCQ 3.3-C `[3.3.1 · Analysis]` A model discounts real (uninflated) cash flows at the sponsor's nominal 9 % hurdle. The result: - A. overstates value, because inflation is counted twice - B. understates value, because inflation is removed from the flows but still charged in the rate - C. is correct if inflation is below 5 % - D. is correct because discount rates are always nominal

Rationale: Real flows with a nominal rate deduct inflation twice — once from the flows, once inside the rate — so value is systematically understated. The consistency rule admits no threshold (C) and no default (D); A reverses the direction of the error.

MCQ 3.3-F `[3.3.1 · Application]` A model shows a nominal cash flow of USD 2,000,000 in year 5; inflation is 3 %. Its value in today's purchasing power (real terms) is: - A. USD 2,000,000 — real and nominal are equal at year 5 - B. USD 1,725,218 - C. USD 1,700,000 - D. USD 2,318,548

Rationale: $2{,}000{,}000 / 1.03^5 = 1{,}725{,}218$ — deflating by the price level, not discounting for time value (that happens separately, at a real rate). A ignores five years of inflation; C deflates simply (`2,000,000 × (1 − 0.03 × 5) = 1,700,000`); D multiplies by 1.03^5 — inflating instead of deflating.

MCQ 3.3-D `[3.3.4 · Application]` A USD 42,000,000 balance accrues at 6 % over a 92-day quarter. Under **actual/360** the interest is: - A. USD 630,000 - B. USD 635,178 - C. USD 644,000 - D. USD 620,548

Rationale: `42,000,000 × 0.06 × 92/360 = 644,000`. A is 30/360 (90/360); B is actual/365; D uses 90/365 — mixing the two conventions' halves.

MCQ 3.3-E `[3.3.2 · Analysis]` A 25-year operating model escalates O&M at "4 % simple" to be "conservative". The effect is: - A. conservative — simple escalation always overstates cost - B. anti-conservative — compound escalation exceeds simple by an amount that grows every year, so late-life costs are materially understated - C. neutral — the choice only redistributes cost between years - D. correct — operating contracts always escalate simply

Rationale: Escalation compounds (contracts index on last year's indexed price): by year 25 the compound multiple $1.04^{25} \approx 2.67$ far exceeds the simple 2.00. A claims the reverse; C ignores the widening gap; D asserts a contractual universal that KA 3.3.2's indexation discipline contradicts.

SELF-CHECK — KA 3.3

1. State the consistency rule. — Nominal flows with nominal rates, real flows with real rates; never mixed in one calculation.
2. Why does a 25-year concession make the Fisher shortcut dangerous? — The 17 bp error (at 9 %/3 %) compounds across every one of 25 discount factors.
3. Is a forward rate a forecast? — No: it is the no-arbitrage consequence of today's spot and the two interest rates.

Advanced topics — Domain 3

3.A.1 Continuous compounding

As compounding frequency $m \to \infty$, $(1 + r/m)^{mn} \to e^{rn}$. USD 100,000 at 8 % for 3 years compounds continuously to $100{,}000 \times e^{0.24}$ = **USD 127,125** — against 125,971 annually. In project work continuous compounding appears mainly in derivative pricing and some academic appraisal literature; its practical value here is as a bound: no compounding frequency can push growth beyond e^{rn}.

3.A.2 Irregular timing and period conventions

Real cash flows do not arrive on anniversary dates. Three conventions bridge the gap: **exact-date discounting** (each flow discounted by its actual day count — the XNPV approach, standard in transaction models); the **mid-period convention** (flows treated as arriving mid-period, appropriate for continuous operations revenue); and **period-end** (conservative for receipts, default in lender base cases). The convention is an assumption — it belongs in the assumption register, it changes results by up to half a period's discounting, and model audits (Domain 13, KA 13.2) check that one convention is applied everywhere.

WORKED EXAMPLE

Worked example 3.A.2 — the payment on day 500.

1. **Setup.** A USD 1,000,000 completion payment falls on **day 500** from the valuation date; the discount rate is 9.0 % annual. Compare exact-date discounting with the period-end habit of "call it year 2".
2. **Formula.** Exact-date: $PV = x / (1 + r)^{(days/365)}$.
3. **Substitution.** $1{,}000{,}000 / 1.09^{(500/365)} = 1{,}000{,}000 / 1.09^{1.36986}$.
4. **Result. USD 888,650**. "Year 2" rounding gives 841,680 — **USD 46,970 low**; "year 1" gives 917,431 — USD 28,781 high.
5. **Interpretation.** Bucketing a mid-period flow to the nearer anniversary mis-states value by up to half a period's discounting — nearly 5 % here. Transaction models discount on dates; period models declare mid-period or period-end explicitly and apply it everywhere. The convention chosen matters less than its disclosure and consistency.

3.A.3 Term structures — when one rate is not enough

A single flat r assumes money has the same price at every horizon. Markets disagree: rates form a **term structure**, and precise valuation discounts each period's flow at that period's rate (equivalently, its own $DF(t)$ from a curve). Project models typically justify a flat rate as a deliberate simplification of a curve — acceptable when documented and stress-tested (Domain 6, KA 6.4), a silent error when inherited unexamined from a template.

WORKED EXAMPLE

Worked example 3.A.3 — flat rate versus the curve.

1. **Setup.** A contractor is owed USD 1,000,000 at the end of each of years 1 and 2. The spot curve prices 1-year money at 5.0 % and 2-year money at 7.0 %; the model uses a flat 6.0 %. How far wrong is the flat rate?
2. **Formula.** Curve: $PV = x/(1 + r_1) + x/(1 + r_2)^2$. Flat: $PV = x/(1+r) + x/(1+r)^2$.
3. **Substitution.** Curve: $1{,}000{,}000/1.05 + 1{,}000{,}000/1.07^2 = 952{,}381 + 873{,}439$. Flat: $943{,}396 + 889{,}996$.
4. **Result.** Curve **USD 1,825,820**; flat **USD 1,833,393** — the flat model overstates value by **USD 7,573** (0.4 %).
5. **Interpretation.** The flat 6 % "averages" the curve but misprices both flows — too harsh on the near one, too kind on the far one — and the errors do not cancel unless the stream is symmetric around the average tenor. For a two-flow example the drift is small; on a 25-year concession against a steep curve it moves tens of millions. The audit question is never "is the rate reasonable?" but "which curve does this rate summarise, and has the summary been stress-tested?"

3.A.4 The reviewer's TVM eye

Experienced reviewers do not recompute everything; they run invariants. The domain's collected set: factors decline; $DF(1)(1+r) = 1$; annuity factor × $r \to 1 - DF(n)$; schedules zero out; Σ principal = P; effective ≥ nominal, with equality only at annual compounding; real < nominal whenever $\pi > 0$; forward off-spot in the direction of the interest differential. Any violated invariant is a defect somewhere — find it before anyone downstream builds on it.

Industry variations — Domain 3

The machinery is universal; its parameters are sectoral, and a project finance leader reads the sector before trusting any inherited rate, tenor or convention:

- **Power and renewables.** Long contracted tenors (20–35 years) make value acutely curve-sensitive (3.A.3) and escalation-sensitive (3.3.2); availability and capacity payments are near-annuities, so the annuity family does most of the valuation work. Debt sculpting to seasonal cash flows (Domain 10) starts from monthly, not annual, discounting.
- **Transport concessions.** Patronage risk pushes discount rates up and makes single-rate valuation least defensible — scenario-based rates and revenue stress tests (Domain 7, KA 7.4) ride directly on this domain's factors.

- **Water and regulated utilities.** Regulatory reset cycles (often 5-yearly) partition the horizon: within-period flows discount at financing rates; across resets, the real/nominal discipline (3.3.1) dominates because regulators commonly work in real terms while lenders live in nominal.
- **Digital infrastructure.** Shorter economic lives and refresh capex compress tenors: bullet and mini-perm shapes (3.2.2) with deliberate refinancing (Domain 15) are standard, so the balloon-risk arithmetic of Case B is daily practice, not a cautionary tale.
- **Oil, gas and mining.** Commodity-linked revenue makes escalation assumptions (3.3.2) the loudest value driver, and multi-currency operations make the day-count and FX conventions (3.3.3–3.3.4) contractual battlegrounds rather than back-office detail.
- **Social infrastructure PPPs.** Availability payments in advance are common — the annuity-due reading (WE 3.2.1b) is the difference between winning and mispricing a bid.

Case study — Domain 3: choosing between an upfront grant and an availability stream (water / PPP)

Situation. Kestrel Water SPC's grantor offers two support packages for the desalination concession, and the board must choose one at financial close:

- **Package U:** a single upfront capital grant of **USD 58,000,000** at completion.
- **Package S:** an availability supplement of **USD 5,600,000 per year for 25 years**, paid annually in arrears from completion.

Kestrel evaluates support at its 8.0 % project discount rate. The grantor's advisers privately evaluate at 10 %.

Analysis. The stream's value to Kestrel: `AF(0.08, 25) = (1 − 1.08⁻²⁵)/0.08 = 10.674776`; `PV = 5,600,000 × 10.674776 =` **USD 59,778,747** — worth **USD 1,778,747 more** than the upfront grant. To the grantor's advisers at 10 %: `AF(0.10, 25) = 9.077040`; `PV = 5,600,000 × 9.077040 =` **USD 50,831,424** — worth **USD 7,168,576 less** than the grant they would otherwise pay today.

The decision. Both sides prefer a different package from the same facts — a discount-rate wedge, not a disagreement about arithmetic. Kestrel takes the stream (worth more at 8 %); the grantor is glad to give it (cheaper at 10 %); the deal closes with both counterparties better off by their own measure. The board minute records the rate, the convention (annual, in arrears) and the sensitivity: the stream's advantage to Kestrel disappears if its discount rate rises past ≈ 8.36 % — the **breakeven rate** at which `AF × 5,600,000 = 58,000,000`, i.e. `AF = 10.3571`.

What the domain teaches here. Valuation is always at a rate, for a party; streams and lumps are only comparable through `PV(x)`; and the negotiation surface between two parties' rates is where structuring value lives (Domain 7 revisits this as tariff design; Domain 9 as the grant-versus-support decision).

Case study B — Domain 3: the bullet that had to be refinanced (energy / refinancing)

Situation. A 30 MW peaking-plant SPV borrowed **USD 30,000,000** for 7 years. The sponsor chose a **bullet** (interest-only at 7.5 %, principal at maturity) over the lenders' preferred **annuity** shape, to keep early cash for distributions.

The two prices. Annuity instalment: $A = 30{,}000{,}000 \times 0.075/(1 - 1.075^{-7})$ = **USD 5,664,009** per year; total paid 39,648,066, of which interest **USD 9,648,066**. Bullet: interest $30{,}000{,}000 \times 0.075 =$ 2,250,000 per year — total interest **USD 15,750,000**, plus the full USD 30,000,000 due in year 7. The bullet pays **USD 6,101,934 more interest** for the privilege of deferral — and retains the entire principal as **refinancing risk**.

What happened. At year 7 the credit market had repriced: the refinancing cleared at 9.8 %, and the new 7-year annuity instalment on the rolled principal became $30{,}000{,}000 \times 0.098/(1 - 1.098^{-7}) = 6{,}121{,}646$ — against 5,664,009 had the original annuity simply been running to zero. Distributions locked up under the new facility's covenants for two years (Domain 10, KA 10.4).

What the domain teaches here. A schedule shape is a risk allocation, not a formatting choice: the annuity retires rate risk continuously; the bullet stores it at maturity, priced at a rate nobody today knows. Deferral has a computable minimum cost (the interest differential) and an uncomputable tail (the refinancing market) — leadership means pricing the first honestly and sizing the second deliberately (Domain 15, KA 15.3).

Executive perspective — Domain 3

What a project finance director cannot delegate in this domain:

- **The rate.** Analysts compute with a discount rate; directors own it. Whoever sets r sets the answer — the choice is a governance act (Domain 4 gives it method; Domain 3 makes its power visible).
- **The world.** Nominal or real, once, for the whole model — and the director asks which, out loud, at the first model review.
- **The conventions.** In-arrears vs in-advance, compounding frequency, day counts, escalation indices: each is small, each moves millions at scale, and together they are why the assumption register (3.T.1) is a board-visible artefact, not analyst hygiene.
- **The verification culture.** The invariants of 3.A.4 cost minutes; unverified TVM in a signed bid costs careers. The director's question is never "who computed this?" but "who recomputed this?" — and, where AI drafted it, the answer must name a human.

Calculation exercises — Domain 3

> **Exercise 3.1** A land payment of USD 850,000 falls due in 8 years; the applicable discount rate is 11 %. Value it today. Solution. $PV = 850{,}000 / 1.11^{8} = 850{,}000 / 2.304538 =$ **USD 368,838** (368,837.52). Common error: using n = 7 gives 409,410 — an off-by-one from counting "in 8 years" as 7 compounding periods.

> **Exercise 3.2** A USD 2,500,000 equipment loan is repaid monthly over 20 years at a 5.0 % nominal annual rate compounded monthly. Find the monthly payment. Solution. Periodic rate $0.05/12 = 0.004167$; $n = 240$; $A = 2{,}500{,}000 \times 0.004167/(1 - 1.004167^{-240})$ = **USD 16,499** per month (16,498.89). Common error: using the EAR (5.116 %) with 20 annual periods misstates debt service materially.

> **Exercise 3.3** Convert a 12 % nominal rate compounded monthly to its effective annual rate. Solution. $(1 + 0.01)^{12} - 1 = 1.126825 - 1 =$ **12.68 %**. Common error: reporting 12 % — the nominal — as "the rate" in a comparison against an annually-compounded 12.5 % offer, which is in fact cheaper.

> **Exercise 3.4** A transmission wayleave pays USD 1,200,000 per year indefinitely; the discount rate is 8.5 %. Value the perpetuity. Solution. `PV = 1,200,000 / 0.085 =` **USD 14,117,647**. Common error: applying a long annuity factor (say 30 years, `AF = 10.694`) "as approximately forever" understates value by ≈ 9 %.

> **Exercise 3.5** A market forecast promises a 12 % nominal return with 5 % inflation. State the real return, exactly and by the shortcut, and the shortcut's error. Solution. Exact: `1.12/1.05 − 1 =` **6.67 %** (6.6667 %). Shortcut: `12 − 5 = 7.00 %`. The shortcut overstates the real return by **33 basis points** — material over any long horizon.

Practitioner's toolkit — Domain 3

Adoption-ready artefacts; adapt headings to your organisation, then keep them stable.

Toolkit 3.T.1 — TVM assumption register (one row per modelled stream)

Field	Discipline it enforces
Stream name & source document	Every number traces to a contract or estimate
Currency	Mismatches surface (KA 3.3.3)
Nominal or real	The consistency rule (3.3.1), stated not assumed
Rate & its owner	The director owns `r` (Executive perspective)
Compounding frequency & day count	Effective-rate honesty (3.2.3)
Timing convention (arrears/advance/mid-period)	Half-period errors caught (3.A.2)
Escalation index, lag, cap/floor	Contractual, not invented (3.3.2)

Toolkit 3.T.2 — Schedule QA checklist

- [ ] Payment recomputed independently of the model (different implementer or method).
- [ ] Final balance exactly zero; Σ principal = original loan.
- [ ] Periodic rate = nominal / frequency; period count = years × frequency.
- [ ] Discount factors monotone; `DF(1) × (1 + r) = 1`.
- [ ] Display rounding only — full precision beneath every printed figure.
- [ ] AI-drafted content marked, and its human verifier named (Domain 16, KA 16.4).

Toolkit 3.T.3 — Escalation register

For each escalating line: base amount and base date · index (publisher, series, definition) · lag · formula (full/partial indexation, weightings) · cap/floor · compounding basis · the clause reference in the contract. A model whose escalation register is complete can survive a Domain 13 model audit; one whose escalation is "4 % because last year" cannot.

Exam preparation — Domain 3

The calculation traps. Off-by-one periods ("in year 5" vs "after 5 years") · simple-for- compound substitution · rounded discount factors in the arithmetic · annuity-due priced as ordinary · EAR used as the periodic rate · nominal/real mixing · escalation applied simply · tenor mismatch in annuity factors (the 3.2-A distractor D) · treating a forward as a forecast.

Reflection questions. 1. Your CFO asks for "the value of the concession". What three questions must you answer before any number is defensible? (At what rate; in which world — nominal or real; under which timing conventions.) 2. A lender and a sponsor disagree about the value of the same payment stream. Under what conditions are both right — and where does that create negotiating room? (Case study A.) 3. Which invariant checks would have caught the last TVM error you saw in the wild — and why didn't they run? (3.A.4; toolkit 3.T.2.)

Domain 3 summary

Money's value is a function of time, and this domain built the complete conversion machinery: compounding and discounting (`FV(x)`, `PV(x)`, `DF(t)`); the annuity family that prices every level stream from availability payments to debt service (`AF(r,n)`, perpetuities, the three loan shapes and their risk meanings); frequency and effective-rate honesty; and the three real-world adjustments — Fisher-consistent inflation treatment, contract-grade escalation, and no-arbitrage currency forwards. Its discipline is as important as its formulae: one world per model, full precision beneath displayed rounding, conventions in the assumption register, invariants run on every schedule, and machine-drafted arithmetic verified by a named human before anyone relies on it. Every discounted number in the thirteen domains that follow stands on this one.

04

DOMAIN 4

Investment Appraisal and Capital Budgeting (quantitative)

Group: Foundations (Domain 4 of 4 in Part One). **Target:** ~70 pages. **Binds to:** the PCI Book Pattern Specification and the shared registries (`docs/books/registries/`). This domain is the home of the appraisal symbols — `NPV`, `IRR`, `MIRR`, `PI`, `EAV` — and builds directly on Domain 3's machinery (`PV(x)`, `DF(t)`, `AF(r, n)`): every formula here is a disciplined arrangement of those parts. British English; USD (+SAR where useful, indicative `USD 1 ≈ SAR 3.75`).

Why this domain exists

Domain 3 taught how to move money across time; this domain teaches how to decide with it. Investment appraisal is where analysis meets commitment: the techniques here — net present value, internal rates of return, payback, profitability, equivalent annual value — are how a project finance leader turns forecast cash flows into a defensible yes, no, or not-this-one. The measures are deliberately plural because each answers a different question: NPV measures value created; IRR measures the return embedded in the flows; payback measures exposure; the profitability index measures value per scarce dollar; equivalent annual value compares unlike lifetimes. Leadership in this domain means knowing which question the decision actually asks — and refusing to let a single seductive percentage (usually IRR) answer all of them. Domain 6 industrialises these calculations in the financial model; Domain 10 replays them from the lender's side of the table.

Learning objectives. After this domain a candidate can: compute and interpret NPV and explain why it is the primary measure of value; compute IRR, recognise its pathologies (multiple roots, scale-blindness, reinvestment assumption) and know when it misleads; compute MIRR and state what it fixes and what it does not; apply payback and discounted payback as exposure measures, not value measures; rank with the profitability index under capital rationing; compare unequal lives with equivalent annual value; resolve NPV–IRR conflicts on mutually exclusive projects; and subject any machine-produced appraisal to the family's verification rule.

The master appraisal. The Kestrel Water SPC financing of Domain 3 now faces its investment decision. Building the plant requires **I_0 = USD 60,000,000** today; the operating model forecasts **net cash inflows of USD 8,900,000 per year for 15 years**; the board's discount rate remains **8.0 %**. Every KA below interrogates this one decision from a different angle.

KNOWLEDGE AREA 4.1

The discounted measures: NPV, IRR and MIRR

Topics: 4.1.1 net present value · 4.1.2 IRR and its pathologies · 4.1.3 MIRR.

4.1.1 Net present value

Definition. NPV discounts every cash flow to today and nets off the investment:

$$NPV = \Sigma\ CF_t / (1 + r)^t - I_0$$

A positive NPV means the project returns more than the capital's required return r — it creates value; a negative NPV destroys it. NPV's authority rests on three properties no rival measure shares: it is **additive** (portfolio NPV is the sum of project NPVs), it is **scale-aware** (twice the project, twice the NPV), and it embeds the **correct reinvestment assumption** (cash thrown off is assumed to earn r, the opportunity cost — not the project's own return).

WORKED EXAMPLE

Worked example 4.1.1 — should Kestrel build?

1. **Setup.** I_0 = USD 60,000,000; net inflows USD 8,900,000 per year for 15 years (in arrears); r = 8.0 %.
2. **Formula.** For a level stream, $NPV = A \times AF(r, n) - I_0$ (Domain 3, KA 3.2).
3. **Substitution.** $AF(0.08, 15) = 8.559479$; $NPV = 8{,}900{,}000 \times 8.559479 - 60{,}000{,}000 = 76{,}179{,}360 - 60{,}000{,}000$.
4. **Result. NPV = +USD 16,179,360** (≈ SAR 60.7 million indicatively).
5. **Interpretation.** At the board's 8 % the plant is worth USD 16.2 million more than it costs — build, on these forecasts. The phrase carrying the professional weight is on these forecasts: NPV is only as honest as the cash flows and the rate beneath it, which is why Domain 6's model discipline and this domain's sensitivity habits (KA 4.3.3) are part of the appraisal, not decoration around it.

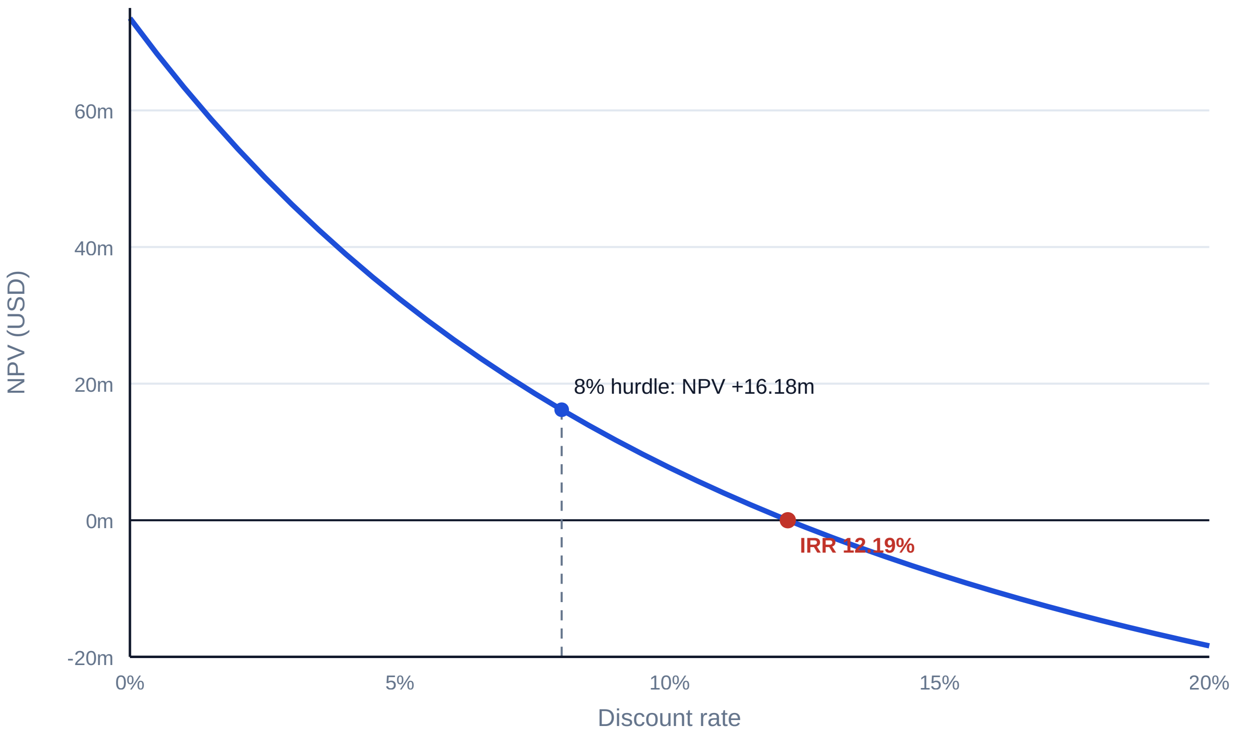

Fig 4.1.1 — Kestrel's NPV profile

4.1.2 IRR and its pathologies

Definition. The internal rate of return is the discount rate at which NPV is zero — the break-even price of capital:

```
NPV(IRR) = 0
```

For Kestrel: solve `8,900,000 × AF(r, 15) = 60,000,000` → `AF = 6.741573` → **IRR = 12.19 %**. Read against an 8 % hurdle, the project clears with 4.19 points to spare — the same verdict as NPV, expressed as a margin of safety rather than a value.

Why decision-makers love it — and where it lies. A percentage needs no context, which is precisely its danger. The three standing pathologies:

1. **Multiple roots.** IRR is a polynomial root; flows that change sign more than once can have more than one. The classic mining/decommissioning shape — invest, harvest, pay to restore — `(−1,000,000, +2,300,000, −1,320,000)` — satisfies NPV = 0 at **both 10 % and 20 %**; between them NPV is positive (about +1,890 at 15 %), outside, negative. Quoting "the IRR" of such a project is meaningless; NPV at the actual cost of capital remains well defined.
2. **Scale-blindness.** A 40 % IRR on USD 100,000 creates less value than 15 % on USD 20,000,000. IRR ranks intensity, NPV ranks money; mutually exclusive choices need money (KA 4.3.1).
3. **The reinvestment fiction.** IRR mathematically assumes interim cash is reinvested at the IRR itself — flattering for high-IRR projects, since no treasury desk reinvests at 28 %. MIRR (4.1.3) exists to repair exactly this.

Common pitfall — comparing IRRs across different tenors. A 3-year 18 % and a 15-year 13 % are not ordered: the short project leaves twelve years of reinvestment risk the percentage hides. The honest comparison is NPV over a common horizon, or EAV (KA 4.2.3).

4.1.3 MIRR — the modified internal rate of return

Definition. MIRR discounts outflows at the finance rate and compounds inflows to the terminal date at an explicit reinvestment rate, then reads the single rate connecting the two:

```
MIRR = ( FV(inflows at r_reinvest) / PV(outflows at r_finance) )^(1/n) − 1
```

WORKED EXAMPLE

Worked example 4.1.3 — Kestrel's honest percentage.

1. **Setup.** The master appraisal, with both finance and reinvestment rates set to the 8 % opportunity cost.
2. **Formula.** Terminal value of the inflows `TV = A × FVAF(r, n)` where `FVAF(0.08, 15) = (1.08¹⁵ − 1)/0.08 = 27.152114`; then `MIRR = (TV/I₀)^(1/15) − 1`.
3. **Substitution.** `TV = 8,900,000 × 27.152114 = 241,653,814`; `MIRR = (241,653,814 / 60,000,000)^(1/15) − 1 = 4.027564^(1/15) − 1`.
4. **Result. MIRR = 9.73 %** — against an IRR of 12.19 %.
5. **Interpretation.** Two and a half points of Kestrel's IRR were the reinvestment fiction. MIRR is single-valued even for sign-changing flows and states its reinvestment assumption in the open — which is why credit committees increasingly ask for it alongside IRR. It remains a rate: still scale-blind, still not additive. The order of authority stands: **NPV decides; rates explain.**

AI in this KA

Appraisal is where optimisation pressure meets a suggestible tool: an assistant asked to "get the IRR above 12 %" will find assumptions that do. The governed workflow inverts the prompt — AI is asked to attack the appraisal (which assumptions move NPV most? where does the IRR become multiple?), the analyst reruns the golden checks (NPV at the stated rate recomputed independently; IRR verified by substitution back into the NPV equation; MIRR's terminal value re-derived), and the professional owns the recommendation. **AI proposes; the professional verifies, decides and remains accountable.**

KEY TERMS — KA 4.1

Term	Meaning
`NPV`	Σ discounted cash flows − investment; the primary value measure.
`IRR`	The rate at which NPV = 0; the flows' embedded return.
NPV profile	NPV plotted against the discount rate; IRR is its zero-crossing.
Multiple roots	Two or more IRRs when flow signs change more than once.
Reinvestment assumption	NPV assumes interim cash earns `r`; IRR assumes it earns the IRR.
`MIRR`	Single rate from explicit finance and reinvestment rates.

SAMPLE MCQS — KA 4.1

MCQ 4.1-A `[4.1.1 · Application]` A project costs USD 60,000,000 and returns USD 8,900,000 per year for 15 years; the discount rate is 8 % (`AF = 8.559479`). Its NPV is: - A. +USD 16,179,360 - B. +USD 73,500,000 - C. −USD 16,179,360 - D. +USD 76,179,360

Rationale: `8,900,000 × 8.559479 − 60,000,000 = +16,179,360`. B is the undiscounted surplus (8.9 × 15 − 60); C reverses the sign (investment minus value); D forgets to deduct the investment at all.

MCQ 4.1-B `[4.1.2 · Analysis]` A project's cash flows are −1,000,000, +2,300,000, −1,320,000. Which statement is correct? - A. its IRR is 10 % - B. its IRR is 20 % - C. it has two IRRs (10 % and 20 %), so decision by IRR is indeterminate and NPV at the cost of capital must decide - D. it has no IRR, so it must be rejected

Rationale: Two sign changes admit two roots — both 10 % and 20 % zero the NPV, which is positive between them. A and B are each half the truth and therefore wrong as "the" IRR; D confuses indeterminate ranking with non-viability — at a 15 % cost of capital the NPV is (slightly) positive.

MCQ 4.1-C `[4.1.3 · Application]` Using an 8 % finance and reinvestment rate, the master appraisal's terminal value of inflows is USD 241,653,814 on an investment of USD 60,000,000 over 15 years. MIRR is closest to: - A. 12.19 % - B. 9.73 % - C. 8.00 % - D. 26.85 %

Rationale: `(241,653,814/60,000,000)^(1/15) − 1 = 9.73 %`. A is the unmodified IRR; C is the reinvestment rate itself; D divides the 4.03× money multiple by 15 as if returns were simple.

MCQ 4.1-D `[4.1.1 · Analysis]` Which property justifies NPV's primacy over IRR for accept/reject and ranking decisions? - A. NPV is easier to compute - B. NPV is expressed as a percentage - C. NPV is additive and scale-aware, and assumes reinvestment at the opportunity cost - D. NPV never requires a forecast

Rationale: The three structural properties of 4.1.1. A is irrelevant and untrue at scale; B describes IRR, not NPV; D is false — NPV consumes the same forecasts as every measure.

SELF-CHECK — KA 4.1

1. State the master appraisal's three verdicts and their meanings. — NPV +16.18m (value created at 8 %); IRR 12.19 % (break-even rate, 4.19-point margin); MIRR 9.73 % (return with honest 8 % reinvestment).
2. Why does a mining-with-restoration cash flow break IRR? — Two sign changes → up to two roots; "the" IRR does not exist.
3. What single question decides between NPV and IRR when they disagree? — Which project adds more money at the actual cost of capital — NPV's answer, by construction.

KNOWLEDGE AREA 4.2

The complementary measures

Topics: 4.2.1 payback and discounted payback · 4.2.2 profitability index · 4.2.3 equivalent annual value.

4.2.1 Payback and discounted payback

Definitions. Payback is the time until cumulative cash inflows repay the investment; **discounted payback** repeats the question in present-value terms. Neither measures value — both measure **exposure**: how long the capital is at risk before it has come home.

WORKED EXAMPLE

Worked example 4.2.1 — how long is Kestrel's money in the ground?

1. **Setup.** The master appraisal: I_0 = 60,000,000; inflows 8,900,000 per year; r = 8 %.
2. **Formula.** Payback = I_0 / A for a level stream. Discounted payback: the smallest n with $A \times AF(r, n) \geq I_0$, interpolated within the crossing year.
3. **Substitution.** Simple: `60,000,000 / 8,900,000 = 6.74`. Discounted: after year 10 the cumulative PV is `8,900,000 × 6.710081 = 59,719,724` — USD 280,276 short; year 11's discounted flow is `8,900,000 × 0.428883 = 3,817,057`; fraction `280,276 / 3,817,057 = 0.07`.
4. **Result.** Payback **6.74 years**; discounted payback **10.07 years**.
5. **Interpretation.** The gap between the two numbers — three and a third years — is the price of time value: nominal repayment by year 7 does not return the capital's worth until year 10. For a 15-year concession, ten years of exposure is a risk fact the board should see beside the +16.2m NPV, not instead of it. The standing rule: **payback screens; NPV decides.** Payback's known biases — it ignores everything after the cut-off and penalises long-life infrastructure precisely for being long-lived — make it a veto-check, never a ranking.

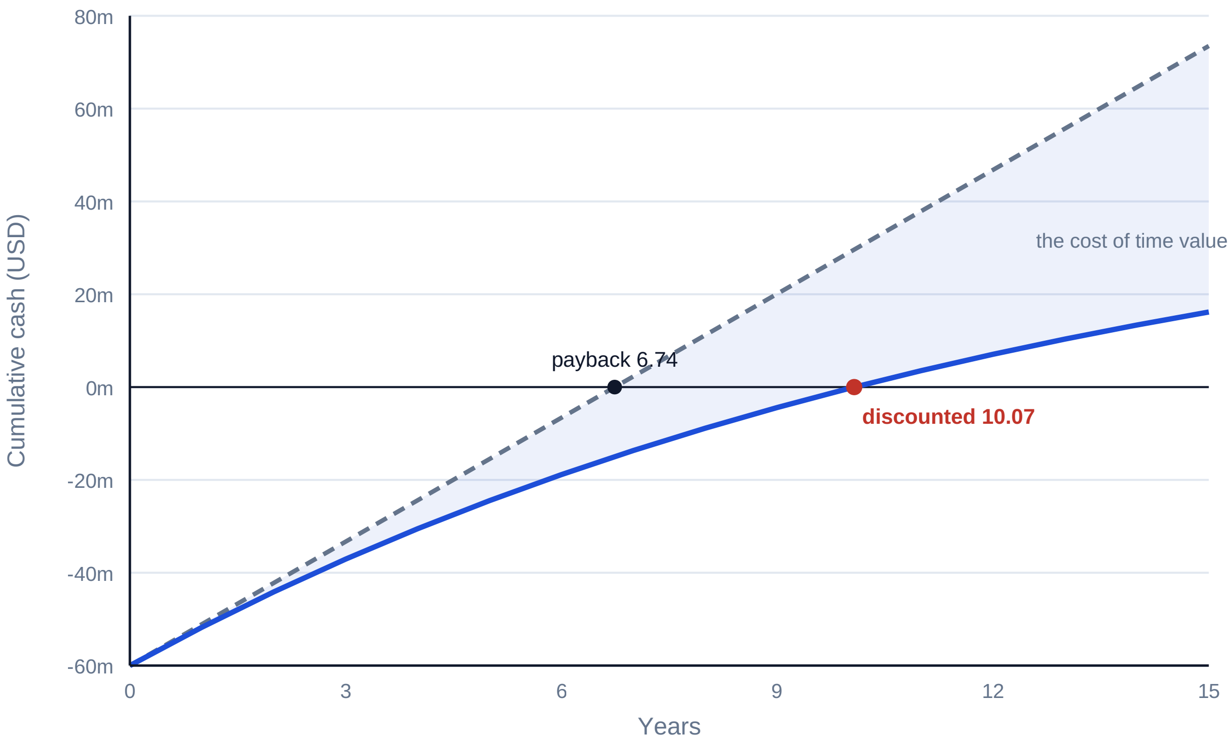

Fig 4.2.1 — Two paybacks: cumulative cash versus cumulative present value

4.2.2 The profitability index

Definition. Value created per unit of scarce capital:

```
PI = PV(future cash flows) / I₀
```

Kestrel: `76,179,360 / 60,000,000 = **1.270**` — each invested dollar buys USD 1.27 of present value. PI ranks efficiency and earns its keep in exactly one situation: **capital rationing** (KA 4.3.2), where the question is not "is this project good?" but "which portfolio of good projects fits the budget?". Unconstrained, PI adds nothing to NPV and can mis-rank mutually exclusive projects of different scale — the same scale-blindness as IRR, in ratio form.

4.2.3 Equivalent annual value

Definition. EAV converts an NPV over a life of `n` years into the level annual amount with the same present value:

```
EAV = NPV / AF(r, n)
```

Its natural habitat is **unequal lives**: comparing a 3-year asset with a 5-year asset by raw NPV smuggles in the assumption that the world ends when the shorter one does.

WORKED EXAMPLE

Worked example 4.2.3 — two pumps, unequal lives.

1. **Setup.** Kestrel must choose a dosing-pump system. **System A**: costs 5,000,000, runs 3 years, operating cost 800,000 per year. **System B**: costs 7,600,000, runs 5 years, operating cost 500,000 per year. Same duty; `r` = 8 %. (Costs, so lower is better.)
2. **Formula.** PV of cost = capex + opex × `AF(r, n)`; equivalent annual cost `EAC = PV / AF(r, n)`.
3. **Substitution.** A: `5,000,000 + 800,000 × 2.577097 = 7,061,678`; `EAC = 7,061,678 / 2.577097`. B: `7,600,000 + 500,000 × 3.992710 = 9,596,355`; `EAC = 9,596,355 / 3.992710`.
4. **Result.** A: **USD 2,740,168 per year**. B: **USD 2,403,469 per year** — B is cheaper by USD 336,699 every year, despite the 52 % higher purchase price.
5. **Interpretation.** Raw PV comparison (7.06m vs 9.60m) points the wrong way because it prices three years against five. EAV puts both on a per-year footing under the standard assumption that each system is replaced in kind at the end of its life. When that assumption fails — technology shifts, the duty ends — model the actual replacement chain instead (Domain 8's lifecycle costing).

AI in this KA

Screening portfolios is high-volume, formulaic work — the natural first place an organisation automates appraisal, and the first place systematic error industrialises. Two governed habits: the screening tool's formulae are validated once against this domain's golden examples (the registry pattern, `_build/verify_formulas.py`), and every auto-screened rejection above a materiality line gets a human review — a portfolio can lose its best project to one wrong sign convention, silently, forever.

KEY TERMS — KA 4.2

Term	Meaning
Payback / discounted payback	Years for cumulative (discounted) inflows to repay I_0; exposure measures.
PI	PV of inflows / I_0; value per scarce dollar, for rationing.
EAV / **EAC**	NPV (cost PV) converted to a level annual equivalent via AF(r, n).
Unequal lives	The comparison EAV exists to make honest.
Replacement chain	The explicit alternative when like-for-like replacement fails.

SAMPLE MCQS — KA 4.2

MCQ 4.2-A [4.2.1 · Application] For the master appraisal (60m in, 8.9m/yr, 8 %), the discounted payback is closest to: - A. 6.74 years - B. 10.07 years - C. 8.00 years - D. 15.00 years

Rationale: Cumulative PV reaches 59.72m after year 10; the 0.28m shortfall is 7 % of year 11's 3.82m discounted flow → 10.07. A is the simple payback; C confuses the discount rate with a duration; D is the whole life.

MCQ 4.2-B [4.2.3 · Application] System A (PV of costs 7,061,678 over 3 years, AF = 2.577097) versus System B (PV 9,596,355 over 5 years, AF = 3.992710). The correct comparison and choice is: - A. raw PV: A is cheaper, choose A - B. equivalent annual cost: A 2,740,168 vs B 2,403,469 — choose B - C. equivalent annual cost: A 2,353,893 vs B 1,919,271 — choose B - D. purchase price: A is cheaper, choose A

Rationale: Unequal lives require EAC: 7,061,678/2.577097 = 2,740,168 vs 9,596,355/3.992710 = 2,403,469. A and D compare unlike horizons; C divides each PV by its raw life in years (÷3 and ÷5), annualising without discounting — the right instinct with the wrong arithmetic.

MCQ 4.2-C [4.2.1 · Analysis] A board adopts "payback under 5 years" as its sole investment criterion. The predictable portfolio distortion is: - A. none — payback is conservative, so the portfolio is safe - B. systematic bias against long-lived infrastructure and toward short-cycle projects, regardless of value created - C. excessive investment in high-NPV projects - D. elimination of all risk

Rationale: Payback ignores everything beyond its cut-off, so a 15-year concession with NPV +16m loses to a 4-year project with NPV +1m. That is a value distortion, not conservatism (A); C reverses the effect; D confuses shorter exposure with no risk.

MCQ 4.2-D [4.2.2 · Recall] The profitability index earns its place in appraisal when: - A. projects are mutually exclusive and differ in scale - B. capital is rationed and the question is which portfolio of positive-NPV projects fits the budget - C. cash flows change sign more than once - D. lives are unequal

Rationale: PI ranks value per scarce dollar — the rationing question exactly. A is where PI (like IRR) mis-ranks by scale; C is MIRR's territory; D is EAV's.

SELF-CHECK — KA 4.2

1. Why is discounted payback always later than simple payback? — Discounting shrinks every inflow, so the cumulative line climbs more slowly (equal only if r = 0).
2. State Kestrel's PI and its meaning. — 1.270: each dollar invested buys 1.27 dollars of present value.

3. What assumption does EAV smuggle in, and when must you drop it? — Like-for-like replacement in perpetuity; drop it when the duty ends or technology shifts, and model the real chain.

KNOWLEDGE AREA 4.3

Decision contexts: exclusivity, rationing and judgment

Topics: 4.3.1 mutually exclusive investments · 4.3.2 capital rationing · 4.3.3 the limits of the numbers.

4.3.1 Mutually exclusive investments

The conflict. Choosing one project instead of another is where NPV and IRR famously disagree. Kestrel's board weighs two intake designs (same risk class, r = 8 %):

Design	I_0	Net inflow × 5 yrs	NPV @ 8 %	IRR
P (compact)	5,000,000	2,000,000	**+2,985,420**	**28.65 %**
Q (full-scale)	20,000,000	6,000,000	**+3,956,260**	**15.24 %**

IRR shouts P; NPV says Q adds USD 970,840 more money. The tie-breaker is the **incremental project** Q−P: invest a further 15,000,000 for a further 4,000,000 per year — an incremental IRR of **10.42 %**, above the 8 % cost of capital. The extra capital earns its keep, so take the bigger project. **Rule: when exclusivity forces a choice, rank by NPV; use incremental IRR only to narrate the same answer.** (At any hurdle above 10.42 % the ranking genuinely flips — the crossover rate is where the two NPV profiles intersect.)

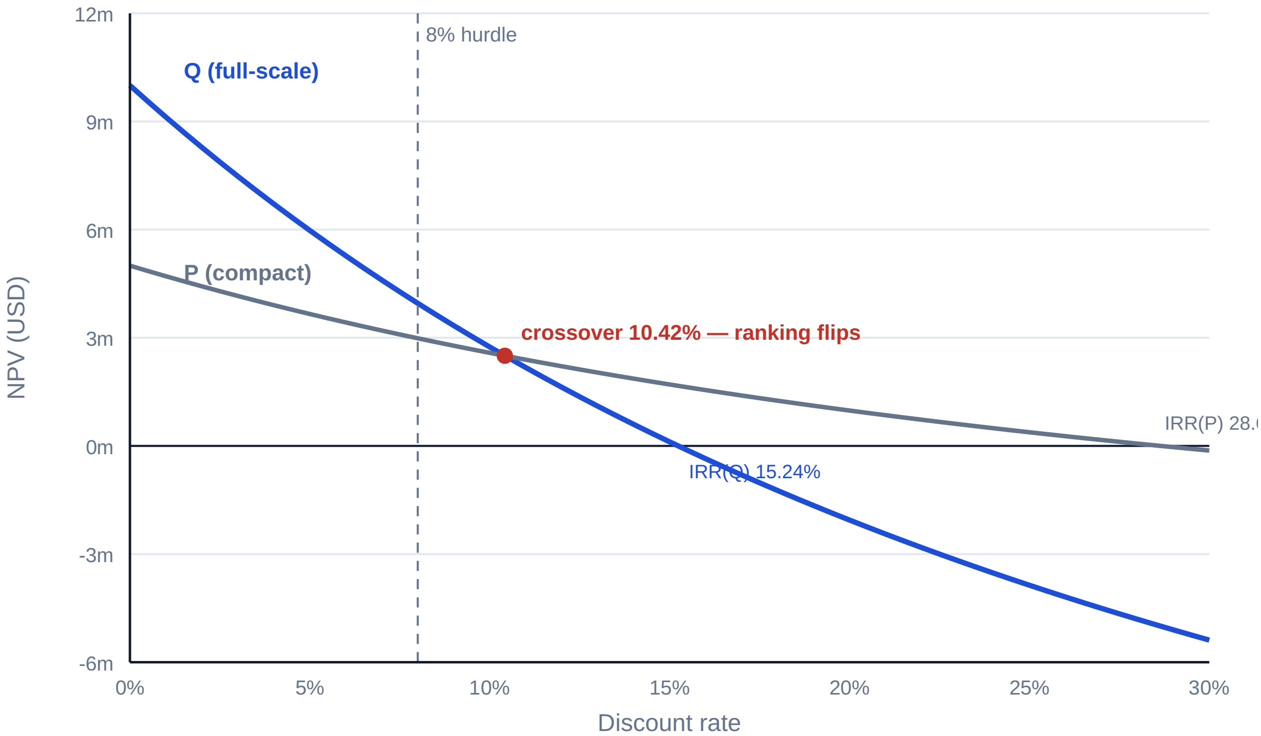

Fig 4.3.1 — Two NPV profiles and the crossover

4.3.2 Capital rationing

The problem. When the budget cannot fund every positive-NPV project, maximise **NPV per budget dollar** — rank by PI and pack the budget:

Project	I_0 (m)	NPV (m)	PI	Cumulative I_0
W	8	2.40	1.300	8
X	12	3.00	1.250	20 ← budget
Y	10	2.20	1.220	—
Z	6	1.02	1.170	—

With a USD 20m budget the PI ranking funds **W + X for NPV +5.40m**; the tempting "biggest NPV first, then fill" (X, then Y won't fit, then W) lands on the same 5.40m here, but the next-best feasible set (W + Y, +4.60m) shows what a wrong packing costs. Discipline points: PI packing is exact only when projects are divisible or the budget binds once — lumpy multi-period rationing is an optimisation problem (Domain 6's model does it honestly); and rationing itself deserves challenge, because turning away a 1.17-PI project is a financing failure as often as a screening success (Domain 9).

4.3.3 The limits of the numbers

Every measure in this domain consumes forecasts and a rate, and both are judgments wearing decimals. The professional frame: **appraisal quantifies; it does not decide strategy.** Numbers cannot see option value (the pilot that buys the right to scale), strategic foreclosure (the market position lost by not building), or the asymmetry of forecast error (Domain 8's ranges belong beside every point NPV). The board paper this domain endorses shows: NPV at the owned rate; the sensitivity of that NPV to the two or three assumptions that dominate it; IRR/MIRR as narrative; exposure via discounted payback; and the explicit statement of what the numbers cannot price. Anything less is theatre with spreadsheets.

AI in this KA

Ranking under constraints is combinatorial — machine territory — and constraint errors are silent. The governed pattern from Domain 6 applies unchanged: the optimiser proposes the portfolio; the analyst verifies the constraint set against reality (is the budget truly annual? are W and X really independent?); golden checks re-verify the NPVs being packed; and the leader owns the strategic overrides the model cannot see (4.3.3) — recorded as decisions, not adjustments smuggled into assumptions.

KEY TERMS — KA 4.3

Term	Meaning
Mutually exclusive	Choosing one forecloses the other; rank by NPV.
Incremental IRR	IRR of the difference project; narrates the NPV ranking.
Crossover rate	Rate where two NPV profiles intersect and rankings flip.
Capital rationing	Budget binds before value runs out; pack by PI.
Option value	Value of flexibility the static NPV cannot see.

SAMPLE MCQS — KA 4.3

MCQ 4.3-A `[4.3.1 · Application]` P: NPV +2,985,420, IRR 28.65 %. Q: NPV +3,956,260, IRR 15.24 %. Cost of capital 8 %; only one may proceed. The correct choice is: - A. P — higher IRR - B. Q — higher NPV, confirmed by an incremental IRR of 10.42 % above the hurdle - C. P — lower investment is always safer - D. both, split the budget

Rationale: Exclusive choices rank by money added: Q adds USD 970,840 more, and the extra 15m earns 10.42 % > 8 %. A is the scale-blindness pathology; C prices fear, not value; D violates the premise.

MCQ 4.3-B `[4.3.2 · Application]` Budget USD 20m. Projects (I_0, NPV): W (8, 2.4), X (12, 3.0), Y (10, 2.2), Z (6, 1.02). The value-maximising funded set is: - A. X and Y - B. W and X: NPV +5.40m - C. W, Y and Z - D. X and Z

Rationale: PI order W (1.300), X (1.250) packs the budget exactly for +5.40m. A needs 22m; C needs 24m; D fits (18m) but yields +4.02m — a 1.38m sacrifice for leaving W unfunded.

MCQ 4.3-C `[4.3.3 · Analysis]` A pilot plant shows NPV −800,000, but building it creates the option to deploy at 20× scale if the technology proves. The appraisal-sound treatment is: - A. reject — negative NPV is disqualifying - B. approve by overriding the NPV silently - C. present the static NPV alongside an explicit valuation (or structured judgment) of the scaling option, and decide on the combined case, recorded as such - D. raise the forecast cash flows until NPV turns positive

Rationale: Static NPV cannot see option value; the remedy is to price or judge the option in the open. A discards real value; B and D are the same dishonesty at different altitudes — one hides the judgment, the other disguises it as a forecast.

MCQ 4.3-D `[4.3.1 · Recall]` The crossover rate of two NPV profiles is: - A. the rate at which both projects' NPVs equal zero - B. the rate at which the two NPVs are equal — above it, the ranking flips - C. the average of the two IRRs - D. the cost of capital

Rationale: Crossover is the intersection of profiles (equivalently the incremental project's IRR). A describes two separate IRRs; C is arithmetic superstition; D is a property of the firm, not of the pair.

SELF-CHECK — KA 4.3

1. Why does incremental IRR always agree with NPV on a pairwise choice? — It tests whether the difference project clears the hurdle, which is exactly what the NPV difference measures.
2. When does PI packing fail as a rationing rule? — Lumpy projects and multi-period budget constraints; then it's an optimisation, not a sort.
3. Name three things a positive NPV cannot tell the board. — Exposure duration; sensitivity concentration; the value of flexibility and strategic position outside the modelled flows.

Advanced topics — Domain 4

4.A.1 Reading NPV profiles like a professional

The profile (Fig 4.1.1) compresses the whole appraisal into one curve: its height at 0 % is the undiscounted surplus; its slope measures duration-sensitivity (long-dated flows steepen it); its zero-crossing is the IRR; and the distance between the hurdle and the crossing is the margin for rate error. Two profiles per decision (Fig 4.3.1) add the crossover — the complete geometry of a mutually exclusive choice. A reviewer who asks for the profile instead of the point estimate has converted an argument about numbers into an inspection of shape.

4.A.2 Inflation-consistent appraisal

The Fisher discipline of Domain 3 (KA 3.3.1) binds hardest here: nominal flows with the nominal rate, or real flows with the real rate — and the same world for every line, including the terminal value. The classic appraisal defect is a nominal hurdle (board-set, inflation-inclusive) discounting real flows (an engineer's "today's money" forecast): NPV is systematically understated and good projects die politely. The assumption register names the world; the reviewer checks one line item end to end.

4.A.3 The reviewer's appraisal eye

The invariants: NPV at 0 % equals the raw flow sum; NPV at the IRR equals zero (substitute it back — the cheapest IRR audit that exists); MIRR lies between the reinvestment rate and the IRR; PI > 1 ⇔ NPV > 0 at the same rate; EAV × `AF(r, n)` reproduces the NPV; incremental IRR above the hurdle ⇔ the bigger project's NPV is higher. Any violated line is a defect somewhere — the appraisal analogue of Domain 3's factor-table checks, and wired into this programme's golden-answer harness.

Industry variations — Domain 4

- **Regulated utilities and availability PPPs.** Contracted revenue narrows the forecast range, so appraisal battles concentrate on the rate (and the regulator's allowed return); NPV margins are thin and Fisher errors (4.A.2) are decision-changing.
- **Merchant power and commodities.** The forecast is the battlefield: point NPVs are close to meaningless without the scenario set (Domain 7's stress tests), and payback recovers status as a survival measure — how long must prices hold?
- **Oil, gas and mining.** Decommissioning liabilities put the sign-change pathology (4.1.2) on almost every appraisal: MIRR and NPV are standing practice, "the IRR" is treated with suspicion by convention.
- **Technology and corporate transformation.** Option value (4.3.3) dominates: staged investments are the norm, and static NPV is a floor, not an answer. Rationing is annual and lumpy — the optimisation caveat of 4.3.2 is the daily reality.
- **Public-sector appraisal.** The discount rate is policy (a social time preference rate, set centrally), distributional effects sit beside NPV, and appraisal guidance is published — the numbers travel with a governance file, which is the direction this domain pushes every sector.

Case study — Domain 4: the intake decision, taken properly (water / desalination)

Situation. Kestrel's board must (1) confirm the plant investment, (2) choose between intake designs P and Q, and (3) allocate the sponsor group's residual USD 20m development budget across four unrelated feeder projects (W–Z of KA 4.3.2). One meeting, three different appraisal questions.

The decisions. Build: NPV +16.18m at 8 %, IRR 12.19 %, MIRR 9.73 %, discounted payback 10.07 years — build, with the exposure noted and the two dominating assumptions (tariff escalation, availability) sensitivity-tested per KA 4.3.3. Intake: Q over P — +970,840 more NPV, incremental IRR 10.42 % over the extra 15m; the 28.65 % on P is recorded as the intensity it is, not the ranking it isn't. Budget: W + X for +5.40m, with the un-funded 1.17-PI project Z referred to Domain 9's financing question rather than silently killed.

What the domain teaches here. Three questions, three measures, one hierarchy: NPV decided all three calls; IRR, MIRR, PI and payback each explained a facet the board needed narrated. The minute records rates, worlds (nominal), sensitivity owners and the option-value judgments — the appraisal file is the governance file.

Case study B — Domain 4: the fund that bought percentages (infrastructure fund)

Situation. An infrastructure fund with a 9 % cost of capital repeatedly preferred short-tenor, high-IRR deals — the P-shape — over long-tenor concessions at 13–15 % IRR. Five years on, its realised portfolio return trailed its own reported deal IRRs by nearly four points.

What happened. The gap was the reinvestment fiction industrialised: each 25–30 % IRR assumed its distributions would earn the same, but the fund redeployed at market rates — when it could redeploy at all. MIRR at honest reinvestment rates (the 4.1.3 arithmetic) had flagged the true economics of every deal; it was computed, and filed. The long concessions it passed over — lower IRR, far higher NPV per dollar and decade — went to competitors who now hold them.

What the domain teaches here. A portfolio of intensities is not a portfolio of value. IRR's flattery compounds at fund level because short deals recycle the fiction; NPV and MIRR discipline is not academic tidiness — over a fund's life it is the difference between the return promised and the return delivered.

Executive perspective — Domain 4

What a project finance director cannot delegate in this domain:

- **The hurdle.** As in Domain 3: whoever sets `r` sets the answer. The director owns the rate, its world (nominal or real), and the standing sensitivity band around it.
- **The measure hierarchy.** NPV decides; rates and ratios explain. The director enforces this in every paper — and personally reads the profile, not the point.
- **The exclusivity discipline.** Incremental thinking (Q−P) is the director's question in the room: "what does the extra money earn?" beats "which percentage is bigger?" every time.
- **The honesty of rejections.** Under rationing, what was not funded and why is board information; a rejected 1.17-PI project is a financing gap wearing a screening verdict.
- **The unpriced.** Option value, strategic position, forecast asymmetry — the director's overrides are legitimate exactly when they are explicit, minuted and owned (4.3.3).

Calculation exercises — Domain 4

> **Exercise 4.1** I_0 = 12,000,000; net flows 3,000,000 · 4,000,000 · 5,000,000 · 5,000,000 in years 1–4; `r` = 10 %. Compute the NPV. Solution. PV = $3/1.10 + 4/1.10^2 + 5/1.10^3 + 5/1.10^4$ (millions) =

> 2,727,273 + 3,305,785 + 3,756,574 + 3,415,067 = 13,204,699; **NPV = +1,204,699**. Common error: discounting year 1 at `t` = 0 (no discount) shifts every factor one period and overstates NPV by ≈ 1.19m.

> **Exercise 4.2** I_0 = 9,000,000; level inflow 1,500,000 for 10 years. Find the IRR. Solution. `AF(r, 10) = 9/1.5 = 6.0`; solving gives **IRR = 10.56 %** (AF(10 %) = 6.145, AF(11 %) = 5.889 bracket it). Common error: reading the payback reciprocal (1.5/9 = 16.7 %) as the IRR — that shortcut holds only for perpetuities.

> **Exercise 4.3** I_0 = 10,000,000; inflows 2,600,000 for 6 years; reinvestment and finance rate 7 %. Compute MIRR. Solution. `FVAF(0.07, 6) = 7.153291`; TV = 18,598,556; `MIRR = (1.859856)^(1/6) − 1 =` **10.90 %**. Common error: compounding the inflows for 6 periods each (rather than to a common terminal date) inflates TV and the rate.

> **Exercise 4.4** Machine C1: 3,000,000 capex, 4-year life, 250,000/yr opex. Machine C2: 4,200,000, 6-year life, 150,000/yr. Same duty; `r` = 9 %. Choose. Solution. C1: PV = 3,000,000 + 250,000 × 3.239720 = 3,809,930; EAC = **1,176,006**. C2: PV = 4,200,000 + 150,000 × 4.485919 = 4,872,888; EAC = **1,086,263**. **Choose C2** — cheaper by 89,743 per year. Common error: comparing PVs across unequal lives (which picks C1).

> **Exercise 4.5** Budget 25,000,000. Projects (I_0; NPV): A (10; 2.60) · B (15; 3.30) · C (8; 1.84) · D (7; 1.40). Fund the best set. Solution. PIs: A 1.260 · C 1.230 · B 1.220 · D 1.200. **Greedy PI packing fails here**: A + C + D fits (25) for +5.84m, but **A + B (25 exactly) yields +5.90m**. With lumpy projects the rule is: use PI to shortlist, then enumerate feasible sets (or optimise). This is the 4.3.2 caveat as arithmetic.

Practitioner's toolkit — Domain 4

Adoption-ready artefacts; adapt headings to your organisation, then keep them stable.

Toolkit 4.T.1 — The appraisal one-pager (per decision)

Decision question (accept/reject · exclusive choice · rationing) · cash-flow source and world (nominal/real, per 3.T.1) · rate and its owner · **NPV at the owned rate** · profile sketch or IRR/MIRR · discounted payback · PI (if rationing) · EAV (if unequal lives) · two-way sensitivity on the dominating assumptions · unpriced factors, stated · recommendation and the named accountable approver.

Toolkit 4.T.2 — IRR pathology checklist

- [ ] Sign changes counted; more than one → MIRR + NPV only, "the IRR" banned from the paper.
- [ ] Scale stated beside every percentage (I_0 and NPV in money).
- [ ] Tenors aligned before any cross-project rate comparison (else EAV).
- [ ] Reinvestment assumption named; MIRR computed when IRR > hurdle + 5 points.
- [ ] IRR substituted back: NPV(IRR) recomputes to zero.
- [ ] Incremental IRR computed for every exclusive pair; agrees with NPV ranking.

Toolkit 4.T.3 — Rationing worksheet

Columns: project · I_0 · NPV · PI · strategic notes/dependencies. Steps: rank by PI → shortlist to ~2× budget → enumerate feasible sets (lumpy) → record funded set, rejected set with reasons, and the financing referral for high-PI rejects (Domain 9).

Exam preparation — Domain 4

The calculation traps. Forgetting to deduct I_0 (MCQ 4.1-A distractor D) · discounting from $t = 0$ (Exercise 4.1) · payback reciprocal as IRR (Exercise 4.2) · ranking exclusives by IRR (4.3.1) · greedy PI with lumpy budgets (Exercise 4.5) · comparing PVs across unequal lives (Exercise 4.4) · quoting one IRR for sign-changing flows (4.1.2) · MIRR terminal value mis-compounded (Exercise 4.3) · nominal/real mixing imported from Domain 3.

Reflection questions. 1. Your board paper shows one number: "IRR 22 %". List the five questions this domain obliges you to ask before anyone votes. (Scale? Sign changes? Tenor? Reinvestment? NPV at our rate?) 2. When is a negative-NPV decision defensible — and what must the record contain? (4.3.3: explicit option/strategic value, owned and minuted.) 3. Which invariant of 4.A.3 would have caught the last appraisal error you saw? (Most often: NPV(IRR) ≠ 0 on substitution, or EAV × AF ≠ NPV.)

Domain 4 summary

Appraisal turns Domain 3's machinery into decisions, under one hierarchy: **NPV decides; everything else explains.** NPV earns primacy by additivity, scale-awareness and the honest reinvestment assumption; IRR narrates a margin of safety but carries three standing pathologies — multiple roots, scale-blindness, the reinvestment fiction — of which MIRR repairs only the last; payback and its discounted form measure exposure, never value; PI ranks value per scarce dollar under rationing (with the lumpy-budget caveat proven in Exercise 4.5); EAV makes unequal lives commensurable. Mutually exclusive choices resolve by NPV, narrated by incremental IRR and the crossover geometry; rationing resolves by disciplined packing plus the financing question for what was turned away; and the limits of the numbers — options, strategy, forecast asymmetry — are stated in the open, as owned judgments. The Kestrel thread continues into Domain 5 (is the project bankable, not merely valuable?) and Domain 6 (the model that industrialises every calculation this domain taught).

02

PART TWO

Structuring and modelling

Domains 5–9 — development and bankability, financial modelling, revenue and demand, cost and contingency, and funding structure: how a project becomes financeable.

05

DOMAIN 5

Project Development and Bankability

Group: Structuring and modelling (Domain 5 of 5 in Part Two). **Target:** ~75 pages. **Binds to:** the PCI Book Pattern Specification and the shared registries (`docs/books/registries/`). This domain is the home of **bankability**, the **special-purpose vehicle** and the development option premium. It consumes Domain 4's appraisal verdict and hands Domain 6 a structure to model. Coverage machinery (`DSCR`, `LLCR`, max debt capacity) is **cited from Domain 10, never re-derived here.** British English; USD (+SAR where useful, indicative `USD 1 ≈ SAR 3.75`). Tax, accounting and legal treatments described here are **illustrative and jurisdiction-specific**; none is presented as universal. In particular, whether a liquidated-damages rate or cap, a dilution or conversion mechanic, or a stated liability basis is enforceable as drafted is a matter for qualified counsel in the governing jurisdiction, and nothing in this domain is legal advice.

Why this domain exists

Domain 4 established that Kestrel Water SPC is **valuable** — `NPV` +USD 16,179,360 at 8 %, `IRR` 12.19 %, `PI` 1.270. It left open the question that decides whether the project happens: **is it financeable?** A valuable project with an unsigned offtake, an unregistered corridor, an unproven technology or an unfundable minority sponsor does not get built, and no positive `NPV` repairs any of those. Bankability is not a stronger form of value but a different test, and it is **conjunctive** — every condition must pass, because lenders lending against project cash alone have no balance sheet to fall back on and therefore no appetite for one unresolved fatal condition.

That structure is the domain's central claim, and it has three computable consequences. **The weakest condition governs the whole:** six conditions each around 90 % likely give a joint probability of close near 55 %, so effort on the strongest is nearly worthless while effort on the weakest is worth several times as much (KA 5.3). **Development spend is an option premium, not a project cost:** it buys the right to continue, is mostly spent on projects that never close, and must be judged across a portfolio and killed early (KA 5.1) — the finance twin of the gate economics PML-AI Domain 3 (KA 3.3.1) priced for delivery, differing only in what holding the gate costs. And **bankability is a set of contracts and dates, not an opinion:** the sponsor group and its vehicle (KA 5.2) and the construction and readiness package (KA 5.4) either deliver the conditions on the timetable or impose a cost this domain quantifies to the day.

Learning objectives. After this domain a candidate can: place development kill gates where irreversibility steps up; compute cost per closed project, close rate and breakeven close rate across a development portfolio; compute a feasibility gate's net value, breakeven detection probability and the bid-window risk at which it stops paying; explain what an SPV achieves and what its ring-fence does not; compute per-sponsor equity, several-liability support exposure and the dilution consequence of a funding default; value an equity bridge and show it is neutral at the bridge rate; compute a condition set's joint probability, the marginal gain from lifting each condition and the breakeven close probability; price a technology premium as lost debt capacity using Domain 10's sizing rule; compute the full cost of a slip in the commercial operations date and size a performance buy-down that restores lenders' coverage; and govern AI use across screening, document review and readiness assessment.

The master project, at its development stage. Kestrel Water SPC continues from Domains 1–4 and 10. Capital cost **USD 60,000,000**, funded **70/30** as **USD 42,000,000** of senior debt at **6.0 % over 12 years** (annual instalment **USD 5,009,635.23**; `AF(0.06, 12) = 8.383844`) plus **USD 18,000,000** of equity. Operating life **25 years**; documented first-year `CFADS` **USD 6,384,000** (6,984,000 before working-capital movements) on revenue of **12,000,000** and cash operating costs of **4,500,000**. Coverage at close (Domain 10): `DSCR` **1.2743** = `LLCR` **1.2743**, `PLCR` **1.9431**, covenant cash trigger **6,011,562**, annual headroom **372,438**. This domain adds three development facts earlier chapters did not need: the **portfolio** that produced Kestrel (40 screened opportunities, USD 14,800,000 of programme spend, two closes), the **sponsor group** behind the 18,000,000 (water operator 55 %, infrastructure fund 35 %, industrial partner 10 %, with a several 6,000,000 overrun support), and the **two-year construction window** whose commercial operations date the whole structure is priced against.

KNOWLEDGE AREA 5.1

Concept and feasibility

Topics: 5.1.1 the development lifecycle and where its gates belong · 5.1.2 development spend as an option premium · 5.1.3 the feasibility gate and the kill decision.

5.1.1 The development lifecycle and where its gates belong

Definition. Project development is the pre-financial-close work that converts an opportunity into a financeable transaction — identifying the payer, securing site and consents, selecting technology and contractors, negotiating the revenue contract, structuring the financing, satisfying every condition precedent. It runs through origination and screening, concept and prefeasibility, feasibility, transaction and bid, and close, with spend intensity rising by roughly an order of magnitude at each step while information improves only gradually. That asymmetry — **commitment rising faster than knowledge** — is the economics of the whole phase, and everything before close is spent at sponsors' risk with no financing to reimburse it.

Gates belong where **irreversibility steps up** — before feasibility spend, before bid submission, before signature — not on the calendar; the design rule and its failure mode (gates that have never stopped anything) are PML-AI Domain 3's, KA 3.3.1. What differs in finance is the price of holding the gate: in delivery it is elapsed time against a cost of delay, while in development it is a **forgone option** — a bid window that closes, a site optioned by a competitor, a procurement that runs without you. The budget must also separate **pursuit costs** (expensed, belonging to the programme), **study and adviser costs** (often capitalisable at close where the framework and the facility permit — Domain 2's rules, jurisdiction-sensitive) and **at-risk commitments** (option premiums, bid bonds, long-lead reservations: cash forfeited if the project dies), because without that split a kill decision cannot answer its only question — how much of this is already gone?

5.1.2 Development spend as an option premium

Definition. Each stage of development spend buys a **real option**: the right, not the obligation, to spend the next and larger tranche. It must therefore be evaluated as a **portfolio** — the closes pay for the abandonments — and the correct unit is the **cost per closed project across everything pursued**, not the cost of the deal that closed.

WORKED EXAMPLE

Worked example 5.1.2 — what did Kestrel's financing actually cost to originate?

1. **Setup.** Kestrel's sponsor group runs a four-stage water and utilities funnel. In the year of Kestrel's origination it **screened 40** opportunities at USD 25,000 each; **12** advanced to concept at USD 250,000 each; **5** entered feasibility and bid at USD 1,200,000 each; **2** were carried to close at a further USD 2,400,000 each. Value per close is Domain 4's Kestrel NPV, **USD 16,179,360** (cited, not re-derived).
2. **Formula.** Stage spend = count × unit cost; programme spend = Σ stage spend; cost per close = programme spend ÷ closes; close rate = closes ÷ screened; breakeven closes = programme spend ÷ value per close.
3. **Substitution.** `40 × 25,000 = 1,000,000`; `12 × 250,000 = 3,000,000`; `5 × 1,200,000 = 6,000,000`; `2 × 2,400,000 = 4,800,000`; per close `= 14,800,000/2`; breakeven `= 14,800,000/16,179,360`.
4. **Result.** Programme spend **USD 14,800,000**; **cost per closed project USD 7,400,000**; close rate **5.0 %** (stage conversions 30.0 %, 41.67 %, 40.0 %); portfolio value **32,358,720**, net **+17,558,720**, value multiple **2.1864×**; **breakeven at 0.9147 closes — a breakeven close rate of 2.29 %**.
5. **Interpretation.** The honest cost of Kestrel's financing is **7,400,000, not the 2,400,000** on its own charge code, and the difference is the premium paid for having found it at all; a sponsor that measures only the winners under-prices development, under-resources screening and is annually surprised by the expense line. The **breakeven close rate of 2.29 % against an achieved 5.0 %** gives the programme **2.19× of margin** — it could halve its hit rate and still create value — which converts "development is expensive" into "development stops paying below one close in 44 screenings", the sentence a budget is actually defended with (the exact breakeven is one close in 43.73). And sensitivity runs through value per close and late-stage conversion, not screening cost: the 1,000,000 of screening is 6.8 % of programme spend and buys the whole funnel, while one lost late-stage project costs 2,400,000 outright — so the governance conclusion is counter-intuitive and consistent, **screen more widely and kill earlier**. The caution: value per close is a forecast (Domain 4, KA 4.3.3), so a programme justified on optimistic deal NPVs has hidden its true breakeven and must be re-tested against **realised** value.

Fig 5.1.1 — The development funnel as an option premium

5.1.3 The feasibility gate and the kill decision

Definition. A feasibility gate is the point at which a project must demonstrate that its **fatal conditions** are capable of being satisfied before the transaction tranche is committed. Its function is not to improve the project but to **stop the unfinanceable ones while stopping is cheap.** The characteristic pathology is the reverse: a project with one unresolvable condition is carried late because everything else about it is attractive, and dies in diligence having consumed the whole transaction budget plus abortive adviser fees.

WORKED EXAMPLE

Worked example 5.1.3 — the gate that pays, and the bid window that kills it.

1. **Setup.** Of the five projects entering feasibility, the sponsor group's own post-mortem record on comparable pursuits puts at **40 %** the share carrying a fatal bankability condition (a pipeline statistic, not a market one). A conditions review costs **USD 180,000** per project and detects such a flaw with probability **0.75**. A flaw surviving the gate is found in diligence, by which point the project has spent its **2,400,000** transaction tranche plus **900,000** of abortive external fees — **3,300,000** wasted. The gate adds **8 weeks**, and in a competitive concession procurement that carries a **10 %** probability of missing the bid window, forgoing a project worth **16,179,360**.
2. **Formula.** The gate-net-value structure of PML-AI Domain 3 (KA 3.3.1), applied to development waste rather than build rework: waste without the gate `= P(flaw) × waste`; with the gate `= gate cost + P(flaw) × P(miss) × waste`; net value is the difference. Breakeven detection solves `gate + P(flaw)(1 − p) × waste = P(flaw) × waste`. Option cost of delay `= P(window missed) × value per close`.
3. **Substitution.** Without: `0.40 × 3,300,000`. With: `180,000 + 0.40 × 0.25 × 3,300,000`. Breakeven `p = 1 − (1,320,000 − 180,000)/(0.40 × 3,300,000)`. Window: `0.10 × 16,179,360`.
4. **Result.** Expected waste **1,320,000** without the gate, **510,000** with it — **gate net value USD 810,000 per project entering feasibility**, or **4,050,000** across five. The gate pays down to a **detection probability of 13.64 %**. But the bid-window option cost is **1,617,936**, so net value becomes **−USD 807,936**, and the **breakeven window-miss probability is 5.01 %**.
5. **Interpretation.** The same gate is worth +810,000 in a bilateral negotiation and −807,936 in a competitive tender, and nothing about the gate changed — only what the delay costs. That is the finance-specific lesson: **in development, elapsed time is priced as a lost option, not as a carrying cost**, and the option can be worth nine times the study fee. The **13.64 % breakeven detection rate** is a very low bar, which is why conditions reviews almost always pay when they run in parallel: a reviewer who can read a title register, a grid-queue position and an offtaker's credit standing inside the existing timetable is close to free money. The **5.01 % breakeven window-miss probability** is what the gate's design is negotiated with — run the review concurrently, stage it behind a two-week fatal-flaw screen, or bid conditionally; a gate whose delay risk is left unquantified will be abolished by the first commercial team that misses a deadline, taking the 810,000 with it. The caution: the 40 % flaw rate and 0.75 detection rate are **empirical claims about your own pipeline**, so a programme that has never recorded why its abandoned projects died cannot populate them — which makes the abandoned-project post-mortem the first artefact to build.

AI in this KA

Screening is the strongest legitimate machine application here: assembling opportunity long lists, extracting site and consent facts from public registers, tabulating comparables, first-pass filtering. Two boundaries. **A machine must never own the kill decision** — a kill destroys an option irreversibly and the model cannot see the strategic value the funnel exists to build; automated rejections above a materiality line get a named human reviewer. And **a model will report a condition as satisfied when it has found a document that mentions it**: the 40 % flaw rate of 5.1.3 is populated by exactly those conditions — an easement that stops 200 metres short, a permit issued to a predecessor entity, an offtaker whose credit sits in another group company. Verification runs to the primary source on every fatal condition, sampled and signed. **AI proposes; the professional verifies, decides and remains accountable.**

KEY TERMS — KA 5.1

Term	Meaning
Financial close	The point at which funding documents become effective and drawdown is available.
Option premium (development)	Stage spend buying the right, not the obligation, to spend the next tranche.
Development cost per close	Programme spend ÷ closes; the honest cost of one financing.
Breakeven close rate	Programme spend ÷ (value per close × opportunities screened).
At-risk commitment	Option premiums, bid bonds, reservations — cash forfeited if the project dies.
Fatal condition	A condition whose failure ends the project regardless of everything else.

SAMPLE MCQS — KA 5.1

MCQ 5.1-A `[5.1.2 · Application]` A programme spends 1,000,000 screening, 3,000,000 on concept, 6,000,000 on feasibility and bid and 4,800,000 carrying two projects to close. Development cost per closed project is: - A. USD 2,400,000 - B. USD 7,400,000 - C. USD 4,800,000 - D. USD 14,800,000

Rationale: `14,800,000/2 = 7,400,000`. A is the closing-stage unit cost, excluding the portfolio that produced the winners; C is the whole closing stage undivided; D attributes the entire programme to one project.

MCQ 5.1-B `[5.1.2 · Analysis]` A programme spends 14,800,000 across a funnel of 40 screened opportunities and delivers 16,179,360 of value per close. Its breakeven close rate is closest to: - A. 5.00 % - B. 2.29 % - C. 45.74 % - D. 91.47 %

Rationale: breakeven closes `= 14,800,000/16,179,360 = 0.9147; ÷ 40 = 2.29 %`. A is the achieved close rate; C divides by the 2 closes achieved rather than the 40 screened; D quotes the breakeven closes as if it were a percentage.

MCQ 5.1-C `[5.1.3 · Application]` A fatal flaw is present with probability 0.40 and costs 3,300,000 if it survives to diligence. A gate costing 180,000 detects it with probability 0.75. The gate's net value, ignoring elapsed time, is: - A. USD 1,320,000 - B. USD 810,000 - C. USD 990,000 - D. USD 330,000

Rationale: `1,320,000 − [180,000 + 0.40 × 0.25 × 3,300,000] = 810,000`. A is the expected waste without the gate; C omits the gate's 180,000 cost; D is the residual expected waste after it.

MCQ 5.1-D `[5.1.3 · Analysis]` A gate worth **+810,000** per project before elapsed time is counted adds 8 weeks and so carries a 10 % chance of missing a bid window on a project worth 16,179,360. The correct conclusion is: - A. the gate still pays, since 810,000 is positive - B. as designed it destroys 807,936 of value, so it should be run concurrently or staged rather than abolished - C. it should be abolished, since delay always dominates in competitive procurement - D. bid-window risk is not a financial cost and is excluded

Rationale: `810,000 − 0.10 × 16,179,360 = −807,936` against a 5.01 % breakeven window-miss probability — the design, not the review, is at fault. A ignores the option cost; C forfeits 810,000 of detection value; D is the omission the arithmetic exists to prevent.

SELF-CHECK — KA 5.1

1. Why measure development spend per closed project? — The spend buys options across a portfolio, so closes must repay abandonments: 7,400,000, not 2,400,000, is Kestrel's origination cost.
2. What replaces "cost of delay" as the price of a development gate? — The forgone option: 5.01 % of value per close is Kestrel's breakeven window-miss probability.
3. Where do development gates belong? — Where irreversibility steps up: before feasibility spend, before bid submission, before signature.

KNOWLEDGE AREA 5.2

Sponsors and special-purpose vehicles

Topics: 5.2.1 sponsors and what each brings · 5.2.2 the SPV and the limits of its ring-fence · 5.2.3 the shareholders' agreement: shares, support and dilution · 5.2.4 funding the equity and the equity bridge.

5.2.1 Sponsors and what each brings

Definition. A **sponsor** in this book's project-finance sense is an equity investor promoting the project — distinct from PML-AI's use of the word for the accountable owner of a business case (terminology registry). Sponsors contribute four separable things, and a group is assembled to assemble them: **capital**, **capability** (operating, technical or construction competence lenders will underwrite), **access** (market position, offtake relationships, host-country standing) and **credit** (a balance sheet able to stand behind the support obligations of 5.2.3). Each archetype supplies some and lacks others: an **industrial** sponsor brings technology and O&M capability but raises conflict questions where it also supplies the EPC or O&M contract; a **fund** brings capital and structuring discipline with a finite life and no operating capability; a **contractor** brings completion commitment and the suspicion that its equity is a device to win the works; a **host-country partner** brings consents and durability and is usually the smallest and least creditworthy holder (Case study B). The task at group formation is to state in the shareholders' agreement **what each sponsor is relied on for**, because that is what is tested at close — and a group whose only creditworthy member is also the contractor will find its completion support discounted for correlation, since the party guaranteeing completion is the party whose failure causes the claim.

5.2.2 The SPV and the limits of its ring-fence

Definition. A **special-purpose vehicle** is the ring-fenced legal entity created to own, finance and operate a single project; Domain 1 (KA 1.1.3, Fig 1.1.2) established it as the hub of the contract structure. It achieves risk containment (a failed project's creditors reach the vehicle, not the sponsors, beyond agreed support), security (lenders take security over the whole of a single-purpose entity, so enforcement delivers a working project rather than assets scattered through a group), clean cash (one set of flows, so `CFADS` is measurable and the waterfall enforceable) and credit separation (the project stands on its own contracts, not the weakest sponsor's rating). Its costs are extra documentation layers, standalone audit, tax and secretarial functions, and lender restrictions on the vehicle's own decisions.

What it does not do, each of which has surprised a sponsor:

- It does not survive **contractual reach-through**: completion guarantees, cost-overrun support, equity commitment letters, O&M performance guarantees and EPC parent guarantees each puncture the fence deliberately, and their aggregate is the sponsor's real exposure (5.2.3).
- It does not decide **accounting consolidation**, which turns on control and the applicable reporting framework — a jurisdiction-specific question for the sponsor's own auditors — nor simplify **tax**, since an added entity brings withholding questions and thin-capitalisation limits that can cap deductible interest, all for qualified tax counsel and all capable of changing after-tax cash.
- It does not remove **reputational exposure**, nor by itself deliver **bankruptcy remoteness**, which is engineered by restrictions in the constitutional and finance documents.

5.2.3 The shareholders' agreement: shares, support and dilution

Definition. The shareholders' agreement fixes the vehicle's economics and control: each sponsor's **equity share**, the **funding obligation** for base equity and support commitments, the **liability basis** for those commitments, reserved matters, transfer restrictions, deadlock resolution and the **default and dilution mechanics** applied when a shareholder fails to fund. The most consequential term is the liability basis: under **several** liability each sponsor owes its own share and no more, while under **joint and several** liability each may be pursued for the whole. Lenders prefer the latter because it converts a group of commitments into one strong one; sponsors resist it because it makes each of them the backstop for the least creditworthy member. Most limited-recourse structures land on several liability with **credit support** — a bank letter of credit or acceptable parent guarantee — required from any sponsor whose own credit falls below the lenders' threshold.

WORKED EXAMPLE

Worked example 5.2.3 — what each Kestrel sponsor has actually committed.

1. **Setup.** Kestrel's USD 60,000,000 capital cost is funded **70/30**: USD 42,000,000 senior debt and **USD 18,000,000** equity, held by a **water operator at 55 %**, an **infrastructure fund at 35 %** and an **industrial partner at 10 %**. The facility requires a **several cost-overrun support of 10 % of capital cost**, subscribed pro rata.
2. **Formula.** `D/E` = debt ÷ equity. Per sponsor: equity = total equity × share; support = support pool × share; committed = equity + support.
3. **Substitution.** `42,000,000/18,000,000`; pool `= 60,000,000 × 0.10 = 6,000,000`; `18,000,000 × 0.55` and `6,000,000 × 0.55`; and so on.
4. **Result.** Gearing **2.3333 : 1** (debt 70.0 % of capex).

Sponsor	Share	Base equity	Several support	Total committed
Water operator	55 %	9,900,000	3,300,000	**13,200,000**
Infrastructure fund	35 %	6,300,000	2,100,000	**8,400,000**
Industrial partner	10 %	1,800,000	600,000	**2,400,000**
Group	100 %	**18,000,000**	**6,000,000**	**24,000,000**

Group committed capital is **40.0 % of capital cost**, not the 30 % headline; every sponsor is committed **33.3 %** beyond its equity share. 5. **Interpretation.** The gap between the **30 % headline and the 40 % commitment** is where sponsor boards are most reliably surprised, and it matters three ways. For **capital planning**, the fund's board approved 6,300,000 and is exposed to 8,400,000 — a difference that belongs in its own commitment register, not a footnote. For **credit**, the partner's 2,400,000 is the smallest number in the table and the likeliest to fail: a 10 % holder is frequently the least creditworthy member, so lenders will require its support to be backed by a letter of credit as a condition precedent — a **timetable risk** outside the sponsors' control, which Case study B prices at eleven weeks. For **negotiation**, each point of support costs the group 600,000, so a lender's move from 10 % to 15 % is a 3,000,000 increase worth pricing against a coverage or tenor concession (Domain 10's four levers). The professional caution is the liability basis: make the support **joint and several** and each sponsor's worst case becomes its own equity plus the **whole** 6,000,000 pool, so the partner's exposure rises from 2,400,000 to **7,800,000** — a factor of **3.25** — while the operator's rises only from 13,200,000 to **15,900,000**, or **20.5 %**. Two words in one clause transfer most of the small holder's protection to the large ones, which is why lenders ask for them and why a small holder must not sign them lightly. It is a question for qualified counsel in the governing jurisdiction, not a modelling assumption.

5.2.4 Funding the equity, and the equity bridge

Equity may be funded **pro rata** with debt, **front-ended** (lenders' preference — sponsors' money at risk before theirs), **back-ended** (sponsors' preference, permitted only against a firm commitment), or through an **equity bridge loan** that funds the equity portion during construction and is repaid by the sponsors at completion against their commitment letters.

Price the bridge on Kestrel. Construction runs **two years**; pro rata the sponsors would contribute **9,000,000 at close** and **9,000,000 a year later**; a bridge at **5.5 %** funds both and is repaid at t = 2. Bridge interest is `9,000,000 × (1.055² − 1) = 1,017,225` plus `9,000,000 × 0.055 = 495,000` = **1,512,225**, so the repayment is **19,512,225**. Discounted at the bridge rate the two profiles are **identical at USD 17,530,806** (`19,512,225/1.055²` against `9,000,000 + 9,000,000/1.055`) — a difference of **zero**. At the sponsors' **12 %** equity requirement — an indicative sponsor hurdle used for illustration here, not a derived cost of equity; Domain 9 (KA 9.1.3) builds Kestrel's `k_e` up from its components — the bridge profile is worth **15,555,026** against **17,035,714**, a **saving of USD 1,480,688**, or **8.69 %**.

The identity is the point: **an equity bridge creates no project value.** It is an arbitrage between the bridge rate and the sponsors' required return — a 6.5-point spread over an average deferral of about eighteen months — so the saving is **a financing return, not a project return**, and the equity `IRR` uplift it produces is exactly the flattery Domain 4 (KA 4.1.2) warned about, since the money simply went in later. Two further consequences: the bridge is **only available against firm commitments**, converting an equity-timing preference into a credit-quality test the group may fail; and it **adds 1,512,225 to project cost**, which if capitalised into senior debt consumes coverage by the mechanism 5.4.2 quantifies.

AI in this KA

Strong: extracting funding obligations, liability bases, reserved matters, transfer restrictions and default mechanics from a shareholders' agreement into a structured commitment register — the artefact 5.2.3 shows most sponsor boards lack — and reconciling it to the finance documents' conditions precedent so no support obligation is discovered late; and maintaining per-sponsor exposure across a portfolio of vehicles, where one sponsor's aggregate several commitments across ten SPVs is a number no single project team can see. **Not here:** liability basis and default consequences are **legal conclusions**, and a summary reporting "several" where the clause says otherwise misstates a balance sheet by a factor. Every such reading is verified against the executed document by qualified counsel, with verifier and date recorded.

KEY TERMS — KA 5.2

Term	Meaning
Sponsor (project finance)	Equity investor promoting the project; brings capital, capability, access and credit.
Special-purpose vehicle (SPV)	The ring-fenced single-purpose entity that owns, finances and operates the project.
Several liability	Each sponsor owes only its own share; joint and several exposes each to the whole.
Cost-overrun support	A sponsor commitment to fund construction cost above budget, up to a cap.
Committed capital	Base equity + support commitments; the real exposure (40.0 % of capex for Kestrel).
Equity bridge loan	Construction-period facility funding equity, repaid by sponsors at completion.

SAMPLE MCQS — KA 5.2

MCQ 5.2-A `[5.2.3 · Application]` Equity of 18,000,000 is split 55/35/10, with a several cost-overrun support of 10 % of a 60,000,000 capital cost subscribed pro rata. The 35 % sponsor's total committed capital is: - A. USD 6,300,000 - B. USD 8,400,000 - C. USD 2,100,000 - D. USD 12,300,000

Rationale: `18,000,000 × 0.35 = 6,300,000` plus `6,000,000 × 0.35 = 2,100,000`. A is base equity only, **25.0 % below** the committed figure — the omission the example corrects, and the same gap seen the other way round as the 33.3 % that support adds to every sponsor's equity share; C is the support alone; D reads the support as joint and several, adding the whole 6,000,000 pool.

MCQ 5.2-B `[5.2.3 · Analysis]` The 55 % sponsor holds 9,900,000 of an 18,000,000 equity ticket and subscribes pro rata to a 6,000,000 cost-overrun support pool. The agreement is amended from several to joint and several liability for that support. The 55 % sponsor's worst-case exposure becomes: - A. unchanged at 13,200,000 - B. 15,900,000 — its own 9,900,000 of equity plus the whole 6,000,000 pool - C. 24,000,000 - D. 6,000,000

Rationale: the sponsor becomes pursuable for the entire support commitment: `9,900,000 + 6,000,000`, a 20.5 % rise against a 3.25× rise for the 10 % holder. A is the several answer; C is the group's total committed capital, applicable only if equity subscriptions were also joint and several; D omits the sponsor's own equity.

MCQ 5.2-C `[5.2.4 · Analysis]` An equity bridge at 5.5 % replaces pro-rata equity of 9,000,000 at t = 0 and t = 1 with 19,512,225 at t = 2. At 5.5 % both profiles are worth 17,530,806. The correct conclusion is: - A. the bridge creates 1,512,225 of value - B. it creates no value at the bridge rate; its benefit is entirely the spread between the bridge rate and the sponsors' required return - C. it destroys 1,512,225 of value - D. it is value-neutral at every discount rate

Rationale: identical present values prove neutrality at that rate; at 12 % the saving is 1,480,688. A treats accrued interest as a gain; C treats it as a pure cost and ignores the deferral; D generalises the identity beyond the rate at which it holds.

MCQ 5.2-D `[5.2.2 · Analysis]` A sponsor states that the SPV caps its exposure at its equity subscription. The most accurate correction is: - A. correct — that is the purpose of the ring-fence - B. the fence is punctured by every support obligation given, and consolidation, tax and reputational consequences are decided outside it - C. incorrect — sponsors are always liable for all project debt - D. correct, provided the vehicle is bankruptcy-remote

Rationale: committed capital, not subscribed equity, is the exposure (5.2.3), and consolidation and tax turn on control and framework. C describes full recourse, which limited-recourse structures exist to avoid; D confuses insolvency engineering with the scope of contractual support.

SELF-CHECK — KA 5.2

1. State Kestrel's per-sponsor committed capital and the group total. — 13,200,000 / 8,400,000 / 2,400,000, totalling 24,000,000 — 40.0 % of capital cost against a 30 % headline.
2. Name three things an SPV does not achieve. — It does not defeat contractual support, does not decide accounting consolidation, and does not remove tax or reputational exposure.
3. What is an equity bridge worth, and to whom? — Nothing to the project; to the sponsors, 1,480,688 — 8.69 % — the spread between the 5.5 % bridge rate and their 12 % requirement over the deferral.

KNOWLEDGE AREA 5.3

The bankability test

Topics: 5.3.1 bankability as a conjunction · 5.3.2 the revenue model and the offtake · 5.3.3 permits, land and consents · 5.3.4 technology and its price.

5.3.1 Bankability as a conjunction

Definition. Bankability is the degree to which a project's contracts, risks and cash flows support limited-recourse financing on acceptable terms (terminology registry). It is not a score and not a probability of success: it is the state in which **every** condition a lender requires is satisfied, on the timetable, in documents. The structural consequence is that bankability composes **multiplicatively**, and that one fact reorganises how a development programme is run.

WORKED EXAMPLE

Worked example 5.3.1 — six conditions, one probability, and where the effort belongs.

1. **Setup.** Kestrel's lenders require six conditions, each with an assessed probability of being satisfied on the target close timetable: **offtake / revenue model 0.92**; **permits and consents 0.90**; **land and site tenure 0.95**; **technology (proven and guaranteed) 0.88**; **EPC wrap (fixed price, date certain) 0.93**; **financing market and credit approval 0.85**. Remaining development spend to close is **2,400,000**; value on close is **16,179,360**.
2. **Formula.** Assuming independence, joint probability $= \Pi\ p_i$. Marginal gain from lifting condition i from p to $p' = \Pi\ p_i \times (p'/p - 1)$. Expected value of continuing `= joint probability × value − remaining spend`. Breakeven close probability `= remaining spend ÷ value`. Uniform probability needed for a joint target J over k conditions $= J^{(1/k)}$.
3. **Substitution.** `0.92 × 0.90 × 0.95 × 0.88 × 0.93 × 0.85`; weakest lift `× 0.95/0.85`; strongest lift `× 0.98/0.95`; continue test `0.547190 × 16,179,360 − 2,400,000`; uniform requirement `0.90^(1/6)`.
4. **Result.** Joint probability of close **0.5472 (54.72 %)** — against an **arithmetic mean of the six conditions of 90.5 %**. Lifting the **weakest** (financing, 0.85 → 0.95) gives **61.16 %**, a gain of **6.4375 points**; lifting the **strongest** (land, 0.95 → 0.98) gives **56.45 %**, a gain of **1.7280 points** — the weakest link is worth **3.7255×** more. Six conditions all at 0.98 still give only **88.58 %**; a **90 % joint probability requires every condition at 98.26 %**. Expected value of continuing `= 8,853,191 − 2,400,000 =` **+6,453,191**; the **breakeven close probability is 14.83 %**.
5. **Interpretation.** The 54.72 % changes behaviour because almost every development team reports the 90.5 %: asked how the project is looking, they average their conditions and answer "about 90 per cent", wrong by 36 points. Three professional uses follow. **Resource allocation inverts:** the ranking of conditions by marginal gain — not by importance or by who is shouting — is the work plan, and 3.7× more value sits in the financing condition than in the land condition. **The continue/kill test separates from the condition test:** at a breakeven close probability of only **14.83 %**, expected value says "continue" on almost anything, which is precisely why a development programme cannot be governed by expected value alone and needs the fatal condition rule of 5.1.3. And the **uniform requirement of 98.26 %** explains what practitioners observe without deriving: bankable projects are not projects with good conditions, they are projects with **no open condition** — what conditions precedent (5.A.2) exist to enforce. The professional caution is independence: these conditions are correlated, usually positively (a supportive host government helps permits, land and financing together), which makes the true joint probability **higher** than the product, while clustered failures fatten the downside. Report the **ranking**, which is robust, and treat 54.72 % as a disciplined lower bound with its assumption stated — never as a forecast.

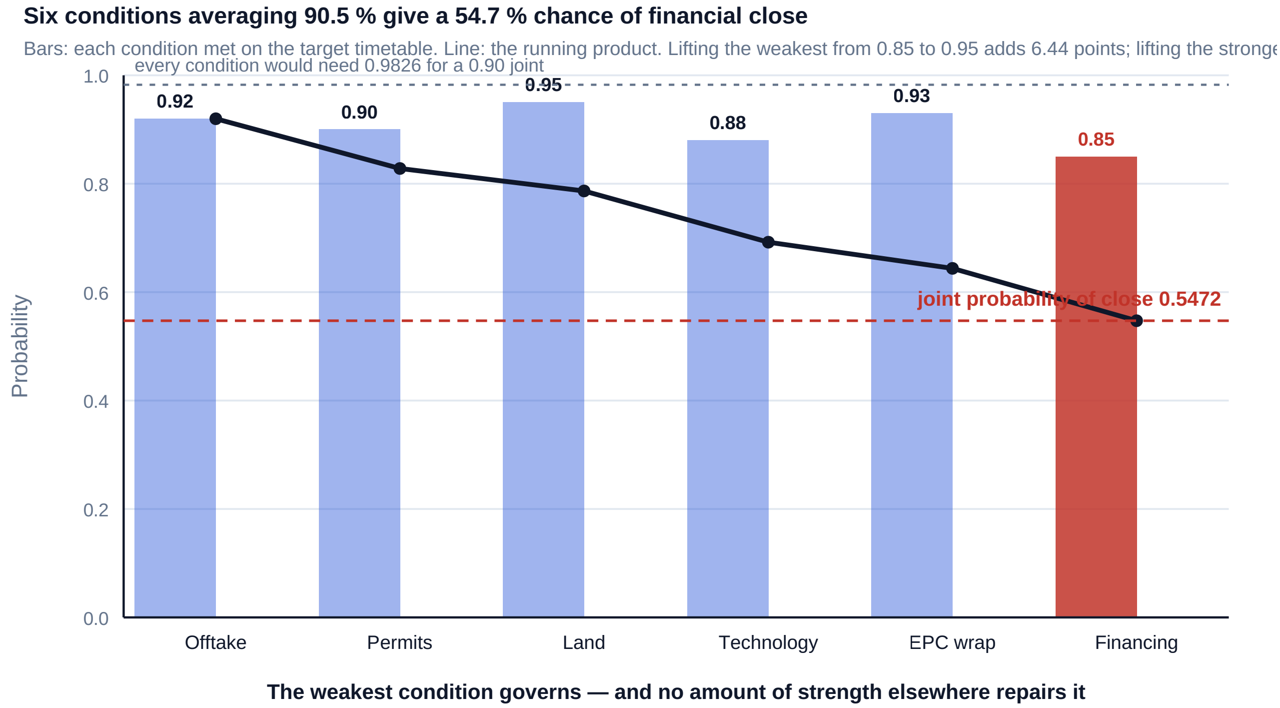

Fig 5.3.1 — Bankability as a conjunction

5.3.2 The revenue model and the offtake

Definition. The **revenue model** is the mechanism by which the project is paid, and it is diligenced first because every other condition is contingent on it. The bankability question is not "how much revenue?" but **"who is contractually obliged to pay, how much, for how long, and what can lawfully stop them?"** Domain 7 builds the taxonomy; the development-stage test has five components.

Component	The test	Why it is a bankability condition
Volume or availability	Does the payer pay for availability (Kestrel's structure) or for what is sold?	Merchant risk moves required coverage by 0.2× or more, and debt capacity by millions (Domain 10)
Tenor	Does contracted revenue extend beyond loan maturity with margin?	A 12-year loan against a 7-year offtake has five unfunded years; Kestrel's 25-year offtake is why its PLCR is 1.9431
Counterparty credit	Can the payer pay, from what source, and is the obligation supported?	An offtake is worth the offtaker's ability to pay; **creditworthy** signature, not signature, is the condition
Indexation	Is every material cost either indexed in the tariff or fixed in a contract of matching tenor?	An unindexed O&M cost inside an indexed tariff is a slow structural margin loss (Domain 3, KA 3.3.2)
Termination and change in law	Does termination compensation repay outstanding debt in non-default scenarios?	Debt sized on that assumption but documented without it is unbankable at close, not at signature

The recurring error sits in the third row: teams treat signature as satisfaction of the revenue condition when the lenders' condition is a creditworthy payer. The remedy is structural — a guarantee, letter of credit or escrow, or resizing to what the weaker credit supports — and it must be established **before** debt is sized, because every ratio in Domain 10 assumes it.

5.3.3 Permits, land and consents

Definition. The **consent set** is the complete list of governmental and third-party permissions the project needs to be built, to operate and to be financed; the **land and tenure set** is the complete list of rights to occupy, cross and use land for the asset and all its corridors. Both are bankability conditions, because security over a project that may not lawfully operate or remain where it is built is worthless.

Two artefacts organise the work. **The consent register** lists every permission with issuing authority, statutory basis, lead time, dependency, expiry, transferability to the SPV and attached conditions — because a permit with unsatisfiable conditions is a refusal with a delay — and identifies which consents drive the critical path, since consents are the development activities least susceptible to acceleration by money. **The land register** does the same for freehold, leasehold, easements, rights of way, wayleaves and access, and its characteristic failure is the **linear asset**: the plant sits on one parcel, but the pipeline, cable, access road or outfall crosses many, and a single unregistered easement over a few hundred metres can stop an otherwise complete project (Case study A).

Environmental and social consent deserves separate naming. Lenders on limited-recourse infrastructure commonly require environmental and social risk to be assessed and managed to a defined standard, and many financial institutions have voluntarily adopted the **Equator Principles**, a lender framework under which participating institutions apply agreed environmental and social requirements to the projects they finance and which in turn refers to the **IFC Performance Standards**. Whether either applies, in what version and with what categorisation is a matter for the specific lenders and project; both are named here for identification only, neither is reproduced or summarised as a source of requirements, and neither body is associated with this book. The duty is to establish the applicable requirement early, because retrofitting a stakeholder-engagement or resettlement process to a completed design is the most expensive rework in infrastructure development.

5.3.4 Technology, the bankable track record and its price

Definition. A technology is **bankable** when lenders will lend against its performance without recourse to sponsors — which requires operating references at comparable scale and duty, a supplier able to stand behind performance guarantees, and an independent technical adviser's opinion the lenders accept. **First-of-a-kind** is not a technical description but a financing category, and it has a price.

Price it on Kestrel. The proven process was sized at a target `DSCR` of **1.30×** on `CFADS` of **6,384,000** over 12 years at 6.0 %. A variant using a novel membrane arrangement would cut operating cost, but the technical adviser cannot supply operating references at scale and the credit committee would require **1.45×** on unchanged cash. Using Domain 10's sizing rule (max debt service = `CFADS` ÷ target `DSCR`; max debt = that × `AF(0.06, 12) = 8.383844`): at 1.30×, `6,384,000/1.30 = 4,910,769` supports **USD 41,171,123**; at 1.45×, `6,384,000/1.45 = 4,402,759` supports **USD 36,912,041**. The **capacity loss is USD 4,259,082**, gearing falls from 68.6 % to **61.5 %** of capital cost, and the 4,259,082 must be found as **additional equity**.

The variant must therefore save more than the cost of 4,259,082 of equity displacing debt, and since equity is the more expensive money the threshold is high: at a 12 % equity requirement against 6 %

debt the annual pre-tax cost of that substitution is about **255,545** (4,259,082 × 6 points of spread), which the operating saving must beat before any allowance for the technology's own performance risk. That is the arithmetic behind a maxim — **project finance is a poor place to innovate** — and two refinements keep it honest. The coverage premium is not the only lever: an extended supplier guarantee, a larger maintenance reserve, output insurance, or a first-loss contribution from a development-finance or concessional lender (Domain 9, KA 9.3–9.4) can each buy the ratio back down, and pricing those against 4,259,082 is the negotiation. And the premium is **not permanent** — the same technology at its fourth installation is a different credit. The caution runs the other way too: bankable is not the same as good, and a leader who lets the lenders' comfort choose the engineering has outsourced a 25-year lifecycle decision to a credit committee whose horizon is 12 years (Domain 8).

AI in this KA

Earns: building and maintaining the consent and land registers from source documents; extracting defined terms, tenors, indexation formulae, termination provisions and conditions precedent into a structured condition set; and re-running the conjunction of 5.3.1 whenever an assessed probability changes, so the marginal-gain ranking stays current instead of being computed once for a board paper. **Must not go:** the probabilities themselves are professional judgment informed by counterparties and counsel, and a model asked to supply them produces confident numbers with no evidential basis — a false precision that propagates into a 54.72 % that looks computed and is invented. Nor may a model conclude that a condition is **satisfied**: satisfaction is a legal state, evidenced by a document, verified by counsel and recorded. Every condition in an AI-built register is traced to a primary document by a named person, probabilities carry the assessor's name and date, and extraction is sampled against source.

KEY TERMS — KA 5.3

Term	Meaning
Bankability	The degree to which contracts, risks and cash flows support limited-recourse financing on acceptable terms.
Conjunctive test	Composition by multiplication: every condition must pass, so the weakest governs.
Marginal gain (condition)	Joint probability × (p′/p − 1); the correct ranking of development effort.
Breakeven close probability	Remaining development spend ÷ value on close (14.83 % for Kestrel).
Consent register	Every permission with authority, lead time, dependency, expiry, transferability and conditions.
Linear-asset tenure risk	Corridor rights whose single gap can stop an otherwise complete project.
First-of-a-kind premium	The coverage, reserve or insurance cost of unproven technology, payable in equity.

SAMPLE MCQS — KA 5.3

MCQ 5.3-A `[5.3.1 · Application]` Six bankability conditions have probabilities 0.92, 0.90, 0.95, 0.88, 0.93 and 0.85. The joint probability of close, assuming independence, is closest to: - A. 90.5 % - B. 54.72 % - C. 85.0 % - D. 43.0 %

Rationale: the product is 0.5472. A is the arithmetic mean — the error the example corrects; C quotes the weakest condition as though the others were certain; D sums the six shortfalls (0.57) and subtracts the total from one, over-counting the failures by treating them as mutually exclusive.

MCQ 5.3-B `[5.3.1 · Analysis]` In a six-condition set whose joint probability is 0.5472, one week of effort can lift either the financing condition from 0.85 to 0.95 or the land condition from 0.95 to 0.98. The value-maximising choice is: - A. land, because 0.98 is the higher absolute probability - B. financing, which adds 6.44 points against land's 1.73 — a factor of 3.73 - C. either, since both raise one condition by a similar amount - D. neither, because correlation makes the calculation meaningless

Rationale: marginal gain is the joint probability times the proportional lift: `0.5472 × (0.95/0.85 − 1) = 6.4375` points against `0.5472 × (0.98/0.95 − 1) = 1.7280`. A ranks by level rather than gain; C ignores that a proportional lift on a low base moves the product much more; D discards a ranking correlation does not reverse.

MCQ 5.3-C `[5.3.4 · Application]` A credit committee raises the target `DSCR` from 1.30× to 1.45× for lack of operating references. On `CFADS` of 6,384,000 over 12 years at 6 % (`AF` = 8.383844), the additional equity required is closest to: - A. USD 4,259,082 - B. USD 6,300,000 - C. USD 508,011 - D. USD 4,910,769

Rationale: capacity falls from 41,171,123 to 36,912,041. B applies the 0.15 ratio increase to the 42,000,000 of debt as though ratio points were percentages of principal; C is the fall in annual debt service; D is the maximum debt service at 1.30×, a per-period figure mistaken for a capital sum.

MCQ 5.3-D `[5.3.2 · Analysis]` A project has a signed 20-year offtake with a counterparty whose payment obligations are unsupported and whose credit lenders assess as weak. The correct conclusion is: - A. bankable — a signed long-tenor offtake is the strongest possible condition - B. the revenue condition fails on counterparty credit; it is repaired by credit support or by resizing, not by the contract's length - C. bankable if the tariff is indexed - D. unbankable permanently

Rationale: an offtake is worth the offtaker's ability to pay, so tenor and indexation do not cure credit (A, C). D overstates: credit support and resizing are the standard structural remedies.

SELF-CHECK — KA 5.3

1. Why does averaging bankability conditions mislead? — Conditions compose multiplicatively: six averaging 90.5 % give a 54.72 % joint probability, a 36-point error.
2. What uniform per-condition probability does a 90 % joint probability require over six conditions? — 0.90^(1/6) = **98.26 %**, which is why bankable projects have no open condition rather than good ones.
3. State Kestrel's first-of-a-kind premium in money. — 1.45× instead of 1.30× costs 4,259,082 of debt capacity, payable as equity, at an annual substitution cost of about 255,545.

KNOWLEDGE AREA 5.4

Construction and operational readiness

Topics: 5.4.1 completion risk and the EPC wrap · 5.4.2 the cost of a slip in the commercial operations date · 5.4.3 completion tests, performance guarantees and buy-down · 5.4.4 operational readiness.

5.4.1 Completion risk and the EPC wrap

Definition. Completion risk is the risk that the project is not delivered on time, to cost and to the performance the financing was sized on. It dominates limited-recourse structures because during construction the project has **all** the debt and **none** of the cash flow: nothing absorbs a shock, and every day of delay is financed.

The standard mitigation is the **EPC wrap**: one contractor takes **fixed-price, date-certain, turnkey** responsibility for the whole works, so interface risk sits with the contractor rather than the SPV. Its components are each a bankability condition — a fixed lump-sum price with defined change mechanics; a date certain with **delay liquidated damages** (5.4.2); **performance guarantees** with performance damages or **buy-down** (5.4.3); performance security sized to the residual exposure; a defects-liability period; and a parent guarantee where the contracting entity is thinly capitalised (Domain 12 drafts these; Domain 11 allocates the risks behind them). Whether a given damages rate, cap or security instrument is enforceable as drafted is jurisdiction-specific and belongs to counsel, not to the model; what this domain computes is the **cash** each provision would have to deliver to be adequate. The wrap's price is the **wrap premium** — a contractor pricing interface and schedule risk it does not control charges for it, and the charge is material enough that a multi-package alternative must be compared against it on interface exposure and price together, never on price alone — and its two failure modes are **multi-package delivery without a wrap**, where the SPV holds every interface dispute, and **a wrap exceeding the contractor's capacity to stand behind it**, since damages and guarantees are worth only the guarantor's balance sheet.

5.4.2 The cost of a slip in the commercial operations date

Definition. The **commercial operations date (COD)** is the contractual date from which the project is treated as operating: revenue begins, the operating-period covenant regime starts, and construction debt converts to term debt. A slip costs money in two distinct and frequently confused ways — **extra interest during construction** on debt already drawn, and **forgone** `CFADS` — and the whole delay-damages negotiation is an attempt to price them.

WORKED EXAMPLE

Worked example 5.4.2 — Kestrel's COD slips 180 days.

1. **Setup.** At the scheduled COD the facility is fully drawn at **42,000,000** at **6.0 %**; operating `CFADS` would have been **6,384,000** per year; the EPC contract carries **delay liquidated damages of USD 20,000 per day** capped at **10 % of the 48,000,000 EPC price**. Interest and daily rates use a **30/360** basis (a common convention; the applicable basis is a negotiated term, and actual/360 or actual/365 would shift these figures slightly). The concession's expiry was fixed at award — **27 years from scheduled financial close** — so a slip **shortens operations rather than extending the term.**
2. **Formula.** Daily extra interest = drawn debt × rate ÷ 360; daily forgone `CFADS` = annual `CFADS` ÷ 360; damages coverage = daily damages ÷ daily economic cost; cap-binding day = cap ÷ daily damages. Coverage consequence: new instalment = new debt ÷ `AF(0.06, 12)`, then `DSCR` and the covenant trigger per Domain 10.
3. **Substitution.** `42,000,000 × 0.06/360 = 7,000`; `6,384,000/360 = 17,733.33`; `20,000/24,733.33`; `4,800,000/20,000`; `43,260,000/8.383844`.
4. **Result.**

Item	Per day	180 days	360 days
Extra interest on drawn debt	7,000.00	1,260,000	2,520,000
Forgone CFADS	17,733.33	3,192,000	6,384,000
Total economic cost	**24,733.33**	**4,452,000**	**8,904,000**
Delay damages at 20,000/day	20,000.00	3,600,000	4,800,000 (capped)
Uncovered, borne by the SPV	**4,733.33**	**852,000**	**4,104,000**

Damages recover **80.86 %** of the daily cost; the **cap binds at day 240**. If the 1,260,000 of extra interest is **capitalised**, debt becomes 43,260,000, the instalment rises to **5,159,924.29**, DSCR falls from 1.2743 to **1.2372**, the covenant cash trigger rises to **6,191,909**, and annual headroom falls from **372,438 to 192,090.85 — 51.6 %** of what it was. Funded as equity instead, equity rises to **19,260,000** and gearing moves from 70.0/30.0 to **68.6/31.4** (D/E 2.1807). 5. **Interpretation.** Three conclusions live in that table. **The damages rate is mis-calibrated, by a knowable amount.** A rate of 20,000 against a daily cost of 24,733.33 leaves equity carrying 4,733.33 per day, and the negotiating position is not "higher damages" but "damages calibrated to interest plus forgone CFADS" — a computation the contractor can check and therefore argue about honestly. The commonest calibration error is to size on the **forgone CFADS alone**, which covers only 71.7 % of the cost; the omitted 7,000 per day is 28.3 % and the most certain component of all, because it accrues whether or not the plant would have run well. **The cap is where the structure actually breaks.** Below day 240 damages absorb most of the pain; beyond it every further day costs the SPV the full 24,733.33, so a 360-day slip leaves **4,104,000** uncovered — which is why lenders test the delay scenario against the cap rather than the rate, and why cost-overrun support (5.2.3) and contingency (Domain 8, KA 8.3) exist for the tail the cap does not reach. **A construction event becomes a permanent operating constraint:** capitalising 1,260,000 of interest halves the covenant headroom Domain 10 measured, for the whole 12-year loan life, on a project otherwise exactly as forecast. The structural choice — capitalise into debt or fund with equity — is therefore a choice about **where the slip lands**, on coverage or on equity return, and it belongs to the sponsors before the event, in the funding documents.

The value view answers a different question. The cash table is the right basis for calibrating damages; it is not the sponsors' loss. Discounting quarterly at Domain 4's 8 % (quarterly rate **1.942655 %**, CFADS 1,596,000 per quarter, concession 108 quarters from close), the operating stream is worth **60,150,401** at close on time and **57,491,504** if COD slips 180 days — a **present-value loss of 2,658,897**, less than the naive 3,192,000 of "six months of CFADS", because what is truly lost is the final half-year of a 25-year stream discounted 27 years, while the rest is merely deferred by two quarters. Adding the 1,260,000 of interest gives a total value loss of **3,918,897 — 24.2 %** of Domain 4's entire NPV of 16,179,360, destroyed by six months. Against that, 3,600,000 of damages received at the delayed COD is worth **2,969,909** at close, leaving a net loss of **948,988**, or **5.9 %** of the project NPV. That pair — 24.2 % gross, 5.9 % net — is the most useful single statement of what an EPC delay-damages regime is for.

5.4.3 Completion tests, performance guarantees and buy-down

Definition. Completion is not a date but a **test**. Financing documents distinguish **mechanical completion** (built and safe to commission), **provisional or substantial completion** (defined performance demonstrated, triggering COD and often conversion to term debt) and **final completion** (punch list cleared, reliability run passed, documentation delivered). Lenders' test is usually the strictest, and it commonly includes a **reliability run** — sustained output over a continuous period — because a single-point test can be passed by a plant that cannot hold it. Where the plant completes but **underperforms**, the remedy is **performance liquidated damages** or a **buy-down**: a payment calibrated so that coverage returns to the sized level, usually applied to prepay debt.

WORKED EXAMPLE

Worked example 5.4.3 — sizing the buy-down for a 3 % output shortfall.

1. **Setup.** Kestrel completes at **97 %** of guaranteed output. Revenue is **12,000,000** at full output; cash operating costs are **4,500,000**, of which **85 % is fixed** (3,825,000) and 15 % varies with output (675,000). Depreciation is **2,400,000**, interest **2,520,000**, cash tax **20 %**, and the working-capital movement is **600,000** as in Domain 2's `CFADS` derivation. The sized `DSCR` was **1.2743** against debt service of **5,009,635.23**.
2. **Formula.** Revenue and variable cost scale with output, fixed cost does not; `EBITDA` = revenue − cash operating cost; `CFADS` = `EBITDA` − cash tax − Δworking capital (Domain 2, KA 2.3.1). At constant target coverage debt is proportional to `CFADS`, so buy-down = debt × (`CFADS` shortfall ÷ base `CFADS`).
3. **Substitution.** `12,000,000 × 0.97 = 11,640,000`; `3,825,000 + 675,000 × 0.97 = 4,479,750`; `EBITDA = 7,160,250`; `7,160,250 − 2,400,000 − 2,520,000 = 2,240,250`; tax `× 0.20 = 448,050`; `CFADS = 7,160,250 − 448,050 − 600,000`; buy-down `= 42,000,000 × (6,384,000 − 6,112,200)/ 6,384,000`.
4. **Result.** `EBITDA` falls to **7,160,250** — a fall of **339,750**, or **4.53 %**, on a 3 % output shortfall (**operating leverage 1.510×**). `CFADS` falls to **6,112,200** and `DSCR` to **1.2201**; covenant headroom collapses from 372,438 to **100,638**, so a 3 % shortfall consumes **73.0 %** of it. The buy-down restoring sized coverage is **USD 1,788,158**, reducing debt to **40,211,842**.
5. **Interpretation.** The **1.510× operating leverage** is what sponsors most often fail to carry into a performance-guarantee negotiation: because most of a plant's cash operating cost is fixed, a small output shortfall produces a proportionately larger cash shortfall, so a guarantee negotiated in **output percentage** must be converted into **cash** before anyone agrees it is adequate. The **73 % of headroom consumed by a 3 % shortfall** is the sentence for the board paper: a shortfall well inside most people's intuition of "close enough" leaves the project 100,638 of annual cash from a covenant breach, with all the lock-up consequences of Domain 10 (KA 10.4.2). And the **1,788,158 buy-down** demonstrates an identity worth carrying: at constant coverage **debt scales linearly with** `CFADS`, so the buy-down is simply debt times the proportional `CFADS` shortfall — computable in a negotiation, on one line, without a model. Two cautions: a buy-down restores the lenders' position, not the sponsors' — equity has permanently lost 271,800 of annual cash and gained only a smaller loan — and performance damages are **capped** like delay damages, so establish the shortfall at which the cap is exhausted and check the project is still bankable there.

5.4.4 Operational readiness

Definition. Operational readiness is the state in which the project can be operated to the standard the revenue contract requires from the first day of the operating period. It is a bankability condition that is systematically under-managed, because the construction contract has an owner with a deadline and readiness usually does not. The readiness set, each item with an owner, a date and evidence: an **operating organisation** (O&M contract with a creditworthy operator, or an in-house team recruited and trained); **permits to operate**, distinct from permits to construct, frequently issued later and occasionally conditional on demonstrated performance, so the consent register must separate the two; **spares, consumables and supply** with tenor matching the offtake; **metering** on which the revenue contract pays, calibrated and accepted by both parties, since a tariff paying on measured availability is only as good as the meter and the dispute process behind it; **insurance transition**, construction cover replaced by operating cover with lender endorsements in force on day one, a gap being both an uninsured exposure and an event of default; **reporting readiness** — model updated to actuals, first covenant test date and compliance certificate scheduled, reserve accounts funded (Domain 10, KA 10.3), because a first covenant test failed for want of a report is a breach on a performing project; and **handover of the record** — as-built documentation, warranties, manuals and a defect list with owners and dates.

The financing consequence is direct: the covenant regime starts at COD, so a project reaching COD operationally unready spends its first test period generating exactly the shortfall 5.4.3 priced, with no track record and no goodwill. Readiness therefore runs as a **separate workstream with its own gate**, held before COD is declared — and the uncomfortable implication is accepted: **declaring COD early to start revenue can be the most expensive decision in the project**, because it converts a construction problem the contractor owns and damages cover into an operating problem equity owns and nothing covers.

AI in this KA

The strong application is **readiness and completion evidence assembly**: tracking hundreds of completion-test, permit-to-operate, insurance, spares and documentation items against owners and dates, reconciling the punch list to defects-liability obligations, and flagging items whose evidence is missing before COD is declared — list discipline at a scale humans do badly, and cheap to verify because each item resolves to a document. Machine-assisted **COD forecasting** is legitimate with one non-negotiable control: the forecast COD drives the arithmetic of 5.4.2, so a model-produced date entering a lender report has become a representation and must carry the named person who owns it. **Where it must not go:** declaring a completion test passed, or the project ready to operate, is a professional certification with contractual and safety consequences — a machine may assemble the evidence and must never make the call; and attributing a 3 % shortfall to design, construction, operation or feed conditions decides who pays under which contract. **AI proposes; the professional verifies, decides and remains accountable.**

KEY TERMS — KA 5.4

Term	Meaning
Completion risk	Risk of not delivering on time, to cost and to sized performance — all debt, no cash flow.
EPC wrap	Single-contractor fixed-price, date-certain, turnkey responsibility for the whole works.
Commercial operations date (COD)	Contractual start of operations: revenue, term debt and the operating covenant regime.
Interest during construction	Interest accruing on drawn debt before revenue; 7,000 per day for Kestrel.
Delay liquidated damages	Daily payment for late completion; calibrate to interest **plus** forgone `CFADS`.
Damages cap	The ceiling beyond which every further day is borne by the SPV (day 240 for Kestrel).
Buy-down	Payment applied to prepay debt so sized coverage is restored after a performance shortfall.
Operating leverage (plant)	`EBITDA` shortfall ÷ output shortfall; 1.510× for Kestrel at 85 % fixed cost.
Operational readiness	Capability to operate to contract standard from the first day of the operating period.

SAMPLE MCQS — KA 5.4

MCQ 5.4-A `[5.4.2 · Application]` Debt of 42,000,000 is fully drawn at 6.0 %; annual `CFADS` would be 6,384,000; 30/360 applies. The daily economic cost of a COD slip is: - A. USD 17,733.33 - B. USD 24,733.33 - C. USD 7,000.00 - D. USD 26,400.00

Rationale: `42,000,000 × 0.06/360 = 7,000` of interest plus `6,384,000/360 = 17,733.33` of forgone `CFADS`. A is the forgone-`CFADS` side alone — the calibration error leaving 28.3 % uncovered; C is the interest alone; D uses the pre-working-capital `CFADS` of 6,984,000 (Domain 2's other definition, 19,400 per day).

MCQ 5.4-B `[5.4.2 · Analysis]` The daily economic cost of a COD slip is 24,733.33. Delay damages are 20,000 per day, capped at 10 % of a 48,000,000 EPC price. For a 360-day slip the SPV bears: - A. nothing — the damages cover the delay - B. USD 4,104,000, because the cap binds at day 240 against an economic cost of 8,904,000 - C. USD 1,704,000 - D. USD 8,904,000

Rationale: cap `= 4,800,000`, binding at `4,800,000/20,000 = 240` days; cost `= 360 × 24,733.33 = 8,904,000`; uncovered `= 4,104,000`. A ignores the cap; C computes damages as though all 360 days were payable (7,200,000) and subtracts; D omits recovery altogether.

MCQ 5.4-C `[5.4.2 · Analysis]` A facility of 42,000,000 over 12 years at 6 % (`AF` = 8.383844) carries a 1.20× cash covenant against `CFADS` of 6,384,000, leaving annual headroom of 372,438. 1,260,000 of extra construction interest is capitalised, taking debt to 43,260,000 at the same tenor and rate. The most important consequence for the operating period is: - A. none — the debt is repaid over the same period - B. annual covenant headroom falls from 372,438 to 192,090.85, roughly halving it for the whole loan life - C. the loan tenor extends - D. `CFADS` falls by 1,260,000

Rationale: the instalment rises to `43,260,000/8.383844 = 5,159,924.29`, `DSCR` falls to 1.2372 and the 1.20× trigger rises to 6,191,909. A ignores that debt service rises; C is a possible structural response, not an automatic consequence; D confuses a financing cost with cash generation, which is unchanged.

MCQ 5.4-D [5.4.3 · Application] A 3 % output shortfall reduces CFADS from 6,384,000 to 6,112,200 against debt of 42,000,000. The buy-down restoring the originally sized coverage is: - A. USD 271,800 - B. USD 1,788,158 - C. USD 1,260,000 - D. USD 4,259,082

Rationale: at constant coverage debt is proportional to CFADS, so the buy-down is 42,000,000 × 271,800/6,384,000. A is the annual CFADS shortfall, not the debt adjustment; C is the COD-slip interest of 5.4.2; D is the technology premium of 5.3.4 — correct numbers in the wrong place.

SELF-CHECK — KA 5.4

1. What two costs make up a COD slip, and which is more often omitted? — Extra interest on drawn debt (7,000 per day) and forgone CFADS (17,733.33 per day); the interest is omitted, and it is the more certain of the two.
2. Why does a 3 % output shortfall cut *EBITDA* by 4.53 %? — Operating leverage: 85 % of cash operating cost is fixed, so EBITDA falls faster than output (1.510×), and CFADS with it — by 4.26 % once tax and working capital are taken.
3. Why can declaring COD early be the most expensive decision in a project? — It converts a construction problem the contractor owns and damages cover into an operating problem equity owns and nothing covers, at the moment the covenant regime begins.

Advanced topics — Domain 5

5.A.1 Development as a portfolio of real options, and the discipline of abandonment

Volatility increases option value: a project whose eventual value is highly uncertain is worth more to hold than one with the same expected value and no uncertainty, because the downside can be abandoned and the upside cannot be capped — the honest financial reason to hold some genuinely speculative pursuits, valid only while the sponsor can still walk away cheaply. More important, **the option dies as commitment accrues:** a bid bond posted, an equipment slot reserved, a jurisdictional reputation staked, each converts option into obligation, so the abandonment decision belongs before each of them, exactly where 5.1.1 puts the gates. The pathology to name is the **sunk-cost carry**: a project retained because 10,000,000 has been spent. Spent money is relevant only to the post-mortem; the question is whether the remaining spend is justified by the remaining probability — the 14.83 % breakeven of 5.3.1, so low that a disciplined programme needs a second, categorical rule: **kill on a fatal condition regardless of expected value.**

5.A.2 Conditions precedent: the bankability list made contractual

The conditions of KA 5.3 become **conditions precedent** (CPs) in the finance documents — the itemised, evidenced list satisfied before first drawdown — and reading a CP schedule as the bankability test in contractual form converts judgment into a critical path with named owners. **CPs are ordered by dependency, not importance:** a permit requiring a land instrument, requiring a board resolution, requiring a shareholders' agreement amendment is a four-link chain whose duration is the sum, so the close date is set by the longest chain rather than the hardest item. **The categories behave differently:** sponsor CPs (corporate authorisations, equity commitments, credit support) are within the group's control and therefore forgivable in timetable terms, while third-party CPs (permits, offtaker approvals, letters of credit from banks that are not the lenders) are not — each is the kind of timetable risk that made Case study B's eleven weeks. And **waived CPs are not**

satisfied CPs: a condition waived to permit close, or converted into a post-close undertaking, remains open with a deadline and a consequence, and the register must track it past close, because Domain 10's covenant regime will test it and Domain 13's close checklist will be audited against it.

5.A.3 The reviewer's bankability eye

Invariants to test on any development or bankability paper. **The conjunction is computed, not averaged**, with its independence assumption stated (5.3.1). **The weakest condition is identified and resourced** — the marginal-gain ranking exists, is current, and matches the work plan. **Development cost is stated per closed project across the portfolio**, with the breakeven close rate on the page (5.1.2). **Every gate's option cost is quantified**, not just its study cost (5.1.3). **Committed capital, not subscribed equity, is the exposure figure**, with the liability basis in the document's own words (5.2.3). **Contracted revenue tenor exceeds loan tenor with margin**, and the offtaker's credit — not merely its signature — has been assessed (5.3.2). **The consent and land registers cover corridors, not only the site**, and permits to operate are separated from permits to construct (5.3.3, 5.4.4). **Delay damages are calibrated to interest plus forgone CFADS on the facility's day-count convention**, with the cap-binding day stated (5.4.2). **Performance guarantees are converted from output percentages into cash through operating leverage**, and the buy-down is computed as debt × the proportional CFADS shortfall (5.4.3). **Every capitalised construction cost is traced to its coverage consequence** — headroom after the event, not before. And **every condition has an owner, a date, an evidence reference and a named verifier**, the single discipline separating a bankability assessment from an opinion.

Industry variations — Domain 5

- **Water and desalination (Kestrel's sector).** Availability-based offtakes with a public or utility payer make the revenue condition strong and move the battle to **consents, water source and discharge rights, and linear assets** — intake, outfall and pipeline corridors, where Case study A's tenure failure lives. Technology is usually proven, so the first-of-a-kind premium is rare and expensive when it appears.
- **Contracted power and renewables.** The binding conditions are **grid connection and curtailment terms** — a queue position is a development asset with a value and a date — and increasingly storage and shaping obligations that turn a simple offtake into an availability product. Portfolios are large and cheap per pursuit, so the funnel of 5.1.2 runs at far higher screened counts and lower close rates.
- **Transport concessions.** Land assembly and route consent dominate the timetable; **traffic and demand studies** substitute for the offtake and are diligenced as such; and public procurement makes the **bid window a hard date** — the case in which 5.1.3's option cost most often defeats a good gate.
- **Digital infrastructure.** Power availability and grid capacity are the critical consents, tenant credit is the offtake condition, and **short asset lives** thin the tail Domain 10's PLCR measures. Sponsor groups are frequently three-party with a thin local holder — Case study B's shape.
- **Oil, gas and mining.** Resource definition replaces the offtake as the primary condition, revenue is usually **merchant**, and social licence and closure obligations extend the consent set decades beyond the loan. Development spend per pursuit is an order of magnitude larger and close rates correspondingly lower, making the portfolio discipline of 5.1.2 the whole business rather than an overhead on it.

- **Social infrastructure PPPs.** Conditions are relatively standardised and the payer is usually a public authority, so bankability turns on **contract precedent and affordability approval** rather than technology; the risk is political and budgetary, and termination-compensation provisions do more of the credit work than anything in the physical asset.

Case study — Domain 5: the easement nobody registered (water)

Situation. Kestrel's feasibility study cleared the plant site — a 14-hectare coastal parcel on a long lease — and reported the **land and site tenure condition satisfied at 0.95**. The project's 4.6 km product-water pipeline crossed eleven third-party parcels under easements granted to the host municipality in an earlier scheme. Nine were registered. Two, covering **1.8 km through a single agricultural holding**, existed only as a decades-old exchange of letters, and the holding had since been sold twice. The bid was submitted, preferred-bidder status awarded, and the defect surfaced in **lender legal diligence eleven weeks before scheduled close.**

What happened. The transaction stopped: lenders would not treat an unregistered corridor right as security-worthy, and no title insurance was available on terms the credit committee would accept. Three options existed — negotiate fresh easements with an owner who now knew the project's position, pursue a statutory route through the municipality in an election period, or **re-route** the 1.8 km along an existing road reserve. The sponsors re-routed, at an additional capital cost of **1,400,000** and a **nine-month** delay to close and therefore to COD. The concession's expiry had been fixed at award, so the operating period shortened rather than shifting.

The arithmetic. Discounting quarterly at 8 % (quarterly rate 1.942655 %, `CFADS` 1,596,000 per quarter, concession 108 quarters from the original scheduled close), the operating stream was worth **60,150,401** on the original timetable and **56,199,935** with COD nine months later — a present-value loss of **3,950,466**, or **6.57 %** of the stream. The 3,600,000 of feasibility and transaction spend already at risk was carried nine months at the sponsors' 12 %, costing **319,368**. With the re-route, the defect cost **USD 5,669,834 — 35.0 %** of Domain 4's entire project `NPV`. A title-and-consents review at the feasibility gate, of exactly the kind priced in Worked example 5.1.3, would have cost **180,000**: the defect cost **31.5 times** the review that would have found it, and would have found it while re-routing was a design choice rather than a crisis.

How it resolved. Close occurred nine months late on the original 42,000,000 / 18,000,000 structure, the 1,400,000 re-route being absorbed by construction contingency inside the unchanged 60,000,000 capital budget, so neither the debt nor the equity ticket moved. Lenders required as a condition precedent a **complete registered-title schedule for every corridor parcel** with counsel's opinion. The sponsor group changed two things permanently: the feasibility gate acquired a mandatory corridor-tenure review for every linear asset, run **concurrently** with technical feasibility so that it adds no elapsed time (the 5.1.3 remedy), and the bankability register stopped recording "land" as one condition, splitting it into **site tenure** and **corridor tenure** with separate probabilities and owners.

What the domain teaches here. A condition assessed at 0.95 was two conditions, one of them near zero, and the conjunction (5.3.1) could not reveal that because the register's granularity was wrong. Bankability failures are rarely failures of judgment about a known condition; they are failures to have decomposed a condition into the things that must independently be true. And the gate arithmetic of 5.1.3 is vindicated as such arithmetic usually is — not by the gates held, but by the 5,669,834 paid for the one that was not.

Case study B — Domain 5: the ten per cent that could not fund (digital infrastructure)

Situation. A **240,000,000** hyperscale data-centre campus was developed by a three-sponsor consortium: an anchor operator at **55 %**, an infrastructure fund at **35 %** and a local partner at **10 %** contributing land assembly, grid liaison and permitting. Funding was **70/30** — **168,000,000** of senior debt and **72,000,000** of equity — with a **several cost-overrun support of 10 % of capital cost** subscribed pro rata, so committed capital was **52,800,000 / 33,600,000 / 9,600,000**, the local partner's comprising 7,200,000 of base equity and **2,400,000** of support.

What happened. Grid works and a mid-construction shift to higher-density cooling, to meet the anchor tenant's revised rack plan, produced a **16,000,000** overrun. Support was called pro rata: **8,800,000** from the operator, **5,600,000** from the fund, **1,600,000** from the local partner. The partner could not fund; its balance sheet was committed elsewhere and its 2,400,000 support had been accepted at close on a parent undertaking rather than a bank letter of credit. Under the shareholders' agreement the operator funded the partner's share, credited at a **punitive 1.25× conversion**.

The arithmetic. Post-overrun equity became **50,400,000** (operator: 39,600,000 + 8,800,000 + 1,600,000 × 1.25), **30,800,000** (fund) and **7,200,000** (partner), totalling **88,400,000** — so shares moved from 55/35/10 to **57.0136 % / 34.8416 / 8.1448**. The overrun bought no additional revenue, so equity value (the present value of distributions, **96,000,000**) was unchanged by it. Had the partner funded, it would hold 10 % of 96,000,000 = 9,600,000 having paid 1,600,000, a net **8,000,000**; having declined, it holds 8.1448 % = **7,819,005** and kept its cash. **Declining therefore cost it 180,995 — 11.3 % of the 1,600,000 it withheld**; at a par conversion the cost would have been **145,455**. For dilution to price the default at the full 1,600,000 the multiplier would have to be **13.50×** — far outside the range such clauses use, and at that level a conversion whose enforceability counsel would have to test before anyone relied on it. The conclusion is uncomfortable and general: **in a small-stake structure, dilution cannot price a funding default.** It is a discount, not a deterrent.

How it resolved. The operator funded, took the dilution credit and — because dilution was inadequate — invoked the agreement's alternative remedy, converting the shortfall into a **shareholder loan at 15 %**, so the partner owed **1,840,000** after one year, **2,116,000** after two and **2,433,400** after three, subordinated to senior debt and repayable from its distributions. The lenders, exposed to the support commitment rather than to the sponsors' internal arrangements, required for the remainder of construction that the partner's undrawn support be **backed by a bank letter of credit**; arranging it took **eleven weeks**, during which the next drawdown was blocked — and that delay was the practical cost of the whole episode.

What the domain teaches here. The 10 % holder is the structural weak point of a sponsor group (5.2.3): smallest number, likeliest to be least creditworthy, able to stop a drawdown for a sum trivial against the project. **Credit support belongs at close, not at default** — the eleven weeks were available at close for a letter-of-credit fee and were paid for in full later. And **remedies must be priced, not assumed:** a group that had computed the 180,995 against 1,600,000 before signing would have known its dilution clause was decorative, and would have negotiated the shareholder-loan remedy, a drag-along or a call option on the defaulting share instead.

Executive perspective — Domain 5

What a project finance director cannot delegate in this domain:

- **The conjunction, and the ranking that follows.** The joint probability of close, computed and not averaged, with the marginal-gain ranking driving the work plan. A director who accepts "about 90 per cent" from six 90 per cent conditions has accepted a 36-point error (5.3.1).
- **The fatal-condition kill.** Expected value justifies continuing almost anything at a 14.83 % breakeven; only a categorical rule kills a project with an unresolvable condition (5.A.1).
- **The development portfolio's economics.** Cost per closed project (7,400,000, not 2,400,000), the close rate and the breakeven close rate — the numbers a development budget is defended with (5.1.2).
- **Committed capital and the liability basis.** What the group has actually promised (24,000,000, 40 % of capex, not the 30 % headline), on what basis, and whether the weakest sponsor's share is credit-supported at close (5.2.3, Case study B).
- **The damages calibration.** Whether delay and performance damages are calibrated to interest plus forgone `CFADS` and converted from output percentages into cash, and where each cap binds (5.4.2, 5.4.3).
- **The COD declaration.** It is the director's, taken against evidence of operational readiness rather than a revenue start date, because the covenant regime begins the same day (5.4.4).

Calculation exercises — Domain 5

Exercise 5.1 A programme screens 60 opportunities at 30,000 each; 15 advance to concept at 200,000 each; 6 enter feasibility and bid at 900,000 each; 3 are carried to close at 1,800,000 each. Each close delivers sponsor value of 11,000,000. Compute programme spend, cost per close, the close rate, the value multiple and the breakeven close rate. Solution. Stage spend `1,800,000 + 3,000,000 + 5,400,000 + 5,400,000 =` **15,600,000**; per close `15,600,000/3 =` **5,200,000**; close rate `3/60 =` **5.0 %**; portfolio value **33,000,000**, net **+17,400,000**, multiple **2.1154×**; breakeven closes `15,600,000/11,000,000 = 1.4182`, breakeven close rate `1.4182/60 =` **2.36 %**. Common error: dividing only the closing-stage spend by the closes (`5,400,000/3 = 1,800,000`), which prices the winners and ignores the portfolio that produced them — a **65 %** understatement.

Exercise 5.2 Five bankability conditions have probabilities 0.95, 0.90, 0.80, 0.96 and 0.92. Compute the joint probability of close, identify the weakest, and compare the gain from lifting it to 0.92 with the gain from lifting the 0.90 condition to 0.96. Solution. Joint `= 0.95 × 0.90 × 0.80 × 0.96 × 0.92 =` **0.6041 (60.41 %)**, against an arithmetic mean of **90.6 %**. The weakest is 0.80; lifting it to 0.92 multiplies the product by 1.15 for **69.47 %**, a gain of **9.0616 points**. Lifting 0.90 to 0.96 multiplies by 1.0667 for **64.44 %**, a gain of **4.0274 points** — so the weakest link is worth **2.250×** more. Common error: reporting the arithmetic mean (90.6 %) as the probability of close, a 30-point overstatement; the next commonest is ranking by absolute lift (0.12 against 0.06) rather than by proportional lift applied to the whole product.

Exercise 5.3 A 90,000,000 project is funded with 25,000,000 of equity held 40/30/20/10, and the facility requires a several cost-overrun support of 12 % of capital cost subscribed pro rata. State each sponsor's committed capital, the group total as a percentage of capex, and the uplift over headline equity. Solution. Support pool `90,000,000 × 0.12 =` **10,800,000**.

Share	Equity	Support	Committed
40 %	10,000,000	4,320,000	**14,320,000**
30 %	7,500,000	3,240,000	**10,740,000**
20 %	5,000,000	2,160,000	**7,160,000**
10 %	2,500,000	1,080,000	**3,580,000**

Group committed **35,800,000**, or **39.78 %** of capital cost against a 27.8 % equity headline; the uplift is a uniform **43.2 %** for every sponsor, because support is subscribed pro rata. Common error: quoting the 10 % sponsor's exposure as its 2,500,000 of equity and omitting the 1,080,000 of several support — **30.2 % below** the 3,580,000 a lender will nonetheless call, the same gap that reads as a 43.2 % uplift when taken the other way round.

> **Exercise 5.4** Debt of 50,000,000 is fully drawn at 7.0 %; annual `CFADS` would be 8,000,000; delay damages are 25,000 per day capped at 8 % of a 60,000,000 EPC price; 30/360 applies. For a 150-day COD slip compute the daily economic cost, the total cost, damages recovered, the shortfall, and the cap-binding day. Solution. Daily interest `50,000,000 × 0.07/360 =` **9,722.22**; daily forgone `CFADS 8,000,000/360 =` **22,222.22**; total **31,944.44 per day**. Over 150 days: interest **1,458,333.33**, forgone `CFADS` **3,333,333.33**, total **4,791,666.67**. Damages `150 × 25,000 =` **3,750,000**, leaving **1,041,666.67**; coverage `25,000/31,944.44 =` **78.26 %**. The cap `60,000,000 × 0.08 = 4,800,000` binds at `4,800,000/25,000 =` **192 days**. Common error: calibrating the damages rate on the forgone `CFADS` alone (22,222.22 per day) and omitting the interest, which is **30.43 %** of the daily cost and its most certain component.

> **Exercise 5.5** A credit committee will size debt at a target `DSCR` of 1.30× on proven technology or 1.50× on a first-of-a-kind configuration. `CFADS` is 8,000,000; the loan runs 14 years at 6.5 % (`AF(0.065, 14) = 9.013842`). Compute debt capacity under each and the equity consequence. Solution. At 1.30×: service `8,000,000/1.30 = 6,153,846.15`, capacity `× 9.013842 =` **55,469,799**. At 1.50×: service **5,333,333.33**, capacity **48,073,826**. The capacity loss is **7,395,973**, to be found as equity — exactly **13.33 %** of the 1.30× capacity, because at constant `CFADS`, tenor and rate the ratio is `1 − 1.30/1.50`. Common error: treating the 0.20 increase in the target ratio as a 20 % reduction in debt, which overstates the loss by roughly 3.7 million; the reduction is 13.33 %.

Practitioner's toolkit — Domain 5

Adoption-ready artefacts; adapt headings to your organisation, then keep them stable.

Toolkit 5.T.1 — The bankability condition register

One row per condition, decomposed to the level at which something must independently be true (Case study A's lesson: "land" is at least two rows — site tenure and corridor tenure). Columns: condition · what must be true, in evidential terms · the document that will prove it · owner · target date · dependency · **assessed probability of satisfaction on the timetable**, with assessor and date · the corresponding condition precedent once drafted · status and verifier. Footer, recomputed at every update: **the product of the probabilities** (never the mean), the joint probability of close, and the conditions ranked by **marginal gain** — the next period's work plan. Rule: a condition with no evidential test is not a condition but a hope, and is recorded as such.

Toolkit 5.T.2 — Sponsor commitment and support schedule

One row per sponsor. Columns: equity share · base equity commitment · each support commitment separately (cost overrun, completion, equity commitment letter, O&M or supply guarantee, debt-service undertaking) with its cap and its **liability basis in the words of the document** · **total committed capital** · sunset or release trigger for each obligation · credit support required by lenders and whether it is **in place at close** · the default and dilution mechanic, with **the cost of declining a call computed** for a representative call size (Case study B's 180,995 against 1,600,000). Footer: group committed capital as a percentage of capital cost, and the identity of the weakest credit in the group. Rule: no board approves an equity share without seeing the committed-capital column.

Toolkit 5.T.3 — COD-slip and readiness pack

Two parts, maintained from twelve months before COD. **The slip calculator:** drawn debt × rate ÷ day-count (state the convention) = daily interest; annual `CFADS` ÷ day-count = daily forgone cash; the sum = **daily economic cost**; the damages rate and its **coverage percentage**; the **cap and the day it binds**; the coverage consequence of capitalising the extra interest (new instalment, `DSCR`, covenant trigger, **headroom before and after**); and the structural decision — capitalise or fund with equity — recorded in advance. **The readiness gate checklist**, held before COD is declared: O&M contract executed and operator mobilised · permits **to operate** issued, listed separately from construction permits · initial spares and supply contracts with matching tenor · revenue metering calibrated and accepted by the payer · operating insurance in force with lender endorsements · reserve accounts funded · model updated to actuals, first covenant test date and compliance certificate scheduled · as-built documentation, warranties and punch list with owners and dates. Rule: **COD is declared against this checklist, by a named person, or it is not declared.**

Exam preparation — Domain 5

What is assessed. The conjunctive nature of bankability and its arithmetic; development spend as an option premium measured across a portfolio; gate economics adapted to development, where holding costs a forgone option rather than a carry; the SPV's purposes and the limits of its ring-fence; per-sponsor equity, several-liability support and dilution mechanics; the equity bridge's neutrality at the bridge rate; the components of a bankable revenue model; the consent and tenure sets including corridors; the price of unproven technology in lost debt capacity; and the full cost of a COD slip and a performance shortfall, including the coverage consequence.

The calculations to do under time pressure. The product of a condition set and the marginal gain from lifting one condition (5.3.1) · programme spend, cost per close and breakeven close rate (5.1.2) · gate net value and breakeven detection probability (5.1.3) · committed capital per sponsor and the group total as a share of capex (5.2.3) · debt capacity at two target coverage ratios using Domain 10's rule (5.3.4) · daily and total COD-slip cost, damages coverage and the cap-binding day (5.4.2) · the new instalment, `DSCR` and headroom after capitalising construction interest (5.4.2) · `EBITDA` and `CFADS` under an output shortfall with a fixed/variable split, and the buy-down that restores sized coverage (5.4.3).

The traps.

- Averaging condition probabilities instead of multiplying them — a 36-point error (5.3.1, MCQ 5.3-A).
- Ranking condition effort by absolute probability rather than marginal gain (5.3.1, MCQ 5.3-B).
- Quoting development cost per deal rather than per close (5.1.2, MCQ 5.1-A).

- Counting a gate's study cost but not its option cost (5.1.3, MCQ 5.1-D).
- Treating subscribed equity as sponsor exposure and omitting several support (5.2.3, MCQ 5.2-A).
- Reading several liability where the document says joint and several, or the reverse (5.2.3, MCQ 5.2-B).
- Presenting the equity bridge's saving as project value when it is neutral at the bridge rate (5.2.4, MCQ 5.2-C).
- Treating a signed offtake as a satisfied revenue condition without testing the offtaker's credit (5.3.2, MCQ 5.3-D).
- Applying a change in target `DSCR` to the principal as though ratio points were percentages (5.3.4, MCQ 5.3-C).
- Calibrating delay damages on the forgone `CFADS` alone and omitting interest during construction — 28.3 % of the daily cost on Kestrel, 30.43 % in Exercise 5.4 (5.4.2, MCQ 5.4-A).
- Ignoring the damages cap, which is where the exposure actually sits (5.4.2, MCQ 5.4-B).
- Forgetting that capitalised construction interest is a permanent coverage cost (5.4.2, MCQ 5.4-C).
- Sizing a buy-down on the annual cash shortfall rather than on debt × the proportional `CFADS` shortfall (5.4.3, MCQ 5.4-D).

How the domain connects. Domain 4 supplied the value this domain tests for financeability, and Domain 1 the recourse spectrum and SPV that make limited-recourse structures possible. Forward: Domain 6 models the structure assembled here; Domain 7 builds the revenue models 5.3.2 only classifies; Domain 8 supplies the estimate classes, contingency and delay arithmetic behind the development budget and the overrun support; Domain 9 supplies the capital sources, including the concessional money that can buy down a first-of-a-kind premium; Domain 10 sizes and covenants the debt these conditions permit; Domain 11 allocates the risks the conditions represent and Domain 12 documents them; Domain 13 converts the condition register into conditions precedent and closes; Domain 14 monitors the construction whose slip 5.4.2 prices. PML-AI Domain 3 (KA 3.3.1) is the delivery twin of this domain's gate economics, and the difference between them — elapsed time against forgone option — is the most useful single idea a finance leader can carry between the two disciplines.

Domain 5 summary

Bankability is a **conjunction**, and that is the domain's whole argument. Kestrel's six conditions — offtake 0.92, permits 0.90, land 0.95, technology 0.88, EPC wrap 0.93, financing 0.85 — average **90.5 %** and multiply to a **54.72 %** probability of close; lifting the weakest to 0.95 adds **6.4375 points** while lifting the strongest to 0.98 adds **1.7280**, a **3.7255×** difference that reorders the whole work plan; and a 90 % joint probability would require every condition at **98.26 %**, which is why bankable projects are projects with no open condition rather than projects with good ones. The spend that buys those conditions is an **option premium** measured across a portfolio: 40 screened opportunities, USD 14,800,000 of programme spend and two closes make Kestrel's honest origination cost **7,400,000**, not the 2,400,000 on its own charge code, against a **breakeven close rate of 2.29 %** on an achieved 5.0 %. A feasibility gate costing 180,000 against a 40 % fatal-flaw rate and a 3,300,000 late-discovery waste is worth **810,000 per project** and pays down to a **13.64 %** detection rate — but an 8-week delay carrying a 10 % chance of missing a bid window costs **1,617,936**, turning the same gate **negative by 807,936** and setting a **5.01 %** breakeven window-miss probability: in development, elapsed time is priced as a lost option, not a carrying cost. The group behind Kestrel's 18,000,000 has committed **24,000,000 — 40.0 % of capital cost** — as 13,200,000 / 8,400,000 / 2,400,000 on a several basis, every sponsor exposed **33.3 %** beyond its equity share and each exposed to the whole 6,000,000 pool if the basis becomes joint and several; its equity bridge at 5.5 % is **exactly value-neutral at the bridge rate** (17,530,806 either way), worth **1,480,688** only because the sponsors discount at 12 %. Unproven technology has a price: **1.45× instead of 1.30× costs 4,259,082** of debt capacity, payable in equity. A slip in the commercial operations date costs **24,733.33 per day** — 7,000 of interest on drawn debt and 17,733.33 of forgone `CFADS` — so a 180-day slip costs **4,452,000** against damages of 3,600,000, leaves **852,000** with equity, and if its interest is capitalised cuts covenant headroom from **372,438 to 192,090.85** for the whole loan life, with the cap binding at **day 240** beyond which every day is the SPV's; in value terms the same slip destroys **3,918,897**, 24.2 % of the project `NPV`, and 948,988 after damages. A 3 % output shortfall cuts `CFADS` to **6,112,200** through **1.510×** operating leverage, consuming **73.0 %** of covenant headroom, and is repaired by a buy-down of **1,788,158** — debt times the proportional `CFADS` shortfall. Case study A paid **5,669,834**, 35.0 % of the project's `NPV`, for an unregistered 1.8 km easement a 180,000 review would have found; Case study B proved a 1.25× dilution clause prices a 1,600,000 funding default at **180,995** and would need a **13.50×** multiplier to price it properly — so credit support belongs at close, not at default. Domain 6 now models the structure this domain assembled; Domain 10 sizes the debt its conditions permit.

06

DOMAIN 6

Financial Modelling and Model Governance (quantitative)

Group: Structuring and modelling (Domain 2 of 5 in Part Two). **Target:** ~78 pages. **Binds to:** the PCI Book Pattern Specification and the shared registries (`docs/books/registries/`). This domain is the home of the model — the artefact on which Domains 3, 4, 5 and 10 all silently depended. It consumes Domain 3's discounting and amortisation machinery (`AF(r, n)`, `DF(t)`), Domain 4's appraisal measures (`NPV`, `IRR`, `MIRR`, `PI`), Domain 2's three statements and `CFADS` definition, and Domain 10's coverage ratios (`DSCR`, `LLCR`, `PLCR`), and it takes responsibility for whether any of them is computed correctly. British English; USD (+SAR where useful, indicative `USD 1 ≈ SAR 3.75`). Tax and accounting treatments are described in principle and are jurisdiction-specific: the arithmetic here is transferable, the treatments are not, and neither is a substitute for professional tax or accounting advice.

Why this domain exists

Every number in Domains 3 to 5 and 10 came out of a model, and not one of those domains examined the model. Domain 4 decided to build on an `NPV` of **+16,179,360**. Domain 10 negotiated debt capacity against a `CFADS` of **6,384,000** and a `DSCR` of **1.2743**. Domain 5 priced a commercial-operations-date slip at **24,733.33** a day. Each is the output of one workbook, produced by one analyst, reviewed by whoever had time — and each has been quoted without its basis, its horizon or its checks attached. That is the gap this domain closes.

The central claim is worth stating plainly: **a financial model's authority comes from its architecture and its checks, not from its answers.** Two competent analysts modelling the same project on the same contracts will produce different numbers, and the difference will almost never be arithmetic. It will be basis (pre-tax or post-tax, accrual or cash, before or after working capital), horizon (fifteen years of evaluation or twenty-five of concession), convention (annual or quarterly periods, interest on opening or average balances) and case (the sponsor's escalation or the lender's flat line). On Kestrel's own figures one project supports five arithmetically correct net present values spanning **USD 29,545,516** — which is why a model output quoted without its basis and horizon is not information but decoration.

Learning objectives. After this domain a candidate can: design a three-block architecture with a single timeline and defend each convention chosen; build a sources-and-uses statement in which capitalised interest is computed rather than assumed, identify its balancing line and test that line against policy; quantify how period length and interest convention change capitalised interest, the depreciation base and after-tax value; distinguish accounting depreciation from tax allowances and compute the coverage and debt-capacity consequences; build a debt schedule that closes to zero, run a cash waterfall through reserve funding to distributions, and compute an equity return and an equity payback; state and test the six arithmetic invariants a project model must satisfy, explain why a balancing balance sheet is necessary and not sufficient, and localise an error from the check that fails; reconcile two models of one project into a basis bridge; build a one-at-a-time sensitivity table with elasticities, state honestly what it cannot see, and convert ratio headroom into input breakevens; compute the net value, breakeven fee, breakeven error rate and breakeven detection rate of a model audit and show why timing matters more than price; and govern AI-assisted modelling with controls specific enough to fail.

The master model. Kestrel Water SPC continues from Domains 1–5 and 10. Capital cost **USD 60,000,000** funded **70/30** as **USD 42,000,000** of senior debt at **6.0 % over 12 years** (annual instalment **USD 5,009,635.23**; year-one interest **2,520,000**, principal **2,489,635**) plus **USD 18,000,000** of equity. Operating life **25 years**; revenue **12,000,000**, cash operating costs **4,500,000**, `EBITDA` **7,500,000**, depreciation **2,400,000**, `EBIT` **5,100,000**, cash tax **516,000**, documented first-year `CFADS` **USD 6,384,000** (6,984,000 before working-capital movements). Coverage at close (Domain 10): `DSCR` **1.2743** = `LLCR` **1.2743**, `PLCR` **1.9431**, covenant cash trigger **6,011,562**, six-month DSRA **2,504,818**. Appraisal (Domain 4): `NPV` **+16,179,360** at 8 %, `IRR` **12.19 %**, `MIRR` **9.73 %**, `PI` **1.270**. This domain adds the assumptions those chapters did not need and could not do without: the **eight-quarter construction drawdown** funded 70/30 against certified spend, the **EPC price of 48,000,000** (Domain 5) inside a **60,000,000 envelope** whose balancing line is contingency at **3,645,403**, capitalised interest of **2,114,597**, the **2.967 % escalation** that reconciles Domain 4's appraisal to Domain 2's `CFADS`, the DSRA funded from operating cash over two years, and the two cases — sponsor and bank — that every subsequent number has to be labelled against.

KNOWLEDGE AREA 6.1

Model architecture

Topics: 6.1.1 the three-block rule · 6.1.2 the timeline as the model's spine · 6.1.3 basis, horizon and case — the labels without which an output means nothing.

6.1.1 The three-block rule

Definition. A project financial model is organised into three strictly separated blocks: **inputs** (every assumption, each entered exactly once, in one place, with a source), **calculations** (formulae only, no constants), and **outputs** (presentation only, no calculation). Nothing flows backwards: outputs never feed calculations, calculations never contain assumptions, inputs never contain formulae.

Why the separation carries the weight. It is not tidiness; it is what makes a model auditable by a stranger in the time a stranger has. An assumption entered once can be changed once, so a scenario is a switch rather than a search-and-replace across sheets — which is why models without the separation end up with scenarios that differ in ways nobody intended. A calculation block containing no constants can be **read**: any number inside a formula is by construction a defect, so review reduces to scanning for numerals. And an output block that computes nothing cannot disagree with the model, which removes the most damaging class of audit finding — the summary page that no longer ties to the engine behind it.

The cardinal defect is the hard-coded constant. A tax rate typed into a formula, a tariff escalator embedded in a growth row, a debt margin buried in an interest calculation: each is invisible to a reviewer, survives every scenario switch, and produces a model that is stable in the wrong answer. Two further rules earn their place. **One row, one formula, copied across**: a row whose formula changes mid-timeline is invisible in the printed output and usually marks the point where somebody patched a symptom. And **sign discipline stated once and enforced everywhere** — either convention works, and mixing them produces a model that adds a cost to a revenue and balances anyway.

6.1.2 The timeline as the model's spine

Definition. The timeline is a single row of periods with, for each period, its start and end dates, its day count on the stated convention, and a set of mutually exclusive **flags** — construction, operations, loan life, post-loan tail — that every calculation multiplies by instead of testing dates itself.

The flags convert scattered date logic into arithmetic that can be checked by addition. Kestrel's annual model runs **27 periods** — two construction and twenty-five operating — and its quarterly construction model runs **eight**. The invariants are trivial to state and therefore trivial to test: construction flags sum to the construction period, operating flags to the operating life, no period carries two flags, and all flags sum to the model length. A model in which those four additions hold cannot have a period that is silently both building and operating, which is the defect that puts revenue into a construction quarter or capitalised interest into an operating year.

Periodicity is an economic choice, not a presentation one. Coarser periods do not merely round; they systematically misstate anything that compounds within a period. Worked example 6.2.1 prices this on Kestrel: the same drawdown profile at the same 6.0 % returns capitalised interest of **2,114,597** quarterly and **1,247,352** annually — an understatement of **867,245**, or **41.01 %**, before a single assumption has been challenged.

The conventions that must be stated once, in the inputs block: day count (30/360, actual/360, actual/365 — Domain 5's slip calculation used 30/360 and said so); whether flows arise at period end, start or mid-period; whether interest accrues on the opening or the average balance; and whether the first period is a stub. None is a matter of taste; each changes printed results, and a model that does not state them cannot be reproduced.

6.1.3 Basis, horizon and case

Definition. Every model output carries three labels without which it is meaningless: its **basis** (pre-tax or post-tax; accrual or cash; before or after working capital; levered or unlevered), its **horizon** (the number of periods valued, and what is assumed about value beyond them), and its **case** (which set of assumptions was switched on).

WORKED EXAMPLE

Worked example 6.1.3 — one project, five correct net present values.

1. **Setup.** Kestrel, on the master model. Domain 4 appraised net inflows of **8,900,000** a year for **15 years** against I_0 = 60,000,000 at **8.0 %**. Domain 2 documented a first-year `CFADS` of **6,384,000**. The concession's operating life is **25 years**. Reconcile the two figures and value the project on each defensible basis and horizon.
2. **Formula.** Domain 4's present value of inflows is `8,900,000 × AF(0.08, 15)`. An `EBITDA` stream starting at 7,500,000 and escalating at `g` has present value `Σ 7,500,000 (1+g)^(t−1)/(1.08)^t`; solve for the `g` that reproduces it. Post-tax unlevered free cash flow, with revenue, cash costs and working capital all indexed at the same `g` and depreciation fixed at 2,400,000, reduces to `FCF(t) = 5,400,000 (1+g)^(t−1) + 480,000` (the second term is the depreciation shield, `0.20 × 2,400,000`).
3. **Substitution.** `8,900,000 × 8.559479 = 76,179,360`. Solving the escalating `EBITDA` stream over 15 years at 8 % gives `g` **= 2.967 %**, reproducing a present value of 76,180,021 — **661** above Domain 4's 76,179,360, or 0.001 %. Then value `FCF(t)` at 8 % over 15 and 25 years, at `g` = 2.967 % and at `g` = 0.
4. **Result.**

#	Basis	Horizon	Case	`NPV` at 8 %
1	Pre-tax operating cash (Domain 4)	15 years	escalating	**+16,179,360**
2	Post-tax, unlevered, after working capital	15 years	`g` = 2.967 %	**−1,041,835**
3	Post-tax, unlevered, after working capital	25 years	`g` = 2.967 %	**+19,875,251**
4	Post-tax, unlevered, after working capital	25 years	flat (bank case)	**+2,767,684**
5	Post-tax, unlevered, after working capital	15 years	flat	**−9,670,265**

The widest defensible spread is **29,545,516** — from −9,670,265 to +19,875,251 — on one asset, one contract set and one discount rate. 5. **Interpretation.** Not one of those five figures is an error, and the reconciliation proves it: Domain 4's 8,900,000 is precisely what Kestrel's year-one `EBITDA` of 7,500,000 becomes as a fifteen-year level equivalent at 2.967 % escalation, so the two models agree on the asset and disagree only about what to measure. The spread decomposes cleanly and each component is a decision somebody must own. **Basis** moves the fifteen-year answer from +16,179,360 to −1,041,835: cash tax and working capital cost the project **17,221,195** of present value, and whoever approves an appraisal on a pre-tax basis has approved that omission. **Horizon** moves the post-tax answer from −1,041,835 to +19,875,251: ten further years of a twenty-five-year concession are worth **20,917,086**, so a fifteen-year evaluation of a twenty-five-year asset is not conservatism, it is a valuation of a different asset. **Case** moves the twenty-five-year answer from +19,875,251 to +2,767,684: the lender's refusal to give credit for escalation is worth **17,107,567** of the sponsor's case, which is why escalation is negotiated as hard as margin. The professional consequence is a rule, not a caution: **no `NPV`, `IRR` or coverage ratio leaves a model without its basis, horizon and case attached to it in the same sentence.** And the reviewer's habit that follows is the **basis bridge** — a reconciliation from the appraisal number the board remembers to the financing number the lenders enforce, showing every step between them. Kestrel's bridge is four lines long and worth 29.5 million; a project without one is carrying that exposure undocumented.

AI in this KA

Where it earns its place. Architectural conformance is machine-checkable and tedious, the ideal combination: scanning a calculation block for embedded numerals, finding rows whose formula changes mid-timeline, listing inputs referenced from more than one place, testing the flag additions, and diffing two versions into a change list a human can read. A large model has tens of thousands of cells and a reviewer has two days; this is the part of review that should never be done by a person.

Where it must not go. An assistant must not choose the model's conventions, and in practice it will: asked to "build a construction model" it picks an annual timeline with opening-balance interest, because that is the simplest defensible construction — and Worked example 6.2.1 prices that choice at 867,245. Nor may it decide the basis, horizon or case; those are governance decisions with owners, and a model whose horizon was chosen by a tool is a model nobody has approved.

Verification, concretely. Require the tool to output the conventions it assumed as an explicit list, and check that list against the term sheet and the assumption register line by line. Re-derive one period of the timeline by hand — dates, day count, flags — and confirm the four flag additions. For any machine-generated architecture review, confirm a sample of at least ten flagged cells against the workbook before the findings are reported: a false positive in an architecture review costs more credibility than the defect it replaced. **AI proposes; the professional verifies, decides and remains accountable.**

KEY TERMS — KA 6.1

Term	Meaning
Three-block architecture	Inputs, calculations, outputs — strictly separated; no back-flow.
Hard-coded constant	An assumption typed inside a formula; invisible to review and immune to scenarios.
Timeline flags	Mutually exclusive period markers (construction, operations, loan life, tail) that calculations multiply by.
Periodicity	Period length; an economic choice because compounding happens within periods.
Basis	Pre/post-tax, accrual/cash, before/after working capital, levered/unlevered.
Horizon	Periods valued, and the treatment of value beyond them.
Case	The named set of assumptions switched on (sponsor case, bank case, downside).
Basis bridge	The documented reconciliation between two models of one project.

SAMPLE MCQS — KA 6.1

MCQ 6.1-A `[6.1.3 · Analysis]` Kestrel's appraisal shows `NPV` +16,179,360 (pre-tax operating cash, 15 years) and its financing model shows +2,767,684 (post-tax unlevered, flat, 25 years). The correct conclusion is:

- A. the financing model contains an error of 13,411,676
- B. the appraisal is optimistic and should be discarded
- C. both are correct on their stated basis, horizon and case, and the required deliverable is the bridge between them
- D. the difference is the interest tax shield

Rationale: Worked example 6.1.3 reconciles them exactly — the appraisal's 8,900,000 is the fifteen-year level equivalent of an EBITDA stream escalating at 2.967 %, and the gap is basis, horizon and case. A and B assume one number must be wrong; D names a term that is absent from both, since each figure here is unlevered.

MCQ 6.1-B [6.1.2 · Application] Kestrel's construction spend is modelled with the same profile and the same 6.0 % rate on an annual rather than a quarterly timeline, interest accruing on the opening balance in both. Capitalised interest changes from 2,114,597 to:

- A. 2,114,597 — periodicity does not affect a total
- B. 1,247,352, an understatement of 867,245
- C. 2,427,554
- D. 8,458,388

Rationale: Coarse periods ignore intra-period draws, so the opening balance on which interest accrues is far too small (6.1.2, Worked example 6.2.1). A assumes periodicity is presentational; C is the quarterly figure on the average-balance convention; D multiplies the quarterly figure by four, confusing periods with rates.

MCQ 6.1-C [6.1.1 · Analysis] A reviewer finds a 20 % tax rate typed inside three calculation formulae rather than referenced from the inputs block. The most serious consequence is:

- A. the model is harder to read
- B. the tax rate cannot be changed
- C. scenario switches silently leave the tax rate unchanged, so every case in the model is internally inconsistent in a way no output reveals
- D. the model will not balance

Rationale: The hard-coded constant survives the scenario switch, so the downside case is run at the base-case tax rate and nothing on any output page says so (6.1.1). A understates it; B is false — it can be changed, three times, which is the problem; D is wrong, because a consistent wrong number balances perfectly (6.4.1).

SELF-CHECK — KA 6.1

1. State the four additions that test a timeline. — Construction flags sum to the construction period; operating flags to the operating life; no period carries two flags; all flags sum to the model length.
2. Why is a hard-coded constant worse than a wrong constant? — A wrong constant is visible in the inputs and can be challenged; a hard-coded one is invisible, immune to scenarios and therefore stable in the wrong answer.
3. State the three labels every model output must carry. — Basis, horizon, case (6.1.3); without them the spread on Kestrel is 29,545,516 and the number conveys nothing.

KNOWLEDGE AREA 6.2

Construction and operating models

Topics: 6.2.1 sources and uses, and capitalised interest · 6.2.2 the operating model and the articulation check · 6.2.3 depreciation, tax allowances and the line most often modelled wrongly.

6.2.1 Sources and uses, and capitalised interest

Definition. The **sources-and-uses statement** lists everything the project must pay for between mandate and commercial operations (uses) against everything that funds it (sources), and the two totals are identical by construction. **Interest during construction (IDC)**, or capitalised interest, is the interest accruing on debt drawn before the asset earns revenue; it is a use of funds like any other, is itself funded, and enters the depreciable asset base.

The identity and the balancing line. Sources equal uses is an identity, not a check — a model can always be made to satisfy it. What makes it informative is knowing **which line balances**. The funding envelope is fixed first, by what sponsors will commit and lenders will lend, and one use absorbs the residual: usually contingency, sometimes owner's costs. The discipline has three steps: let one line balance; **test that line against policy** (is the resulting contingency inside the band lenders require for this technology and contract structure?); and if it is not, the envelope is wrong, not the table. One corollary catches plugged tables at a glance: **the balancing line is never a round number.**

WORKED EXAMPLE

Worked example 6.2.1 — Kestrel's construction funding, and what the timeline costs.

1. **Setup.** A **60,000,000** envelope funded **70/30** — 42,000,000 senior debt at 6.0 %, 18,000,000 equity — with every use, capitalised interest included, funded in that proportion against certified spend. Committed uses: EPC price **48,000,000** (Domain 5), owner's costs and land **3,600,000**, capitalised development costs **1,800,000**, arrangement and financing fees at 2.0 % of the facility, **840,000**. Construction runs **eight quarters** with a certified spend profile of **6, 9, 13, 16, 17, 15, 13 and 11 per cent**. Interest accrues quarterly at 6.0 %/4 on the **opening** debt balance (draws treated as made at period end). Fees and development costs are funded at financial close. Compute capitalised interest, the balancing contingency, and the cost of running the same model annually.
2. **Formula.** For each quarter: `interest = opening debt balance × 0.015`; `funding requirement = certified spend + interest`; `debt draw = 0.70 × requirement`; `equity draw = 0.30 × requirement`; closing balance = opening + debt draw. Contingency solves `fees + development + EPC + owner's costs + contingency + IDC(contingency) = 60,000,000`, and the profile applies to the **certified-spend base** — EPC plus owner's costs plus contingency.
3. **Substitution.** Close: requirement 2,640,000, debt draw 1,848,000. The certified-spend base solves to **55,245,403**, so quarter 1's 6 % is 3,314,724. Quarter 1: interest `1,848,000 × 0.015 = 27,720`; requirement `3,314,724 + 27,720 = 3,342,444`; debt draw 2,339,711; closing balance 4,187,711. Iterating to quarter 8 gives a closing balance of exactly 42,000,000 and cumulative equity of exactly 18,000,000.
4. **Result.**

Uses	USD	Sources	USD
EPC contract price	48,000,000	Senior debt drawn	42,000,000
Owner's costs and land	3,600,000	Equity contributed	18,000,000
Capitalised development costs	1,800,000		
Arrangement and financing fees (2.0 %)	840,000		
Contingency (balancing line)	**3,645,403**		
Interest during construction	**2,114,597**		
Total uses	**60,000,000**	**Total sources**	**60,000,000**

Capitalised interest is **3.52 %** of the envelope. Contingency is **7.59 %** of the EPC price. On an **average**-balance interest convention IDC rises to **2,427,554** (+312,957); on an **annual** timeline with opening-balance interest it collapses to **1,247,352** (−867,245, or −41.01 %); on an annual timeline with average-balance interest it is **2,481,619**. 5. **Interpretation.** Three results leave this example. **The convention is worth more than most of the assumptions people argue about.** A 41 % swing in capitalised interest from period length alone dwarfs a 10-basis-point margin negotiation, and it is invisible in every output the model prints. It also propagates: 867,245 of understated IDC understates the depreciable base by the same amount, annual depreciation by **34,690**, and the present value of the tax it shelters — 20 % over 25 years at 8 % — by **74,061**. Small, but silent: a reviewer finds it in one line and a board never sees it. **The pairing of a coarse timeline with an opening-balance convention is the defect, not either choice alone.** An annual model on average balances lands within 367,022 of the quarterly answer, so the 41 % error requires both mistakes together — which is why convention lists must be read as a set. **And the balancing line is evidence.** Kestrel's 3,645,403 of contingency is 7.59 % of the EPC price, inside the band a lender would expect for proven water-treatment technology under a full EPC wrap (Domain 5, KA 5.4.1; Domain 8, KA 8.1.2 tests that percentage against estimate class), so the envelope is credible. Had the residual come out at 2 %, the correct response would not have been to accept a thin contingency but to conclude that a 60,000,000 envelope does not fund this project — a financing conversation, not a modelling one. **Contingency is what is left, and what is left must still be defensible**; the arithmetic cannot tell you which of those two is binding.

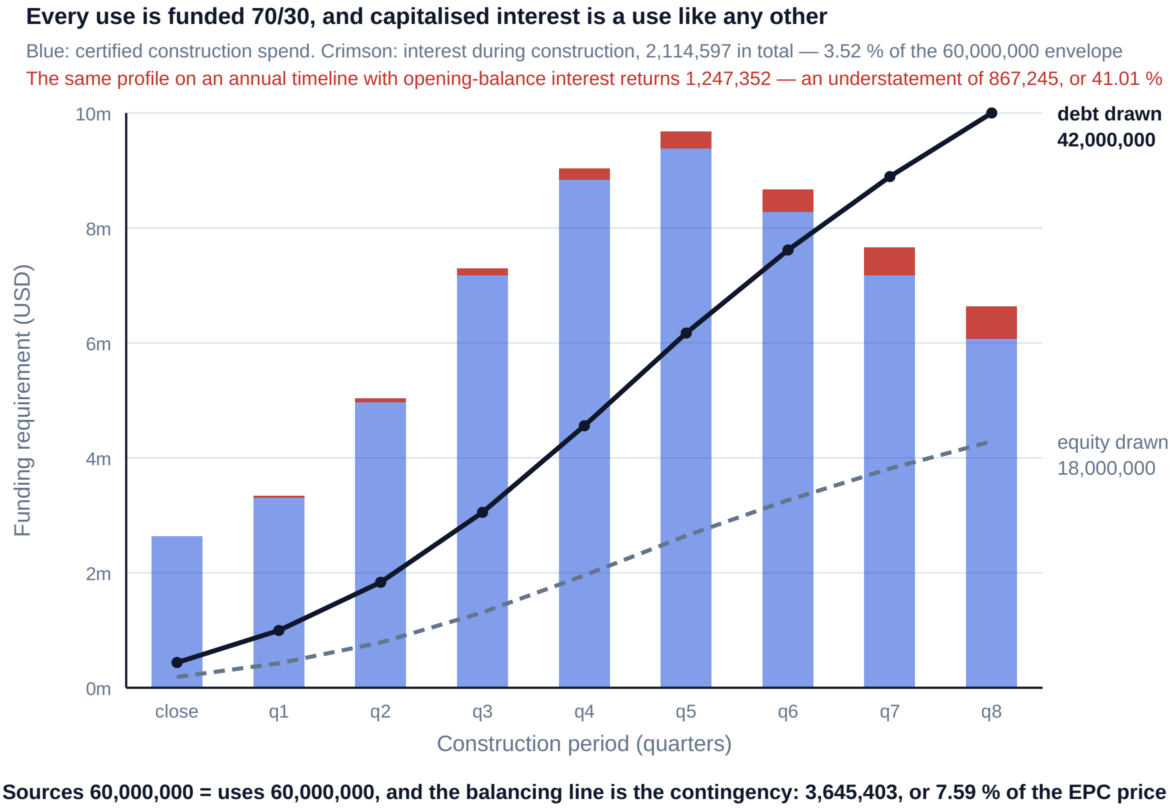

Fig 6.2.1 — Construction funding and interest during construction

6.2.2 The operating model and the articulation check

Definition. The operating model produces, for every operating period, an income statement, a balance sheet and a cash-flow statement that **articulate** — each derived from the same calculations, so that the three cannot disagree — and from that cash-flow statement it derives `CFADS` on the facility's documented definition.

Domain 2 built the three statements and Domain 10 built the ratios; the modelling content here is the linkage, and one identity does most of the work. Where interest paid is classified in operating cash flow, `CFADS` **equals operating cash flow plus interest paid** — because `CFADS` is struck before all debt service and operating cash flow is struck after the interest half of it. That single line converts `CFADS` from an assertion into something a reviewer can tie to a statement, which is the whole point of building the statements at all.

WORKED EXAMPLE

Worked example 6.2.2 — Kestrel's first operating year, proved three ways.

1. **Setup.** Opening balance sheet at the commercial operations date: plant 60,000,000, cash nil, debt 42,000,000, equity 18,000,000. Year one: revenue 12,000,000, cash operating costs 4,500,000, depreciation 2,400,000, interest 2,520,000, tax at 20 % of taxable profit, receivables up 900,000, payables up 300,000. Debt service 5,009,635.23 of which principal 2,489,635.23. The facility requires a six-month DSRA of **2,504,818**, funded from operating cash in two equal annual instalments of **1,252,409**, ranking above distributions (Domain 10, KA 10.3.3). Distributions are the residual, subject to the 1.20× test.
2. **Formula.** `EBITDA` = revenue − cash costs; `EBIT` = `EBITDA` − depreciation; taxable profit = `EBIT` − interest; operating cash flow = net income + depreciation − Δreceivables + Δpayables; `CFADS` = operating cash flow + interest paid; distributable cash = `CFADS` − debt service − reserve funding; closing cash = operating cash flow − principal − distributions.
3. **Substitution.** `EBITDA` 7,500,000; `EBIT` 5,100,000; taxable profit 2,580,000; tax 516,000; net income 2,064,000. Operating cash `2,064,000 + 2,400,000 − 900,000 + 300,000`. `CFADS` `3,864,000 + 2,520,000`. Distributions `6,384,000 − 5,009,635.23 − 1,252,408.81`.
4. **Result.** Operating cash flow **3,864,000**; `CFADS` **6,384,000** — reconciling to the documented figure exactly; `DSCR` **1.2743**; distributable cash **121,956**.

Closing balance sheet	USD		USD
Plant, net	57,600,000	Payables	300,000
Receivables	900,000	Senior debt	39,510,365
Cash (restricted, DSRA)	1,252,409	Equity	19,942,044
Total assets	**59,752,409**	**Total**	**59,752,409**

1. **Interpretation.** The balance sheet balances to the cent, closing cash equals the DSRA balance, and `CFADS` ties to the cash-flow statement through the interest line — three independent confirmations of internal consistency. What the year says about the project is more interesting than any of them. Equity receives **121,956** in its first operating year: **0.68 %** of the 18,000,000 contributed, on a project with a healthy `DSCR` and a positive `NPV`. That is not distress, it is the ordinary arithmetic of a leveraged project with a reserve to build, and it is the number sponsors most reliably fail to model before close — Domain 10's Case study B is the same error with a board attached. Three consequences follow. **Reserve funding is a claim on early cash, and early cash is all the equity has**: a distribution forecast struck before reserve funding is an aspiration. **The residual is where modelling risk concentrates** — the distribution's elasticity to revenue is **73.8** against the `CFADS` elasticity of 1.4098, so the equity line is about **52 times** more revenue-sensitive than the coverage ratio, and a **1.36 %** revenue miss (162,608 of revenue) erases the first-year distribution entirely. And **the restricted-cash label is load-bearing**: the 1,252,409 in the DSRA is on the balance sheet, is not available to the business, and a liquidity statement showing it as cash is wrong in exactly the way that matters.

6.2.3 Depreciation, tax allowances and the line most often modelled wrongly

Definitions. Accounting depreciation allocates the cost of an asset over its useful life under the applicable financial-reporting framework — Kestrel's 60,000,000 over 25 years, straight line, **2,400,000** a year. **Tax depreciation**, or capital allowances, is what the tax authority permits to be deducted, on its own profile and its own base. **Cash tax** is what is actually paid. They are three different numbers, and only the last enters `CFADS`.

Domain 2 established the principle (KA 2.A.1) and named the standard finding: modelling accounting tax as if it were cash tax. This is where it is priced.

WORKED EXAMPLE

Worked example 6.2.3 — what a tax assumption is worth.

1. **Setup.** Kestrel's base model, following Domain 2, assumes tax depreciation equals accounting depreciation — a simplification, chosen for exposition and stated as one. Suppose instead the jurisdiction grants **declining-balance capital allowances at 15 %** on the 60,000,000 base, with tax losses carried forward without limit. Compute cash tax for the first five operating years and the consequences for coverage and debt capacity. (Rates, profiles, loss-carry rules and the availability of any allowance are jurisdiction-specific and change; the arithmetic below is transferable, the treatment is not, and the treatment is a matter for qualified tax advice in the relevant jurisdiction.)
2. **Formula.** Allowance in year `t` = 15 % of the written-down value; taxable profit = `EBITDA` − allowance − interest − losses brought forward; cash tax = 20 % of taxable profit if positive, otherwise nil with the loss carried forward.
3. **Substitution and result.**

Year	Allowance	Interest	Taxable profit	Loss carried forward	Cash tax
1	9,000,000	2,520,000	(4,020,000)	4,020,000	**nil**
2	7,650,000	2,370,622	(6,540,622)	6,540,622	**nil**
3	6,502,500	2,212,281	(7,755,403)	7,755,403	**nil**
4	5,527,125	2,044,440	(7,826,968)	7,826,968	**nil**
5	4,698,056	1,866,528	(6,891,552)	6,891,552	**nil**

Year-one `CFADS` becomes `7,500,000 − nil − 600,000 =` **6,900,000**, a `DSCR` of **1.3773** against 1.2743 — **0.1030** of coverage from a tax assumption alone. At Domain 10's 1.30× sizing target, debt capacity rises from **41,171,123** to **44,498,864**: an extra **3,327,741** of debt, and therefore 3,327,741 less equity. 4. **Interpretation.** The comparison is not between a right and a wrong model but between two assumptions about a jurisdiction, and the gap between them is **more than four times the 828,877 debt-capacity shortfall Domain 10's entire negotiation was fought over**. That is why the tax line is both the most commonly mis-modelled row in project finance and the most consequential: technically difficult, jurisdiction-specific, liable to change during the loan's life, and sitting directly in `CFADS`, so every coverage ratio and sizing calculation inherits it. Four disciplines follow. **The tax line traces to a written opinion**, not to a modeller's understanding, and the opinion's date and scope go in the assumption register. **Cash tax and accounting tax are separate rows**, always, even where they happen to be equal — a model that conflates them cannot represent the year in which they diverge. **Tax losses carry in a stated balance with a stated expiry rule**, since losses that vanish or persist wrongly misstate cash tax for a decade. And **the sponsor case must not bank an allowance regime the lenders will not credit**: sizing debt on 44,498,864 because a fifteen-per-cent allowance exists today embeds a legislative forecast in a twelve-year loan — a risk for Domain 11's register, not a benefit. The honest presentation shows both, states which case governs which decision, and puts the 3,327,741 of capacity difference in front of the credit committee rather than inside the model.

AI in this KA

Where it earns its place. Building a first-pass construction and operating model from a term sheet, a spend curve and stated conventions is fast machine work, and so is the surrounding apparatus: the sources-and-uses skeleton, the loss-carry-forward logic, the articulation checks, and the assumption register drafted from the model's own input block so that register and model cannot drift apart.

Where it must not go. It must not select the tax treatment. Worked example 6.2.3 shows that choice moving debt capacity by 3,327,741, and an assistant asked about capital allowances will produce a fluent, jurisdiction-neutral answer that is nobody's tax opinion. The same prohibition covers the useful life behind accounting depreciation, whether a cost is capitalised or expensed (Domain 2, KA 2.3.3), and whether an allowance survives a change of control — advice questions with named professional owners.

Verification, concretely. Recompute one construction quarter and one operating year by hand and tie all three statements; it is a dozen operations and it catches structural error immediately. Confirm the implied effective tax rate period by period against the statutory rate and the explained differences — the check that catches Worked example 6.4.1's failure. Require every tax input to carry the reference and date of the opinion supporting it, and reject the model, not the input, where that reference is absent.

KEY TERMS — KA 6.2

Term	Meaning
Sources and uses	The funding requirement against its funding; identical totals by construction.
Interest during construction (IDC)	Interest accruing before revenue; a funded use, and part of the depreciable base.
Balancing line	The use that absorbs the residual of a fixed envelope; must be tested against policy.
Articulation	The three statements derived from one calculation set, so they cannot disagree.
`CFADS` **tie**	`CFADS` = operating cash flow + interest paid, where interest is classified as operating.
Capital allowance	Tax depreciation on the tax authority's profile and base; not accounting depreciation.
Restricted cash	Reserve balances on the balance sheet that are not available to the business.

SAMPLE MCQS — KA 6.2

MCQ 6.2-A `[6.2.1 · Application]` Kestrel's committed uses are 48,000,000 of EPC, 3,600,000 of owner's costs, 1,800,000 of development costs and 840,000 of fees, inside a 60,000,000 envelope, with capitalised interest computed at 2,114,597. The contingency is:

- A. 3,600,000
- B. 3,645,403
- C. 5,760,000
- D. 6,000,000

Rationale: `60,000,000 − 48,000,000 − 3,600,000 − 1,800,000 − 840,000 − 2,114,597 = 3,645,403` (6.2.1). A is a plausible round number and therefore the tell-tale of a plugged table; C omits capitalised interest from the deduction; D applies a 10 % rule of thumb to the envelope instead of computing the residual.

MCQ 6.2-B `[6.2.2 · Application]` Operating cash flow is 3,864,000 and interest paid, included in operating cash flow, is 2,520,000. `CFADS` is:

- A. 1,344,000
- B. 3,864,000
- C. 6,384,000
- D. 8,904,000

Rationale: `CFADS` = operating cash flow + interest paid (6.2.2). A deducts interest a second time; B forgets that operating cash flow is already struck after interest; D adds interest a second time to a `CFADS` already struck before debt service, a figure no definition produces — and principal, a financing flow, never enters operating cash flow at all.

MCQ 6.2-C `[6.2.3 · Analysis]` A model of a project with flat revenue and level annuity debt service shows cash tax of 516,000 against taxable profit of 2,580,000 in year one and, in year eight, cash tax of 766,771 against taxable profit of 3,833,856. The statutory rate is 20 %. A reviewer should conclude:

- A. the tax line is wrong, because cash tax has risen while revenue is flat
- B. the effective rate is 20.0 % in both years, so the tax line is consistent; cash tax rises because the interest deduction falls as the loan amortises
- C. the model has omitted deferred tax
- D. the loss carry-forward has been applied incorrectly

Rationale: `516,000/2,580,000 = 766,771/3,833,856 = 20.0 %`; the rise is the amortising interest deduction, the mechanism behind the year-12 minimum of 6.4.1 (6.2.3, KA 6.4.1). A mistakes a correct behaviour for an error; C and D name plausible defects for which there is no evidence here.

MCQ 6.2-D `[6.2.3 · Analysis]` Under 15 % declining-balance allowances Kestrel pays no cash tax for five years, lifting year-one `DSCR` from 1.2743 to 1.3773 and debt capacity at 1.30× from 41,171,123 to 44,498,864. The sound treatment is:

- A. size the debt at 44,498,864, since the allowance exists
- B. ignore the allowance, since lenders will not credit it
- C. model both, state which case governs which decision, and put the 3,327,741 of capacity difference — and the legislative risk in it — in front of the credit committee
- D. average the two capacities

Rationale: The difference is an assumption about a jurisdiction over a twelve-year loan, so it is disclosed and owned rather than banked or suppressed (6.2.3). A embeds a legislative forecast in a financing; B discards a real economic benefit; D is arithmetic without meaning.

SELF-CHECK — KA 6.2

1. State the *`CFADS`* tie and why it matters. — `CFADS` = operating cash flow + interest paid; it converts `CFADS` from an assertion into a figure tied to a statement.
2. What does Kestrel's first-year distribution of 121,956 tell a sponsor? — That reserve funding consumes almost all first-year equity cash, and that the residual is far more sensitive to revenue than the coverage ratio is.

3. Why must cash tax and accounting tax be separate rows even when equal? — Because a model that conflates them cannot represent the year in which they diverge, and it will diverge.

KNOWLEDGE AREA 6.3

Debt schedules, the cash waterfall and returns

Topics: 6.3.1 the debt schedule and its circularity · 6.3.2 the waterfall in the model · 6.3.3 project return and equity return.

6.3.1 The debt schedule and its circularity

Definition. The debt schedule is the period-by-period record of opening balance, drawings, interest, scheduled principal, prepayments and closing balance, for each facility separately. It is the denominator of every coverage ratio (Domain 10, KA 10.3.3) and therefore load-bearing for covenant compliance.

Its arithmetic is Domain 3's amortisation table and needs no re-derivation; its **invariants** are what a modeller owns. The closing balance at maturity is zero; total scheduled principal equals total drawings; each period's interest equals the balance on which it accrues times the rate for the days in the period, on the stated convention; and debt service equals interest plus scheduled principal, tying to the schedule the facility agreement annexes. Kestrel's twelve years satisfy all four: principal rises from **2,489,635** in year one to **4,726,071** in year twelve, interest falls from **2,520,000** to **283,564**, the closing balance is **zero**, and the principal column sums to **42,000,000** exactly.

Circularity, and the convention that removes it. Capitalised interest depends on the debt balance; the balance depends on drawings; drawings include the funding of capitalised interest. In a workbook that is a circular reference, with three honest resolutions. **Charge interest on the opening balance** and the circularity disappears, because the period's interest is fixed before its draw — Worked example 6.2.1's choice, at a cost of 312,957 against the average-balance answer. **Iterate to convergence**, which is more accurate and needs either the workbook's iterative calculation or a documented macro. **Solve algebraically**, which for simple structures is best because it is deterministic. What is not acceptable is an undocumented iteration whose convergence nobody has tested, or a switch left off: both produce results that depend on the order in which someone pressed the keys. **The circularity resolution is a documented convention with a named owner, and the model states which one it uses.**

Two structural features belong in the schedule explicitly rather than as adjustments: a **cash sweep** applies a defined share of surplus cash to prepayment, so the schedule carries a prepayment row that changes future interest; and **sculpted** service (Domain 10, KA 10.1.3) sets each period's debt service from that period's `CFADS`, which is itself circular through the tax deduction — the same three resolutions apply, and the same requirement to say which.

6.3.2 The waterfall in the model

Definition. The cash waterfall is the contractual priority order in which each period's cash is applied. In a model it is a column of sequential deductions, each with a **test** and each producing a balance that can be nil but never negative.

Kestrel's order, per Domain 10 KA 10.3.3: cash operating costs, then cash tax, then **senior debt service**, then **reserve funding and top-ups**, then subordinated debt if any, then — only if every distribution test passes — **distributions to equity**. Three requirements make the difference between a waterfall and a subtraction.

Every tier is floored at zero and the shortfall is carried, not lost. A tier that goes negative has silently netted its shortfall against a later one, which is the defect that makes a distressed case look survivable. The correct construction pays what is available, records the shortfall, and applies the documented consequence — reserve drawing, lock-up, event of default.

Tests are modelled, not assumed. The distribution test is a formula with a threshold, so the model must show cash trapped when it fails. Kestrel's lock-up bites at a `DSCR` of 1.15×, a `CFADS` of **5,761,081**; a model whose distribution row is simply the residual after debt service has modelled an arithmetic identity, not a facility.

Reserve movements are two-directional and ranked. Funding a reserve is a use above distributions; drawing on it is a source; releasing it at maturity is a source. Kestrel's DSRA is funded at 1,252,409 in each of the first two operating years, sits at 2,504,818 for the rest of the loan, and **releases in year twelve** — which is why year twelve's distribution is 3,431,895 against year eleven's 980,580, and why a model that omits the release understates the equity return without touching any coverage ratio.

6.3.3 Project return and equity return

Definitions. The **project return** values the asset before financing: unlevered post-tax free cash flow against total capital cost. The **equity return** values the sponsors' position: equity drawn during construction against distributions actually received after every prior claim in the waterfall. They answer different questions and are routinely conflated.

WORKED EXAMPLE

Worked example 6.3.3 — what Kestrel actually returns, to whom, and when.

1. **Setup.** The master model on the **bank case**: revenue flat at 12,000,000, cash tax computed on the real interest schedule so that `CFADS` declines as the interest deduction amortises, DSRA funded over two years and released in year twelve. Equity is drawn 30 % of each period's funding requirement — **8,203,951** in construction year one and **9,796,049** in year two. Compare with the **sponsor case**, identical but with revenue, cash costs and working capital escalating at the 2.967 % of Worked example 6.1.3.
2. **Formula.** Unlevered free cash flow = `EBITDA` − 20 % × (`EBITDA` − depreciation) − Δworking capital; project `NPV` = `PV` at 8 % − 60,000,000; project `IRR` solves `NPV` = 0. Distribution in period `t` = `CFADS(t)` − debt service − reserve funding + reserve release; equity `IRR` solves the present value of draws and distributions to zero.
3. **Substitution.** Unlevered free cash flow `7,500,000 − 0.20 × 5,100,000 − 600,000 = 5,880,000`, level for 25 years; `× AF(0.08, 25) = 5,880,000 × 10.674776`. Equity flows: −8,203,951, −9,796,049, then 121,956 · 92,080 · 1,312,821 · … · 980,580 · 3,431,895, then 5,880,000 for thirteen years.
4. **Result.**

Measure	Bank case	Sponsor case
Project (unlevered, post-tax) NPV at 8 %	**+2,767,684**	**+19,875,251**
Project IRR (unlevered, post-tax)	**8.54 %**	—
Equity IRR	**9.83 %**	**13.52 %**
Total distributions over 25 years	**90,507,502**	**151,536,729**
Money multiple on 18,000,000	**5.028×**	**8.419×**
Equity payback from financial close	**14.67 years**	—

1. **Interpretation.** Set these beside Domain 4's appraisal — IRR **12.19 %**, discounted payback **10.07 years** — and the gap is the lesson. On a post-tax, twenty-five-year, flat-revenue basis the asset earns **8.54 %**, barely above the 8 % hurdle; the equity earns **9.83 %**, and the difference is leverage doing what leverage does. Domain 4's 12.19 % is neither: a pre-tax return over a fifteen-year horizon, sitting above both. Anyone who remembers "the project returns 12 %" has remembered a number describing no party's position. **Leverage pushes equity cash to the far side of the loan.** The 14.67-year equity payback against Domain 4's 10.07-year discounted payback is a different question, not a contradiction: equity waits for the loan to be repaid and the reserve released, and the first two operating years return **121,956** and **92,080** on 18,000,000 contributed. A sponsor whose investment committee expects distributions in year one has misread the waterfall, not the project. **The case, not the asset, produces most of the return.** The 3.69-point difference between 9.83 % and 13.52 % comes entirely from a 2.967 % escalation assumption — which is why escalation is contested in offtake negotiation (Domain 7) and why an equity IRR without its case is worthless. The habit that follows: state the escalation at which the equity return meets the sponsor's hurdle, and ask whether it is contracted, indexed to a published index, or hoped for. Finally, **the multiple and the rate tell different stories on purpose** — 5.028× over twenty-five years is 9.83 % a year, and a board shown only the multiple has been shown a long horizon dressed as a return.

AI in this KA

Where it earns its place. Solving a sculpted schedule, testing a cash sweep across hundreds of cases and computing rates of return over a scenario grid is root-finding and iteration at volume — work at which a human is slow and no more accurate. Generating the waterfall's test logic from the drafted priority order and then attacking that logic with adversarial cash flows is equally strong: the failure cases are enumerable, and a machine will enumerate them.

Where it must not go. It must not decide the waterfall's order or the tests' thresholds, which are terms of the finance documents, read by lawyers rather than inferred from a model. It must not choose the circularity resolution silently — an assistant picks whichever converges, and Worked example 6.2.1 prices that choice. And it must not be the source of a quoted return: an equity IRR is a number with a case attached, and the case is a governance object.

Verification, concretely. Substitute any machine-produced IRR back into the present-value equation and confirm it returns zero (Domain 4, KA 4.A.3 — the cheapest audit that exists). Run the waterfall with a CFADS low enough to fail every tier and confirm that no tier goes negative, that the shortfall is carried, and that the lock-up traps cash rather than the model netting it away. Re-derive one period's interest and principal by hand and tie to the annexed schedule. And require the tool to state, in words, which case it ran.

KEY TERMS — KA 6.3

Term	Meaning
Debt schedule invariants	Closing balance nil at maturity; Σ principal = drawings; interest ties to balance, rate and days.
Circularity	Interest depends on balance depends on drawings depends on interest; resolved by convention, iteration or algebra.
Waterfall tier	A ranked deduction with a test, floored at zero, with the shortfall carried.
Reserve release	The return of a funded reserve at maturity; a source that lifts the final period's distribution.
Project return	Unlevered post-tax cash against total capital cost; the asset's return.
Equity return	Draws against distributions after every prior claim; the sponsors' return.
Money multiple	Total distributions ÷ equity contributed; a horizon-blind measure.

SAMPLE MCQS — KA 6.3

MCQ 6.3-A `[6.3.3 · Analysis]` Kestrel's unlevered post-tax `IRR` is 8.54 %, its bank-case equity `IRR` is 9.83 %, and Domain 4's appraisal reported 12.19 %. The correct reading is:

- A. the appraisal was wrong by 3.65 points
- B. the asset earns 8.54 %, the equity earns 9.83 % because of leverage, and 12.19 % is a pre-tax fifteen-year figure describing neither party's position
- C. the equity return should exceed the project return by the debt margin
- D. the three should be equal once tax is removed

Rationale: Each measure is correct on its own basis and horizon; conflating them is the defect (6.3.3, 6.1.3). C invents a relationship — the gap depends on gearing, tenor and the shape of distributions; D is false, since horizon alone separates them.

MCQ 6.3-B `[6.3.2 · Application]` Kestrel's year-twelve distribution is 3,431,895 against year eleven's 980,580, on `CFADS` that falls between the two years. The explanation is:

- A. a cash sweep
- B. the DSRA of 2,504,818 releasing at final repayment
- C. an error, since `CFADS` fell
- D. the final principal instalment being smaller

Rationale: The reserve is released when the debt it secures is repaid, a source in the waterfall (6.3.2). C mistakes a modelled contractual event for an inconsistency; D is the opposite of the truth — year twelve's principal, 4,726,071, is the largest of the twelve.

MCQ 6.3-C `[6.3.1 · Analysis]` A construction model charges interest on the average debt balance and the workbook's iterative calculation is switched off. The consequence is:

- A. interest is understated by a known amount
- B. the model returns a stale or unconverged figure whose value depends on calculation order, so the result is not reproducible
- C. the model will not open
- D. capitalised interest becomes zero

Rationale: Average-balance interest is genuinely circular; without a resolution the answer is whatever the last pass left behind (6.3.1). A describes the opening-balance convention, which is deliberate and quantified at 312,957 on Kestrel; C and D are not how circular references behave.

MCQ 6.3-D [6.3.3 · Application] Kestrel returns 90,507,502 of distributions on 18,000,000 of equity over 25 years — a 5.028× multiple — at an equity IRR of 9.83 %. A board shown only the multiple has been shown:

- A. a complete picture, since the multiple includes every distribution
- B. a long horizon dressed as a return, because the multiple is blind to when the cash arrives
- C. an understatement, since multiples ignore reinvestment
- D. the same information as the IRR

Rationale: A multiple has no time dimension; 5.028× over 25 years is 9.83 % a year (6.3.3, and Domain 4's insistence that rates and ratios explain rather than decide). C reverses the bias; D is false — the same multiple over ten years would be a materially higher rate.

SELF-CHECK — KA 6.3

1. Name the four debt-schedule invariants. — Closing balance nil at maturity; Σ principal = drawings; interest ties to balance, rate and day count; debt service = interest + scheduled principal, tying to the annexed schedule.
2. Why must a waterfall tier never go negative? — A negative tier has silently netted a shortfall against a later claim, which makes a distressed case look survivable.
3. State Kestrel's first two operating-year distributions and what they teach. — 121,956 and 92,080 on 18,000,000: reserve funding consumes almost all early equity cash, so distribution forecasts must be struck after reserve funding and lock-up tests.

KNOWLEDGE AREA 6.4

Checks, sensitivity, model audit and AI controls

Topics: 6.4.1 the check block and the six invariants · 6.4.2 sensitivity, scenario and the breakeven translation · 6.4.3 model audit, governance and its economics · 6.4.4 AI-assisted modelling controls.

6.4.1 The check block and the six invariants

Definition. A **check block** is a dedicated output area in which every arithmetic invariant the model must satisfy is computed as a difference that should be nil, with a single aggregate flag that turns red if any check fails. It is built first, not last, and no output is quoted while it is red.

The six invariants a project model must satisfy:

#	Invariant	What its failure localises
1	The balance sheet balances, every period	An omitted flow, or a flow posted once
2	Sources equal uses, every construction period	A funding gap or a double-funded use
3	Closing debt is nil at maturity, for every facility	A schedule error or a missing prepayment
4	Σ scheduled principal = total drawings	A rounding accumulation or a mis-specified profile
5	Cash is never negative in any period or account	A liquidity failure the model has netted away
6	`CFADS` = operating cash flow + interest paid	A `CFADS` line built outside the statements

To these belong three ratio-level checks inherited from Domain 10 (KA 10.A.3): `LLCR` = `DSCR` where cash is level and service is an annuity at the loan rate; `PLCR` ≥ `LLCR` wherever a tail exists; and the **minimum** `DSCR` over the loan life reported alongside the average. And one that belongs to this domain: the **implied effective tax rate** reconciles, period by period, to the statutory rate plus explained differences.

WORKED EXAMPLE

Worked example 6.4.1 — the error that balances perfectly.

1. **Setup.** Kestrel's first operating year. A modeller computes tax as 20 % of `EBIT` rather than 20 % of taxable profit — a one-cell error, omitting the interest deduction. Every other formula is correct and consistent. Trace the consequences and identify which check catches it.
2. **Formula.** Correct: tax = 20 % × (`EBIT` − interest). As modelled: tax = 20 % × `EBIT`. Then propagate through net income, operating cash flow, `CFADS`, `DSCR`, distributable cash, the balance sheet and debt capacity at Domain 10's 1.30× target.
3. **Substitution.** Tax `0.20 × 5,100,000 = 1,020,000` against the correct 516,000; net income `2,580,000 − 1,020,000 = 1,560,000`; operating cash flow `1,560,000 + 2,400,000 − 600,000`; `CFADS 3,360,000 + 2,520,000`.
4. **Result.** `CFADS` **5,880,000** against 6,384,000; `DSCR` **1.1737** against 1.2743 — **below the 1.20× covenant**; distributable cash **−382,044**, so the DSRA cannot be funded on schedule; debt capacity at 1.30× falls from **41,171,123** to **37,920,771**, a shortfall of **3,250,352**. And the balance sheet **balances to the cent**, because the error is propagated consistently: net income is lower, equity is lower by the same amount, and assets tie.
5. **Interpretation.** This is the single most important result in the domain, and it is a negative one: **invariant 1 is necessary and not sufficient.** A consistent wrong number balances perfectly, and a modeller who reports "the model balances" has reported nothing about accuracy. What catches this error is the **effective-rate check**: `1,020,000/2,580,000 =` **39.53 %** against a statutory 20 %, a discrepancy no reader can rationalise. The negative distributable cash catches it too, through invariant 5 — but only if the waterfall is floored rather than allowed to run negative, which is precisely why 6.3.2 insists on the floor. The consequences scale badly in both directions. Understated, as here, the model reports a covenant breach that does not exist, kills 3,250,352 of debt capacity, and sends the sponsors to find equity they do not need. Substituting 3,250,352 of equity for debt at a 6-point spread (a 12 % equity requirement against the 6.0 % achieved on debt) costs 195,021 a year, a present value of **1,469,694** over the twelve years at 8 %. Overstated, the same class of error sizes debt the project cannot service, which is Worked example 6.4.3's illustrative error. The discipline is therefore not "check that it balances" but **"know what each check can and cannot see"**: the six invariants test structure, the effective-rate and ratio checks test economics, and only a reconciliation to the documents tests definitions (Domain 10, Toolkit 10.T.1).

The other half of the check block: the minimum, not the average. Domain 10 held Kestrel's `CFADS` level "for this illustration" and derived `DSCR` = `LLCR` = 1.2743. The modelled profile does not behave that way, and the reason is 6.2.3's: as the loan amortises the interest deduction shrinks, cash tax rises, and `CFADS` falls against a level instalment.

WORKED EXAMPLE

Worked example 6.4.1b — the minimum the level line hides.

1. **Setup.** Kestrel's bank case, revenue flat at 12,000,000, cash tax computed on the actual interest schedule. Compute `DSCR` for each of the twelve loan years and compare with the level assumption.
2. **Formula.** `CFADS(t) = 7,500,000 − 0.20 × (5,100,000 − interest(t)) − 600,000`; `DSCR(t) = CFADS(t)/5,009,635.23`.
3. **Substitution.** Year 1: interest 2,520,000, cash tax 516,000, `CFADS` 6,384,000. Year 12: interest 283,564, cash tax 963,287, `CFADS` 5,936,713.
4. **Result.** `DSCR` falls monotonically from **1.2743** in year one to **1.1851** in year twelve. The **minimum is 1.1851 — a breach of the 1.20× covenant** — and the average over the loan life is **1.2340**, comfortably above it. Present values move too: `PV` of the modelled `CFADS` at 8 % over 25 years is **65,315,883** against **68,147,771** on the level assumption, so the level line overstates value by **2,831,888**, or **4.34 %**. `LLCR` becomes **1.2395** and `PLCR` **1.8555**, against Domain 10's level-case 1.2743 and 1.9431 — the identity breaks, as Domain 10 said it must when cash is not level.
5. **Interpretation.** A model that holds `CFADS` level reports 1.2743 for twelve years and never shows the year in which the project breaches. The mechanism is unglamorous and entirely predictable — the interest tax shield amortises with the loan — and it is missed constantly because it requires the tax line to be modelled period by period rather than assumed. Three professional consequences. **Report the minimum and the year it occurs**, which is Domain 10's rule now earned rather than asserted: the average of 1.2340 passes and the project still breaches. **The escalation assumption decides where the minimum lies.** On the sponsor case, with revenue escalating at 2.967 %, `DSCR` rises from 1.2743 to **1.5940** and the minimum is year one — which is why lenders insist on the flat case: not because they believe it, but because it is the case that finds this defect. **And the fix belongs to sizing, not reporting.** At Domain 10's properly sized **41,171,123** the instalment is 4,910,769 and the year-twelve `DSCR` is **1.2087** — still inside the covenant. The debt level at which the year-twelve minimum is exactly 1.20× is **41,472,081**, so the structure needed **527,919** less debt than the 42,000,000 requested. Domain 10's 828,877 gap, argued on year-one coverage, was already protecting against a year-twelve problem nobody in that negotiation had modelled. That is the most valuable thing a properly built model does: it makes a conservative constraint legible as the specific risk it was guarding.

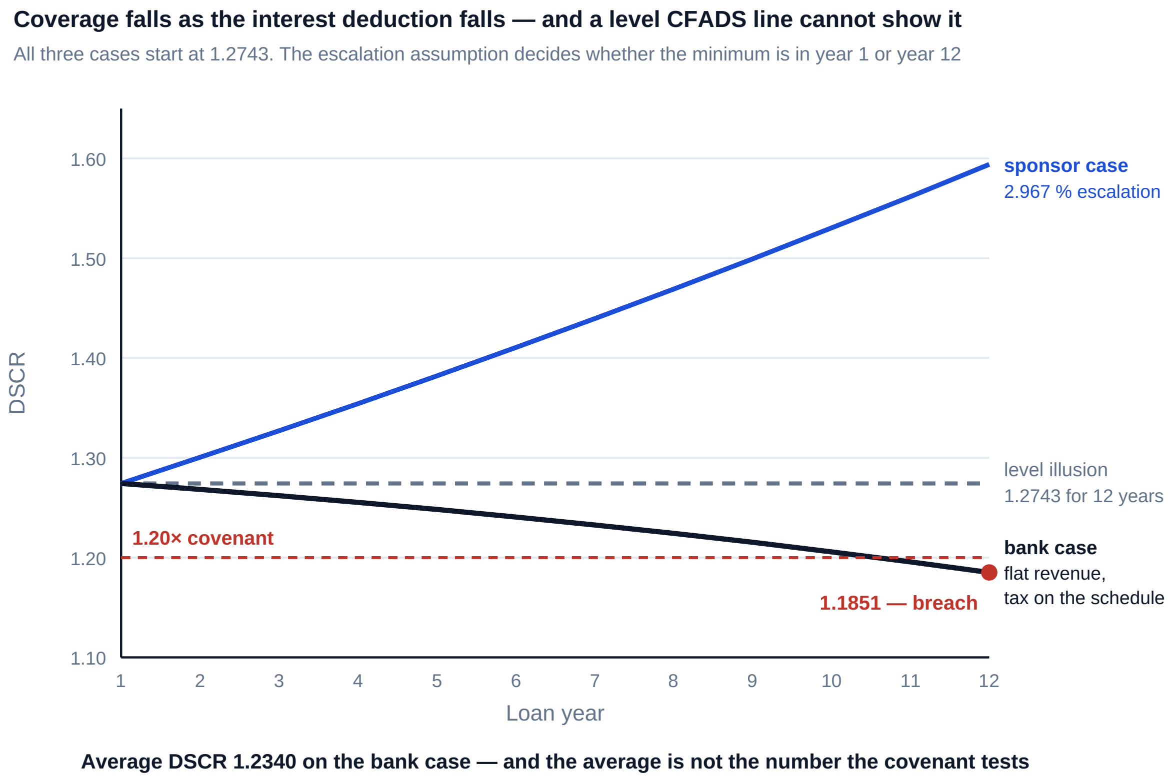

Fig 6.4.1 — The minimum coverage a level line hides

6.4.2 Sensitivity, scenario and the breakeven translation

Definitions. Sensitivity analysis moves one input at a time and records the effect on outputs. **Scenario analysis** moves a coherent set of inputs together to represent a state of the world. **Elasticity** expresses a sensitivity as a ratio: the percentage change in an output per one per cent change in an input, so that inputs measured in different units can be ranked.

WORKED EXAMPLE

Worked example 6.4.2 — Kestrel's sensitivity table, honestly labelled.

1. **Setup.** Two cases run in parallel: the **sponsor case** (2.967 % escalation, `NPV` **+19,875,251**) and the **bank case** (flat, `NPV` **+2,767,684**). `NPV` is unlevered, post-tax, over 25 years at 8 %; `DSCR` figures are from the bank case with cash tax on the actual interest schedule. Each input is moved ±10 %; the capital-cost case assumes the overrun is funded by equity, the lenders' commitment being fixed.
2. **Formula.** Elasticity = `(ΔNPV/NPV) ÷ (Δinput/input)`.
3. **Result.**

Input	Sponsor NPV at −10 % / +10 %	Elasticity	Bank NPV at −10 % / +10 %	Elasticity	Bank DSCR yr 1 at +10 %	Bank minimum DSCR at +10 %
Revenue	7,416,691 / 32,333,810	**6.27**	−6,839,615 / 12,374,983	**34.71**	1.4540	1.3647
Cash operating cost	24,858,674 / 14,891,827	**−2.51**	6,610,603 / −1,075,235	**−13.88**	1.2025	1.1132
Capital cost	25,362,861 / 14,387,640	**−2.76**	8,255,295 / −2,719,927	**−19.83**	1.2839	1.1946
Interest rate	19,875,251 / 19,875,251	**0.00**	2,767,684 / 2,767,684	**0.00**	1.2432	1.1485
Tax rate	21,439,288 / 18,311,213	**−0.79**	3,856,511 / 1,678,857	**−3.93**	1.2640	1.1658
Escalation	17,852,671 / 21,978,907	**1.02 / 1.06**	not applicable	—	1.2743	1.1851
Discount rate	26,469,750 / 14,022,003	**−3.32 / −2.94**	7,305,980 / −1,294,646	**−16.40 / −14.68**	1.2743	1.1851

1. **Interpretation.** Five readings, in order of what a practitioner is paid for.

Revenue dominates, and operating leverage is why. An elasticity of 6.27 means a one per cent revenue miss costs 6.27 % of NPV: cash operating costs are largely fixed, so CFADS amplifies revenue at an elasticity of **1.4098**, and the discount factors and the fixed depreciation shield do the rest. Diligence effort should be allocated in the order of this column, which is the table's real purpose.

Elasticity is a property of the case, not of the project. The same revenue move is 6.27 on the sponsor case and **34.71** on the bank case, because the bank case's base NPV is thin. A tornado drawn on a marginal case looks terrifying and one drawn on a generous case looks reassuring, for the same asset. Never compare elasticities across cases; always label the case.

The interest-rate row is the most instructive line in the table. Unlevered NPV does not move at all — elasticity **0.00** — because financing does not change the asset's cash. But the year-one DSCR falls to **1.2432** and the minimum to **1.1485**, below the lock-up as well as the covenant. A table reporting only NPV would rank interest rate last among Kestrel's risks. **Sensitivity must be run on the outputs the decision turns on**, and for a financing that means coverage, not only value.

The capital-cost row is counterintuitive. A 10 % overrun raises the year-one DSCR to **1.2839**, because the depreciation shield grows with the asset base and cash tax falls while the instalment is unchanged; NPV falls sharply, as it must. A model showing only coverage would report an overrun as good news — which is why no single output is a control.

The asymmetry in the discount-rate row is convexity, not error. −3.32 downward against −2.94 upward is the curvature of the present-value function (Domain 4, KA 4.A.1); a symmetric elasticity reported for a discount rate means somebody has linearised.

What one-at-a-time cannot see — stated precisely, because it is usually stated wrongly. The common objection is that one-at-a-time "misses interactions". On Kestrel it does not: the model is linear in revenue and cash cost, so a joint move of revenue −10 % and cash cost −5 % gives an NPV of **−4,918,155**, exactly the sum of the two individual effects, to the cent. What it genuinely misses is more dangerous. **It says nothing about correlation** — the joint probability of two moves occurring together. Revenue and cost are commonly indexed to related measures, so the realistic downside is a correlated bundle rather than "revenue −10 %", and one-at-a-time never assigns it a likelihood; that is what scenarios and, where the exposure justifies the parameter work, probabilistic simulation are for. **And it breaks at thresholds.** At revenue −10 % Kestrel's year-one `DSCR` is **1.0947**, below both the 1.20× covenant and the 1.15× lock-up, so distributions are trapped and the equity return collapses by far more than the `NPV` column suggests. Discontinuities — breach, lock-up, cash sweep, tax losses starting or exhausting, a reserve emptying — are where linear intuition fails and a sensitivity table must be replaced by a case.

The breakeven translation. A ratio conveys no magnitude; an input level does. Converting each threshold into the input that crosses it is the most useful single page a model produces:

Threshold	CFADS	Revenue	Fall from base
1.30× sizing target	6,512,526	12,171,368	revenue must **rise 1.43 %**
1.20× covenant	6,011,562	11,503,416	**4.14 %**
1.15× lock-up	5,761,081	11,169,441	**6.92 %**
1.00× (payment fails)	5,009,635	10,167,514	**15.27 %**

Two of those lines change a conversation. Domain 10 reported covenant headroom as **372,438** of annual cash, **5.83 %** of `CFADS`; in revenue terms the headroom is only **4.14 %**, because `CFADS` amplifies revenue by 1.4098. Quoting headroom as a percentage of `CFADS` to an operations team that manages revenue overstates their room by 41 %. And the first line closes the loop with Domain 10's negotiation: revenue would have to be **1.43 % higher** for the requested 42,000,000 to deliver the 1.30× the credit committee wanted. On the cost side the covenant breaks at cash operating costs of **4,965,547**, **10.35 %** above base, and on the financing side at an interest rate of **7.48 %** against the 6.0 % achieved.

6.4.3 Model audit, governance and its economics

Definition. A **model audit** is an independent review of a financial model against the transaction documents, the model's own logic and its arithmetic, reported as findings by severity. It is a standard condition precedent in limited-recourse financing (Domain 13, KA 13.2) and it is the lenders' control, not the sponsor's convenience.

Model governance is the surrounding apparatus that makes the audit possible and its findings durable: a **version register** (one authoritative file, a version number in the file name and on every printed page, a change log naming the author and the reason); **input provenance** (every external input traced to a source document with its version and date, so that a superseded traffic forecast cannot be silently retained — Case study B); **input lock and change control** after a declared freeze; **a named model owner** and a named reviewer per version; and a **regression suite** of golden answers that the model must still reproduce after any edit.

The economics: is the audit worth its fee and its elapsed time? The shape is Domain 3's gate economics from PML-AI (PML-AI KA 3.3.1 — reviewed here in the same form, on financing rather than delivery parameters), and the answer is more interesting than "yes".

WORKED EXAMPLE

Worked example 6.4.3 — the model audit priced.

1. **Setup.** Illustrative parameters an organisation must estimate from its own record; the arithmetic, not the parameters, is transferable. Probability that an unaudited model of this class carries a **material** error, `p` = **0.35**. Probability the audit detects it, `d` = **0.85**. Audit fee **180,000**. The audit adds **two weeks** to financial close; at Kestrel's forgone `CFADS` of **17,733.33** a day on a 30/360 basis (Domain 5, KA 5.4.2, where the concession's expiry is fixed so a slip shortens operations), that elapsed cost is **248,267**. Correcting a detected error pre-close costs 60,000 of model rework plus one further week, **184,133**. The illustrative material error is an **omitted 500,000 annual maintenance provision**: true year-one `CFADS` **5,984,000**, a `DSCR` of **1.1945**, and debt capacity at 1.30× of **38,591,479** against 42,000,000 signed — a resize of **3,408,521**.
2. **Formula.** Cost if the error reaches close = amendment fee + advisers + duress premium on the resized equity + reopening delay. Expected cost without audit = `p × C`. Expected cost with audit = fee + elapsed + `p × [d ×` correction `+ (1 − d) × C]`. Net value = the difference.
3. **Substitution.** `C` = amendment fee at 0.30 % of 42,000,000 = 126,000, plus advisers 320,000, plus a 200-basis-point duress premium on 3,408,521 of equity raised at short notice for twelve years (`3,408,521 × 0.02 × AF(0.08, 12) =` 513,738), plus a ten-week reopening at 24,733.33 a day (forgone `CFADS` plus interest on drawn debt, Domain 5's daily figure) = 1,731,333. Without audit `0.35 × 2,691,071`. With audit `180,000 + 248,267 + 0.35 × [0.85 × 184,133 + 0.15 × 2,691,071]`.
4. **Result.** `C` = **2,691,071**. Expected cost without the audit **941,875**; with it **624,328**. **The audit is worth 317,547.** Its **breakeven fee** is **497,547 — 2.76×** what it costs. Its **breakeven error rate** is `p*` = **20.10 %**: below that, the audit destroys value. Its **breakeven detection rate** is `d*` = **48.81 %**.
5. **Interpretation.** The headline is not 317,547 but what the breakevens do to the argument. **Elapsed time, not fee, decides whether the audit pays.** Run the same audit early, in parallel with other conditions precedent so that it adds no time to close and a detected error costs only its 60,000 of rework, and the net value nearly doubles to **602,744** while the breakeven error rate collapses from 20.10 % to **8.05 %**. The fee is 180,000 and the elapsed cost is 248,267: **the delay costs more than the auditor.** A leader arguing about the fee is arguing about the smaller number, and the negotiation that matters is the timetable. **A weak audit is worse than none**, because below `d*` = 48.81 % it consumes the fee and the delay and returns less than it costs — so the auditor's competence on this asset class and this jurisdiction's tax is a commercial question, not a procurement formality. **And the honest caveat sits in `p`.** An organisation whose material-error rate is genuinely below 20 % cannot justify a late audit on expected value alone, and should not pretend otherwise; what justifies it anyway is that the audit is the lenders' condition precedent, that the distribution of `C` has a long tail this arithmetic averages away, and that the 20.10 % breakeven is itself an estimate nobody has ever tested on a sample of one project. The defensible position is therefore: audit early, audit competently, and use the arithmetic to argue about timing and scope rather than about whether to audit at all.

6.4.4 AI-assisted modelling controls

The specific opportunity. Four uses are strong enough to change how the work is staffed. **Formula and structure scanning** at whole-workbook scale — embedded constants, mid-row formula changes, inconsistent references, orphaned cells, flag arithmetic: the 6.1.1 rules, enforced mechanically. **Version diffing** into a human-readable change list, which is what makes a change log trustworthy rather than aspirational. **Adversarial test generation**: the cases that break a waterfall — a `CFADS` failing every tier, a reserve emptying, tax losses exhausting mid-loan — which humans under time pressure reliably fail to imagine. And **documentation from the model**, drafting the assumption register from the input block so the two cannot drift apart.

The specific prohibitions. No AI output is a tax treatment, an accounting policy, a covenant definition, a waterfall order or a legal consequence — each has a named professional owner and a document behind it. No AI-generated case becomes a reported case without a human declaring its basis, horizon and assumptions. No AI edit reaches the authoritative file without appearing in the version diff and the change log with a named human author. And nothing produced by a model whose check block is red is quoted, whatever produced it.

The controls that make this auditable, in the order they bind: a **golden-answer regression suite** — this domain's verified figures (`CFADS` 6,384,000; `DSCR` 1.2743; minimum `DSCR` 1.1851 in year twelve; IDC 2,114,597; contingency 3,645,403; closing balance sheet 59,752,409) rerun after every edit, machine or human, with any change explained before it is accepted; **provenance for every input**, source document and version, because Case study B's failure was a paste and not a formula; **a named human owner per model version**; **sampled manual recomputation** of one construction period and one operating year per version; and **a stated verification sample size** for any machine-produced finding list, every finding confirmed by a human before it is reported.

The arithmetic of an AI pre-check, and why it does not settle the question. Suppose a machine scan costs **8,000**, adds no elapsed time and detects **55 %** of material errors. On Worked example 6.4.3's parameters its net value is **498,481** — above the late audit's 317,547, purely because it consumes no time. As a pre-check before an early audit it cuts the residual error probability from 0.35 to **0.1575** and yields **670,716**, the best of the four options.

Control	Expected cost	Net value
Nothing	941,875	—
Late model audit (2 weeks of delay)	624,328	**317,547**
AI pre-check alone	443,394	**498,481**
Early model audit	339,131	**602,744**
AI pre-check + early model audit	271,159	**670,716**

And here the arithmetic must be allowed to lose an argument. Expected value ranks the pre-check above the late audit, and a careless reader would conclude that a tool can replace a review. Three reasons it cannot, all about the parameters rather than the sums. The audit is a condition precedent; a sponsor does not get to substitute for it. The detection rate `d` is not constant across error types: a scanner finds structural defects and systematically misses the definitional ones — whether `CFADS` matches the facility's clause, whether the tax treatment is the jurisdiction's, whether the waterfall follows the drafted priority — which are precisely the errors carrying the largest `C`, so treating `d` as independent of error class is itself a model error, and one that biases the comparison toward the cheap control. And nobody has validated 0.55 on this asset class at all. The conclusion the table supports and the arithmetic alone does not: **machine checks are additive to review and never substitutive**, and the honest use of these numbers is to fund the pre-check and move the audit earlier.

KEY TERMS — KA 6.4

Term	Meaning
Check block	A dedicated output area computing every invariant as a difference that must be nil.
Elasticity	Percentage change in an output per one per cent change in an input.
Tornado	Sensitivities ranked by magnitude; meaningless without its case labelled.
Breakeven translation	Converting a ratio threshold into the input level that crosses it.
Model audit	Independent review against documents, logic and arithmetic; a lenders' condition precedent.
Input provenance	Every external input traced to a source document, version and date.
Golden-answer regression suite	Verified results rerun after every edit; unexplained change blocks release.
Breakeven fee / error rate / detection rate	The three points at which a review stops adding value.

SAMPLE MCQS — KA 6.4

MCQ 6.4-A `[6.4.1 · Analysis]` A modeller taxes `EBIT` of 5,100,000 at 20 % instead of taxable profit of 2,580,000. The balance sheet still balances. The check that catches it is:

- A. the balance-sheet check, once the period is recalculated
- B. sources equal uses
- C. the implied effective tax rate — 1,020,000/2,580,000 = 39.53 % against a statutory 20 %
- D. closing debt nil at maturity

Rationale: A consistently propagated error balances, so invariant 1 is necessary and not sufficient (6.4.1). B and D test construction funding and the debt schedule, neither of which the error touches.

MCQ 6.4-B `[6.4.1 · Application]` Kestrel's bank case shows `DSCR` of 1.2743 in year one, 1.1851 in year twelve and an average of 1.2340 against a 1.20× covenant. The reportable position is:

- A. compliant, since the average exceeds the covenant
- B. compliant, since year one exceeds the covenant
- C. a breach in year twelve; the minimum and its year must be reported, because covenants are tested in periods
- D. indeterminate without the sponsor case

Rationale: Domain 10's rule, earned here: an average conceals the period that breaches (6.4.1b, KA 10.A.3). D inverts the logic — the sponsor case, in which coverage rises to 1.5940, is the case that hides the problem.

MCQ 6.4-C `[6.4.2 · Analysis]` Kestrel's unlevered `NPV` has an elasticity of 0.00 to the interest rate, while a 10 % rate rise takes the minimum `DSCR` to 1.1485 — below the 1.15× lock-up. The correct inference is:

- A. interest-rate risk is immaterial
- B. the model has an error, since a higher rate must reduce value
- C. sensitivity must be run on the outputs the decision turns on: financing does not change the asset's cash, but it changes coverage, which is what the covenant tests
- D. the discount rate and the interest rate should be equal

Rationale: Unlevered cash is financing-independent by construction (6.4.2). A reads only the NPV column, which is exactly the mistake; B misunderstands what unlevered means; D confuses the opportunity cost of capital with the cost of debt.

MCQ 6.4-D [6.4.2 · Application] Domain 10 reported covenant headroom of 372,438 of annual CFADS, 5.83 % of base case. Expressed as revenue, the headroom is:

- A. 5.83 %
- B. 4.14 %
- C. 8.22 %
- D. 15.27 %

Rationale: CFADS amplifies revenue by 1.4098, so the covenant breaks at revenue of 11,503,416, a 4.14 % fall (6.4.2). A carries the CFADS percentage across as if the two were interchangeable — overstating the operations team's room by 41 %; C multiplies rather than divides by the elasticity; D is the fall at which payment itself fails.

MCQ 6.4-E [6.4.3 · Evaluation] A model audit costs 180,000 and two weeks of delay worth 248,267; p = 0.35, d = 0.85, C = 2,691,071, pre-close correction 184,133. Moving the audit early, so it adds no delay and correction costs only 60,000, changes its net value and breakeven error rate to:

- A. 317,547 and 20.10 %
- B. 602,744 and 8.05 %
- C. 941,875 and nil
- D. 497,547 and 20.10 %

Rationale: 941,875 − [180,000 + 0.35 × (0.85 × 60,000 + 0.15 × 2,691,071)] = 602,744, and 180,000/(0.85 × (2,691,071 − 60,000)) = 8.05 % (6.4.3). A is the late-audit answer; C is the expected cost of no control at all; D pairs the late audit's breakeven fee with its breakeven error rate.

SELF-CHECK — KA 6.4

1. Why is "the model balances" not evidence of accuracy? — A consistently propagated error balances to the cent; invariant 1 tests structure, not economics (6.4.1).
2. What does one-at-a-time sensitivity genuinely miss? — Correlation, and behaviour at thresholds; on a linear model it does not miss the arithmetic of combining moves, which is additive to the cent.
3. What decides whether a model audit pays? — Its elapsed time, not its fee: moving it early lifts net value from 317,547 to 602,744 and cuts the breakeven error rate from 20.10 % to 8.05 %.

Advanced topics — Domain 6

6.A.1 Probabilistic modelling, and when it earns its cost

Where a one-at-a-time table cannot assign likelihoods, a simulation can — drawing correlated inputs from stated distributions and reporting a distribution of outputs. The attraction is real: it yields the probability of covenant breach over the loan life, which is what a credit committee wants and no sensitivity table can give. The cost is usually understated. A simulation requires **distributions** and a **correlation matrix**, assumptions with far less evidence behind them than the point estimates they replace, and a plausible-looking distribution of NPV built on invented correlations is more dangerous than a point estimate because its shape implies knowledge nobody has. The proportionate rule: simulate where the exposure justifies defensible parameter work, report the assumptions as prominently as the results, and never let a distribution's smoothness stand in for evidence. Used well, the deliverable is not a mean NPV — roughly what the base case already gave — but a **probability of breach** and the **inputs that drive it**.

6.A.2 Model reuse, templates and the drift problem

Templates are the right answer for organisations that finance repeatedly: they carry conventions, check blocks and a tested waterfall, and they make a new model reviewable on day one. They carry two risks that must be managed rather than hoped away. **Silent inheritance**: a template's tax logic, day-count convention or reserve mechanic is this transaction's only once someone confirms it, and the confirmation is a task with an owner. **Drift**: each transaction improves its copy, none of the improvements returns, and after five deals there are five templates and no standard. The governance answer is a template owner, a versioned template with a change log, a deviation register per transaction ("this model departs from template v4.2 in these six respects, for these reasons") and a post-close pass that promotes genuine improvements back. The reuse argument survives being disciplined; it does not survive being informal.

6.A.3 The reviewer's model eye

Invariants and habits to test on any project model before relying on a single output. The **six invariants** of 6.4.1 all compute to nil, and the check block is visible on the output page rather than hidden in a working sheet. The **implied effective tax rate** reconciles period by period to the statutory rate plus explained differences (Worked example 6.4.1). The **flag additions** of 6.1.2 hold, and no period is both construction and operating. **Sources equal uses in every construction period**, not merely in total. **Capitalised interest** is computed from a drawdown profile, and the interest convention and periodicity are stated (Worked example 6.2.1 — the 41.01 % swing). **CFADS = operating cash flow + interest paid**, and the CFADS line reconciles to the facility's definition clause by clause (Domain 10, Toolkit 10.T.1). **Minimum DSCR is reported with the year in which it occurs**, and the level-cash shortcut is not doing the work (Worked example 6.4.1b — 1.1851 in year twelve). LLCR **=** DSCR only where cash is level and service is an annuity at the loan rate; where cash is not level, the divergence is explained and not smoothed. **Every waterfall tier is floored at zero** and shortfalls are carried; the model has been run with a CFADS that fails every tier. **Cash tax, not accounting tax, feeds CFADS**, and tax losses carry with a stated expiry rule. **No calculation cell contains a numeral** other than a period counter. **Every external input has a source document with a version and a date.** And the model file bears a version number that appears on every printed page, so that the paper on the table can be tied to the workbook that produced it. Any violated line is a defect, and the violated line localises it — which is the entire value of building the checks before the answers.

Industry variations — Domain 6

- **Contracted power and availability PPPs.** Revenue is nearly deterministic, so modelling effort concentrates on the **availability and deduction mechanism** — a formula-heavy translation of the performance regime into cash — and on the tax line. Coverage is thin by design, which makes the minimum-DSCR discipline of 6.4.1b decisive rather than academic.
- **Merchant power, commodities and mining.** The price deck is the model, and it arrives as a third-party forecast with versions, so 6.4.3's provenance control is the first-order defence (Case

study B is this failure in another sector). Lenders size on stressed decks, so the bank case must be genuinely separate from the sponsor case rather than a switch changing one row.

- **Transport concessions.** Patronage ramps make the level-cash shortcut untenable from period one, so sculpted service (Domain 10, KA 10.1.3) is close to mandatory and the model carries 6.3.1's sculpting circularity openly. Traffic models are separate models, and the interface is where the errors live.
- **Water and regulated utilities.** Reset cycles create step-changes in revenue at known dates, so the timeline carries a **reset flag** and the model shows coverage across the reset rather than assuming continuity. Regulatory and statutory asset values coexist and must never be conflated.
- **Digital infrastructure.** Short useful lives and heavy refresh capex make the capital-expenditure profile an operating-model problem: lifecycle capex competes with debt service in the waterfall, and its position relative to `CFADS` is negotiated.
- **Oil, gas and heavy industry.** Decommissioning provisions and their funding create an end-of-life obligation most templates handle badly, and Domain 4's sign-change pathology (KA 4.1.2) follows directly — a further reason these sectors report `NPV` and treat "the `IRR`" with suspicion.

Case study — Domain 6: the year the level line hid (water)

Situation. Kestrel's financing model went to the lenders' model auditor three weeks before the scheduled close, showing a base-case `DSCR` of **1.2743** held constant across all twelve loan years, a `CFADS` of **6,384,000**, and an `NPV` of **+16,179,360** carried forward from the sponsors' appraisal. The senior facility was **42,000,000** with a **1.20×** covenant and a **1.15×** lock-up. The audit's first substantive finding was not an arithmetic error.

What happened. The auditor asked why `CFADS` was constant when the interest deduction was not, and rebuilt the tax line period by period. Cash tax rose from **516,000** in year one to **963,287** in year twelve as interest fell from **2,520,000** to **283,564**, and `CFADS` fell correspondingly from 6,384,000 to **5,936,713**. The `DSCR` profile declined monotonically to **1.1851** — a **covenant breach in year twelve** — while the twelve-year average stayed at **1.2340**, comfortably compliant. Three further findings followed from the same source: the present value of `CFADS` at 8 % was **65,315,883**, not the 68,147,771 the level line implied, an overstatement of **2,831,888** or **4.34 %**; the `LLCR` was **1.2395**, not the 1.2743 the level-cash identity had produced; and the appraisal `NPV` of +16,179,360 was a **pre-tax** figure over a **fifteen-year** horizon that had been carried into a financing paper without a bridge — the same asset on a post-tax, twenty-five-year, flat-revenue basis is **+2,767,684**.

How it resolved. The sponsors first proposed reporting the average, which the auditor rejected in one line: covenants are tested in periods. They then proposed an equity cure in the affected years, which is arithmetically small — restoring 1.20× in years eleven and twelve costs **21,347** and **74,849**, **96,196** in total — and was rejected for the right reason: a structure that is designed to need a cure has consumed an option it should be holding in reserve (Domain 10, KA 10.4.3). The structure closed instead at Domain 10's independently derived **41,171,123**, at which the instalment is **4,910,769** and the year-twelve `DSCR` is **1.2087**, inside the covenant. The debt level at which the year-twelve minimum is exactly 1.20× is **41,472,081**, so the binding constraint at close remained Domain 10's year-one 1.30× sizing test — 828,877 below the request — and the year-twelve problem was already covered by it. The sponsors funded the difference with equity, as Domain 10's case recorded, and the model went forward with a period-by-period tax line, a minimum-`DSCR` output with its year, and a four-line basis bridge from the appraisal to the financing case.

What the domain teaches here. A model can be arithmetically perfect and still hide the year in which the project breaches, because the hiding is done by an assumption — level cash — and not by an error. The mechanism was entirely predictable: the interest tax shield amortises with the loan. Two habits would have caught it before the auditor did, and both are cheap: report the minimum `DSCR` with its year, and never hold a line constant when a driver underneath it is not. The third lesson is about conservatism. Domain 10's 1.30× requirement looked like a bank being difficult; it was in fact protecting against a year-twelve exposure nobody in that negotiation had modelled. **A well-built model makes a conservative constraint legible as the specific risk it was guarding** — and that, rather than any single output, is what makes it worth its cost.

Case study B — Domain 6: the forecast that was pasted, not linked (transport)

Situation. A toll-road concession, capital cost **420,000,000**, sought **294,000,000** of senior debt — 70 % gearing — at **6.5 % over 18 years**, on a sizing target of **1.35×** and a **1.30×** covenant. The annuity factor `AF(0.065, 18)` is **10.432466**, giving level debt service of **28,181,255**. The financing model showed steady-state `CFADS` of **38,000,000** and a `DSCR` of **1.3484** — comfortably inside the sizing target. The traffic forecast underneath it came from the technical adviser's separate demand model, imported as **pasted values** into a tab labelled "traffic input v3".

What happened. Eleven weeks before close the technical adviser issued a revised forecast **7 % lower**, reflecting a change to a competing route's toll policy. The revision was circulated, acknowledged and filed. Nobody re-pasted it, because the financing model's traffic tab was an input sheet and the model's change log recorded only formula changes. The lenders' model auditor found it by doing the one thing the sponsor's team had not: checking each external input's document version against the current issue. Rebuilt on the revised forecast, `CFADS` was **35,340,000** and the `DSCR` **1.2540 — below the 1.30× covenant on the first test date**, and 0.096 below the sizing target the facility had been structured against.

How it resolved. Debt capacity at 1.35× on the revised forecast is `35,340,000/1.35 × 10.432466 =` **273,098,787**, so the facility had to fall by **20,901,213** and gearing from 70/30 to **65.0/35.0**. The sponsors funded it. Substituting 20,901,213 of equity for debt, at a 12 % equity requirement against 6.5 % debt over the 18-year tenor, costs `20,901,213 × 0.055 × AF(0.08, 18) = 20,901,213 × 0.055 × 9.371887 =` **10,773,610** of present value — the true price of a paste. The resize and re-diligence took six weeks; at forgone `CFADS` of **105,556** a day (38,000,000/360) the delay cost **4,433,333**, and re-cutting the schedule, re-running credit approval and re-issuing the information memorandum cost a further **900,000**: **5,333,333** in total, over and above the equity substitution the truth required in any event. Had the error reached close, the reopening would have run twelve weeks against drawn debt at **158,639** a day (forgone `CFADS` plus interest on 294,000,000), with a 0.30 % amendment fee of 882,000 and 1,400,000 of advisers — **15,607,667**. **The audit's timing avoided 10,274,333.**

What the domain teaches here. Not one formula in the model was wrong. The defect was a **model boundary** with no provenance control across it, and the change log's scope — formula changes only — guaranteed that an input change would be invisible. Three controls would each have caught it independently: input provenance recording every external input's source document, version and date; a live link or a documented import with a version check rather than a paste; and a golden-answer regression suite, which would have flagged that `CFADS` had not moved when the forecast underneath it had. The economic lesson generalises beyond the sector: **the cost of a model error is set by how late it is found, and the expensive errors are almost never arithmetic.** Domain 13 makes this a diligence stream; here it is a modelling discipline that costs a spreadsheet column.

Executive perspective — Domain 6

What a project finance director cannot delegate in this domain:

- **The basis, horizon and case of every number quoted upward.** Kestrel supports five correct NPVs spanning 29,545,516 (6.1.3). The director owns which one appears in a board paper and insists on the bridge to the others — because the number that survives in an organisation's memory is the one nobody labelled.
- **The tax line's provenance.** It moves debt capacity by 3,327,741 on one allowance assumption (6.2.3), it is jurisdiction-specific, it changes during a loan's life, and it must trace to a dated written opinion. This is the one input a director should be able to name the author of.
- **The minimum, with its year.** Kestrel's average DSCR of 1.2340 complies and its year-twelve minimum of 1.1851 breaches (6.4.1b). Averages in covenant reporting are a governance failure wearing a statistic.
- **The audit's timetable, not its fee.** Two weeks of delay cost 248,267 against a fee of 180,000; moving the audit early nearly doubles its value and cuts its breakeven error rate from 20.10 % to 8.05 % (6.4.3). The director's intervention is on the plan, and it must happen months before the auditor is appointed.
- **The distribution profile after reserve funding and lock-up.** 121,956 in operating year one on 18,000,000 of equity, and a 14.67-year equity payback from close (6.2.2, 6.3.3). Boards that have not seen this before close are boards that will be surprised after it.
- **One authoritative file, one named owner, one version on every page.** Case study B cost 5,333,333 and could have cost 15,607,667 because a superseded input survived a change log that did not cover inputs. Model governance is not a modelling matter; it is a control the director signs for.

Calculation exercises — Domain 6

Exercise 6.1 A project has a **40,000,000** envelope funded **65/35** — 26,000,000 of debt at **7.0 %**, 14,000,000 of equity — with every use funded in that proportion. Committed uses: EPC **33,000,000**, owner's costs **2,500,000**, capitalised development **1,200,000**, arrangement fees at **1.75 %** of the facility. Construction runs **five quarters** with a spend profile of **10, 20, 25, 25 and 20 per cent**; interest accrues quarterly at 7.0 %/4 on the opening debt balance. Compute the fees, capitalised interest and the balancing contingency, and test the contingency against a 5–10 % of EPC policy band. Solution. Fees `26,000,000 × 0.0175 =` **455,000**. Iterating the quarterly draw and interest calculation gives capitalised interest of **849,752**. Contingency balances: `40,000,000 − 33,000,000 − 2,500,000 − 1,200,000 − 455,000 − 849,752 =` **1,995,248**, which is **6.05 %** of the EPC price — inside the band, so the envelope is credible. Total uses 40,000,000; debt drawn 26,000,000; equity 14,000,000. Common error: treating contingency as a policy percentage and letting capitalised interest balance instead — which hides an IDC figure nobody has computed and produces a model whose construction interest is a plug.

Exercise 6.2 Net income 3,680,000; depreciation 2,500,000; receivables up 1,100,000; payables up 400,000; interest paid 2,100,000, classified in operating cash flow. Compute operating cash flow and CFADS. Solution. Operating cash flow `3,680,000 + 2,500,000 − 1,100,000 + 400,000 =` **5,480,000**; `CFADS = 5,480,000 + 2,100,000 =` **7,580,000**. Common error: deducting interest again to reach CFADS (5,480,000 − 2,100,000 = 3,380,000) — the interest has already been paid inside operating cash flow, and CFADS is struck before all debt service.

Exercise 6.3 A facility of **30,000,000** at **5.5 %** amortises over **10 years** on a level annuity. `EBITDA` is **6,000,000** flat, accounting and tax depreciation both **1,800,000**, the tax rate **25 %**, and the working-capital absorption **250,000** a year. Compute the instalment, the `DSCR` in years 1, 5 and 10, and state the minimum and its year. Solution. `AF(0.055, 10) = 7.537626`; instalment `30,000,000/7.537626 =` **3,980,033**. Year 1: interest 1,650,000, cash tax `0.25 × (6,000,000 − 1,800,000 − 1,650,000) =` 637,500, `CFADS` **5,112,500**, `DSCR` **1.2845**. Year 5: interest 1,093,531, tax 776,617, `CFADS` 4,973,383, `DSCR` **1.2496**. Year 10: interest 207,490, tax 998,128, `CFADS` 4,751,872, `DSCR` **1.1939**. Minimum **1.1939 in year 10** — the final year, as it must be when revenue is flat and the interest deduction amortises. Common error: computing year one and assuming coverage improves as debt falls; the debt service is level, so the only moving part is cash tax, and it moves the wrong way.

Exercise 6.4 On Kestrel's bank case (`CFADS` 6,384,000; instalment 5,009,635.23; covenant 1.20×; `CFADS` = 0.75 × revenue − 2,616,000), compute the covenant's cash trigger, the revenue at which it is crossed, the fall as a percentage of revenue, and reconcile that percentage with Domain 10's headroom of 5.83 % of `CFADS`. Solution. Trigger `5,009,635.23 × 1.20 =` **6,011,562**. Revenue: `(6,011,562 + 2,616,000)/0.75 =` **11,503,416**, a fall of **4.14 %**. The reconciliation is the elasticity: `CFADS` responds to revenue at `0.75 × 12,000,000/6,384,000 =` **1.4098**, so `5.83 %/1.4098 = 4.14 %`. Common error: quoting the `CFADS` percentage to a team that manages revenue, overstating their room by 41 %.

Exercise 6.5 A model audit costs **210,000** and adds delay worth **300,000**. The material-error probability is **0.28**, the detection probability **0.80**, the pre-close correction cost **90,000**, and the cost if the error reaches close **3,400,000**. Compute the audit's net value and its breakeven error rate. Solution. Without the audit `0.28 × 3,400,000 =` **952,000**. With it `210,000 + 300,000 + 0.28 × (0.80 × 90,000 + 0.20 × 3,400,000) = 510,000 + 0.28 × 752,000 =` **720,560**. Net value **231,440**. Breakeven error rate `(210,000 + 300,000)/(0.80 × (3,400,000 − 90,000)) =` **19.26 %** — below that, the audit destroys value at this timing. Common error: omitting the elapsed cost, which reports a net value of 531,440 and a breakeven error rate of 7.93 %, flattering the control by more than twice.

Practitioner's toolkit — Domain 6

Adoption-ready artefacts; adapt headings to your organisation, then keep them stable.

Toolkit 6.T.1 — The model conventions sheet (one page, in the inputs block)

One line per convention, each with the value chosen, the alternative rejected, and the reason: period length (and why); flow timing (period end, start, mid-period); day-count basis; interest accrual base (opening or average balance) **and the circularity resolution chosen**; first-period stub treatment; currency and units; sign convention; rounding, and where it is applied; the basis (pre/post-tax, levered/unlevered, before/after working capital); the horizon and the treatment of value beyond it; and the named case list with the single input that switches between them. Rule: a convention absent from this sheet may not be relied on in a printed output, and the sheet's version matches the model's.

Toolkit 6.T.2 — The check block (built before the first answer)

- [ ] Balance sheet balances, every period — difference nil.
- [ ] Sources = uses, every construction period, not only in total.

- [ ] Closing debt nil at maturity, per facility; Σ scheduled principal = total drawings.
- [ ] Cash ≥ 0 in every period and every account; every waterfall tier floored, shortfalls carried.
- [ ] `CFADS` = operating cash flow + interest paid.
- [ ] Implied effective tax rate = statutory rate + explained differences, period by period.
- [ ] Timeline flags: four additions hold; no period double-flagged.
- [ ] Minimum `DSCR` **and its year** reported beside the average; `LLCR` = `DSCR` only where cash is level and service is an annuity at the loan rate; `PLCR` ≥ `LLCR` where a tail exists.
- [ ] No numeral in any calculation cell other than a period counter.
- [ ] Golden-answer regression suite reruns clean, or every change is explained and accepted.
- [ ] Aggregate flag visible on the output page; **no output quoted while it is red.**

Toolkit 6.T.3 — Model governance register (per model, per version)

Columns: version number (appearing on every printed page) · date · author · reviewer · reason for the version · change list (from the version diff, inputs **and** formulae) · check-block status · golden-answer status · **input provenance table** (each external input: source document, its version, its date, who confirmed it against the current issue) · freeze status and change-control authority after freeze · basis, horizon and case list as released · outstanding audit findings by severity with owners and dates · AI-assisted edits, with the tool, the prompt intent, the human author and the diff reference. Front line: **which file is authoritative, who owns it, and what its check block says today.**

Exam preparation — Domain 6

What is assessed. The three-block rule and the flag invariants; sources and uses with computed capitalised interest and an identified balancing line; the effect of periodicity and interest convention; accounting depreciation against tax allowances, and their coverage and capacity consequences; the three-statement articulation and the `CFADS` tie; debt-schedule invariants and circularity resolutions; waterfall construction, reserve funding and release, and the distribution test; project return against equity return; the six check-block invariants and what each can and cannot see; sensitivity, elasticity, the breakeven translation and the limits of one-at-a-time; and the economics of model audit and AI-assisted controls.

The calculations to do under time pressure. Capitalised interest from a drawdown profile on a stated convention, and the balancing line that follows (6.2.1, Exercise 6.1). Operating cash flow to `CFADS` via the interest tie (6.2.2, Exercise 6.2). Cash tax from taxable profit, and the `DSCR` in a late loan year when the interest deduction has shrunk (6.2.3, 6.4.1b, Exercise 6.3). Minimum `DSCR` and its year. Elasticity from two `NPV`s, and a ratio threshold translated into an input level (6.4.2, Exercise 6.4). Review net value and its three breakevens (6.4.3, Exercise 6.5).

The traps. Quoting an `NPV` without its basis, horizon and case — the 29,545,516 spread (6.1.3, MCQ 6.1-A) · an annual timeline with opening-balance interest, understating capitalised interest by 41.01 % (6.1.2, MCQ 6.1-B) · letting capitalised interest balance the sources-and-uses table instead of contingency (Exercise 6.1) · a round balancing line, which means the table was plugged (6.2.1, MCQ 6.2-A) · deducting interest twice on the way to `CFADS` (6.2.2, MCQ 6.2-B) · taxing `EBIT` instead of taxable profit, which balances perfectly (6.4.1, MCQ 6.4-A) · treating a balancing balance sheet as evidence of accuracy (6.4.1) · reporting average rather than minimum `DSCR` (6.4.1b, MCQ 6.4-B) · holding `CFADS` level when the interest deduction is amortising (6.4.1b, Case study A) · reading only the `NPV` column of a sensitivity table and concluding interest-rate risk is immaterial (6.4.2, MCQ 6.4-C) · comparing elasticities across cases (6.4.2) · carrying a `CFADS` percentage across as a revenue percentage, overstating headroom by 41 % (6.4.2, MCQ 6.4-D) · omitting elapsed time from review economics (6.4.3, Exercise 6.5) · treating an AI detection rate as constant across error classes (6.4.4) · a change log that covers formulae but not inputs (Case study B).

How the domain connects. It industrialises Domain 3's discounting and Domain 4's appraisal and audits both — Worked example 6.1.3 reconciles Domain 4's NPV to the financing model — and it implements Domain 2's statements, CFADS definition and cash-versus-accounting tax distinction. It supplies the drawdown Domain 5's slip arithmetic priced and the model whose coverage Domain 10 negotiated, earning Domain 10's minimum-DSCR rule rather than asserting it. Domains 7, 8 and 9 supply the revenue, cost and funding blocks this architecture consumes; Domain 13 makes the model audit a diligence stream and a condition precedent; Domain 14 runs the construction model against actual drawdowns; Domain 15 operates the waterfall of 6.3.2; Domain 16 generalises 6.4.4's controls into model-risk governance.

Domain 6 summary

A financial model's authority comes from its architecture and its checks, not from its answers, and Kestrel proves it four times over. **One project supports five arithmetically correct net present values spanning USD 29,545,516** — from −9,670,265 to +19,875,251 — differing only in basis, horizon and case; Domain 4's **+16,179,360** is the pre-tax, fifteen-year member of that family, reconciled to Domain 2's CFADS by an implied escalation of **2.967 %** to within 661 of present value, and the required deliverable is the bridge rather than any one number. **Convention is worth more than most contested assumptions**: the same drawdown at the same 6.0 % returns capitalised interest of **2,114,597** quarterly on opening balances and **1,247,352** annually — a 41.01 % understatement that propagates silently into the depreciation base — while the sources-and-uses table closes at **60,000,000** on a balancing contingency of **3,645,403**, 7.59 % of the EPC price, and a balancing line that is never round. **The tax line is the most consequential row in the model**: a 15 % declining-balance allowance regime would take year-one CFADS from 6,384,000 to **6,900,000**, DSCR from 1.2743 to **1.3773**, and debt capacity at 1.30× from 41,171,123 to **44,498,864** — a 3,327,741 swing, four times the gap Domain 10's whole negotiation was fought over. **Balancing is necessary and not sufficient**: taxing EBIT instead of taxable profit gives a CFADS of 5,880,000, a DSCR of 1.1737, a debt-capacity shortfall of **3,250,352** — and a balance sheet that balances to the cent, caught only by an implied effective tax rate of **39.53 %** against a statutory 20 %. And **the level-cash shortcut hides a breach**: modelled period by period, Kestrel's coverage declines from 1.2743 to **1.1851 in year twelve** against a 1.20× covenant, while the twelve-year average of **1.2340** complies — which is why the minimum and its year are reported, and why Domain 10's 1.30× sizing test, at **41,171,123** with a year-twelve DSCR of 1.2087, was already protecting against a risk nobody in that negotiation had modelled. The first operating year articulates completely — operating cash flow **3,864,000**, CFADS **6,384,000** through the interest tie, a closing balance sheet of **59,752,409** on both sides, and a distribution to equity of **121,956**, 0.68 % of the 18,000,000 contributed — and the returns it leads to are honest about who earns what: the asset **8.54 %**, the equity **9.83 %** on the bank case and **13.52 %** with escalation, with equity payback **14.67 years** from close. Sensitivity ranks revenue first at an elasticity of **6.27** on the sponsor case and **34.71** on the bank case — proof that elasticity is a property of the case — reports **0.00** elasticity of unlevered value to the interest rate while the same move drives the minimum DSCR to 1.1485, and translates coverage into the language operations use: the covenant breaks at revenue **4.14 %** below base, not the 5.83 % of CFADS Domain 10 quoted. Review economics close the domain: a model audit worth **317,547** late and **602,744** early, with breakeven fee **497,547**, breakeven error rate falling from **20.10 %** to **8.05 %** on timing alone, and an AI pre-check worth **498,481** that still cannot substitute for the audit because its detection rate is not independent of the error class it misses. Domain 7 builds the revenue block this architecture consumes, Domain 8 the cost and contingency block, Domain 9 the sources side, Domain 13 the audit as a condition precedent, and Domain 16 the model-risk governance this domain's controls anticipate.

07

DOMAIN 7

Revenue, Demand and Commercial Models

Group: Structuring and modelling (Domain 3 of 5 in Part Two). **Target:** ~75 pages. **Binds to:** the PCI Book Pattern Specification and the shared registries (`docs/books/registries/`). This domain owns the revenue side of every ratio the book computes: it supplies the `CFADS` numerator that Domain 10 divides, the revenue line that Domain 6 models, and the third bankability component Domain 5 (KA 5.3.2) diligenced first and deferred to here. It adds no new coverage symbol; it explains why the same expected `CFADS` supports very different debt. British English; USD (+SAR where useful, indicative `USD 1 ≈ SAR 3.75`).

Why this domain exists

Domains 5 and 6 left one question open. Domain 5 established that a bankable project needs a creditworthy payer under a contract of adequate tenor (KA 5.3.2); Domain 6 built the model that turns a revenue forecast into coverage; Domain 10 sized debt from `CFADS` and a target coverage ratio and treated both as given. None of them addressed how the **shape** of the revenue promise — its architecture, not its expectation — determines how much debt exists to be sized. This domain shows where `CFADS` and the target ratio come from, and the answer changes the negotiation.

The central claim is a single sentence with expensive consequences: **debt is sized on the worst period a lender will underwrite, so the distribution of revenue matters more than its mean, and two commercial structures with identical expected cash can differ in debt capacity by a quarter of the facility.** Everything in the domain is a corollary. The contracted-to-merchant spectrum (KA 7.1) is a spectrum of dispersion, priced by lenders in required coverage before it is priced in a tariff. Demand-based and concession structures (KA 7.2) put the dispersion inside a payment mechanism whose banded, floored and shared forms are risk-transfer instruments in tariff clothing. Escalation and volume risk (KA 7.3) are the two ways a structure that looked adequate at close stops being adequate later — one slowly and arithmetically, the other quickly and non-linearly. And counterparty credit and stress testing (KA 7.4) make the rest conditional: a contracted stream is worth the payer's ability to pay, and concentration is invisible to the expected-loss calculation a credit committee will nevertheless be shown.

Learning objectives. After this domain a candidate can: place a revenue structure on the contracted–merchant spectrum and state the coverage consequence of its position; distinguish capacity, energy, availability, take-or-pay and merchant mechanisms and compute revenue under an availability-deduction formula; demonstrate that two structures with identical expected `CFADS` support materially different debt, and size each correctly; distinguish sponsor, bank and downside cases and model a ramp; compute revenue under banded, floored and shared mechanisms and value the risk transfer they effect; compute effective escalation rates on partly indexed tariffs and cost bases and quantify the resulting margin drift; compute the degree of operating leverage and the `CFADS` elasticity to price and to volume and explain why they differ; convert a coverage threshold into a tolerance in each driver a business manages; compute expected loss as probability of default times exposure times loss given default, and demonstrate why it is blind to concentration while the loss distribution is not; price credit enhancement against the coverage it unlocks; design and read a stress matrix and a reverse stress test; and govern AI-assisted demand forecasting, contract extraction and counterparty monitoring.

The master thread. Kestrel Water SPC continues. Its first operating year, established in Domain 2 and modelled in Domain 6 (KA 6.2.2), is revenue **USD 12,000,000**, cash operating costs **4,500,000**, `EBITDA` **7,500,000**, depreciation 2,400,000, interest 2,520,000, cash tax 516,000 and a working-capital absorption of 600,000, giving `CFADS` **6,384,000** against debt service of **5,009,635.23** — a `DSCR` of **1.2743** (Domain 10, KA 10.2.1). This domain opens that revenue line for the first time. The plant's nameplate capacity is **30,000,000 m³ a year** (about 82,192 m³ a day); expected despatch is **24,000,000 m³**, an 80 % load factor; the water-purchase agreement pays **USD 0.50 per m³** of expected despatch, and the cost base splits into **3,600,000 of fixed** cash operating cost and **USD 0.0375 per m³ variable**. Every figure in the domain is built from those five inputs, and one closed form recurs throughout:

```
CFADS = 0.75 × revenue − 0.03 × volume(m³) − 1,896,000
```

— the `CFADS` bridge of KA 6.2.2 written as a function of the two commercial variables, absorbing the fixed cost, the 20 % tax rate applied to `EBITDA` less depreciation and interest, and the working-capital absorption of 5 % of revenue. Substituting revenue 12,000,000 and volume 24,000,000 returns **6,384,000** exactly. It is the domain's arithmetic engine, and every result below is reproducible from it.

KNOWLEDGE AREA 7.1

Contracted and merchant revenue, tariffs and availability payments

Topics: 7.1.1 the revenue-risk spectrum · 7.1.2 tariff architecture · 7.1.3 availability payments and the deduction mechanism.

7.1.1 The revenue-risk spectrum

Definition. A project's revenue structure sits on a spectrum defined by **who bears the risk that the asset earns less than forecast**. At the contracted end a creditworthy payer is obliged to pay a defined amount whether or not it takes the output, provided the asset is available; at the merchant end the project sells into a market at prices and volumes nobody has promised. Intermediate forms — take-or-pay, capacity plus energy, minimum revenue guarantees, volume bands — allocate the risk in specified proportions.

Why the spectrum, not the mean, governs debt. Domain 10 sized debt as `CFADS` divided by the target `DSCR`, times the annuity factor. Both numerator and divisor move with position on this spectrum, and they move in the same direction, so the effect compounds: merchant revenue lowers the case a lender will underwrite and raises the coverage it requires. That double movement is the domain's first quantitative result.

WORKED EXAMPLE

Worked example 7.1.1 — two structures, one expected cash flow, and 10.7 million of debt.

1. **Setup.** Kestrel's offtaker will accept either of two structures for the same plant. **Structure A (availability payment):** a capacity charge of **12,000,000 a year** for making 30,000,000 m³ of capacity available, subject to deduction for unavailability; the offtaker takes all volume risk. **Structure B (volume tariff): USD 0.50 per m³** of water actually taken, with no minimum. Independent demand study gives three despatch outcomes: **28,800,000 m³** with probability 0.25, **24,000,000 m³** with probability 0.50 and **19,200,000 m³** with probability 0.25. Cost base and tax as the master thread. Compute `CFADS` and `DSCR` in each outcome, the expected values, and the debt each structure supports.
2. **Formula.** `CFADS = 0.75R − 0.03V − 1,896,000`; `DSCR = CFADS ÷ 5,009,635.23`; expected value = Σ(probability × outcome); debt capacity = (`CFADS` ÷ target `DSCR`) × `AF(0.06, 12)`, with `AF(0.06, 12) = 8.383844` (Domain 3).
3. **Substitution and result.**

Structure B outcome	Volume (m³)	Revenue	EBITDA	Cash tax	Δ working capital	CFADS	DSCR
High (p 0.25)	28,800,000	14,400,000	9,720,000	960,000	720,000	**8,040,000**	**1.6049**
Base (p 0.50)	24,000,000	12,000,000	7,500,000	516,000	600,000	**6,384,000**	**1.2743**
Low (p 0.25)	19,200,000	9,600,000	5,280,000	72,000	480,000	**4,728,000**	**0.9438**
Expected	**24,000,000**	**12,000,000**	—	—	—	**6,384,000**	**1.2743**

Structure A delivers `CFADS` of **6,384,000** with certainty subject to availability. Debt capacity: Structure A at the credit committee's 1.30 × target supports **41,171,123** (Domain 10's figure, unchanged). Structure B, underwritten on the low case at the same 1.30 ×, supports debt service of 3,636,923 and debt of **30,491,396**; underwritten on expected `CFADS` at a merchant-appropriate 1.50 × it supports **35,681,640**. 4. **Interpretation.** The expected `CFADS` is **identical to the dollar** — 6,384,000 in both structures — and so is the expected `DSCR`, at 1.2743, because `DSCR` is linear in `CFADS` when debt service is fixed. An adviser who reports expected values has reported that the two structures are the same. They differ by **10,679,727 of debt capacity**, **25.94 %** of the contracted structure's, and by **11,508,604** against the sponsors' 42,000,000 request. Every dollar of that gap is equity, and Domain 10 showed what a mere 828,877 of it did to the sponsors' return.

Three readings deserve to leave this example. **First, the mean is not a coverage ratio.** Coverage is tested in a period, and lenders underwrite the period they are prepared to be wrong about — commonly a P90 or an explicitly stressed case, never the mean. In the low outcome Structure B's `DSCR` is **0.9438**: debt service is not merely uncovered by covenant, it is unpaid from operating cash. A structure with a one-in-four chance of failing to pay is not a 1.2743 × project with a caveat; it is a different credit. **Second, the case moves as well as the ratio**, and a leader who negotiates only the ratio while letting the case be chosen unopposed has conceded the larger of the two variables. **Third, the risk has a computable price.** Structure A transfers volume risk to the offtaker, who will charge for it in tariff, tenor or termination terms. The question is never "which structure is better" but "is the offtaker's price for taking volume risk less than 10,679,727 of equity plus the return on it?" Asking that question is the professional act.

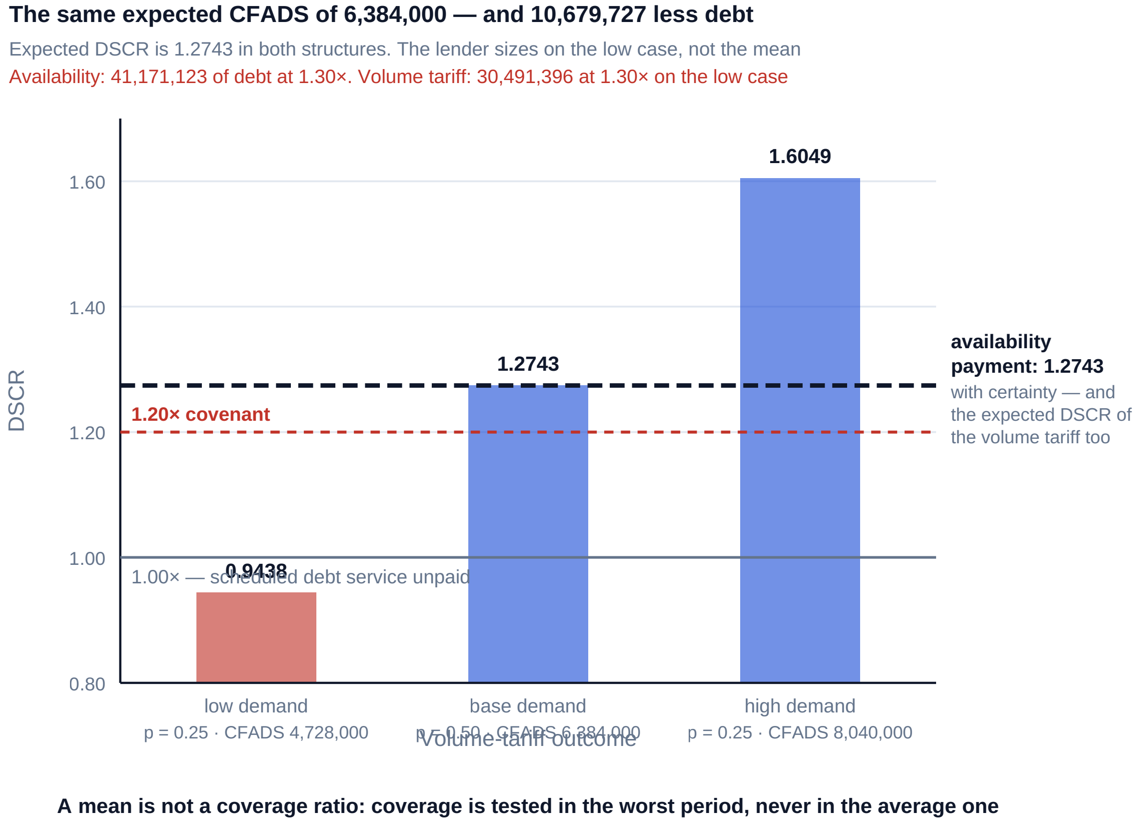

Fig 7.1.1 — Identical expected cash, different bankability

7.1.2 Tariff architecture

Definition. A **tariff** is the formula by which the payer's obligation is computed; its architecture, not its headline level, determines which party bears which risk. Four building blocks recur across sectors under different names.

Block	What it pays for	Risk it leaves with the project
Capacity / availability charge	Making the asset available, whether or not output is taken	Availability, and therefore reliability and maintenance
Energy / variable / usage charge	Output actually taken, usually at a pass-through or indexed unit rate	Efficiency against a contracted rate; sometimes none
Take-or-pay minimum	A floor volume, paid whether taken or not	Volume above the floor only
Merchant tail	Output sold outside the contract	Price and volume, entirely

The **two-part tariff** — capacity plus variable — is the most common bankable form because it separates the recovery of fixed costs and debt service (which cannot flex) from the recovery of variable costs (which can). Its discipline is a matching test: **the capacity charge should recover fixed cash costs, debt service and the return on capital; the variable charge should recover variable costs at contracted efficiency, no more.** A tariff that recovers fixed costs through a volume-linked charge has converted a fixed obligation into a variable receipt, which is the structural error behind much demand-risk distress. Kestrel's Structure B is that error in pure form: 3,600,000 of fixed cost and 5,009,635 of debt service, recovered per cubic metre.

Take-or-pay and its limits. A floor is worth what the party writing it is good for — Domain 5's creditworthiness condition, with a number attached. Two cautions. Take-or-pay volumes are frequently **make-up** rights rather than pure obligations: the payer pays for volume not taken but may take it later at no further charge, which protects cash timing far less than a summary suggests. And **the floor's tenor must reach the loan's**; an eight-year take-or-pay against a twelve-year loan is a merchant tail with a delay (KA 7.A.2).

7.1.3 Availability payments and the deduction mechanism

Definition. An **availability payment** pays for the asset being available to a defined standard rather than for output, reduced by **deductions** when the standard is not met. The deduction formula, not the headline payment, is where the exposure lives: it converts an operational measure into a cash loss, usually at a multiple, and is routinely modelled as though the multiple were one.

WORKED EXAMPLE

Worked example 7.1.3 — the four points of availability that breach a covenant.

1. **Setup.** Kestrel adopts Structure A. The capacity charge is **12,000,000**; guaranteed availability is **95 %** of nameplate hours; the deduction is the capacity charge multiplied by the availability shortfall in decimal terms and by a **1.5 × liquidated-damage multiplier**. Output falls in proportion to the shortfall applied to expected despatch, so variable cost falls with it. Compute revenue, `CFADS` and `DSCR` at 93 % and 91 % availability, and find the availability at which the 1.20 × covenant, the 1.15 × lock-up and payment itself fail.
2. **Formula.** Let `s` be the shortfall in decimal availability points. Deduction `= 12,000,000 × 1.5 × s`; revenue `= 12,000,000 − 18,000,000 s`; volume `= 24,000,000 × (1 − s)`. Substituting into the `CFADS` bridge gives the linear form `CFADS = 6,384,000 − 12,780,000 s`.
3. **Substitution.** At `s = 0.02`: deduction 360,000; revenue 11,640,000; volume 23,520,000. At `s = 0.04`: deduction 720,000; revenue 11,280,000; volume 23,040,000. Thresholds solve `6,384,000 − 12,780,000 s = 5,009,635.23 × threshold`.
4. **Result.**

Availability	Shortfall s	Deduction	Revenue	CFADS	DSCR
95.000 %	—	—	12,000,000	6,384,000	1.2743
93.000 %	2.000 pts	360,000	11,640,000	6,128,400	1.2233
92.086 %	2.914 pts	524,560	11,475,440	**6,011,562**	**1.2000** ← covenant
91.000 %	4.000 pts	720,000	11,280,000	5,872,800	1.1723
90.126 %	4.874 pts	877,351	11,122,649	**5,761,081**	**1.1500** ← lock-up
84.246 %	10.754 pts	1,935,725	10,064,275	**5,009,635**	**1.0000** ← payment fails

1. **Interpretation.** The operational number a plant manager recognises is availability, and the sentence this table produces belongs on an operations dashboard: **Kestrel breaches its covenant at 92.09 % availability — 2.91 points of headroom against a 95 % guarantee.** That is not comfortable. It is the same headroom Domain 10 reported as 372,438 of annual cash and Domain 6 as a 4.14 % revenue fall, expressed in the unit the people who control it actually manage.

The mechanism is where the arithmetic surprises people. A single point of availability costs **127,800 of** `CFADS` — the slope of the linear form — of which 180,000 is lost revenue and 52,200 is recovered through lower variable cost, tax and working-capital absorption. **The 1.5 × multiplier is doing most of the damage.** The general slope is `9,000,000 × multiplier − 720,000`, so at a multiplier of 1.0 it falls to 8,280,000 and the covenant does not break until **90.502 %** — the negotiated multiplier costs **1.584 percentage points** of operating tolerance. (At a multiplier of 1.0 that slope is exactly the volume term of KA 7.3.2, because a pro-rata deduction is economically identical to losing the output.) Multipliers, caps on cumulative deductions, cure periods, planned-outage allowances and the treatment of force majeure and offtaker-caused unavailability are therefore first-order commercial terms, not schedules for the technical adviser. The caution is blunt: **an availability structure is not a risk-free structure, it is one whose risk has been converted from demand into operations.** Kestrel's revenue is now a function of its own reliability — a risk it controls but must then actually control, with Domain 10's maintenance reserve funded and the outage plan built around the deduction formula rather than engineering convenience.

AI in this KA

Machine assistance earns its place in three tasks. **Extracting the payment mechanism** — charge formulae, deduction multipliers, caps, outage allowances, indexation clauses and their cross-references — is high-volume reading that models do well and humans do inconsistently, and the output is a specification a modeller can implement (Toolkit 7.T.1). **Reconciling the modelled tariff to the contractual tariff** by recomputing a historical invoice from the extracted formula is a genuine test with a right answer. And **monitoring deductions** against availability data as it accrues turns a quarterly surprise into a weekly signal.

Where it must not go: **the deduction formula must never be paraphrased into the model.** A summary rendering "1.5 times the availability shortfall applied to the capacity charge" as a pro-rata reduction understates Kestrel's exposure by a third and moves the covenant breakpoint by 1.584 availability points — an error no downstream check catches, because the model is internally consistent and wrong. And a model must not be asked to choose the structure: Worked example 7.1.1 is arithmetic, but judging what the offtaker's price for volume risk is worth is a commercial decision with an accountable owner. Verification is concrete: recompute one period's revenue from the clause text by hand, at full and at reduced availability, and confirm both against the model before quoting any coverage ratio derived from it. **AI proposes; the professional verifies, decides and remains accountable.**

KEY TERMS — KA 7.1

Term	Meaning
Contracted / merchant revenue	Revenue under an obligation to pay / revenue sold into a market with no obligation.
Capacity (availability) charge	Payment for making the asset available, independent of output taken.
Two-part tariff	Capacity charge plus variable charge; separates fixed-cost recovery from output.
Take-or-pay	Contractual floor volume paid whether taken or not; often subject to make-up rights.
Availability deduction	Reduction in the capacity charge for failing the availability standard, commonly at a multiplier.
Deduction multiplier	The factor applied to the availability shortfall; converts one operational point into more than one point of cash.

SAMPLE MCQS — KA 7.1

MCQ 7.1-A `[7.1.1 · Analysis]` Two structures for the same plant have identical expected `CFADS` of 6,384,000 and identical expected `DSCR` of 1.2743. Structure A pays that amount with certainty; Structure B's outcomes are 4,728,000 / 6,384,000 / 8,040,000 with probabilities 0.25 / 0.50 / 0.25. The soundest conclusion is: - A. the structures are equivalent, since expected coverage is the same - B. Structure B supports materially less debt, because a lender underwrites a stressed or low case and requires higher coverage for dispersion - C. Structure B supports more debt, because its upside outcome is higher - D. the difference can be resolved by raising the target `DSCR` alone

Rationale: Sizing on the low case at 1.30 × gives 30,491,396 against 41,171,123 — a 10,679,727 gap (7.1.1). A treats a mean as a coverage ratio, which coverage tests never are; C mistakes an upside the lender has no claim on for capacity; D captures only part of the effect, since the case moves as well as the ratio.

MCQ 7.1-B `[7.1.3 · Application]` A capacity charge of 12,000,000 carries a 95 % availability guarantee and a deduction equal to the charge multiplied by the shortfall and by 1.5. At 91 % availability the deduction is: - A. USD 480,000 - B. USD 720,000 - C. USD 1,620,000 - D. USD 600,000

Rationale: `12,000,000 × 1.5 × 0.04 = 720,000`. A omits the 1.5 × multiplier; C applies the multiplier to total unavailability (9 points, `1 − 0.91`) instead of the shortfall against the 95 % guarantee; D applies a 5 % reduction, reading the guarantee as the deduction base.

MCQ 7.1-C `[7.1.2 · Analysis]` A project recovers all of its fixed cash costs and its entire debt service through a per-unit volume charge with no floor. The structural defect is: - A. the unit rate will be too low to be commercial - B. fixed obligations are funded by a variable receipt, so any volume shortfall falls entirely on coverage - C. the tariff cannot be indexed - D. it prevents the use of an availability standard

Rationale: Matching discipline requires fixed-cost and debt-service recovery in a charge that does not flex with volume (7.1.2). A is a pricing observation, not a structural one; C and D are simply untrue of volume tariffs.

MCQ 7.1-D `[7.1.3 · Analysis]` Kestrel breaches its 1.20 × covenant at 92.086 % availability. Reducing the negotiated deduction multiplier from 1.5 to 1.0 would move that breakpoint to 90.502 %. The correct reading is: - A. the multiplier is a technical schedule with no financial consequence -

B. the multiplier costs 1.584 percentage points of operating tolerance and is therefore a first-order commercial term - C. the multiplier only matters if availability falls below 90 % - D. the breakpoint depends on the tariff level, not the multiplier

Rationale: The multiplier scales the slope of the CFADS line against availability, so it scales headroom directly (7.1.3). C inverts the logic — the multiplier is what brings the breakpoint closer; D ignores that the deduction is computed on the charge and the multiplier together.

SELF-CHECK — KA 7.1

1. Why does an expected *DSCR* convey nothing about bankability? — Coverage is tested in a period; lenders underwrite a low or stressed case, and dispersion also raises the required ratio, so the mean moves neither variable.
2. State Kestrel's covenant breakpoint in availability terms and in cash terms. — 92.086 % availability; CFADS of 6,011,562, which is 372,438 of annual cash below base (Domain 10).
3. What has an availability structure actually done to a project's risk? — Converted demand risk into operational risk, which the project controls but must then genuinely manage.

KNOWLEDGE AREA 7.2

Demand models, concessions and service revenue

Topics: 7.2.1 demand forecasting and the ramp · 7.2.2 concession payment mechanisms · 7.2.3 subscription and service revenue.

7.2.1 Demand forecasting and the ramp

Definitions. A **demand model** forecasts the quantity of output the market will take at a given price over the asset's life. The **ramp** is the period after commercial operations during which realised demand approaches its mature level as users discover and reorganise around the asset. **Optimism bias** is the tendency of forecasts prepared by parties interested in the project proceeding to exceed outturn — a documented pattern in demand-risk infrastructure, and the reason lenders commission their own market adviser rather than reviewing the sponsors'.

The professional content here is not forecasting technique, which belongs to market specialists, but the **discipline of using a forecast in a financing**. Three practices carry almost all the value.

Separate the cases and name their owners. The **sponsor case** is the sponsors' central expectation; the **bank case** is what lenders will lend against, typically the sponsor case with specified haircuts, a slower ramp, no unindexed growth and a conservative price deck; the **downside case** is the stress the structure must survive. Domain 6 (KA 6.1.3) made this a modelling rule; here it is a commercial one, because the gap between the cases is the gap between the sponsors' return and the lenders' credit — Domain 6 priced it on Kestrel at **17,107,567 of** NPV from the escalation assumption alone.

Model the ramp explicitly and let the debt schedule respond. Level debt service against a rising cash profile wastes coverage late and breaches early, which is the case for sculpting (Domain 10, KA 10.1.3). The recurring error is to model the ramp in the revenue line and then size level debt on the mature year: the ratio computed on year five is met, and the covenant tested in year two is not.

State the forecast's elasticity, not only its level. A demand forecast at a given tariff is an implicit price-elasticity assumption. Where the project sets price — a toll road, a car park, a data-centre re-let — a shortfall can be met by raising price only if elasticity permits; where price is contractual it cannot be met at all. A demand model presented without its elasticity has withheld the one property that determines whether management has a lever.

7.2.2 Concession payment mechanisms

Definition. A **concession** grants a private party the right and obligation to build, finance and operate a public asset for a defined term, with revenue arriving through a payment mechanism the grant defines and the asset handed back at the end. The mechanism is the risk allocation, and the standard family is worth naming because the names travel badly between jurisdictions and sectors.

Mechanism	Who pays	Volume risk sits with	Typical use
Availability payment	The grantor, from budget	The grantor	Social infrastructure, non-tolled roads, some water
Real toll / user charge	Users, directly	The project	Toll roads, ports, car parks
Shadow toll	The grantor, per unit of measured usage	Shared: the project bears volume, the grantor bears price	Roads where user charging is politically unavailable
Volume band	Either, per unit within defined bands at declining rates	Shared by construction	Water, waste, transport with uncertain demand
Minimum revenue guarantee with revenue sharing	Grantor tops up below a floor; shares above a ceiling	Shared symmetrically	Long concessions with contested forecasts

WORKED EXAMPLE

Worked example 7.2.2 — what a volume band is worth.

1. **Setup.** Kestrel's offtaker will not accept Structure A's full volume risk but offers a **banded tariff**: **USD 0.55 per m^3** on the first 18,000,000 m^3, **USD 0.35** on the next 6,000,000 m^3, and **USD 0.10** on anything above 24,000,000 m^3. At expected despatch of 24,000,000 m^3 the band structure pays exactly 12,000,000, so the headline tariff is unchanged. Using the same three demand outcomes as Worked example 7.1.1, compute revenue, `CFADS` and `DSCR`, the expected values, and the debt capacity on the low case at 1.30 ×.
2. **Formula.** Banded revenue = Σ(volume in band × band rate); `CFADS` and `DSCR` as before.
3. **Result.**

Outcome	Volume (m^3)	Banded revenue	Flat-tariff revenue	CFADS banded	DSCR banded	DSCR flat
High (p 0.25)	28,800,000	12,480,000	14,400,000	6,600,000	**1.3175**	1.6049
Base (p 0.50)	24,000,000	12,000,000	12,000,000	6,384,000	**1.2743**	1.2743
Low (p 0.25)	19,200,000	10,320,000	9,600,000	5,268,000	**1.0516**	0.9438
Expected	24,000,000	**11,700,000**	12,000,000	**6,159,000**	1.2294	1.2743

Debt capacity on the low case at 1.30 ×: **33,973,915** banded against 30,491,396 flat — a gain of **3,482,520**. The expected revenue given up is **300,000 a year**, 2.50 % of base revenue, whose present value over the twelve loan years at 6 % is **2,515,153**. 4. **Interpretation.** The band is a risk-transfer instrument dressed as a price list, and its economics are explicit once computed: the sponsors surrender 2,515,153 of present value and receive 3,482,520 of debt capacity, a **net gain of 967,367** before the return on the equity released. Whether that is a good trade depends on the cost of equity against the cost of debt (Domain 9), but the method matters more than the answer: a negotiating team that has not computed it is arguing about band rates without knowing which way it wants them to move.

Two further readings matter more than the headline. **The band compresses the distribution from both ends, and the lender pays for only one end.** `DSCR` dispersion narrows from 0.9438–1.6049 to 1.0516–1.3175: the low tail improves by 0.1078 of coverage and the high tail is surrendered entirely. Since lenders price the low tail and equity owns the high one, a band transfers value **from equity's upside to debt capacity** — which is why sponsors confident in demand resist bands and sponsors who need leverage accept them, and why the same band is a good deal for one shareholder and a poor one for another in the same consortium. **And the low case still breaches.** At 1.0516 the banded structure pays its debt service but fails both the 1.20 × covenant and the 1.15 × lock-up, so the band has bought bankability, not compliance. A band should be sized against the lock-up, not against zero.

7.2.3 Subscription and service revenue

Definition. In **subscription and service models** revenue arrives as recurring payments from a population of customers under contracts materially shorter than the financing, so the revenue risk is **re-contracting risk** rather than volume or price risk. Three measures govern: **annual recurring revenue** (`ARR`, the annualised value of contracts in force), **net revenue retention** (next year's revenue from this year's customers after churn and expansion, as a percentage), and the **weighted-average remaining contract term** (`WARCT`), the contract-weighted mean of remaining tenors. Digital infrastructure, managed services and waste all finance on this shape.

WORKED EXAMPLE

Worked example 7.2.3 — the run-off case, and why a 7-year loan was refused.

1. **Setup.** Halyard Connect Networks (a fictitious SPV) operates a metropolitan fibre network. At financial close, `ARR` is **24,000,000**; cash operating costs are wholly fixed at **13,200,000**, so `EBITDA` is 10,800,000; maintenance capex is **1,800,000** and cash tax is nil in the period by reason of capital allowances (stated as an assumption of the illustration, not a general treatment). Net revenue retention is **93 %**. The contract book is 40 % of `ARR` with five years remaining, 35 % with three years and 25 % with one year. The sponsors seek **40,000,000** of senior debt at **7 % over seven years**. Compute the lenders' run-off case, the coverage it delivers against level debt service, the `WARCT`, and the debt the run-off case actually supports at a 1.30 × target.
2. **Formula.** Run-off revenue in year `t` = `ARR × 0.93^(t−1)` — contracted revenue only, with no credit for unsigned new sales. `CFADS` = revenue − 13,200,000 − 1,800,000. Level instalment = 40,000,000 ÷ `AF(0.07, 7)`, with `AF(0.07, 7) = 5.389289`. Sculpted capacity = Σ (`CFADS(t)` ÷ 1.30) × `DF(0.07, t)` (Domain 10, KA 10.1.3). `WARCT` = Σ(share × remaining term).
3. **Result.**

Year	Run-off revenue	EBITDA	CFADS	DSCR on level service
1	24,000,000	10,800,000	9,000,000	**1.2126**
2	22,320,000	9,120,000	7,320,000	**0.9862**
3	20,757,600	7,557,600	5,757,600	0.7757
4	19,304,568	6,104,568	4,304,568	0.5800
5	17,953,248	4,753,248	2,953,248	0.3979
6	16,696,521	3,496,521	1,696,521	0.2286
7	15,527,764	2,327,764	527,764	0.0711

Level instalment **7,422,128.79**. WARCT **3.30 years — 47.14 %** of the seven-year tenor. Sculpted debt capacity on the run-off case at 1.30 × is **20,271,839**, **50.68 %** of the 40,000,000 sought. 4. **Interpretation.** The structure fails in **year two**, and the mechanism is worth stating precisely because it is the characteristic failure of service-revenue financings. A 7 % annual revenue decline against a wholly fixed cost base takes EBITDA down **15.56 %** in the first year of run-off and **17.13 %** in the second, and CFADS down **18.67 %** and then **21.34 %** — the operating-leverage amplification of KA 7.3.2 applied to attrition rather than to demand, and accelerating, because the fixed cost base is a growing share of a shrinking revenue. Net revenue retention of 93 % is in many service markets a perfectly respectable number; against a wholly fixed cost base and a seven-year loan it is a sixth of EBITDA in the first year and more thereafter.

The professional consequences are three. **The lender's case is the run-off case, and arguing otherwise is arguing to be lent against unsigned revenue.** Sponsors routinely present a forecast including new sales; the honest response is to present both, with debt sized on the contracted book and the new-sales case shown as the equity story. **The binding metric is WARCT against tenor, not ARR.** At 3.30 years against a seven-year loan, less than half the loan's life is contracted, and the disciplined responses are a shorter tenor, a re-contracting reserve, a cash sweep that retires debt while the contracted book still exists, or anchor extensions as a condition precedent. **And 20,271,839 is the honest answer, so the negotiation is about the other 19.7 million** — a shorter tenor, a lower amount, extensions before close, or more equity. The sponsors are better served by a leader who computes 50.68 % early than by one who discovers it in credit committee.

AI in this KA

Demand forecasting is the most attractive and most dangerous machine application in this domain. It earns its place in **pattern extraction from usage data** — seasonality, elasticity estimation from observed price and volume, cohort-level churn and expansion behaviour no spreadsheet summary reveals — in **churn and re-contracting prediction** for service models, and in **scenario generation**, producing the coherent bundles a stress matrix needs faster and more consistently than a workshop.

Where it must not go. **A model trained on comparable assets' ramps inherits their optimism bias**, because the training data is forecasts and outturns from projects that were financed — a selected population — and the assistant will not say so. The output must be tested against the base rate that matters: outturn against forecast for comparable assets including those that underperformed, which is precisely the data least likely to be available. **A forecast must never be presented without its case label and its owner**, and a machine-generated forecast has no owner until a named professional adopts it. **And no assistant should select the bank case**, which is a negotiating position rather than an estimate. Verification: reconstruct the forecast's implied elasticity and load factor by hand, check them against the asset's physical and contractual capacity, and require that any forecast used in a financing carry the market adviser's name.

KEY TERMS — KA 7.2

Term	Meaning
Ramp	The period during which realised demand approaches its mature level after commercial operations.
Sponsor / bank / downside case	Central expectation · the case lenders will lend against · the stress the structure must survive.
Shadow toll	Grantor pays the project per unit of measured usage; volume risk stays with the project, price risk with the grantor.
Volume band	Declining unit rates across volume tranches; compresses revenue dispersion from both ends.
`ARR`	Annual recurring revenue: the annualised value of contracts in force.
Net revenue retention	Next year's revenue from this year's customers, after churn and expansion, as a percentage.
`WARCT`	Weighted-average remaining contract term; the contracted horizon to be compared with loan tenor.
Run-off case	Contracted revenue only, with no credit for unsigned new business.

SAMPLE MCQS — KA 7.2

MCQ 7.2-A `[7.2.2 · Application]` A banded tariff pays 0.55 per m³ on the first 18,000,000 m³, 0.35 on the next 6,000,000 and 0.10 above 24,000,000. At despatch of 19,200,000 m³ revenue is: - A. USD 9,600,000 - B. USD 10,320,000 - C. USD 10,560,000 - D. USD 12,000,000

Rationale: `18,000,000 × 0.55 + 1,200,000 × 0.35 = 9,900,000 + 420,000 = 10,320,000`. A applies the flat 0.50 tariff; C applies 0.55 to the whole volume; D is the base-case revenue, which the band does not guarantee.

MCQ 7.2-B `[7.2.3 · Analysis]` An SPV has `ARR` of 24,000,000, net revenue retention of 93 % and wholly fixed cash costs of 13,200,000. In the first year of run-off, `EBITDA` falls by approximately: - A. 7 % - B. 15.6 % - C. 18.7 % - D. it does not fall, because retention is above 90 %

Rationale: `EBITDA` falls from 10,800,000 to 9,120,000, a fall of **15.56 %** — the 7 % revenue decline amplified by the fixed cost base. A mistakes the revenue decline for the earnings decline; C is the `CFADS` decline (9,000,000 to 7,320,000, −18.67 %), the right amplification applied to the wrong measure; D misreads a retention rate as a growth rate.

MCQ 7.2-C `[7.2.3 · Analysis]` A service-revenue SPV has a `WARCT` of 3.30 years and seeks a seven-year loan. The lender's most likely structural response is: - A. to accept the tenor, since `ARR` covers debt service in year one - B. to size on the contracted run-off case and require a shorter tenor, a re-contracting reserve, a cash sweep or contract extensions before close - C. to size on the sponsor case including new sales, discounted by 10 % - D. to require an availability guarantee

Rationale: Less than half the loan's life is contracted, so the loan is being asked to bridge uncontracted years (7.2.3). A relies on year one alone; C lends against unsigned revenue; D applies a mechanism from a different revenue architecture.

MCQ 7.2-D [7.2.2 · Analysis] A volume band raises the low-case DSCR from 0.9438 to 1.0516 and lowers the high-case DSCR from 1.6049 to 1.3175, reducing expected revenue by 300,000 a year. The correct characterisation is: - A. the band destroys value, since expected revenue falls - B. the band transfers value from equity's upside to debt capacity, and the trade is computable: 2,515,153 of present value surrendered for 3,482,520 of capacity - C. the band has no effect on debt capacity, since expected CFADS still exceeds debt service - D. the band removes the covenant risk

Rationale: Lenders price the low tail and equity owns the high one, so compressing both is a transfer, not a loss (7.2.2). A ignores the capacity gain; C treats expected cash as the sizing basis; D is false — the low case at 1.0516 still breaches the 1.20 × covenant.

SELF-CHECK — KA 7.2

1. What distinguishes a bank case from a downside case? — The bank case is what lenders will lend against; the downside case is the stress the structure must survive. They have different purposes and different owners, and conflating them hides both.
2. Why is *WARCT* more informative than *ARR* for a service-revenue financing? — ARR measures the size of the contracted book; WARCT measures how much of the loan's life it covers, which is the risk being financed.
3. Which tail does a volume band buy, and who pays for it? — It buys the low tail, which lenders price; equity pays for it by surrendering the high tail.

KNOWLEDGE AREA 7.3

Price escalation and volume risk

Topics: 7.3.1 indexation architecture and margin drift · 7.3.2 volume risk and operating leverage · 7.3.3 turning revenue risk into a tolerance.

7.3.1 Indexation architecture and margin drift

Definition. Indexation is the contractual escalation of a price by reference to a published index. Domain 3 (KA 3.3.2) established the compounding arithmetic and the register of index mechanics — publisher, series, definition, lag, cap, floor, compounding basis. This topic addresses the commercial structure those mechanics sit inside: the **indexation architecture**, meaning which proportion of each revenue and cost line escalates, on which index, and what the mismatch is worth.

Each line has three parameters and only the first two are usually negotiated with care: the index, the rate it is expected to run at, and the **indexed share**. A tariff described as "CPI-indexed" may escalate 60 %, 80 % or 100 % of its value; the unindexed remainder is a fixed nominal amount eroding in real terms for the whole concession. Because revenue-side and cost-side shares are set in different negotiations, by different people, at different times, they are almost never matched — and the mismatch compounds.

WORKED EXAMPLE

Worked example 7.3.1 — the 36 basis points that cost 3.27 margin points.

1. **Setup.** Kestrel's water-purchase agreement indexes **80 %** of the tariff to a consumer price index assumed to run at **2.5 %** a year; the remaining 20 % is fixed for the 25-year concession. On the cost side, **70 %** of the 4,500,000 cash operating cost — labour, chemicals and power — escalates at **3.2 %** under a separate index in the O&M contract; the remaining 30 % is fixed under long-term agreements. Compute revenue, cost, `EBITDA` and the `EBITDA` margin in years 1, 12 (loan maturity) and 25, the effective compound rates, and the value of the unindexed tariff slice.
2. **Formula.** Revenue in year `t = 12,000,000 × [0.80 × 1.025^(t−1) + 0.20]`; cost in year `t = 4,500,000 × 0.70 × 1.032^(t−1) + 4,500,000 × 0.30`. Effective rate = (year-25 value ÷ year-1 value)^(1/24) − 1.
3. **Result.**

Year	Revenue	Cash operating cost	EBITDA	Margin
1	12,000,000	4,500,000	7,500,000	**62.50 %**
5	12,996,604	4,922,970	8,073,634	62.12 %
12	14,996,032	5,804,380	9,191,652	**61.29 %**
25	19,763,769	8,058,467	11,705,302	**59.23 %**

Effective compound rates: revenue **2.101 %**, cost **2.457 %** — a gap of **35.7 basis points**. Under full tariff indexation at the same 2.5 % the year-25 margin would be **62.87 %** and `EBITDA` **13,646,244**; the present value at 8 % of the `EBITDA` forgone across 25 years by indexing only 80 % is **6,204,143**, of which **2,619,217** falls inside the twelve loan years discounted at 6 %. Under a fully indexed cost base at 3.2 % the year-25 margin would be **51.51 %**. 4. **Interpretation. Neither contractual index is the rate that matters.** The headline gap between a 2.5 % tariff index and a 3.2 % cost index is 70 basis points; the effective rates that partial indexation actually produces are 2.101 % and 2.457 %, a gap of **35.7 basis points**. Half the apparent mismatch is neutralised by the 30 % of cost that does not escalate, so a negotiator working from headline indices has overstated the exposure twofold. The corrective is the same in both directions: **compute effective rates from shares and indices, and negotiate on those.**

The decomposition is where the money is, and it is counter-intuitive. Of the 3.27 margin points lost by year 25, the unindexed 20 % of the tariff accounts for **3.65 points** (the year-25 margin under full tariff indexation is 62.87 %) while the fixed 30 % of the cost base saves **7.72 points** (a fully indexed cost base would end at 51.51 %). The single most valuable term in Kestrel's commercial package is therefore not the tariff index but the fixed-price portion of the O&M contract, worth more than twice what the unindexed tariff slice costs — an insight invisible unless the architecture is decomposed.

The drift is slow, which is why it is dangerous. Nothing breaks: year-12 coverage is a comfortable **1.4253**, because the interest deduction has fallen and the tariff has risen faster than the covenant needs. The damage lands after the loan matures — in refinancing, handback economics and the equity return — so the party most exposed to indexation drift is the party least present in the financing negotiation. Price the whole concession, not the loan life: the tariff index a lender is indifferent to is the one shareholders live with for thirteen years after the lender has gone. **And an index the model assumes is not an index the contract grants.** CPI at 2.5 % is a forecast;

entitlement to 80 % of CPI is a term. The first belongs in the sensitivity table, the second in the contract review, and confusing them is how projects end up with an escalation forecast where they needed an escalation right.

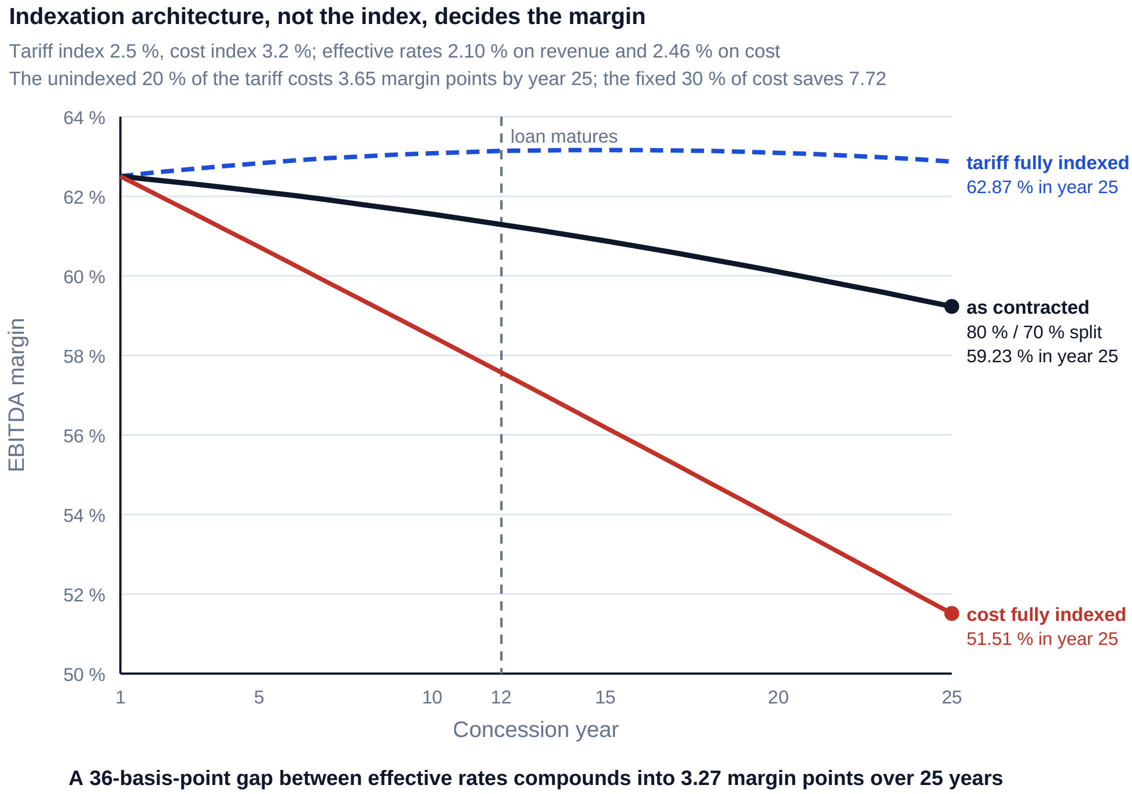

Fig 7.3.1 — Margin drift over the concession

7.3.2 Volume risk and operating leverage

Definition. Operating leverage is the amplification of a change in volume into a larger proportional change in earnings, caused by costs that do not fall when volume does. The **degree of operating leverage** is the ratio of contribution to `EBITDA`:

```
DOL = contribution ÷ EBITDA = (unit price − unit variable cost) × volume ÷ EBITDA
```

and the corresponding `CFADS` **elasticity** to a driver is the proportional change in `CFADS` per one per cent change in that driver. Domain 6 (KA 6.4.2) computed Kestrel's `CFADS` elasticity to **revenue** as **1.4098** and used it to convert coverage headroom into a revenue breakeven. This topic supplies the mechanism and one distinction Domain 6 did not need.

WORKED EXAMPLE

Worked example 7.3.2 — why a price miss hurts more than a volume miss.

1. **Setup.** Kestrel's master thread: price 0.50 per m³, variable cost 0.0375 per m³, volume 24,000,000 m³, fixed cash cost 3,600,000, `EBITDA` 7,500,000, `CFADS` 6,384,000. Compute the degree of operating leverage, the `CFADS` elasticity to price and to volume, and verify each against a finite move.
2. **Formula.** Contribution = `(0.50 − 0.0375) × 24,000,000`. From the bridge `CFADS = 0.75R − 0.03V − 1,896,000`: elasticity to price (revenue moving alone) = `0.75R ÷ CFADS`; elasticity to volume (revenue and variable cost and working capital moving together) = `(0.75 × 0.50 − 0.03) × V ÷ CFADS`.
3. **Substitution.** Contribution `11,100,000`; `DOL = 11,100,000 ÷ 7,500,000`. Price elasticity `9,000,000 ÷ 6,384,000`; volume elasticity `8,280,000 ÷ 6,384,000`.
4. **Result.** `DOL` **1.4800**. `CFADS` elasticity to **price 1.4098** (reproducing Domain 6's figure exactly); `CFADS` elasticity to **volume 1.2970**. Verification: a 20 % volume fall takes `CFADS` to 4,728,000, a fall of **25.94 %** — 20 × 1.2970; a 10 % price fall takes it to 5,484,000, a fall of **14.10 %** — 10 × 1.4098.
5. **Interpretation. A price cut and a volume shortfall of the same revenue magnitude are not the same event.** Losing 10 % of revenue through price costs 14.10 % of `CFADS`; losing the same 10 % through volume costs 12.97 %, because lost volume brings relief on variable cost, cash tax and working-capital absorption. The 1.13-point difference is small at Kestrel because only 20 % of its cash cost is variable; where the variable share is 60 % the two elasticities diverge sharply, and a matrix treating "revenue down 10 %" as one scenario has merged two materially different ones. **Always stress the driver, not the revenue line.**

The generalisation is the useful part, and Domain 10 supplies its test case. Domain 10's Case study B reported a toll road whose 12 % patronage shortfall produced a **17.9 %** fall in `CFADS` — an implied elasticity of **1.4931**. Since a road's cash costs are substantially fixed, elasticity in that limiting case is simply revenue divided by `CFADS`, so the reported outcome implies base revenue of **35,833,333** against cash costs and tax of **11,833,333**. The relationship generalises to a table every practitioner should be able to reconstruct, whose last column answers the only question a board asks:

`CFADS` as % of revenue	`CFADS` elasticity to volume (fixed costs)	Volume fall that takes a 1.30 × base to a 1.20 × covenant
80 %	1.2500	**6.15 %**
67 %	1.4925	**5.15 %**
60 %	1.6667	**4.62 %**
40 %	2.5000	**3.08 %**
20 %	5.0000	**1.54 %**

Reaching a 1.20 × covenant from a 1.30 × base requires a `CFADS` fall of **7.6923 %** in every row; the demand tolerance that produces it ranges from **6.15 % to 1.54 %** purely on cost structure. **Two projects with the same coverage ratio and the same covenant can have demand tolerances that differ fourfold, and the ratio does not show it.** That is the most important thing this domain has to say to a credit committee, and it is why cost structure belongs in a revenue-risk discussion. High operating leverage is not a defect — it is what makes infrastructure profitable when demand holds — but it must be known, disclosed and matched by a structure that tolerates it: lower gearing, a larger reserve, a floor mechanism (KA 7.A.1) or sculpting.

7.3.3 Turning revenue risk into a tolerance

The principle. A ratio conveys no magnitude and an elasticity conveys no threshold; together they convert a covenant into a sentence an operating team can act on. Domain 6 (KA 6.4.2) established the breakeven translation as a modelling output; the commercial extension is to express it in **each driver the business actually manages**, because different teams control different drivers.

Kestrel's tolerance set, in four units. From a base DSCR of 1.2743, the 1.20 × covenant requires CFADS of 6,011,562 — a fall of 372,438, or **5.8339 %** (Domain 10). Dividing that CFADS tolerance by each elasticity gives:

Driver	Elasticity	Tolerance	Level at which the covenant fails
Price / tariff	1.4098	**4.14 %**	revenue 11,503,416 (Domain 6)
Volume / despatch	1.2970	**4.50 %**	22,920,470 m³ — a loss of 1,079,530 m³
Availability	—	**2.91 points**	92.086 % availability (KA 7.1.3)
CFADS	1.0000	**5.83 %**	6,011,562

Four numbers, one covenant, each right for a different conversation: the commercial team hears 4.14 % on tariff, operations hears 4.50 % on despatch or 2.91 points on availability, finance hears 5.83 % on CFADS, and the board hears whichever is smallest. **Quoting the CFADS tolerance to a team that manages volume overstates their room by 30 %** — the same error Domain 6 identified when it observed that a CFADS percentage overstates revenue room by 41 %. The lock-up bites earlier still, at **22,194,436 m³**, a **7.52 %** volume fall; since lock-up is what a sponsor actually feels, that is the threshold for the shareholder dashboard.

AI in this KA

Escalation and elasticity work is arithmetically simple and definitionally treacherous — the profile in which machine output is most confidently wrong. The legitimate applications are narrow: **reconciling every escalating line to its index clause** (publisher, series, lag, cap, floor, indexed share), which is Domain 3's escalation register populated by extraction rather than transcription; **recomputing effective compound rates** across a whole revenue and cost base; and **producing the tolerance table** for every covenant and driver combination on demand.

Two failure modes. **An assistant asked for "the escalation rate" returns the headline index, not the effective rate**, because the headline is what the document says while the effective rate requires the indexed share — the parameter most often omitted from a contract summary. Kestrel's 70-basis-point headline against a 35.7-basis-point effective gap is that error in one comparison. **And an elasticity is a property of a case, not of a project** (Domain 6, KA 6.4.2): elasticities computed on a generous case and quoted against a marginal one are meaningless, and machine-generated sensitivity tables are frequently unlabelled as to case. Verification: recompute one escalating line for one year from the clause text by hand; confirm the effective rate against first- and final-year values; and require every elasticity to carry its case label and base value.

KEY TERMS — KA 7.3

Term	Meaning
Indexation architecture	Which proportion of each line escalates, on which index — the negotiated structure, distinct from the index itself.
Indexed share	The proportion of a price that escalates; the remainder is fixed nominal and erodes in real terms.
Effective escalation rate	The compound rate a partly indexed line actually achieves; computable only from shares and indices together.
Margin drift	The slow change in margin caused by a mismatch between effective revenue and cost escalation.
Operating leverage (`DOL`)	Contribution ÷ `EBITDA`; the amplification of volume changes into earnings changes.
`CFADS` **elasticity**	Proportional change in `CFADS` per one per cent change in a driver; differs by driver.
Tolerance	The fall in a named driver that takes coverage to a threshold; the operational form of headroom.

SAMPLE MCQS — KA 7.3

MCQ 7.3-A `[7.3.1 · Application]` A tariff indexes 80 % of its value at 2.5 % a year; the remaining 20 % is fixed. Over 24 years the tariff's **effective** compound escalation rate is closest to: - A. 2.500 % - B. 2.101 % - C. 2.000 % - D. 3.125 %

Rationale: Year-25 revenue is `12,000,000 × (0.80 × 1.025²⁴ + 0.20) = 19,763,769`, giving `(19,763,769/12,000,000)^(1/24) − 1 = 2.101 %`. A is the headline index applied as though the whole tariff escalated; C is the naive `0.80 × 2.5 %`, which is an approximation valid only over one period; D divides the index by the indexed share instead of multiplying.

MCQ 7.3-B `[7.3.2 · Application]` Price 0.50 per m³, variable cost 0.0375 per m³, volume 24,000,000 m³, `EBITDA` 7,500,000. The degree of operating leverage is: - A. 1.4800 - B. 1.6000 - C. 1.2970 - D. 0.6757

Rationale: `contribution 11,100,000 ÷ EBITDA 7,500,000 = 1.4800`. B uses revenue rather than contribution (`12,000,000/7,500,000`); C is the `CFADS` elasticity to volume, a different measure computed after tax and working capital; D inverts the ratio.

MCQ 7.3-C `[7.3.2 · Analysis]` Two projects each report a base `DSCR` of 1.30 × against a 1.20 × covenant. Project M's `CFADS` is 80 % of revenue; Project N's is 40 %. Their demand tolerances are: - A. identical, since the coverage ratio and covenant are identical - B. M 6.15 %, N 3.08 % — cost structure halves the tolerance at identical coverage - C. M 3.08 %, N 6.15 % - D. indeterminate without the debt amount

Rationale: Both need a 7.6923 % `CFADS` fall; dividing by elasticities of 1.25 and 2.50 gives 6.15 % and 3.08 % (7.3.2). A is the error the table exists to correct; C reverses the relationship — a thinner `CFADS` margin means higher elasticity and less tolerance; D is wrong because the tolerance is a ratio property, independent of scale.

MCQ 7.3-D `[7.3.3 · Analysis]` Kestrel's `CFADS` tolerance to its covenant is 5.83 %. The figure that should be given to the operations team, which manages despatch, is: - A. 5.83 %, the `CFADS` tolerance - B. 4.50 %, the volume tolerance, being 5.83 % divided by the `CFADS` elasticity to volume of 1.2970 - C. 4.14 %, the revenue tolerance - D. 7.52 %, the lock-up tolerance

Rationale: Each team needs the tolerance in the driver it controls; quoting the CFADS figure to a volume-managing team overstates their room by about 30 % (7.3.3). C is the correct figure for the commercial team, which sets tariff, not for despatch; D is the correct volume figure for the lock-up, a different and later threshold.

SELF-CHECK — KA 7.3

1. Why is the headline index not the rate that matters? — Because the indexed share determines the effective rate; Kestrel's 70-basis-point headline gap is a 35.7-basis-point effective gap.
2. Why does a price miss cost more *CFADS* than a volume miss of equal revenue? — Lost volume brings relief on variable cost, cash tax and working capital; a price cut brings none.
3. State the one sentence that converts a covenant into an operational instruction. — "Coverage breaks at a 4.50 % fall in despatch, or 1,079,530 m³ of the 24,000,000 forecast."

KNOWLEDGE AREA 7.4

Counterparty credit quality and revenue stress testing

Topics: 7.4.1 counterparty credit and expected loss · 7.4.2 concentration and credit enhancement · 7.4.3 revenue stress testing and the bank case.

7.4.1 Counterparty credit and expected loss

Definition. A contracted revenue stream is a claim on a counterparty, and its value is bounded by that counterparty's ability and willingness to pay. **Expected loss** decomposes it into three estimable parts:

```
Expected loss = probability of default × exposure at default × loss given default
```

where **probability of default** (PD) is the likelihood of failure to perform over a stated horizon, **exposure at default** (EAD) is the amount at risk when it happens, and **loss given default** (LGD) is the proportion of that exposure not recovered. In project finance the exposure that matters is rarely the outstanding receivable: it is the **present value of the contracted stream that would be lost**, because the loss event is termination of the revenue, not non-payment of an invoice.

WORKED EXAMPLE

Worked example 7.4.1 — what Kestrel's single offtaker is worth as a credit.

1. **Setup.** Kestrel's water-purchase agreement runs 25 years with a single regional water authority. Mapping its published rating to an internal grade gives an **annual** `PD` **of 0.60 %**. If the offtaker fails, Kestrel can re-sell water into a regional merchant market at a discount and expects to recover 55 % of the contracted value, so `LGD` is **45 %**. Exposure is taken as the present value of `CFADS` over the loan life at the loan rate — the amount the lenders are relying on. Compute the twelve-year cumulative `PD`, the exposure, the expected loss, and the coverage ratio net of an annualised credit charge. (Rating mappings, recovery assumptions and regulatory capital treatments are institution- and jurisdiction-specific; the arithmetic transfers, the parameters do not.)
2. **Formula.** Cumulative `PD` over `n` years = `1 − (1 − PD)ⁿ`; `EAD = CFADS × AF(0.06, 12)`; `EL = cumulative PD × EAD × LGD`; annualised credit charge = `EL ÷ AF(0.06, 12)`.
3. **Substitution.** `1 − 0.994¹²`; `6,384,000 × 8.383844`; `0.069671 × 53,522,460 × 0.45`; `1,678,031 ÷ 8.383844`.
4. **Result.** Twelve-year cumulative `PD` **6.9671 %**; `EAD` **USD 53,522,460**; expected loss **USD 1,678,031**, which is **3.135 %** of exposure; annualised credit charge **200,151**, reducing `CFADS` to 6,183,849 and the `DSCR` to **1.2344**.
5. **Interpretation.** The last line is the one worth carrying: **charging the counterparty's expected loss against coverage costs Kestrel 0.0400 of** `DSCR` **— 53.7 %** of its total covenant headroom of 0.0743 — and the project is still compliant. That is a defensible position, and stating it that way converts an opinion ("the offtaker is investment grade") into a quantified one ("the credit is worth four hundredths of a coverage ratio, against 0.0743 of headroom").

Three cautions belong with the arithmetic, each a way the calculation is routinely abused. `PD` **is not a fact.** It is a mapping from a rating or a model to a number, on a horizon, in a state of the economy; a public-sector offtaker's `PD` in particular is often taken from sovereign or sub-sovereign proxies that assume a support relationship the documents may not create. Present it with its source and horizon, and move it in the sensitivity table like any other assumption. `EAD` **depends on what the loss event is**, and choosing it is a judgment: the receivable (2,958,904 on 90-day terms — trivial), the present value of contracted `CFADS` over the loan life (53,522,460 — the lenders' exposure, used above), or the present value over the whole concession (much larger — the shareholders'). Three exposures, three expected losses, one correct answer per decision, and quoting the wrong one is how a concentration problem gets presented as a working-capital matter. **And** `LGD` **embeds a market and a legal outcome.** Kestrel's 45 % assumes a merchant market exists to re-sell into and that termination compensation, step-in rights and the security package work as drafted — matters for Domain 12 and, in a real transaction, for qualified counsel in the relevant jurisdiction. An `LGD` not traced to those documents is a guess wearing a decimal point.

7.4.2 Concentration and credit enhancement

The result that matters. Expected loss is a linear function of exposure, so **it cannot see concentration**. Splitting one counterparty into four of equal size and equal `PD` leaves expected loss exactly unchanged, while changing the loss distribution beyond recognition. A credit committee shown only expected loss has been shown a number that is blind to the risk the committee exists to control.

WORKED EXAMPLE

Worked example 7.4.2 — one offtaker or four, and what enhancement is worth.

1. **Setup.** Compare Kestrel's single offtaker (annual `PD` 0.60 %, twelve-year cumulative 6.9671 %, exposure 53,522,460, `LGD` 45 %) with a hypothetical structure in which four offtakers each take 25 % of the output on identical terms, with independent defaults. Compute expected loss in each, and the probability of losing at least half of the revenue and of losing all of it. Then price a sovereign guarantee that reduces the effective annual `PD` to **0.20 %** for a fee of **0.35 % a year on the outstanding debt balance**, against both its expected-loss benefit and its effect on required coverage — the lenders having indicated that a guaranteed structure would be sized at **1.20 ×** rather than **1.30 ×**.
2. **Formula.** Four-counterparty default counts follow the binomial distribution with `n = 4` and `p = 0.069671`; losing at least half the revenue requires at least two defaults. Guarantee fee present value = Σ (0.35 % × opening balance in year `t`) × `DF(0.06, t)` over the amortisation schedule of Domain 3. Debt capacity as in Domain 10.
3. **Result.**

Measure	One offtaker	Four offtakers at 25 %
Expected loss	**1,678,031**	**1,678,031** — identical
P(no default)	93.0329 %	74.9111 %
P(exactly one default)	—	22.4399 %
P(losing ≥ 50 % of revenue)	**6.9671 %**	**2.6489 %**
P(losing all revenue)	**6.9671 %**	**0.0024 %**

Splitting the offtake reduces the probability of a revenue loss of half or more by **61.98 %** and the probability of total loss by a factor of about 2,957 — with **no change whatever in expected loss**. On the guarantee: expected loss falls from 1,678,031 to **571,726**, a reduction of **1,106,304**, against a fee whose total is 1,056,745 and whose present value at 6 % is **805,901** — an expected-loss gain of only **300,403**. But the coverage step from 1.30 × to 1.20 × raises debt capacity from 41,171,123 to **44,602,050**, a gain of **3,430,927**, or **2,625,026** net of the fee's present value. 4. **Interpretation. Two structures with identical expected loss are not equally creditworthy, and that difference is the whole subject of concentration risk.** The four-way structure is far more likely to suffer a small loss — a 22.4 % chance of one default against a 6.97 % chance of any default at all in the single-offtaker case — and far less likely to suffer a fatal one. Since a project can absorb a quarter of its revenue disappearing for a period but not all of it, the four-way structure is materially better credit while reporting the same expected loss. **Report the loss distribution, or at minimum the probability of crossing the thresholds the structure cannot survive, and never let expected loss stand alone.** Kestrel's largest revenue risk is not the offtaker's credit quality — 0.60 % a year is respectable — it is that the offtaker is one entity, and diversification is unavailable because a single regional authority is the only buyer of the water. Where concentration cannot be diversified away it must be **structured** against: guarantees, letters of credit, escrow, step-in rights, termination compensation that repays debt, or lower gearing.

And credit enhancement is bought for the coverage it unlocks, not the expected loss it removes. On expected-loss grounds the guarantee barely earns its fee — 300,403 of net benefit is inside the noise of the `PD` assumption. On coverage grounds it releases **2,625,026** of equity net of cost, because it changes the case the lender underwrites and therefore the divisor in Domain 10's sizing arithmetic. That is how experienced sponsors evaluate every mitigant: a letter of credit, a parent guarantee, an escrow account and a termination-compensation regime are all priced by asking what they do to the required ratio and the underwritten case, then comparing that with the cost of the equity they release. Expected loss is the wrong metric for the decision, though it remains the right metric for the provision.

7.4.3 Revenue stress testing and the bank case

Definition. Revenue stress testing moves the commercial drivers to levels the structure must survive and reports the outputs the decision turns on — for a financing, coverage and liquidity, not value. **Reverse stress testing** inverts the question, asking **what combination of driver movements reaches a defined failure point** — the form that produces an answer management can monitor.

WORKED EXAMPLE

Worked example 7.4.3 — Kestrel's stress matrix and its reverse.

1. **Setup.** Move tariff (price) and despatch (volume) jointly across the ranges a market adviser considers credible, and report year-one `DSCR`. Then find the volume required to hold the 1.20 × covenant at a 5 % tariff reduction.
2. **Formula.** `CFADS = 0.75R − 0.03V − 1,896,000` with `R = 0.50 × price factor × V`; `DSCR = CFADS ÷ 5,009,635.23`. Reverse: solve `V × (0.375 × price factor − 0.03) = 6,011,562 + 1,896,000`.
3. **Result.** Year-one `DSCR`:

Tariff \ Despatch	−20 %	−10 %	base	+10 %
+5 %	1.0156	1.1899	1.3642	1.5384
base	0.9438	1.1091	**1.2743**	1.4396
−5 %	0.8719	1.0282	1.1845	1.3408
−10 %	0.8001	0.9474	1.0947	1.2420

At a 5 % tariff cut, despatch must **rise 0.9906 %** — to 24,237,739 m³ — to hold 1.20 ×. 4. **Interpretation.** Read the matrix by its boundaries, not its cells. **Only three of the sixteen cells clear the 1.20 × covenant**, and the covenant contour runs diagonally from just below the base cell to the top-right corner — the visual statement that Kestrel has almost no joint tolerance. **Five cells fall below 1.00 ×**, where debt service cannot be paid from operating cash at all and the debt-service reserve (2,504,818, Domain 10 KA 10.3.2) is the difference between a difficult year and a payment default. Two cells tie the matrix to the rest of the book: the base-despatch, −10 % tariff cell is **1.0947**, exactly Domain 6's revenue-down-10 % figure, and extending the tariff row to +10 % reproduces Domain 6's **1.4540**.

The reverse stress is the more useful instrument, and the one usually missing. A 5 % tariff reduction cannot be recovered by any volume response: 0.99 % more despatch is attainable in principle, but Kestrel has no right to more despatch, because volume is the offtaker's choice — so the recovery lever does not exist. That is what a reverse stress test is for. It identifies not just the failure point but **whether the project holds the lever that would avoid it**, and a breakeven reported without asking who controls the driver is arithmetic without decision content.

Three disciplines complete the practice. **Stress drivers, not outputs** — "revenue −10 %" is two different stresses with different elasticities (KA 7.3.2). **Stress in correlated bundles**, because a demand recession arrives with a price response and a delayed indexation catch-up, and Domain 6 (KA 6.4.2) showed that one-at-a-time analysis is additive on this model and therefore says nothing about joint probability. **And report the minimum over the loan life** — this matrix is a year-one snapshot, and Domain 6's Fig 6.4.1 showed the bank-case minimum falling to 1.1851 in year 12 with no revenue stress at all, so every cell here is optimistic as a statement about the loan's worst year.

AI in this KA

Counterparty and stress work divides cleanly. **Machines should**: monitor counterparty credit signals continuously — rating actions, payment behaviour, filings, published statements — and raise exceptions, which is surveillance no human team performs consistently across a portfolio; generate and run large stress grids including the correlated bundles a workshop would never enumerate; and extract guarantee, letter-of-credit and termination-compensation terms into the structured form Toolkit 7.T.3 requires.

They must not: assign a `PD`, `LGD` or recovery assumption that becomes an input to a financing decision without a named professional adopting it, because each embeds a legal and market judgment a model states with unwarranted confidence; conclude whether facts constitute a counterparty default or trigger termination compensation, which is a matter for qualified counsel; or select the bank case. One failure mode deserves naming: asked to assess a concentrated revenue structure, a model readily computes expected loss — that is the formula in its training data — and does not volunteer that expected loss is blind to concentration, which is the entire question. **Ask for the distribution, not the expectation**, and check that the threshold used is one the structure cannot survive rather than an arbitrary confidence level.

KEY TERMS — KA 7.4

Term	Meaning
`PD` / `EAD` / `LGD`	Probability of default over a stated horizon · exposure when it occurs · proportion not recovered.
Expected loss	`PD × EAD × LGD`; linear in exposure, and therefore blind to concentration.
Cumulative `PD`	$1 - (1 - PD)^n$; the horizon matters and single-year `PD` understates a twelve-year exposure.
Concentration risk	The exposure created by dependence on few counterparties; visible in the loss distribution, not in expected loss.
Credit enhancement	Guarantee, letter of credit, escrow or reserve that improves the credit; priced against the coverage it unlocks.
Reverse stress test	Solving for the driver movements that reach a defined failure point, rather than testing given movements.

SAMPLE MCQS — KA 7.4

MCQ 7.4-A `[7.4.1 · Application]` Annual `PD` 0.60 %, exposure 53,522,460, `LGD` 45 %, over a twelve-year loan life. Expected loss is closest to: - A. USD 144,511 - B. USD 1,678,031 - C. USD 3,728,957 - D. USD 24,085,107

Rationale: Twelve-year cumulative `PD` is $1 - 0.994^{12} = 6.9671$ %, so `0.069671 × 53,522,460 × 0.45 = 1,678,031`. A applies the single-year `PD`, understating the twelve-year exposure by a factor of eleven; C omits `LGD`, treating default as total loss; D omits `PD` entirely.

MCQ 7.4-B [7.4.2 · Analysis] A single offtaker taking 100 % of output is replaced by four independent offtakers taking 25 % each, with identical PD and LGD. Expected loss: - A. falls by 75 % - B. is unchanged, while the probability of losing at least half the revenue falls from 6.9671 % to 2.6489 % - C. falls to one quarter of its previous value - D. rises, because there are four counterparties who might default

Rationale: Expected loss is linear in exposure, so splitting it changes nothing; the loss distribution changes profoundly (7.4.2). A and C confuse per-counterparty exposure with total expected loss; D confuses the probability of some default — which does rise, to 25.09 % — with expected loss.

MCQ 7.4-C [7.4.2 · Analysis] A sovereign guarantee reduces expected loss by 1,106,304 at a fee with a present value of 805,901, and would allow the lenders to size at 1.20 × rather than 1.30 ×, raising debt capacity from 41,171,123 to 44,602,050. The correct basis for the decision is: - A. reject it: the expected-loss saving of 300,403 net is immaterial - B. accept it: it releases 3,430,927 of debt capacity, or 2,625,026 net of the fee's present value, which is what credit enhancement is bought for - C. accept it because guarantees always improve bankability - D. the two effects cannot be compared

Rationale: Enhancement is priced against the coverage and case it unlocks, not the provision it reduces (7.4.2). A applies the wrong metric to the decision, though it is the right metric for the provision; C is an unsupported generality; D is false — both effects are in present-value terms.

MCQ 7.4-D [7.4.3 · Analysis] A stress matrix shows that only three of sixteen tariff and despatch combinations clear the 1.20 × covenant, and that a 5 % tariff cut requires despatch 0.99 % above forecast to remain compliant. The most valuable observation for management is: - A. the matrix should be widened until more cells comply - B. the project has almost no joint tolerance, and it does not hold the lever — despatch is the offtaker's choice — that would recover a tariff cut - C. the year-one figures understate the risk, so the matrix should be discarded - D. the covenant should be renegotiated to 1.00 ×

Rationale: A reverse stress test must identify both the failure point and whether the project controls the driver that would avoid it (7.4.3). A is presentational dishonesty; C is half right — year-one figures are optimistic, so the matrix should be extended to the minimum year, not discarded; D mistakes a covenant for the problem.

SELF-CHECK — KA 7.4

1. Why does expected loss understate a concentrated exposure? — It is linear in exposure, so it is identical whether one counterparty or four carry the same total; only the loss distribution shows the difference.
2. On what basis is credit enhancement priced? — On the coverage and underwritten case it unlocks, and therefore the equity it releases — 2,625,026 net for Kestrel's guarantee — not on the expected loss it removes.
3. What must a reverse stress test establish beyond the breakeven? — Whether the project controls the driver that would avoid the failure.

Advanced topics — Domain 7

7.A.1 The collar: a minimum revenue guarantee with revenue sharing

A **minimum revenue guarantee** (`MRG`) is a floor the grantor writes; **revenue sharing** is a ceiling it buys back. Together they form a **collar**, which can be structured to cost the grantor nothing in expectation while transforming bankability — making it the most efficient instrument in this domain.

Take Kestrel's Structure B distribution and add an `MRG` at **10,800,000** (90 % of base revenue) with **100 % revenue sharing above 13,200,000** (110 %). In the low outcome the grantor tops up by 1,200,000; in the high outcome it recovers 1,200,000; in the base outcome nothing happens. The **expected net transfer is exactly nil** and **expected `CFADS` is unchanged at 6,384,000**. What changes is the distribution: `CFADS` outcomes narrow from 4,728,000–8,040,000 to **5,628,000–7,140,000**, `DSCR` outcomes from 0.9438–1.6049 to **1.1234–1.4253**, and debt capacity sized on the low case at 1.30 × rises from 30,491,396 to **36,295,595** — a gain of **5,804,200 for nothing in expectation.**

Three professional points. The collar is **not free to the grantor**, whose exposure has moved from an expectation to a contingency: a 25 % chance of paying 1,200,000 is a real budgetary and accounting matter in its own jurisdiction and framework, and it is why guarantees of this kind are increasingly recognised as contingent liabilities rather than treated as costless policy. It is **not free to equity** either: the shared upside is the shareholders', and 5,804,200 of extra debt is 5,804,200 of leverage risk. And **it does not fix compliance** — the collared low case at 1.1234 still breaches both the 1.20 × covenant and the 1.15 × lock-up, so a floor set at the lock-up level is worth far more than a floor at a round percentage of base revenue, and costs the grantor only slightly more.

7.A.2 The merchant tail and the contracted horizon

The tenor a project can support is set by its **contracted horizon**, not its asset life. Suppose Kestrel's offtake ran eight years rather than twenty-five, with output thereafter sold merchant at 70 % of the contracted tariff: merchant-year revenue is 8,400,000, `CFADS` **3,684,000** and `DSCR` on the existing instalment **0.7354** — the loan cannot be paid in those years. Three structural responses, each with a computed price:

- **Shorten the tenor to the contracted horizon.** Eight years at 1.30 × on contracted `CFADS`, with `AF(0.06, 8) = 6.209794`, supports **30,494,864** — 10,676,259 less than the twelve-year contracted structure, and it creates refinancing risk at year eight.
- **Keep twelve years with level service.** The weakest period governs: 3,684,000 ÷ 1.30 = 2,833,846 of service, supporting only **23,758,524**. Level service against a cliff is the worst of the three.
- **Sculpt, with a higher target in the merchant years.** At 1.30 × through the contracted years and 1.80 × through the merchant years, sculpted capacity is **34,944,420** — the best answer, and the reason sculpting (Domain 10, KA 10.1.3) is close to mandatory where a contract cliff sits inside the tenor.

The leadership point is that the second option, which looks like the conservative one, destroys **11,185,896** of capacity by pricing every year at the worst year's rate. And all three answers depend on a merchant price forecast for years nine to twelve — a market forecast embedded in a debt structure, which is Domain 10's discipline on refinancing assumptions applied to revenue.

7.A.3 The reviewer's revenue eye

Invariants to test on any revenue model or commercial structure:

- Base-case revenue reconciles to the tariff formula applied to base-case volume, clause by clause, including every deduction, cap and multiplier.
- The capacity or availability charge covers fixed cash costs, debt service and return; the variable charge covers variable costs at contracted efficiency and no more (7.1.2).
- Expected revenue and expected `CFADS` are reported **with** the low case and the coverage each delivers; a case labelled "expected" never appears as a sizing basis (7.1.1).
- Every escalating line carries an index, a lag and an **indexed share**, and the model's effective compound rate reconciles to first-year and final-year values (7.3.1).
- Revenue and cost effective escalation rates are compared, and the margin at loan maturity and at concession end are both reported (7.3.1).
- `CFADS` elasticity is stated separately for price and for volume, with its case label and base value (7.3.2).
- Every covenant is expressed as a tolerance in each driver the business manages, and the smallest tolerance is identified (7.3.3).
- Counterparty exposure is stated as the present value of the stream at risk, on a named horizon, with cumulative rather than single-year `PD` (7.4.1).
- Revenue concentration is reported as a distribution or as the probability of crossing the thresholds the structure cannot survive — never as expected loss alone (7.4.2).
- Contracted revenue tenor is compared with loan tenor, and any uncontracted years inside the tenor are separately sized and separately priced (7.A.2).
- Stress is applied to drivers, in correlated bundles, and reported on the **minimum** period over the loan life (7.4.3).

Industry variations — Domain 7

- **Contracted power and renewables.** Two-part tariffs and capacity payments dominate; the specific difference is **curtailment and deemed generation** — whether the offtaker pays for output it instructs the plant not to produce, a drafting question worth several points of availability. Under a contract-for-differences the project sells merchant and settles the difference against a strike: economically contracted, operationally merchant, and modelled wrongly whenever the two are conflated.
- **Merchant power and commodities.** There is no contracted case, so the **price deck is the commercial structure**: lenders size on a conservative deck, require coverage well above 1.5 ×, shorten tenors and often demand a hedge over a defined share of output. The specific difference is that the hedge is itself a counterparty exposure, so KA 7.4 applies to the mitigant as well as to the revenue.
- **Transport concessions.** Patronage risk with a ramp; the specific difference is that **the project often controls price but faces elasticity**, so a toll increase to recover volume may reduce revenue. Operating leverage is extreme — Domain 10's Case study B implies an elasticity of 1.4931 — so tolerance is thin at any coverage ratio, and ramp reserves and sculpting are standard rather than optional.
- **Water and regulated utilities.** Availability or take-or-pay with a public counterparty; the specific difference is the **regulatory reset**, a scheduled discontinuity no escalation formula

describes. Covenant testing and reserve sizing must straddle resets, and the indexation architecture applies only between them.

- **Digital infrastructure.** Contracted revenue with short tenors relative to asset life; the specific difference is that **re-contracting risk replaces volume risk**, so `WARCT` against tenor is the binding metric (7.2.3), tenant concentration is usually severe, and required coverage is driven by re-letting assumptions and tenant credit rather than demand forecasts.
- **Social infrastructure PPP.** Pure availability payments from a public budget; the specific difference is that **the deduction and performance regime is the entire revenue risk**. With no demand exposure at all, diligence and management attention belong on the performance-measurement system, the multiplier, the cumulative-deduction cap and the cure regime — which KA 7.1.3 showed can move a covenant breakpoint by 1.584 points of availability.

Case study — Domain 7: the concession that was worth less than it looked (water)

Situation. Kestrel's tariff negotiation reached indexation last, as it usually does. The offtaker's position was **80 % of the tariff indexed to CPI**, assumed at 2.5 %, with 20 % fixed for the 25-year concession. The sponsors' commercial director, uncomfortable with an unindexed slice, asked for **full indexation**; the offtaker agreed — at **CPI less 50 basis points**, presented as a fair exchange. The finance director asked for the calculation before the term sheet was initialled.

What happened. The exchange was value-destroying. Present value of revenue over 25 years at 8 %: 80 % at 2.5 % gives **152,913,886**; 100 % at 2.0 % gives **152,088,430** — the "full indexation" offer was worth **825,456 less**, and 166,192 less over the twelve loan years at 6 %. The mechanism is not obvious, and that is the point: the unindexed 20 % compounds at nothing, but the indexed 80 % compounds at the higher rate, so by year 25 the partly indexed tariff reaches 19,763,769 against the fully indexed one's 19,301,247. Minimum `DSCR` over the loan life was 1.2743 in year one under both and year-12 coverage was 1.4253 against 1.4140, so **no coverage test would have detected the loss** — a covenant-compliant structure can be materially worse than the alternative and nothing in the coverage model will say so.

The second finding was larger. Decomposing the architecture (Worked example 7.3.1) showed the year-25 margin falling from 62.50 % to **59.23 %**, and the **effective** rate gap at 35.7 basis points rather than the headline 70 — half the apparent mismatch already neutralised by the 30 % fixed portion of the O&M contract. The lever nobody had pulled was on the cost side: the O&M escalation index, negotiated separately by the technical team, ran at 3.2 % against the tariff's 2.5 % for no reason beyond the contractor having proposed it.

How it resolved. Two changes, negotiated together. The tariff moved to **90 % indexed at full CPI**, and the O&M contract was re-tendered with its escalating 70 % **matched to the tariff's index**. Present value of `EBITDA` over 25 years at 8 % rose from **93,938,399** to **99,836,527** — a gain of **5,898,128**, of which **3,102,071** came from the extra ten points of tariff indexation and **2,796,057** from index matching, the two effects being exactly additive. The year-25 margin rose to **66.01 %**, above the year-1 62.50 %, because matched indices against a partly fixed cost base produce margin expansion. The offtaker took a tighter availability regime in exchange, which KA 7.1.3 priced separately.

What the domain teaches here. Indexation is an **architecture**, not a rate, and it must be valued rather than preferred. The sponsors' instinct — more indexation is better — was wrong by 825,456 as offered, and the valuable change was on a line nobody had connected to the revenue negotiation.

Two disciplines follow: compute the present value of every indexation proposal before responding to it, and negotiate revenue and cost indices **in the same room**, because the exposure is the gap between them and neither team can see it alone.

Case study B — Domain 7: the tenant they gave a discount to (digital infrastructure)

Situation. Northwind Data Campus (a fictitious SPV) was a 200,000,000 colocation development fully pre-let to a single investment-grade anchor tenant on a ten-year lease: revenue **34,000,000**, fixed cash operating costs **12,000,000**, maintenance capex **1,500,000**, so `CFADS` **20,500,000**, with cash tax nil in the early years under capital allowances. The sponsors regarded one strong tenant as their best credit feature. The lead arranger, pricing a ten-year facility at **6.5 %** (`AF(0.065, 10) = 7.188830`), required **1.60 ×** coverage precisely because 100 % of revenue depended on one counterparty and one lease: 20,500,000 ÷ 1.60 = 12,812,500 of service, supporting **92,106,887** and leaving equity of **107,893,113**, **53.95 %** of capital, against a business plan built on 40 %.

What happened. The sponsors first argued the ratio, which failed for the reason Domain 10's Case study A established — a coverage requirement is an output of the credit, not a negotiating variable in isolation. Their second response worked. They asked the anchor, whose own growth forecast had softened, to release **25 % of the capacity**, and re-let it to three independent tenants at an average **6 % below** the anchor rate. Revenue fell to **33,490,000**, down 510,000 or **1.50 %**; operating costs rose 400,000 on the additional billing and service load; `CFADS` fell to **19,590,000**, down **910,000**.

How it resolved. With four tenants and none above 75 % of revenue, the arranger reduced the requirement to **1.35 ×**: debt service of 14,511,111 supporting **104,317,914** — **12,211,027** more debt on **910,000 less** `CFADS`. Equity fell to **95,682,086**, from 53.95 % to **47.84 %** of capital. The arranger then imposed a condition the sponsors had not anticipated and that was entirely correct: because the three new leases ran three years against the anchor's ten, `WARCT` fell from **10.00 to 8.25 years** against a ten-year loan, so a **2,000,000 re-letting reserve** was required, reducing the net capacity gain to **10,211,027**.

What the domain teaches here. Concentration, not expected value, priced the debt. The sponsors gave up 910,000 a year of `CFADS` — **4.44 %** — and received twelve million of debt, because what the arranger charged 0.25 × of coverage for was not the level of the cash flow but the fact that all of it came from one signature (KA 7.4.2). The second lesson is the reserve: reducing concentration by adding shorter leases traded counterparty risk for **re-contracting risk**, and a lender who accepts the first while ignoring the second has not improved its position. Every diversification must be assessed on both dimensions — how many payers, and for how long — which is the `WARCT`-against-tenor test of KA 7.2.3.

Executive perspective — Domain 7

What a project finance director cannot delegate in this domain:

- **The choice of revenue structure, priced.** The decision between an availability payment and a volume tariff is worth 10,679,727 of debt capacity on Kestrel's numbers (7.1.1). It is a capital-structure decision wearing commercial clothes, and it cannot be settled by the commercial team alone.

- **The deduction and performance regime.** Multipliers, caps, cure periods and outage allowances set the operational tolerance the whole project then lives inside — 2.91 availability points for Kestrel, of which a point and a half is the negotiated multiplier (7.1.3).
- **The indexation architecture, on both sides.** Indexed shares and indices for revenue and cost, negotiated together, with effective rates computed. Case study A's 5,898,128 of present value came from two terms nobody had connected.
- **The case that debt is sized on.** The bank case is a negotiating position, and conceding it unopposed concedes more than the coverage ratio does (7.2.1).
- **The concentration statement.** Not the counterparty's rating and not the expected loss, but the probability of losing more revenue than the structure can survive, and what has been structured against it (7.4.2).
- **The tolerance in each team's own unit.** 4.14 % on tariff, 4.50 % on despatch, 2.91 points on availability, 5.83 % on `CFADS` — the same covenant, translated into four dashboards, with the smallest one owned (7.3.3).

Calculation exercises — Domain 7

Exercise 7.1 A capacity charge of 20,000,000 a year carries a 97 % availability guarantee and a deduction equal to the charge multiplied by the availability shortfall and by a 1.25 × multiplier. Availability outturns at 94 %. Compute the deduction and the revenue received. Solution. Shortfall `0.97 − 0.94 = 0.03`; deduction `20,000,000 × 1.25 × 0.03 =` **750,000**, which is **3.75 %** of the charge; revenue **19,250,000**. Common error: omitting the multiplier (600,000), or applying the multiplier to total unavailability of 6 % rather than to the 3 % shortfall against the guarantee (1,500,000) — the first understates exposure by a fifth, the second doubles it.

Exercise 7.2 A project earns 0.80 per unit with variable cost 0.10 per unit and fixed cash cost 4,000,000; there is no tax or working-capital movement. Volume is 18,000,000 units with probability 0.60 and 10,500,000 with probability 0.40. Debt service is 6,000,000 a year and the loan runs ten years at 7 % (`AF(0.07, 10) = 7.023582`). Compute `CFADS` and `DSCR` in each outcome, the expected values, and debt capacity at a 1.25 × target sized (a) on expected `CFADS` and (b) on the low case. Solution. High: revenue 14,400,000, `CFADS 14,400,000 − 4,000,000 − 1,800,000 =` **8,600,000**, `DSCR` **1.4333**. Low: revenue 8,400,000, `CFADS` **3,350,000**, `DSCR` **0.5583**. Expected volume 15,000,000; expected `CFADS` **6,500,000**; expected `DSCR` **1.0833**. Capacity on expected `CFADS`: `6,500,000/1.25 × 7.023582 =` **36,522,624**; on the low case: `3,350,000/1.25 × 7.023582 =` **18,823,199** — a difference of **17,699,425**. Common error: sizing on the expected case and reporting the expected `DSCR` of 1.0833 as the project's coverage; the low outcome does not pay debt service at all, and a 40 % probability of that is not a rounding matter.

Exercise 7.3 A concession's revenue is 30,000,000 with 90 % indexed at 2.2 %; its operating cost is 18,000,000 with 60 % indexed at 3.5 %. Compute the `EBITDA` margin in year 1 and year 20, and the effective compound rates. Solution. Year 1: `EBITDA` **12,000,000**, margin **40.00 %**. Year 20: revenue $30{,}000{,}000 \times (0.90 \times 1.022^{19} + 0.10) =$ **43,825,432**; cost $18{,}000{,}000 \times 0.60 \times 1.035^{19} + 7{,}200{,}000 =$ **27,963,014**; `EBITDA` **15,862,417**, margin **36.19 %** — a drift of **3.81 points**. Effective rates: revenue **2.015 %**, cost **2.346 %**. Common error: comparing the headline 2.2 % and 3.5 % and concluding a 130-basis-point exposure; the effective gap is 33.1 basis points, because both sides are only partly indexed.

> **Exercise 7.4** Revenue 50,000,000; variable costs are 30 % of revenue; fixed cash costs 18,000,000. Compute the degree of operating leverage, and the volume fall that takes a 1.40 × base coverage ratio to a 1.25 × covenant. Solution. Contribution `50,000,000 × 0.70 =` **35,000,000**; `EBITDA` **17,000,000**; `DOL 35,000,000/17,000,000 =` **2.0588**. Required earnings fall `1 − 1.25/1.40 =` **10.7143 %**; volume fall `10.7143 % ÷ 2.0588 =` **5.2041 %**. Verification: volume down 5.2041 % gives `EBITDA` of **15,178,571**, which is **89.29 %** of base — and `1.40 × 0.8929 = 1.25`. Common error: reading the coverage gap as the volume tolerance (a 10.71 % volume fall would take coverage to 1.09 ×, well below the covenant); operating leverage means the tolerance is always smaller than the earnings tolerance.

> **Exercise 7.5** A single offtaker has an annual `PD` of 1.0 %. Exposure over a ten-year loan is 80,000,000 and `LGD` is 50 %. Compute the expected loss, then compute the probability of losing at least half the revenue if the same output were sold instead to **two** independent offtakers of 50 % each on identical terms. Comment. Solution. Ten-year cumulative `PD` $1 - 0.99^{10} =$ **9.5618 %**; expected loss `0.095618 × 80,000,000 × 0.50 =` **3,824,717**. With two offtakers, losing at least half the revenue requires **at least one** default: $1 - (1 - 0.095618)^2 =$ **18.2093 %**, against **9.5618 %** with a single offtaker — the probability has almost **doubled**, while the probability of total loss falls from 9.5618 % to $0.095618^2 =$ **0.9143 %**. Comment: splitting a revenue base into parcels at or above the survival threshold does not reduce the risk of crossing it; diversification only helps when each parcel is **strictly smaller** than the loss the structure can absorb. Common error: assuming that any diversification reduces concentration risk — the four-way split of Worked example 7.4.2 reduces the probability of a ≥ 50 % loss to 2.6489 %, but a two-way split increases it.

Practitioner's toolkit — Domain 7

Adoption-ready artefacts; adapt headings to your organisation, then keep them stable.

Toolkit 7.T.1 — Revenue mechanism specification

One row per component of the payment mechanism, populated from the agreement and signed off before the model is built. Columns: component (capacity charge · variable charge · take-or-pay floor · band · deduction · bonus · merchant) · **clause reference** · the formula in full, with every rate, threshold, multiplier and cap · the units and the measurement source · the billing period and payment terms · indexation reference (cross-refer to 7.T.2) · the model line that implements it · the person who reconciled model to clause, and the date. Rule: **no revenue figure is reportable until a historical or hypothetical period has been computed from the clause text by hand and agreed with the model** (7.1.3). The deduction row carries its multiplier, its cap and its cure regime explicitly, because a summary that omits them understates exposure by a third.

Toolkit 7.T.2 — Indexation mismatch map

Domain 3's escalation register (Toolkit 3.T.3) catalogues each line's index mechanics. This artefact pairs them. One row per **cost** line: amount · indexed share · index and expected rate · **effective rate** · the revenue line that funds it · that revenue line's indexed share, index and effective rate · **the gap in basis points** · the annual and present-value exposure the gap represents over the concession, and separately over the loan life. Footer rows: total unindexed revenue exposure, total unindexed cost protection, net effective gap, and margin at loan maturity and at concession end. Rule: **negotiate on effective rates, never on headline indices**, and review the map whenever either side of a pair is re-tendered — Case study A's second finding came from an O&M re-tender nobody had connected to the tariff.

Toolkit 7.T.3 — Revenue stress and counterparty pack

Section A, **counterparty**: each payer's share of revenue · rating or internal grade with source and date · annual and horizon-cumulative `PD` · exposure defined as the present value of the stream at risk, on a stated horizon · `LGD` with its recovery assumption traced to the security and termination provisions · expected loss · enhancement in place, its cost, and **the coverage step it purchased**. Section B, **concentration**: revenue share by payer · the loss the structure can absorb · the probability of exceeding it · what has been structured against it. Section C, **stress**: the driver list with elasticities and case labels · the two-way matrix on the drivers, not on revenue · the correlated bundles with their rationale · the **minimum** coverage over the loan life in each · the reverse stress result for every covenant, expressed as a tolerance in each driver, with a note on **who controls that driver**. Front page: the smallest tolerance, its driver, its owner, and the date it was last recomputed.

Exam preparation — Domain 7

What is assessed. Placing a structure on the revenue-risk spectrum and stating its coverage and capacity consequence; computing revenue under two-part, banded, floored and deduction-based mechanisms; demonstrating and quantifying the gap between expected value and bankable value; computing effective escalation rates and margin drift from indexed shares; computing `DOL` and `CFADS` elasticities and converting a coverage threshold into a driver tolerance; computing expected loss with a cumulative `PD` and explaining its blindness to concentration; and pricing credit enhancement against coverage.

The calculations to do under time pressure. Revenue from a banded tariff at a given volume. Deduction from a shortfall, a multiplier and a charge. `CFADS` and `DSCR` from the domain's bridge. Expected value across a discrete distribution, and debt capacity from a low case (`CFADS` ÷ target × `AF`). Effective compound rate from an indexed share, an index and a horizon. `DOL` as contribution ÷ `EBITDA`. A driver tolerance as the `CFADS` tolerance divided by that driver's elasticity. Cumulative `PD` as $1 - (1 - PD)^n$, and expected loss as the product of three factors.

The traps. - Treating expected `CFADS` or expected `DSCR` as a sizing basis (Worked example 7.1.1; Exercise 7.2) — the single most consequential error in the domain. - Omitting the deduction multiplier, or applying it to total unavailability rather than to the shortfall against the guarantee (MCQ 7.1-B; Exercise 7.1). - Recovering fixed costs and debt service through a purely volume-linked charge (7.1.2). - Using the headline index instead of the effective rate on a partly indexed line (MCQ 7.3-A; Exercise 7.3) — and its mirror, comparing headline indices across revenue and cost and overstating the gap twofold (Case study A). - Assuming more indexation is better without computing the present value (Case study A). - Treating "revenue −10 %" as one stress when price and volume have different elasticities (7.3.2, 7.4.3). - Quoting a `CFADS` tolerance to a team that manages volume or availability (7.3.3). - Using a single-year `PD` for a multi-year exposure (MCQ 7.4-A), or omitting `LGD`. - Letting expected loss stand alone on a concentrated revenue base (7.4.2; MCQ 7.4-B). - Assuming that any diversification reduces concentration risk, when parcels at or above the survival threshold make matters worse (Exercise 7.5). - Reporting a year-one stress matrix as though it described the loan's worst year (7.4.3). - Sizing a tenor on asset life rather than on the contracted horizon (7.A.2).

How the domain connects. Domain 5 (KA 5.3.2) named the five components of the revenue bankability test and deferred the taxonomy here; Domain 6 built the model this domain feeds and supplied the elasticity and breakeven machinery it extends; Domain 8 escalates the cost side with the same indexation arithmetic; Domain 9 prices the equity that a weaker revenue structure makes necessary; Domain 10 divides this domain's CFADS by the coverage this domain's structure earns; Domain 11 places demand, price and counterparty risk in the allocation framework; Domain 12 drafts the mechanisms described here; and Domain 15 operates the covenant tests whose tolerances KA 7.3.3 computed.

Domain 7 summary

Revenue structure, not revenue level, decides how much debt a project can carry. Kestrel's two offers — a 12,000,000 availability payment and a 0.50 per m³ volume tariff — have **identical expected CFADS of 6,384,000** and identical expected DSCR of **1.2743**, and differ by **10,679,727 of debt capacity**, 25.94 %, because a lender underwrites the low case (CFADS 4,728,000, DSCR **0.9438** — debt service unpaid) and raises the required ratio for dispersion as well. An availability structure does not remove volatility, it converts demand risk into operational risk: Kestrel breaches its 1.20 × covenant at **92.086 %** availability against a 95 % guarantee, **2.91 points** of headroom, of which about a point and a half is the negotiated 1.5 × deduction multiplier. Banded, floored and shared mechanisms are risk-transfer instruments priced in tariff form — a volume band costing 2,515,153 of present value bought 3,482,520 of capacity, and a value-neutral collar with an MRG at 90 % and full sharing above 110 % bought **5,804,200 for nothing in expectation** — while service-revenue structures fail on re-contracting rather than volume: at 93 % net revenue retention against a fixed cost base, Halyard Connect's run-off case supports **20,271,839**, half the 40,000,000 sought, and its WARCT of 3.30 years covers 47 % of the tenor. Escalation is an architecture: 80 % of Kestrel's tariff at 2.5 % against 70 % of its cost at 3.2 % produces effective rates of **2.101 %** and **2.457 %** — a 35.7-basis-point gap, half the headline — and **3.27 margin points** of drift by year 25, all of it landing after the lender has gone; the unindexed tariff slice costs 3.65 points and the fixed cost slice saves 7.72, which is why Case study A's most valuable term was in the O&M contract. Operating leverage explains why a 12 % patronage miss became Domain 10's 17.9 % cash miss — an elasticity of **1.4931** — and generalises: at identical 1.30 × coverage and an identical 1.20 × covenant, demand tolerance ranges from **6.15 % to 1.54 %** on cost structure alone. Kestrel's own tolerance is **4.14 %** on tariff, **4.50 %** on despatch (1,079,530 m³) and **5.83 %** on CFADS, and quoting the last to a team that manages the second overstates their room by 30 %. Counterparty quality bounds all of it: an annual PD of 0.60 % over twelve years is **6.9671 %** cumulative, giving an expected loss of **1,678,031** on exposure of **53,522,460** and a credit charge worth **0.0400** of DSCR — but expected loss is identical whether one offtaker or four carry the revenue, while the probability of losing half of it falls from **6.9671 % to 2.6489 %**, so concentration must be reported as a distribution and never as an expectation. And credit enhancement is bought for the coverage it unlocks: Kestrel's guarantee saves 300,403 of net expected loss and releases **2,625,026** of equity. Domain 8 escalates the cost side, Domain 9 prices the equity a weak revenue structure demands, and Domain 10 divides this domain's CFADS by the coverage this domain's structure has earned.

08

DOMAIN 8

Cost, Schedule and Contingency Integration (quantitative — the project-controls bridge)

Group: Structuring and modelling (Domain 4 of 5 in Part Two). **Target:** ~76 pages. **Binds to:** the PCI Book Pattern Specification and the shared registries (`docs/books/registries/`). This is the **bridge domain between the two books**: it consumes PML-AI Domain 7's earned-value machinery (`BAC`, `CPI`, `SPI`, the `EAC` family) and PML-AI Domain 8's `EMV` and confidence-level arithmetic, and converts both into the two questions a financing asks that a project control account never does — is the remaining funding sufficient, and what does the coverage ratio do? It uses Domain 3's compounding and `AF(r, n)`, Domain 4's `EAV`, Domain 5's completion-risk and delay-damages structure, Domain 6's sources-and-uses and capitalised-interest arithmetic, and Domain 10's `DSCR` machinery. British English; USD (+SAR where useful, indicative `USD 1 ≈ SAR 3.75`). Tax, accounting and legal treatments described here are **illustrative and jurisdiction-specific**; none is presented as universal. In particular, whether a delay-damages rate, cap or stepped structure, a contingency-recalculation clause or a handback obligation is enforceable as drafted is a matter for qualified counsel in the governing jurisdiction, and whether a cost is capitalised into the depreciable base is a matter for the sponsor's own auditors under the applicable framework. Nothing in this domain is legal, tax or accounting advice. Kestrel Water SPC, Project Auriga and both case studies are illustrative constructs, not accounts of identifiable projects or organisations.

Why this domain exists

Domain 6 built a model in which the balancing line was contingency, and reported it: **USD 3,645,403**, 7.59 % of the EPC price, "inside the band a lender would expect" (KA 6.2.1). It did not say where that band comes from, whether 7.59 % is generous or reckless, or what would happen if the number were wrong. Domain 5 priced a slip in the commercial operations date but only at COD, where remaining spend is zero. Neither domain touched the fact that a project under construction produces a monthly cost and schedule report — earned value, forecasts, variances — that a lender receives, reads, and acts on differently from the project controller who wrote it.

This domain closes all three gaps, and its central claim is a single sentence: **a contingency percentage is meaningless without the estimate class it was struck against and the basis it was sized on.** Everything else here is that claim worked out. A cost estimate carries an accuracy range that narrows as scope definition matures, and the contingency implied by that range at a stated confidence is computable (KA 8.1). A schedule converts a cost estimate into a spend profile, and the shape of that profile prices two things in opposite directions — capitalised interest and escalation — with a breakeven between them (KA 8.2). Contingency sized from a quantified risk register states what confidence it buys; contingency sized as a percentage of base cost does not, and is wrong in ways this domain computes exactly (KA 8.3). And when the build slips, the cost of a month is three components, not one, and its consequence lands on a coverage ratio that will be tested for the next twelve years (KA 8.4).

The professional discipline the domain teaches is the discipline of the interface. A project controller and a lender's technical adviser read the same monthly report and extract different facts from it. The controller asks what the project will cost; the lender asks whether the money still in the facility will finish it. Those are not the same question, and a finance leader who cannot translate between them is dependent on whoever can.

Learning objectives. After this domain a candidate can: separate the four cost families a financing must fund and place each in the sources-and-uses statement or the operating model; state an estimate's accuracy class and derive the contingency its range implies at a nominated confidence; explain why a fixed-price wrap changes the estimate class and therefore the defensible contingency; compute a whole-life maintenance charge and size the reserve that converts it from a covenant cliff into a distribution reduction; build a schedule-driven spend profile and compute the capitalised interest it generates using the area rule; compute the escalation the same profile generates and find the breakeven escalation rate at which the two effects cancel; size contingency from a quantified register to a stated confidence level and reconcile it against the estimate range; demonstrate in figures why a percentage-of-base contingency fails; convert an earned-value forecast into a cost to complete and run the funds sufficiency test a lender runs; price a month of construction slip as escalation plus extra interest plus deferred revenue and compute its permanent `DSCR` consequence; and govern AI-assisted estimating, simulation and forecasting.

The master thread. Kestrel Water SPC continues. Its **USD 60,000,000** envelope is funded 70/30 — **42,000,000** senior debt at **6.0 %** over 12 years (annual instalment **5,009,635.23**) and **18,000,000** equity — against an EPC contract price of **48,000,000**, owner's costs and land of 3,600,000, capitalised development costs of 1,800,000, fees of 840,000, capitalised interest of **2,114,597** and the balancing contingency of **3,645,403** (Domain 6, KA 6.2.1). Construction runs **eight quarters** on the certified spend profile **6, 9, 13, 16, 17, 15, 13, 11 per cent**, with cumulative debt drawn reaching 42,000,000 exactly at COD. Operating `CFADS` is **6,384,000** and the base-case `DSCR` **1.2743** against a 1.20× covenant, giving annual headroom of **372,438** (Domain 10, KA 10.2.1). Construction cost escalation is assumed at **3.6 % per annum** on unwrapped scope. From PML-AI: **Project Auriga**, `BAC` **4,000,000**, `CPI` **0.905660** and `SPI` **0.923077** at the week-13 data date, with the `EAC` family **4,200,000 / 4,416,667 / 4,608,056** (PML-AI D7, KA 7.3.3) and a sanction-date risk register whose mean exposure is **278,000**, standard deviation **252,642** and P80 **490,624** (PML-AI D8, KA 8.2.4). This domain does not re-derive any of those figures. It uses them.

KNOWLEDGE AREA 8.1

Development, capital, operating and lifecycle cost; estimate classes

Topics: 8.1.1 the four cost families and where each lands · 8.1.2 estimate classes and the contingency a range implies · 8.1.3 lifecycle cost, major maintenance and the reserve that smooths it.

8.1.1 The four cost families and where each lands

Definition. A financing must fund four distinct families of cost, and each enters the structure in a different place, on a different basis, under different control.

Development cost is everything spent between concept and financial close: studies, surveys, legal and advisory fees, permit applications, land options, the sponsors' own staff time. It is spent at risk, before any facility exists, and its treatment at close is a negotiation — Kestrel capitalised 1,800,000 of it into the funding envelope (Domain 6), which means the equity that funded it is reimbursed and — subject to the accounting and tax framework that applies, which is a question for the sponsor's own auditors (Domain 2) — the amount enters the depreciable base. Development cost that is not accepted into the envelope is sunk equity, and Domain 5 (KA 5.1.2) priced it per close rather than per deal for exactly that reason.

Capital cost is the asset: the EPC contract price, owner-supplied equipment, owner's costs, land, connection charges, insurance during construction. It is what the estimate classes of 8.1.2 measure, and it is the base against which contingency percentages are quoted.

Operating cost is the recurring cash cost of running the asset — labour, energy, chemicals, routine maintenance, insurance, administration, the operator's fee. It never appears in the sources-and-uses statement; it appears as a deduction inside `CFADS`, which makes it a coverage driver rather than a funding requirement. The distinction has teeth: an error of 500,000 per year in operating cost is an error of 500,000 per year in `CFADS`, which at Kestrel's 1.20× covenant is 1.34 times the entire annual headroom of 372,438.

Lifecycle cost is the periodic, non-routine expenditure that keeps the asset capable of performing for its whole concession — membrane replacement, major overhauls, control-system refresh, and at the end handback or decommissioning. It is lumpy, it falls in specific years, and it is the family most often omitted from a first-pass model because it is invisible in year one (KA 8.1.3).

The professional point. These four families are not four lines of one budget; they are four different financial objects. Development cost is at-risk equity that may or may not be reimbursed. Capital cost is funded, drawn and capitalised. Operating cost reduces `CFADS` every period. Lifecycle cost creates a reserve obligation. A cost report that presents them as one "project cost" figure has destroyed the information a financing needs.

8.1.2 Estimate classes and the contingency a range implies

Definition. An **estimate class** records how well the scope was defined when the estimate was made, and carries with it a stated **accuracy range** — the band within which the estimate's author expects the outturn to fall. The range is not a confidence interval in the statistical sense; it is an expression of definitional maturity, and it narrows as engineering, site investigation, procurement and contracting progress. Professional estimating bodies and large owner organisations publish classification frameworks of their own; the principle is common practice, the specific bands are always organisation-specific, and the table below is the PCI illustrative ladder used throughout this book — it reproduces no other body's classification and should not be read as one.

Stage	Scope definition	Typical basis	Accuracy range	Implied contingency to the upper bound on a 48,000,000 base
A — screening	Concept only; capacity and location	Capacity-factored from analogues	−30 % / +50 %	24,000,000
B — concept	Process selected; block layout	Factored equipment costs	−20 % / +40 %	19,200,000
C — feasibility	Basic engineering; major equipment listed	Semi-detailed, some quotations	−15 % / +30 %	14,400,000
D — definition	Detailed engineering substantially complete	Quotations for most packages	−10 % / +18 %	8,640,000
E — control	Contract awarded; scope fixed	Contract prices, priced bill	−5 % / +8 %	3,840,000

Read the last column before reading anything else. **The same 48,000,000 base estimate implies a contingency of 24,000,000 or 3,840,000 depending only on how well the scope was known when it was priced** — a factor of 6.25. Any conversation about "how much contingency is normal" that has not first established the estimate class is a conversation about nothing.

WORKED EXAMPLE

Worked example 8.1.2 — is Kestrel's 7.59 % contingency defensible?

1. **Setup.** Domain 6's balancing line gave contingency of **USD 3,645,403** on a base of **48,000,000**. Establish the percentage, identify which estimate class that percentage is a provision for, and test whether Kestrel's contracting structure justifies it.
2. **Formula.** Contingency as a percentage of base = contingency ÷ base. Implied class = the ladder row whose upper bound is nearest above that percentage. Coverage of the class band = contingency ÷ (base × upper-bound percentage).
3. **Substitution.** `3,645,403 / 48,000,000`; compare with the upper bounds 8 %, 18 %, 30 %, 40 %, 50 %; `3,645,403 / (48,000,000 × 0.08)`.
4. **Result.** Contingency is **7.59 %** of base. That is a **Stage E** provision — it sits inside the −5 %/+8 % control-estimate band and covers **94.93 %** of it, falling **194,597** short of the full +8 % position of 3,840,000. Against a Stage C feasibility estimate the same money covers only **25.32 %** of the +30 % band; the Stage C provision would be **14,400,000**, or **3.950 times** what is funded.
5. **Interpretation.** The 7.59 % is defensible, and the reason it is defensible is **not** that 7.59 % is a normal number. It is that Kestrel let a **fixed-price, date-certain EPC contract with a full wrap** (Domain 5, KA 5.4.1) before the envelope was fixed, which converted a Stage C estimate into a Stage E position. The wrap is what makes a thin contingency honest: it transfers the base-estimate uncertainty — the quantity growth, the productivity assumption, the unpriced package — to a contractor who has accepted a price. What remains with the owner is a short list of retained risks (KA 8.3), which is why the owner's provision can be small. Reverse the contracting decision and the arithmetic reverses with it. **The same project, procured as six packages against a Stage C estimate with owner-managed interfaces, needs a provision of the order of 14,400,000 rather than 3,645,403 —** and a 60,000,000 envelope does not fund it. The professional caution is therefore that the contingency line and the contracting strategy are one decision taken twice, and the two halves are usually taken by different people in different months. Where a lender sees a thin contingency and a weak wrap together, it is looking at a structure whose funding assumption has already failed; where it sees a thin contingency behind a full wrap, it is looking at a risk transfer it can price. **The breakeven question to ask of any contingency is: which estimate class does this percentage belong to, and did we buy the contract that earns it?**

Common pitfall — the contingency inside the contract price. A contractor's price contains the contractor's own contingency, which the owner cannot see and does not control. Adding an owner's contingency on top is correct — they cover different risks — but describing the total as "the project's contingency" is not, because only one of the two is available to the owner on a draw request (KA 8.4.1). Where a cost report shows a single contingency figure, ask whose it is.

8.1.3 Lifecycle cost, major maintenance and the reserve that smooths it

Definition. Lifecycle cost (also whole-life cost) is the total cost of ownership across the asset's economic life: capital cost, operating cost, periodic major maintenance, and terminal handback or decommissioning obligations. In a financing its distinctive property is timing: it falls in a few specific years, in amounts large relative to a single year's cash, which is exactly the profile a coverage covenant handles worst.

WORKED EXAMPLE

Worked example 8.1.3 — Kestrel's whole-life charge, and the year-seven cliff.

1. **Setup.** Kestrel's 25-year operating life requires **membrane replacement in years 7, 14 and 21** at **USD 3,200,000** each at base-date prices, and **high-lift pump refurbishment in years 12 and 24** at **USD 1,400,000** each. Costs escalate at **3.6 %** per annum; the sponsor's appraisal rate is **8.0 %** (Domain 4). Operating `CFADS` of 6,384,000 is struck **before** major-maintenance funding; annual debt service is **5,009,635.23**. Compute the present value of the lifecycle programme, the level annual charge equivalent to it, and what happens in year seven with and without a funded maintenance reserve.
2. **Formula.** Nominal cost at year `t` = real cost × $(1 + 0.036)^t$; present value = nominal ÷ $(1 + 0.08)^t$ (Domain 3, `DF(t)`); level annual charge = PV ÷ `AF(0.08, 25)` (Domain 4's `EAV`); year-seven `DSCR` without a reserve = (`CFADS` − nominal cost) ÷ debt service (Domain 10).
3. **Substitution.**

Year	Item	Real (USD)	Nominal (USD)	Present value (USD)
7	Membrane replacement	3,200,000	4,098,909	2,391,674
12	Pump refurbishment	1,400,000	2,140,154	849,885
14	Membrane replacement	3,200,000	5,250,329	1,787,533
21	Membrane replacement	3,200,000	6,725,194	1,335,999
24	Pump refurbishment	1,400,000	3,271,615	515,931
	Total	**12,400,000**	**21,486,202**	**6,881,021**

`AF(0.08, 25) = 10.674776`; level charge `= 6,881,021 / 10.674776`. 4. **Result.** Present value of the lifecycle programme **USD 6,881,021**; **level annual charge USD 644,606**, which is **10.10 %** of `CFADS`. Without a reserve, year seven's `CFADS` falls to `6,384,000 − 4,098,909 =` **2,285,091** and the `DSCR` to **0.4561**. With the charge funded into a maintenance reserve account below `CFADS` and above distributions (Domain 10, KA 10.3.3), `CFADS` and therefore the `DSCR` stay at **1.2743** in every year, and the cost lands on distributions instead. 5. **Interpretation.** A `DSCR` of 0.4561 is not a covenant breach; it is a **payment default** — the project cannot pay debt service from cash in the year it replaces its membranes, having been entirely healthy in the six years before and the six years after. That is the whole case for a maintenance reserve in one number, and it explains why lenders commonly require one on an asset with a material overhaul cycle and why they generally require the reserve to be **forward-looking**: funded against the next overhaul, not accumulated at a policy percentage. Three professional qualifications belong beside the arithmetic. **The level charge is not the reserve deposit schedule.** 644,606 per year is the economically equivalent annual cost; the actual deposit schedule must have the money in the account before year seven, which with only six years of deposits means a higher early contribution or an opening balance funded at close — and the difference is a real funding requirement that belongs in the sources and uses. **The escalation assumption is load-bearing.** At 3.6 % the year-21 membrane costs 6,725,194 against 3,200,000 today; assume 2.0 % instead and the whole-life PV falls to a materially smaller number, which is why the lifecycle escalation rate must be the same rate used for operating cost and stated once in the assumption register (Domain 6, KA 6.1). **And the reserve converts a covenant event into a distribution event.** That is the trade: 644,606 per year of distributions forgone buys the removal of a payment default. Sponsors who resist it are choosing to hold a 0.4561 `DSCR` in year seven, which no lender will accept and no board should.

Handback and decommissioning. Where a concession requires the asset to be returned in a specified condition, the handback obligation is a lifecycle cost with a legal deadline, usually secured by a dedicated reserve funded in the final years. Where the asset must instead be removed, the decommissioning provision is both an accounting matter (Domain 2, KA 2.3.4) and a cash matter, and its cash timing sits after the loan matures — which is why it reduces the tail that Domain 10's `PLCR` of 1.9431 measures without appearing in any coverage ratio the lender tests. A reviewer's habit worth forming: read the `PLCR` and then ask what the project owes at the end of the life the `PLCR` counts.

AI in this KA

Estimate benchmarking is genuinely good machine work — comparing a new estimate's unit rates, quantities and factored allowances against a library of prior estimates and outturns, and flagging lines outside the historical distribution — and so is lifecycle-schedule construction from equipment documentation into a dated, costable programme. Both are high-volume comparison tasks with verifiable outputs. What an assistant must **not** do is assign the estimate class, because the class is a statement about how well the scope is defined and a model reading the estimate cannot know what engineering has not been done. The characteristic failure is confident class inflation: an estimate presented in the format of a control estimate is classified as one, and a 30 % range is reported as an 8 % range — a misclassification worth 10,560,000 of contingency on Kestrel's base that nothing downstream will catch.

Verification is therefore three specific requirements. The class assignment carries a named human owner and a one-line justification referencing the engineering deliverables actually complete. The benchmark comparison is re-run on a held-out sample of the organisation's own outturns, with both flagged and unflagged items reviewed — a benchmark that flags nothing is not evidence of a good estimate. And the lifecycle programme is reconciled to the supplier's written maintenance schedule item by item, because a missed overhaul is invisible until the year it happens. **AI proposes; the professional verifies, decides and remains accountable.**

KEY TERMS — KA 8.1

Term	Meaning
Development cost	Pre-close spend at risk; capitalised into the envelope only if accepted.
Capital cost	The asset: contract price, owner's costs, land, connection, insurance in construction.
Operating cost	Recurring cash cost; a deduction inside `CFADS`, never a funding line.
Lifecycle (whole-life) cost	Periodic major maintenance plus handback or decommissioning.
Estimate class	The definitional maturity of an estimate, carrying a stated accuracy range.
Accuracy range	The band the estimator expects the outturn to fall in; narrows with definition.
Level annual charge	Lumpy lifecycle PV converted to a level equivalent via `AF(r, n)` (Domain 4's `EAV`).
Maintenance reserve (MRA)	Funded account that converts a lifecycle spike into a distribution reduction.

SAMPLE MCQS — KA 8.1

MCQ 8.1-A `[8.1.2 · Application]` A 48,000,000 base estimate carries a funded contingency of 3,645,403. Which statement is defensible? - A. 7.59 % is a normal contingency, so the provision is adequate - B. 7.59 % is a Stage E (control-estimate) provision, adequate only because a fixed-price wrap moved the estimate to that class - C. 7.59 % is adequate for any estimate class, since contingency is a policy percentage - D. the provision is inadequate at every class, because contingency should be 10 %

Rationale: 3,645,403/48,000,000 = 7.59 %, which sits inside the −5/+8 control band and covers 94.93 % of it; the same money covers 25.32 % of a Stage C +30 % band. A appeals to a norm that does not exist; C is the error the whole KA exists to remove; D substitutes a different unreasoned percentage for the first one.

MCQ 8.1-B `[8.1.2 · Analysis]` A project is procured as six separate packages against a Stage C feasibility estimate of 48,000,000 carrying a stated accuracy range of −15 % / +30 %, with the owner managing interfaces. The contingency implied by the estimate's own upper bound is: - A. 3,840,000 - B. 8,640,000 - C. 14,400,000 - D. 24,000,000

Rationale: Stage C's +30 % upper bound on 48,000,000 is 14,400,000. A is the Stage E provision (the answer only a full wrap earns); B is Stage D's +18 %; D is Stage A's +50 %.

MCQ 8.1-C `[8.1.3 · Application]` Membranes costing 3,200,000 at base-date prices are replaced in year 7; escalation is 3.6 % per annum. `CFADS` is 6,384,000 and debt service 5,009,635.23. With no maintenance reserve, the year-seven `DSCR` is closest to: - A. 1.2743 - B. 0.4561 - C. 0.6356 - D. 1.1457

Rationale: Nominal cost `3,200,000 × 1.036⁷ = 4,098,909`; `(6,384,000 − 4,098,909)/5,009,635.23 = 0.4561`. A is the ratio with a funded reserve; C forgets to escalate (using 3,200,000, giving 3,184,000/5,009,635.23 = 0.6356); D deducts only the level annual charge of 644,606 rather than the actual spend (`5,739,394/5,009,635.23 = 1.1457`).

MCQ 8.1-D `[8.1.1 · Analysis]` An operating-cost forecast understates annual cost by 500,000. Against Kestrel's 1.20× covenant with annual headroom of 372,438, the consequence is: - A. a funding shortfall in the sources-and-uses statement - B. no consequence, since operating cost is not a funded use - C. a covenant breach — the error is 1.34 times the entire annual headroom, and it recurs every year - D. a one-off reduction in the contingency line

Rationale: Operating cost is a deduction inside `CFADS`, so a 500,000 understatement removes 500,000 of `CFADS` annually against 372,438 of headroom (500,000/372,438 = 1.34). A and D put an operating error in a capital line; B confuses "not funded" with "no effect".

■ SELF-CHECK — KA 8.1

1. Why is "10 % contingency" not an answer? — Because contingency is only meaningful against a stated estimate class; 10 % is generous at Stage E and reckless at Stage C, and the ladder spans a factor of 6.25 on the same base.
2. What did Kestrel buy that makes 7.59 % defensible? — A fixed-price, date-certain EPC wrap, which transferred base-estimate uncertainty and moved the estimate to a Stage E position.
3. State the year-seven consequence of omitting a maintenance reserve. — `CFADS` 2,285,091, `DSCR` 0.4561 — a payment default, not merely a breach, on an otherwise healthy project.

KNOWLEDGE AREA 8.2

Schedule-driven cash flow and escalation

Topics: 8.2.1 from schedule to spend profile · 8.2.2 the shape of the S-curve and the area rule for capitalised interest · 8.2.3 escalation, base dates and the breakeven that links the two.

8.2.1 From schedule to spend profile

Definition. A **spend profile** (informally an **S-curve**) is a cost estimate distributed over time according to the schedule that will execute it. It is produced by loading cost onto activities or work packages and summing by period, and its characteristic shape — slow, then steep, then slow — arises because mobilisation and commissioning are cheap per unit time while the middle of a build is not.

The profile is the single most consequential output the schedule hands to the financing, and it is handed over in a specific direction: **the schedule owns the timing and the financing owns the consequence.** PML-AI Domain 6 builds the network, the critical path and the float that determine when work happens; this domain takes that timing as given and prices it. Three distinctions matter at the handover. **Certified spend is not incurred cost**, because a draw request is paid against certified progress, and the certification lag shifts the funding profile relative to the cost profile. **Committed is not spent**, because a purchase order creates a commitment that will appear in a cost report long before it appears in a payment. And **the profile must be at a stated price basis** — base-date money or then-current money — because the whole of 8.2.3 depends on knowing which.

Kestrel's certified spend profile of **6, 9, 13, 16, 17, 15, 13 and 11 per cent** over eight quarters is the profile Domain 6 funded. Note what it is not: it is not a straight line, and the difference between it and a straight line is worth money.

8.2.2 The shape of the S-curve and the area rule

The result to internalise. Capitalised interest depends on the shape of the drawdown, not only on its total. Two profiles that spend the same money over the same duration produce different interest during construction, and the difference is computable in one line.

The area rule. With draws made at period end and interest accruing on the opening balance at the periodic rate `r_q`, on total spend `S` funded at gearing `g`:

```
IDC = r_q × g × S × Σ cum(t−1)
```

where `cum(t−1)` is the cumulative fraction of spend completed before period `t`. The summation is the discrete area under the cumulative drawdown curve — so **capitalised interest is proportional to the area under the S-curve**, and "shape" is precisely the thing that area measures.

WORKED EXAMPLE

Worked example 8.2.2 — three shapes, one total, three interest bills.

1. **Setup. USD 48,000,000** of certified spend over **eight quarters**, funded **70 %** by debt at **6.0 % per annum** (1.5 % per quarter) on the opening balance, draws at period end. For this example escalation is set to zero and interest is accrued outside the debt balance, to isolate the shape effect; Domain 6 (KA 6.2.1) runs the full circular capitalisation. Three profiles: **A**, Kestrel's S-curve (6, 9, 13, 16, 17, 15, 13, 11); **B**, front-loaded (14, 16, 16, 14, 12, 11, 9, 8); **C**, back-loaded (5, 7, 9, 12, 15, 17, 18, 17). All sum to 100 %.
2. **Formula.** The area rule above; `r_q × g × S = 0.015 × 0.70 × 48,000,000 = 504,000` per unit of accumulated area.
3. **Substitution.** Σ cum(t−1) for A: `0 + 6 + 15 + 28 + 44 + 61 + 76 + 89 = 319` per cent, i.e. **3.1900**. For B: `0 + 14 + 30 + 46 + 60 + 72 + 83 + 92 = 397` → **3.9700**. For C: `0 + 5 + 12 + 21 + 33 + 48 + 65 + 83 = 267` → **2.6700**. Then `504,000 × area`.
4. **Result. A: USD 1,607,760 · B: USD 2,000,880 · C: USD 1,345,680.** The spread between front-loaded and back-loaded is **655,200**, or **40.75 %** of the S-curve's own interest bill, on identical total spend over an identical duration.
5. **Interpretation.** 655,200 is more than the arrangement fee on this facility would move in any plausible negotiation, and it is produced by a decision nobody in the room thinks of as financial: the sequence in which the contractor plans to build. That is the first professional consequence — **the construction programme is a financing variable, and the person who sets it is usually not consulted about interest.** The second is diagnostic. Because the area rule is linear in the area, a reviewer can test any construction model in one calculation: recompute `r_q × g × S × Σ cum(t−1)` from the profile and compare with the model's IDC. Agreement validates the whole interest calculation; disagreement localises the defect immediately (wrong periodic rate, average rather than opening balance, draws at period start, gearing applied to the wrong base). The third is a caution against the obvious inference. It does **not** follow that back-loading is good: delaying spend delays completion unless the duration is held, and holding the duration while back-loading the money usually means accepting a steeper, riskier finish. And it costs escalation, which is 8.2.3 — where the two effects meet.

8.2.3 Escalation, base dates and the breakeven that links the two

Definitions. An estimate is stated at a **base date** — the price level at which its rates were compiled. **Escalation** is the movement in input prices between that base date and the date the money is actually spent; it is a real cost, and it is not the same as **inflation** in the general price level, which is why construction cost indices for labour, steel, cement and specialist plant move differently from consumer price indices and from each other. Escalation applies to the timing of spend, so it cannot be computed from a total: it must be computed profile-period by profile-period.

```
Escalated spend = Σ  base spend(t) × (1 + e)^(t / periods per year)
```

Under a fixed-price contract the escalation on the contracted scope is the contractor's risk — which is a large part of what a fixed price is for. The owner retains escalation on owner-retained scope, on variations priced at then-current rates, and on any scope let after the base date.

WORKED EXAMPLE

Worked example 8.2.3 — the two prices of shape, and where they cancel.

1. **Setup.** The three profiles of 8.2.2, now with **3.6 % per annum** construction escalation on the full 48,000,000 (the unwrapped variant — the same works procured as packages, so the owner carries escalation), quarterly escalation factor `1.036^(1/4) = 1.00888099`. Debt draws are 70 % of escalated spend; interest as before. Compute escalated spend, IDC and the total for each profile, and find the escalation rate at which the S-curve and the back-loaded profile cost the same.
2. **Formula.** Escalated spend as above; IDC by accumulation of `opening balance × 0.015`; total funded construction cost = escalated spend + IDC. Breakeven `e*` solves `total(C, e*) = total(A, e*)`.
3. **Substitution and result.**

Profile	Escalated spend (USD)	Escalation over base (USD)	IDC (USD)	Total (USD)
B front-loaded	49,750,365	1,750,365	2,052,036	**51,802,401**
A S-curve	50,093,393	2,093,393	1,658,953	**51,752,346**
C back-loaded	50,324,512	2,324,512	1,391,209	**51,715,720**

Escalation spread B→C **574,146**; IDC spread C→B **660,827**; total spread **86,681**. Breakeven escalation between A and C: **4.1659 %**; between A and B: **4.1147 %**. 4. **Result.** At 3.6 % escalation the back-loaded profile is cheapest, but only by **36,626** against the S-curve and **86,681** against the front-loaded one — **0.17 %** of a 51.7 million spend. **Above an escalation rate of about 4.17 % the ranking reverses** and front-loading wins. 5. **Interpretation.** The near-cancellation is the most useful result in this KA, and it is not a coincidence. Deferring a dollar of spend by a period costs escalation at rate `e` on the whole dollar and saves interest at rate `r` on the geared fraction `g` of it, so shape is price-neutral when `e ≈ g × r` — here `0.70 × 6.0 % = 4.20 %`, against the computed 4.1659 %, the 3.4-basis-point gap being the discrete quarterly conventions. That heuristic is worth carrying: **at gearing of 70 % and a 6 % debt rate, construction escalation above roughly 4.2 % makes early spending cheaper and below it makes late spending cheaper**, and the further the actual rate is from that line, the more the programme sequence is worth arguing about. Three professional cautions. **The components are far larger than the net**, so a model that gets both wrong in the same direction produces a plausible total and two wrong lines — which is why the escalation and IDC lines are reviewed separately, never as a "construction cost" total. **The rates are not interchangeable**: using the revenue escalation assumption (2.967 % on Kestrel, Domain 6) for construction cost, or one construction index for all trades, is a silent error, and where an offtake escalates at one rate while costs escalate at another the difference compounds across the whole concession (Domain 7). **And the base date is a document, not an assumption.** An estimate whose base date nobody can name has an escalation exposure nobody can compute; the first question of any cost review is "as at what date are these rates?"

Common pitfall — escalating the total. Multiplying a 48,000,000 total by `1.036²` for a two-year build gives 51,518,208, against the profile-correct 50,093,393 for the S-curve — an overstatement of **1,424,815**, because it escalates all the money as though every dollar were spent on the last day. The mirror error, escalating nothing because "the contract is fixed price", understates the owner's retained exposure. Both are avoided by the same discipline: escalate the profile, and state which scope is wrapped.

AI in this KA

Cost-loading a schedule and pricing every plausible sequence for escalation and interest is a deterministic search problem and proper machine work; so is index research — assembling and documenting the published construction cost indices relevant to a project's trade mix with sources and publication dates recorded. Two boundaries. **It must not select the escalation rate**, which is a forecast of input prices over a specific build in a specific market; an assistant asked for "a reasonable construction escalation assumption" supplies a plausible number with no provenance, and it then propagates into the funding envelope, the depreciable base and the delay arithmetic of KA 8.4. **And it must not re-shape the profile as an optimisation** — the area rule makes it trivially easy to cut IDC by pushing spend later, and the constraint that makes that legitimate lives in the construction logic, not in the cost model.

Verification: recompute IDC independently from the area rule and require agreement to the dollar; confirm escalation on wrapped scope is zero and on retained scope is not; re-derive one period's escalated spend by hand from the base rate and the stated base date; and require every escalation assumption to carry a named index, source and human owner in the assumption register before the model is run.

KEY TERMS — KA 8.2

Term	Meaning
Spend profile / S-curve	A cost estimate distributed over time by the schedule that executes it.
Certified spend	Progress certified for payment; the basis of a draw request, lagging incurred cost.
Area rule	`IDC = r_q × g × S × Σ cum(t−1)`; capitalised interest is proportional to the area under the drawdown curve.
Base date	The price level at which an estimate's rates were compiled.
Escalation	Input-price movement between base date and spend date; trade-specific, not general inflation.
Shape neutrality	Profile timing is cost-neutral when escalation ≈ gearing × debt rate (4.20 % on Kestrel; 4.1659 % computed).

SAMPLE MCQS — KA 8.2

MCQ 8.2-A `[8.2.2 · Application]` 48,000,000 of spend over eight quarters, 70 % debt-funded at 1.5 % per quarter on opening balances. The profile's cumulative-before-period fractions sum to 3.9700. Capitalised interest is: - A. USD 1,607,760 - B. USD 2,000,880 - C. USD 2,858,400 - D. USD 504,000

Rationale: `0.015 × 0.70 × 48,000,000 × 3.9700 = 2,000,880`. A is the S-curve's area of 3.1900; C omits the gearing, applying interest to full spend (`0.015 × 48,000,000 × 3.9700 = 2,858,400`); D is the per-unit-of-area coefficient mistaken for the answer.

MCQ 8.2-B `[8.2.3 · Analysis]` A model escalates a 48,000,000 two-year construction estimate by multiplying it by `1.036²`, giving 51,518,208, where the profile-correct figure for the same S-curve is 50,093,393. The error is: - A. 1,424,815 of understatement - B. 1,424,815 of overstatement, because escalating a total prices every dollar as though spent on the final day - C. immaterial, since the rate is correct - D. offset by the interest calculation

Rationale: 51,518,208 − 50,093,393 = 1,424,815 too much; escalation must be applied period-by-period to the profile. C mistakes a correct rate for a correct method; D is wrong in direction — a higher spend also raises draws and therefore IDC.

MCQ 8.2-C `[8.2.3 · Analysis]` At 70 % gearing and a 6.0 % debt rate, the construction escalation rate above which front-loading spend becomes cheaper than back-loading it is closest to: - A. 6.0 % - B. 4.2 % - C. 3.6 % - D. 1.8 %

Rationale: Deferral costs escalation on 100 % of spend and saves interest on the geared 70 %, so neutrality is at `e ≈ g × r = 4.20 %` — computed as **4.1352 %** between Kestrel's front- and back-loaded profiles on the quarterly convention (8.2.3 quotes 4.1659 % for the same effect measured between the S-curve and the back-loaded profile; every pairwise breakeven sits just below `g × r`). A ignores gearing; C is the assumed escalation rate, not the breakeven; D halves the rate for no stated reason.

MCQ 8.2-D `[8.2.2 · Analysis]` A reviewer wants one calculation that validates a construction model's entire capitalised-interest line. The best choice is: - A. confirm the total spend equals the contract price - B. recompute IDC from the area rule and require agreement to the dollar - C. confirm the closing debt balance equals the facility amount - D. compare the IDC percentage against other projects

Rationale: The area rule reproduces IDC from the profile, rate and gearing, so agreement validates all four inputs and disagreement localises the defect. A and C are necessary but pass with a wrong interest convention; D is benchmarking, not verification.

SELF-CHECK — KA 8.2

1. State the area rule and what it proves. — `IDC = r_q × g × S × Σ cum(t−1)`; capitalised interest is proportional to the area under the cumulative drawdown curve, so shape has a price.
2. How much did shape alone move Kestrel's interest bill? — 655,200 between front- and back-loaded profiles, 40.75 % of the S-curve's own 1,607,760, on identical spend and duration.
3. Why is the total cost of the three profiles nearly equal at 3.6 % escalation? — Escalation and interest run in opposite directions and cancel near `e = g × r = 4.20 %`; the two components differ by 574,146 and 660,827 while the totals differ by 86,681.

KNOWLEDGE AREA 8.3

Contingency and management reserve

Topics: 8.3.1 the reserve family in a financing, and who controls the draw · 8.3.2 sizing contingency from quantified risk · 8.3.3 why the percentage method fails, computably.

8.3.1 The reserve family in a financing, and who controls the draw

Definition. Contingency is funded provision for identified risks within the agreed scope. **Management reserve** is provision for what the register does not contain — in delivery language, unknown-unknowns and scope change. PML-AI Domain 8 (KA 8.3.2) sets out that distinction as a delivery-governance matter, with contingency controlled by the project manager under a published draw protocol and management reserve controlled by the sponsor through change control. In a financing, three things change.

Contingency becomes a funded line with an external gatekeeper. It sits in the sources-and-uses statement (Domain 6) as money the facility will lend and the equity will match, and a draw against it typically requires the lenders' technical adviser to certify both that the cost is a proper contingency item and that the remaining contingency is still adequate for the remaining risk (KA 8.4.1). The project manager's draw protocol still applies internally; it is no longer the only gate.

Management reserve mostly leaves the envelope. Lenders do not fund unknown scope, so the financing equivalent is **contingent support outside the base case**: cost-overrun undertakings, standby equity, a standby debt tranche, or sponsor several-liability support (Domain 5, KA 5.2.3). The economic difference is important — funded contingency is drawn on certification, contingent support is called on a defined trigger, and the second is worth less than the first to a lender precisely because it depends on someone's willingness and ability to pay when called.

Owner's contingency and contractor's contingency are different money. The contractor's sits inside the contract price, invisible and unavailable; the owner's is the only provision the project can actually draw (KA 8.1.2's pitfall).

WORKED EXAMPLE

Worked example 8.3.1 — sizing Kestrel's contingent support.

1. **Setup.** Kestrel's funded contingency is **3,645,403**. The lender requires that the retained construction risk be covered to **P95** by funded contingency plus contingent equity support. The retained register's mean exposure is **2,690,000** with a standard deviation of **1,848,973** (derived in 8.3.2).
2. **Formula.** P95 ≈ mean + 1.6449 × σ (the 95th percentile of a normal approximation); standby requirement = P95 − funded contingency.
3. **Substitution.** `2,690,000 + 1.6449 × 1,848,972.69 = 2,690,000 + 3,041,375.17`; then `5,731,375 − 3,645,403`.
4. **Result.** P95 exposure **USD 5,731,375**; contingent support required **USD 2,085,972**.
5. **Interpretation.** The number to hold on to is the ratio: covering from P80 to P95 costs another 1,485,280 of committed support on a project whose whole funded contingency is 3,645,403. Tail cover is expensive, which is why it is provided as a commitment rather than as funded cash — the sponsors post an obligation, not money, and pay for it only in the futures where it is called. That is the correct structure and it carries a specific professional caution: a commitment is only as good as the entity behind it, so 2,085,972 of support from a well-rated sponsor and the same figure from a thinly capitalised project company are not the same credit, and Domain 5 (KA 5.2.3) showed the distinction between joint-and-several and several liability deciding which. **A lender's contingency question is never only "how much?"; it is "how much, funded by whom, callable on what trigger, and enforceable against what balance sheet?"**

8.3.2 Sizing contingency from quantified risk

The method. Take the register, quantify each item as probability × impact, aggregate to a mean and a variance, and read the amount at the confidence level policy requires. The machinery is PML-AI Domain 8's (KA 8.2.2 and 8.2.4) and is not re-derived here: `EMV = p × impact`; for independent items mean = Σ `EMV` and variance = Σ `p(1 − p) × impact²`; a P80 amount ≈ mean + 0.8416 σ.

WORKED EXAMPLE

Worked example 8.3.2 — Kestrel's retained construction risk, and two P80s that disagree.

1. **Setup.** Under the fixed-price wrap the owner retains six quantified items on a 48,000,000 base. Compute the mean, the standard deviation and the P80; separately, translate the Stage C accuracy range (−15 %/+30 %) into a P80 using a triangular distribution with minimum 40,800,000, mode 48,000,000 and maximum 62,400,000; then reconcile the two answers.
2. **Formula.** As above. For a triangular distribution with parameters $a < m < b$, the cumulative probability above the mode is $F(x) = 1 - (b - x)^2 / ((b - a)(b - m))$, its mean is $(a + m + b)/3$, and the P80 solves $(b - x)^2 = 0.20\,(b - a)(b - m)$.
3. **Substitution and result.**

ID	Retained risk	p	Impact (USD)	EMV (USD)
C1	Ground conditions worse than surveyed (owner-retained geotechnical basis)	0.40	2,400,000	960,000
C2	Utility diversion and third-party interface scope growth	0.30	1,800,000	540,000
C3	Membrane supply price above the contract indexation formula	0.35	1,400,000	490,000
C4	Marine intake weather standby beyond the allowance	0.50	900,000	450,000
C5	Permit condition requiring additional monitoring works	0.20	2,000,000	400,000
C6	Early-completion rebate (opportunity)	0.25	(600,000)	**(150,000)**
	Mean exposure			**2,690,000**

Variance $= 0.40\times0.60\times2{,}400{,}000^2 + 0.30\times0.70\times1{,}800{,}000^2 + 0.35\times0.65\times1{,}400{,}000^2 + 0.50\times0.50\times900{,}000^2 + 0.20\times0.80\times2{,}000{,}000^2 + 0.25\times0.75\times600{,}000^2 = 3{,}418{,}700{,}000{,}000$; σ = **1,848,973**; P80 $= 2{,}690{,}000 + 0.8416 \times 1{,}848{,}973 =$ **4,246,095**. Worst-case sum of threats **8,500,000** (17.71 % of base). Triangular: mean **50,400,000** (contingency 2,400,000, 5.00 %); P80 total **54,512,795**, contingency **6,512,795** (**13.57 %**). 4. **Result. Register P80 contingency USD 4,246,095 (8.85 % of base); range-based P80 contingency USD 6,512,795 (13.57 %).** The funded 3,645,403 sits at the **69.7th percentile** of the register and the **62.81st percentile** of the Stage C range. 5. **Interpretation.** Two defensible methods have produced answers 2,266,700 apart, and the resolution is the most valuable idea in this KA: **they are not competing estimates of the same thing.** The accuracy range measures systemic uncertainty in the base estimate — quantities, productivity, unpriced scope, the things a better-defined design would resolve. The register measures discrete risk events that either happen or do not. Adding them double-counts, because a Stage C range already contains an allowance for the sort of surprise the register enumerates; taking the lower of them under-provides. The defensible convention is to **take the higher, and then ask which of the two the contracting structure has actually eliminated.** On Kestrel the fixed-price wrap transfers the base-estimate uncertainty to the contractor, so the range-based number is the right measure of a risk the owner no longer carries, and the register-based 4,246,095 is the owner's provision. On the unwrapped variant of 8.1.2 the range governs and 6,512,795 is the floor. **The funded 3,645,403 is therefore 600,692 short of the P80 the retained register supports** — a small number with a permanent consequence: capitalise it and debt becomes 42,600,692, the instalment 5,081,284.04, the DSCR **1.2564** and annual headroom **286,459**, down 85,979 or **23.1 %** from the 372,438 Domain 10 measured, for the whole twelve-year loan life. Two standing caveats carry over from PML-AI Domain 8 unchanged and must be stated in any paper using this arithmetic: **independence is an optimistic assumption** (C1 and C2 both concern ground and third parties, and if they are correlated the variance and the true P80 are higher), and **the normal approximation is a convenience** for a handful of Bernoulli items where a simulation over the register and the schedule is the proper instrument. Neither caveat weakens the method against the alternative in 8.3.3; both bound what may be claimed for it.

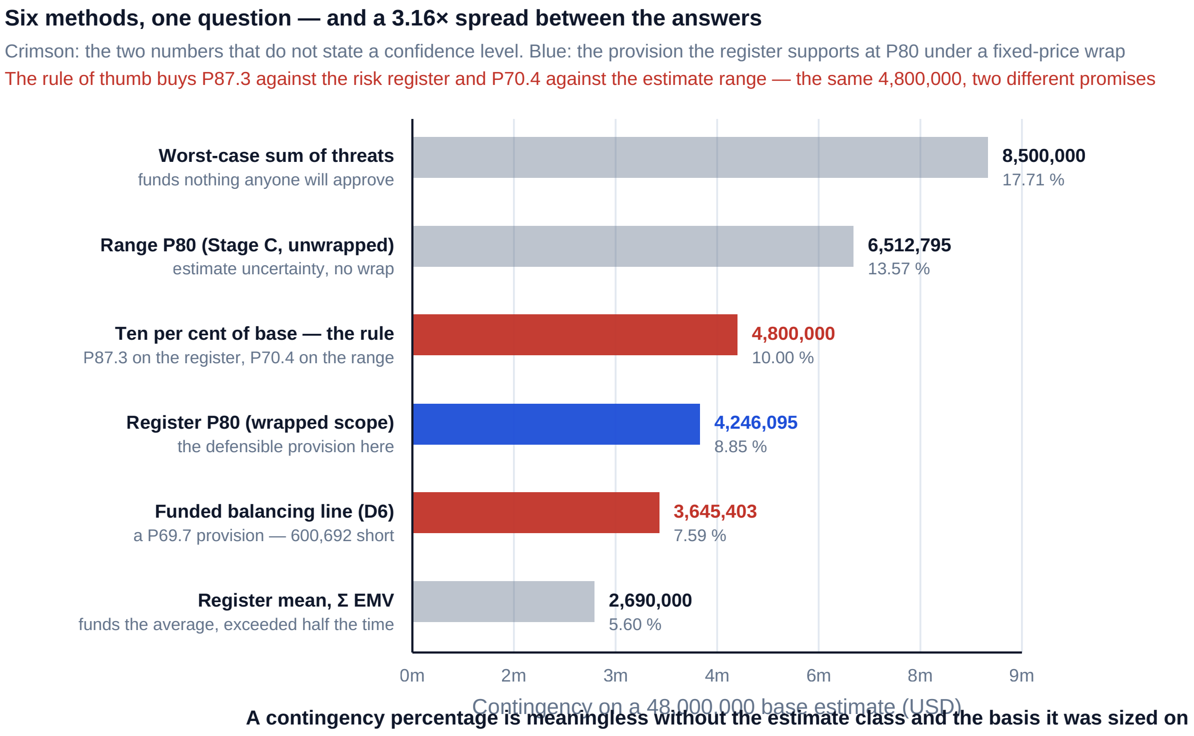

Fig 8.3.1 — Six answers to one question: how much contingency?

8.3.3 Why the percentage method fails, computably

The practice. Contingency is set at a policy percentage of base cost — 5 %, 10 %, 15 % — chosen by convention, precedent or negotiation. It is fast, it is universally understood, and it has two defects that can be stated as arithmetic rather than as opinion.

Defect one: it does not know what confidence it buys. Kestrel's ten-per-cent rule gives **4,800,000**. Against the retained register (mean 2,690,000, σ 1,848,973) that is $z = (4{,}800{,}000 - 2{,}690{,}000)/1{,}848{,}973 = 1.1412$, a **P87.3** provision. Against the Stage C accuracy range it is $F(52{,}800{,}000) = 1 - (9{,}600{,}000)^2/((21{,}600{,}000)(14{,}400{,}000)) = 0.7037$, a **P70.4** provision. **The same 4,800,000 is a 87.3 % promise on one basis and a 70.4 % promise on the other**, and the rule cannot tell which it is making. A provision whose confidence level is unknown cannot be compared with a covenant, a support commitment or another project, which is the entire practical purpose of sizing it.

Defect two: it does not move when the risk moves. Suppose C1, the ground-conditions risk, is retired — the additional site investigation is complete, the conditions are as surveyed, the risk window has closed with no impact. The register responds immediately: mean falls to **1,730,000**, σ to **1,426,990**, and the P80 provision to **2,930,955** — a release of **1,315,141**. The percentage rule responds not at all: 10 % of base is still 4,800,000, because base cost has not changed. The organisation is now holding **1,869,045** of provision against risks that no longer exist, and that holding has a price in two stages. **While it is undrawn** it is committed capital: at a 0.60 % per annum commitment fee the excess costs **11,214** a year — small, and the reason the excess is rarely challenged. **If it is drawn**, the price becomes permanent. Funded 70/30, the excess adds **1,308,332** to senior debt, raising the annual instalment by **156,054** to 5,165,689.12, taking the `DSCR` from 1.2743 to **1.2358**, lifting the covenant cash trigger to **6,198,827** and cutting annual headroom from 372,438 to **185,173 —** a fall of **187,265**, or **50.3 %**, for the whole twelve-year loan life. That is the financing translation of the slush-fund failure PML-AI Domain 8 (KA 8.3.2) warns about: **an unreleased provision that gets spent because it is there halves the project's covenant headroom for a decade**, and the percentage method guarantees there will always be some.

The symmetric error is worse. A percentage rule that happens to under-provide is invisible in exactly the same way. Nothing in "10 % of base" reacts when a new risk is added to the register, so a project can accumulate exposure all through detailed design while its contingency line stays constant, and the first evidence anyone sees is a draw request that cannot be certified (KA 8.4.1).

What survives of the percentage method. Two legitimate uses. As a **screening figure** at Stage A or B, where no register exists and none could — a factored allowance is the honest instrument for an estimate that is itself factored. And as a **cross-check** on a register-based number: if quantified risk produces 1 % of base at Stage C, the register is incomplete, and if it produces 40 %, the project is being asked to fund a scope that is not defined. The rule of thumb is a plausibility test on the arithmetic, never a substitute for it.

AI in this KA

Monte Carlo simulation over the register and the schedule is the correct instrument that 8.3.2's normal approximation stands in for, and running it — with correlation structures, which are what make the approximation optimistic — is machine work. Correlation detection is a further genuine use: mining an organisation's own historical registers and outturns for items that have moved together is beyond manual analysis and directly improves the variance estimate. So is register hygiene — duplicates, items double-counted between register and base estimate, impacts stated in inconsistent price bases. Three things a model must not do: **set probabilities**, which are expert judgments about a specific site, contractor and market, and which if invented give the P80 a false precision a board cannot see; **choose the confidence level**, a risk-appetite decision belonging to governance (PML-AI D8, KA 8.2.4); and **reconcile the two P80s of 8.3.2**, which is a judgment about what the contract transferred and therefore a reading of a document.

Verification: recompute the mean by hand and require it to equal Σ `EMV` — a simulated mean that differs from the analytic mean means the model is not implementing the register; recompute one percentile with the normal approximation and require the simulation to be in the same region, a large divergence being either documented correlation or an error; and require every probability and impact to carry a named owner and a dated basis, because nothing in a simulation adds credibility its inputs do not have.

KEY TERMS — KA 8.3

Term	Meaning
Contingency	Funded provision for identified risks within agreed scope; drawn on certification.
Management reserve	Provision for unidentified risk and scope change; in a financing, largely replaced by contingent support.
Contingent support	Cost-overrun undertaking, standby equity or standby tranche called on a defined trigger.
Confidence level (P50 / P80 / P95)	The probability that the provision covers the aggregate outcome; a policy choice.
Owner's vs contractor's contingency	Only the owner's is drawable by the project; the contractor's is inside the price.
Percentage method	Contingency as a policy share of base cost; states no confidence and does not respond to risk retirement.
Contingency release	Reduction in the provision when a risk retires; the percentage method cannot produce one.

SAMPLE MCQS — KA 8.3

MCQ 8.3-A `[8.3.2 · Application]` A retained register has mean exposure 2,690,000 and standard deviation 1,848,973. The P80 contingency is closest to: - A. USD 2,690,000 - B. USD 4,246,095 - C. USD 8,500,000 - D. USD 5,731,375

Rationale: `2,690,000 + 0.8416 × 1,848,973 = 4,246,095`. A is the mean, which by construction is exceeded about half the time; C is the worst-case sum of threats; D is the P95 (z = 1.6449), a different policy choice.

MCQ 8.3-B `[8.3.3 · Analysis]` A 10 %-of-base contingency of 4,800,000 is a P87.3 provision against the risk register and a P70.4 provision against the estimate's accuracy range. The correct conclusion is: - A. the rule is validated, since both figures exceed P50 - B. the rule states no confidence level, so the provision cannot be compared with a covenant, a support commitment or another project - C. the register must be wrong, since the two bases disagree - D. the average of the two, P78.9, is the provision's true confidence

Rationale: The two percentiles measure different uncertainties (discrete events versus systemic estimate error), so a single percentage of base cannot express a confidence at all — which is the defect. A mistakes "above the median" for "sized"; C misreads a difference in basis as an error; D averages two probabilities that are not on the same scale.

MCQ 8.3-C `[8.3.3 · Application]` The largest register item (p 0.40, impact 2,400,000) retires with no impact. Register P80 falls from 4,246,095 to 2,930,955 while the 10 % rule stays at 4,800,000. If the resulting excess is drawn and funded 70/30, the effect on annual covenant headroom of 372,438 is a fall of: - A. nil — contingency is not carried at a cost - B. 187,265, or 50.3 %, because 1,308,332 of extra senior debt raises the instalment by 156,054 - C. 11,214, the commitment fee on the excess - D. 1,869,045, the excess itself

Rationale: `4,800,000 − 2,930,955 = 1,869,045`; `× 0.70 = 1,308,332`; `/8.383844 = 156,054` of extra instalment; the 1.20× trigger rises to 6,198,827 and headroom falls to 185,173. A ignores that drawn contingency is debt serviced for the loan life; C is the cost while the excess is undrawn, not drawn; D confuses the provision with its coverage effect.

MCQ 8.3-D `[8.3.1 · Analysis]` Why do lenders treat 2,085,972 of contingent equity support as worth less than 2,085,972 of funded contingency? - A. it is a smaller amount in present-value terms - B. it is only drawable on certification - C. it depends on a sponsor's willingness and ability to pay when called, which is a credit exposure rather than cash in the structure - D. it cannot be documented

Rationale: Funded contingency is money the facility will lend against certification; contingent support is a commitment whose value is the obligor's credit (8.3.1, and Domain 5 KA 5.2.3 on several versus joint-and-several liability). B describes funded contingency; A confuses timing with credit; D is false.

SELF-CHECK — KA 8.3

1. Why can two correct P80 contingencies differ by 2,266,700 on one project? — They measure different uncertainties: the register measures discrete events, the accuracy range measures systemic base-estimate error; take the higher, then ask which the contract transferred.
2. State the two computable defects of the percentage method. — It states no confidence level (4,800,000 is P87.3 on one basis, P70.4 on the other), and it does not respond to risk retirement: 1,869,045 of excess, costing 11,214 a year undrawn and 50.3 % of covenant headroom if drawn.
3. What does the P80-to-P95 step cost on Kestrel, and how is it provided? — 1,485,280 more of cover, provided as a 2,085,972 contingent commitment rather than funded cash, because tail cover as cash is prohibitively expensive.

KNOWLEDGE AREA 8.4

Delay impact, cost-to-complete, and the interface with project controls

Topics: 8.4.1 from earned value to cost to complete and the funds sufficiency test · 8.4.2 the cost of a month of slip during construction · 8.4.3 one data spine, two questions.

8.4.1 From earned value to cost to complete and the funds sufficiency test

Definition. Cost to complete (`CTC`) is the money still required to finish: `CTC = EAC − AC` on any chosen `EAC` method. The **funds sufficiency test** — often written into a facility as an "in balance" condition — asks whether the funds remaining available, from the undrawn facility, unused contingency and uncalled equity, are at least `CTC`. It is the test a lender runs on a monthly cost report, and it is not a test a project control account produces.

WORKED EXAMPLE

Worked example 8.4.1 — what a lender does with Auriga's earned value.

1. **Setup.** Project Auriga (PML-AI D7, KA 7.3.3) at the week-13 data date: BAC **4,000,000**, AC **2,120,000**, EV **1,920,000**, CPI **0.905660**, SPI **0.923077**, and the three published forecasts EAC **4,200,000** (remaining work at the budgeted rate), **4,416,667** (BAC/CPI) and **4,608,056** (CPI × SPI). Assume the delivery organisation financed it: total funding **4,300,000**, being BAC 4,000,000 plus a funded contingency of **300,000** (7.5 % — the same order as Kestrel's 7.59 %), of which **180,000** has already been drawn. Of the sanction register (PML-AI D8, KA 8.2.2), R2 has materialised and R4 has been retired; R1 (p 0.35, impact 240,000), R3 (p 0.25, 320,000) and the R5 opportunity (p 0.30, −120,000) remain open. Run both lender tests.
2. **Formula.** CTC = EAC − AC; funds available = total funding − AC; surplus/(shortfall) = available − CTC. Contingency adequacy: remaining contingency versus the P80 of the remaining register (mean = Σ EMV, variance = Σ $p(1 - p)$impact^2, P80 ≈ mean + 0.8416 σ).
3. **Substitution.** Available = 4,300,000 − 2,120,000 = 2,180,000. CTC by method: 4,200,000 − 2,120,000; 4,416,667 − 2,120,000; 4,608,056 − 2,120,000. Remaining register mean = 84,000 + 80,000 − 36,000; variance = $0.2275 \times 240{,}000^2 + 0.1875 \times 320{,}000^2 + 0.21 \times 120{,}000^2$.
4. **Result.**

EAC method	EAC (USD)	CTC (USD)	Funds available (USD)	Surplus / (shortfall)
Remaining at budgeted rate	4,200,000	2,080,000	2,180,000	**+100,000**
BAC/CPI	4,416,667	2,296,667	2,180,000	**(116,667)**
CPI × SPI	4,608,056	2,488,056	2,180,000	**(308,056)**

Remaining register: mean **128,000**, σ **187,957**, **P80 286,185** against remaining contingency of **120,000** — a **166,185** shortfall on the contingency-adequacy test. 5. **Interpretation.** This is the domain's bridge, in one table. A project controller's report ends at the EAC column and its honest conclusion is a range: the project will cost between 4.2 and 4.6 million depending on which assumption about the remaining work holds (PML-AI D7's point, that a forecast is a statement about the future's resemblance to the past). **A lender reads the same three numbers and produces a date and an amount:** on the BAC/CPI forecast the project is out of balance by 116,667, so before the next drawdown the sponsor must inject 116,667 of equity or the facility does not fund. The forecast has become a cash call. Three consequences follow for a finance leader. **The choice of EAC method is a commercial negotiation, not a technical one**, because it sets the size of the cash call: the sponsor argues for the budgeted-rate forecast (+100,000, in balance), the technical adviser for BAC/CPI or worse, and the honest position is that a CPI of 0.906 sustained over thirteen weeks is evidence about the remaining work unless something specific has changed — which someone must name. **The contingency test fails independently and earlier.** With 120,000 of contingency left against a remaining P80 exposure of 286,185, a certifier cannot honestly confirm that remaining contingency is adequate for remaining risk, and that conclusion needed no EAC at all — only the register and the draw history. A project can therefore be out of balance on the contingency test while every cost report still shows a forecast inside budget. **And the shortfall was arithmetically predictable at sanction.** The 300,000 funded was a **P53.5** provision against the sanction register (mean 278,000, σ 252,642), while the P80 was **490,624**; the actual overrun on the BAC/CPI forecast is 416,667, a **P70.8** event — comfortably inside a P80 provision and comfortably outside a P53.5 one. Nothing unlikely happened. **The project was funded to a confidence level nobody named, and then behaved normally.**

8.4.2 The cost of a month of slip during construction

The three components. A month of delay declared during construction costs three separable things: **escalation** on the scope not yet bought, because the money will be spent a month later at a month's higher prices; **extra interest** on the balance already drawn, which accrues for a month with no offsetting progress; and **deferred revenue**, because the operating period starts a month later. Domain 5 (KA 5.4.2) priced a slip at COD, where remaining spend is zero and the escalation row vanishes. During the build it does not vanish, and the relative sizes of the three rows change continuously through the programme.

WORKED EXAMPLE

Worked example 8.4.2 — Kestrel declares a three-month slip at the end of quarter six.

1. **Setup.** At the end of construction quarter six, cumulative debt drawn is **31,990,655** (Domain 6's funding profile). The remaining owner-retained scope — owner's costs and land of 3,600,000 spent on the same profile — stands at **922,906** at then-current prices; the EPC scope is fixed-price and date-certain, so its escalation is the contractor's risk. Escalation on retained scope is **3.6 %** per annum, giving a monthly factor of `1.036^(1/12) = 1.00295161`, i.e. **0.2952 %** per month. The debt rate is **6.0 %**; operating `CFADS` would be **6,384,000** per year. Delay damages are **USD 20,000 per day** (Domain 5). Compute the cost of one month, the cost of the three-month slip, the damages coverage, and the coverage consequence if the cost components are capitalised.
2. **Formula.** Escalation row = remaining retained scope at then-current prices × monthly escalation factor − 1. Interest row = debt drawn × rate ÷ 12. Revenue row = annual `CFADS` ÷ 12. Coverage consequence: new debt = 42,000,000 + capitalised cost; instalment = new debt ÷ `AF(0.06, 12)`; `DSCR` = `CFADS` ÷ instalment; covenant trigger = instalment × 1.20 (Domain 10).
3. **Substitution.** `922,906 × 0.00295161`; `31,990,655 × 0.06/12`; `6,384,000/12`; `20,000 × 30`; `(2,724 + 159,953) × 3`; `42,488,032 / 8.383844`.
4. **Result.**

Component	Per month (USD)	Share	Three months (USD)
Escalation on owner-retained scope	2,724	0.39 %	8,172
Extra interest on drawn debt	159,953	23.03 %	479,860
Deferred `CFADS`	532,000	76.58 %	1,596,000
Total economic cost	**694,677**		**2,084,032**
Delay damages at 20,000/day	600,000	86.37 % recovered	1,800,000
Uncovered, borne by the SPV	**94,677**		**284,032**

Capitalising the escalation and interest components (488,032) takes debt to **42,488,032**, the instalment to **5,067,846.24**, the `DSCR` from 1.2743 to **1.2597**, the covenant cash trigger to **6,081,415** and annual headroom to **302,585** — down **69,853**, or **18.8 %**. 5. **Interpretation.** Read the share column first. **Three quarters of the cost of a delay during construction is revenue the project will never earn, and almost none of it is escalation** — which is precisely the reverse of the intuition a cost engineer brings to the meeting, because escalation is the row a cost report shows and deferred revenue is a row it does not contain. The 0.39 % escalation share is not a general result; it is the value of the wrap, and it is measurable: on the unwrapped variant of 8.1.2 the escalation row would be **39,045** per month rather than 2,724, so the fixed-price contract is worth **36,321 per month of slip** on this row alone. The second reading is the damages calibration. At 20,000 per day the regime recovers 86.37 % of a delay declared at quarter six, but its adequacy moves through the programme: at financial close a month of slip costs **552,329** and damages of 600,000 over-recover; at COD it costs **742,000** (reconciling exactly to Domain 5's 7,000 + 17,733.33 per day) and they recover 80.86 %. **The crossing is between quarter two (580,033) and quarter three (604,156)** — after which a single flat damages rate is never again sufficient, because the interest row grows monotonically as the balance builds while the escalation row shrinks as scope is bought. A negotiator who understands that asks for a **stepped** damages rate rather than a higher flat one, which is both cheaper for the contractor to accept and better matched to the exposure. The third reading is the permanence. 488,032 of capitalised delay cost — a rounding error against a 60,000,000 envelope — removes **18.8 %** of the covenant headroom for the entire twelve-year loan life, on a project that is otherwise exactly as forecast. That is the mechanism Domain 5 named and this example generalises: **a construction event becomes an operating constraint the moment it is capitalised**, and whether it is capitalised or funded with equity is a choice about whether the slip lands on coverage or on equity return. It belongs to the sponsors before the event, in the funding documents, not to a project director in the month it happens. Finally, the deferred-revenue row deserves the caution Domain 5 attached to it: 1,596,000 of `CFADS` deferred is a cash fact, not a value loss of that size, because most of it is postponed rather than destroyed — the value calculation (Domain 5, KA 5.4.2) is the right instrument for a damages negotiation and the cash calculation is the right instrument for a funding one.

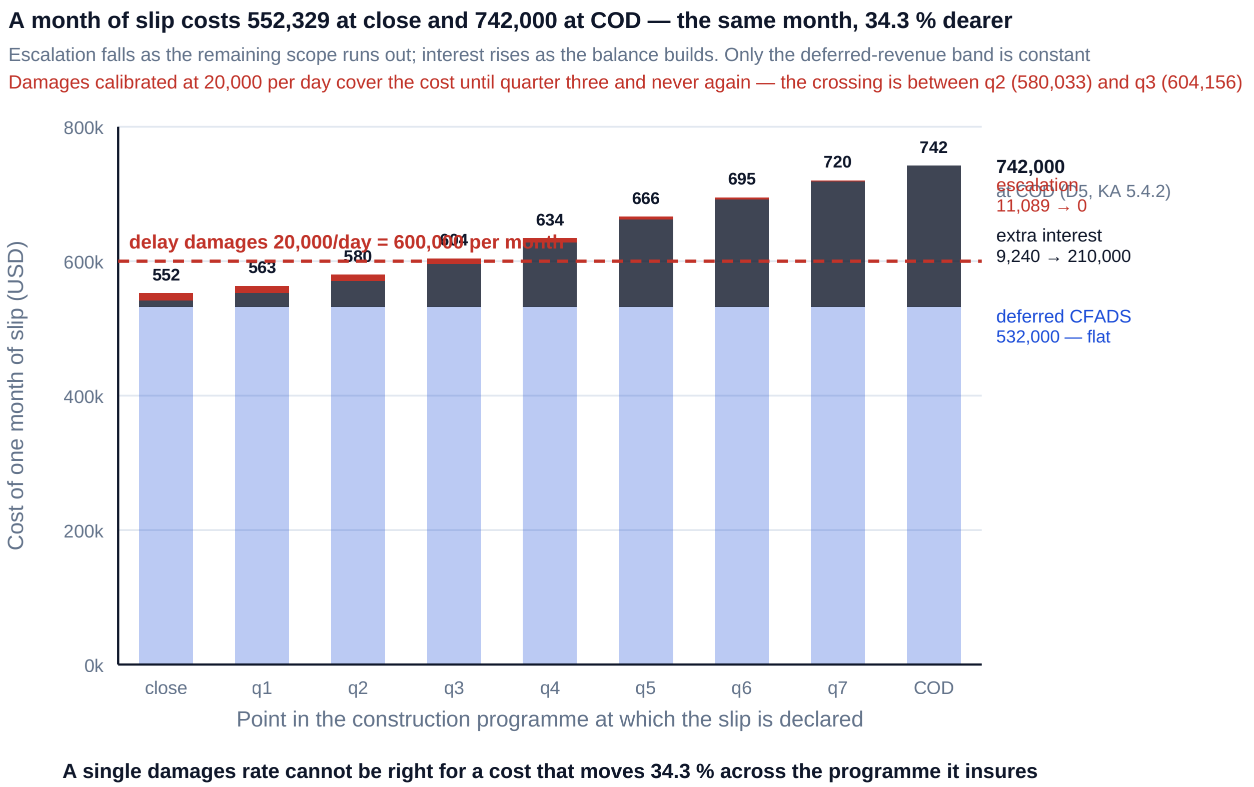

Fig 8.4.1 — The cost of a month of slip rises through the build

8.4.3 One data spine, two questions

The principle. The project controls function and the financing function should read the same numbers. They will always ask different questions of them, and the interface exists to keep the numbers common while letting the questions differ. Where two sets of numbers appear — a cost report for management and a different one for lenders — the organisation has created a reconciliation burden it will fail at exactly the moment it matters.

What each side owns.

Object	Project controls owns	The financing asks
Cost breakdown and control accounts	The structure, the coding, the hundred-per-cent rule (PML-AI D4)	Do the accounts map to the funded uses in the sources and uses?
Earned value at the data date	`PV`, `EV`, `AC`, `CPI`, `SPI` and their integrity	What is `CTC`, and is the facility in balance? (8.4.1)
The `EAC` family	The method and its stated assumption	Which forecast does the technical adviser certify, and what cash call follows?
Schedule and forecast COD	The network, the critical path, the float	What is the cost per month of the slip, and who bears it? (8.4.2)
Risk register	Identification, probability, impact, response	What contingency does it support at P80, and is the remainder adequate? (8.3.2)
Contingency drawn and remaining	The internal draw protocol	Can a draw be certified against the remaining risk?
Change control	The baseline and its integrity	Is this a variation under the contract, a scope change requiring consent, or a cost overrun?

The monthly reporting pack this domain endorses is therefore one document with a financing annexe: certified progress and the draw request; earned value and the three `EAC` forecasts with their assumptions named; `CTC` and the funds sufficiency position against undrawn facility, unused contingency and uncalled equity; contingency drawn to date trended against **risk retired** rather than against time (PML-AI D8, KA 8.3.2 — burning 60 % of contingency while retiring 20 % of the register is the signal); the register's current P80 and the remaining-adequacy test; forecast COD with the cost-per-month figure current at that point in the programme; and the covenant consequence of any capitalised delay or overrun cost. Every one of those lines derives from data the controls function already holds. **What the financing adds is not data collection; it is three transformations — `CTC`, sufficiency, and the coverage consequence — and the discipline of stating which confidence level the provisions represent.**

The governance point. The reconciliation must be scheduled, not discovered. A finance leader who first meets a `CPI` of 0.906 in a certifier's refusal to fund has lost the two months in which the problem was cheap to fix, and the relationship cost of that surprise is the one Domain 10 (KA 10.4.4) priced: a lender who learns of a problem from a compliance certificate prices the discovery into everything afterwards.

AI in this KA

The transformations of 8.4.3 are deterministic and monthly, which makes them ideal automation: `CTC` under every `EAC` method, the sufficiency test against current facility balances, the register's P80 recomputed as items retire, and the cost-per-month-of-slip figure recomputed as the drawn balance moves. Anomaly detection is a second strong use — a `CPI` that steps without an operational cause is usually a measurement change (PML-AI D7, MCQ 7.2-A), and a machine watching the series sees it before a monthly reader does. Three prohibitions. It must not **choose the `EAC` method**, because that choice sets the size of a cash call and is a commercial position requiring an accountable owner. It must not **certify** — whether a cost is a proper contingency item, whether a variation falls within the contract, whether remaining contingency is adequate are certifications with contractual consequences, and Domain 12's boundary applies. And it must not **write the narrative** accompanying a deteriorating forecast, because a fluent explanation of a variance is exactly what a lender must not receive without a human who can be questioned about it.

Verification: recompute `CPI = EV/AC` and one `EAC` independently at each data date and require reconciliation to the automated pack to the dollar; test PML-AI D7's identity (KA 7.3.4) that `TCPI` to the `BAC/CPI` forecast equals the current `CPI`, as a free check on the forecast engine; confirm by hand that the register mean behind the automated P80 equals Σ `EMV`; and tie the funds-sufficiency line to the facility agent's own statement of undrawn commitment, not to the model's memory of it.

KEY TERMS — KA 8.4

Term	Meaning
Cost to complete (`CTC`)	`EAC − AC`; the money still required on a stated forecast method.
Funds sufficiency / in balance	Funds available (undrawn facility + unused contingency + uncalled equity) ≥ `CTC`.
Cash call	The equity injection a sufficiency failure requires before the next drawdown.
Contingency adequacy test	Remaining contingency ≥ P80 of the remaining register; fails independently of `EAC`.
Cost per month of slip	Escalation on unbought scope + extra interest on drawn debt + deferred `CFADS`.
Stepped damages	Delay damages that rise through the programme to match a rising exposure.
Data spine	One set of numbers, two sets of questions; the controls-financing interface.

SAMPLE MCQS — KA 8.4

MCQ 8.4-A `[8.4.1 · Application]` `BAC` 4,000,000, contingency 300,000, `AC` 2,120,000, `EAC` on `BAC/CPI` 4,416,667. The funds sufficiency position is: - A. a surplus of 100,000 - B. a shortfall of 116,667, requiring an injection before the next drawdown - C. a shortfall of 416,667 - D. in balance, since the `EAC` is within the total funding of 4,300,000

Rationale: Available `= 4,300,000 − 2,120,000 = 2,180,000`; `CTC = 4,416,667 − 2,120,000 = 2,296,667`; shortfall 116,667. A is the position on the budgeted-rate forecast; C is the overrun against `BAC`, not against funding; D is false — 4,416,667 exceeds 4,300,000.

MCQ 8.4-B `[8.4.1 · Analysis]` Remaining contingency is 120,000 and the P80 of the remaining register is 286,185. A cost report shows the forecast inside total funding. The correct reading is: - A. no action is required, since the forecast is inside funding - B. the contingency-adequacy test fails by 166,185 and does so independently of any `EAC` forecast - C. the register must be revised downwards to match the contingency - D. the shortfall is 120,000

Rationale: Adequacy compares remaining provision with remaining exposure and needs only the register and the draw history; 286,185 − 120,000 = 166,185. A misses the second test entirely; C is the corruption the test exists to prevent; D subtracts in the wrong direction.

MCQ 8.4-C `[8.4.2 · Application]` Debt drawn 31,990,655 at 6.0 %; remaining owner-retained scope 922,906 escalating at 3.6 % per annum; annual `CFADS` 6,384,000. One month of slip costs closest to: - A. USD 532,000 - B. USD 694,677 - C. USD 162,677 - D. USD 730,998

Rationale: `922,906 × (1.036^(1/12) − 1) = 2,724`; `31,990,655 × 0.06/12 = 159,953`; `6,384,000/12 = 532,000`; total 694,677. A is the deferred revenue alone; C omits the revenue row; D is the unwrapped variant, where escalation applies to the whole remaining scope (39,045).

MCQ 8.4-D `[8.4.2 · Analysis]` Flat delay damages of 20,000 per day recover 86.37 % of the cost of a month's slip at quarter six, over-recover at financial close and recover 80.86 % at COD. The best negotiating response is: - A. accept the rate — it recovers most of the cost - B. seek a higher flat rate calibrated to the COD figure - C. seek a stepped rate rising through the programme, since the exposure rises monotonically as the drawn balance builds - D. remove damages and rely on contingency

Rationale: A flat rate cannot fit a cost that moves 34.3 % across the programme; a stepped rate matches exposure and is cheaper for a contractor to accept than a flat rate set at the maximum. A ignores the tail; B over-recovers early and is resisted for that reason; D abandons the transfer altogether.

SELF-CHECK — KA 8.4

1. What does a lender compute from an *EAC* that a controller does not? — `CTC = EAC − AC`, then funds sufficiency against undrawn facility, unused contingency and uncalled equity — producing a cash call with a date rather than a forecast range.
2. Why was Auriga's shortfall predictable at sanction? — The 300,000 funded was a P53.5 provision against a register whose P80 was 490,624; the 416,667 overrun is a P70.8 event, normal behaviour against a provision nobody had priced.
3. Which row of the slip cost is largest, and which is the wrap worth? — Deferred `CFADS` at 76.58 %; the wrap saves 36,321 per month on the escalation row (2,724 rather than 39,045).

Advanced topics — Domain 8

8.A.1 Correlation, and why the independent P80 is the optimistic one

The aggregation of 8.3.2 assumes independence, and construction risks are systematically not independent: a single adverse ground condition drives geotechnical cost, programme, dewatering and third-party interface simultaneously; a single supply-chain event moves several package prices at once. The consequence is directional and knowable. Variance under positive correlation is $\Sigma \sigma_i^2 + 2 \Sigma_{i<j} \rho_{ij} \sigma_i \sigma_j$, so any positive ρ raises σ and therefore raises the P80, while the mean is unchanged — **correlation never changes what you expect to pay and always changes what you must provide.** On Kestrel's register, treating C1 and C2 as correlated rather than independent raises the provision without touching the 2,690,000 mean, which is why a P80 quoted without a statement about correlation should be read as a lower bound. Practically, three habits: name the common drivers (ground, weather, a shared supplier, a single approval authority) and group the items that share them; run the simulation with a correlation matrix rather than an independence assumption; and where the matrix cannot be defended, state the independence assumption explicitly in the paper rather than letting the reader infer sophistication that is not there.

8.A.2 Contingency drawdown as a monitored curve

A funded contingency should be expected to deplete, and the useful control is not its balance but the relationship between two curves: contingency consumed and register retired. Plot both as percentages of their sanction values on the same axes and the diagnosis is immediate. Consumption tracking retirement is a healthy project spending its provision on the risks it provided for. Consumption ahead of retirement is the classic overrun signature — money going out against events that were not in the register, which means either the register was incomplete or the draws are not proper contingency items. Retirement ahead of consumption is the release case of 8.3.3, worth 1,315,141 on Kestrel when C1 closed, and it is the case organisations handle worst: released contingency is rarely given back, because nobody is rewarded for returning money. The financing answer is to write the release into the documents — a defined recalculation of the required provision at stated milestones, with the excess applied to prepayment or released to distribution. That converts a behavioural problem into a mechanical one.

8.A.3 The reviewer's cost-and-contingency eye

Testable invariants for any construction cost and contingency model:

- Every contingency percentage is accompanied by the **estimate class** of the base it is a percentage of; a percentage without a class is unreviewable (8.1.2).
- The four cost families are separately identifiable, and operating cost appears **only** inside `CFADS`, never in the sources and uses (8.1.1).
- Lifecycle cost is present, dated, escalated at the same rate as operating cost, and matched by a reserve whose deposit schedule funds the **first** overhaul before it falls due (8.1.3).
- Capitalised interest reproduces the **area rule** `r_q × g × S × Σ cum(t−1)` to the dollar; a mismatch localises the defect to rate, balance convention, draw timing or gearing (8.2.2).
- Escalation is applied **period by period to the profile**, never to a total; escalation on fixed-price wrapped scope is zero and on retained scope is not (8.2.3).
- Every escalation rate names an index, a source, a base date and a human owner, and construction and revenue escalation are distinct assumptions; contingency states a **confidence level**, the register's mean equals Σ `EMV`, and the independence assumption is stated or a correlation matrix is documented (8.2.3, 8.3.2, 8.A.1).
- Range-based and register-based provisions are both computed, the higher is taken, and the contracting structure that eliminates one of them is named (8.3.2).
- Contingency consumed is trended against **risk retired**, not against elapsed time (8.A.2).
- `CTC = EAC − AC` on a **named** `EAC` method, and the funds sufficiency line ties to the facility agent's undrawn commitment, not to the model (8.4.1).
- The contingency-adequacy test is run separately from the sufficiency test, because it fails earlier and independently (8.4.1).
- Delay cost has three rows, and the interest row uses the **balance drawn at the date of the slip**, not the full facility (8.4.2).
- Any capitalised overrun or delay cost is carried through to the instalment, the `DSCR`, the covenant trigger in cash and the headroom (8.3.2, 8.4.2, Domain 10).

Industry variations — Domain 8

- **Water and desalination.** Short, well-characterised membrane and dosing cycles make 8.1.3's arithmetic unusually reliable and the maintenance reserve unusually large as a share of `CFADS` — 10.10 % on Kestrel. Intake and marine works carry the ground and weather-standby risks that dominate the retained register, so the contingency negotiation is about who owns the geotechnical basis.

- **Solar and wind generation.** Capital cost is modular and heavily benchmarked, so ranges narrow earlier than in process plant and Stage D provisions are genuinely thin. The estimating battleground moves to grid connection and civil works, where one interface can carry more range than all the generating equipment; lifecycle cost turns on the timing of a few component replacements rather than their price.
- **Transport concessions.** Earthworks and structures make ground risk the largest line and quantity growth against a Stage C estimate the norm, so unwrapped procurement forces 8.3.2's range-based provision rather than the register-based one. Long construction periods magnify both the escalation exposure and the shape effect of 8.2.2.
- **Digital infrastructure.** Fit-out is phased against demand, so the profile is a sequence of S-curves and the area rule must be applied per phase. Short equipment lifecycles compress the tail and make the lifecycle charge a far larger share of operating economics than in civil assets.
- **Process industry and petrochemicals.** Estimate classes matter most here, because scope definition genuinely drives outturn and the ladder's factor of 6.25 is not theoretical. Long-lead equipment pulls escalation exposure early, pushing 8.2.3's breakeven towards front-loading; turnaround and inspection cycles drive lifecycle cost and are regulated in many jurisdictions, so their timing is less discretionary than elsewhere.
- **Social infrastructure and availability PPPs.** Capital cost is comparatively predictable and the lifecycle obligation is the commercial substance: a 25- or 30-year hard-facilities obligation with a handback condition, priced at bid and unalterable afterwards. Getting 8.1.3 wrong here is not a reserve problem; it is a bid that cannot be performed.

Case study — Domain 8: the contingency that was a percentage (transport)

Situation. A sponsor group bid a 34-kilometre highway concession in an emerging market. The capital estimate at bid was a **Stage C** feasibility estimate of **USD 420,000,000**, carrying a stated accuracy range of **−15 % / +30 %**. The financing envelope allowed contingency at the group's standing policy of **8 % of base cost — 33,600,000** — and the works were let as **four separate packages** with the concessionaire managing interfaces, because splitting the packages had produced the lowest bid prices.

What happened. The two decisions were incompatible and the incompatibility was computable at bid. An 8 % provision is a Stage E number (8.1.2); the project had a Stage C estimate and, having declined a wrap, had retained the base-estimate uncertainty the range describes. Translating the range through a triangular distribution with minimum 357,000,000, mode 420,000,000 and maximum 546,000,000 gives a **P80 outturn of 476,986,958 — a contingency requirement of 56,986,958, or 13.57 % of base**. The funded 33,600,000 was a **P64.1** provision against the range (`F = 1 − (92,400,000)²/((189,000,000)(126,000,000)) = 0.6415`). Twenty-one months into a forty-month programme, earthworks quantities on two of the four packages exceeded the bid quantities, the interface between the earthworks and structures packages produced a sequence of compensation events that neither contractor owned, and cumulative certified cost reached **252,000,000** against an earned value of **231,000,000** — a `CPI` of **0.9167**. The `BAC/CPI` forecast was **458,181,818**, an overrun of **38,181,818** against a base of 420,000,000 and **4,581,818** more than the entire funded contingency.

How it resolved. The facility's in-balance test failed at the next certification: funds available were `453,600,000 − 252,000,000 = 201,600,000` against a cost to complete of `458,181,818 − 252,000,000 =`

`206,181,818` — a **4,581,818** shortfall, and the technical adviser could not certify a further contingency draw because, with **26,200,000** of the 33,600,000 already drawn, remaining contingency was **7,400,000** against a remaining register whose P80 exposure was assessed at **19,300,000**. Drawdown stopped. The sponsors funded the shortfall as additional equity of **20,000,000** — deliberately above the 4,581,818 minimum, because a cash call computed to the last dollar fails again on the next report — and accepted a tightened reporting regime and a lenders' quantity surveyor with access to the packages. Against an original equity of 113,400,000 on a 75/25 structure, the injection **increased the sponsors' equity base by 17.6 %** and moved gearing to **71.8/28.2**, diluting the modelled equity return correspondingly.

What the domain teaches here. The contingency was not too small by accident; it was sized by a method that could not see the estimate class or the procurement decision. Both defects of 8.3.3 are visible: the 8 % rule stated no confidence level, so nobody noticed it was a P64.1 provision against a distribution its own estimate had published, and it did not respond when the decision to split the packages transferred the base-estimate uncertainty back to the owner. The cost of the arithmetic nobody did at bid was 20,000,000 of equity — 17.6 % of the sponsors' committed capital — and the whole of it was computable from two documents the sponsors already held: the estimate's stated range and the procurement strategy. **The provision was 23,386,958 short of the P80 its own estimate implied, and the eventual call was 20,000,000.** The arithmetic was not merely available; it was approximately right.

Case study B — Domain 8: the release nobody wanted to take (solar generation)

Situation. A 180 MW solar project reached financial close with a **Stage D** capital estimate of **USD 148,000,000**, a funded contingency of **11,100,000** (7.5 % of base) and a retained risk register of eleven items with a mean exposure of **6,240,000**, a standard deviation of **4,050,926** and a **P80 of 9,649,259**. Two items dominated: a grid-connection energisation risk (p 0.35, impact 6,000,000) and a land-title consolidation risk (p 0.25, impact 4,800,000), together carrying **3,300,000** of the mean and, on their own, **3,536,948** of standard deviation — 76.2 % of the register's total variance from two of eleven lines. The remaining nine carried a mean of 2,940,000 and a standard deviation of **1,974,842**. Contingency was drawable on certification against a **0.60 % per annum** commitment fee on the undrawn balance.

What happened. Fourteen months in, both dominant items retired cleanly: the connection agreement was energised on schedule and the final title parcels were consolidated. Recomputing the register without them gave a mean of **2,940,000**, a standard deviation of **1,974,842** and a **P80 of 4,602,027** — the P80 requirement had fallen by **5,047,232**, more than half, on a single month's news. Against 11,100,000 of contingency of which **1,800,000** had been drawn, the project was holding an undrawn commitment of **9,300,000** against a P80 requirement of 4,602,027 — an excess of **4,697,973**. Nobody proposed releasing it. The project director's position was that contingency should not be reduced while construction was still running; the sponsor's finance function did not challenge it, because the visible cost of the excess was only the commitment fee — **28,188 a year** — which no operating budget felt.

How it resolved. The lenders' technical adviser raised it at the next quarterly review, and the facility's contingency-recalculation clause was invoked at the second construction milestone. The required provision was reset to the P80 of the live register plus a **15 % correlation uplift** — **5,292,331** — and the difference between the undrawn balance and that figure, **4,007,669**, was cancelled from the commitment. The fee saving was modest: **36,069** over the remaining eighteen months. The real prize was what cancellation made impossible. Had the excess instead been drawn

— which is what available money tends to become — it would have added **2,805,368** of senior debt at 70 % gearing, and at the facility's 5.4 % over a 15-year tenor (`AF(0.054, 15) = 10.104624`) that is **277,632** of additional annual debt service for fifteen years, against a project whose entire contingency provision was 11,100,000.

What the domain teaches here. The cost of holding excess contingency while it is undrawn is trivially small, which is exactly why it is never challenged; the cost if it is drawn is permanent, and it lands on the coverage ratio rather than on any construction budget. That asymmetry — 36,069 of fee against 277,632 a year of debt service — is the whole case for a mechanical release. The 15 % correlation uplift is the honest form of 8.A.1: the release was taken, and the independence assumption was not pretended to. And the governance failure is the general one — released contingency requires a mechanism, because no individual is rewarded for handing money back. Write the recalculation into the documents at defined milestones and the release happens; leave it to judgment and the excess is carried to COD every time, and then spent.

Executive perspective — Domain 8

What a project finance director cannot delegate in this domain:

- **The estimate class behind every contingency percentage.** Not the percentage — the class. A 7.59 % provision is prudent at Stage E and negligent at Stage C, and the director is the only person in the room who sees both the estimate and the procurement decision that sets which one applies (8.1.2).
- **The link between the contracting strategy and the contingency line.** They are one decision taken twice, usually by different people in different months. Whoever declines a wrap has just resized the contingency, and someone must say so out loud before the envelope is fixed (8.1.2, Case study A).
- **The confidence level, named.** P80 or P50 is a risk-appetite decision belonging to governance, not to an analyst and never to a model. A provision whose confidence nobody stated is a provision nobody chose — Auriga's 300,000 was P53.5, and behaving normally broke it (8.4.1).
- **The lifecycle obligation and the reserve that funds it.** The director owns the question "what does this asset owe in year seven, and is the money going to be there?" A `DSCR` of 0.4561 in one year is a payment default on an otherwise healthy project, and it is entirely a reserve design question (8.1.3).
- **The release mechanism.** Contingency held after its risk has retired costs almost nothing while it is undrawn — 11,214 a year on Kestrel, 28,188 on Case study B — which is exactly why nobody challenges it, and a great deal if it is then spent: 50.3 % of Kestrel's covenant headroom, or 277,632 a year of debt service for fifteen years in Case study B. The director's job is to have the recalculation written into the documents before anyone has to be brave (8.A.2, Case study B).
- **The interface, scheduled rather than discovered.** One data spine, two questions. A director who first meets a `CPI` of 0.906 in a certifier's refusal to fund has lost the months in which it was cheap, and has paid the relationship cost Domain 10 priced (8.4.3).

Calculation exercises — Domain 8

> **Exercise 8.1** A Stage C estimate of **30,000,000** carries a range of **−15 % / +30 %**. Using a triangular distribution, compute the mean outturn, the P80 outturn and the contingency each

implies; then state the percentile a 10 %-of-base rule buys. Solution. $a = 25{,}500{,}000$, $m = 30{,}000{,}000$, $b = 39{,}000{,}000$. Mean $= (a + m + b)/3 = 31{,}500{,}000$ → contingency **1,500,000 (5.00 %)**. P80 solves $(b - x)^2 = 0.20\ (b - a)(b - m) = 0.20 \times 13{,}500{,}000 \times 9{,}000{,}000 = 2.43 \times 10^{13}$ → $b - x = 4{,}929{,}503$ → $x =$ **34,070,497**, contingency **4,070,497 (13.57 %)**. The 10 % rule gives 3,000,000, i.e. $x = 33{,}000{,}000$; $F = 1 - (6{,}000{,}000)^2/(1.215 \times 10^{14}) = 0.7037$ → **P70.4**. Common error: assuming the percentile depends on project size — it does not. The percentile a percentage rule buys is a function of the range alone, which is why −15/+30 always returns 13.57 % at P80 and P70.4 for a 10 % rule, whatever the base.

Exercise 8.2 **30,000,000** of spend over six quarters, **60 %** debt-funded at **8 %** per annum (2 % per quarter) on opening balances, draws at period end. Profile as planned: 10, 15, 20, 25, 18, 12 per cent. Compute capitalised interest, and the interest under the reversed profile 12, 18, 25, 20, 15, 10. Solution. $r_q \times g \times S = 0.02 \times 0.60 \times 30{,}000{,}000 = 360{,}000$ per unit of area. As planned, Σ cum(t−1) $= 0 + 10 + 25 + 45 + 70 + 88 = 238\ \%$ → 2.3800 → IDC **856,800**. Reversed, $0 + 12 + 30 + 55 + 75 + 90 = 262\ \%$ → 2.6200 → IDC **943,200**. Difference **86,400** on identical total spend and duration. Common error: computing interest on the closing balance, which adds one period of interest to every draw and overstates IDC by $r_q \times g \times S = 360{,}000$ — an error of 42 % on the correct figure, and the single commonest defect in construction models.

Exercise 8.3 A register carries p/impact pairs 0.30/1,500,000 · 0.45/800,000 · 0.20/2,200,000 · 0.25/1,000,000, plus an opportunity 0.20/(500,000), on a base estimate of 22,000,000. Compute the mean, σ and P80, and state the percentile bought by an 8 %-of-base rule. Solution. Mean $= 450{,}000 + 360{,}000 + 440{,}000 + 250{,}000 - 100{,}000 =$ **1,400,000**. Variance $= 0.21\times1{,}500{,}000^2 + 0.2475\times800{,}000^2 + 0.16\times2{,}200{,}000^2 + 0.1875\times1{,}000{,}000^2 + 0.16\times500{,}000^2 = 1{,}632{,}800{,}000{,}000$; σ **1,277,811**; P80 $= 1{,}400{,}000 + 0.8416 \times 1{,}277{,}811 =$ **2,475,405**. The 8 % rule gives **1,760,000**, i.e. $z = 0.2817$ → **P61.1**, leaving a **715,405** shortfall against P80. Common error: omitting the opportunity from the variance. It reduces the mean by 100,000 but adds $0.20 \times 0.80 \times 500{,}000^2 = 40{,}000{,}000{,}000$ to the variance, because an opportunity is an uncertain outcome like any other — dropping it understates σ and therefore the P80.

Exercise 8.4 A project declares a one-month slip with **18,000,000** of debt drawn at **7.0 %**, **2,400,000** of owner-retained remaining scope at then-current prices escalating at **4.2 %** per annum, and annual `CFADS` of **4,200,000**. Delay damages are 12,000 per day (30-day month). Compute the cost of the month, the damages coverage, and the share of cost a revenue-only calibration would have missed. Solution. Monthly escalation factor $1.042^{(1/12)} = 1.00343438$, i.e. 0.3434 %; escalation row $2{,}400{,}000 \times 0.00343438 =$ **8,243**. Interest row $18{,}000{,}000 \times 0.07/12 =$ **105,000**. Revenue row $4{,}200{,}000/12 =$ **350,000**. Total **463,243**. Damages $12{,}000 \times 30 = 360{,}000$ → **77.71 %** coverage; uncovered **103,243**. A calibration on forgone revenue alone targets 350,000 and misses $(8{,}243 + 105{,}000)/463{,}243 =$ **24.45 %** of the cost. Common error: using the full facility amount rather than the balance actually drawn at the date of the slip in the interest row — the error grows through the programme and is largest exactly where the delay exposure is largest.

Exercise 8.5 `BAC` **30,000,000**, funded contingency **2,400,000**, at the data date `AC` **16,800,000**, `EV` **15,600,000**, `PV` **16,000,000**. Compute `CPI`, `SPI`, the three `EAC` forecasts, `CTC` and the funds sufficiency position on each. Solution. $CPI = 15{,}600{,}000/16{,}800{,}000 = 0.928571$; $SPI = 15{,}600{,}000/16{,}000{,}000 = 0.975000$. Available $= 32{,}400{,}000 - 16{,}800{,}000 =$ **15,600,000**. (a) $EAC = 16{,}800{,}000 + 14{,}400{,}000 = 31{,}200{,}000$, `CTC` **14,400,000**, surplus **+1,200,000**. (b) $EAC = 30{,}000{,}000/0.928571 = 32{,}307{,}692$, `CTC` **15,507,692**, surplus **+92,308**. (c) $EAC = 16{,}800{,}000 + 14{,}400{,}000/(0.928571 \times 0.975) = 32{,}705{,}325$, `CTC` **15,905,325**, shortfall **(305,325)**. Common error:

> reporting the (b) result as "in balance" without qualification. A surplus of 92,308 is **0.28 %** of the facility — inside the rounding of a monthly report, not a margin — and on the `CPI × SPI` forecast the same project is 305,325 short. Sufficiency conclusions must be stated with the method that produced them.

Practitioner's toolkit — Domain 8

Adoption-ready artefacts; adapt headings to your organisation, then keep them stable.

Toolkit 8.T.1 — Contingency basis note (one page, per sanction and per recalculation)

One page that must exist before any contingency figure is quoted. **Base estimate**: amount, base date, price basis, preparer. **Estimate class**: stage, accuracy range, and the engineering deliverables complete that justify the class, with a named owner. **Procurement**: wrapped or unwrapped, package count, interface owner — and the explicit statement of which uncertainties the contract transfers. **Range-based provision**: distribution used, parameters, mean, P50, P80, P95. **Register-based provision**: item count, mean = Σ `EMV`, σ, P80, P95, independence assumption or correlation matrix. **The reconciliation**: which of the two governs and why. **The funded figure**: amount, percentage of base, and the percentile it represents on the governing basis. **Contingent support**: amount, form, trigger, obligor, enforceable against what balance sheet. **Approval**: confidence level chosen, by whom, on what date. Rule: no contingency number leaves the organisation without this page attached, and the page is reissued at every recalculation milestone.

Toolkit 8.T.2 — Construction cost model check (before any envelope is fixed)

- [ ] Four cost families separately identified; operating cost appears only inside `CFADS`.
- [ ] Base date stated; every escalation rate names an index, a source and a human owner.
- [ ] Escalation applied period-by-period to the profile; zero on wrapped scope, non-zero on retained.
- [ ] Construction and revenue escalation are distinct assumptions.
- [ ] IDC reproduces the area rule `r_q × g × S × Σ cum(t−1)` to the dollar.
- [ ] Interest convention stated (opening / average balance) and paired with the period length.
- [ ] Balancing line identified; it is not a round number; it is tested against 8.T.1's percentile.
- [ ] Lifecycle programme dated, escalated and matched by a reserve deposit schedule that funds the first overhaul before it falls due.
- [ ] Handback or decommissioning obligation identified, with its cash date relative to loan maturity.
- [ ] Sensitivity run on: escalation rate, spend profile shape, contingency drawn, and COD date.
- [ ] Every capitalised overrun scenario carried through to instalment, `DSCR`, covenant trigger in cash and headroom.

Toolkit 8.T.3 — Monthly construction pack, financing annexe

One annexe to the existing controls report — never a second set of numbers. Rows: certified progress and this month's draw request against the funding profile · `PV`, `EV`, `AC`, `CPI`, `SPI` at the data date · the three `EAC` forecasts with assumptions named, and the method the certifier is using · `CTC` **and the funds sufficiency position** against undrawn facility, unused contingency and uncalled equity, tied to the agent's statement · contingency drawn and remaining, trended **against register retired** rather than against time · the register's current P80 and the remaining-adequacy test · forecast COD and the cost per month of slip current at this point in the programme, in three rows · damages coverage at the current exposure · the DSCR, covenant trigger and headroom consequence of any capitalised cost · exceptions, each with a named accountable person. Front line, always: **is the facility in balance, and is the remaining contingency adequate for the remaining risk?**

Exam preparation — Domain 8

What is assessed. The four cost families and where each enters a financing; estimate classes and the contingency a range implies at a stated confidence; why a fixed-price wrap changes the defensible contingency; whole-life cost, the level annual charge and the reserve that prevents a lifecycle payment default; the area rule and the price of shape; escalation applied to a profile, base dates and the shape-neutrality breakeven; contingency sized from a register and reconciled against a range-based provision; the two computable defects of the percentage method; contingent support and why it is worth less than funded cash; the conversion of an EAC into a CTC, a sufficiency test and a cash call; the contingency-adequacy test as a separate and earlier failure; and the three components of a month's slip, their movement through the programme, and the coverage consequence of capitalising any of them.

The calculations to do under time pressure. Contingency implied by an accuracy range at the upper bound and at P80 via a triangular distribution (8.1.2, 8.3.2, Exercise 8.1) · the percentile a percentage rule buys, on both bases (8.3.3) · present value and level annual charge of a lifecycle programme, and the year-of-overhaul DSCR without a reserve (8.1.3) · IDC from the area rule, and the same profile reversed (8.2.2, Exercise 8.2) · escalated spend period-by-period, and the shape-neutral escalation rate `g × r` (8.2.3) · register mean, variance, σ, P80 and P95, including opportunities (8.3.2, Exercise 8.3) · standby support as P95 (5,731,375) less funded contingency (8.3.1) · CTC and funds sufficiency on all three EAC methods (8.4.1, Exercise 8.5) · remaining-register P80 against remaining contingency (8.4.1) · cost per month of slip in three rows, damages coverage, and the DSCR/headroom consequence of capitalising it (8.4.2, Exercise 8.4).

The traps.

- Quoting a contingency percentage without the estimate class of its base — the domain's central defect (8.1.2, MCQ 8.1-A).
- Applying a Stage E provision to a Stage C estimate after declining a wrap (8.1.2, MCQ 8.1-B, Case study A).
- Putting operating cost in the sources and uses, or forgetting that an operating error recurs every year against a fixed headroom (8.1.1, MCQ 8.1-D).
- Omitting lifecycle cost, or funding it at a policy percentage rather than against the next overhaul's date (8.1.3, MCQ 8.1-C).
- Escalating a total rather than the profile — 1,424,815 of overstatement on Kestrel (8.2.3, MCQ 8.2-B).
- Computing IDC on closing rather than opening balances — a 42 % overstatement in Exercise 8.2 (8.2.2, Exercise 8.2).
- Summing worst cases, or funding the mean, and calling either a contingency (8.3.2, MCQ 8.3-A).
- Dropping opportunities from the variance while keeping them in the mean (Exercise 8.3).
- Presenting an independent P80 as conservative when correlation makes it a lower bound (8.A.1).
- Treating contingent support as equivalent to funded contingency (8.3.1, MCQ 8.3-D).

- Holding contingency after its risk has retired, and pricing it as the commitment fee (11,214 a year on Kestrel) rather than as the coverage loss if it is drawn — 50.3 % of headroom (8.3.3, MCQ 8.3-C, Case study B).
- Running the sufficiency test and not the contingency-adequacy test, which fails earlier and independently (8.4.1, MCQ 8.4-B).
- Pricing a delay on forgone revenue alone — 24.45 % of the cost missed in Exercise 8.4 — or using the full facility rather than the drawn balance in the interest row (8.4.2, MCQ 8.4-C).
- Accepting a flat damages rate against an exposure that moves 34.3 % across the programme (8.4.2, MCQ 8.4-D).
- Reporting a 0.28 %-of-facility surplus as "in balance" without naming the `EAC` method (Exercise 8.5).

How the domain connects. Backward: Domain 3 supplied the compounding and `AF(r, n)` this domain escalates and annuitises with; Domain 4 supplied the `EAV` machinery behind the level lifecycle charge and the appraisal rate; Domain 5 supplied the EPC wrap that makes a thin contingency defensible and the at-COD delay arithmetic this domain generalises to the whole programme; Domain 6 supplied the sources-and-uses statement, the funding profile and the 3,645,403 balancing line this domain finally justifies; Domain 7 supplied the revenue escalation that must not be confused with cost escalation. Forward: Domain 9 supplies the standby tranches and concessional money that fund the contingent support of 8.3.1; Domain 10 tests every coverage consequence computed here; Domain 11 allocates the risks the register enumerates and Domain 12 documents the wrap that transfers them; Domain 13's model audit checks the area rule and the escalation basis; **Domain 14 is this domain in operation** — draw requests, cost-to-complete reports and contingency certification, month by month. Across the suite: PML-AI Domain 6 owns the schedule that produces the S-curve, PML-AI Domain 7 owns the earned value this domain transforms, and PML-AI Domain 8 owns the register and the confidence- level arithmetic. **The single most useful idea to carry between the two books is that a forecast and a funding position are different objects, and the transformation between them — `CTC`, sufficiency, coverage — is the finance leader's contribution to a monthly report they did not write.**

Domain 8 summary

A contingency percentage is meaningless without the estimate class it was struck against. On a 48,000,000 base, the same estimate implies a provision of 24,000,000 at screening and 3,840,000 at control-estimate maturity — a factor of 6.25 — so Kestrel's funded **3,645,403 (7.59 %)** is defensible only because a fixed-price, date-certain EPC wrap moved a Stage C estimate to a Stage E position; procured unwrapped, the same works need of the order of **14,400,000** and a 60,000,000 envelope does not fund them. Lifecycle cost is the family most often missed: Kestrel's membrane and pump programme is worth **6,881,021** in present value, **644,606** as a level annual charge and **10.10 %** of `CFADS`, and without a reserve the year-seven `DSCR` is **0.4561** — a payment default, not a breach, on an otherwise healthy project. The schedule prices the financing through its shape: capitalised interest obeys the area rule `IDC = r_q × g × S × Σ cum(t−1)`, so the same 48,000,000 over the same eight quarters costs **1,345,680** back-loaded, **1,607,760** on the S-curve and **2,000,880** front-loaded — a **655,200** spread — while escalation runs the other way by **574,146**, leaving a total spread of only **86,681** and a shape-neutral escalation rate of **4.1659 %**, almost exactly the heuristic `g × r = 4.20 %`. Contingency sized from the retained register (mean **2,690,000**, σ **1,848,973**) supports **4,246,095** at P80, against **6,512,795** from the Stage C range at P80: two defensible numbers 2,266,700 apart, reconciled by taking the higher and naming the contract that eliminated the other. The funded amount is a **P69.7** provision and **600,692** short of the register's P80, which capitalised would take the `DSCR` to **1.2564** and headroom to **286,459**, down **23.1 %**, for twelve years. The percentage method fails in two computable ways: 10 % of base is **4,800,000**, a **P87.3** promise against the register and a **P70.4** promise against the range — the rule cannot say which — and it does not move when risk retires, leaving **1,869,045** of excess after one item closes, which costs **11,214** a year while undrawn and, if drawn, adds 1,308,332 of senior debt, takes the `DSCR` to **1.2358** and removes **50.3 %** of covenant headroom for twelve years. Finally, the bridge: a controller's `EAC` family becomes a lender's cash call. Auriga's three forecasts give costs to complete of 2,080,000, 2,296,667 and 2,488,056 against 2,180,000 available — in balance, then **116,667** short, then **308,056** short — while the contingency-adequacy test fails independently and earlier, 120,000 remaining against a remaining P80 of **286,185**; and the shortfall was predictable at sanction, because a **P53.5** provision met a **P70.8** event. A month of slip declared at quarter six costs **694,677** — 0.39 % escalation, 23.03 % interest, 76.58 % deferred revenue — rising to **742,000** at COD, so a flat damages rate of 20,000 per day covers the cost only until quarter three; capitalising three months of the cost components removes **18.8 %** of Kestrel's covenant headroom permanently. Domain 9 funds the contingent support this domain sizes; Domain 10 tests every ratio it moves; Domain 14 runs it monthly.

09

DOMAIN 9

Funding Structure and Sources of Capital

Group: Structuring and modelling (Part Two, Domain 5 of 5). **Target:** ~75 pages. **Binds to:** the PCI Book Pattern Specification and the shared registries (`docs/books/registries/`). This domain is the home of `WACC` and of the cost-of-capital components beneath it — after-tax cost of debt, cost of equity, blended cost of a multi-tranche stack — and it is where the 8.0 % that Domain 4 took as **given** is finally derived. It uses Domain 3's `AF(r, n)` machinery and Domain 10's `DSCR`/`LLCR` machinery without re-deriving either. British English; USD (+SAR where useful, indicative `USD 1 ≈ SAR 3.75`). Tax, accounting and legal treatments described here are **illustrative and jurisdiction-specific**; none is presented as universal.

Why this domain exists

Domain 4 discounted Kestrel's cash flows at 8.0 % because the board said 8.0 %. Domain 5 asked whether the project was bankable, Domain 6 built the model, Domains 7 and 8 filled it with revenue and cost. None of them asked the question this domain exists to answer: **where does the money come from, what does each source cost, and who decides the mix?** A hurdle rate inherited rather than derived is the largest single unexamined assumption a project finance leader can carry.

The domain's central claim is deliberately contrarian. In corporate finance, capital structure is chosen by minimising the weighted average cost of capital. **In project finance it is not.** Moving Kestrel from 60 % to 80 % gearing cuts its `WACC` by **20.4 basis points** while lifting the sponsors' equity return by **184.6 basis points** and destroying **0.3716** of `DSCR` — and the 1.20× covenant fails outright above 74.34 % gearing. The structure is chosen by the **coverage constraint**, negotiated between people whose returns move in opposite directions, and the cost of capital is the consequence of that negotiation rather than its objective. Everything here follows from that inversion: equity and shareholder instruments and what they cost (KA 9.1); the debt stack from senior to mezzanine to bonds and the blended arithmetic across it (KA 9.2); the specialist sources — Islamic finance structures, export credit, development finance — that change tenor and price in ways commercial banks cannot (KA 9.3); and the public-sector and market instruments that shift the answer again (KA 9.4).

Learning objectives. After this domain a candidate can: distinguish the forms of project equity and shareholder instruments and state what each does to coverage, tax and control; build a cost of equity from a documented premium build-up and re-lever a beta for a project's own structure; compute an after-tax cost of debt and a project `WACC`, and say precisely why a project `WACC` is not the sponsor's corporate `WACC`; blend a senior/mezzanine/equity stack with correct weights and tax treatment; quantify the leverage trade-off on both faces and identify the gearing at which each covenant threshold binds; describe subordinated, mezzanine and bond instruments including negative arbitrage; describe Islamic finance structures in economic terms and compare them with conventional debt on an all-in basis; compute the all-in cost of an export-credit or development-finance tranche and separate that cost effect from the debt-capacity effect of tenor; compute a grant's effect on WACC, DSCR and equity return and state the treatment choice that drives it; test sustainability-linked pricing against its own verification cost; compute and decompose a refinancing gain and price a gain-share obligation; and govern AI-assisted structuring so that no structure is recommended on an unverified cost of capital.

The master financing. Kestrel Water SPC continues from Domains 1–8 and 10. Capex is **USD 60,000,000**, funded 70/30 as **USD 42,000,000** of senior debt at **6.0 % over 12 years** (annual instalment **USD 5,009,635.23**; year-one interest **2,520,000**, principal **2,489,635**) plus **USD 18,000,000** of equity. Operating life is **25 years**. Documented CFADS is **USD 6,384,000** (6,984,000 before working-capital movements), against EBITDA of **7,500,000** and EBIT of **5,100,000**; the resulting ratios are DSCR **1.2743** = LLCR **1.2743**, PLCR **1.9431** (Domain 10, KA 10.2). Domain 4's appraisal at 8 % gave NPV **+16,179,360**, IRR **12.19 %**, MIRR **9.73 %**, PI **1.270**.

Two Kestrel cash-flow streams appear in this volume and this domain keeps them apart. Domain 4's appraisal stream — net inflows of **USD 8,900,000 a year for fifteen years** — is the pre-financing screening forecast against which a **discount rate** is applied. Domain 10's documented CFADS of **6,384,000**, held level over the 25-year life on that domain's stated illustrative convention, is the stream against which **coverage** is tested and from which **distributions** are paid. This domain applies costs of capital to the first and computes coverage and equity returns on the second, never mixing them inside one calculation; equity returns quoted on the level-CFADS basis are labelled as such. Their divergence is not an inconsistency to be hidden — KA 9.1.4 turns it into the domain's most useful breakeven.

The tax rate, derived rather than assumed. Every after-tax cost below depends on it, and the thread fixes it. EBITDA **7,500,000** less CFADS-before-working-capital **6,984,000** leaves cash tax of **516,000**; taxable profit is EBIT **5,100,000** less interest **2,520,000** = **2,580,000**; the implied rate is `516,000 / 2,580,000 =` **20.0 %** exactly, and depreciation reconciles as `EBITDA − EBIT =` **2,400,000**. Kestrel's year-one interest tax shield is `2,520,000 × 20 % =` **504,000**. A tax rate is a jurisdictional fact and an interest deduction a jurisdictional privilege: both come from the project's actual tax position (Domain 2), never from a modelling convention.

KNOWLEDGE AREA 9.1

Equity and shareholder instruments

Topics: 9.1.1 equity in a limited-recourse structure · 9.1.2 shareholder instruments · 9.1.3 the cost of equity, built up · 9.1.4 the leverage trade-off, computed on both faces.

9.1.1 Equity in a limited-recourse structure

Definition. Project equity is the capital that ranks last in the cash waterfall, absorbs the first losses, carries no contractual return, and in exchange holds the residual and the votes. In a limited-recourse structure it is also the only capital whose provider is normally expected to lose the whole of it without the transaction failing — which is why lenders size it as a risk buffer rather than as a funding source.

Three practical distinctions decide how equity behaves in a financing. **Committed versus contributed:** equity is committed at close through a subscription agreement and contributed when called, and lenders care about the enforceability of the commitment far more than its size — a commitment supported by a letter of credit from an investment-grade bank is worth more, in credit terms, than a larger promise from a thinly-capitalised sponsor. **Pro-rata versus back-ended**

funding: equity drawn alongside debt in the agreed ratio, or debt drawn first with equity plugging the final gap; sponsors prefer the latter because it defers the outflow, lenders resist it because it leaves them exposed while the sponsor has nothing at risk, and where they concede it they require a standby letter of credit for the undrawn balance. **The equity bridge loan (EBL):** a construction-period facility secured on the equity commitment rather than on the project, funding the sponsor's contribution and repaid by it at commercial operations date. Because a rate of return is sensitive to timing, this is one of the cheapest available uplifts to a sponsor's headline number — and one of the least understood.

WORKED EXAMPLE

Worked example 9.1.1 — what an equity bridge is actually worth.

1. **Setup.** For this example only, Kestrel's two-year construction period is modelled explicitly: equity of **18,000,000** at t = 0, operations from year 3, `CFADS` **6,384,000**, senior debt service **5,009,635.23** for twelve operating years, then `CFADS` alone to the end of the 25-year operating life. An EBL is offered at **5.0 %**, interest capitalised, repaid from equity at COD.
2. **Formula.** Equity `IRR` is the rate at which the equity cash-flow stream has zero present value. The EBL's capitalised cost is `E × ((1 + r_EBL)^2 − 1)`.
3. **Substitution.** Without the EBL the stream is `(−18,000,000; 0; 0; 1,374,364.77 × 12; 6,384,000 × 13)`. The EBL's cost is `18,000,000 × (1.05² − 1) = 18,000,000 × 0.1025`; the contribution at t = 2 becomes `18,000,000 + 1,845,000`.
4. **Result.** Base equity `IRR` **10.6696 %**. With the EBL, the sponsor injects **19,845,000** at t = 2 and the equity `IRR` rises to **11.6157 %** — an uplift of **94.61 basis points**, bought by paying **1,845,000** of interest.
5. **Interpretation.** The uplift is real but it is timing, not value: the project generates exactly the same cash. The governing test is a clean identity — **an equity bridge is accretive if and only if its rate is below the equity `IRR` it defers**, and at a rate exactly equal to that `IRR` the uplift is precisely zero; solving for the indifference rate here returns **10.6696 %**, the base equity `IRR` itself, to every decimal tested. Three cautions follow. The bridge is **debt**, and a sponsor reporting an EBL-enhanced `IRR` without disclosing it has presented a financing artefact as a project return. It is **recourse to the sponsor**, so it consumes corporate credit capacity the treasury may value above 95 basis points. And a bridge that cannot be repaid at COD — because the commitment has weakened or COD has slipped past the bridge's maturity — converts a timing benefit into a refinancing event at the worst moment in the project's life (Domain 8's delay arithmetic, in funding form).

9.1.2 Shareholder instruments

Equity is rarely a single instrument. The standard family, and what each is for:

Instrument	Economic character	Why it is used
Ordinary share capital	Residual claim, votes, no contractual return	The irreducible risk layer lenders require
Subordinated shareholder loan	Contractual interest and repayment, ranked below all third-party debt	Interest may be deductible; repayment is easier than a capital reduction
Preference shares	Fixed or participating dividend, priority over ordinary equity, usually non-voting	Sizing a passive investor's return without giving control
Sponsor support / contingent equity	An obligation to inject on a defined trigger (cost overrun, ratio breach)	Bridges lender risk without funding it at close
Deferred consideration / development-cost roll-in	Development spend credited as equity at close	Recognises promoter value without cash

The shareholder loan is the instrument that repays study, because it changes the after-tax arithmetic: part of a sponsor's contribution structured as subordinated debt can create a deductible interest expense while remaining, in ranking terms, indistinguishable from equity to third-party lenders.

WORKED EXAMPLE

Worked example 9.1.2 — the shareholder-loan shield, and the ranking that destroys it.

1. **Setup.** Of Kestrel's 18,000,000 of equity, **6,000,000** is provided as a subordinated shareholder loan at **12.0 %**, ranked below the senior facility and serviceable only from distributions. Cash tax rate **20.0 %** (derived above). Discount the shield at the board's 8.0 % over the 25-year life.
2. **Formula.** Annual shield = interest × tax rate. `PV = shield × AF(0.08, 25)`.
3. **Substitution.** Interest `6,000,000 × 12.0 % = 720,000`; shield `720,000 × 20 % = 144,000`; `AF(0.08, 25) = 10.674776`; `144,000 × 10.674776`.
4. **Result.** Annual shield **144,000**; present value **USD 1,537,168 — 25.62 %** of the 6,000,000 tranche, created by a documentation choice rather than by any change in the project.
5. **Interpretation.** A quarter of the tranche's face value in present-value terms explains why shareholder loans are ubiquitous. It is also the most jurisdiction-dependent number in this domain: thin-capitalisation limits, interest-deduction caps expressed as a share of `EBITDA`, transfer-pricing constraints on the rate, and withholding tax on cross-border interest can each reduce the shield to nothing. **No shareholder-loan shield may be modelled without a written tax opinion on the specific structure.** The structural caution is sharper still: the shield exists only while the loan is genuinely subordinated. If the interest is drafted to rank **above** `CFADS` rather than being paid out of distributions, Kestrel's `DSCR` falls from **1.2743** to `6,384,000 / (5,009,635.23 + 720,000) = ` **1.1142** — straight through the 1.20× covenant and the 1.15× lock-up — and the tax saving has bought a default. Serviced correctly, out of distributions of **1,374,364.77**, the same interest is covered **1.9088** times and the senior ratio is untouched. The instrument is safe; the ranking clause is where it becomes dangerous.

9.1.3 The cost of equity, built up

Definition. The cost of equity `k_e` is the return an equity provider requires for bearing the project's residual risk. It is not observed and never will be; it is **constructed**, and the professional obligation is to construct it in a way that is documented, decomposable and challengeable. The build-up form used throughout this volume:

```
k_e = r_f + β_e × ERP + CRP + SP
β_e = β_a × (1 + (1 − T) × D/E)
```

where `r_f` is the risk-free rate (a long-dated government yield in the cash flows' currency), `ERP` the equity risk premium, `β_a` the asset (unlevered) beta of the project's activity, `β_e` the equity beta after re-levering for the project's own capital structure, `CRP` a country risk premium and `SP` a project-specific premium for single-asset concentration and illiquidity. Each term is a judgment; stating them separately is what makes the judgment reviewable.

WORKED EXAMPLE

Worked example 9.1.3 — Kestrel's cost of equity.

1. **Setup.** Illustrative inputs, all to be replaced with the transaction's own: `r_f` **4.10 %** (long-dated sovereign yield in USD), `ERP` **6.00 %**, `β_a` **0.60** (contracted water utility — low systematic exposure because revenue is availability-based, per Domain 7), `CRP` **0.50 %**, `SP` **0.50 %**. Capital structure `D/E = 42,000,000 / 18,000,000`; `T` **20.0 %**.
2. **Formula.** As above.
3. **Substitution.** `D/E = 2.333333`; `(1 − 0.20) × 2.333333 = 1.866667`; `β_e = 0.60 × (1 + 1.866667) = 0.60 × 2.866667`; `β_e × ERP = 1.72 × 6.00`; `k_e = 4.10 + 10.32 + 0.50 + 0.50`.
4. **Result.** `β_e` **1.72**; `k_e` **15.42 %**.
5. **Interpretation.** Read the decomposition, not the total. Of the 15.42 %, **4.10 points** is the time value of money, **10.32 points** is systematic risk amplified by leverage, and **1.00 point** is country and single-asset risk. The leverage term is the largest single component and it is **created by the financing decision** — the first hint of this domain's central claim, that the sponsors' required return is a property of the structure they chose rather than of the plant. Two disciplines follow. `k_e` is **not constant across structures**: re-lever it whenever gearing changes, or every comparison in KA 9.2 is invalid. And the non-beta terms are not re-levered in this convention — a modelling choice with visible consequences in 9.A.1, to be stated and applied uniformly. Sponsors also carry a **target** return, commonly a round 15 % or 16 % set by committee policy rather than by build-up; the gap between a derived 15.42 % and a policy 16 % is a negotiation to be minuted, not a discrepancy to be resolved by adjusting a beta until the two agree.

9.1.4 The leverage trade-off, computed on both faces

Gearing is the only structural lever that moves the sponsor's return and the lender's protection in opposite directions at the same time, which is why it is negotiated last and hardest. The two faces must be computed together or the conversation is not honest.

WORKED EXAMPLE

Worked example 9.1.4 — five gearings, three consequences.

1. **Setup.** Kestrel's 60,000,000 capex funded at 60 %, 65 %, 70 %, 75 % and 80 % senior gearing; loan rate **6.0 %**, tenor **12 years**, `AF(0.06, 12) = 8.383844`; level `CFADS` **6,384,000** over the 25-year life; `k_e` re-levered at each gearing per 9.1.3; after-tax cost of debt **4.80 %** (KA 9.2.1). Equity cash flow is `CFADS` less senior debt service for twelve years, then `CFADS` for thirteen.
2. **Formula.** `DSCR = CFADS ÷ (D ÷ AF(r, n))` (Domain 10, KA 10.2.1); equity `IRR` from the equity stream; `WACC = g × k_d(1 − T) + (1 − g) × k_e`.
3. **Substitution.** At 70 %: `42,000,000 / 8.383844 = 5,009,635.23`; `6,384,000 / 5,009,635.23`; equity stream `(−18,000,000; 1,374,364.77 × 12; 6,384,000 × 13)`; `0.70 × 4.80 + 0.30 × 15.42`.
4. **Result.**

Gearing	Senior debt	Equity	Debt service	DSCR	Distributable yrs 1-12	k_e	Equity IRR	WACC
60 %	36,000,000	24,000,000	4,293,973.06	**1.4867**	2,090,026.94	13.02 %	**11.7685 %**	**8.0880 %**
65 %	39,000,000	21,000,000	4,651,804.15	**1.3724**	1,732,195.85	14.05 %	**12.1206 %**	**8.0370 %**
70 %	42,000,000	18,000,000	5,009,635.23	**1.2743**	1,374,364.77	15.42 %	**12.5311 %**	**7.9860 %**
75 %	45,000,000	15,000,000	5,367,466.32	**1.1894**	1,016,533.68	17.34 %	**13.0193 %**	**7.9350 %**
80 %	48,000,000	12,000,000	5,725,297.41	**1.1151**	658,702.59	20.22 %	**13.6146 %**	**7.8840 %**

1. **Interpretation.** Across twenty points of gearing the sponsors gain **184.61 basis points** of equity return, the project saves **20.40 basis points** of `WACC`, and the lenders lose **0.3716** of coverage. The ratio is the finding: **the equity return gain is 9.05 times the `WACC` gain**, which means gearing in a project financing is overwhelmingly a transfer to equity rather than an efficiency for the project. Priced at the margin, moving from 70 % to 75 % buys **48.82 basis points** of equity `IRR` for **8.50** points of `DSCR` — about **5.75 basis points of equity return per hundredth of coverage surrendered** — and that exchange rate, computed for the actual structure, is the only defensible basis on which to argue a gearing. Now impose Domain 10's covenants and the table stops being a menu. At **75 %** the 1.20× covenant requires `CFADS` of `5,367,466.32 × 1.20 =` **6,440,960**, which exceeds the base-case 6,384,000 — the covenant **fails on the base case, before any stress**. At **80 %** even the 1.15× lock-up requires **6,584,092** and also fails. Precisely: the 1.30× target binds at **41,171,123** of debt (**68.6185 %** gearing, Domain 10's Case A), the 1.20× covenant at **44,602,050** (**74.3367 %**), and the 1.15× lock-up at **46,541,269** (**77.5688 %**). The cost-minimising structure is unavailable and the constraint is not price, it is coverage. That is the domain's central claim, in a table.

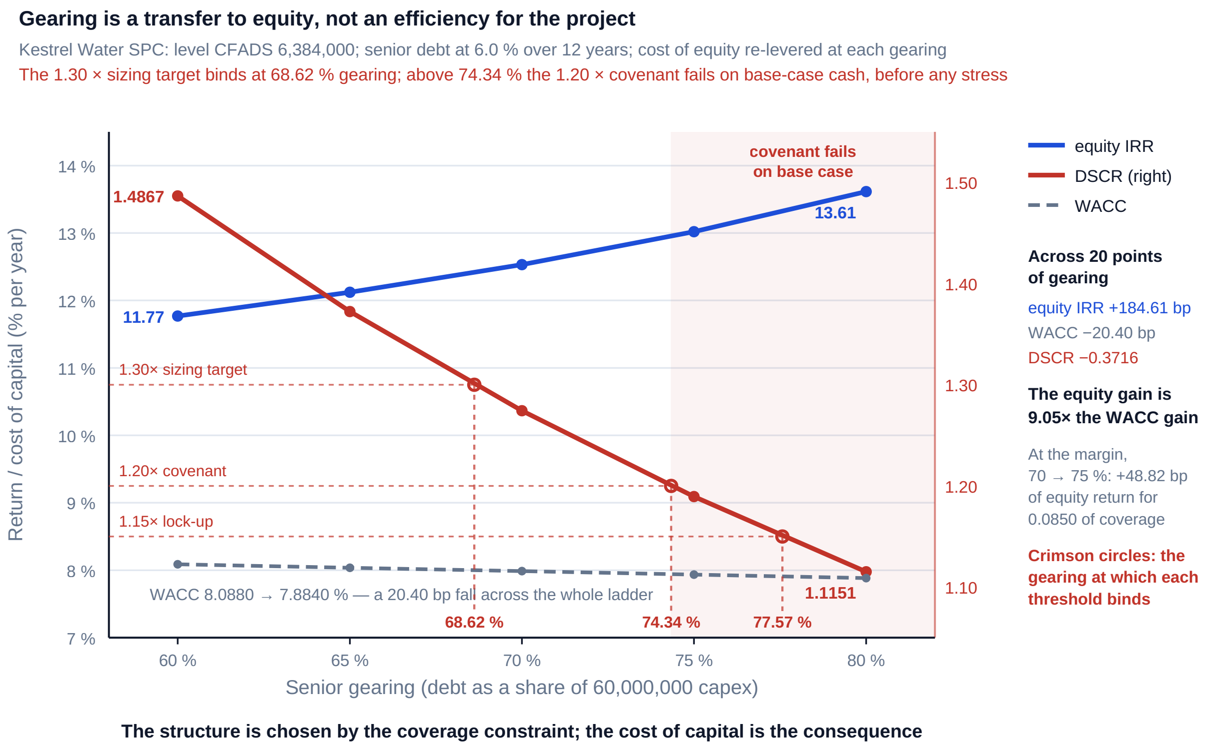

Fig 9.1.4 — The leverage ladder: what gearing buys and what it costs

The two cases, and where the equity return really comes from. On the level-`CFADS` basis the base structure's equity `IRR` is **12.5311 %** — below the **15.42 %** cost of equity derived in 9.1.3. That is not an arithmetic error; it is the difference between a lender's flat "bank case" and a sponsor's escalating "sponsor case", and it is where equity's return actually lives. Solving for the constant `CFADS` escalation that lifts the equity `IRR` to exactly 15.42 % gives **1.7403 %** a year: `CFADS` of 6,384,000 in year one becomes 6,495,101 in year two, 6,608,136 in year three, and `DSCR` improves from **1.2743** to **1.5407** by year twelve. The whole equity case therefore rests on **174 basis points of annual escalation** — a tariff-indexation clause, not a financial-engineering choice. State it that way in the board paper: the equity return is a contracted escalation assumption, and Domain 7's indexation drafting is the asset that produces it.

AI in this KA

Where it earns its place. Re-levering a beta across a gearing ladder, recomputing equity `IRR` for twenty candidate structures, and solving for the escalation breakeven above are mechanical, high-volume and error-prone by hand — exactly the work to delegate.

Where it must not go. It must not choose the premiums. `ERP`, `β_a`, `CRP` and `SP` determine `k_e`, they are unobservable, and a model asked to "estimate the cost of equity" returns a confident figure assembled from whatever premium conventions dominated its training data, with no disclosure of which convention, which market or which date. A cost of equity without a named, dated, owned source for each component is not an input, it is an opinion wearing decimals. Nor may it decide the gearing: that trades a lender's protection against a sponsor's return and belongs to accountable people.

Verification, concretely. Re-lever one case by hand and confirm `β_e`. Confirm the `WACC` weights sum to one and that the debt weight uses the same debt figure as the coverage test. Test the equity-bridge identity — set the bridge rate equal to the base equity `IRR` and confirm the uplift computes to zero; a model reporting a gain there has a timing error. Confirm every quoted equity IRR names its cash-flow case. **AI proposes; the professional verifies, decides and remains accountable.**

KEY TERMS — KA 9.1

Term	Meaning
Committed vs contributed equity	Enforceable obligation to subscribe, versus cash actually injected.
Pro-rata / back-ended funding	Equity drawn alongside debt, versus equity plugging the final gap.
Equity bridge loan (EBL)	Construction facility against the equity commitment; accretive only below the equity IRR it defers.
Subordinated shareholder loan	Sponsor funding as ranked debt; may create a tax shield, dangerous if mis-ranked.
Contingent / sponsor support	Obligation to inject on a defined trigger rather than at close.
k_e	Cost of equity, built up as r_f + β_e × ERP + CRP + SP.
β_a / β_e	Asset (unlevered) beta; equity beta re-levered as β_a(1 + (1 − T)D/E).
Bank case / sponsor case	Lender's conservative cash forecast versus sponsor's escalating forecast; equity's return lives in the gap.

SAMPLE MCQS — KA 9.1

MCQ 9.1-A [9.1.3 · Application] With β_a = 0.60, D/E = 42,000,000 / 18,000,000, T = 20 %, r_f = 4.10 %, ERP = 6.00 %, CRP = 0.50 % and SP = 0.50 %, the cost of equity is: - A. 15.42 % - B. 8.70 % - C. 17.10 % - D. 14.42 %

Rationale: β_e = 0.60 × (1 + 0.80 × 2.333333) = 1.72; k_e = 4.10 + 1.72 × 6.00 + 0.50 + 0.50 = 15.42 %. B uses the **unlevered** beta throughout (4.10 + 0.60 × 6.00 + 1.00 = 8.70) — the error of not re-levering at all. C re-levers **without** the tax adjustment (β_e = 0.60 × 3.333333 = 2.00 → 4.10 + 12.00 + 1.00 = 17.10). D omits the country and single-asset premiums (4.10 + 10.32 = 14.42) — building the beta term correctly and then forgetting the rest of the build-up.

MCQ 9.1-B [9.1.4 · Analysis] Moving Kestrel from 70 % to 80 % gearing changes WACC from 7.9860 % to 7.8840 % and equity IRR from 12.5311 % to 13.6146 %, while DSCR falls from 1.2743 to 1.1151 against a 1.20× covenant. The correct conclusion is: - A. gear to 80 % — the WACC is lower and WACC minimisation is the objective - B. the 80 % structure is not financeable: the 1.20× covenant fails on the base case, and the 10.2 basis points of WACC saved are irrelevant beside a breach - C. gear to 80 % and negotiate a 1.10× covenant, since the equity gain is large - D. the two structures are equivalent because total capital is unchanged

Rationale: At 80 % the covenant requires CFADS of 5,725,297.41 × 1.20 = 6,870,357 against 6,384,000 available — a base-case breach, so the structure does not exist to be optimised (9.1.4). A applies the corporate-finance objective to a constrained problem; C proposes a covenant no lender sizing at 1.30× would grant and ignores that the lock-up also fails; D confuses the funding total with its risk allocation.

MCQ 9.1-C [9.1.1 · Analysis] A sponsor's equity IRR on Kestrel is 10.6696 % without an equity bridge. An EBL is offered at exactly 10.6696 %, interest capitalised over the two-year build. The effect on the reported equity IRR is: - A. it rises, because the equity outflow is deferred - B. it is unchanged: a bridge priced exactly at the equity IRR it defers is value-neutral - C. it falls, because the sponsor pays interest it would otherwise not have paid - D. it rises by 94.61 basis points

Rationale: Deferring an outflow at exactly the rate at which that outflow discounts leaves the IRR identical — the indifference identity of 9.1.1, which the bisection confirms to the tested precision. A states the general direction of a cheap bridge but ignores its pricing; C confuses a cash cost with a rate effect, since the capitalised interest is exactly compensated by the deferral; D quotes the uplift computed at a **5.0 %** bridge rate and applies it as though the bridge rate were irrelevant.

MCQ 9.1-D `[9.1.2 · Analysis]` A 6,000,000 shareholder loan at 12 % is drafted so that its interest ranks **above** CFADS rather than being paid from distributions. The consequence is: - A. an additional tax shield of 144,000 a year and no other effect - B. DSCR falls from 1.2743 to 1.1142, breaching both the 1.20× covenant and the 1.15× lock-up - C. the senior lenders are unaffected because the loan is subordinated in name - D. the loan becomes senior debt for all purposes

Rationale: `6,384,000 / (5,009,635.23 + 720,000) = 1.1142` (9.1.2). A is the shield without the ranking consequence; C mistakes a label for a waterfall position; D overstates — the ranking clause changes the coverage calculation without converting the instrument.

SELF-CHECK — KA 9.1

1. When is an equity bridge accretive? — Only when its rate is below the equity IRR it defers; at equality the uplift is exactly zero.
2. Why must *k_e* be re-levered before any structure comparison? — Because the leverage term is the largest component of k_e (10.32 of Kestrel's 15.42 points); holding it constant across gearings compares incomparable things.
3. State Kestrel's leverage exchange rate at the margin. — About 5.75 basis points of equity IRR per 0.01 of DSCR surrendered, moving from 70 % to 75 %.

KNOWLEDGE AREA 9.2

Senior, subordinated and mezzanine debt; bonds

Topics: 9.2.1 the senior tranche and the after-tax cost of debt · 9.2.2 subordination and mezzanine · 9.2.3 the blended cost of a multi-tranche stack and the project *WACC* · 9.2.4 the capital-markets route.

9.2.1 The senior tranche and the after-tax cost of debt

Definition. Senior debt ranks first in the cash waterfall, holds the security package, sets the covenants, and is therefore the cheapest and largest tranche in almost every project financing. Its cost to the borrower has three layers: the **base rate** (a floating reference rate, or a fixed rate achieved by swapping), the **credit margin** (basis points over base, stepping up or down with project phase and sometimes with coverage), and **fees** — arrangement, commitment on undrawn amounts, agency, and any hedging cost. The headline rate is never the cost; 9.3 makes that point quantitatively.

The after-tax cost of debt is what enters WACC, because interest is (in many but not all jurisdictions) deductible:

```
k_d(after tax) = k_d × (1 − T)
```

For Kestrel: `6.00 % × (1 − 0.20) =` **4.80 %**. The shield is worth **504,000** in year one (2,520,000 × 20 %) and declines with the interest profile, which matters because a project's amortising debt front-loads the shield — a level after-tax rate is a simplification, and where tax losses, carry-forwards or interest-limitation rules bite it is a material simplification that the model must handle explicitly (Domain 6, KA 6.2).

Three properties distinguish senior project debt from corporate borrowing: it is **drawn progressively** against certified construction spend, so commitment fees and interest during construction are real costs (Domain 14); it is **covenanted on cash** rather than on accounting measures (Domain 10, KA 10.4.1); and its **tenor is bounded by the concession or offtake term**, which is why Domain 10's sizing levers include tenor but not appetite.

9.2.2 Subordination and mezzanine

Definition. Subordinated debt ranks behind senior debt in the waterfall and in enforcement, and is compensated for that position with a materially higher return. **Mezzanine** describes instruments sitting between senior debt and ordinary equity: subordinated loans, high-yield notes, payment-in-kind instruments, and hybrids with equity participation (warrants, conversion rights, or a return that steps up if the project outperforms).

Mezzanine exists because there is a gap between what senior lenders will advance and what sponsors will fund with equity, and it is priced in that gap — above senior margins, below the equity return, at a level reflecting a genuine risk of total loss with no upside beyond the coupon. Two placements matter enormously and are routinely confused. **Project-level (SPV) mezzanine** sits inside the borrowing entity, so its debt service appears in the project's own coverage calculation; senior lenders normally prohibit it or confine it to a defined basket, precisely because it consumes the `CFADS` their covenant measures. **HoldCo mezzanine** sits above the SPV in the sponsors' holding company and is serviced from **distributions**: invisible to the SPV's `DSCR`, which is why sponsors prefer it, but fully exposed to the distribution lock-up of Domain 10, KA 10.4.2 — the single most under-appreciated risk in the instrument, which Case study A quantifies.

9.2.3 The blended cost of a multi-tranche stack, and the project `WACC`

Definition. The weighted average cost of capital is the weighted mean of each tranche's cost, weighted by its share of total capital, with debt-like tranches taken **after tax**:

$$\mathrm{WACC} = \Sigma\ (V_i\ /\ V) \times k_i \qquad \text{with } k_i = k_i(\text{pre-tax}) \times (1 - T) \text{ for deductible tranches}$$

Two rules do all the damage when broken. **Weights are market or funding values of the capital actually deployed**, not book equity or authorised capital. And **only genuinely deductible costs are tax-adjusted** — equity never is, and mezzanine may or may not be, depending on the instrument and the jurisdiction.

WORKED EXAMPLE

Worked example 9.2.3 — three structures priced, and the reconciliation of Domain 4's 8 %.

1. **Setup.** Kestrel's 60,000,000 funded three ways. **A (base):** senior 42,000,000 at 6.00 %, equity 18,000,000. **B (mezzanine displacing senior):** senior 36,000,000 at 5.70 % — the margin tightens because 9,000,000 of junior capital now sits beneath it — mezzanine 9,000,000 at 11.50 % amortising over 15 years, equity 15,000,000. **C (mezzanine displacing equity):** senior 42,000,000 at 6.00 %, mezzanine 6,000,000 at 11.50 % over 15 years, equity 12,000,000. Tax 20.0 %; mezzanine interest assumed deductible (a jurisdictional assumption, flagged); `k_e` re-levered at each structure per 9.1.3 with all debt-like tranches counted as debt; `AF(0.057, 12) = 8.523470`, `AF(0.115, 15) = 6.996708`.
2. **Formula.** `WACC` as above; `DSCR` per Domain 10, KA 10.2.1, computed both on senior debt service alone and on total debt service; equity `IRR` from the residual stream.
3. **Substitution.** B: senior service `36,000,000 / 8.523470 = 4,223,631.91`; mezzanine `9,000,000 / 6.996708 = 1,286,319.25`; `k_e` at `D/E = 45,000,000/15,000,000 = 3.0` gives `β_e = 0.60 × (1 + 0.80 × 3.0) = 2.04` and `k_e = 5.10 + 2.04 × 6.00 = 17.34 %`; `WACC = (36 × 4.56
4. 9 × 9.20 + 15 × 17.34) / 60`.
5. **Result.**

	A — base 70/30	B — mezz displaces senior	C — mezz displaces equity
Senior	42,000,000 @ 6.00 %	36,000,000 @ 5.70 %	42,000,000 @ 6.00 %
Mezzanine	—	9,000,000 @ 11.50 %	6,000,000 @ 11.50 %
Equity	18,000,000	15,000,000	12,000,000
`k_e` (re-levered)	15.42 %	17.34 %	20.22 %
Senior debt service	5,009,635.23	4,223,631.91	5,009,635.23
Mezzanine service	—	1,286,319.25	857,546.17
`DSCR` — senior only	**1.2743**	**1.5115**	**1.2743**
`DSCR` — total service	**1.2743**	**1.1586**	**1.0881**
Equity `IRR`	**12.5311 %**	**12.0824 %**	**12.7378 %**
`WACC`	**7.9860 %**	**8.4510 %**	**8.3240 %**

1. **Interpretation. Mezzanine raises the blended cost in both directions** — by 46.50 basis points in B and 33.80 in C — and the reason is structural: mezzanine's after-tax cost of **9.20 %** sits far above senior's **4.80 %**, and the equity left underneath it is more levered and therefore dearer (17.34 % and 20.22 % against 15.42 %). Anyone selling mezzanine as a route to cheaper capital is selling arithmetic that does not exist. What it actually buys depends on **which tranche it displaces**. In B it displaces senior debt and buys the senior lenders comfort — `DSCR` **1.5115** against 1.2743, inside the 1.30× target Domain 10's Case A could not reach — at the price of 46.50 basis points of `WACC` and 44.87 basis points of equity return, because 11.50 % money replaced 6.00 % money. In C it displaces equity and buys the sponsors 20.67 basis points of equity `IRR`, but total coverage collapses to **1.0881**, below any covenant in the facility, so C exists only if the mezzanine sits at HoldCo level and is invisible to the SPV test. **Mezzanine is bought for**

coverage relief or for equity uplift, never for a lower cost of capital, and the tranche it displaces decides which.

Now the reconciliation this volume has owed since Domain 4. Substitute Kestrel's re-levering into the `WACC` definition and the whole ladder collapses to a straight line:

```
WACC(g) = [F + ERP × β_a] − g × [F + ERP × β_a × T − k_d(1 − T)]
        = 8.70 % − 1.02 % × g          with F = r_f + CRP + SP = 5.10 %
```

Every entry in the 9.1.4 table satisfies it exactly: 8.0880 at g = 0.60, 7.9860 at 0.70, 7.8840 at 0.80. The slope tells the real story. Its two parts are `F − k_d(1 − T) = 5.10 − 4.80 = 0.30` and the tax-shield term `ERP × β_a × T = 3.60 × 0.20 = 0.72`. **Because the non-systematic part of the equity build-up (5.10 %) barely exceeds the after-tax cost of senior debt (4.80 %), leverage buys the project almost nothing** — 1.02 basis points of `WACC` per point of gearing, most of it supplied by the tax shield. The benefit of gearing accrues to equity as return, not to the project as cheaper capital.

Now evaluate the line where the project is actually financeable. Domain 10 established that a 1.30× target caps senior debt at **41,171,123 — 68.6185 %** gearing, with equity of **18,828,877**. At that gearing `β_e` is **1.649565**, `k_e` is **14.9974 %**, and

```
WACC = 8.70 % − 1.02 % × 0.686185 = 8.0001 %
```

Domain 4's given 8.0 % was exactly right — and for a reason nobody had written down. It is not a policy round number: it is the project's own weighted average cost of capital at the maximum gearing its coverage requirement permits. The board's rate and the credit committee's ratio were describing the same structure from opposite sides of the table. Two consequences. The rate is right **only for that structure**: the actual 70/30 proposal prices at **7.9860 %**, raising the appraisal `NPV` from 16,179,360 to **16,244,525** (**+65,164**) — a harmless difference, but only because it was checked. And the cost-minimising 80 % structure prices at **7.8840 %**, so the coverage constraint costs **11.61 basis points** of `WACC`. That is the honest price of bankability, and it is small — the best argument available for not fighting the coverage requirement.

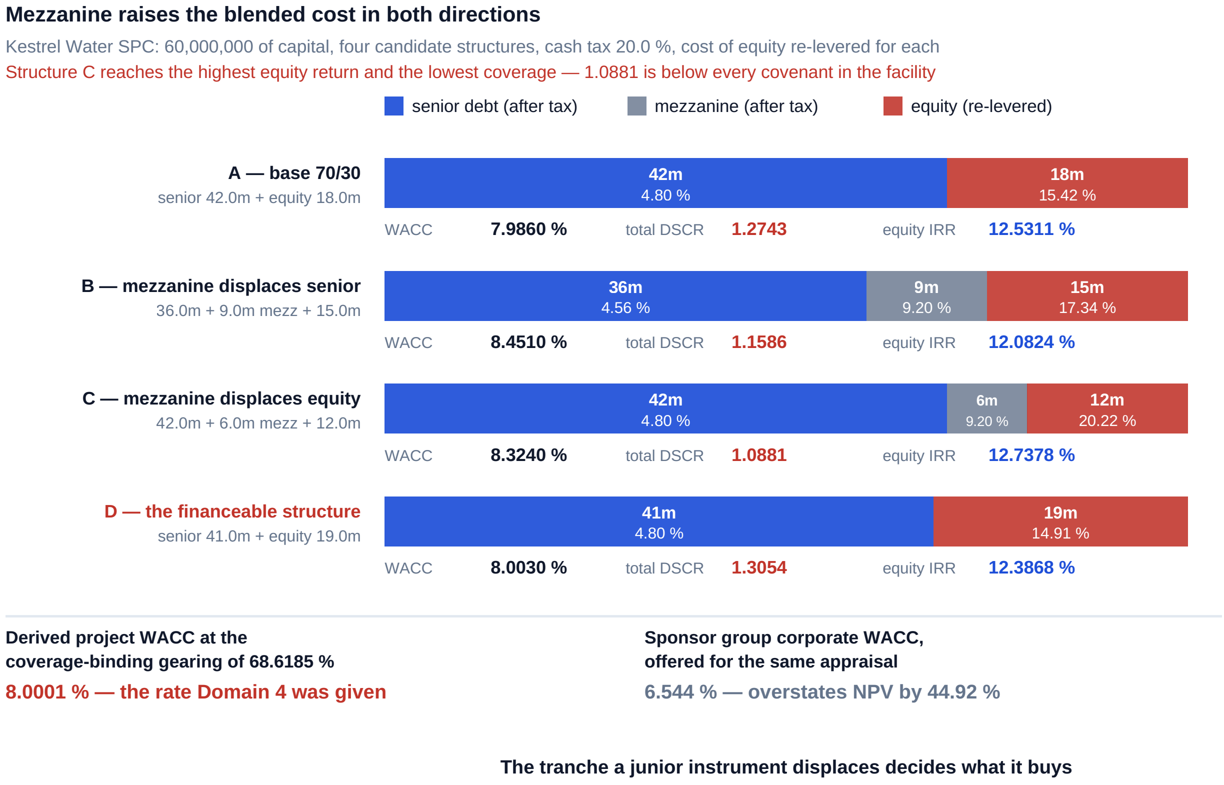

Fig 9.2.3 — Four capital structures, priced

9.2.4 The capital-markets route

Definition. Instead of a bank loan, a project may issue **bonds** — notes sold to institutional investors, publicly or by private placement, with or without a rating and with or without a third-party guarantee ("wrapped"). Three axes drive the choice. **Tenor:** insurers and pension funds matching long liabilities will hold 20- to 30-year paper that few banks will, and for a long concession that is decisive (9.3.3 quantifies why tenor beats rate). **Price and rating:** rated investment-grade paper can price below bank margins and sub-investment grade well above, and the rating process imposes disclosure and a published opinion on the project's credit — valuable or unwelcome depending on circumstances. **Negative arbitrage:** bonds are normally drawn in full at close while construction spends over years, so idle proceeds earn a deposit rate far below the coupon — a real cost that progressively-drawn bank debt does not incur.

WORKED EXAMPLE

Worked example 9.2.4 — the price of drawing early.

1. **Setup.** Kestrel's 42,000,000 raised as a bond at a 6.0 % coupon, fully drawn at close, spent evenly over a two-year construction period so the average idle balance is half the issue. Deposit rate on idle proceeds **3.0 %**. Bank alternative: progressive drawdown with a **0.60 %** commitment fee on the average undrawn balance.
2. **Formula.** Negative arbitrage = average idle balance × (coupon − deposit rate) × years. Commitment cost = average undrawn × fee rate × years.
3. **Substitution.** `21,000,000 × (6.0 % − 3.0 %) × 2` against `21,000,000 × 0.60 % × 2`.
4. **Result.** Negative arbitrage **1,260,000**; commitment fees **252,000**; the bank route is **1,008,000** cheaper on this dimension alone. Spread over the 12-year facility, the negative arbitrage is worth **35.78 basis points** a year on the 42,000,000.
5. **Interpretation.** Thirty-six basis points is the same order as the entire `WACC` benefit of twenty points of gearing (9.2.3), which puts the drawdown mechanism in its proper place: a first-order structuring decision, not an administrative detail. It also explains the standard market answer — a **bank facility during construction, refinanced into bonds at or after completion** — capturing progressive drawdown while the spend is uncertain and long institutional tenor once the risk profile has changed. That is not a compromise but the correct sequencing, and KA 9.4.4 prices it. Two asymmetries complete the comparison. **Flexibility:** a bank syndicate of six can approve a waiver in weeks (Domain 10, KA 10.4.3), while a dispersed bondholder base cannot easily be assembled at all — making bonds a poor choice where amendments are foreseeable. **Prepayment:** bonds typically carry make-whole or spens provisions capturing the lender's lost yield, so the refinancing gain computed in 9.4.4 may simply not be available.

AI in this KA

Where it earns its place. Normalising a dozen indicative term sheets — rate, tenor, amortisation profile, fee schedule, prepayment mechanics, covenant thresholds — into one comparable table is genuinely strong machine work and the foundation of Toolkit 9.T.2, as is generating the candidate structures whose arithmetic 9.2.3 tabulates.

Where it must not go. It must not decide the tax treatment of a tranche. Whether mezzanine interest is deductible, whether a shareholder loan survives thin-capitalisation rules and whether withholding applies are questions of law in a specific jurisdiction at a specific date, and a wrong answer moves `WACC` by hundreds of basis points while looking entirely plausible. Nor may it assert a market price: an indicative margin generated by a model is not a quote, and presenting one as though it were misrepresents to whoever relies on it.

Verification, concretely. Recompute one structure's `WACC` by hand and confirm the weights sum to one. Confirm every tax adjustment traces to a written opinion rather than a default. Confirm any all-in cost was solved from the actual stream rather than assembled by adding fees to a headline rate. And test the closed form: under the re-levering convention of 9.1.3 a `WACC` ladder that is not linear in gearing contains an error.

KEY TERMS — KA 9.2

Term	Meaning
k_d **(after tax)**	k_d × (1 − T); the debt cost that enters WACC.
WACC	Σ(tranche share × tranche cost), debt-like tranches after tax.
Mezzanine	Capital between senior debt and ordinary equity; priced in the gap, never cheap.
SPV vs HoldCo mezzanine	Inside the borrower and visible to DSCR, versus above it and exposed to lock-up.
Negative arbitrage	Cost of holding fully drawn bond proceeds at deposit rates below the coupon.
Wrapped bond	Notes credit-enhanced by a third-party guarantor.
Make-whole / spens	Prepayment compensation that can eliminate a refinancing gain.

SAMPLE MCQS — KA 9.2

MCQ 9.2-A [9.2.3 · Application] A 150,000,000 project is funded with senior 80,000,000 at 5.2 %, mezzanine 20,000,000 at 10.8 % (both deductible) and equity 50,000,000 at 14.5 %; tax 30 %. The WACC is: - A. 7.7827 % - B. 9.0467 % - C. 8.5667 % - D. 6.3327 %

Rationale: (80 × 3.64 + 20 × 7.56 + 50 × 14.5)/150 = 7.7827 %. B omits the tax shield on both debt tranches; C takes a simple average of the three costs, ignoring the weights; D tax-adjusts the **equity** cost as well, which no jurisdiction permits.

MCQ 9.2-B [9.2.3 · Analysis] Adding 9,000,000 of 11.50 % mezzanine in place of senior debt raises Kestrel's senior-only DSCR from 1.2743 to 1.5115 and its WACC from 7.9860 % to 8.4510 %. The correct reading is: - A. the structure is superior because coverage improved - B. the structure is inferior because WACC rose - C. mezzanine bought 23.72 points of senior coverage for 46.50 basis points of WACC and 44.87 basis points of equity IRR; whether that is worth paying depends on whether the senior tranche is otherwise unavailable - D. WACC and DSCR cannot both be affected by one tranche

Rationale: Both faces move and the decision is the exchange rate between them (9.2.3). A and B each optimise one number in isolation — the specific error this domain exists to prevent; D is simply false, since the mezzanine changes both the weighted cost and the debt service.

MCQ 9.2-C [9.2.4 · Application] A 42,000,000 bond at a 6.0 % coupon is drawn in full at close and spent evenly over two years; idle proceeds earn 3.0 %. The negative arbitrage is: - A. USD 2,520,000 - B. USD 1,260,000 - C. USD 630,000 - D. nil, because the proceeds are invested

Rationale: Average idle balance 21,000,000 × 3.0 % spread × 2 years = 1,260,000. A applies the spread to the **full** 42,000,000 for two years, ignoring the drawdown profile; C covers one year only; D confuses earning a return with earning enough of one.

MCQ 9.2-D [9.2.2 · Analysis] Why do senior lenders usually prohibit mezzanine debt at SPV level while tolerating it at HoldCo level? - A. HoldCo debt is cheaper - B. SPV mezzanine consumes the CFADS their covenant measures, while HoldCo mezzanine is serviced from distributions that already rank behind every senior test - C. HoldCo debt carries no security - D. accounting standards require it

Rationale: The placement determines whether the junior service appears in the coverage calculation (9.2.2). A is not generally true — HoldCo debt is usually dearer, being further from the cash; C is incidental; D confuses drafting practice with reporting requirements.

SELF-CHECK — KA 9.2

1. State the closed form for Kestrel's *WACC* and what its slope means. — `WACC(g) = 8.70 % − 1.02 % × g`; the slope is small because the non-systematic equity premium (5.10 %) barely exceeds after-tax senior debt (4.80 %), so gearing transfers return to equity rather than cheapening the project.
2. What is the project *WACC* at the coverage-binding gearing, and why does it matter? — **8.0001 %** at 68.6185 % gearing: it is Domain 4's given rate, derived.
3. What does mezzanine buy? — Coverage relief or equity uplift, depending on the tranche it displaces; never a lower cost of capital.

KNOWLEDGE AREA 9.3

Islamic finance concepts, export credit and development finance

Topics: 9.3.1 Islamic finance structures in economic terms · 9.3.2 export credit · 9.3.3 development finance · 9.3.4 assembling a multi-source stack.

9.3.1 Islamic finance structures in economic terms

Scope statement, and it binds. This topic describes, in economic and cash-flow terms, structures used in Islamic finance markets, because a project finance leader must be able to price them, model them and compare them. Whether any particular structure is compliant with Shariah is a determination for the relevant Shariah supervisory board, guided by the standards of bodies such as the Accounting and Auditing Organisation for Islamic Financial Institutions; **it is outside the scope of this book, and nothing here expresses, implies or substitutes for any religious or legal ruling.**

The structures share one economic feature: the financier's return arises from a transaction in an asset — a sale, a lease, a partnership, an agency — rather than from lending money at interest. The principal forms encountered in project financing:

Structure	Economic mechanics	Typical project use
Istisna'a	Commissioned manufacture or construction: the financier procures the asset to specification and takes delivery risk during the build	The construction-period facility
Ijara	Lease of a completed asset; the lessee pays rentals comprising a capital and a return element; ownership sits with the lessor	The operating-period facility
Ijara mawsufah fi al-dhimmah (forward lease)	Rentals commence on a described future asset, bridging construction to operation	The standard istisna'a-to-ijara conversion
Murabaha	Cost-plus sale with deferred payment; the mark-up is agreed and fixed at inception	Working capital, procurement of specific inputs
Wakala	Agency: the financier appoints an agent to invest funds for an expected return	Portfolio and liquidity tranches
Sukuk	Certificates representing undivided beneficial ownership in an asset, usufruct or business, with returns generated by that asset	The capital-markets tranche, often alongside a bank ijara

Three consequences a leader must handle. **The asset must exist and be identifiable**, which constrains what can be financed and makes security and title arrangements materially different from a conventional mortgage-and-charge package. **Ownership carries obligations** — a lessor's responsibility for major maintenance and insurance in an ijara is usually passed back to the lessee through a service agency agreement, and the drafting of that pass-back is where economic equivalence is achieved or lost. And **intercreditor arrangements between Islamic and conventional tranches are the hardest documentation in the deal**, because both must share security and enforcement proceeds on a genuinely equal footing without either being subordinated in substance.

WORKED EXAMPLE

Worked example 9.3.1 — comparing an ijara tranche with a conventional tranche, properly.

1. **Setup.** Kestrel's 42,000,000 is to be raised half conventionally and half through an ijara tranche, each **21,000,000 over 12 years**. The conventional tranche carries **6.00 %** with a **1.20 %** arrangement fee deducted from proceeds. The ijara carries a **6.15 %** profit rate with a **0.60 %** structuring fee. The sponsor's question is which is dearer.
2. **Formula.** The only defensible comparison is the **all-in effective cost**: the rate `r` solving `net proceeds = periodic payment × AF(r, n)`. Headline rates and fees cannot be added.
3. **Substitution.** Conventional: payment `21,000,000 / 8.383844 = 2,504,817.62`; fee `21,000,000 × 1.20 % = 252,000`; net proceeds **20,748,000**; solve `20,748,000 = 2,504,817.62 × AF(r, 12)`. Ijara: `AF(0.0615, 12) = 8.315327`; rental `21,000,000 / 8.315327 = 2,525,456.84`; fee **126,000**; net proceeds **20,874,000**; solve likewise.
4. **Result.** Conventional all-in **6.2209 %**; ijara all-in **6.2604 %**. The ijara is dearer by **3.95 basis points** — not by the 15 basis points the headline rates suggest. Annual payments differ by **20,639.22**, worth **173,036** in present value at 6 % over the twelve years.
5. **Interpretation.** The headline gap of 15 basis points shrank to 4 once the lower structuring fee was counted, and that reversal is the whole lesson: **a tranche is compared on the effective rate solved from its own cash-flow stream, never on its stated rate plus its fees.** Four basis points on 21,000,000 is 173,036 of present value — real money, but small enough that it should not decide the question. What should decide it is what the arithmetic cannot show: whether the ijara tranche reaches **investors the conventional tranche cannot**, widening the pool and reducing concentration risk; whether the extra documentation and intercreditor complexity are worth **173,036**; and whether the service-agency pass-back leaves the SPV with a residual ownership obligation the model has not priced. A leader who compares only the rates decides on the wrong axis in either direction.

9.3.2 Export credit

Definition. An export credit agency (ECA) is a state-backed institution that supports its country's exporters by guaranteeing, insuring or directly lending against the purchase of that country's goods and services. The effect for a project is a tranche whose credit risk is largely sovereign rather than commercial, producing a lower margin and a longer tenor than the commercial market would offer — in exchange for three conditions. **Sourcing (content) requirements** tie support to eligible goods and services from the supporting country, usually as a minimum share of contract value; this constrains procurement, and Case study B prices what that constraint can cost. **An exposure premium** reflecting buyer-country risk and tenor is charged, commonly **capitalised into the loan**, so it does not reduce cash proceeds but increases the amount repaid — precisely the structure that makes headline-rate comparison worthless. And participating agencies operate within the **OECD Arrangement on Officially Supported Export Credits**, an inter-governmental understanding constraining minimum premium rates, maximum repayment terms, starting points and local-content treatment; its terms are revised periodically and vary by sector, so they must be checked as at the transaction date rather than assumed.

WORKED EXAMPLE

Worked example 9.3.2 — the true cost of a cheap ECA tranche.

1. **Setup. 15,000,000** of eligible equipment is financed by an ECA-supported tranche at **3.80 %** over **14 years**, with an exposure premium of **6.0 %** of the tranche **capitalised** into the loan. The commercial comparator is 15,000,000 at **6.00 %** over 12 years.
2. **Formula.** All-in cost is the rate solving `cash proceeds = instalment × AF(r, n)`, where the instalment is computed on the **grossed-up** loan and the proceeds are the amount actually available to the project.
3. **Substitution.** Premium `15,000,000 × 6.0 % = 900,000`; loan **15,900,000**; `AF(0.038, 14) = 10.703972`; instalment `15,900,000 / 10.703972 = 1,485,429.84`; solve `15,000,000 = 1,485,429.84 × AF(r, 14)`.
4. **Result.** All-in cost **4.6895 % — 88.95 basis points** above the 3.80 % headline, and still **131 basis points** below the commercial alternative. Annual debt service falls from `15,000,000 / 8.383844 = 1,789,155.44` to **1,485,429.84**, a saving of **303,725.60** a year.
5. **Interpretation.** Both halves of that result matter and they are usually conflated. The **cost** advantage is 131 basis points, which is genuine but not transformational. The **coverage** advantage is 303,726 of annual debt service released, and that is decisive: substituting the ECA tranche for 15,000,000 of Kestrel's commercial senior debt leaves total debt service of `27,000,000/8.383844 + 1,485,429.84 = 3,220,479.79 + 1,485,429.84 =` **4,705,909.63**, so `DSCR` rises from **1.2743** to **1.3566 — 8.23 points**, straight through the 1.30× sizing target the all-commercial structure could not reach. The two effects have different causes — the cost advantage from the sovereign-backed margin, the coverage advantage mostly from the **two extra years of tenor** — and separating them is the discipline of 9.3.3. The cautions are procedural and expensive when missed: ECA processes run to their own timetable and can add months to financial close; documentation is agency-specific and in places non-negotiable; and the eligible-content test is verified against invoices after the fact, so a procurement change during construction can retrospectively disqualify a tranche.

9.3.3 Development finance

Definition. Development finance institutions (DFIs) — multilateral and bilateral development banks and their private-sector arms — lend and invest to advance development objectives, contributing what commercial lenders often cannot: **tenor** well beyond the bank market, **availability** through periods when commercial appetite has withdrawn, political-risk comfort that can reduce other lenders' margins, and structures such as an A/B loan in which the institution is lender of record and commercial banks take participations behind its umbrella.

The price is **conditionality**: environmental and social performance requirements — the IFC Performance Standards and the Equator Principles are the frameworks most commonly referenced by lenders in this market — together with procurement, disclosure, integrity and reporting obligations. Meeting them costs money and time, and that cost belongs in the tranche's economics rather than in an overhead line where nobody sees it.

WORKED EXAMPLE

Worked example 9.3.3 — separating the cost effect from the capacity effect.

1. **Setup.** A DFI offers **12,000,000** at **5.25 %** over **18 years**, with a **1.00 %** front-end fee, an incremental environmental and social advisory cost of **350,000** at close, and **120,000** a year of monitoring and reporting cost through the loan life.
2. **Formula.** All-in economic cost solves `net proceeds = (instalment + annual compliance cost) × AF(r, n)`. Debt capacity is Domain 10's `max debt = (CFADS ÷ target DSCR) × AF(r, n)`, computed on the **contractual** rate and tenor.
3. **Substitution.** `AF(0.0525, 18) = 11.464588`; instalment `12,000,000 / 11.464588 = 1,046,701.34`; net proceeds `12,000,000 − 120,000 − 350,000 = 11,530,000`; annual outflow `1,046,701.34 + 120,000 = 1,166,701.34`; solve for `r`. Capacity: `6,384,000 / 1.30 = 4,910,769.23`, then `× 11.464588` against `× 8.383844`.
4. **Result.** All-in economic cost **7.2465 %** — **199.65 basis points** above the 5.25 % headline, and **125 basis points above the commercial 6.00 %**. Yet on the contractual terms, debt capacity at a 1.30× target rises from Domain 10's **41,171,123** to **56,299,948**, an uplift of **15,128,825**; and Kestrel's existing 41,171,123 of debt, repaid over 18 years at 5.25 %, would carry debt service of only **3,591,155.80** and a `DSCR` of **1.7777**.
5. **Interpretation.** This is the domain's most counter-intuitive result and it must not be collapsed into one verdict. On **cost** the DFI tranche is expensive: the headline 5.25 % is a fiction once the front-end fee, the advisory spend and 120,000 a year of monitoring are counted, and at 7.2465 % it is dearer than the commercial market. On **capacity** it is transformational: six extra years of tenor support **36.7 % more debt** at the same coverage, more than any plausible margin negotiation could deliver. **Tenor beats rate**, because coverage is the binding constraint and tenor is the lever that moves it most — Domain 10, KA 10.1.2 lists four levers, and this is the arithmetic showing which one pays. Three cautions. The compliance cost is **not** waste: it buys an environmental and social management system a well-run project would want anyway, and other lenders price the comfort it provides, so charging all of it against the DFI tranche overstates its cost. The monitoring obligation runs for the full 18 years and must be budgeted for 18 years, not three. And DFI processes are slow: putting a DFI tranche on the critical path to financial close without allowing for that creates a schedule risk Domain 8's arithmetic prices at hundreds of thousands a month.

9.3.4 Assembling a multi-source stack

Combining commercial banks, an ECA, a DFI and an Islamic tranche in one financing is ordinary in large infrastructure, and it creates a documentation problem often larger than the financing problem. Four recurring issues. **Common terms:** one common terms agreement carries the representations, covenants, events of default and the definition of `CFADS`, each facility agreement then carrying only its own commercial terms — without it, coverage is tested on different definitions by different lenders, the defect Domain 10, KA 10.1.1 warns against, multiplied by the number of tranches. **Intercreditor:** voting thresholds, instructing-group mechanics, sharing of enforcement proceeds and which decisions require unanimity; a tranche holding a blocking position it did not pay for is a structural defect visible only in a workout. **Availability and drawdown order:** tranches whose conditions precedent mature at different times cannot be drawn pro rata, and whichever is drawn first bears the most construction risk and prices accordingly. **Currency:** a tranche in a currency other than the revenue currency introduces an exposure that must be hedged or accepted, and the cheapest tranche is frequently the one that creates it — a project with local-currency availability payments and a hard-currency ECA tranche has borrowed a devaluation risk in exchange for a margin saving. The natural hedge, borrowing in the currency of revenue, is worth paying for and its price is exactly that margin difference (Domain 3, KA 3.3.3; Domain 11, KA 11.3).

AI in this KA

Where it earns its place. Normalising heterogeneous tranche offers onto a common all-in basis is arithmetic with many steps and one method, and it is where analysts most often err under time pressure. Solving effective rates across a dozen structures, and re-solving when a fee schedule changes, should be automated — with the method (solve from the cash-flow stream) fixed in the tool rather than left to the user.

Where it must not go. It must not opine on Shariah compliance: that belongs to the relevant supervisory board and to no model. It must not assert current ECA premium rates, maximum tenors or content thresholds from memory — those come from a periodically revised inter-governmental framework, and a stale figure presented confidently is worse than no figure. And it must not draft or interpret intercreditor mechanics, where a misread voting threshold shows up only in a default.

Verification, concretely. For each tranche, confirm the all-in cost was solved from the stream and that the proceeds figure excludes capitalised premiums and deducted fees. Confirm the compliance-cost assumption runs for the full tenor. Confirm any framework constraint cited was checked against the current published text at a stated date by a named person. And keep cost and capacity in separate columns — a model reporting one number per tranche has already lost the 9.3.3 distinction.

KEY TERMS — KA 9.3

Term	Meaning
Istisna'a / ijara / forward lease	Commissioned-construction, lease, and described-future-asset lease structures used in Islamic finance markets.
Murabaha / wakala / sukuk	Cost-plus deferred sale; agency investment; certificates of undivided beneficial ownership.
All-in effective cost	The rate solving `net proceeds = payment × AF(r, n)`; the only valid basis for tranche comparison.
ECA / exposure premium	State-backed export-credit support; the risk premium it charges, usually capitalised.
Content requirement	Minimum share of contract value sourced from the supporting country.
DFI conditionality	Environmental, social, procurement and reporting obligations attached to development finance.
Common terms agreement	One document carrying shared covenants and definitions across all tranches.
Natural hedge	Borrowing in the currency of revenue; its price is the margin forgone.

SAMPLE MCQS — KA 9.3

MCQ 9.3-A `[9.3.2 · Application]` A 40,000,000 tranche carries 4.10 % over 12 years with a 5.5 % exposure premium capitalised into the loan. The all-in cost on the 40,000,000 of proceeds is closest to: - A. 4.10 % - B. 5.0378 % - C. 4.5583 % - D. 9.60 %

Rationale: Loan `42,200,000`; `AF(0.041, 12) = 9.330854`; instalment `4,522,630.17`; solve `40,000,000 = 4,522,630.17 × AF(r, 12)` → **5.0378 %**. A ignores the premium entirely; C spreads the premium straight-line over the tenor and adds it to the rate (`4.10 + 5.5/12`), which understates because it ignores that the premium is also financed; D adds the premium to the rate as though it were annual.

MCQ 9.3-B `[9.3.3 · Analysis]` A DFI tranche at 5.25 % over 18 years has an all-in economic cost of 7.2465 % against a commercial market at 6.00 % over 12 years. The soundest conclusion is: - A. reject the DFI tranche — it is more expensive - B. accept it — the headline rate is lower - C. the cost comparison favours the commercial market by about 125 basis points, while the six extra years of tenor raise debt capacity at a 1.30× target from 41,171,123 to 56,299,948; the decision turns on whether coverage or cost is the binding constraint - D. the two are equivalent because tenor and rate offset

Rationale: Cost and capacity are different effects with different causes (9.3.3). A optimises cost while ignoring the constraint that actually binds; B is the headline-rate error the worked example exists to destroy; D asserts an offset without computing either side.

MCQ 9.3-C `[9.3.1 · Application]` A conventional tranche of 21,000,000 at 6.00 % over 12 years carries a 1.20 % fee; an ijara tranche of the same size and tenor carries a 6.15 % profit rate and a 0.60 % fee. The difference in all-in cost is closest to: - A. 15 basis points in favour of the conventional tranche - B. 4 basis points in favour of the conventional tranche - C. 60 basis points in favour of the ijara tranche - D. nil, since the structures are economically identical

Rationale: All-in 6.2209 % against 6.2604 % — **3.95 basis points** (9.3.1). A is the headline difference before fees; C confuses the fee saving with a rate advantage; D asserts an equivalence the arithmetic disproves, small though the gap is.

MCQ 9.3-D `[9.3.4 · Recall]` The primary purpose of a common terms agreement in a multi-source financing is to: - A. reduce legal fees - B. ensure every tranche tests coverage on the same definitions and shares one covenant and default architecture - C. give the DFI a veto - D. permit tranches to be drawn in any order

Rationale: One set of shared definitions and covenants prevents the same project being measured differently by different lenders (9.3.4). A is a by-product; C describes an intercreditor outcome to be negotiated, not the purpose; D is a drawdown question the agreement constrains rather than liberates.

SELF-CHECK — KA 9.3

1. How is a tranche's cost compared, and why not by rate plus fees? — By solving the effective rate from its own cash-flow stream; fees, capitalised premiums and tenor interact and cannot be added.
2. State the two distinct effects of a longer-tenor tranche. — A cost effect (usually adverse once fees and conditionality are counted) and a capacity effect (36.7 % more debt at the same coverage, in the 9.3.3 case) — reported separately.
3. What does this book decline to determine about Islamic finance structures? — Their compliance with Shariah, which rests with the relevant supervisory board; the book describes only economics and cash flows.

KNOWLEDGE AREA 9.4

Government support, grants, sustainable finance and refinancing

Topics: 9.4.1 forms of government support · 9.4.2 grants and concessional tranches · 9.4.3 green and sustainability-linked finance · 9.4.4 refinancing.

9.4.1 Forms of government support

Definition. Government support is any intervention that improves a project's financeability without the grantor becoming an ordinary equity investor. Every form transfers value, and every form has a fiscal price — either cash now or a contingent liability later — which is why the discipline of this topic is to name the price alongside the benefit.

Form	What it does for the project	What it costs the grantor
Capital grant / viability gap funding	Reduces the amount to be financed	Cash, at the point of greatest budget pressure
Availability or capacity payment	Removes demand risk from the revenue line (Domain 7)	A long-dated committed expenditure line
Minimum revenue guarantee	Caps downside on a demand-based revenue	A contingent liability, valued as an option
Sovereign or agency guarantee of debt	Converts project credit into sovereign credit	Full contingent exposure to the debt
Concessional or subordinated public loan	Fills the gap between senior debt and equity cheaply	The margin forgone, plus subordination risk
Tax incentives, allowances, exemptions	Raises `CFADS` by reducing cash tax	Forgone revenue, often unbudgeted and unmeasured
Land, permits, connections in kind	Reduces development cost and timeline	Opportunity cost of the asset, rarely quantified

Two professional points cut across the table. **A guarantee is not free because no cash moves** — it is a written option whose value can be estimated, and a grantor that does not estimate it is accumulating unmeasured liabilities. And **support that improves coverage is worth more to a project than support of equal value that improves return**, because coverage is the binding constraint (9.1.4). A grantor seeking maximum financeability per unit of fiscal cost should therefore direct support at the coverage face — which is exactly what 9.4.2 quantifies.

9.4.2 Grants and concessional tranches

WORKED EXAMPLE

Worked example 9.4.2 — one grant, two structures, two entirely different projects.

1. **Setup.** A **6,000,000** capital grant (10 % of capex) is awarded to Kestrel. The funding plan must place it. **Case G1:** the grant **displaces equity** — senior stays at 42,000,000, equity falls to 12,000,000. **Case G2:** the grant **displaces senior debt** — senior falls to 36,000,000, equity stays at 18,000,000. In both, the grant is treated as zero-cost capital that absorbs risk alongside equity, so `β_e` is re-levered on debt against equity **plus grant**.
2. **Formula.** `WACC = Σ(share × cost)` with the grant at zero cost; `DSCR` per Domain 10; equity `IRR` from the residual stream.
3. **Substitution.** G1: `WACC = (42 × 4.80 + 12 × 15.42 + 6 × 0)/60`; `DSCR` unchanged; equity stream `(−12,000,000; 1,374,364.77 × 12; 6,384,000 × 13)`. G2: senior service `36,000,000 / 8.383844 = 4,293,973.06`; `β_e = 0.60 × (1 + 0.80 × 36/24) = 1.32` so `k_e = 13.02 %`; `WACC = (36 × 4.80 + 18 × 13.02)/60`; equity stream `(−18,000,000; 2,090,026.94 × 12; 6,384,000 × 13)`.
4. **Result.**

	Base (no grant)	G1 — grant displaces equity	G2 — grant displaces debt
Senior debt	42,000,000	42,000,000	36,000,000
Equity	18,000,000	12,000,000	18,000,000
`k_e`	15.42 %	15.42 %	13.02 %
`WACC`	**7.9860 %**	**6.4440 %**	**6.7860 %**
`DSCR`	**1.2743**	**1.2743**	**1.4867**
Equity `IRR`	**12.5311 %**	**16.8231 %**	**14.9940 %**

1. **Interpretation.** The same 6,000,000 produces two different projects, and the difference is the allocation question every grant negotiation is really about. **G1 is the sponsors' case:** equity return jumps **429.20 basis points**, from 12.5311 % to 16.8231 % — clearing the 15.42 % cost of equity for the first time on the flat-`CFADS` basis, without a single change to the project — while lenders gain **nothing**, because `DSCR` is untouched at 1.2743. **G2 is the lenders' case:** coverage rises **21.24 points** to 1.4867, comfortably above the 1.30× sizing target that Domain 10's Case A could not reach, while equity gains 246.29 basis points rather than 429.20. A grantor who does not specify which face the grant is intended to strengthen has funded whichever the sponsors preferred, and the answer will be G1. The tie-break should follow the public purpose: if the objective is **financeability** — getting a project financed that otherwise could not be — the grant must reduce debt (G2), because coverage is the binding constraint; if the objective is **attracting private capital to a thin sector**, raising equity returns (G1) is defensible, and should be said out loud. **The treatment caution is as large as the allocation question.** The re-levering convention above places the grant in the risk-bearing base with equity. Treat it instead as outside that base — so `D/E` in G1 becomes 42/12 = 3.5, `β_e` = 2.28 and `k_e` = **18.78 %** — and G1's `WACC` rises from 6.4440 % to **7.1160 %**, a difference of **67.20 basis points** on identical cash flows. Both treatments are defensible; only one can be used; and it must be documented before the numbers are quoted, because 67 basis points is more than the entire `WACC` effect of twenty points of gearing.

Concessional tranches and the grant element. Support often arrives as cheap debt rather than cash, and its subsidy content can be measured precisely. Take **12,000,000 at 2.0 % over 25 years with 7 years' grace** (interest-only, then level amortisation), against a **6.0 %** market rate. Interest during grace is `12,000,000 × 2.0 % =` **240,000** a year; the amortising instalment is `12,000,000 / AF(0.02, 18) = 12,000,000 / 14.992031 =` **800,425.23**. Discounting the whole service stream at 6.0 % gives a present value of **7,103,613**, so the **grant element** is `1 − 7,103,613/12,000,000 =` **40.80 %**, and the subsidy is worth **4,896,387** in present-value terms. A concessional tranche of 12,000,000 is therefore 7.1 million of debt and 4.9 million of grant, and reporting it as 12 million of borrowing misstates both the leverage and the support received.

9.4.3 Green and sustainability-linked finance

Two mechanisms, routinely confused. A **use-of-proceeds** instrument — a green bond or green loan — commits the borrower to spend the money on defined eligible assets, with reporting and usually an external review; the pricing benefit, where it exists, comes from demand. A **sustainability-linked** instrument leaves the use of proceeds unrestricted and instead ties the **margin** to performance against key performance indicators through a ratchet, so the borrower pays less for hitting targets and more for missing them. The first constrains what the money does; the second prices what the borrower does.

Both rest on voluntary market frameworks — the principles published by international market associations for green bonds and green loans, and their sustainability-linked counterparts — together with jurisdiction-specific taxonomies and disclosure regimes that differ materially between jurisdictions and change frequently. None of them is a global standard, and treating one jurisdiction's taxonomy as universal is the characteristic error in this area.

WORKED EXAMPLE

Worked example 9.4.3 — does the ratchet pay for itself?

1. **Setup.** Kestrel's 42,000,000 senior facility is offered as a sustainability-linked loan with a **±15 basis point** margin ratchet tied to two KPIs (specific energy consumption per cubic metre produced, and a verified brine-discharge metric). Meeting both targets earns the full 15 basis point reduction. The apparatus required — metering, data assurance, annual limited-assurance verification, a second-party opinion refreshed periodically, and reporting — costs **85,000** a year. Assess over the 12-year facility at 6 %.
2. **Formula.** Annual benefit = tranche × ratchet. Compare present values using `AF(0.06, 12) = 8.383844`.
3. **Substitution.** `42,000,000 × 0.15 % = 63,000` a year against 85,000 a year; `63,000 × 8.383844` against `85,000 × 8.383844`.
4. **Result.** Present value of the margin saving **528,182**; present value of the compliance cost **712,627**; **net −184,445**. On margin alone the ratchet **destroys value**, even assuming both KPIs are met every year. The breakeven is a ratchet of **20.24 basis points**, or equivalently a base-margin reduction of **5.24 basis points** alongside the 15.
5. **Interpretation.** This is the number the market conversation usually omits, and stating it is not scepticism about sustainability — it is honesty about where the value comes from. A ratchet of 15 basis points on a facility of this size cannot pay for a credible verification apparatus, so **if the case for the label rests on the ratchet, there is no case**. The case rests elsewhere, and it is quantifiable: if the label genuinely widens the lender pool enough to cut the **base** margin by **10 basis points** — 42,000 a year, **352,121** in present value — the combined position turns **positive by 167,677**. That reframes the negotiation correctly: ask the arranger what the label does to the **base** margin and to the size of the club, not what the ratchet is worth. Three further cautions. The ratchet is **symmetric** in most drafting, so missing targets increases the margin, and a KPI set without headroom converts an ESG initiative into a cost. Targets must be **material and measurable** — a KPI the project would have met anyway invites the accusation of greenwashing, which is a reputational and, increasingly, a regulatory exposure. And **the verification cost is permanent** for the life of the facility, while the ratchet benefit is contingent on performance; budget them asymmetrically.

9.4.4 Refinancing

Definition. Refinancing replaces existing debt with new debt on better terms, and in project finance the opportunity is structural rather than opportunistic: construction risk, ramp-up risk and counterparty uncertainty all resolve in the first operating years, so the project that emerges is a materially better credit than the one that was financed. Better terms mean a lower margin, a longer tenor, a looser covenant package, or all three.

WORKED EXAMPLE

Worked example 9.4.4 — Kestrel's refinancing gain, decomposed.

1. **Setup.** At the end of **year 6**, six of twelve instalments of 5,009,635.23 have been paid. The market offers **4.75 %**. Transaction costs — arrangement on the new facility plus a prepayment fee on the old — total **1.75 %** of the refinanced balance. Two structures are available: refinance the remaining **6 years**, or refinance over **12 years**, extending maturity from year 12 to year 18 within the 25-year operating life.
2. **Formula.** Outstanding balance = instalment × `AF(r_old, remaining)`; new instalment = balance ÷ `AF(r_new, new tenor)`; the equity gain is the present value of the change in the equity cash-flow stream, discounted at the cost of equity.
3. **Substitution.** Balance `5,009,635.23 × AF(0.06, 6) = 5,009,635.23 × 4.917324`. Rate-only: `balance / AF(0.0475, 6)`. Extended: `balance / AF(0.0475, 12) = balance / 8.989557`. Transaction cost `balance × 1.75 %`.
4. **Result.** Outstanding balance **24,634,001.18** (principal repaid to date **17,365,998.82**). Transaction cost **431,095.02**.

	Rate only, 6 years	Rate + extension, 12 years
New instalment	**4,814,595.21**	**2,740,290.88**
Annual saving, years 7–12	**195,040.02**	**2,269,344.35**
New `DSCR`	**1.3260**	**2.3297**
New `LLCR`	1.3260	**2.3297**
Tail (project life beyond maturity)	13 years	**7 years**
PV of equity gain at `k_e` 15.42 %	**298,757**	**3,723,616**
PV of equity gain at 8.00 %	**470,552**	**2,076,800**
PV of equity gain at 4.75 %	—	**566,832**

1. **Interpretation.** The extension appears to create **3.72 million** of equity value against 299 thousand from the rate reduction alone, and the decomposition shows why that comparison is misleading: of the 3.72 million, only **298,757** is the rate saving and **3,424,859** is the **extension**, which is not a gain at all but a **deferral** — six years of debt service moved from years 7–12 into years 13–18. Its apparent value is almost entirely an artefact of the discount rate: at 15.42 % the deferral is worth 3.42 million, at 8.00 % it is worth 1.61 million, and at the new loan rate of 4.75 % the whole package is worth only **566,832**. **A refinancing gain that shrinks by 85 % when the discount rate falls to the loan rate is a financing artefact, not value creation**, and a leader who presents the 3.72 million without the sensitivity has misled the board. What is unambiguously real: the rate saving of 195,040 a year; the coverage improvement to **2.3297**, which creates genuine covenant headroom; and the reduction of the **tail from 13 years to 7**, which is a genuine loss — `PLCR` at the refinancing date rises from 2.8917 to 3.1968 only because the new rate discounts more gently, while the contractual cushion behind the lenders has almost halved. Two further disciplines. **Costs are real and immediate:** 431,095 is paid at once against a gain earned over years, and at the rate-only structure the costs consume 59 % of the discounted benefit at 15.42 %. **Refinancing gain-share is common in concession-based structures**, requiring the grantor's consent and a defined share of the gain — at 50 % of the 15.42 % figure,

1,861,808 returns to the grantor, which changes the economics enough that the clause must be read before the refinancing is modelled, not after (Domain 12, KA 12.2).

AI in this KA

Where it earns its place. Refinancing screening across a portfolio — outstanding balances, candidate rates and tenors, opportunities ranked by gain net of costs — is repetitive arithmetic on data the organisation already holds, and it surfaces what manual review misses. Grant-element and sustainability-linked breakeven calculations are similarly mechanical.

Where it must not go. It must not conclude what a grant, subsidy or state guarantee means legally or fiscally: state-aid and subsidy-control regimes, the accounting treatment of grants and the tax character of concessional support are jurisdiction-specific and consequential. It must not generate sustainability claims or KPI narratives — statements about environmental performance carry regulatory and reputational exposure, and model-written text published as a project's own assertion is the shortest route to a greenwashing allegation. And it must not settle a refinancing recommendation, which turns on the discount-rate judgment 9.4.4 shows dominates the answer.

Verification, concretely. Recompute the outstanding balance independently — instalment × `AF(r, remaining)` must equal the schedule's closing balance (Domain 3's check). Confirm every gain is reported at two or more discount rates, one of them the new loan rate, and net of transaction costs and any gain-share. Confirm the grant treatment used in a `WACC` is stated. And confirm a sustainability-linked benefit is compared against the **full-tenor** verification cost, not the first year's.

KEY TERMS — KA 9.4

Term	Meaning
Viability gap funding	Capital grant sized to make an otherwise unfinanceable project viable.
Minimum revenue guarantee	Grantor-provided floor under demand-based revenue; a written option.
Grant element	`1 − PV(debt service at a market rate) ÷ face value`; the subsidy content of concessional debt.
Use-of-proceeds instrument	Green bond or loan; proceeds restricted to eligible assets.
Sustainability-linked loan	Margin ratchet tied to KPI performance; usually symmetric.
Refinancing gain	PV of improved terms, net of costs; decomposed into rate and extension components.
Gain-share	Contractual obligation to pass a defined share of a refinancing gain to the grantor.
Tail	Project life beyond debt maturity; reduced by tenor extension.

SAMPLE MCQS — KA 9.4

MCQ 9.4-A `[9.4.2 · Analysis]` A 6,000,000 grant applied to reduce **equity** leaves `DSCR` at 1.2743 and lifts equity `IRR` from 12.5311 % to 16.8231 %; applied to reduce **debt** it lifts `DSCR` to 1.4867 and equity `IRR` to 14.9940 %. If the grantor's stated objective is to make an otherwise unfinanceable project financeable, it should: - A. apply the grant to reduce equity, maximising private-sector return - B. apply the grant to reduce debt, because coverage is the binding constraint on financeability and 1.4867 clears the 1.30× sizing target - C. split the grant evenly, as a neutral position - D. either — the `WACC` reduction is what matters

Rationale: Financeability is a coverage question (9.1.4, 9.4.2). A serves a different objective and should be stated as such; C is a decision avoided rather than made; D optimises the number that this domain shows is not binding — and `WACC` actually falls further under the equity-displacing case (6.4440 % against 6.7860 %), which is exactly why `WACC` is the wrong test here.

MCQ 9.4-B `[9.4.3 · Application]` A 42,000,000 facility offers a 15 basis point sustainability ratchet; the verification and reporting apparatus costs 85,000 a year; the facility runs 12 years and `AF(0.06, 12) = 8.383844`. On margin alone the arrangement is: - A. value-positive by 528,182 - B. value-negative by 184,445 - C. value-neutral - D. value-positive by 63,000 a year

Rationale: `63,000 × 8.383844 = 528,182` of benefit against `85,000 × 8.383844 = 712,627` of cost → **−184,445**. A counts the benefit and omits the cost; D quotes the annual benefit gross, also omitting cost; C asserts an offset that the arithmetic contradicts.

MCQ 9.4-C `[9.4.4 · Analysis]` A refinancing that extends maturity shows an equity gain of 3,723,616 at a 15.42 % cost of equity, 2,076,800 at 8.00 % and 566,832 at the new 4.75 % loan rate. The correct board disclosure is: - A. a gain of 3,723,616 - B. a gain of 3,723,616, since the cost of equity is the correct discount rate for equity cash flows - C. that most of the reported gain is deferral rather than saving — the rate component is 298,757 and the extension component 3,424,859 at 15.42 % — and that the package is worth only 566,832 discounted at the loan rate - D. no gain, since the total cash paid over the loan's life increases

Rationale: The decomposition and the rate sensitivity are the disclosure (9.4.4). A and B report a correct arithmetic result while concealing that it is an artefact of the discount rate; D swings to the opposite error, ignoring the genuine 195,040 a year of rate saving and the coverage improvement.

MCQ 9.4-D `[9.4.2 · Application]` A 12,000,000 concessional loan at 2.0 % over 25 years with 7 years' grace has a debt-service stream worth 7,103,613 at a 6.0 % market rate. Its grant element is: - A. 40.80 % - B. 4.00 % - C. 59.20 % - D. nil — it is a loan, not a grant

Rationale: `1 − 7,103,613/12,000,000 = 40.80 %`, a subsidy worth 4,896,387 (9.4.2). B quotes the interest-rate difference as though it were the subsidy; C is the complement — the debt content, not the grant content; D confuses legal form with economic substance.

SELF-CHECK — KA 9.4

1. A grant of fixed size: which face should it strengthen and why? — The coverage face, if the objective is financeability, because coverage binds; the return face only if attracting equity to a thin sector is the stated objective.
2. When does a sustainability-linked ratchet pay for itself? — Only above about 20 basis points on a facility of this size, or when the label also reduces the base margin — 5.24 basis points is the breakeven addition here.
3. How must a refinancing gain be reported? — Net of costs and gain-share, decomposed into rate and extension components, at no fewer than two discount rates including the new loan rate.

Advanced topics — Domain 9

9.A.1 Why capital-structure irrelevance fails in a project SPV

The classical result — that in frictionless markets the value of a firm is independent of how it is financed — is a useful diagnostic precisely because every one of its assumptions fails in a project financing, and naming the failures tells a leader where the value actually sits. **Taxes** are the first: the deductibility of interest supplies 0.72 of Kestrel's 1.02 `WACC` slope, so most of the apparent benefit of leverage is a transfer from the tax authority, contingent on a jurisdictional rule that can change. **Bankruptcy and distress costs** are the second, and in a single-asset SPV they are severe: there is no diversified balance sheet to absorb a bad year, which is why lenders demand coverage rather than accepting the theory's indifference. **Contracting frictions** are the third — covenants, lock-up, reserve accounts, cure limits — and they impose a hard boundary on the feasible set that no cost calculation can cross (9.1.4). The fourth is a modelling artefact worth confronting: the build-up of 9.1.3 re-levers only the beta term, leaving `r_f`, `CRP` and `SP` un-levered, which is what makes `WACC` linear in gearing and gives the slope its value. Re-lever the country and single-asset premiums as well and the slope changes; leave the beta un-levered and `WACC` rises with gearing. The convention is therefore not innocent, and the honest presentation states it, applies it uniformly, and reports how much of the conclusion depends on it.

9.A.2 Concessionality, blending, and measuring what the public sector gave

The grant-element calculation of 9.4.2 — **40.80 %** on the illustrative 12,000,000 tranche — generalises into the discipline of **blended finance**: combining concessional public capital with commercial capital so that a project unfinanceable on market terms becomes financeable. Three disciplines make it defensible. **Measure the subsidy**, using the grant element at an explicit market comparator rate, and report it as subsidy rather than as debt; a portfolio described as "5 billion of development lending" with an unstated 30 % grant element has misreported both its leverage and its fiscal cost. **Test additionality**: concessional capital that displaces commercial capital that would have come anyway has funded a transfer, not a project. And **price the subordination separately** — public capital that ranks behind commercial debt is providing first-loss protection whose value is not captured by the interest-rate difference at all, and which should be valued as the credit enhancement it is.

9.A.3 The reviewer's funding-structure eye

Invariants to test on any funding structure or cost-of-capital model:

- Sources equal uses, at every drawdown date and in total (Domain 14's discipline, applied at close).
- `WACC` weights sum to one, and the debt weight uses the same debt figure as the coverage test.
- Every debt-like tranche is tax-adjusted and equity is not; each adjustment traces to a written opinion.
- `k_e` is re-levered for each candidate structure; no two structures share a cost of equity unless they share a gearing.
- Under the re-levering convention of 9.1.3, `WACC` is **linear in gearing**; a non-linear ladder contains an error.
- The `WACC` at the coverage-binding gearing is reported alongside the `WACC` at the proposed gearing (**8.0001 %** and **7.9860 %** for Kestrel) — the gap is the price of bankability.
- Every quoted equity `IRR` names its cash-flow case (bank case or sponsor case) and its treatment of any equity bridge.
- An equity bridge priced at exactly the equity `IRR` it defers produces **zero** uplift; a model reporting a gain there has a timing defect.
- Every tranche's cost is an **all-in effective rate solved from its own stream**, with capitalised premiums added to the loan and deducted fees excluded from proceeds.
- Compliance and monitoring costs run for the tranche's full tenor.

- The grant treatment used in WACC is stated; the alternative treatment's effect is disclosed (**67.20 basis points** for Kestrel).
- Every refinancing gain is net of costs and gain-share, decomposed, and reported at two or more discount rates.
- Cost effects and capacity effects are reported in separate columns and never netted.

Industry variations — Domain 9

- **Contracted power and renewables.** Availability or fixed-price offtake supports high gearing and a deep pool of infrastructure debt funds; the distinctive feature is a large, jurisdiction-specific incentive layer (allowances, credits, feed-in mechanisms) whose value is often monetised through a specialist tranche with its own investor base and its own tax opinion. Green-labelled instruments are near-universal, so the label carries little pricing benefit and the 9.4.3 analysis matters more, not less.
- **Merchant power and commodities.** Coverage requirements of 1.5× and above compress feasible gearing, so structures lean on mezzanine and on private credit funds that will price risk banks will not; reserve-based and borrowing-base facilities re-size against periodic technical redeterminations, making debt capacity a variable rather than a term.
- **Transport concessions.** Ramp-up risk pushes structures toward long-tenor institutional debt, substantial capital grants or viability gap funding, and staged refinancing after patronage stabilises — which is why refinancing gain-share clauses (9.4.4) are most common in this sector and why the grantor's consent regime is a financing term, not a legal footnote.
- **Water and regulated utilities.** Regulatory reset cycles cap the tenor over which revenue is contractually visible, so debt maturities are often set to reset dates and refinancing is structural. Public grant support is common and the G1/G2 allocation question of 9.4.2 is a standing negotiation with the regulator as well as the grantor.
- **Digital infrastructure.** Shorter asset lives compress tails, so lenders rely on contracted tenant credit rather than on PLCR; capital arrives fast and in size from private credit and from vendor financing, and gearing is constrained by lease-term coverage rather than by concession length.
- **Social infrastructure PPPs.** Availability payments from a public counterparty support the thinnest equity in the market — gearing of 90 % is ordinary — which makes the leverage exchange rate of 9.1.4 extreme: tiny coverage movements dominate the equity return, and a lock-up is catastrophic rather than inconvenient.

Case study — Domain 9: three ways to close 828,877 (water)

Situation. Domain 10's Case A left Kestrel with an arithmetic gap. Against base-case CFADS of 6,384,000 and a 1.30× requirement, sustainable senior debt service is **4,910,769** supporting **41,171,123** — **828,877** short of the 42,000,000 requested. The bank would not move on coverage. The sponsors, whose investment committee had approved 18,000,000 of equity and no more, asked their adviser for options.

Option one: more equity. Senior 41,000,000 (debt service `41,000,000/8.383844 = ` **4,890,358**, DSCR **1.3054**) with equity of **19,000,000**. Distributions run at **1,493,642** a year, the equity IRR on the flat-

CFADS basis is **12.3868 %**, and the structure's WACC is **8.0030 %**. This is Domain 10's answer, and it works.

Option two: mezzanine at SPV level. Senior 41,000,000, mezzanine 4,000,000 at 11.50 % over 15 years (debt service 4,000,000/6.996708 = **571,697**), equity 15,000,000. The bank rejected it in one line: total debt service becomes 5,462,056 and the project's DSCR falls to **1.1688**, below both the covenant and the lock-up. Mezzanine inside the borrower consumes the CFADS the covenant measures (9.2.2), and no amount of subordination language changes the arithmetic.

Option three: mezzanine at HoldCo level. The same 4,000,000 raised by the sponsors' holding company, serviced from **distributions**. The SPV's DSCR stays at **1.3054** because the SPV has no additional obligation; the mezzanine is covered 1,493,642/571,697 = **2.6126** times out of distributions; and the sponsors' equity falls to 15,000,000 with an IRR of **12.4934 %** — 10.66 basis points better than option one, on 4,000,000 less cash committed. The bank consented, subject to the mezzanine having no recourse to the SPV and no security over its shares.

How it resolved, and what nearly went wrong. The sponsors took option three. Six months into operations the adviser ran the stress that should have been run first. The distribution lock-up triggers when CFADS falls below 4,890,358.20 × 1.15 = **5,623,912**, which is **11.9061 %** below base case. Immediately above that line the mezzanine still looks serviceable: at an **11.5 %** shortfall CFADS is **5,649,840**, the SPV's DSCR is **1.1553** (a covenant breach, but payment made in full), distributions are **759,482** and HoldCo cover is **1.3285** — yet that case sits only **25,928** above the trigger. Four-tenths of a percentage point further down — a **12 %** shortfall, CFADS **5,617,920**, a further **31,920** of cash gone — the lock-up trips, the SPV retains cash it would otherwise have distributed, and **HoldCo coverage goes from 1.3285 to zero in a single step** while the senior lenders are paid on time. The mezzanine's 2.6126× cover in the base case was therefore worth almost nothing in the one scenario that mattered: it stood 2.6 times covered against a cliff **11.9 %** away, with no gradual deterioration to warn anyone. The sponsors renegotiated a 12-month HoldCo interest reserve of **571,697** and a standstill on HoldCo enforcement while an SPV lock-up subsisted, and the structure held.

What the domain teaches here. Mezzanine's placement, not its price, is its defining characteristic. HoldCo mezzanine is invisible to the SPV's coverage test and entirely dependent on the distribution it sits behind, so its true coverage is not the base-case multiple but the distance between base case and the **lock-up trigger** — for Kestrel, **11.9061 %**. Any junior instrument serviced from distributions must be stress-tested against the lock-up rather than the covenant, because the exposure is a cliff and not a slope: the number that matters is the shortfall at which the cash stops, not the multiple by which it is covered today.

Case study B — Domain 9: the cheap money that bought expensive rolling stock (transport)

Situation. A metropolitan rail authority's concessionaire needed a **120,000,000** rolling-stock and signalling package. Supplier N, in a country with an active export credit agency, quoted **120,000,000**. Supplier S quoted **111,111,111 — 8.0 %** less, a difference of **8,888,889** — but with no ECA support available. The ECA offered **85 %** of the contract value at **3.90 %** over **14 years**, with an exposure premium of **6.5 %** capitalised. The commercial alternative for Supplier S was 85 % at **6.30 %** over **12 years** with a 1.10 % fee. The concessionaire's finance team recommended Supplier N: "the money is 240 basis points cheaper."

The arithmetic done properly. All-in costs first. Route N: ECA loan 120,000,000 × 85 % = **102,000,000**, premium **6,630,000**, financed loan **108,630,000**, instalment 108,630,000/

`AF(0.039,14) = 108,630,000/10.633202 =` **10,216,114**; solving on the 102,000,000 of proceeds gives an all-in of **4.8656 %** — a full point above the 3.90 % headline. Route S: commercial loan **94,444,444**, fee **1,038,889**, proceeds **93,405,556**, instalment `94,444,444/8.247657 =` **11,451,063**, all-in **6.5041 %**. So the money really is **164 basis points** cheaper, not 240.

But the money and the equipment must be decided together. Discounting each route's whole cash profile — the own-funds outflow at close plus the debt service — at the project's 8.0 % cost of capital: Route N costs **102,224,065** in present value against Route S's **104,001,662**. **Route N wins by 1,777,597** on a 120,000,000 package — **1.5 %**, and the finance team's recommendation was right, for reasons it had not calculated.

How it resolved. The team then computed the number that should have opened the analysis: the **breakeven equipment premium**, the price differential at which the two routes are indifferent. At an 8.0 % discount rate it is **9.8780 %**. The ECA route was worth paying up to about a **9.9 %** premium on the equipment, and the actual premium was **8.0 %** — a margin of **1.9 percentage points**, against a sole-source negotiation still in progress and a discount-rate assumption that moved the breakeven to **8.7179 %** at 6 % and **10.8612 %** at 10 %. The decision was live, not comfortable. Two months later the ECA re-rated the host country and raised its exposure premium from 6.5 % to **9.0 %**, lifting the financed loan to 111,180,000, the instalment to **10,455,929**, the all-in to **5.2291 %** and the route's present value to **104,201,155** — **199,494 worse** than Route S. The flip point on the exposure premium was **8.7477 %**. The concessionaire switched to Supplier S, having preserved the option by refusing to sign a sole-source commitment while the premium was un-fixed.

What the domain teaches here. A financing decision that is tied to a procurement decision is one decision, and it must be evaluated on the present value of the combined cash flows rather than on a comparison of rates. The output a leader should demand is not "which tranche is cheaper" but **the breakeven premium** — the price differential at which the cheap money stops being worth its conditions — because that number is a negotiating position and a monitoring trigger, while a rate comparison is neither. Route N also carried a real coverage advantage worth naming: annual debt service of 10,216,114 against 11,451,063, **1,234,949** a year lower, almost entirely from two extra years of tenor (9.3.3). Had coverage been the binding constraint rather than cost, the answer would have survived the premium re-rating.

Executive perspective — Domain 9

What a project finance director cannot delegate in this domain:

- **The hurdle rate, derived.** Not inherited, not a policy round number. The director owns the build-up — `r_f`, `ERP`, `β_a`, `CRP`, `SP`, tax rate, each with a named source and a date — and owns the reconciliation. Kestrel's 8.0001 % at the coverage-binding gearing was Domain 4's given 8.0 %; that agreement was luck until somebody checked it.
- **The refusal to use the group's cost of capital.** Kestrel's project `WACC` is **8.0001 %** against the sponsor's corporate **6.544 %**; discounting the project at the group rate raises its `NPV` from 16,179,360 to **23,447,722** — an overstatement of **7,268,362**, or **44.92 %**. The gap is structural, not a rounding difference: a diversified investment-grade group carries lower gearing, cheaper debt, a lower asset beta, and neither a country premium nor a single-asset illiquidity premium. **The rate belongs to the project, not to the balance sheet that sponsors it**, and a director who allows the group rate into a project appraisal has authorised the systematic acceptance of value-destroying projects.

- **The gearing decision, on both faces.** The exchange rate — 5.75 basis points of equity return per hundredth of DSCR at the margin — and the gearing at which each covenant binds (68.62 %, 74.34 %, 77.57 %). These are the director's numbers because they trade a lender's protection against a shareholder's return, and no analyst should be asked to make that trade.
- **Where a grant lands.** The same 6,000,000 either lifts equity return 429 basis points or lifts coverage 21 points. The director states the objective **before** the allocation is negotiated, and states the treatment convention before any WACC is quoted.
- **The all-in discipline.** No tranche enters a comparison on its headline rate. The DFI tranche at 5.25 % costs 7.2465 %; the ECA tranche at 3.80 % costs 4.6895 %; the ijara's 15 basis point premium is really 4. The director enforces the method, because the method is where the errors are.
- **The honesty of a refinancing.** Report gains net of cost and gain-share, decomposed into rate and extension, at more than one discount rate — and never let a deferral be presented as value created.

Calculation exercises — Domain 9

Exercise 9.1 A 150,000,000 project is geared 70/30. Senior debt costs 5.5 % and the tax rate is 25 %. Build the cost of equity from `r_f` 3.80 %, `ERP` 5.5 %, `β_a` 0.55, `CRP` 1.20 % and `SP` 0.40 %, then compute the `WACC`. Solution. `D/E = 105,000,000/45,000,000 = 2.333333`; `(1 − 0.25) × 2.333333 = 1.75`; `β_e = 0.55 × 2.75 =` **1.5125**; `β_e × ERP = 8.31875`; `k_e = 3.80 + 8.31875 + 1.20 + 0.40 =` **13.71875 %**. After-tax debt `5.5 × 0.75 =` **4.125 %**; `WACC = 0.70 × 4.125 + 0.30 × 13.71875 =` **7.0031 %**. Common error: using the pre-tax cost of debt, giving 7.9656 % — a 96 basis point overstatement that would kill marginal projects. A second common error is failing to re-lever at all (`β_e = β_a = 0.55`), giving `k_e` 8.425 % and a `WACC` of 5.4150 % — a 159 basis point understatement, which approves projects that destroy value.

Exercise 9.2 A 100,000,000 project generates level `CFADS` of 14,000,000 over a 22-year life; debt costs 6.5 % over 15 years (`AF(0.065, 15) = 9.402669`). Compare 65 % and 75 % gearing on `DSCR` and equity `IRR`. Solution. At 65 %: debt 65,000,000, equity 35,000,000, debt service `65,000,000/9.402669 =` **6,912,930.89**, `DSCR` **2.0252**, distributable 7,087,069.11, equity `IRR` **20.7836 %**. At 75 %: debt 75,000,000, equity 25,000,000, debt service **7,976,458.72**, `DSCR` **1.7552**, distributable 6,023,541.28, equity `IRR` **24.8151 %**. Ten points of gearing buys **403.15 basis points** of equity return for **0.2700** of coverage. Common error: computing the equity `IRR` on the whole `CFADS` stream rather than on `CFADS` less debt service, which produces the project return and makes gearing look irrelevant.

Exercise 9.3 Senior 80,000,000 at 5.2 %, mezzanine 20,000,000 at 10.8 %, equity 50,000,000 at 14.5 %; tax 30 %; both debt tranches deductible. Compute the `WACC`. Solution. After-tax senior `5.2 × 0.7 =` **3.64 %**; mezzanine `10.8 × 0.7 =` **7.56 %**; `WACC = (80 × 3.64 + 20 × 7.56 + 50 × 14.5)/150 = (291.2 + 151.2 + 725)/150 =` **7.7827 %**. Common error: tax-adjusting the equity cost as well, giving 6.3327 % — a 145 basis point understatement. Also common: a simple average of the three costs (8.5667 %), which ignores that senior debt is more than half the capital.

Exercise 9.4 A 40,000,000 tranche carries 4.10 % over 12 years with a 5.5 % exposure premium capitalised into the loan. Compute the all-in cost on the proceeds actually available. Solution. Premium **2,200,000**; loan **42,200,000**; `AF(0.041, 12) = 9.330854`; instalment `42,200,000/9.330854`

> `= `**4,522,630.17**; solve `40,000,000 = 4,522,630.17 × AF(r, 12)` → **5.0378 %**, an uplift of **93.78 basis points** over the headline. Common error: spreading the premium straight-line across the tenor and adding it to the rate (`4.10 + 5.5/12 = 4.5583 %`), which ignores that the premium is itself financed and repaid with a return on it.

> **Exercise 9.5** A 120,000,000 project has `CFADS` of 13,000,000 and debt at 6.2 % over 13 years (`AF(0.062, 13) = 8.750167`); tax 25 %. An 18,000,000 grant is awarded. Case (a): the grant displaces equity, leaving debt 84,000,000 and equity 18,000,000, with `k_e` 16.50 %. Case (b): the grant displaces debt, leaving debt 66,000,000 and equity 36,000,000, with `k_e` 13.50 %. Compute `DSCR` and `WACC` for each. Solution. After-tax debt `6.2 × 0.75 =` **4.65 %**. Case (a): debt service `84,000,000/8.750167 =` **9,599,817.02**, `DSCR` **1.3542**; `WACC = (84 × 4.65 + 18 × 16.50 + 18 × 0)/120 =` **5.7300 %**. Case (b): debt service `66,000,000/8.750167 =` **7,542,713.37**, `DSCR` **1.7235**; `WACC = (66 × 4.65 + 36 × 13.50)/120 =` **6.6075 %**. The equity-displacing case has the **lower** `WACC` and the **worse** coverage — the 9.4.2 result, in a different project. Common error: omitting the grant from the denominator of the `WACC` weights, which inflates the debt and equity shares to sum to more than one and produces a meaningless figure.

Practitioner's toolkit — Domain 9

Adoption-ready artefacts; adapt headings to your organisation, then keep them stable.

Toolkit 9.T.1 — Cost-of-capital derivation record (one per project, per structure)

One page, signed. Rows: `r_f` — instrument, tenor, currency, quotation date, source · `ERP` — value, source, date, market · `β_a` — value, comparator set named, source of the unlevering · re-levering formula and whether non-beta premiums are re-levered · `D/E` used, and confirmation it matches the funding plan · `CRP` and `SP` — value and basis · resulting `k_e` · tax rate and the written opinion supporting deductibility, per tranche · each tranche's pre- and post-tax cost · weights and their sum · resulting `WACC` · **the `WACC` at the coverage-binding gearing** and the gap to the proposed structure · the sponsor's corporate `WACC`, stated for contrast, with a note that it is **not** the project rate. Rule: no appraisal in the organisation uses a discount rate without a current signed record, and the record is re-signed whenever the funding plan changes.

Toolkit 9.T.2 — Tranche comparison sheet (all-in, like-for-like)

One column per candidate tranche. Rows: amount · currency · base rate and margin, or profit rate · fee schedule (arrangement, commitment, agency, structuring) · capitalised premiums · tenor, availability period and amortisation profile · grace · prepayment mechanics and any make-whole · security and ranking · conditions precedent and expected time to satisfy · compliance and monitoring cost, for the full tenor · currency of repayment against currency of revenue · **cash proceeds actually available** · **all-in effective rate, solved from the stream** · annual debt service · `DSCR` contribution · **debt capacity supported at the target coverage** · grant element, where concessional. Two mandatory footers: the method statement (all-in rates solved, not added) and the separation rule (cost effects and capacity effects reported in different rows, never netted).

Toolkit 9.T.3 — Capital-structure decision record

For each candidate structure, one row: tranche amounts and costs · re-levered `k_e` · `WACC` · senior-only `DSCR` · total `DSCR` · minimum `DSCR` over the loan life · gearing at which each covenant and the lock-up binds · equity `IRR` on the bank case **and** on the sponsor case, with the escalation assumption that separates them · treatment of any grant, stated · junior instruments and their placement (SPV or HoldCo), with the lock-up shortfall at which HoldCo service stops · refinancing assumption, if any, and whether a gain-share applies. Front line: **which constraint binds, at what gearing, and what the structure costs in `WACC` to stay inside it.** The record is minuted with the decision, and the rejected structures stay in it.

Exam preparation — Domain 9

What is assessed. Building a cost of equity from stated components with correct re-levering; computing an after-tax cost of debt and a multi-tranche `WACC` with correct weights and tax treatment; quantifying the leverage trade-off on both faces and identifying the gearing at which a covenant binds; distinguishing a tranche's cost effect from its debt-capacity effect; computing an all-in effective cost from a stream containing fees and capitalised premiums; computing a grant's effect on `WACC`, `DSCR` and equity return, and a concessional loan's grant element; testing sustainability-linked pricing against its verification cost; and computing and decomposing a refinancing gain.

The calculations to have automatic under time pressure. `k_d(1 − T)`. `β_e = β_a(1 + (1 − T)D/E)` and the build-up onto it. `WACC` as a weighted sum with the grant at zero cost and in the denominator. `DSCR = CFADS ÷ (D ÷ AF(r, n))` and its inversion to a binding gearing (`max debt = CFADS/target × AF`, then ÷ capex). All-in cost as the rate solving `proceeds = payment × AF(r, n)`. Grant element as `1 − PV(service at market rate)/face`. Outstanding balance as `instalment × AF(r, remaining)`.

The traps.

- Using the pre-tax cost of debt in `WACC` — Exercise 9.1, 96 basis points.
- Tax-adjusting the cost of equity — Exercise 9.3, 145 basis points.
- Failing to re-lever `k_e` when comparing structures — 9.1.3, 9.2.3; every comparison in KA 9.2 is invalid without it.
- Re-levering **without** the tax adjustment — MCQ 9.1-A, distractor C.
- Omitting a grant from the `WACC` denominator so the weights exceed one — Exercise 9.5.
- Comparing tranches on headline rate plus fees instead of the solved all-in rate — 9.3.1, 9.3.2, Exercise 9.4.
- Treating a capitalised premium as a one-off percentage added to the rate — Exercise 9.4.
- Netting a cost effect against a capacity effect — 9.3.3, MCQ 9.3-B.
- Placing mezzanine at SPV level and expecting the senior `DSCR` to be unaffected — 9.2.2, Case A option two.
- Stress-testing HoldCo junior debt against the covenant instead of the lock-up — Case A, 11.9061 %.
- Reporting a refinancing extension as value created — 9.4.4, MCQ 9.4-C.
- Discounting a project at the sponsor's corporate `WACC` — Executive perspective, 44.92 % overstatement.
- Quoting an equity `IRR` without naming its cash-flow case, or without disclosing an equity bridge — 9.1.1, 9.1.4.
- Modelling a shareholder-loan tax shield without a tax opinion, or with the interest mis-ranked — 9.1.2, MCQ 9.1-D.
- Valuing a sustainability ratchet against a single year's compliance cost rather than the full tenor — 9.4.3.

How the domain connects. Domain 3 supplied `AF(r, n)`, which does almost all the arithmetic here, and Domain 4 the appraisal this domain's `WACC` finally justifies. Domain 7's indexation drafting supplies the 1.7403 % escalation on which the equity case rests; Domain 8's contingency and delay arithmetic sets the funding headroom. Domain 10 supplies the coverage machinery that constrains everything here and the four sizing levers this domain prices; Domain 11 allocates the risks that set `CRP`, `SP` and the required coverage; Domain 12 documents the intercreditor, security and gain-share terms; Domain 13 audits the model; Domain 14 draws the tranches; and Domain 15 lives with the structure and executes the refinancing this domain sized.

Domain 9 summary

Funding structure is where a project's economics are finally decided, and this domain's central claim is that the decision is **not** made by minimising the cost of capital. Kestrel's cost of equity, built up from a 4.10 % risk-free rate, a 6.00 % equity risk premium on an asset beta of 0.60 re-levered to **1.72**, and one point of country and single-asset premium, is **15.42 %**; its after-tax cost of debt, on the 20.0 % cash tax rate the thread's own numbers imply, is **4.80 %**; and its `WACC` at the proposed 70/30 structure is **7.9860 %**. The whole ladder collapses to `WACC(g) = 8.70 % − 1.02 % × g`, whose slope is small precisely because the non-systematic equity premium of 5.10 % barely exceeds after-tax senior debt at 4.80 % — so across twenty points of gearing the project saves **20.40 basis points** while the sponsors gain **184.61** and the lenders lose **0.3716** of `DSCR`. Gearing is a transfer, not an efficiency, and its exchange rate at the margin is about **5.75 basis points of equity return per 0.01 of coverage surrendered**. The constraint that decides the structure is coverage: the 1.30× sizing target binds at **68.6185 %** gearing, the 1.20× covenant fails on the base case above **74.3367 %**, and the 1.15× lock-up above **77.5688 %**. Evaluated at the coverage-binding gearing the project `WACC` is **8.0001 %** — which is Domain 4's given 8.0 %, derived at last, and right only for that structure; the sponsor's corporate `WACC` of **6.544 %** would have overstated the appraisal `NPV` by **7,268,362**, or **44.92 %**, and is never the project's rate. Within the stack, mezzanine raises the blended cost in every direction — **8.4510 %** displacing senior, **8.3240 %** displacing equity — and is bought for coverage relief (senior `DSCR` **1.5115**) or equity uplift, never for cheaper capital; its **placement** decides everything, and HoldCo mezzanine covered **2.6126** times in the base case receives nothing once `CFADS` falls **11.9061 %** to the lock-up trigger. Every tranche is compared on an all-in rate solved from its own stream: the ijara's 15 basis point headline premium is really **3.95**; the ECA's 3.80 % is **4.6895 %** and releases **303,726** a year of debt service; the DFI's 5.25 % is **7.2465 %** all-in yet its six extra years of tenor lift debt capacity from **41,171,123** to **56,299,948** — because tenor, not rate, moves the binding constraint. Public support must be aimed: the same 6,000,000 grant either lifts equity return **429.20 basis points** or coverage **21.24 points**, and the treatment convention alone moves `WACC` by **67.20 basis points**; a 12,000,000 concessional loan at 2 % carries a **40.80 %** grant element. A 15 basis point sustainability ratchet worth **528,182** cannot pay for **712,627** of verification, so the case for a label must rest on the base margin and the investor pool, not on the ratchet. And a refinancing gain of **3,723,616** is **298,757** of rate saving and **3,424,859** of deferral, worth only **566,832** discounted at the loan rate and halved again by a gain-share — which is why every gain is reported net, decomposed, and at more than one rate. Domain 10 tests the coverage this domain has just priced; Domain 13 audits the model behind it; Domain 15 operates the structure and executes the refinancing.

03

PART THREE

Executing the transaction

Domains 10–13 — debt sizing and covenants, risk allocation, contracts and transaction structure, and due diligence to financial close: turning an appraised project into a funded one.

10

DOMAIN 10

Debt Sizing, Covenants and Credit Metrics (quantitative flagship)

Group: Executing the transaction (Part Three). **Target:** ~80 pages. **Binds to:** the PCI Book Pattern Specification and the shared registries (`docs/books/registries/`). This domain is the home of `CFADS`, `DSCR`, `LLCR`, `PLCR` and `ICR`, and it is where Domains 2, 3 and 4 converge: Domain 2 defined `CFADS` and warned that it is a defined term, Domain 3 built the discounting and the amortisation schedule, Domain 4 built the appraisal. British English; USD (+SAR where useful, indicative `USD 1 ≈ SAR 3.75`).

Why this domain exists

Everything before this point valued a project. This domain decides **how much debt it can carry, and on what conditions** — which is the question that determines whether the financing exists at all. It is the lender's side of the table, and a project finance leader who cannot compute it is negotiating blind against people who can.

The logic is inverted from corporate finance and that inversion is the domain's central idea. A company borrows against its balance sheet; a project's debt is sized **from its cash flow outwards**: forecast `CFADS`, divide by the coverage the lender requires, and the result is the debt service the project can afford — from which the debt amount follows. Debt capacity is therefore not a negotiation about appetite but an arithmetic consequence of cash flow and required coverage (KA 10.1). The coverage ratios themselves — `DSCR`, `LLCR`, `PLCR`, `ICR` — measure different things and are routinely confused (KA 10.2). Reserve accounts and the debt-service schedule convert the arithmetic into liquidity that survives a bad year (KA 10.3). And covenants, default, cure rights and distribution lock-up are the machinery that gives lenders control before, not after, a project fails (KA 10.4).

Learning objectives. After this domain a candidate can: define `CFADS` and explain why its documented definition governs every ratio built on it; size debt from a `CFADS` forecast and a target `DSCR`; compute and interpret `DSCR`, `LLCR`, `PLCR` and `ICR`, and state what each sees that the others do not; explain and apply debt sculpting; stress a coverage position and identify the covenant that binds first; explain reserve accounts and size a debt-service reserve; describe covenant types, the lock-up mechanism, events of default and cure rights; explain why lenders prefer cash-based to accounting-based covenants; and govern AI-assisted covenant and model analysis.

The master financing. Kestrel Water SPC continues from Domains 1–4. Its senior loan is **USD 42,000,000 at 6.0 % over 12 years** (Domain 3's schedule: annual instalment **USD 5,009,635**, year-one interest **USD 2,520,000**, year-one principal **USD 2,489,635**). Domain 2 derived its first-year `CFADS` **of USD 6,384,000** on the facility's documented definition — after working-capital movements. Equity at close was **USD 18,000,000**, giving 70/30 gearing. The project's operating life is **25 years**; the loan's is 12.

KNOWLEDGE AREA 10.1

Debt capacity and sizing

Topics: 10.1.1 *CFADS* — the term everything rests on · 10.1.2 sizing from coverage · 10.1.3 sculpting and the shape of debt service.

10.1.1 CFADS — the term everything rests on

Definition. CFADS is the cash a project generates that is available to service debt, in a period, **before** debt service and after everything that must be paid first. Its construction runs:

```
Revenue collected
  − cash operating costs
  − cash taxes
  − maintenance capex (usually; sometimes below debt service)
  ± movements in working capital
  ± movements in reserve accounts (as defined)
  = CFADS
```

Every one of those lines is negotiated. Domain 2 (KA 2.3.1) demonstrated the consequence with Kestrel: CFADS of **6,984,000** before working-capital movements and **6,384,000** after, on the same year's trading. That is a DSCR of 1.39 versus 1.27 — the difference between comfort and near breach — from a definitional choice.

The professional discipline follows directly. CFADS **is whatever the finance documents say it is**, and the ratio a lender enforces is computed on their definition, not on textbook practice. The leader's obligations are to know the definition, to ensure the model implements it (Domain 13's model audit checks precisely this), and never to quote a coverage ratio without being able to state the CFADS definition underneath. The recurring negotiation points worth naming: whether maintenance capex sits above or below CFADS; whether cash taxes or accounting tax are used (Domain 2, KA 2.A.1 — cash, always, and the model must implement it); whether working capital is included; and how reserve-account movements are treated.

10.1.2 Sizing from coverage

The principle. Debt is sized so that forecast cash covers debt service by the required margin:

```
Maximum debt service per period = CFADS ÷ target DSCR
Maximum debt                    = maximum debt service × AF(r, n)
```

The second line is Domain 3's annuity factor doing the work: level debt service discounted at the loan rate over the loan tenor gives the principal it can support.

WORKED EXAMPLE

Worked example 10.1.2 — how much can Kestrel actually borrow?

1. **Setup.** `CFADS` **USD 6,384,000** per year (level, for this illustration), loan rate **6.0 %**, tenor **12 years**, lender's target `DSCR` **1.30×**. The sponsors have asked for USD 42,000,000.
2. **Formula.** Max debt service = `CFADS`/target `DSCR`; max debt = max debt service × `AF(0.06, 12)`.
3. **Substitution.** `6,384,000 / 1.30 = 4,910,769`; `AF(0.06, 12) = 8.383844`; `4,910,769 × 8.383844`.
4. **Result.** Maximum debt service **USD 4,910,769**; maximum debt **USD 41,171,123** — so the requested 42,000,000 is **USD 828,877 too much** at a 1.30× target. At 42,000,000 the actual `DSCR` is **1.2743**, below the target.
5. **Interpretation.** This single calculation is the centre of most financing negotiations, and notice what it does not depend on: the sponsors' preference, the project's NPV (Domain 4's +16.2m), or the asset's cost. Debt capacity is a function of **cash flow, required coverage, rate and tenor** — nothing else. That gives the leader four levers and no others: raise `CFADS` (revenue or cost), argue the coverage requirement down (usually by reducing revenue risk — Domain 7's contracted structures, Domain 11's allocation), lengthen the tenor, or lower the rate. The fifth option is to **contribute more equity**, which is what happens when the first four fail — and the 828,877 gap is exactly the additional equity Kestrel's sponsors would need to find.

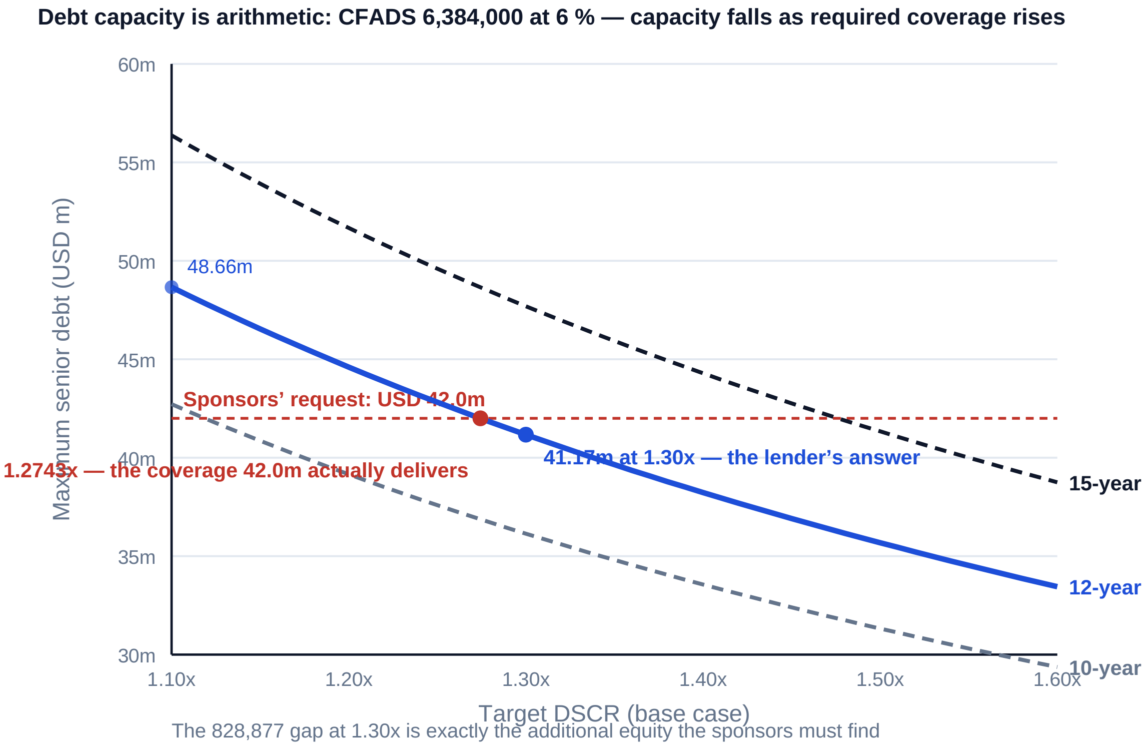

Fig 10.1.1 — Debt capacity as a function of coverage and tenor

10.1.3 Sculpting and the shape of debt service

The problem with level debt service. An annuity assumes cash flow is level. Projects rarely oblige: availability payments may escalate, merchant revenue may ramp, and major maintenance may fall in specific years. Level debt service against uneven cash produces **wasted coverage** in good years and **breach risk** in weak ones.

Sculpting sets each period's debt service so that coverage is constant by design:

```
Debt service in period t = CFADS(t) ÷ target DSCR
```

The principal profile then follows from what is left after interest — irregular by construction, and that is the point. Sculpted debt supports **more** total debt than level debt against the same uneven cash flow, because no period is over-covered merely to protect the weakest one. The trade is complexity: an irregular repayment profile has to be documented, modelled and monitored, and a sculpted schedule computed on a forecast must be re-cut when the forecast changes (which the facility agreement provides for, or does not).

Two related shapes worth naming. A **cash sweep** applies a defined share of surplus cash above the required coverage to accelerated repayment — protecting lenders and shortening the effective tenor while reducing distributions. A **balloon or bullet** leaves principal at maturity, cutting periodic debt service and creating refinancing risk, exactly as Domain 3 (Case study B) priced it.

AI in this KA

Model-building and scenario generation are legitimate machine work at this scale, and there is a specific failure mode here worth naming: an assistant asked to "size the debt" will apply the textbook `CFADS` definition rather than the facility's, and produce a defensible-looking number that the lender's own model will contradict. Verification is therefore definitional before it is arithmetical — check that the model's `CFADS` line implements the documented definition clause by clause, then recompute one period's ratio by hand. **AI proposes; the professional verifies, decides and remains accountable.**

KEY TERMS — KA 10.1

Term	Meaning
`CFADS`	Cash available for debt service in a period, on the facility's documented definition.
Debt capacity	`CFADS`/target `DSCR` × `AF(r, n)`; a function of cash, coverage, rate and tenor only.
Target `DSCR`	The coverage the lender requires; the divisor in sizing.
Sculpting	Setting each period's debt service to hold coverage constant against uneven cash.
Cash sweep	Mandatory prepayment from a defined share of surplus cash.
Balloon / bullet	Principal deferred to maturity; lower periodic service, refinancing risk.

SAMPLE MCQS — KA 10.1

MCQ 10.1-A `[10.1.2 · Application]` `CFADS` is 6,384,000 per year; the lender requires a 1.30× `DSCR`; the loan runs 12 years at 6 % (`AF` = 8.383844). Maximum debt is closest to: - A. USD 42,000,000 - B. USD 41,171,123 - C. USD 53,522,460 - D. USD 36,143,689

Rationale: `6,384,000/1.30 = 4,910,769`; `× 8.383844 = 41,171,123`. A is the requested amount, which the calculation rejects; C omits the coverage divisor (sizing on full `CFADS`, `6,384,000 × 8.383844`); D uses a 10-year tenor (`AF = 7.360087`) instead of the 12-year tenor.

MCQ 10.1-B [10.1.1 · Analysis] Two advisers compute Kestrel's DSCR as 1.39 and 1.27 from the same audited year. The most likely explanation is: - A. one has made an arithmetic error - B. they are applying different CFADS definitions — one before and one after working-capital movements — and only the facility's definition governs - C. they are using different interest rates - D. one has used accounting profit instead of cash

Rationale: Domain 2's demonstration exactly: 6,984,000 versus 6,384,000 of CFADS on the same trading. The documented definition decides which is enforceable (10.1.1). D would produce a much larger discrepancy and is a different error.

MCQ 10.1-C [10.1.3 · Analysis] A project's cash flow ramps over its first five years. Against the same forecast, sculpted debt service compared with level debt service will: - A. support less total debt, being more complex - B. support more total debt, because no period is over-covered merely to protect the weakest one - C. support the same debt, since total cash is unchanged - D. eliminate refinancing risk

Rationale: Level service is constrained by the weakest period; sculpting holds coverage constant and so uses the stronger periods (10.1.3). C ignores that capacity is set period by period, not in total; D confuses sculpting with tenor structure.

SELF-CHECK — KA 10.1

1. Name the only four variables debt capacity depends on. — CFADS, target coverage, rate, tenor. (Equity contribution is the residual, not a driver.)
2. Why can two correct *DSCR* figures differ for one year? — Different CFADS definitions; only the facility's governs.
3. What does sculpting buy and what does it cost? — More debt against uneven cash; complexity in documentation, modelling and re-cutting.

KNOWLEDGE AREA 10.2

The coverage ratios

Topics: 10.2.1 *DSCR* · 10.2.2 *LLCR* and *PLCR* · 10.2.3 *ICR* and leverage · 10.2.4 reading a ratio set together.

10.2.1 DSCR — the period test

```
DSCR = CFADS ÷ debt service          (debt service = interest + scheduled principal)
```

DSCR is a **period** measure: it asks whether this period's cash covers this period's obligations. It is the ratio lenders covenant on, test on defined dates, and lock up distributions against, because liquidity failures happen in periods, not in averages.

WORKED EXAMPLE

Worked example 10.2.1 — Kestrel's year-one position, and what breaks it.

1. **Setup.** `CFADS` **6,384,000**; debt service **5,009,635** (Domain 3's instalment). The facility carries a **1.20× covenant** and a **1.15× lock-up**. Compute the `DSCR`, the cash headroom to each threshold, and the position under a 20 % `CFADS` shortfall.
2. **Formula.** `DSCR` = `CFADS`/debt service. Threshold cash = debt service × threshold.
3. **Substitution.** `6,384,000 / 5,009,635`. Covenant cash `5,009,635 × 1.20`; lock-up cash `× 1.15`. Stress: `6,384,000 × 0.80 = 5,107,200`, then `÷ 5,009,635`.
4. **Result.** `DSCR` **1.2743**. Covenant is breached below `CFADS` of **6,011,562**; distributions lock up below **5,761,081**; the project fails to cover debt service at all below **5,009,635**. Under a 20 % `CFADS` shortfall, `DSCR` falls to **1.0195** — a **covenant breach**, though the project still pays.
5. **Interpretation.** The headroom is **USD 372,438** of annual cash (6,384,000 − 6,011,562) — 5.8 % of `CFADS`. That is the sentence that belongs in a board paper, not "`DSCR` 1.27": it says how much cash may be lost before a covenant is breached, which is what management can actually monitor. And the stress case makes the crucial distinction between **breach and default on payment**: at 1.0195 the lenders are paid in full and the project is nonetheless in breach, with all the consequences of KA 10.4. Covenants bite long before cash runs out — by design, because that is when intervention still works.

10.2.2 `LLCR` and `PLCR` — the horizon tests

```
LLCR = PV(CFADS over the remaining loan life, discounted at the loan rate) ÷ debt outstanding
PLCR = PV(CFADS over the remaining project life)                          ÷ debt outstanding
```

Where `DSCR` asks about a period, these ask about a **horizon**: is there enough total discounted cash ahead to repay what is outstanding? `LLCR` looks to the loan's maturity; `PLCR` looks to the end of the project's economic life, and therefore counts the cash that exists after the loan is due — which is why `PLCR` exceeds `LLCR` whenever the project outlives the loan.

WORKED EXAMPLE

Worked example 10.2.2 — Kestrel's three ratios, and an identity worth knowing.

1. **Setup.** `CFADS` 6,384,000 per year (level), debt outstanding 42,000,000, loan rate 6 %, loan life 12 years, project life 25 years.
2. **Formula.** As above, with `AF(0.06, 12) = 8.383844` and `AF(0.06, 25) = 12.783356`.
3. **Substitution.** `LLCR = (6,384,000 × 8.383844)/42,000,000`; `PLCR = (6,384,000 × 12.783356)/42,000,000`.
4. **Result.** `DSCR` **1.2743** · `LLCR` **1.2743** · `PLCR` **1.9431**.
5. **Interpretation.** `LLCR` equals `DSCR` **exactly** — and that is not a coincidence but an **identity**: when `CFADS` is level and debt service is an annuity at the discount rate used, the two ratios must coincide, because both are the same cash divided by the same present value. A reviewer should use it as a check: level-cash models whose `DSCR` and `LLCR` differ contain an inconsistency (usually a discount rate that is not the loan rate, or a `CFADS` line that differs between the two calculations). Where cash is not level the ratios diverge, and the divergence is informative — `LLCR` above `DSCR` means the weak period is early and later cash is stronger. `PLCR`'s 1.9431 says something different again: substantial value exists beyond the loan's maturity, which is what makes a **tail** and supports refinancing (Domain 15). Lenders nevertheless rely on `PLCR` least, because cash beyond their maturity is cash they have no contractual claim on.

10.2.3 `ICR` and leverage

`ICR` = `EBIT` (or `EBITDA`) ÷ interest — an **accounting** coverage measure that ignores principal entirely. Domain 2 computed Kestrel's `EBIT`-based interest cover at 2.02× and, on `EBITDA`, **2.98×**. Leverage measures — debt/`EBITDA`, gearing — describe the balance sheet: Kestrel's 42,000,000 debt against 18,000,000 equity is **2.33:1**, the 70/30 structure of Domain 1.

Why project lenders covenant on `DSCR` rather than `ICR`: `ICR` can look comfortable while principal goes unpaid, and it is computed on accounting figures exposed to the classification judgments Domain 2 (Case study B) showed can flip a covenant without any change in cash. `ICR` appears mainly in corporate facilities and as a secondary project covenant.

10.2.4 Reading a ratio set together

No single ratio is sufficient, and the four answer different questions:

Ratio	Question	Blind to
`DSCR`	Can this period pay?	Everything outside the period
`LLCR`	Is there enough cash to loan maturity?	Timing within the loan life
`PLCR`	Is there value beyond the loan?	The lender's lack of claim on it
`ICR`	Do earnings cover interest?	Principal; and exposed to accounting judgment

Kestrel's set — 1.27 / 1.27 / 1.94 / 2.98 — describes a project with adequate but unspectacular period coverage, no timing distortion, a substantial tail, and comfortable interest cover. The story a lender reads is: repayable, thin in a bad year, refinanceable. That reading, not any single number, is what credit committees actually decide on.

AI in this KA

Computing four ratios across a 25-year model is exactly what machines should do, and their outputs are dangerously plausible because the arithmetic is simple and the definitions are not. The invariants of 10.A.3 are the defence, and the `LLCR` = `DSCR` identity of 10.2.2 is the cheapest single check available on any level-cash model. Where an AI produces a covenant-compliance summary across a portfolio of facilities, the verification duty is to test it against the documents on a sample — because the failure will not be arithmetic, it will be a definition read from the wrong agreement.

KEY TERMS — KA 10.2

Term	Meaning
`DSCR`	`CFADS` ÷ debt service; the period test lenders covenant on.
`LLCR`	PV of `CFADS` to loan maturity ÷ debt outstanding; the loan-horizon test.
`PLCR`	PV of `CFADS` to end of project life ÷ debt outstanding; counts the tail.
Tail	Project life beyond loan maturity; supports refinancing.
`ICR`	Earnings ÷ interest; ignores principal, exposed to accounting judgment.
Headroom	Cash that can be lost before a covenant threshold is crossed.

SAMPLE MCQS — KA 10.2

MCQ 10.2-A `[10.2.1 · Application]` `CFADS` 6,384,000; debt service 5,009,635. The `DSCR` is: - A. 1.2743 - B. 0.7847 - C. 1.3941 - D. 2.5333

Rationale: `6,384,000/5,009,635 = 1.2743`. B inverts the ratio; C uses the pre-working-capital `CFADS` of 6,984,000 (Domain 2's other definition); D divides by interest only.

MCQ 10.2-B `[10.2.1 · Analysis]` With a 1.20× covenant and `DSCR` at 1.2743 on `CFADS` of 6,384,000, the most useful figure for a board is: - A. the `DSCR` of 1.2743 - B. the annual cash headroom of USD 372,438 — 5.8 % of `CFADS` — before the covenant is breached - C. the debt outstanding of 42,000,000 - D. the loan's remaining tenor

Rationale: Headroom states what may be lost before consequence, which management can monitor (10.2.1). The ratio alone does not convey magnitude, and C and D are facts, not exposures.

MCQ 10.2-C `[10.2.2 · Analysis]` A model with level `CFADS` and annuity debt service reports `DSCR` 1.27 and `LLCR` 1.41. The soundest conclusion is: - A. the project has a strong tail - B. there is an inconsistency: with level cash and annuity service at the loan rate the two must be equal - C. the `LLCR` is correct and the `DSCR` is stale - D. this is normal and requires no investigation

Rationale: The identity of 10.2.2 makes divergence a defect indicator — typically a discount rate that is not the loan rate, or differing `CFADS` lines. A describes `PLCR`, not `LLCR`.

MCQ 10.2-D `[10.2.3 · Analysis]` Why do project lenders covenant primarily on `DSCR` rather than `ICR`? - A. `ICR` is harder to compute - B. `ICR` ignores principal repayment and rests on accounting figures exposed to classification judgment - C. `ICR` is only used for equity investors - D. `DSCR` is required by accounting standards

Rationale: `ICR` can look healthy while principal is unpaid, and Domain 2 showed accounting classification flipping a profit-based covenant with no cash change. D confuses covenant drafting with accounting requirements.

■ **SELF-CHECK — KA 10.2**

1. When must *LLCR* equal *DSCR*? — Level CFADS with annuity debt service discounted at the loan rate; divergence signals an inconsistency.
2. Why do lenders rely least on *PLCR*? — It counts cash beyond their maturity, on which they have no contractual claim.
3. State Kestrel's headroom to covenant in cash terms. — USD 372,438 per year, 5.8 % of CFADS.

KNOWLEDGE AREA 10.3

Reserve accounts and the debt-service schedule

Topics: 10.3.1 the reserve family · 10.3.2 sizing a debt-service reserve · 10.3.3 the schedule and the waterfall's top.

10.3.1 The reserve family

Coverage ratios measure adequacy on a forecast; reserves provide liquidity when reality departs from it. The standard family:

Reserve	Purpose	Typical sizing
Debt service reserve (DSRA)	Bridge a short cash shortfall so scheduled service is paid	3–12 months of debt service
Maintenance reserve (MRA)	Fund lumpy major maintenance without a cash spike	Forward-looking schedule of major overhauls
Insurance / proceeds account	Hold claim proceeds for application per the documents	Event-driven
Distribution / lock-up account	Hold cash that fails a distribution test	Event-driven (KA 10.4)

A reserve is not a covenant substitute: it buys **time**, which converts a liquidity failure into a negotiation. Its cost is that funded cash is not distributed — which is precisely why sponsors argue for smaller reserves and lenders for larger ones, and why the negotiation is really about how long the lenders want before they must act.

10.3.2 Sizing a debt-service reserve

WORKED EXAMPLE

Worked example 10.3.2 — how much time does Kestrel's DSRA buy?

1. **Setup.** Annual debt service **USD 5,009,635**. The facility requires a **six-month** DSRA. `CFADS` is 6,384,000 in the base case. How much must be funded, and what shortfall does it absorb?
2. **Formula.** DSRA = debt service × months/12. Absorbable shortfall = DSRA ÷ debt service, as a fraction of a period's obligation.
3. **Substitution.** `5,009,635 × 6/12`.
4. **Result.** DSRA **USD 2,504,818**. It covers **half** of one year's debt service — meaning the project can absorb a `CFADS` collapse to as low as **2,504,818** (39 % of base case) in a single year and still pay in full.
5. **Interpretation.** Express the reserve as the shortfall it survives, not as a number of months: six months of debt service sounds procedural, while "we can lose 61 % of one year's cash and still pay" is a risk statement a board can weigh. Note what it does not do — it does not prevent the **covenant breach** (`DSCR` would be far below 1.20 in that year), so the reserve buys payment continuity and time to negotiate, not compliance. That distinction is the practical content of KA 10.4.

10.3.3 The schedule and the waterfall's top

The **debt-service schedule** is Domain 3's amortisation table with a purpose: it fixes the obligations each coverage test is measured against. Kestrel's year one — interest 2,520,000, principal 2,489,635, total 5,009,635 — is the denominator of the `DSCR`, and its accuracy is therefore load-bearing for covenant compliance.

The schedule sits inside the **cash waterfall** (built fully in Domain 15), whose ordering is the whole security architecture in miniature: operating costs, then taxes, then **senior debt service**, then reserve top-ups, then subordinated debt, then — only if every test passes — distributions to equity. Two consequences follow for a leader. **Equity is paid last, and only by permission** — which is the economic meaning of subordination. And **the reserve top-up sits above distributions**, so a depleted reserve is refilled before shareholders see anything; a sponsor forecasting distributions without modelling reserve replenishment has forecast cash that is contractually unavailable.

KEY TERMS — KA 10.3

Term	Meaning
DSRA	Debt service reserve; buys payment continuity and time, not compliance.
MRA	Maintenance reserve; smooths lumpy major maintenance.
Debt-service schedule	The obligation profile that coverage tests are measured against.
Cash waterfall	The contractual priority order for applying project cash.
Subordination	Junior claims paid only after senior tests pass.

SAMPLE MCQS — KA 10.3

MCQ 10.3-A `[10.3.2 · Application]` Annual debt service is 5,009,635 and the facility requires a six-month DSRA. The amount to be funded is: - A. USD 5,009,635 - B. USD 2,504,818 - C. USD 1,252,409 - D. USD 417,470

Rationale: `5,009,635 × 6/12 = 2,504,818`. A is twelve months; C is three; D is one month.

MCQ 10.3-B `[10.3.2 · Analysis]` In a year when `CFADS` falls to 3,000,000 against debt service of 5,009,635, a fully funded six-month DSRA means: - A. no covenant breach occurs, since the reserve covers the gap - B. scheduled debt service is paid in full from cash plus reserve, but the `DSCR` covenant is still breached - C. the lenders must accelerate the loan - D. distributions may continue, since payment was made

Rationale: The reserve preserves payment (gap 2,009,635, within the 2,504,818 reserve) but the ratio is computed on `CFADS`, so the covenant fails (10.3.2). D is wrong because a breach triggers lock-up (KA 10.4), and C overstates the automatic consequence.

MCQ 10.3-C `[10.3.3 · Recall]` In the cash waterfall, reserve-account top-ups rank: - A. below distributions to equity - B. above distributions to equity - C. above senior debt service - D. at the same level as operating costs

Rationale: Reserves are replenished before equity is paid (10.3.3); senior service ranks above the top-up, and operating costs above both.

SELF-CHECK — KA 10.3

1. What does a DSRA actually buy? — Payment continuity and time to negotiate; not covenant compliance.
2. How should a reserve be expressed to a board? — As the shortfall it survives, not as a number of months.
3. Why must distribution forecasts model reserve replenishment? — Top-ups rank above equity, so unreplenished reserves make forecast distributions contractually unavailable.

KNOWLEDGE AREA 10.4

Covenants, default and cure

Topics: 10.4.1 covenant types · 10.4.2 distribution lock-up · 10.4.3 events of default and cure rights · 10.4.4 living with covenants.

10.4.1 Covenant types

Financial covenants are tested ratios — `DSCR` (historic, forward-looking, or both), `LLCR`, sometimes leverage or `ICR` — measured on defined dates with defined inputs. **Information covenants** require reporting: management accounts, compliance certificates, model updates, budgets. **Positive covenants** require action (maintain insurance, comply with law, operate to standard). **Negative covenants** prohibit action without consent (no additional debt, no asset disposals, no material contract amendments, no change of control).

The design intent runs through all four: **give lenders visibility and control while the project is still fixable.** A covenant set that only bites at payment failure is worthless, because by then the options have gone (Domain 1's irreversibility point, in credit form).

Two mechanical distinctions matter in practice. **Historic versus forward-looking tests**: a backward test measures what happened; a forward test (often required for distributions) measures the projection, which introduces the question of whose projection and on what assumptions. **Cash-based versus accounting-based**: as Domain 2 established, cash-based tests are indifferent to classification judgments that can flip an accounting covenant with no change in economics — which is why project facilities lean on `DSCR`.

10.4.2 Distribution lock-up

The **lock-up** is the mechanism that makes covenants effective without triggering default: if a distribution test fails, cash that would have gone to equity is trapped — held in a blocked account, applied to prepayment, or simply retained. Kestrel's structure has a **1.20× covenant** and a **1.15× lock-up trigger**, and the sequencing is deliberate: cash is retained before a breach becomes an event of default, so the lenders' exposure reduces while the project stabilises.

The economics for a sponsor are severe and often underappreciated: distributions deferred are equity returns deferred, and because equity returns are the residual (Domain 1's leverage arithmetic), a lock-up hits the equity IRR far harder than the ratio shortfall suggests. This is why lock-up thresholds are negotiated as hard as the covenant itself, and why a sponsor's model must show the distribution profile **after** lock-up tests, not before.

10.4.3 Events of default and cure rights

An **event of default** is a defined breach that entitles lenders to remedies — typically acceleration (all sums due), enforcement of security, or step-in. In practice lenders rarely accelerate a fundamentally sound project: enforcement destroys value and leaves them owning an asset. The realistic path is renegotiation from a much stronger position, which is why the consequences of default are usually commercial before they are legal.

Cure rights are the sponsors' contractual ability to fix a breach — most commonly an **equity cure**, injecting cash treated as `CFADS` (or applied to prepayment) so the ratio is restored. Facilities limit them: a maximum number of cures over the loan life, a maximum in consecutive periods, and rules on whether cure cash counts in the ratio or reduces debt. The negotiation matters because cure rights convert a technical breach into a funding decision the sponsor controls, and that is a meaningful option in a downside — Domain 8's optionality, in financing form.

Waivers and amendments are the ordinary resolution of a breach: lenders consent, usually for a fee, often with tightened terms. Domain 15 handles the restructuring end of this spectrum.

10.4.4 Living with covenants

The leadership content of this domain is not the arithmetic but the operating posture it implies:

- **Know which covenant binds first.** Kestrel's binds at `CFADS` 6,011,562 — 5.8 % below base case. That number, not the ratio, is the operational trigger, and it should sit on the management dashboard.
- **Forecast the tests, do not just report them.** A covenant tested next quarter on a forecast that is already visibly deteriorating is a conversation to have with lenders now, from a position of candour, rather than in three months from a position of breach.
- **Never surprise a lender.** The single most valuable asset in a covenant negotiation is a track record of early, accurate disclosure; a lender who learns of a problem from a compliance certificate will price that discovery into everything afterwards. (Domain 1's honesty asymmetry, applied to credit relationships.)

- **Model the lock-up, not just the covenant.** Sponsors are hurt by trapped cash long before default, and boards are routinely surprised by a distribution profile nobody tested.

AI in this KA

Covenant extraction and monitoring is a genuinely strong application: pulling defined terms, test dates and thresholds out of long agreements, and tracking actuals against them across a portfolio. Two boundaries. **Definitions are where models fail** — a covenant summary is only as good as its reading of the definitions clause, and that reading must be verified against the document before anyone relies on it (Domain 1's document-against-summary check). And **legal consequence is not decision support**: whether a set of facts constitutes an event of default, and what remedies follow, is a question for qualified counsel. Use AI to ensure no test is missed; never to conclude what a breach means.

KEY TERMS — KA 10.4

Term	Meaning
Financial / information / positive / negative covenant	Tested ratios · reporting · required acts · prohibited acts.
Historic vs forward-looking test	Measured on what happened vs on projection.
Distribution lock-up	Trapping equity cash when a test fails, short of default.
Event of default	Defined breach entitling lenders to remedies including acceleration.
Equity cure	Sponsor cash injected to restore a ratio; limited by the facility.
Waiver / amendment	Consented departure from terms, usually for a fee and tighter conditions.

SAMPLE MCQS — KA 10.4

MCQ 10.4-A `[10.4.2 · Analysis]` A facility has a 1.20× `DSCR` covenant and a 1.15× lock-up trigger. Why is the lock-up set below the covenant? - A. it is a drafting convention with no economic effect - B. so that cash is trapped only after a breach has occurred - C. so that cash begins to be retained as coverage deteriorates, reducing exposure before a breach becomes an event of default - D. because lock-up and covenant tests use different `CFADS` definitions

Rationale: Graduated triggers act early while the project is fixable (10.4.2). B misreads the ordering; a lock-up below the covenant level means it engages as the ratio falls through it.

MCQ 10.4-B `[10.4.3 · Application]` A `DSCR` covenant is breached and the sponsors have an unused equity cure. The realistic sequence is: - A. lenders accelerate and enforce security - B. sponsors decide whether to inject cure cash; failing that, waiver or amendment negotiations follow, with acceleration a remedy of last resort - C. the breach is disregarded if payment was made - D. the loan converts automatically to equity

Rationale: Acceleration destroys value for lenders too, so cure, waiver and amendment are the normal path (10.4.3). C ignores that breach and payment failure are distinct (10.2.1).

MCQ 10.4-C `[10.4.4 · Analysis]` Which figure best belongs on a management dashboard for covenant management? - A. the current `DSCR` - B. the `CFADS` level at which the binding covenant fails — 6,011,562, or 5.8 % below base case - C. the debt outstanding - D. the loan's maturity date

Rationale: The operational trigger is the cash level, which management can influence and monitor (10.4.4). The ratio alone conveys no magnitude of headroom.

SELF-CHECK — KA 10.4

1. What does a lock-up achieve that a covenant alone does not? — It retains cash as coverage deteriorates, short of default, reducing exposure while the project is still fixable.
2. Why do lenders rarely accelerate a sound project? — Enforcement destroys value and leaves them owning the asset; renegotiation from strength is better.
3. What is the most valuable asset in a covenant negotiation? — A track record of early, accurate disclosure.

Advanced topics — Domain 10

10.A.1 Forward-looking tests and whose forecast counts

A forward DSCR test measures a projection, which raises three questions the documents must answer: whose model, on what assumptions, and reviewed by whom. Facilities typically require the borrower's model updated to an agreed basis, sometimes with the lenders' technical adviser's assumptions prevailing on specified inputs. The practical consequence is that **assumption control becomes a covenant matter** — a sponsor who cannot defend an assumption cannot pass a forward test on it — and it is why Domain 13's model audit and Domain 6's model governance are contractual, not merely prudent.

10.A.2 Refinancing, tails and mini-perms

Kestrel's PLCR of 1.9431 against an LLCR of 1.2743 quantifies its **tail** — the 13 years of project life beyond loan maturity. Tails are what make refinancing possible, and structures deliberately exploit them: a **mini-perm** (a short facility priced and documented in the expectation of refinancing, sometimes with punitive step-up margins if it is not) trades cheap early money for refinancing risk, which is Domain 3's Case study B arithmetic at facility scale. The leader's discipline is to state the refinancing assumption explicitly and stress it, because a plan that requires a functioning credit market at a specific date has embedded a market forecast in a financing structure.

10.A.3 The reviewer's coverage eye

Invariants to test on any coverage model: CFADS reconciles line-by-line to the facility's definition; the same CFADS line feeds DSCR, LLCR and lock-up tests; LLCR **equals** DSCR **where cash is level and service is an annuity at the loan rate** (10.2.2); PLCR ≥ LLCR whenever a tail exists; debt service equals interest plus scheduled principal and ties to the amortisation schedule; the schedule's closing balance is zero at maturity and total principal equals the loan (Domain 3's checks); cash tax, not accounting tax, feeds CFADS; reserve movements are treated as the documents specify; every covenant has a modelled test date; and the model's minimum DSCR over the loan life is reported, not merely the average — because covenants are tested in periods, and an average conceals the one that breaches.

Industry variations — Domain 10

- **Contracted power and availability PPPs.** Revenue certainty supports the lowest coverage requirements (often 1.15–1.25× base case), long tenors and high gearing; `DSCR` is tested against a tightly defined `CFADS`.
- **Merchant power and commodities.** Coverage requirements rise sharply (1.5× and above), tenors shorten, and lenders size on stressed rather than base-case cash — often on a bank case using conservative price decks.
- **Transport concessions.** Patronage risk produces ramp-up profiles that make sculpting close to mandatory, and lenders commonly require larger reserves through the ramp.
- **Water and regulated utilities.** Regulatory reset cycles create step-changes in cash, so covenant testing and reserve sizing must straddle resets rather than assume continuity.
- **Digital infrastructure.** Shorter asset lives compress tails, so `PLCR` provides less comfort and structures rely more on contracted tenant credit and refinancing at a defined point.

Case study — Domain 10: the 828,877 that changed the structure (water)

Situation. Kestrel's sponsors sought USD 42,000,000 of senior debt against a base-case `CFADS` of 6,384,000. The lead bank's credit committee required a **1.30×** base-case `DSCR`.

The arithmetic. At 1.30×, sustainable debt service is 4,910,769, supporting `4,910,769 × AF(0.06, 12) = 41,171,123` — **828,877 short** of the request. At 42,000,000 the `DSCR` is 1.2743 and the `LLCR`, on level cash, is identically 1.2743.

The four levers, worked. Raise *`CFADS`*: the sponsors first proposed adding back the 600,000 working-capital movement to reach 6,984,000 and a 1.39× ratio — rejected, because the facility's own definition was struck after working capital (Domain 2's point, now with a price attached). Reduce required coverage: the bank agreed **1.25×** in exchange for a longer offtake term and a tightened lock-up at 1.15×, which on base-case cash lifts capacity to `6,384,000/1.25 × 8.383844 =` **42,817,968** — apparently enough. But the bank sizes on its own **stressed case**, and at a `CFADS` 5 % lower (6,064,800) the same 1.25× supports only **40,677,069**, which is less than the 1.30× base-case answer. The concession was therefore worth less than it appeared, and this is the recurring trap in coverage negotiations: **a lower ratio applied to a lower cash case is not a concession.** Lengthen tenor: 15 years was unavailable against a 25-year concession with a required tail. Lower the rate: not negotiable in the market of the day.

The outcome. A structure at **41,000,000** senior debt — debt service `41,000,000/8.383844 =` **4,890,358**, a base-case `DSCR` of **1.3054**, comfortably inside the 1.30× requirement — with the residual **1,000,000 funded as additional equity**, a six-month DSRA (2,504,818), sculpted service through the two-year ramp, and a 1.15× lock-up. Equity rose from 18,000,000 to 19,000,000, gearing from 70/30 to **68.3/31.7** (debt/equity 2.16:1), and the sponsors' modelled IRR fell by roughly 40 basis points — the true price of the coverage requirement, paid in equity rather than argued away.

What the domain teaches here. Debt capacity is arithmetic, and the negotiation is about its four inputs — not about the number itself. Every attempt to move the answer without moving an input (redefining `CFADS`, quoting a more flattering ratio) fails at the first competent review, and the attempt costs credibility that the later, genuine negotiation needs.

Case study B — Domain 10: paid in full and in breach (transport)

Situation. A toll-road SPV entered its third operating year with patronage **12 % below forecast**. Because most of a road's operating cost is fixed, CFADS fell further than revenue: from a base-case 24,000,000 to **19,700,000**, a fall of **17.9 %** — the operating-leverage effect a sponsor should model rather than discover. Against debt service of 18,600,000 that is a DSCR of **1.0591**, versus a **1.25× covenant** and a **1.15× lock-up**. The DSRA (six months, 9,300,000) was fully funded and untouched.

What happened. Debt service was paid in full and on time, from operating cash alone. The project was nonetheless in breach of covenant from the first test date, distributions locked up automatically at the 1.15× trigger, and the sponsors — who had modelled distributions against the covenant rather than the lock-up — had budgeted dividends that were contractually unavailable. The board's initial position, that "the lenders are being paid, so there is no problem", was precisely wrong and cost two months of credibility before the finance director corrected it internally.

How it resolved. The sponsors injected an equity cure of **3,700,000**, lifting cure-adjusted CFADS to 23,400,000 and the ratio to **1.2581×** — back above the covenant, and deliberately above the bare minimum of `18,600,000 × 1.25 − 19,700,000 =` **3,550,000**, because a cure computed to the last dollar breaches again on any further slippage. They used one of their two permitted cures, and agreed an amended covenant profile stepping from 1.15× to 1.25× over three years in exchange for a fee, a cash sweep of 50 % of surplus above 1.30×, and monthly rather than quarterly reporting. Patronage recovered to within 6 % of forecast by year five and the sweep retired debt ahead of schedule.

What the domain teaches here. Breach and payment failure are different events, and the covenant is designed to bite first — while cure, waiver and amendment are all still available. The sponsors' avoidable error was modelling the covenant and not the lock-up, which is the specific discipline of KA 10.4.2.

Executive perspective — Domain 10

What a project finance director cannot delegate in this domain:

- **The CFADS definition.** Read it in the document, clause by clause, and confirm the model implements it. Every ratio the project will be judged on is built on that one term (Case study A).
- **The binding trigger in cash.** Not the ratio — the CFADS level at which the first covenant fails (6,011,562 for Kestrel, 5.8 % below base case), on the dashboard, owned.
- **The distribution profile after lock-up tests.** Boards are surprised by trapped cash, and the surprise is always avoidable (Case study B).
- **The minimum, not the average.** Minimum DSCR across the loan life is the covenant-relevant number; averages conceal the period that breaches.
- **The refinancing assumption.** If the structure needs a market at a date, say so explicitly and stress it (10.A.2).
- **The relationship.** Early, accurate disclosure is the asset that determines the terms of every future waiver — and it is built before it is needed.

Calculation exercises — Domain 10

Exercise 10.1 CFADS 9,200,000; target DSCR 1.35×; loan 10 years at 7 %. Compute maximum debt service and maximum debt. Solution. Max debt service 9,200,000/1.35 = **6,814,815**; AF(0.07, 10) = 7.023582; max debt 6,814,815 × 7.023582 = **USD 47,864,408**. Common error: sizing on full CFADS (9,200,000 × 7.023582 = 64,616,950), which omits the coverage requirement entirely.

Exercise 10.2 Debt outstanding 47,864,408; CFADS 9,200,000 level; loan life 10 years at 7 %; project life 20 years. Compute LLCR and PLCR, and verify the DSCR identity. Solution. LLCR = (9,200,000 × 7.023582)/47,864,408 = **1.3500**; AF(0.07, 20) = 10.594014, PLCR = (9,200,000 × 10.594014)/47,864,408 = **2.0363**. DSCR = 9,200,000/6,814,815 = **1.35** — equal to LLCR, as the identity requires with level cash and annuity service. Common error: discounting CFADS at the project's equity discount rate rather than the loan rate, which breaks the identity and the comparability.

Exercise 10.3 A facility has debt service of 6,814,815, a 1.20× covenant and a 1.10× lock-up. State the CFADS levels at which each triggers, and the headroom from a base case of 9,200,000. Solution. Covenant at 6,814,815 × 1.20 = **8,177,778**; lock-up at × 1.10 = **7,496,297**. Headroom to covenant 9,200,000 − 8,177,778 = **1,022,222**, or **11.1 %** of base-case CFADS. Common error: quoting headroom in ratio points (0.15×), which conveys no magnitude.

Exercise 10.4 CFADS falls 25 % from 9,200,000. Compute the DSCR against debt service of 6,814,815 and state whether the 1.20× covenant holds and whether payment is made. Solution. CFADS **6,900,000**; DSCR = 6,900,000/6,814,815 = **1.0125**. The covenant is **breached** (1.0125 < 1.20) but **scheduled debt service is still paid** from cash alone (6,900,000 > 6,814,815) — the distinction of KA 10.2.1. Common error: treating breach and payment default as the same event.

Practitioner's toolkit — Domain 10

Adoption-ready artefacts; adapt headings to your organisation, then keep them stable.

Toolkit 10.T.1 — CFADS definition reconciliation

Column 1: each line of the facility's CFADS definition, with its clause reference. Column 2: the model line implementing it. Column 3: included/excluded and on what basis (cash or accrued; above or below CFADS). Column 4: the person who confirmed the match, and the date. Rule: no coverage ratio is reportable until every definition line has a confirmed model line (Domain 2's defined-terms sheet, made specific to CFADS).

Toolkit 10.T.2 — Covenant dashboard (per facility, per test date)

Per covenant: test name and clause · test date and frequency · historic or forward-looking · threshold · current/forecast value · **the CFADS level at which it triggers** · cash headroom in currency and as a percentage of base case · lock-up threshold and its own trigger level · reserve balances against required levels · cures used and remaining · reporting obligations due. Front line: **which covenant binds first, and by how much cash.**

Toolkit 10.T.3 — Coverage model check (before any ratio is quoted)

- [ ] CFADS reconciles to the documented definition (10.T.1 complete).
- [ ] The same CFADS line feeds DSCR, LLCR and every distribution test.
- [ ] LLCR = DSCR where cash is level and service is an annuity at the loan rate.
- [ ] PLCR ≥ LLCR wherever a tail exists.
- [ ] Debt service = interest + scheduled principal, tied to the amortisation schedule.
- [ ] Schedule closes at zero; Σ principal = loan drawn.
- [ ] Cash tax, not accounting tax, in CFADS.
- [ ] **Minimum** DSCR over the loan life reported alongside the average.
- [ ] Every covenant has a modelled test date; lock-up modelled as well as covenant.
- [ ] AI-produced covenant summaries sampled against the documents; verifier named.

Exam preparation — Domain 10

The traps. Sizing debt on full CFADS without the coverage divisor (Exercise 10.1) · quoting a DSCR without its CFADS definition (10.1.1) · treating covenant breach and payment default as the same event (Exercise 10.4, Case study B) · discounting CFADS at an equity rate in LLCR (Exercise 10.2) · reporting average rather than minimum DSCR · modelling the covenant but not the lock-up (10.4.2) · assuming a DSRA prevents breach (MCQ 10.3-B) · using accounting rather than cash tax in CFADS · inverting DSCR · relying on PLCR as a lender comfort.

Reflection questions. 1. For your facility: state the CFADS definition from memory, then check it against the document. How close were you, and what would the difference have done to your reported ratio? 2. At what CFADS level does your first covenant fail, and is that number on anyone's dashboard? 3. Does your distribution forecast pass the lock-up tests, or only the covenant tests?

Domain 10 summary

Project debt is sized from cash flow outwards, and the arithmetic admits only four inputs: CFADS, required coverage, rate and tenor. Kestrel's request for 42,000,000 against level CFADS of 6,384,000 supports only **41,171,123** at a 1.30× target — an 828,877 gap closed by equity, because every attempt to close it by redefining CFADS fails at the first competent review. CFADS itself is a **defined term** whose documented construction governs every ratio built on it, as Domain 2 proved by moving Kestrel's DSCR from 1.39 to 1.27 with one working-capital treatment. The ratios answer different questions: DSCR tests a period (1.2743, with covenant headroom of **372,438** of annual cash — the number that belongs on a dashboard); LLCR tests the loan horizon and, with level cash and annuity service, is **identically equal to** DSCR — an invariant that catches model inconsistency; PLCR (1.9431) counts the tail that makes refinancing possible but that lenders have no claim on; and ICR covers interest while ignoring principal and inheriting accounting judgment. Reserves buy payment continuity and time rather than compliance — Kestrel's six-month DSRA of **2,504,818** survives a collapse to 39 % of base-case cash while the covenant still breaches — and they rank above distributions in the waterfall. Covenants exist to give lenders control while the project is fixable, with lock-up trapping cash short of default, cure rights giving sponsors a funding option, and waiver or amendment as the ordinary resolution; breach and payment failure are different events, and confusing them is the error Case study B cost two months of credibility. Domain 11 allocates the risks these ratios are stressed against; Domain 13 audits the model that computes them; Domain 15 operates the waterfall and handles the restructuring end.

11

DOMAIN 11

Risk Identification and Allocation

Group: Executing the transaction (Domain 2 of 4 in Part Three). **Target:** ~75 pages. **Binds to:** the PCI Book Pattern Specification and the shared registries (`docs/books/registries/`). This domain prices the allocation decisions that Domain 12 then documents. It uses `EMV` on the PML-AI Domain 8 definition, and it reads its consequences in the coverage machinery Domain 10 built (`CFADS`, `DSCR`, covenant and lock-up triggers) — neither is re-derived here. British English; USD (+SAR where useful, indicative `USD 1 ≈ SAR 3.75`).

Why this domain exists

Domain 10 established what a project's cash flow can carry and on what conditions. It stress-tested that cash flow without ever asking the prior question: **who bears each of the things that could reduce it, and what did that allocation cost?** Domains 5 to 9 named the risks; Domain 10 measured the consequence of them materialising. Neither decided the allocation, and the allocation is where most of a project financing's value is created or quietly destroyed.

The domain's central claim is that **risk allocation is a price, not a preference.** Practitioners talk as though risk were transferred by assertion — "the contractor takes ground risk", "the offtaker takes volume risk" — but no counterparty absorbs an exposure for nothing. It charges, and its charge is its own expected cost plus a loading for capital, uncertainty and margin. The value of a transfer is therefore the transferor's expected cost minus the price paid, and that difference can be negative. It is reliably negative in one identifiable circumstance: when the party being asked to take the risk can neither influence its probability nor its consequence. Such a party prices its ignorance, and a project that buys the transfer anyway pays more than the risk costs — sometimes enough to break its own coverage. The five items of Kestrel's construction register that were offered to the contractor and declined would have cost **USD 4,620,000** in premium against an owner's expected cost of **USD 2,840,000**, and buying them at 70/30 gearing would have pushed the project's year-one `DSCR` to **1.1832** — below its own covenant, before the plant produced a litre of water.

The domain then works outwards through the four risk families. Construction and completion risk is where the pricing test is sharpest, because counterparty and price are both explicit (KA 11.1). Market and operating risk is allocated through indexation and pass-through formulae rather than liability clauses, and a **mismatch** between the index on revenue and the driver of cost is a slow covenant breach nobody tests for (KA 11.2). The financial and event risks are where the arithmetic is most tractable and most often left undone: 74 basis points of reference rate, a 5.06 % devaluation and a 21-day outage each sit between Kestrel and a breach, all three computable in advance (KA 11.3). And the modern families — environmental and social, technology, cybersecurity, AI model risk — are where the register is newest, thinnest and most aggressively re-cut by lenders (KA 11.4).

Learning objectives. After this domain a candidate can: build a risk register at the level of granularity an allocation decision requires; state the two legitimate grounds for transferring a risk and distinguish them from bargaining power; compute the net value of a transfer as expected cost less loaded premium, and the breakeven premium and breakeven loading at which a transfer stops paying; demonstrate arithmetically why transferring an uncontrollable risk destroys value even at a zero margin; translate an input-cost pass-through share into a coverage tolerance multiplier; detect and quantify an indexation mismatch between revenue escalation and cost drivers; compute unhedged, partially hedged and fully hedged debt service and the `DSCR` range each produces; derive the minimum hedge ratio that survives a specified rate shock; quantify the coverage consequence of a currency mismatch and compute the exchange-rate-indexed revenue share that neutralises debt service; test an insurance waiting period against covenant headroom in days; aggregate a risk register to a mean, a standard deviation and a percentile on the PML-AI Domain 8 method, explain why a lender re-cuts it and what correlation does to it; convert a register into a debt-capacity number; and govern AI use in risk identification while treating AI itself as a register line.

The master thread. Kestrel Water SPC continues. Capital cost **USD 60,000,000** funded **70/30** as **USD 42,000,000** of senior debt at **6.0 % over 12 years** — annual instalment **USD 5,009,635.23**, year-one interest **2,520,000**, year-one principal **2,489,635.23**, `AF(0.06, 12) = 8.383844` (Domain 3) — plus **USD 18,000,000** of equity. Operating life 25 years; documented first-year `CFADS` **USD 6,384,000** (6,984,000 before working-capital movements) on revenue of 12,000,000 and cash operating costs of 4,500,000. Coverage at close (Domain 10): `DSCR` **1.2743** = `LLCR` **1.2743**, `PLCR` **1.9431**, covenant cash trigger **6,011,562.28**, lock-up trigger **5,761,080.51**, annual headroom **372,437.72**. Domain 8 provisioned against a retained construction register with a mean of **2,690,000** and a P80 of **4,246,095**. This domain supplies the fact those chapters took as given: **where that retained register came from**, and what the allocation that produced it cost.

KNOWLEDGE AREA 11.1

Construction and completion risk

Topics: 11.1.1 the register at allocation granularity · 11.1.2 the two grounds for transfer · 11.1.3 pricing the transfer · 11.1.4 completion risk and the residue.

11.1.1 The register at allocation granularity

Definition. A **risk register** for financing purposes is a list of identified events, each with a probability, a monetary impact, an owner and — the line most registers omit — **the mechanism by which that ownership is effected.** Registers written for project-management purposes stop at the owner's name; a financing register must name the clause, the guarantee, the reserve or the insurance policy that makes the ownership real, because an unmechanised allocation is an intention, not a transfer.

Granularity is the practical craft. A register line reading "construction risk — contractor" is useless for allocation because construction risk is not one thing: it decomposes into workmanship and plant performance, programme and productivity, procurement and logistics, ground conditions, third-party interfaces, input prices and permit conditions, and those seven items belong to different parties on different grounds. The test of adequate granularity is operational: **a register is granular enough when every line could in principle be allocated separately.** Aggregating two risks with different natural owners into one line guarantees that one of them is mis-allocated, and the mis-allocation is invisible because the line has only one owner column.

Two consequences follow. Identification must run through the **cash-flow model**, not around it: a risk that cannot be expressed as a change to a modelled line — revenue, an operating cost, capex, a timing — cannot be quantified, priced or allocated on any basis but instinct. And identification must be **adversarial**, because a sponsor's register supports a case while a lender's supports a doubt, so the sponsor's is systematically the shorter of the two. A leader who has not written the lender's register before the lender does (Domain 13) will negotiate from a document already discredited.

11.1.2 The two grounds for transfer

The principle. There are exactly two defensible reasons to move a risk from one party to another, and one common indefensible one.

Control. The transferee can change the probability or the impact. A contractor can supervise welding, sequence a programme and expedite a supplier; when it holds the consequence, it does those things better. Transfer on control grounds creates value because it **changes the underlying distribution**, not merely who observes it.

Capacity. The transferee cannot change the distribution but has a lower cost of bearing it — because it is diversified across many such exposures (an insurer), because the exposure is small relative to its balance sheet (a state offtaker facing a devaluation that would destroy an SPV), or because it can hedge in a market the transferor cannot access. Transfer on capacity grounds creates value without touching the distribution, by moving the exposure to a cheaper holder of it.

Bargaining power is the indefensible ground. A party with a weak negotiating position can be made to accept a risk it neither controls nor can cheaply bear. The transfer looks free because it is not priced in the agreement, but it is priced somewhere: in a claims strategy, in a contingency the counterparty holds privately, in a bid that fails at the diligence stage, or — most expensively — in a counterparty that fails when the risk crystallises and hands the exposure back at the worst possible moment (11.3.3). **A risk allocated to a party that can neither control nor absorb it has not been transferred; it has been hidden.**

11.1.3 Pricing the transfer

The arrangement. Nothing new is needed beyond `EMV` (PML-AI Domain 8, KA 8.2.2: `EMV` = probability × impact), arranged three ways:

```
Expected cost if retained          = EMV(transferor's own p and impact)
Loaded premium the transferee wants = EMV(transferee's own p and impact) × (1 + loading)
Net value of transfer               = expected cost if retained − loaded premium
Breakeven premium                   = expected cost if retained
Breakeven loading                   = (transferor's EMV ÷ transferee's EMV) − 1
```

The **breakeven loading** is the number that carries the professional insight, because it separates the two grounds for transfer arithmetically. Where the transferee controls the risk, its own `EMV` is much lower than the transferor's, so the breakeven loading is large and a generous margin still leaves value on the table. Where it does not control the risk, its `EMV` equals or exceeds the transferor's, the breakeven loading is zero or negative, and **no negotiation over margin can rescue the transfer.**

WORKED EXAMPLE

Worked example 11.1.3 — Kestrel's construction register, priced item by item.

1. **Setup.** Before award, Kestrel's sponsors hold eight quantified construction threats plus one opportunity. The preferred bidder is asked to price each item into a fixed-price wrap. It quotes each on **its own** probability and impact estimates, with a **40 % loading** for capital, uncertainty and margin (a stated assumption, not a market constant: the loading reflects the transferee's cost of capital and its own uncertainty, and is itself negotiable). Determine which items should transfer.
2. **Formula.** As above, per item, then aggregated by decision.
3. **Substitution and result.**

ID	Item	Owner p	Owner impact	Owner EMV	Bidder p	Bidder impact	Bidder EMV	Premium ×1.40	Net value
A1	Workmanship, plant defects, performance shortfall	0.30	6,000,000	1,800,000	0.12	4,000,000	480,000	672,000	**+1,128,000**
A2	Construction programme and labour productivity	0.35	5,200,000	1,820,000	0.20	3,400,000	680,000	952,000	**+868,000**
A3	Equipment procurement and logistics	0.25	3,200,000	800,000	0.15	2,400,000	360,000	504,000	**+296,000**
A4	Ground conditions beyond the disclosed geotechnical baseline	0.40	2,400,000	960,000	0.45	2,900,000	1,305,000	1,827,000	**(867,000)**
A5	Utility diversion and third-party interface scope growth	0.30	1,800,000	540,000	0.31	2,000,000	620,000	868,000	**(328,000)**
A6	Membrane supply price above the contract indexation formula	0.35	1,400,000	490,000	0.35	1,500,000	525,000	735,000	**(245,000)**
A7	Marine intake weather standby beyond the allowance	0.50	900,000	450,000	0.50	900,000	450,000	630,000	**(180,000)**
A8	Permit condition requiring additional monitoring works	0.20	2,000,000	400,000	0.20	2,000,000	400,000	560,000	**(160,000)**
A9	Early-completion rebate (opportunity, retained)	0.25	(600,000)	**(150,000)**	—	—	—	—	—

A1–A3 transfer: owner EMV 4,420,000 against premiums of **2,128,000** — value created **+2,292,000**. **A4–A8 do not:** owner EMV 2,840,000 against premiums of **4,620,000** — value destroyed **(1,780,000)**. 4. **Result.** The wrap is drawn round A1–A3 for a priced risk premium of **USD 2,128,000** inside the 48,000,000 EPC price (an un-risked scope of **45,872,000** plus that premium). The retained residue is **2,840,000 − 150,000 = USD 2,690,000** — precisely the register Domain 8 sized contingency against. Gross expected cost of the risk position falls from **7,110,000** wholly retained to **4,818,000** after allocation. 5. **Interpretation.** Four things in that table are worth more than the arithmetic that produced them. **The premium is not the cost; the net is.** Every dollar of the 2,128,000 premium bought 2.0771 dollars of expected-cost reduction, and an organisation that judges a wrap by its price rather than by its net value will reject the transfers that pay for themselves twice over. **Where control exists, price barely matters.** The breakeven loading on A1–A3 is **190.79 %** — the bidder could charge nearly three times its own expected cost and the transfer would still create value, because the transfer changes the distribution: 6,000,000 of impact at 0.30 becomes 4,000,000 at 0.12 when the party doing the welding carries the consequence. **Where control does not exist, no price works.** The breakeven loading on A4–A8 is **−13.94 %**: the bidder would have to accept the bundle at 86.06 % of its own expected cost before the transfer merely broke even. Stripping the margin out entirely still destroys **460,000**, because the bidder's own expected cost on those items (3,300,000) exceeds the owner's (2,840,000) — it must assume the worst about a ground investigation it did not commission, a utility corridor it does not own and a permit condition it cannot influence. **And the asymmetry has a shape.** A4 is the extreme case and the most instructive: the owner commissioned the geotechnical survey, holds the data and set the baseline, so the owner is the better-informed party. Transferring a risk to the less-informed party converts an information advantage into a price disadvantage. The standing professional caution is that the arithmetic depends entirely on the honesty of the two EMV columns: a sponsor who understates its own retained probabilities will "prove" that every transfer destroys value, and a bidder who overstates its own will justify any premium. **The register's probabilities are the negotiation**, which is why they must be evidenced (Domain 8's estimate basis, PML-AI Domain 8's elicitation discipline) before they are argued.

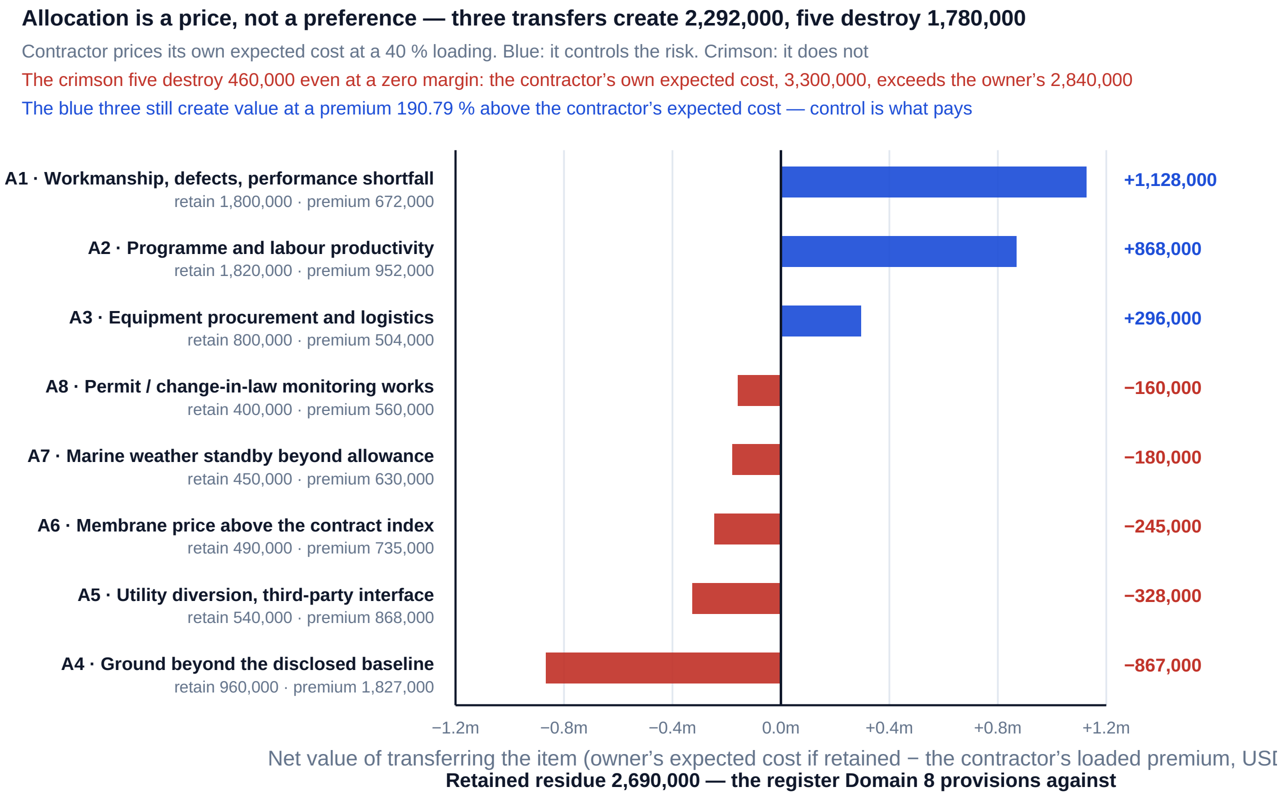

Fig 11.1.1 — The allocation price test

11.1.4 Completion risk and the residue

Definition. Completion risk is the risk that the project does not reach commercial operation on time, on budget, or to the performance the financing was sized against. It is the one construction risk that cannot be wholly transferred, because its consequences fall on parties the EPC contract does not reach: the lenders lose their repayment source, and the equity loses its return.

Three residues survive even a well-priced wrap, and Domain 5 measured each of them, so they are cited rather than re-derived. **Timing residue:** a slip costs 24,733.33 per day in extra interest and forgone CFADS, against delay damages of 20,000 per day capped at day 240 — so damages recover 80.86 % of the cost and nothing at all beyond the cap (Domain 5, KA 5.4.2). **Performance residue:** completion at 97 % of guaranteed output requires a buy-down that restores lenders' coverage, not merely compensation for lost revenue (Domain 5, KA 5.4.3). **Credit residue:** the whole wrap is worth the contractor's balance sheet, priced in 11.3.3.

The professional discipline is to state the residue in the same units as the covenant. Kestrel's retained register mean of 2,690,000, spread across the two-year construction period, is not a coverage number; **capitalised into debt it becomes one.** Domain 8 demonstrated the mechanism precisely: the 600,692 by which the funded contingency fell short of the register P80, capitalised, moved the DSCR from 1.2743 to 1.2564 and cut annual headroom by 23.1 % for the whole twelve-year loan life. That is the sentence a construction risk register should end with: not "the exposure is 2.69 million" but "the exposure, if funded by debt, costs us this much covenant headroom for twelve years."

AI in this KA

Risk identification is one of the strongest genuine applications of language models in project finance, and one of the most dangerous places to accept an output unedited. **Where it earns its place:** extracting risk allocation from a long contract suite into a matrix of item, clause, mechanism and cap; cross-checking a sponsor's register against the risk taxonomies implied by comparable transactions and the lender's diligence scopes, and reporting what is missing — the omission is the failure mode registers have, and a machine that has read a thousand registers is good at noticing an absent line; and normalising register wording so that two advisers' items can be compared at all. **Where it must not go:** the probability and impact columns. Those are elicited judgments with an evidentiary basis, and a model asked to supply them will produce plausible central values with no basis, which then propagate through EMV, the P80, the contingency and the debt sizing. A model may challenge a probability ("this is inconsistent with the ground investigation extent you described"); it may not supply one. **Verification, concretely:** for an extracted allocation matrix, sample at least ten lines back to the clause text and confirm the cap and the mechanism, because the recurring error is a correct owner attached to a wrong cap; for a machine-suggested register addition, require that it be expressed as a change to a named model line before it is accepted, which filters the plausible-but-unquantifiable; and never let a model perform the arithmetic of 11.1.3 without an independent recomputation, since a single transposed probability flips a transfer decision. **AI proposes; the professional verifies, decides and remains accountable.**

KEY TERMS — KA 11.1

Term	Meaning
Risk register (financing)	Identified events with probability, impact, owner **and mechanism**; granular enough for each line to be allocated separately.
Grounds for transfer	Control (changes the distribution) and capacity (cheaper holder); bargaining power is neither.
Loaded premium	Transferee's own EMV × (1 + loading) — what the transfer actually costs.
Net value of transfer	Transferor's EMV − loaded premium; negative means retain.
Breakeven loading	(transferor EMV ÷ transferee EMV) − 1; negative means no price works.
Retained residue	The register left after allocation — the thing contingency and reserves are sized against.
Completion risk	Risk of not reaching commercial operation on time, on budget or to guaranteed performance; never wholly transferable.

SAMPLE MCQS — KA 11.1

MCQ 11.1-A [11.1.3 · Application] An owner's EMV on an item is 960,000. The contractor assesses the same item at p 0.45 and impact 2,900,000 and applies a 40 % loading. The net value of transferring is: - A. +573,000 - B. −345,000 - C. −867,000 - D. −1,827,000

Rationale: Premium = 0.45 × 2,900,000 × 1.40 = 1,827,000; net = 960,000 − 1,827,000 = −867,000. B is the net at a **zero** loading (960,000 − 1,305,000) and understates the destruction; D is the premium alone, ignoring the retained cost avoided; A compares the owner's gross **impact** (2,400,000) with the premium, omitting probability from the retained side.

MCQ 11.1-B [11.1.3 · Analysis] A bundle of items has an owner EMV of 2,840,000 and a contractor EMV of 3,300,000. The correct conclusion is: - A. transfer if the loading can be negotiated below 40 % - B. transfer, because the contractor is better placed to manage construction - C. do not transfer: the breakeven loading is −13.94 %, so even at a zero margin the transfer destroys 460,000 - D. transfer, and recover the premium through the contingency

Rationale: The transferee's own expected cost exceeds the transferor's, so the loading is not the problem — the distribution is (11.1.3). A treats a structural result as a pricing negotiation; B asserts control where the items are ground, interface, index and permit risks the contractor does not control; D funds a value destruction twice.

MCQ 11.1-C [11.1.2 · Analysis] An SPV with weak negotiating position accepts a risk it neither controls nor can absorb. The most accurate description of what has happened is: - A. an efficient transfer, since the price was zero - B. the risk has not been transferred but hidden — it will reappear as a claim, a private contingency, a failed bid or a counterparty default - C. a transfer on capacity grounds - D. a transfer on control grounds

Rationale: Neither ground applies, and an unpriced allocation is not a costless one (11.1.2). C and D name grounds that are absent by the stem's own terms.

SELF-CHECK — KA 11.1

1. State the test for adequate register granularity. — Every line could in principle be allocated separately; if two risks in one line have different natural owners, one is mis-allocated invisibly.

2. Why does the breakeven loading separate the two grounds for transfer? — Control lowers the transferee's EMV below the transferor's, making the breakeven loading large; without control it is zero or negative, and no margin negotiation helps.
3. State Kestrel's allocation outcome in one line. — 2,128,000 of premium bought 4,420,000 of expected-cost reduction (net +2,292,000); 4,620,000 of offered premium against 2,840,000 of retained EMV was declined; the residue is 2,690,000.

KNOWLEDGE AREA 11.2

Market risks and operations

Topics: 11.2.1 why market risk resists transfer · 11.2.2 pass-through as a coverage multiplier · 11.2.3 the indexation mismatch · 11.2.4 operating risk and the O&M interface.

11.2.1 Why market risk resists transfer

Demand, price and supply risks are the hardest to allocate because **no party to the transaction controls them.** The pricing test of 11.1.3 therefore predicts what practice confirms: transfers of market risk are either expensive or illusory. Domain 7 built the machinery — contracted versus merchant structures, tariff architecture, minimum revenue guarantees, offtaker credit — and this KA adds only the allocation view of it, which reduces to three findings.

Transfer to the offtaker is a capacity transfer, not a control transfer. An availability-based structure moves volume risk to a payer that is diversified or sovereign-adjacent where the SPV is neither, and it is bought with a lower tariff — Domain 7 priced that trade at 10,679,727 of debt capacity relative to a volume tariff.

Transfer to a guarantor is a capacity transfer with a credit tail. A minimum revenue guarantee is worth the guarantor's balance sheet in the state of the world where it is called — by construction, the state in which the guarantor's other exposures are also stressed. Domain 7 valued Kestrel's guarantee at 2,625,026 of released equity; the discipline is to value it for the coverage it unlocks and then discount it for that correlation.

Transfer to the contractor or operator is usually a mis-allocation. Asking an O&M contractor to carry input-price risk it cannot hedge produces exactly the A6 result of 11.1.3: a premium above the retained cost, because a counterparty that can only insure itself by assuming the worst case prices the worst case.

11.2.2 Pass-through as a coverage multiplier

Definition. A **pass-through** is a contractual term under which a defined movement in a defined input cost is recovered in revenue, in whole or in a stated share. Its financial effect is not to remove the exposure but to **divide** it, and the arithmetic of that division is more powerful than practitioners expect.

If an input has a base annual cost C and a share φ of its price movement is passed through, the residual cash effect of a 1 % rise in that input's price is $C \times 0.01 \times (1 - \varphi)$. Because covenant tolerance is fixed headroom divided by residual exposure per unit, **the tolerance is multiplied by $1/(1 - \varphi)$**.

WORKED EXAMPLE

Worked example 11.2.2 — what a 70 % power pass-through is worth to Kestrel.

1. **Setup.** Of Kestrel's 4,500,000 of cash operating costs, **1,800,000 is electricity** — the dominant input in reverse-osmosis desalination. The water-purchase agreement passes through **70 %** of movements in the reference power tariff. Annual covenant headroom is **372,437.72** (Domain 10). Compute the tolerance to a power-price rise with and without the pass-through, and the position after a 25 % rise.
2. **Formula.** Residual per 1 % rise = `1,800,000 × 0.01 × (1 − φ)`; tolerance = headroom ÷ residual per 1 %; multiplier = `1/(1 − φ)`.
3. **Substitution.** With `φ` = 0: `1,800,000 × 0.01 = 18,000`; `372,437.72 / 18,000`. With `φ` = 0.70: `1,800,000 × 0.01 × 0.30 = 5,400`; `372,437.72 / 5,400`. At a 25 % rise: residual `1,800,000 × 0.25 × 0.30 = 135,000`.
4. **Result.** Unprotected, the covenant fails on a **20.6910 %** power-price rise. With 70 % pass-through it fails on a **68.9699 %** rise — a tolerance multiplier of **3.3333×**. A 25 % rise costs **135,000** of `CFADS`, leaving 6,249,000 and a `DSCR` of **1.2474**; unprotected the same rise costs 450,000, leaving 5,934,000 and a `DSCR` of **1.1845** — a breach.
5. **Interpretation.** The teachable content is the leverage. Negotiating the pass-through share from zero to 70 % triples the project's tolerance to its single largest cost driver, and the marginal value of each increment is increasing: moving from 0 to 50 % doubles tolerance, but moving from 70 % to 85 % doubles it again from a much higher base. The professional consequence is that a pass-through share is a **coverage term, not a commercial detail**, and it belongs in the same negotiating tranche as the covenant level. Two cautions bound the result. First, a pass-through is only as good as its **reference index and its reset frequency**: a share of 70 % reset annually leaves eleven months of exposure inside each year, and a share indexed to a published tariff the plant does not actually pay leaves a basis risk that no share removes. Second, the tolerance figures assume the rise is permanent and everything else holds; combined with any other adverse movement the tolerances share the same 372,437.72 of headroom and are **not additive** — which is the single most common misuse of a sensitivity table (Domain 7, KA 7.4.3 built the joint matrix for exactly this reason).

11.2.3 The indexation mismatch

Definition. An **indexation mismatch** exists when the index that escalates revenue is not the driver that escalates cost. It is the most under-detected structural risk in project finance because it is invisible at close — year one is unaffected by construction — and because it does not appear in any single-year sensitivity. It appears as a **trend in coverage**, and a model that reports only the base-case minimum `DSCR` over the loan life will report it, while a model that reports the year-one ratio will not.

WORKED EXAMPLE

Worked example 11.2.3 — Kestrel's CPI revenue against power-driven cost.

1. **Setup.** Domain 7 established that **80 % of Kestrel's tariff is indexed to a consumer price index**, assumed at 2.5 % a year. On the cost side, 2,700,000 of operating cost tracks that same index, but the **1,800,000 of power escalates at 8.0 %** on the sponsors' own forecast. Cash tax (516,000) and the working-capital movement (600,000) are held constant, a simplifying assumption stated so the mismatch is isolated. Trace `CFADS` and `DSCR` across the loan life against the 1.20× covenant.
2. **Formula.** Revenue(t) = `12,000,000 × (1 + 0.80 × 0.025)^(t−1)`; CPI cost(t) = `2,700,000 × 1.025^(t−1)`; power(t) = `1,800,000 × 1.08^(t−1)`; `CFADS(t)` = revenue − CPI cost − power − 1,116,000.
3. **Substitution and result.**

Year	Revenue	CPI-linked cost	Power	CFADS	DSCR
1	12,000,000	2,700,000	1,800,000	**6,384,000**	**1.2743**
3	12,484,800	2,836,688	2,099,520	6,432,593	1.2840
5	12,989,186	2,980,295	2,448,880	**6,444,011**	**1.2863**
8	13,784,228	3,209,452	3,084,884	6,373,893	1.2723
10	14,341,111	3,371,930	3,598,208	6,254,972	1.2486
12	14,920,492	3,542,634	4,196,950	**6,064,908**	**1.2106**
15	15,833,745	3,815,029	5,286,949	5,615,767	1.1210

Coverage **improves** to year five, then deteriorates continuously. The first covenant breach falls in **year 13** — one year after the loan matures. The **breakeven power escalation** that would put year twelve exactly on the covenant is **8.1241 %**. 4. **Result.** The structure survives its own indexation mismatch across the loan life, but by **12.41 basis points of assumed power escalation** and with **85.68 %** of its covenant headroom consumed: year-twelve headroom is **53,345.25** against 372,437.72 at close. 5. **Interpretation.** This is the most important arithmetic in the KA and the least likely to be found in a base-case pack. **Coverage improving early conceals a deteriorating structure.** For the first five years the 2 % effective revenue escalation outruns a power cost that is still small in absolute terms; by year twelve power has grown from 1,800,000 to 4,196,950 while the same input escalated at CPI would have reached only 2,361,756 — a wedge of **1,835,194** a year, which is 4.93 times the entire original headroom. **The structure is not robust; it is lucky.** A power forecast of 8.1241 % rather than 8.0 % breaches in year twelve, and no one in the transaction would claim twelve basis points of precision on a twelve-year commodity forecast. That is the honest characterisation to put in front of a credit committee, and it points to three interventions, each priceable: raise the power pass-through share (11.2.2 shows the leverage), index part of the tariff to the power reference rather than to CPI, or size a reserve that funds the late-life wedge. **And the diagnostic is cheap.** Any level-escalation model can be tested in one line: compute the revenue-weighted escalation rate and the cost-weighted escalation rate, and if the cost rate is higher the structure has a mismatch whose only question is when it bites. Kestrel's are 2.00 % and 4.70 % respectively — a 270-basis-point gap that a single-year sensitivity cannot see. The standing caution is that the arithmetic is only as good as the differential forecast: the level of power prices matters far less here than the **spread** between power escalation and the revenue index, and a model that stresses both at the same rate will show no exposure at all while destroying none of it.

11.2.4 Operating risk and the O&M interface

Operating risk allocation is a smaller-scale replay of 11.1.3 with one structural difference: the O&M contractor is a **long-lived counterparty with a thin balance sheet relative to the exposure**, so transfers to it are capped early and often. The practical allocation set is: availability and performance to the operator, within a liability cap and against a fee at risk; input volumes to the operator, which controls consumption; input **prices** to the offtaker or retained, since nobody in the structure controls them (11.2.2); major maintenance to a reserve rather than to a party (Domain 10's MRA); lifecycle replacement to whichever party the handback standard makes accountable.

The interface that produces the most disputes is the boundary between **availability** and **force majeure**, because it decides whether a lost month is the operator's liability or nobody's. Domain 7 computed the coverage sensitivity that makes this boundary financially material: Kestrel breaches its 1.20× covenant at **92.086 %** availability against a 95 % guarantee, so 2.9 percentage points of availability separate compliance from breach, and the definition of what counts as an excusable outage is worth more than the damages rate attached to it.

AI in this KA

Where it earns its place: monitoring. Pass-through and indexation terms are formulae with published inputs, and an automated monthly recomputation of the escalation wedge (cost-weighted escalation less revenue-weighted escalation, and the projected year of covenant breach at current observed rates) is exactly the sort of persistent arithmetic that humans stop doing after financial close. Anomaly detection on input consumption against a plant model is similarly strong, and it is where a model earns operational trust before it is trusted with anything consequential. **Where it must not go:** forecasting the differential. The whole exposure in 11.2.3 turns on the spread between power escalation and a consumer price index over twelve years, and a model trained on recent history will extrapolate a regime rather than a relationship. Use scenarios owned by named people, not a machine point forecast, and note that a plausible-looking generated forecast is more dangerous than an obviously crude one because it survives review. **Verification, concretely:** require any automated escalation monitor to reproduce the year-one figures of the closing model exactly before its later years are believed — Kestrel's 6,384,000 and 1.2743 are the check — and require that any machine-produced sensitivity table state whether its cases are independent or joint, because non-additive tolerances presented additively is the error the tool will not flag.

KEY TERMS — KA 11.2

Term	Meaning
Pass-through share φ	Contractual share of an input-price movement recovered in revenue; multiplies coverage tolerance by 1/(1 − φ).
Basis risk (pass-through)	Exposure remaining because the reference index differs from the price actually paid.
Indexation mismatch	Revenue index differs from the driver of cost escalation; appears as a trend in coverage, not in a single-year sensitivity.
Escalation wedge	Cumulative cash difference between a cost escalating at its own driver and the same cost escalating at the revenue index.
Fee at risk	The portion of an operator's fee forfeited on performance failure — its economic stake in the allocation.
Excusable outage	An unavailability event that does not count against the availability guarantee; the availability/force-majeure boundary.

SAMPLE MCQS — KA 11.2

MCQ 11.2-A [11.2.2 · Application] An input costs 1,800,000 a year; 70 % of its price movement passes through to revenue; covenant headroom is 372,437.72. The input-price rise that breaches the covenant is closest to: - A. 20.7 % - B. 69.0 % - C. 29.5 % - D. 100.0 %

Rationale: Residual per 1 % = 1,800,000 × 0.01 × 0.30 = 5,400; 372,437.72 / 5,400 = 68.97 %. A ignores the pass-through (the φ = 0 answer); C uses the **passed-through** share 0.70 in place of the retained share 0.30 (372,437.72 / 12,600 = 29.56 %) — the commonest sign error here; D assumes only a doubling of the input can breach, which no calculation supports.

MCQ 11.2-B [11.2.3 · Analysis] A project's DSCR is 1.2743 in year one, rises to 1.2863 in year five and falls to 1.2106 in year twelve. The soundest reading is: - A. the structure is robust — coverage never breaches - B. an indexation mismatch is consuming headroom; 85.7 % of it is gone by year twelve and the breach falls just outside the loan life - C. the model contains an error, since coverage cannot both rise and fall - D. the improvement to year five shows revenue growth exceeding costs

Rationale: The shape is diagnostic of a cost driver escalating faster than the revenue index (11.2.3). A reads compliance as robustness and ignores that twelve basis points of forecast separate the two; C mistakes a normal profile for a defect; D is true only for the first five years and misses the trend.

MCQ 11.2-C [11.2.1 · Analysis] Why is asking an O&M contractor to bear input-price risk usually a mis-allocation? - A. O&M contractors are not creditworthy - B. it neither controls the price nor can hedge it, so it prices the worst case and the premium exceeds the retained expected cost - C. input prices are always passed through by law - D. the risk is immaterial

Rationale: This is the 11.1.3 result applied to operations — neither ground for transfer is present. A is a separate (and secondary) objection; C asserts a universal legal position that does not exist; D is contradicted by 11.2.2's arithmetic.

SELF-CHECK — KA 11.2

1. State the pass-through multiplier and Kestrel's two tolerances. — Tolerance is multiplied by 1/(1 − φ); 20.6910 % unprotected, 68.9699 % at φ = 0.70, a 3.3333× multiplier.
2. What single test detects an indexation mismatch? — Compare the revenue-weighted escalation rate with the cost-weighted rate (2.00 % against 4.70 % for Kestrel); a higher cost rate means the only open question is when it bites.
3. Why are separate tolerance percentages not additive? — They all consume the same 372,437.72 of headroom; joint movements require a joint matrix.

KNOWLEDGE AREA 11.3

Counterparty, political, currency, interest-rate and force-majeure risk

Topics: 11.3.1 interest-rate exposure and the hedge ratio · 11.3.2 currency mismatch · 11.3.3 the counterparty risk inside the allocation · 11.3.4 political, regulatory and force majeure.

11.3.1 Interest-rate exposure and the hedge ratio

The structure. Kestrel's senior facility is priced as a floating reference rate plus a **200 basis point margin**. At close the reference stood at **4.00 %**, giving the 6.00 % all-in rate Domains 3 and 10 used. The amortisation schedule fixes **principal** — year-one principal is 2,489,635.23 — so only the interest leg floats, and debt service in any period is `principal + outstanding × all-in rate`. An interest-rate swap is available that fixes the reference at **4.20 %**, above the spot rate because the curve slopes upward; the all-in fixed cost is therefore **6.20 %**.

WORKED EXAMPLE

Worked example 11.3.1 — what an unhedged Kestrel is actually exposed to.

1. **Setup.** Debt 42,000,000; year-one principal 2,489,635.23; `CFADS` 6,384,000; covenant 1.20×, lock-up 1.15×. Compute year-one `DSCR` across a ±200 basis point range in the reference rate, unhedged, 75 % hedged and fully hedged; find the rate at which each threshold is crossed; and find the minimum hedge ratio that survives a 200 basis point shock.
2. **Formula.** Blended rate = `swap all-in × h + floating all-in × (1 − h)`; debt service = `2,489,635.23 + 42,000,000 × blended rate`; `DSCR` = `6,384,000 ÷ debt service`. Breakeven rate: `((CFADS ÷ target DSCR) − principal) ÷ debt`. Minimum hedge ratio: `(shocked rate − breakeven rate) ÷ (shocked rate − swap rate)`.
3. **Substitution and result.**

Reference shift	All-in rate	Interest	Debt service	DSCR unhedged	DSCR at h = 75 %	DSCR at h = 100 %
−200 bp	4.00 %	1,680,000	4,169,635.23	**1.5311**	1.3129	1.2533
−100 bp	5.00 %	2,100,000	4,589,635.23	1.3910	1.2851	1.2533
base	6.00 %	2,520,000	5,009,635.23	**1.2743**	1.2585	1.2533
+50 bp	6.50 %	2,730,000	5,219,635.23	1.2231	1.2456	1.2533
+100 bp	7.00 %	2,940,000	5,429,635.23	**1.1758**	1.2330	1.2533
+200 bp	8.00 %	3,360,000	5,849,635.23	**1.0914**	**1.2085**	1.2533

Unhedged breakeven rates: **6.7390 % (+73.9 bp)** for the 1.20× covenant, **7.2897 % (+129.0 bp)** for the 1.15× lock-up, **9.2723 % (+327.2 bp)** before scheduled debt service cannot be paid at all. Full hedge: interest 2,604,000, debt service 5,093,635.23, `DSCR` **1.2533** at any reference rate, at a year-one cash cost of **84,000**. Minimum hedge ratio surviving +200 bp at the covenant: **70.0576 %**. 4. **Result.** Unhedged, **74 basis points** of reference rate stand between this project and a covenant breach. The swap costs **0.0210 of `DSCR`** and removes **0.4397 of `DSCR` range** — 20.92 units of range eliminated per unit of coverage given up. 5. **Interpretation.** Three professional conclusions, in order of how often they are missed. **Interest-rate exposure is a covenant exposure long before it is a payment exposure.** The distance to a breach is 73.9 basis points; the distance to a payment failure is 327.2. A treasury discussion framed around "can we afford the interest?" is answering the wrong question by a factor of four and a half, and the right question — at what reference rate does our first covenant fail? — has a single computable answer that belongs on the same dashboard as Domain 10's 6,011,562.28 cash trigger. **The hedge is bought with coverage, and the exchange rate is extraordinarily favourable.** Giving up 0.0210 of ratio to remove 0.4397 of range is not a close call, and the reason it is nonetheless argued is that the 84,000 is certain and visible while the range is contingent and invisible. That asymmetry of salience, not the economics, is what produces under-hedged project financings. **And the right answer is a ratio, not a binary.** A 75 % hedge leaves 10,500,000 floating, holds the covenant at +200 bp with **1.2085**, and costs only **63,000** a year — three-quarters of the full hedge cost for the protection that matters, while retaining a quarter of the benefit of falling rates. The **70.0576 %** minimum is why facilities covenant a hedging ratio rather than a full hedge: below it the structure fails the lenders' own rate stress, and above it the incremental protection is bought at a diminishing rate. Two cautions belong with the arithmetic. A swap is not free of risk — it introduces **mark-to-market exposure** (a break cost on prepayment or refinancing, which can be large when rates have fallen) and a **hedge counterparty** whose own credit is now inside the structure (11.3.3). And the figures above are year one; the exposure declines with the outstanding balance, so a hedge profile should amortise with the loan rather than sit flat, or the project will be over-hedged in its later years and paying for protection it no longer needs.

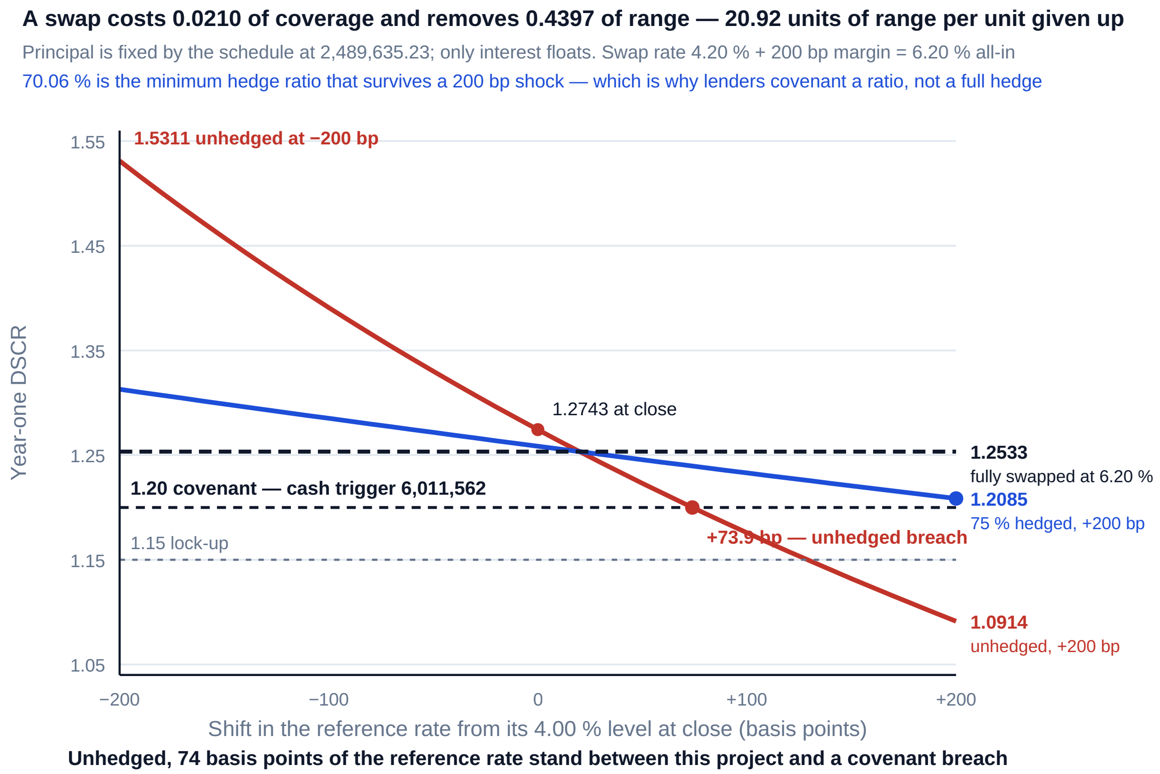

Fig 11.3.1 — Coverage against the reference rate, at three hedge ratios

11.3.2 Currency mismatch

Definition. A **currency mismatch** exists when a project's revenue and its debt service are denominated in different currencies. It is the most destructive of the financial risks because, unlike an interest-rate movement, a devaluation is **unbounded and correlated with everything else that goes wrong in a host economy** — and because it cannot be hedged for a twelve-year tenor in most markets at any price a project can pay.

Kestrel's water-purchase agreement is denominated in USD, which is why Domains 2 to 10 could treat its revenue as a USD figure. That term was not free, and this KA prices the alternative the offtaker pressed for: a tariff denominated wholly in the host currency (`HC`), with the USD debt unchanged.

WORKED EXAMPLE

Worked example 11.3.2 — the tariff term that was worth more than the tariff.

1. **Setup.** Assume the exchange rate at close is `HC` **4.00 = USD 1** (an illustrative rate for a fictitious currency; no real currency or jurisdiction is implied). Under the `HC`-tariff variant, revenue is `HC` 48,000,000; of the 4,500,000 of cash operating costs, **1,350,000 is USD-denominated** (imported membranes, chemicals, specialist spares) and the balance of 3,150,000 is local (`HC` 12,600,000); cash tax `HC` 2,064,000 and the working-capital movement `HC` 2,400,000 are local. Debt service remains USD 5,009,635.23. Compute the coverage consequence of devaluation and the exchange-rate-indexed revenue share that protects it.
2. **Formula.** `CFADS` in USD = `(HC revenue − HC costs − HC tax − HC working capital) ÷ x − USD costs`, where `x` is `HC` per USD. Breakeven `x` for a target ratio: `local numerator ÷ (target × debt service + USD costs)`. Debt-service-matching indexed share = `(debt service + USD operating costs) ÷ revenue`.
3. **Substitution.** Local numerator = `48,000,000 − 12,600,000 − 2,064,000 − 2,400,000 = HC 30,936,000`. At `x` = 4.00: `30,936,000/4.00 − 1,350,000 = 6,384,000` — the master thread reproduced, which is the check that the decomposition is right.
4. **Result.**

HC per USD	Devaluation	CFADS (USD)	DSCR
4.000000	—	6,384,000	**1.2743**
4.202369	**+5.06 %**	6,011,562	**1.2000** — covenant
4.350394	+8.76 %	5,761,081	1.1500 — lock-up
4.400000	+10.00 %	5,680,909	1.1340
4.864430	+21.61 %	5,009,635	**1.0000** — cannot pay
5.000000	+25.00 %	4,837,200	**0.9656**

The **debt-service-matching indexed share** is `6,359,635.23 / 12,000,000 =` **52.997 %**. The **minimum indexed share** that holds the covenant through a 25 % devaluation is **48.9318 %**. At the matching 52.997 % share, a 25 % devaluation leaves `CFADS` of **6,109,128** and a `DSCR` of **1.2195**, and the covenant survives devaluation up to **+37.17 %**. 5. **Interpretation.** The headline is stark enough to be worth stating plainly: **a 5.06 % currency movement breaches this project's covenant, and a 21.61 % movement stops it paying its debt.** For comparison, the interest-rate exposure of 11.3.1 required 73.9 basis points — a rare event in a quiet market — while a 5 % currency move is a routine month in many host economies. Unhedged currency mismatch is therefore not one risk among many; on these numbers it is the **binding** exposure, larger than rates, availability (Domain 7's 2.9 availability points) and input prices combined. **Note also what the arithmetic quietly reveals about "natural hedges".** Local costs do fall in USD terms as the currency weakens, which is why the exposure is 30,936,000 of local numerator rather than the full 48,000,000 of revenue — but the offset is far too small, because debt service (5,009,635.23) and USD operating costs (1,350,000) are **52.997 %** of base-case revenue and do not devalue at all. A project whose costs are mostly local and whose debt is entirely foreign has a natural hedge on the wrong side of the balance sheet. **The remedy is indexation, and it has a number.** Indexing 52.997 % of the tariff to the exchange rate — exactly the share that funds USD outflows — is the structurally clean ask, and it lifts the tolerable devaluation from 5.06 % to 37.17 %. The covenant-preserving minimum against a specified 25 % stress is 48.93 %, only 4.07 points lower, which is a useful negotiating fact: the clean structural ask is barely more expensive than the minimum defensible one, so there is little to be gained by conceding to a partial share. **And two limits must be stated honestly.** Indexation protects **debt service, not equity return**: full `DSCR` neutrality is impossible while any local surplus exists, because that surplus keeps shrinking in USD terms — at the 52.997 % share a 25 % devaluation still costs 274,872 of `CFADS`. And indexation is a **transfer to the offtaker**, so the pricing test of 11.1.3 applies: the offtaker does not control the exchange rate, so the transfer must be justified on **capacity** grounds — a public or utility payer with local-currency revenue and sovereign-adjacent standing bears a devaluation that would destroy the SPV — and it will be paid for in the tariff. That is a legitimate, well-grounded transfer; the illegitimate version is a sponsor accepting an unindexed `HC` tariff because the offtaker insisted, which is 11.1.2's bargaining-power allocation with a five-per-cent trigger attached.

11.3.3 The counterparty risk inside the allocation

The principle. Every risk successfully transferred becomes a **credit exposure** to the party that took it. The allocation table of 11.1.3 removed 4,420,000 of expected cost from Kestrel's balance sheet and replaced it with a claim on the EPC contractor — and that claim is worth the contractor's willingness and ability to pay.

The arithmetic that matters is conditional, not unconditional. Take the transferred bundle at 4,420,000 of avoided expected cost, an assumed recovery of 30 % on the contractor's obligations, and a two-year probability of default of 1.5 % for a contractor of the assumed standing. Expected credit loss on the transfer is `0.015 × 0.70 × 4,420,000 =` **46,410** — 12.46 % of annual covenant headroom, and easy to dismiss. But the unconditional probability is the wrong one. **The contractor most likely to default is the one that has already incurred the losses it was allocated**, so the relevant figure is the probability of default conditional on a claim of the size the allocation contemplates. At a conditional probability of 12 %, expected credit loss becomes `0.12 × 0.70 × 4,420,000 =` **371,280 — 99.69 %** of the entire annual covenant headroom of 372,437.72.

What matters is not the coincidence of the two magnitudes but the **eightfold difference between the two calculations**, and the fact that the conditional one is almost never done. Risk transfer converts a diversifiable operational exposure into a concentrated, correlated credit exposure, and the correlation is structural: the state of the world in which the guarantee is needed is the state in which the guarantor is weakest. Three consequences follow. **Test the guarantor, not the guarantee** — Domain 5's distinction between joint and several liability, and the standing of the entity behind a commitment, decides what a cap is worth. **Require the credit support to sit above the operating entity**, so that a trading counterparty's promise becomes a claim on a balance sheet or a bank. And **count the counterparties**: after allocation Kestrel's coverage depends on the EPC contractor, the operator, the offtaker (Domain 7 priced its credit at 0.0400 of `DSCR`, 53.7 % of its total expected-loss exposure), the hedge counterparty of 11.3.1, the insurers of 11.3.4 and the sponsors behind Domain 5's 24,000,000 of committed capital. Six credit dependencies are six ways to lose the protection the project paid for, and a **counterparty concentration schedule** — exposure, mechanism, cap, credit standing, correlation of the call with that standing — makes them visible (11.T.2).

11.3.4 Political, regulatory and force-majeure risk

Political and regulatory risk covers expropriation, currency inconvertibility and transfer restriction, breach of undertaking by a state counterparty, and adverse change in law, tax or regulated tariff. Its allocation is unusual in that the natural transferee is often not a commercial party at all: political risk insurance and the guarantee products of export credit agencies and multilateral development institutions exist precisely because private parties cannot bear these exposures (Domain 9 covers the instruments; the allocation view here is that they are **capacity transfers**, and the pricing test applies unchanged). Change-in-law risk is the one commonly split rather than allocated: discriminatory change (aimed at the project or its sector) to the state, general change to the project, is a widely used convention — but conventions are not law, and the line between the two categories is where the disputes live.

Force majeure is the allocation of last resort: it allocates a risk to **nobody**, suspending obligations rather than shifting cost. That structure is right for genuinely uncontrollable, uninsurable events, and it has a specific financial consequence that projects consistently under-model: relief from performance obligations does not relieve **debt service**. The lenders' schedule continues, so a force-majeure event that suspends the operator's availability obligation lands squarely on coverage.

WORKED EXAMPLE

Worked example 11.3.4 — the insurance waiting period nobody checks against the covenant.

1. **Setup.** Kestrel carries business-interruption cover with a **60-day waiting period** — the insurer indemnifies lost margin only for the portion of an outage beyond 60 days. Daily `CFADS` is `6,384,000/360 =` **17,733.33** (Domain 5's 30/360 basis). Covenant headroom is 372,437.72. Determine the maximum outage the covenant survives, and the effect of outages of 21, 30 and 60 days.
2. **Formula.** Uninsured days = `min(outage days, waiting period)`; lost `CFADS` = uninsured days × daily `CFADS`; maximum survivable days = headroom ÷ daily `CFADS`.
3. **Substitution.** `372,437.72 / 17,733.33`; then `CFADS − days × 17,733.33`, divided by 5,009,635.23.
4. **Result.**

Outage	Uninsured days	Lost CFADS	CFADS	DSCR
21 days	21	372,400	6,011,600	**1.2000** — exactly at covenant
30 days	30	532,000	5,852,000	**1.1681** — breach
60 days	60	1,064,000	5,320,000	**1.0620** — breach, below lock-up
90 days	60 (capped by the waiting period)	1,064,000	5,320,000	1.0620

The maximum survivable outage is **21.0021 days**, so the maximum tolerable waiting period is **21 days**. 5. **Interpretation.** The insurance programme and the finance documents were negotiated by different teams against different units, and the result is a structural gap nobody owns: **the waiting period is 60 days and the covenant survives 21.** Any outage longer than three weeks breaches, and the insurance — which is doing its job perfectly — pays nothing until the covenant has already failed by a wide margin. The insight generalises beyond insurance: **deductibles, waiting periods, cure periods, grace periods, notice periods and reporting lags are all calibrated in time, while covenants are calibrated in cash, and the translation between the two is almost never performed.** Three responses are available and should be priced against one another: buy down the waiting period towards 21 days (the cheapest fix if the market offers it); size the debt-service reserve to bridge the gap — Domain 10's six-month DSRA of 2,504,818 comfortably funds the 1,064,000 maximum uninsured loss, so payment continuity is not at issue and only compliance is; or negotiate a covenant carve-out for insured events within the waiting period, which is the cleanest answer and the one most often overlooked because it must be asked for before close. Note finally the shape of the exposure: it is **capped** at 1,064,000 by the waiting period itself, so this is a high-probability, bounded exposure — the opposite profile from the currency mismatch of 11.3.2 — and it should be managed with a reserve or a carve-out rather than with capital.

AI in this KA

Where it earns its place: the translation layer. Every number in this KA is a mechanical consequence of terms written in documents — a margin, a waiting period, a hedging covenant, an indexation formula — and the failure mode is that the translation is done once at close and never again. A monitored model that recomputes, every month, the reference rate at which the covenant fails, the exchange rate at which it fails, the outage length at which it fails and the hedge ratio in force against the ratio covenanted is high-value automation with a low error surface, because each output is a closed-form calculation against a stated document term. Extraction of hedging covenants, waiting periods and caps from a document suite is the same strong application named in 11.1. **Where it must not go:** rate, currency and event forecasting. A model asked what the reference rate or the exchange rate will be in year seven will answer, and the answer will carry no information; the governed use is to compute the **breakeven** — the level at which something breaks — because a breakeven is a fact about the structure rather than a prediction about the world. Equally, the question of whether a set of facts constitutes force majeure, or a discriminatory change in law, is a legal determination for qualified counsel and not a model output. **Verification, concretely:** require each monitored breakeven to be reproducible by hand from the closing model on one period — the check values are 6.7390 %, `HC` 4.202369 and 21 days — and re-verify after any amendment, because the recurring failure is a monitor that keeps computing correctly against superseded terms.

KEY TERMS — KA 11.3

Term	Meaning
Hedge ratio	Share of floating debt fixed by swap or cap; lenders covenant a minimum rather than a full hedge.
Breakeven rate / breakeven exchange rate	The level at which a named covenant fails; a fact about the structure, not a forecast.
Mark-to-market (swap)	The break cost of terminating a hedge early — an exposure a swap creates rather than removes.
Currency mismatch	Revenue and debt service in different currencies; unbounded and correlated with host-economy stress.
Exchange-rate-indexed share	The portion of tariff linked to the exchange rate; the debt-service-matching share funds USD outflows.
Conditional probability of default	Default probability given a claim of the size the allocation contemplates — the relevant figure for transferred risk.
Waiting period / deductible	Time or amount an insured bears before indemnity; must be tested against covenant headroom in days.
Force majeure	Allocation to nobody: obligations suspend, debt service does not.

SAMPLE MCQS — KA 11.3

MCQ 11.3-A `[11.3.1 · Application]` Debt 42,000,000; fixed year-one principal 2,489,635.23; `CFADS` 6,384,000; covenant 1.20×. The all-in interest rate at which the covenant fails is closest to: - A. 6.00 % - B. 6.74 % - C. 8.00 % - D. 9.27 %

Rationale: Maximum debt service = 6,384,000/1.20 = 5,320,000; maximum interest = 2,830,364.77; ÷ 42,000,000 = 6.7390 %. A is the rate at close; C is a +200 bp shock, at which the ratio is already 1.0914; D is the rate at which debt service cannot be paid at all — a different and much later threshold.

MCQ 11.3-B [11.3.1 · Analysis] A swap moves year-one DSCR from 1.2743 to 1.2533 and replaces a 1.0914–1.5311 range with a single value. The correct characterisation is: - A. the swap is uneconomic, since coverage falls - B. 0.0210 of coverage is given up to remove 0.4397 of coverage range — 20.92 units of range per unit surrendered - C. the swap eliminates all interest-rate exposure and creates no new exposure - D. the swap is unnecessary because the project can pay debt service up to 9.27 %

Rationale: The trade is certainty for a small, certain cost (11.3.1). A counts the cost and not the benefit; C ignores mark-to-market and hedge-counterparty exposure; D confuses the payment threshold with the covenant threshold.

MCQ 11.3-C [11.3.2 · Application] Local numerator HC 30,936,000, USD operating costs 1,350,000, debt service 5,009,635.23, rate at close HC 4.00 = USD 1, covenant 1.20×. The devaluation at which the covenant fails is closest to: - A. 5.1 % - B. 15.9 % - C. 21.6 % - D. 25.0 %

Rationale: Covenant CFADS = 5,009,635.23 × 1.20 = 6,011,562.28; x = 30,936,000 ÷ (6,011,562.28 + 1,350,000) = 4.202369, i.e. +5.06 %. B divides CFADS by the covenant instead of multiplying debt service by it — the commonest covenant-trigger error; C is the point at which the DSCR reaches 1.00; D is the illustrative stress case, at which the ratio is already 0.9656.

MCQ 11.3-D [11.3.4 · Analysis] Business-interruption cover has a 60-day waiting period; daily CFADS is 17,733.33 and covenant headroom is 372,437.72. The most useful statement for the finance committee is: - A. the cover is adequate, since the maximum uninsured loss of 1,064,000 is within the DSRA - B. the covenant survives 21 days of outage while the waiting period is 60 — so any outage beyond three weeks breaches, and a carve-out or a bought-down waiting period is required - C. the waiting period should be extended to reduce premium - D. force majeure relief will suspend debt service during the outage

Rationale: The gap between a time-calibrated insurance term and a cash-calibrated covenant is the finding (11.3.4). A is true about payment and silent about compliance — the distinction of Domain 10, KA 10.2.1; C widens the gap; D is false, since force majeure suspends performance obligations and not debt service.

SELF-CHECK — KA 11.3

1. State Kestrel's three unhedged interest-rate breakevens. — Covenant at 6.7390 % (+73.9 bp), lock-up at 7.2897 % (+129.0 bp), payment failure at 9.2723 % (+327.2 bp).
2. Why is the unconditional probability of default the wrong input for transferred risk? — The counterparty most likely to default is the one that has already incurred the allocated loss; Kestrel's expected credit loss moves from 46,410 to 371,280 on that correction.
3. State the maximum tolerable insurance waiting period for Kestrel and why. — 21 days: covenant headroom of 372,437.72 divided by daily CFADS of 17,733.33.

KNOWLEDGE AREA 11.4

Environmental and social, technology, cybersecurity and AI model risk

Topics: 11.4.1 the operating-phase register · 11.4.2 why a lender re-cuts it · 11.4.3 AI model risk as a register line · 11.4.4 from register to debt capacity.

11.4.1 The operating-phase register

The construction register of KA 11.1 has a counterparty and therefore a price. The operating register usually has neither: there is no single party to whom environmental compliance, technology underperformance, a cyber intrusion or a model error can be transferred, so these risks are managed, insured in part, reserved against, and otherwise **retained**. That makes their quantification more important, not less, because retained risk is the only kind that shows up directly in coverage.

Definitions worth fixing. Environmental and social risk is the risk of loss from non-compliance with environmental consents, from social-licence failure, or from the conditions attached to lender environmental and social standards — which are contractual obligations in a project financing, not aspirations, and whose breach is an event of default in many facilities. **Technology risk** is the risk that plant performs below its warranted characteristics over life rather than at completion: degradation faster than warranted, consumables consumed faster than modelled, obsolescence of a control system. **Cybersecurity risk**, in an industrial project, is primarily an **availability** risk on the operational-technology network rather than a data risk, and it is therefore a `CFADS` risk with the same shape as any other outage (11.3.4's arithmetic applies). **Model risk** is, on the shared registry's definition, the risk of loss from decisions based on flawed, misused or misunderstood models — financial or AI.

WORKED EXAMPLE

Worked example 11.4.1 — Kestrel's operating register, aggregated on the Domain 8 method.

1. **Setup.** Six retained operating-phase items, each quantified once over the loan life. The aggregation machinery is PML-AI Domain 8's (KA 8.2.2 and 8.2.4) and is not re-derived: $\text{EMV} = p \times \text{impact}$; for independent items mean = ΣEMV and variance = $\Sigma p(1-p) \times \text{impact}^2$; a P80 ≈ mean + 0.8416σ.
2. **Formula.** As above; then the same aggregation with a uniform pairwise correlation ρ, for which variance = $(1-\rho)\Sigma\sigma_i^2 + \rho(\Sigma\sigma_i)^2$.
3. **Substitution and result.**

ID	Retained operating risk	p	Impact (USD)	EMV (USD)
O1	Discharge-consent breach requiring plant modification	0.15	3,200,000	480,000
O2	Community objection interrupting intake maintenance access	0.10	1,500,000	150,000
O3	Membrane flux decline faster than warranted (technology)	0.20	2,600,000	520,000
O4	Cyber intrusion on the control network: outage plus remediation	0.12	4,500,000	540,000
O5	AI dosing-optimisation model error producing off-spec output	0.25	900,000	225,000
O6	Asset-model drift causing deferred-maintenance catch-up	0.30	700,000	210,000
	Mean exposure			**2,125,000**

Variance (independent) **4,982,875,000,000**; σ **2,232,235**; **P80 4,003,649**. At ρ = 0.30, σ rises to **3,227,338** and the **P80 to 4,841,128 —** an uplift of **837,478**, or **20.92 %**. 4. **Result.** The register's mean is **USD 2,125,000** and its P80 **USD 4,003,649** on an independence assumption that raises it to **USD 4,841,128** once a modest correlation is admitted. 5. **Interpretation.** Two observations do most of the work. **The modern risk classes dominate the register.** Cyber (540,000) and the two model-risk lines (435,000 combined) contribute 975,000 of the 2,125,000 mean — 45.9 % — and the cyber item, with the largest single impact, contributes the largest share of the variance and therefore of the tail. A register written five years earlier would have carried none of these lines, and a project whose register still does not is not less exposed, merely less informed. **The independence assumption is expensive.** The 837,478 uplift at $\rho = 0.30$ is more than twice Kestrel's entire annual covenant headroom, and correlation here is not hypothetical: O1 and O2 are both consequences of the same community and regulatory environment, and O5 and O6 are both consequences of the same data pipeline and the same modelling culture. Where items share a cause, independence is a choice to under-provide, and the choice should be stated in the paper rather than embedded in the arithmetic. PML-AI Domain 8's two standing cautions carry over unchanged: the normal approximation is a convenience for a handful of Bernoulli items, and a simulation over the register is the proper instrument where the answer is decision-critical.

11.4.2 Why a lender re-cuts the register

A lender does not accept a sponsor's register, and the reason is structural rather than adversarial: **the two parties have different loss functions.** Equity holds the upside and the downside, so its rational provision is near the middle of the distribution. A lender holds a fixed claim: it captures none of the upside, so every dollar of distribution above its expectation is irrelevant to it while every dollar of shortfall is loss. A rational lender therefore prices the **tail**, not the mean, and it does so by re-cutting the register's inputs rather than by arguing about the output.

WORKED EXAMPLE

Worked example 11.4.2 — the same register, on the lender's basis.

1. **Setup.** The lender's technical and E&S advisers re-cut the register of 11.4.1: probabilities at **1.5×** the sponsor's (the upper end of the plausible range rather than the central estimate), impacts at **1.4×** (a P80 within each item's own impact range rather than its mode), and a uniform pairwise correlation of **0.30** in place of independence. These multipliers are illustrative of the direction and order of magnitude advisers apply; the actual re-cut is item-specific and evidenced.
2. **Formula.** As 11.4.1, on the re-cut inputs.
3. **Result.** Lender EMVs: O1 1,008,000 · O2 315,000 · O3 1,092,000 · O4 1,134,000 · O5 472,500 · O6 441,000. **Lender mean 4,462,500**; σ **5,253,726**; **lender P80 8,884,036**.
4. **Interpretation.** The single most useful comparison in this KA: **the lender's expected case (4,462,500) exceeds the sponsor's 80th percentile (4,003,649) by 458,851 — a ratio of 1.1146 — and the lender's P80 is 4.1807 times the sponsor's mean.** A sponsor who arrives at a credit committee expecting to negotiate a percentile has misread the disagreement: the parties are not arguing about confidence levels, they are working from different input sets, and the only productive negotiation is **item by item on the evidence** — this probability, on this ground investigation; this impact, on this remediation quotation. Two further consequences deserve stating. **The gap is a financeable quantity, not a debating point**: it will be closed by a reserve, a sponsor commitment, a contingent equity undertaking or less debt, and the sponsor's choice among those is a real commercial decision (Domain 8's distinction between funded cash and a posted commitment applies directly). And **the re-cut is where AI-assisted analysis is most tempting and least appropriate**: a model asked to "produce the bank case" will multiply, which is precisely the intellectually empty version of the exercise the arithmetic above is designed to expose. Multipliers illustrate; evidence decides.

11.4.3 AI model risk as a register line

Definition. In a project financing, **AI model risk** is the risk of loss arising from a decision taken, or an action executed, on the output of a machine-learning model — whether because the model was wrong, because it was used outside the conditions it was validated for, or because its output was misunderstood by the person who acted on it. Kestrel's register carries two such lines: **O5**, an optimisation model that sets chemical dosing and can drive output off specification, and **O6**, an asset model whose drift defers maintenance that later has to be caught up.

Three properties distinguish AI model risk from the other register lines. **The impact distribution is fat-tailed relative to the probability**: most model errors are small and self-correcting, but a model with authority over a physical process can produce a large excursion quickly, and a single `p` × impact pair compresses that shape. **It is correlated with everything the model touches**, so one model feeding dosing, energy optimisation and maintenance planning creates a common cause across three otherwise independent register lines — a concrete reason the `ρ` = 0.30 case of 11.4.1 is the honest one. And **its probability is a function of governance rather than of nature**: unlike a membrane's degradation rate, `p` here is set by whether validation, monitoring, drift detection, human approval and rollback exist. That last property makes it the one register line whose probability the project can genuinely change — which by 11.1.2's logic makes the project its own best holder of the risk, and controls worth more than transfer.

For allocation purposes the practical settlement is: the **vendor** takes defect and availability risk on the software within a liability cap that will be small relative to the exposure; the **operator** takes the risk of using the model outside its validated envelope, backed by procedure and training; and the **project retains the consequence**, provisions for it in the register, and manages it with the control set Domain 16 specifies. Two negotiating points are worth naming because they are so often missed: the right to the **model's validation evidence and monitoring outputs** (without which the project cannot demonstrate its own `p`), and the right to **operate without the model** — a documented manual fallback, which converts a dependency into a preference and is the cheapest single mitigation available.

11.4.4 From register to debt capacity

A register is not a document; it is a **debt-capacity input**. The translation is short: convert the register's present-value exposure into an annual equivalent over the loan life using the loan-rate annuity factor, deduct it from `CFADS`, and read the coverage.

WORKED EXAMPLE

Worked example 11.4.4 — what Kestrel's register does to its coverage and its debt.

1. **Setup.** `AF(0.06, 12) = 8.383844`; `CFADS` 6,384,000; debt service 5,009,635.23; covenant 1.20×. Test the five register measures of 11.4.1 and 11.4.2.
2. **Formula.** Annual equivalent = PV exposure ÷ `AF(0.06, 12)`; stressed `CFADS` = 6,384,000 − annual equivalent; `DSCR` = stressed `CFADS` ÷ 5,009,635.23. Covenant-preserving maximum PV exposure = headroom × `AF(0.06, 12)`.
3. **Result.**

Register measure	PV exposure	Annual equivalent	Stressed CFADS	DSCR
Sponsor mean	2,125,000	253,464	6,130,536	**1.2237** — holds
Sponsor P80, independent	4,003,649	477,543	5,906,457	**1.1790** — breach
Sponsor P80, ρ = 0.30	4,841,128	577,435	5,806,565	**1.1591** — breach, near lock-up
Lender mean	4,462,500	532,274	5,851,726	**1.1681** — breach
Lender P80	8,884,036	1,059,661	5,324,339	**1.0628** — breach

The **covenant-preserving maximum PV exposure** is 372,437.72 × 8.383844 = **USD 3,122,460**. On the lender's P80 the sustainable debt at 1.20× falls to (5,324,339/1.20) × 8.383844 = **USD 37,198,687 — 4,801,313** less than the 42,000,000 drawn. 4. **Interpretation.** This table is the domain's closing argument. **Kestrel's covenant survives its own operating risk register only at the mean.** At its own P80 — the confidence level its own contingency policy uses for construction (Domain 8) — the covenant fails, and it fails on every one of the lender's measures. The number to carry away is the **3,122,460 covenant-preserving ceiling**: that, and not a percentage of capex or an industry norm, is how much present-value operating risk this structure can absorb, and it is directly comparable with the register's own 4,003,649. The gap of 881,189 is the honest size of the problem, and it has exactly the same four resolutions as any coverage shortfall (Domain 10, KA 10.1.2): more cash, lower required coverage, longer tenor, or less debt — with the fifth, more equity, as the residual. **The lender's arithmetic is the same arithmetic.** Its re-cut register removes 4,801,313 of debt capacity, which is why a diligence process that appears to be about risk management is in fact a negotiation about the debt quantum, and why a sponsor who cannot reproduce the calculation above will not understand what it is conceding. Two cautions bound the method. The **annuity-equivalent conversion is a simplification**: register events are lumpy and dated, and a proper treatment models each in the year it is assumed to fall, which will produce a worse minimum DSCR than the levelised figure shown here because coverage is tested in periods. And the **register and the estimate range must not be added** — Domain 8's demonstration on the construction side applies unchanged to operations.

AI in this KA

Where it earns its place: three specific, high-value uses. Extracting environmental and social obligations from consents and lender standards into a monitored obligation register, which is document work at a volume humans do badly. Anomaly detection on operational-technology telemetry, where a model that flags a deviation for a human to investigate is doing what machines are for. And **register completeness challenge** — asking what a comparable project's register contains that this one does not, which is where the model's breadth is a genuine advantage over an individual's experience. **Where it must not go:** into the probability and impact columns (11.1's rule, and here it bites harder because there is no counterparty pricing the risk to provide an external check); into autonomous action on a physical process without a human approval step and a tested rollback; and into producing the bank case by multiplication (11.4.2). **Verification, concretely:** treat every model in the operating envelope as a register line with a named owner, a validation date, a monitored drift metric and a documented manual fallback — a model without those four attributes is an unquantified register line, which is the definition of an unmanaged risk. Recompute the register's mean by hand and require it to equal ΣEMV before any percentile is quoted. And require any automated register aggregation to state its correlation assumption explicitly on the face of its output, because independence is the assumption tools default to and the assumption that is worth 837,478 here. **AI proposes; the professional verifies, decides and remains accountable.**

KEY TERMS — KA 11.4

Term	Meaning
Environmental and social risk	Loss from consent breach, social-licence failure or breach of lender E&S standards — contractual, not aspirational.
Technology risk	Under-performance over life (degradation, consumption, obsolescence) rather than at completion.
Cybersecurity risk (project)	Primarily an availability risk on the operational-technology network; a `CFADS` risk.
AI model risk	Loss from decisions or actions taken on model output — wrong model, wrong envelope, or misunderstood output.
Bank case	The lender's re-cut of the register: higher probabilities, higher impacts, correlation admitted.
Correlation uplift	The increase in a percentile from admitting $\rho > 0$; 837,478 at $\rho = 0.30$ on Kestrel's register.
Covenant-preserving exposure ceiling	Headroom × `AF(r, n)` — the present-value risk a structure can absorb (3,122,460 for Kestrel).

SAMPLE MCQS — KA 11.4

MCQ 11.4-A `[11.4.1 · Application]` A register of independent items has a mean of 2,125,000 and a variance of 4,982,875,000,000. Its P80 is closest to: - A. 2,125,000 - B. 4,003,649 - C. 4,357,235 - D. 4,841,128

Rationale: σ = 2,232,235; P80 = 2,125,000 + 0.8416 × 2,232,235 = 4,003,649. A is the mean, which is exceeded roughly half the time; C applies a 1.0-σ (P84) factor instead of 0.8416; D is the P80 once a 0.30 correlation is admitted — right method, different stated assumption.

MCQ 11.4-B `[11.4.2 · Analysis]` A sponsor's register P80 is 4,003,649 and its lender's re-cut mean is 4,462,500. The correct inference is: - A. the lender is applying a higher confidence level than the sponsor - B. the parties are working from different input sets, not different percentiles — the lender's expected case already exceeds the sponsor's 80th - C. the sponsor's arithmetic is wrong - D. the difference is immaterial at 458,851

Rationale: The lender re-cuts probabilities, impacts and correlation; the disagreement is evidential and must be negotiated item by item (11.4.2). A misdiagnoses the disagreement as a percentile choice; D ignores that 458,851 exceeds the whole annual covenant headroom of 372,437.72.

MCQ 11.4-C `[11.4.4 · Application]` Annual covenant headroom is 372,437.72 and `AF(0.06, 12)` is 8.383844. The present-value operating-risk exposure the structure can absorb before its covenant fails is: - A. USD 372,438 - B. USD 3,122,460 - C. USD 4,003,649 - D. USD 44,423

Rationale: Headroom × `AF(0.06, 12)` = 3,122,460 — the present value of losing that much cash every year of the loan life. A is the annual figure, not its present value; C is the register's own P80, which is the exposure being tested against the ceiling, not the ceiling; D divides (44,423) rather than multiplies by the annuity factor.

MCQ 11.4-D `[11.4.3 · Analysis]` Which property makes AI model risk different in kind from membrane degradation risk? - A. its impact is always larger - B. its probability is a function of governance — validation, monitoring, approval and rollback — so the project can genuinely change it - C. it can be transferred to the software vendor in full - D. it is uninsurable

Rationale: 11.4.3: p is set by the control set rather than by nature, which by 11.1.2's logic makes the project the right holder and controls worth more than transfer. A is unsupported; C is false — vendor liability caps are small relative to the exposure; D overstates a market position.

SELF-CHECK — KA 11.4

1. State Kestrel's register on four bases. — Sponsor mean 2,125,000; sponsor P80 independent 4,003,649; sponsor P80 at ρ = 0.30 4,841,128; lender P80 8,884,036.
2. What is the covenant-preserving exposure ceiling and how is it computed? — 3,122,460 = annual headroom 372,437.72 × `AF(0.06, 12)` 8.383844.
3. Why is a lender's mean higher than a sponsor's P80? — Different loss functions produce different input sets: a fixed claim prices the tail, so probabilities, impacts and correlation are all re-cut upwards.

Advanced topics — Domain 11

11.A.1 Bargaining power, priced

The illegitimate ground for transfer (11.1.2) can be quantified, and doing so is the fastest way to stop an organisation using it. When a party is compelled to accept a risk it neither controls nor can absorb, one of four things happens and each has a cost the transferor eventually pays. It prices the risk **privately**, and the premium is paid unseen — the A4 case, where 1,827,000 sits inside a lump sum against 960,000 of retained expected cost. It **adopts a claims strategy**, converting a priced allocation into an unpriced dispute costing legal spend, management time and programme. It **withdraws from the bid**, reducing competitive tension — the most expensive outcome and the hardest to attribute. Or it **accepts, fails and hands the risk back** at the conditional default probabilities of 11.3.3, the 371,280 case. Hence one question in every negotiation: if this counterparty cannot influence this outcome, what is it charging us, and where is that charge?

11.A.2 Correlation, and the two places it hides

Correlation enters a risk position twice, and practitioners usually model neither. **Within the register**, admitting ρ = 0.30 on Kestrel's operating items raises σ from 2,232,235 to 3,227,338 and the P80 by **837,478** — 20.92 %, and more than twice the annual covenant headroom. **Between the register and the counterparties**, the correlation is between the call on a protection and the condition of the party providing it: the eightfold move from 46,410 to 371,280 in 11.3.3 is this second correlation priced. The two compound in the state of the world that matters, and the structural implication is the one to carry: **a risk position is not the sum of its lines**, and every mitigation that depends on a counterparty is weakest exactly when it is needed. The practical mitigations are diversification of counterparties, credit support that sits above the trading entity, and funded reserves in place of commitments where the correlation is highest — each of which is a real cost, honestly incurred, rather than an assumption quietly avoided.

11.A.3 The reviewer's allocation eye

Invariants to test on any risk allocation and its arithmetic. Every register line names a **mechanism** (clause, guarantee, insurance, reserve), not merely an owner. Every transferred line names the **counterparty, the cap and the credit standing behind it**, and the sum of caps is compared with the sum of transferred exposures. Every transfer has a **priced net value**: transferor `EMV` less loaded premium, with the breakeven loading stated; any transfer of a risk the transferee cannot control or cheaply bear is flagged. The register's mean equals **Σ**`EMV`, and its correlation assumption is stated on the face of the output. Register and estimate-range provisions are **not added** (Domain 8). Every exposure is expressed in the **covenant's units**: the `CFADS` it consumes, and the basis points, percentage, days or exchange-rate move that consumes the headroom. Time-calibrated terms (waiting periods, deductibles, grace and cure periods) are **tested against headroom in days**. The hedge ratio in force is compared with the ratio covenanted, and the hedge profile amortises with the loan. Interest-rate, currency and input-price tolerances are reported as **non-additive** unless a joint matrix is provided. The `PLCR`/tail is not used to justify retaining an exposure the lenders have no claim on. And the register is reconciled to the **debt quantum**: the covenant-preserving exposure ceiling (headroom × `AF(r, n)`) is stated alongside the register's own percentile, because a register that exceeds the ceiling has already re-sized the debt whether or not anyone has said so.

Industry variations — Domain 11

- **Water and desalination (Kestrel's sector).** Allocation turns on the availability/force-majeure boundary and on input-cost pass-through, because power is the dominant operating cost; ground and marine-interface risks are typically retained by the owner, which commissions the investigations, and E&S risk concentrates on discharge consents.
- **Contracted power and availability PPPs.** The cleanest allocation set in the market: volume to the payer, availability to the operator, fuel or input price passed through by formula. The residual risks are therefore financial — rate, currency, refinancing — which is why hedging covenants are tightest in this sector.
- **Merchant power and commodities.** Market risk cannot be allocated at all, only hedged for a fraction of the tenor. Lenders respond with higher required coverage and stressed price decks rather than with allocation, and the register's price lines dominate its mean.
- **Transport concessions.** Patronage risk is the defining exposure and is genuinely uncontrollable by any party, which is why minimum revenue guarantees, availability structures and traffic-band mechanisms exist; ramp-up correlation makes the independence assumption of 11.4.1 least defensible here.
- **Digital infrastructure.** Technology obsolescence and cybersecurity move from the tail to the centre of the register, tenant credit substitutes for offtaker credit, and power price and availability become the dominant operating exposures; short asset lives compress the tail that would otherwise absorb a late-life register.
- **Mining and heavy industry.** Environmental and social risk carries the largest single impacts — closure, rehabilitation, community and permitting — and is least transferable, so the allocation question becomes one of funded provision, bonding and completion support rather than of contractual ownership.

Case study — Domain 11: the wrap that could not be afforded (water / desalination)

Situation. Two weeks before Kestrel's EPC award, the preferred bidder offered to convert its partial wrap into a **full wrap**, taking the five items the pricing exercise of 11.1.3 had left with the owner — ground conditions beyond the disclosed baseline, utility diversion and third-party interfaces, membrane price above the contract index, marine weather standby, and permit-driven monitoring works. The price was an additional **USD 4,620,000** on the EPC contract, taking it from 48,000,000 to **52,620,000**. Two of the three sponsors were in favour: the wrap removed 2,840,000 of retained expected cost and, in the words of the meeting, "made the project a single-point-of- responsibility financing".

What happened. The finance director ran the arithmetic of 11.1.3 and then carried it one step further, into coverage. On expectations the transfer destroyed **1,780,000** of value, and it destroyed **460,000** even if the bidder's 40 % loading were negotiated to zero, because the bidder's own expected cost on the bundle (3,300,000) exceeded the owner's (2,840,000) — it had priced a geotechnical baseline it had not set, a utility corridor it did not control and a permit condition it could not influence. The breakeven loading was **−13.94 %**. But the decisive number was the financing consequence. Capital cost rises to **64,620,000**. Held at 70/30 gearing, senior debt becomes **45,234,000**, the annual instalment `45,234,000/8.383844 =` **5,395,377.11**, and the year-one `DSCR` falls from 1.2743 to **1.1832** — below the facility's own 1.20× covenant at financial close and only **0.0332** above its 1.15× lock-up threshold. (The figure is generous: it ignores the additional interest during construction the larger facility would accrue.) Funded with equity instead, the project is financeable but equity rises from 18,000,000 to **22,620,000**, gearing moves from 70.0/30.0 to **65.0/35.0** (`D/E` **1.8568**), and the same distributable cash stream is spread over **25.67 %** more equity.

How it resolved. The wrap was declined and the partial allocation of 11.1.3 was executed: A1–A3 transferred for **2,128,000** of priced risk premium inside the 48,000,000 price, A4–A8 retained, leaving the **2,690,000** register that Domain 8 provisioned against. Two of the five retained items were then attacked on their own terms rather than by transfer: the owner extended the geotechnical investigation before award (reducing its own A4 probability and, more importantly, narrowing the baseline the bidder had been pricing blind), and the utility diversion scope was fixed by a tripartite interface agreement with the utility before contract signature — an allocation to the party that actually controlled the corridor, which is 11.1.2's control ground applied to the right counterparty rather than the convenient one.

What the domain teaches here. A wrap is not a good or a bad thing; it is a price, and the price is comparable with a number the project already knows. Kestrel's full wrap cost **1,780,000** of expected value and **0.0911 of** `DSCR` — more coverage than its entire 372,437.72 of annual headroom represents — and the sponsors who favoured it were not being imprudent, they were reasoning about responsibility rather than about arithmetic. **Single-point responsibility is worth paying for only where the single point can actually influence the outcome.** And the resolution shows the third option that allocation debates routinely omit: where transfer destroys value, the answer is often not to retain the risk passively but to **reduce it at source**, or to allocate it to a party outside the EPC relationship who genuinely controls it.

Case study B — Domain 11: the currency the model held constant (transport / aviation)

Situation. Larkspur Regional Airport Concession Company (a fictitious concessionaire) financed a terminal expansion with **USD 180,000,000** of senior debt at **6.5 % over 14 years** — `AF(0.065, 14) = 9.013842`, annual instalment **USD 19,969,286.50** — against a 20-year concession. Revenue of **USD 60,000,000** equivalent came entirely from regulated charges denominated in the host currency. Cash operating costs of **24,200,000** split **17,600,000 local** and **6,600,000 USD-linked** (imported spares, systems maintenance and expatriate technical services); cash tax of 4,000,000 was local. Base-case `CFADS` was **31,800,000**, a `DSCR` of **1.5924** against a **1.30× covenant** and a 1.20× lock-up — a structure everyone involved described as comfortable. The financial model held the exchange rate constant across all fourteen years, and the sensitivity pack contained traffic, tariff, opex and interest-rate cases but no currency case.

What happened. In the concession's fourth operating year the host currency depreciated such that the cost of a dollar rose **30 %**. Traffic was on forecast; the airport was operating exactly as planned. `CFADS` in USD fell to **22,938,462** — revenue down to 46,153,846, partly offset by local operating costs falling to 13,538,462 and tax to 3,076,923 — and the `DSCR` fell to **1.1487**, breaching both the covenant and the lock-up while debt service continued to be paid in full. The retrospective arithmetic was unforgiving: the covenant had a breakeven at a devaluation of only **17.94 %**, and payment failure at **44.53 %**. Neither number had ever been computed. The natural hedge everyone had relied on — "our costs are local too" — was worth exactly the 21,600,000 of local costs and tax against 60,000,000 of local revenue, leaving a local numerator of **38,400,000** exposed against 6,600,000 of USD costs and 19,969,286.50 of USD debt service that did not move at all.

How it resolved. The lenders waived the breach for four quarters against a fee, a distribution lock-up that was in any case automatic, and a binding obligation on the concessionaire to seek an exchange-rate indexation amendment from the concession grantor. The amendment obtained indexed **40 %** of regulated charges to the exchange rate. At that share, a 30 % devaluation leaves `CFADS` of **28,476,923** and a `DSCR` of **1.4260**; a 60 % devaluation — the lenders' stress case — leaves **26,400,000** and **1.3220**, still above the covenant; and the covenant breakeven moves from a 17.94 % devaluation to **68.22 %**, a **3.804-fold** improvement in tolerance. The 40 % share was chosen, deliberately, above the **38.04 %** minimum that survives the 60 % stress, so that the structure was not calibrated to the last basis point of its own stress case. The grantor accepted the indexation because the analysis demonstrated the alternative: a concessionaire in default on a strategic asset, at a moment when the state's own capacity to intervene was weakest.

What the domain teaches here. Three things. **A model that holds a variable constant has made a forecast, not an assumption** — and the variable held constant in this financing was the largest single exposure in it. **A natural hedge must be measured, not asserted:** Larkspur's was real but covered only 36 % of its local revenue, and the untouched USD debt service was 33.3 % of base-case revenue. And **the remedy is a number, obtainable before it is needed.** The 40 % indexed share, the 38.04 % minimum, the 17.94 % and 68.22 % breakevens are all computable at close from the closing model in an afternoon; obtained at close they cost a tariff negotiation, and obtained in year four they cost a waiver fee, a locked-up distribution and a year of reporting under breach.

Executive perspective — Domain 11

What a project finance director cannot delegate in this domain:

- **The price of every allocation.** Not who owns each risk but what the ownership cost: transferor EMV less loaded premium, with the breakeven loading stated. A transfer nobody has priced is a transfer nobody has evaluated (11.1.3, Case study A).
- **The refusal to use bargaining power as a ground for transfer.** Where a counterparty can neither control nor cheaply bear a risk, the director's job is to say so in the room, because the charge for that transfer is real and will be paid somewhere less visible (11.A.1).
- **The four breakevens.** The reference rate (6.7390 %), the exchange rate (HC 4.202369, +5.06 %), the outage length (21 days) and the input-price rise (68.9699 %) at which the first covenant fails. These are facts about the structure, computable at close, and they belong on one page beside Domain 10's 6,011,562.28 cash trigger.
- **The hedging and indexation policy.** The hedge ratio, its amortisation profile, the pass-through shares and the exchange-rate-indexed share are coverage terms, not treasury detail, and the director owns the decision that trades 0.0210 of ratio for 0.4397 of range.
- **The counterparty concentration after allocation.** Six credit dependencies are six ways to lose the protection the project paid for, and the relevant default probability is the conditional one (11.3.3).
- **The register's correlation and confidence assumptions, stated in the open.** Independence is worth 837,478 of understatement on Kestrel's register, and the covenant-preserving ceiling of 3,122,460 is the number against which any register must be read (11.4.4).

Calculation exercises — Domain 11

Exercise 11.1 Three risks. X1: owner p 0.25, impact 4,000,000; transferee p 0.10, impact 3,000,000. X2: owner p 0.40, impact 1,500,000; transferee p 0.40, impact 1,600,000. X3: owner p 0.30, impact 2,000,000; transferee p 0.15, impact 1,800,000. Loading 35 %. Compute each net value of transfer, each breakeven loading, and the bundle decision. Solution. X1: own EMV **1,000,000**, transferee EMV 300,000, premium 405,000, net **+595,000**, breakeven loading **233.33 %**. X2: own 600,000, transferee 640,000, premium 864,000, net **(264,000)**, breakeven loading **−6.25 %** — no price works. X3: own 600,000, transferee 270,000, premium 364,500, net **+235,500**, breakeven loading **122.22 %**. Transfer X1 and X3 for 769,500 of premium against 1,600,000 of retained expected cost (net **+830,500**); retain X2. Common error: applying the loading to the transferor's EMV rather than the transferee's, which inflates every premium and rejects transfers that create value.

Exercise 11.2 Debt 120,000,000; fixed period principal 6,000,000; CFADS 17,500,000; covenant 1.25×. The facility floats at an all-in 5.50 %; a swap fixes the all-in rate at 5.75 %. Compute the DSCR at 5.50 % and at a 150 basis point shock, the breakeven all-in rate, and the minimum hedge ratio that survives the shock. Solution. At 5.50 %: interest 6,600,000, debt service 12,600,000, DSCR **1.3889**. At 7.00 %: interest 8,400,000, debt service 14,400,000, DSCR **1.2153** — breach. Breakeven: maximum debt service `17,500,000/1.25 =` 14,000,000; maximum interest 8,000,000; ÷ 120,000,000 = **6.6667 %** (+116.7 bp). Minimum hedge ratio `(0.07 − 0.066667)/(0.07 − 0.0575) =` **26.6667 %**. Fully swapped: debt service 12,900,000, DSCR **1.3566**, at a cost of **300,000** a period. Common error: computing the breakeven on total debt service rather than netting the fixed principal first, which understates the tolerable rate and over-buys the hedge.

Exercise 11.3 A project earns 30,000,000 of revenue, 75 % in local currency and 25 % in USD; its operating costs are 11,000,000, 60 % local; debt service is 12,400,000 in USD; the covenant is 1.30×. Compute base CFADS and DSCR, the position after 20 % and 35 % devaluations, and the covenant breakeven. Solution. Local numerator `30,000,000 × 0.75 − 11,000,000 × 0.60 =` **15,900,000**; USD numerator `30,000,000 × 0.25 − 11,000,000 × 0.40 =` **3,100,000**; base CFADS **19,000,000**, DSCR **1.5323**. At +20 %: `15,900,000/1.20 + 3,100,000 =` **16,350,000**, DSCR **1.3185** — holds. At +35 %: **14,877,778**, DSCR **1.1998** — breach. Breakeven: covenant CFADS `12,400,000 × 1.30 =` 16,120,000; devaluation multiple `15,900,000/(16,120,000 − 3,100,000) =` 1.2212, i.e. **+22.12 %**. Common error: applying the devaluation to gross revenue and ignoring that local costs devalue too, which overstates the exposure — here by treating 22,500,000 rather than 15,900,000 as exposed.

Exercise 11.4 Four independent register items: (p 0.20, 2,500,000), (0.10, 4,000,000), (0.30, 1,200,000), (0.15, 2,000,000). Compute the mean, σ and P80; recompute the P80 at ρ = 0.30; and state the covenant-preserving exposure ceiling for a project with 900,000 of annual headroom and `AF(0.06, 10) = 7.360087`. Solution. Mean `500,000 + 400,000 + 360,000 + 300,000 =` **1,560,000**. Variance $0.16 \times 2{,}500{,}000^2 + 0.09 \times 4{,}000{,}000^2 + 0.21 \times 1{,}200{,}000^2 + 0.1275 \times 2{,}000{,}000^2 =$ **3,252,400,000,000**; σ **1,803,441**; P80 `1,560,000 + 0.8416 × 1,803,441 =` **3,077,776**. With ρ = 0.30: the item standard deviations are 1,000,000 · 1,200,000 · 549,909 · 714,143, so $\Sigma\sigma_i$ = **3,464,052** and variance $0.7 \times 3{,}252{,}400{,}000{,}000 + 0.3 \times 3{,}464{,}052^2 =$ **5,876,576,724,325**, giving σ **2,424,165** and P80 **3,600,177** — an uplift of **522,401**. Ceiling `900,000 × AF(0.06, 10) = 900,000 × 7.360087 =` **6,624,078**, comfortably above the correlated P80. Common error: summing the σ_i rather than the variances for the independent case, which produces 3,464,052 in place of 1,803,441 and nearly doubles the provision.

Exercise 11.5 A project has CFADS of 9,600,000, debt service of 7,100,000 and a 1.20× covenant. Its business-interruption cover carries a 45-day waiting period; use a 30/360 basis. Compute annual headroom, the maximum survivable outage, and the DSCR after outages of 45, 30 and 15 days. State the maximum tolerable waiting period. Solution. Covenant trigger `7,100,000 × 1.20 =` 8,520,000; headroom **1,080,000**. Daily CFADS `9,600,000/360 =` **26,666.67**; maximum survivable outage `1,080,000/26,666.67 =` **40.50 days**. At 45 days uninsured: CFADS 8,400,000, DSCR **1.1831** — breach. At 30 days: 8,800,000, **1.2394**. At 15 days: 9,200,000, **1.2958**. Maximum tolerable waiting period **40 days** (40.50 rounded down to a whole day), so the 45-day period leaves a **4.5-day gap** to be closed by a carve-out, a buy-down or a reserve. Common error: computing headroom as CFADS ÷ covenant rather than debt service × covenant — the error that turns a 5 % currency tolerance into a 16 % one in MCQ 11.3-C.

Practitioner's toolkit — Domain 11

Adoption-ready artefacts; adapt headings to your organisation, then keep them stable.

Toolkit 11.T.1 — Risk allocation price sheet (one row per register line)

Columns: ID · risk statement · phase · **our p and impact, with the evidence reference** · our EMV · proposed transferee · **their p and impact as priced** · their EMV · loading applied · loaded premium · **net value of transfer** · **breakeven premium** · **breakeven loading** · ground for transfer (control / capacity / neither) · mechanism (clause, guarantee, insurance, reserve) · liability cap · credit standing behind the cap · decision and decision owner. Rules: a line with "neither" in the grounds column may not be transferred without a written reason from the decision owner; a negative breakeven loading is a stop, not a negotiation; and the sheet totals must reconcile to the retained register the contingency provision is sized against.

Toolkit 11.T.2 — The breakeven page (one side of paper, per facility)

Every material exposure, expressed in the covenant's own units. Rows: covenant and lock-up **cash triggers** and the annual headroom (Domain 10) · the **reference rate** at which the covenant fails, the hedge ratio in force and the ratio covenanted, and the hedge profile against the amortisation profile · the **exchange rate** at which the covenant fails, the exchange-rate-indexed revenue share in force and the debt-service-matching share · the **input-price rise** at which the covenant fails, per material input, with its pass-through share and reset frequency · the **availability or volume** level at which it fails (Domain 7) · the **outage length in days** at which it fails, beside every insurance waiting period and deductible, every grace period and every cure period · the **counterparty concentration schedule**: exposure, mechanism, cap, credit standing, and the correlation of the call with that standing. A footer states, in bold, which single exposure is nearest its breakeven — and the answer is rarely the one the risk register ranks first.

Toolkit 11.T.3 — Register aggregation and re-cut worksheet

Three panels. **Sponsor case:** items with `p`, impact, evidence reference, `EMV`, σ_i; mean = Σ`EMV` (hand-checked); variance and σ; the required percentile; and the **correlation assumption stated on the face of the output** with the uplift it suppresses. **Bank case:** the same items with re-cut `p` and impact, each change carrying its evidential basis — not a multiplier — plus the correlation assumption; the resulting mean and percentile; and the gap to the sponsor case, item by item. **Coverage translation:** annual equivalent of each measure over the loan life at `AF(r, n)`, the stressed `CFADS` and `DSCR`, the **covenant-preserving exposure ceiling** (headroom × `AF(r, n)`), and the debt quantum each measure supports at the covenant ratio. A model register in the operating envelope appears in the item list with an owner, a validation date, a monitored drift metric and a documented manual fallback, or it is recorded as unquantified.

Exam preparation — Domain 11

What is assessed. The pricing of allocation rather than its description: net value of transfer, breakeven premium, breakeven loading, and the identification of transfers that destroy value regardless of margin. The translation of every risk into coverage units — basis points, exchange-rate moves, days, input-price percentages — and the level at which a named covenant fails. Hedge-ratio arithmetic, including the minimum ratio surviving a stated shock. Currency-mismatch decomposition and the indexed share that funds foreign outflows. Register aggregation on the PML-AI Domain 8 method, the effect of correlation, why a lender's inputs differ from a sponsor's, and the conversion of a register into a debt-capacity number.

The calculations to do under time pressure. Net value of transfer and breakeven loading for a three-item bundle. Debt service and `DSCR` at a shocked floating rate with fixed principal, and the breakeven all-in rate. Minimum hedge ratio for a stated shock and covenant. `CFADS` and `DSCR` after a devaluation, and the covenant-breakeven exchange rate. Register mean, σ and P80, independent and correlated. Covenant-preserving exposure ceiling. Maximum survivable outage in days against a waiting period. Pass-through tolerance and its $1/(1-\varphi)$ multiplier.

The traps. Computing a covenant trigger as `CFADS` ÷ ratio rather than debt service × ratio, which turns Kestrel's 5.06 % currency tolerance into 15.9 % (MCQ 11.3-C, Exercise 11.5) · applying the loading to the transferor's `EMV` instead of the transferee's (Exercise 11.1) · treating a negative breakeven loading as a pricing negotiation (MCQ 11.1-B) · computing an interest-rate breakeven on total debt service without netting fixed principal first (Exercise 11.2) · confusing the covenant threshold with the payment threshold — 73.9 bp against 327.2 bp (MCQ 11.3-A) · applying a devaluation to gross revenue and ignoring that local costs devalue too (Exercise 11.3) · summing standard deviations instead of variances (Exercise 11.4) · presenting single-variable tolerances as additive when they share one pool of headroom (11.2.2) · adding a register provision to an estimate-range provision (Domain 8, KA 8.3.2) · using an unconditional default probability for transferred risk (11.3.3) · reporting a year-one ratio for a structure with an indexation mismatch (11.2.3) · and quoting a P80 without stating the correlation assumption underneath it (11.4.1).

How the domain connects. Domain 5 identified the bankability conditions each of these risks threatens; Domain 7 built the revenue and credit machinery this domain allocates; Domain 8 supplied the `EMV` and confidence method and provisioned against the residue this domain derives; Domain 10 supplied the coverage units every exposure is expressed in. Forward, Domain 12 documents the allocations priced here — and a clause that does not implement a priced allocation is the defect that domain exists to catch; Domain 13's diligence streams are where the lender's re-cut register is actually produced; Domain 14 monitors the construction risks A1–A8 as they crystallise or expire; and Domain 15 lives with whatever was retained, including the indexation wedge that only appears in the loan's later years.

Domain 11 summary

Risk allocation is a price, not a preference, and the price is computable. The net value of a transfer is the transferor's expected cost less the transferee's loaded premium, and it is reliably negative where the transferee can neither control the risk nor cheaply bear it: Kestrel's five declined items carried an owner's `EMV` of **2,840,000** against a premium of **4,620,000**, and destroyed **460,000** even at a zero margin because the bidder's own expected cost (**3,300,000**) exceeded the owner's — a breakeven loading of **−13.94 %**, against **+190.79 %** on the three items the bidder genuinely controlled. Those three transferred for **2,128,000** of premium against **4,420,000** of retained expected cost — **2.0771** dollars of expected-cost reduction per dollar of premium — leaving the **2,690,000** register Domain 8 provisions against. Buying the full wrap would have raised capex to **64,620,000** and, at 70/30, pushed the year-one `DSCR` to **1.1832**, below the project's own covenant. Market risk is allocated by formula rather than by clause: a **70 %** power pass-through multiplies Kestrel's tolerance to its largest cost driver by **3.3333×**, from a **20.6910 %** to a **68.9699 %** price rise, while an indexation mismatch — revenue escalating at an effective **2.00 %** against costs at **4.70 %** — consumes **85.68 %** of the covenant headroom by year twelve and breaches in year thirteen, surviving the loan life by **12.41 basis points** of assumed power escalation. The financial exposures are the most tractable and the most neglected: unhedged, **73.9 basis points** of reference rate stand between Kestrel and a covenant breach against **327.2** to a payment failure, and a swap trades **0.0210** of coverage for **0.4397** of coverage range with **70.0576 %** the minimum hedge ratio that survives a 200 basis point shock; an unindexed host-currency tariff would breach the covenant on a **5.06 %** devaluation and stop payment at **21.61 %**, against a debt-service-matching indexed share of **52.997 %** that lifts tolerance to **37.17 %**; every transferred risk becomes a credit exposure whose relevant default probability is conditional, moving Kestrel's expected credit loss from **46,410** to **371,280**; and a **60-day** insurance waiting period sits against a covenant that survives **21 days** of outage. The operating register — where environmental and social, technology, cybersecurity and AI model risk now sit — has a mean of **2,125,000** and a P80 of **4,003,649**, rising **837,478** once a 0.30 correlation is admitted, against a lender's re-cut **mean** of **4,462,500** that already exceeds the sponsor's P80 and a lender P80 of **8,884,036**. Read through the coverage machinery, Kestrel's covenant survives its register only at the mean: the covenant-preserving ceiling is **372,437.72 × 8.383844 = 3,122,460** of present-value exposure, and the lender's P80 removes **4,801,313** of debt capacity. That is the domain's whole argument in one number — a risk register is not a document, it is a debt-capacity input — and it is the argument Domain 12 must now write into contracts.

12

DOMAIN 12

Contracts and Transaction Structure

Group: Executing the transaction (Domain 3 of 4 in Part Three). **Target:** ~75 pages. **Binds to:** the PCI Book Pattern Specification and the shared registries (`docs/books/registries/`). This domain documents the allocations Domain 11 priced and reads their consequences in the coverage machinery Domain 10 built (`CFADS`, `DSCR`, covenant and lock-up triggers) — neither is re-derived here. **Commercial purpose only.** Contract law, enforceability, penalty doctrines, limitation of liability, security perfection, insolvency priority and dispute procedure differ fundamentally between jurisdictions and change over time; every jurisdiction-specific matter in this domain must be referred to qualified counsel in the relevant jurisdiction, and nothing here is legal advice. British English; USD (+SAR where useful, indicative `USD 1 ≈ SAR 3.75`).

Why this domain exists

Domain 11 established that risk allocation is a price and computed what Kestrel's transfers were worth. It stopped one step short of the thing that actually binds anyone: **the document**. An allocation that has been priced but not written is an intention; an allocation that has been written badly is a liability the model does not show. This domain is where the arithmetic of Domains 5 to 11 becomes contractual language, and where the leader discovers whether the protections the financial model assumed are the protections the contracts deliver.

The domain's central claim is that **a contract's commercial value is not its promise but the size, quality and reach of the money standing behind it.** Practitioners read contracts for their obligations — the contractor shall achieve the guaranteed output, the offtaker shall take the water. Lenders read them for their limits: the liquidated-damages rate, the cap on that rate, the aggregate cap above it, the exclusion of consequential loss, the identity and balance sheet of the guarantor, the expiry date of the bond, and what the termination clause pays. Those limits are numbers, they are computable before signature, and they routinely fail to cover the loss they were drafted against. Kestrel's contract stack looks orthodox — delay damages of USD 20,000 per day capped at 10 % of the EPC price, performance damages capped at another 10 %, an aggregate liability cap of 20 % — and yet a 300-day delay combined with a 5 % output shortfall produces exposure of **USD 12,255,674** against nominal cover of **USD 9,600,000** and risk-adjusted cover of **USD 8,160,000**. The uncovered residue of **2,655,674** is 14.75 % of the entire equity cheque, and it was knowable at signature from four numbers on two pages.

The domain works through the stack in the order a transaction is assembled. The construction and operating contracts are where damages, caps and buy-downs are calibrated — or mis-calibrated against a loss that is computable to the day (KA 12.1). The revenue contracts are where the project's cash flow is manufactured, and where the single most under-tested number in project finance lives: the **contracted volume floor**, which for Kestrel must be **95.8892 %** of capacity to hold its own covenant, not the 90 % the commercial team negotiated (KA 12.2). The guarantees, direct agreements and security package convert promises into recoveries, and their worth is the guarantor's credit quality multiplied by the cap, not the cap (KA 12.3). And claims and change are where the allocation is tested in anger, on an arithmetic most organisations never do: on Kestrel's disputed claim, fighting costs a present value of **1,347,115** against a disputed sum of **1,520,000**, so the rational settlement ceiling is **84.68 %** of the claim and the disputed sum would have to exceed **6,901,234** before litigation beat settlement (KA 12.4).

Learning objectives. After this domain a candidate can: read an EPC, O&M, offtake, concession and guarantee package as a set of computable financial limits rather than a set of promises; calibrate a delay liquidated-damages rate against daily carrying cost plus forgone `CFADS`, and identify the day on which the cap binds; compute the uncovered residue of a delay beyond the cap and express it as a share of contract value, equity and project `NPV`; build a cap stack and show where sub-caps exhaust an aggregate cap; calibrate performance liquidated damages on a value basis and distinguish that figure from the coverage-restoring buy-down and from the bare covenant-restoring prepayment, quantifying the gap between all three; explain why the clause governing the application of damages proceeds can leave equity short even when the amount is right; derive the minimum contracted volume that holds a stated `DSCR` covenant and explain why it exceeds the commercially negotiated floor; test a termination-compensation formula against the debt-outstanding profile and against unreturned equity; compute risk-adjusted cover from instrument face amounts and guarantor credit quality, and the equivalent face amount of a conditional guarantee against an unconditional bond; assess the commercial value of a claim including time impact and the cost of the dispute itself; compute a settlement ceiling, the saving from settling and the disputed sum at which litigation becomes rational; and govern AI use in contract review while never delegating a legal conclusion to it.

The master thread. Kestrel Water SPC continues. Capital cost **USD 60,000,000** funded **70/30** as **USD 42,000,000** of senior debt at **6.0 % over 12 years** — annual instalment **USD 5,009,635.23**, year-one interest **2,520,000**, year-one principal **2,489,635.23**, `AF(0.06, 12) = 8.383844` (Domain 3) — plus **USD 18,000,000** of equity. Operating life 25 years; documented `CFADS` **6,384,000** (6,984,000 before working-capital movements); `EBITDA` **7,500,000**, `EBIT` **5,100,000**; appraisal at 8 % gives `NPV` **+16,179,360**, `IRR` 12.19 %, `MIRR` 9.73 %, `PI` 1.270; year-one `DSCR` **1.2743 = LLCR**, `PLCR` **1.9431**, with a **1.20×** covenant (cash trigger **6,011,562**, headroom **372,438**) and a **1.15×** lock-up (cash trigger **5,761,081**). Its contracts are a fixed-price, date-certain EPC wrap at **48,000,000** (Domains 5, 6 and 8), delay damages of **20,000 per day** capped at **10 %** of that price, and a 25-year water offtake inside a 27-year concession. This domain adds the rest of the stack — the performance-damages regime, the aggregate cap, the security package and the claims machinery — and computes what each is worth.

KNOWLEDGE AREA 12.1

EPC and O&M

Topics: 12.1.1 the construction contract as a set of financial limits · 12.1.2 delay damages, caps and the uncovered residue · 12.1.3 performance damages, buy-down and the three make-good numbers · 12.1.4 O&M contracts and the liability asymmetry.

12.1.1 The construction contract as a set of financial limits

Definition. An **EPC contract** is a single agreement under which one contractor engineers, procures and constructs the whole works for a fixed lump-sum price, by a date certain, to defined performance — the **wrap** whose bankability role Domain 5 (KA 5.4.1) established and whose premium Domain 11

(KA 11.1.3) priced. Its obligations are familiar. Its **financial limits** are what a financier reads, and they are these six numbers:

Limit	What it is	Kestrel
Contract price	The fixed sum, with defined change mechanics	48,000,000
Delay damages rate	Payable per day of delay beyond the date certain	20,000 / day
Delay damages cap	Maximum recoverable for delay, whatever its length	10 % = 4,800,000
Performance damages cap	Maximum recoverable for output or efficiency shortfall	10 % = 4,800,000
Aggregate liability cap	Maximum total contractual liability, all heads	20 % = 9,600,000
Performance security	Instruments callable against the above (KA 12.3)	bond 10 % + parent guarantee

Three properties of that table decide whether the financing works. First, **the rate and the cap are independent decisions**: a well-calibrated rate under a low cap protects only the early days of a delay. Second, **the caps interact**, and the aggregate cap is the real limit — Kestrel's two 10 % sub-caps sum to exactly the 20 % aggregate, so any third head of claim (defect rectification after the sub-caps are exhausted, a third-party indemnity that is not carved out) has no room left in the stack. Third, **the exclusion of consequential or indirect loss** removes from recovery precisely the losses a project company actually suffers — forgone revenue, financing cost, loss of profit — which is why liquidated damages exist at all: they are the agreed, recoverable substitute for a loss the contract has excluded.

Whether a particular damages provision is enforceable as agreed compensation, whether it is vulnerable to challenge as a penalty, how "consequential loss" is construed, and whether an aggregate cap survives particular breaches are all **jurisdiction-specific questions of law and drafting**. This domain computes the commercial arithmetic those provisions are meant to deliver; it does not opine on their effect. Refer each such question to qualified counsel in the governing jurisdiction, and require counsel's confirmation before the model relies on a recovery.

12.1.2 Delay damages, caps and the uncovered residue

Definition. Delay liquidated damages are a pre-agreed sum per unit of time by which the contractor misses the date certain. Their commercial purpose is to reimburse the project company for the cost of lateness, and their correct calibration basis is therefore the **daily economic cost of delay**: the carrying cost of debt already drawn plus the `CFADS` the project would have earned. Domain 5 (KA 5.4.2) computed Kestrel's: **7,000 per day** of interest on the fully drawn 42,000,000 at 6.0 % on a 30/360 basis, plus **17,733.33 per day** of forgone `CFADS`, giving **24,733.33 per day**. A rate of 20,000 recovers **80.86 %** of that; a rate calibrated on forgone revenue alone would recover **71.70 %**, omitting the most certain component of the loss, because interest accrues whether or not the plant would have run well.

What Domain 5 did not do — and what a financier must do before signature — is compute the whole cap stack against a combined stress, because delay and underperformance are correlated (a contractor in schedule trouble commissions in a hurry) and the caps are shared.

WORKED EXAMPLE

Worked example 12.1.2 — Kestrel's cap stack against a 300-day delay and a 5 % shortfall.

1. **Setup.** The contract stack of 12.1.1. The stress: commissioning is **300 days** late and the plant completes at **95 %** of guaranteed output. Daily economic cost of delay **24,733.33** (Domain 5, KA 5.4.2, 30/360). The value of the output shortfall is computed in 12.1.3 below as **4,835,673.53**. Compute the total exposure, the recovery the caps permit, and the residue.
2. **Formula.** Delay exposure = daily economic cost × days. Delay recovery = min(rate × days, delay cap). Performance recovery = min(value of shortfall, performance cap). Total recovery is further limited by the aggregate cap. Residue = exposure − recovery.
3. **Substitution.** Delay: `24,733.33 × 300`; recovery `min(20,000 × 300, 4,800,000)`. Performance: `min(4,835,673.53, 4,800,000)`. Aggregate: `min(4,800,000 + 4,800,000, 9,600,000)`.
4. **Result.**

Head	Exposure	Contractual recovery	Uncovered
Delay, 300 days	7,420,000.00	4,800,000.00 (capped at day 240)	2,620,000.00
Output shortfall, 5 %	4,835,673.53	4,800,000.00 (capped)	35,673.53
Total	**12,255,673.53**	**9,600,000.00** (= the aggregate cap, exactly)	**2,655,673.53**

The uncovered residue is **5.53 %** of the EPC price, **14.75 %** of the equity contribution and **16.41 %** of Domain 4's entire project `NPV` of 16,179,360. 5. **Interpretation.** Four things in that table are decision-grade and none of them requires a model. **The cap, not the rate, is where the structure breaks.** Damages of 20,000 per day recover 80.86 % of the daily cost for 240 days and **nothing at all** thereafter; the 300-day case leaves 2,620,000 uncovered and Domain 5's 360-day case leaves 4,104,000. The negotiating priority is therefore not the rate — sponsors habitually spend their leverage there — but the **cap-binding day**, which is the cap divided by the rate and which the sponsor should compare with the credible worst-case delay from the schedule risk analysis. A cap that binds before the P80 delay is a cap that does not cover the risk it was bought for. **The aggregate cap is not an additional protection.** Two 10 % sub-caps summing to a 20 % aggregate means the aggregate binds only when a third head arises, and then it binds at zero: after a full delay and a full performance claim, a defect the contractor refuses to rectify is uncompensated. Structures that intend the aggregate to be meaningful set it above the sum of the sub-caps, or ring-fence specific heads outside it. **The residue is an equity number, and it belongs in the equity case.** 2,655,674 is 14.75 % of the 18,000,000 cheque; a sponsor that has not stated that figure to its investment committee has not described the transaction. **And the contractor's incentive changes at the cap.** Once cumulative damages reach 4,800,000 the contractor's marginal cost of a further day of delay is zero, which converts a financial deterrent into a pure commercial negotiation exactly when the project can least afford one — the reason sophisticated structures pair a cap with a **termination-for-delay right** at a defined long-stop date, so that something other than money still bites.

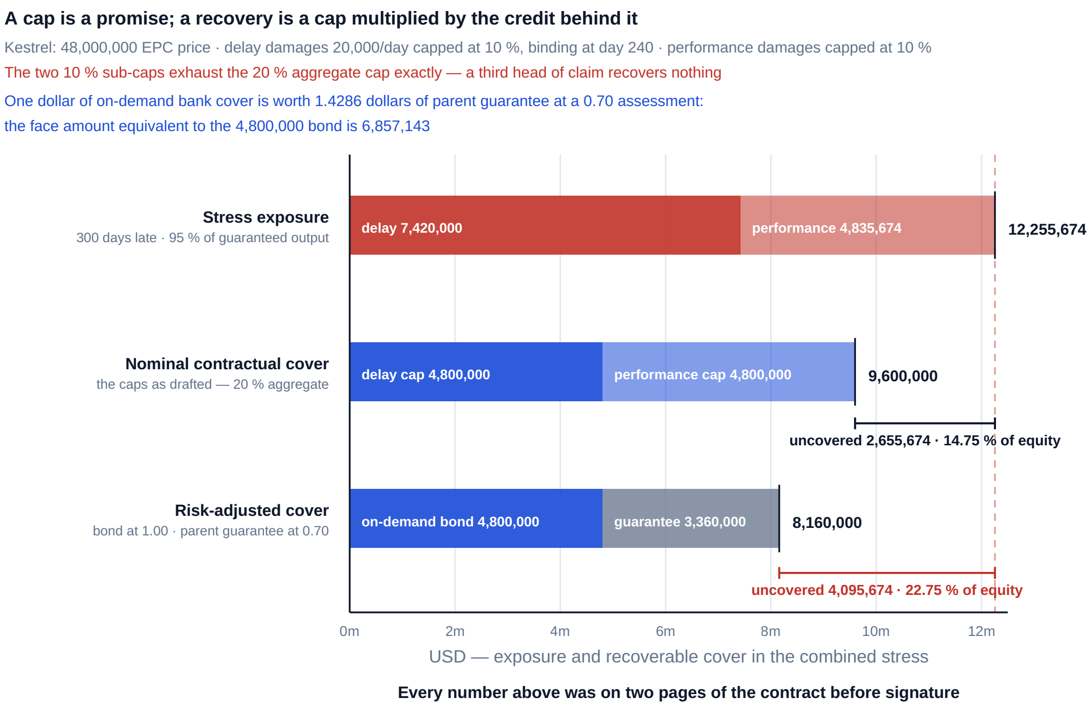

Fig 12.1.1 — The cap stack against the loss it was drafted for

12.1.3 Performance damages, buy-down and the three make-good numbers

Definition. Performance liquidated damages compensate for a plant that completes but underperforms — output, efficiency, availability or consumption below guarantee. Where the shortfall is permanent they are commonly structured as a **buy-down**: a lump sum, usually applied to prepay debt, calibrated so that the financing survives a smaller plant. The question nobody asks early enough is calibrated to restore what — because there are three defensible answers and they differ by an order of magnitude.

Kestrel's cash flow is linear in output, which makes the arithmetic clean. With revenue of 12,000,000 at full output, cash operating costs of 4,500,000 of which **85 % is fixed** (3,825,000), depreciation 2,400,000, year-one interest 2,520,000, cash tax at 20 % and a working-capital movement of 600,000 (Domain 5, KA 5.4.3 established this cost structure and the **1.510× operating leverage** it produces), `CFADS` as a function of output share `x` is:

```
CFADS(x) = 9,060,000 x − 2,676,000
```

so **each percentage point of output is worth 90,600 of annual** `CFADS`. At `x` = 1 the line reproduces the documented 6,384,000; at 0.97 it reproduces Domain 5's 6,112,200.

WORKED EXAMPLE

Worked example 12.1.3 — what should a 5 % output shortfall be worth?

1. **Setup.** Kestrel completes at **95 %** of guaranteed output, permanently. `EBITDA` falls to **6,933,750** and `CFADS` to **5,931,000** — an annual shortfall of **453,000**. Remaining operating life 25 years; the sponsors' appraisal rate is Domain 4's **8 %**; the performance damages cap is **4,800,000**. Compute the value of the shortfall, the damages rate it implies, the shortfall the cap can cover, and the two rival make-good figures.
2. **Formula.** Value of the shortfall = annual `CFADS` shortfall × `AF(0.08, 25)`. Damages rate per point = 90,600 × `AF(0.08, 25)`. Coverage-restoring buy-down = debt × (`CFADS` shortfall ÷ base `CFADS`), first-order as in Domain 5, KA 5.4.3. Covenant-restoring prepayment: reduce debt until `CFADS`(new) ÷ (debt ÷ `AF(0.06, 12)`) = 1.20.
3. **Substitution.** `AF(0.08, 25) = 10.674776`; `453,000 × 10.674776`; `90,600 × 10.674776`; `42,000,000 × 453,000/6,384,000`; `5,931,000/1.20 = 4,942,500`, then `× 8.383844`.
4. **Result.**

Basis	What it restores	Amount
Bare covenant	`DSCR` back to the 1.20 covenant, nothing more	**562,851.03** (debt to 41,437,148.97)
Sized coverage	`DSCR` back to the 1.2743 the debt was sized on	**2,980,263.16** (debt to 39,019,736.84)
Value	The present value of 25 years of lost `CFADS`	**4,835,673.53**

The value basis implies a damages rate of **967,134.71 per percentage point** of shortfall, and the **4,800,000 cap therefore covers a shortfall of 4.9631 %** and not one basis point more. For reference, the covenant itself fails at a shortfall of **4.1108 %** (the point at which 90,600 per point exhausts the 372,438 of headroom): at 5 % the `DSCR` is **1.1839**, breaching the 1.20 covenant by **80,562** of cash. 5. **Interpretation.** The spread between 562,851 and 4,835,674 is **8.591×**, and every party at the table has a favourite end of it. **Lenders are satisfied by the smallest number**, because their interest ends at the covenant — which is why a facility agreement that merely requires "performance damages sufficient to restore the covenant" protects the banks and abandons the equity. **Sponsors need the largest**, because their loss is 25 years of cash, not 12 years of coverage. **The sized-coverage basis, at 2,980,263, is the negotiated middle** and the one most commonly drafted, which means the standard market outcome quietly transfers 1,855,410 of value loss from the contractor to the equity. A sponsor that understands this asks for the value basis and settles for the sized basis knowing what it conceded; a sponsor that does not, discovers the gap in operations.

The application clause matters as much as the amount, and this is the subtler and more expensive point. Suppose the full 4,800,000 is received and, as facility agreements usually require, applied to **mandatory prepayment**. Debt falls to 37,200,000, the instalment falls to **4,437,105.46** — a relief of **572,529.77** per year — and the `DSCR` rises to **1.3367**, better than the position the debt was sized on. Distributions actually rise, from 1,374,365 to **1,493,895**. And yet equity is still worse off: the relief runs for the loan's remaining 12 years and is worth `572,529.77 × AF(0.08, 12) = 7.536078` → **4,314,628.99**, while the loss runs for 25 years and is worth 4,835,673.53. The residual gap is **521,044.53**. Paying a 25-year loss with a 12-year debt prepayment under-compensates by construction, however generous the headline. The remedies are to direct part of the proceeds to a distribution account or a maintenance reserve rather than wholly to prepayment, or to size the damages on the value basis in the first place — a drafting choice worth half a million dollars that costs nothing to make at signature and cannot be made afterwards.

The caution: this arithmetic assumes the shortfall is permanent, the cost structure holds and the 8 % rate is the right measure of the sponsors' loss. Where the shortfall can be engineered out, the honest comparison is the rectification cost against the damages; where the offtake pays for availability rather than output, the loss runs through the deduction regime instead (KA 12.2) and this calculation must be rebuilt on that basis.

12.1.4 O&M contracts and the liability asymmetry

Definition. An **operation and maintenance agreement** engages an operator to run the asset to defined standards for a fee, usually with a fixed element, a variable element and a performance regime of bonuses and deductions. Structurally it is the mirror of the EPC contract, and it carries one systematic defect a financier must look for: **the liability asymmetry**.

The asymmetry is arithmetic. An O&M contractor's annual fee on a project like Kestrel might be 1,200,000 — roughly 27 % of the 4,500,000 cash operating cost, the rest being power, chemicals and insurance. Its liability cap is customarily expressed as a share of that fee: one year's fee, or 50 % of it, is common. But the loss an operator can cause is a revenue loss, measured on the same daily basis as a construction delay. A **30-day** unplanned outage costs Kestrel `24,733.33 × 30 =` **742,000**, which already exceeds a cap set at 50 % of the fee (600,000) by **142,000**; a **60-day** outage costs **1,484,000**, leaving **884,000** uncovered against that cap and **284,000** uncovered even against a full-year-fee cap of 1,200,000. The consequence is that **operating risk is only nominally transferred**: the O&M agreement buys competence, mobilisation and a set of key performance indicators, not indemnity. Lenders know this, which is why they price operating risk in the coverage ratio and the maintenance reserve (Domain 10, KA 10.3.1) rather than relying on the O&M contract, and why an availability guarantee from a thinly capitalised operator adds less bankability than sponsors expect.

Three structural responses are worth naming. A **parent guarantee or performance bond** from the operator's group raises the recoverable amount to something comparable with the loss (KA 12.3). A **deduction regime that bites on the fee before the cap** — availability-linked fee at risk — gives the operator a running incentive rather than a terminal liability. And **a step-in and replacement right**, with a pre-qualified substitute operator and a transition plan, is often worth more than any cap: the value at stake is continuity of revenue, and the fastest route to continuity is a new operator, not a claim against the old one. Where the O&M contractor is an affiliate of the sponsor — very common — the conflict must be disclosed and the agreement tested on arm's-length terms, because a lender is being asked to accept a related-party contract as a bankability item (Domain 1, KA 1.3.2).

AI in this KA

Where it earns its place. Extracting the six financial limits of 12.1.1 from a 400-page contract and its schedules, across a portfolio, is exactly the work machines should do: rate, cap, cap-binding day, aggregate cap, exclusions, security instruments and their expiry dates, assembled into a structured table with clause references. So is the consistency check between that table and the financial model — an assistant that reads both and flags "the model recovers 6,000,000 of delay damages; the contract caps them at 4,800,000" has earned its licence fee in one line. And so is generating the stress grid: exposure and residue across a matrix of delay days and output shortfalls, which is 400 arithmetic cells nobody computes by hand.

Where it must not go. It must not conclude whether a damages provision is enforceable, whether a cap survives a particular breach, what "consequential loss" excludes in a given jurisdiction, or whether a set of facts triggers a termination right. Those are legal conclusions, they are jurisdiction-specific, and they belong to qualified counsel; a model's confident answer on any of them is a liability dressed as a summary. Nor should it draft the damages number: the calibration of 12.1.3 rests on a judgment about whose loss is being restored, which is a commercial decision with a named owner.

Verification, concretely. Sample the extracted limits back to the clause on at least five items per contract, including every cap, and record the human verifier and date. Recompute one cap-binding day and one residue by hand. Confirm that the exclusions list in the extraction matches the contract's own list word for word rather than a normalised paraphrase — the normalising is where the error enters. **AI proposes; the professional verifies, decides and remains accountable.**

KEY TERMS — KA 12.1

Term	Meaning
EPC wrap	Single-contractor fixed-price, date-certain, turnkey responsibility for the whole works.
Delay liquidated damages	Pre-agreed sum per unit of time of late completion; calibrated on carrying cost plus forgone `CFADS`.
Cap-binding day	Delay cap ÷ daily damages rate; the day after which further delay is uncompensated.
Cap stack	The interacting set of sub-caps and the aggregate cap; the aggregate is the real limit.
Performance liquidated damages / buy-down	Payment for permanent underperformance, usually applied to prepay debt.
Uncovered residue	Exposure less contractual recovery; an equity number, and a disclosure item.
Liability asymmetry (O&M)	Operator liability capped on fee while the loss it can cause is revenue-scaled.

SAMPLE MCQS — KA 12.1

MCQ 12.1-A `[12.1.2 · Application]` Delay damages are 20,000 per day, capped at 10 % of a 48,000,000 EPC price, against a daily economic cost of delay of 24,733.33. For a 300-day delay the amount borne by the project company is: - A. USD 1,420,000 - B. USD 2,620,000 - C. USD 1,484,000 - D. nil — the damages regime covers it

Rationale: Economic cost `24,733.33 × 300 = 7,420,000`; recovery is capped at 4,800,000 (the cap binds at day 240), so 2,620,000 is uncovered. A applies the 4,733.33 per day uncovered rate to 300 days and ignores that the cap stops recovery entirely after day 240. C is the 60-day interface figure of 12.2.4. D assumes a cap covers any delay.

MCQ 12.1-B `[12.1.2 · Analysis]` An EPC contract has a 10 % delay-damages sub-cap, a 10 % performance sub-cap and a 20 % aggregate liability cap. The correct reading is: - A. total recoverable liability is 40 % of the contract price - B. the aggregate cap adds protection above the sub-caps - C. the two sub-caps can exhaust the aggregate cap exactly, leaving no room for any third head of claim - D. the aggregate cap applies only to indemnities

Rationale: 10 % + 10 % = 20 %, so once both sub-caps are drawn the aggregate is fully consumed and a later defect or indemnity claim recovers nothing (12.1.1, 12.1.2). A double-counts; B is the common misreading the arithmetic disproves; D asserts a carve-out the structure does not contain.

MCQ 12.1-C `[12.1.3 · Application]` Kestrel's `CFADS` falls by 453,000 per year for 25 years on a 5 % output shortfall; the appraisal rate is 8 % (`AF(0.08, 25) = 10.674776`); debt is 42,000,000 against base `CFADS` of 6,384,000 and an instalment of 5,009,635.23. Which figure is the **value-basis** performance damages amount? - A. USD 562,851 — the prepayment that restores the 1.20 covenant - B. USD

2,980,263 — the buy-down that restores the sized 1.2743 DSCR - C. USD 4,835,674 - D. USD 453,000 — one year of lost CFADS

Rationale: `453,000 × 10.674776 = 4,835,673.53`. A restores only the covenant, B only the sized coverage — both are lender-facing measures, not the sponsors' loss; D omits discounting and the remaining 24 years.

MCQ 12.1-D `[12.1.3 · Analysis]` A 4,800,000 performance damages receipt is applied wholly to mandatory prepayment, cutting debt to 37,200,000 and the instalment to 4,437,105, and lifting the DSCR to 1.3367. The soundest conclusion is: - A. equity has been over-compensated, since the DSCR now exceeds the sized 1.2743 - B. equity remains short by about 521,000, because a 25-year loss has been compensated with 12 years of debt-service relief - C. equity is exactly compensated, since the amount equals the cap - D. the prepayment is irrelevant to equity

Rationale: Relief of 572,529.77 × `AF(0.08, 12) = 7.536078` is 4,314,629 against a loss of 4,835,674 — a residual gap of 521,045 (12.1.3). A confuses a coverage ratio with value; C confuses the cap with the loss; D ignores that prepayment raises distributions.

SELF-CHECK — KA 12.1

1. Which negotiating variable matters more, the damages rate or the cap, and why? — The cap: the rate governs recovery only up to the cap-binding day (day 240 for Kestrel), after which further delay is wholly uncompensated.
2. Name the three make-good bases for a permanent output shortfall and Kestrel's figures. — Bare covenant 562,851; sized coverage 2,980,263; value 4,835,674 — a spread of 8.591×.
3. Why does an O&M liability cap rarely transfer operating risk? — It is scaled to the fee while the loss it can cause is scaled to revenue; the practical protections are guarantees, fee at risk and a step-in replacement right.

KNOWLEDGE AREA 12.2

Offtake, concession, supply and interface agreements

Topics: 12.2.1 the four load-bearing terms of a revenue contract · 12.2.2 the contracted volume floor is a financing parameter · 12.2.3 concession termination compensation · 12.2.4 supply, interface and the hole between packages.

12.2.1 The four load-bearing terms of a revenue contract

Definition. The **offtake agreement** (a power purchase agreement, water purchase agreement, capacity or availability agreement, or a concession's tariff regime) is the contract that manufactures the project's cash flow. Domain 7 built its economics; this domain reads it as a document, and four of its terms carry the entire financing:

The volume or availability commitment. Whether the offtaker must pay for output it does not take (**take-or-pay**) or only for output delivered (**take-and-pay**), and at what level. This is the term of 12.2.2, and it is systematically under-tested.

The price and its indexation. The tariff, its escalation formula and the pass-through of input costs. Domain 11 (KA 11.2.3) demonstrated the cost of an **indexation mismatch** — a revenue index that does not track the cost driver it is meant to fund — and that analysis is not repeated here; the contractual point is that the mismatch is created by the words in the escalation schedule, so the schedule must be read against the operating cost build-up line by line, not accepted as "CPI-linked".

The deduction and abatement regime. What reduces payment: unavailability, quality failures, delivery-point failures, metering disputes. A deduction regime is a liability cap in reverse — uncapped, running annually, and biting on revenue before any of the project's own protections engage. The single most useful test is to compute the **maximum annual deduction** the regime permits and compare it with the covenant headroom: for Kestrel, a deduction exceeding 372,438 in any year breaches the 1.20 covenant regardless of how well the plant ran in every other respect.

Termination and compensation. What is paid, by whom, on whose default (12.2.3). For a lender this is the most important clause in the document, because it is the only one that speaks to the recovery of principal.

12.2.2 The contracted volume floor is a financing parameter

The claim. Commercial teams negotiate the take-or-pay level as a commercial matter — how much volume will the offtaker commit to, how much flexibility does it need. It is not a commercial matter. **The minimum contracted volume is determined by the covenant**, and it is computable from the cash-flow line and the debt service before anyone sits down.

WORKED EXAMPLE

Worked example 12.2.2 — what floor does Kestrel's covenant actually require?

1. **Setup.** `CFADS(x) = 9,060,000x − 2,676,000` from 12.1.3, where `x` is the share of guaranteed output taken and paid for. Debt service **5,009,635.23**; covenant **1.20×**; lock-up **1.15×**. The commercial team has agreed a take-or-pay floor of **90 %** of capacity. Find the `DSCR` at that floor, the floor the covenant requires, the floor the lock-up requires, and the floors at which coverage and cash reach unity and zero.
2. **Formula.** `DSCR(x) = CFADS(x)` ÷ debt service. Inverting for a target ratio `k`: `x = (k × 5,009,635.23 + 2,676,000) ÷ 9,060,000`.
3. **Substitution.** At `x` = 0.90: `(8,154,000 − 2,676,000)/5,009,635.23`. For `k` = 1.20: `(6,011,562.28 + 2,676,000)/9,060,000`. Similarly for 1.15, 1.00 and `CFADS` = 0.
4. **Result.**

Contracted floor	CFADS	DSCR	Status
100.0000 %	6,384,000	**1.2743**	The sized case
95.8892 %	6,011,562	**1.2000**	Covenant exactly met
95.0000 %	5,931,000	1.1839	**Covenant breached**
93.1245 %	5,761,080	**1.1500**	Lock-up trigger
90.0000 %	5,478,000	**1.0935**	Breach and lock-up; debt still paid
84.8304 %	5,009,635	**1.0000**	Cash exactly equals debt service
29.5364 %	0	—	CFADS exhausted
5. **Interpretation. The negotiated 90 % floor does not support the project's own covenant**,			
and the gap is not marginal: 90 % delivers 1.0935 against a 1.20 requirement, a breach and an			
automatic distribution lock-up from the first test date, on a plant performing exactly to			
specification and an offtaker performing exactly to contract. The floor the financing needs is			
95.8892 % — which, expressed the way it must be expressed to a commercial negotiator, means			
the offtaker's flexibility is worth **4.11 percentage points of capacity, not ten**.			

The reason the intuition fails is **operating leverage compounded by fixed debt service**. Because 85 % of cash operating cost is fixed, a 10 % volume reduction cuts CFADS by 14.19 % (906,000 on 6,384,000), and because debt service does not move at all, the whole reduction lands on the ratio. The general form is worth carrying: DSCR is linear in contracted volume with a slope of `90,600 ÷ 5,009,635.23 = 0.0181` of ratio per percentage point, so **each point of contracted volume conceded costs 0.0181 of coverage** — and the 372,438 of headroom buys 4.11 points and no more.

Three professional consequences. **The floor is a financing deliverable, not a commercial preference**, and it should be issued to the commercial team as a constraint with its derivation attached, before negotiation, in the same way a lender issues a target DSCR. **A lower floor has a price, and the price is computable**: if the offtaker will only commit to 90 %, the project needs either more equity (reducing debt service until 90 % clears 1.20 — debt of `5,478,000/1.20 × 8.383844 =` **38,272,248**, so **3,727,752** of additional equity), or a compensating floor price, or a volume-shortfall payment that is economically take-or-pay under another name. **And the two thresholds must both be mapped onto volume**: the covenant fails at 95.8892 % and the lock-up engages **2.76 points of volume lower**, at 93.1245 %, so between those two floors the project is in breach — with all the consequences of Domain 10, KA 10.4 — while distributions are not yet automatically trapped. A sponsor who models only the lock-up will therefore believe it has more contractual room than it has, which is the mirror image of the error Domain 10's Case study B punished.

The caution: this arithmetic is year-one, on a level CFADS assumption, and it takes the cost structure as fixed. A ramping or seasonal offtake requires the same test **period by period** against the sculpted debt-service profile (Domain 10, KA 10.1.3), and the binding period — almost always the first full year or the year of a major maintenance outage — is the one the floor must clear.

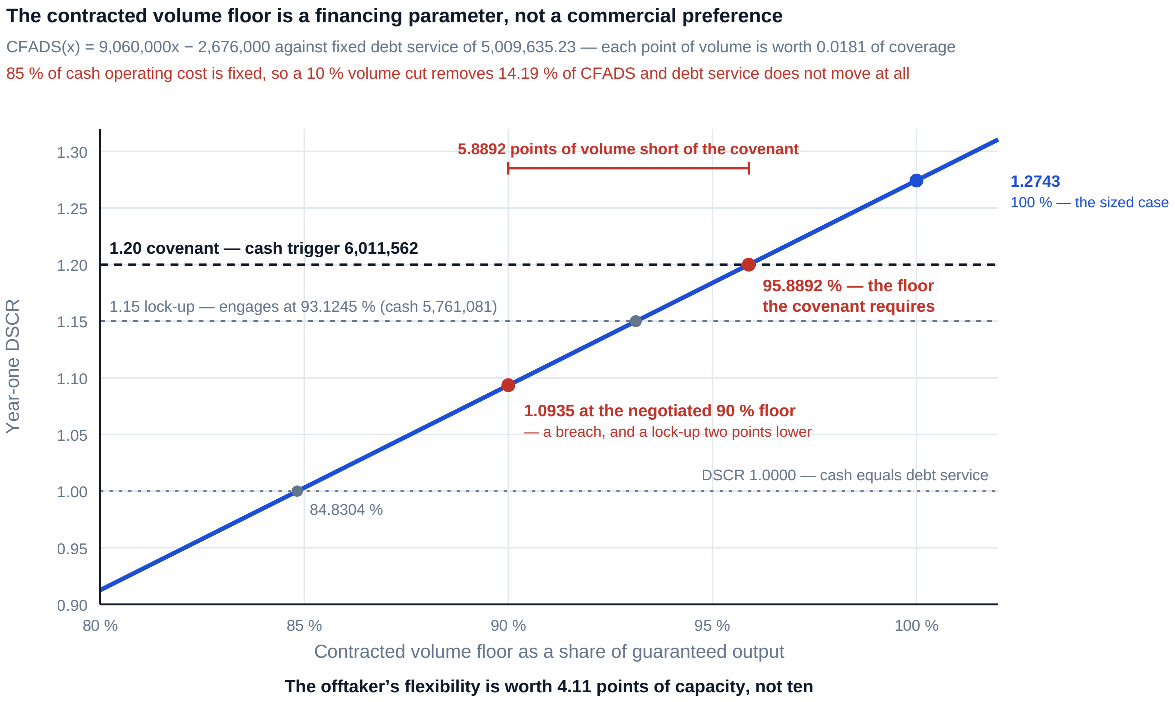

Fig 12.2.1 — The contracted volume floor a covenant requires

12.2.3 Concession termination compensation

Definition. A **concession agreement** grants the project company the right to build, operate and charge for an asset for a defined term, subject to obligations and reversion. Its **compensation-on-termination** provisions state what the grantor pays if the arrangement ends early, and they are conventionally graded by cause: grantor default or voluntary termination (most generous, typically debt plus equity plus a return); force majeure or extended unavailability of the asset (intermediate, often debt-focused); and project-company default (least generous, sometimes a market-value or debt-haircut formula).

The financier's test is arithmetic and takes ten minutes. **Compare the compensation formula, at each year of the term, with the debt outstanding on that date.** Kestrel's 12-year amortisation gives:

End of year	Interest	Principal	Debt outstanding
1	2,520,000.00	2,489,635.23	**39,510,364.77**
3	2,212,281.09	2,797,354.14	**34,073,997.28**
5	1,866,528.11	3,143,107.12	**27,965,694.77**
8	1,266,144.36	3,743,490.87	**17,358,915.21**
12	283,564.26	4,726,070.97	**0**

Two consequences follow. First, **a formula that pays a percentage of debt outstanding fails worst early**: an 85 %-of-debt formula on project-company default leaves a gap of **5,926,555** at the end of year 1 and **4,194,854** at the end of year 5 — and the early years are precisely when default risk is highest, so the haircut bites exactly when it is most likely to be needed. Lenders respond by requiring compensation at least equal to senior debt outstanding plus breakage in the grantor-default and force-majeure cases, and by treating the project-company default case as a real exposure to be sized rather than a theoretical one.

Second, **a formula that covers debt in full makes the lenders whole and leaves equity at zero**, and this is the single most important number in a concession for an equity investor. If Kestrel's concession terminated for force majeure at the end of year 5 on a debt-outstanding formula, the lenders recover 27,965,695 and the sponsors recover nothing against 18,000,000 contributed less the **6,871,824** of nominal distributions received in five years (1,374,365 per year) — an unreturned **11,128,176**. That figure, not the elegance of the clause, is what an investment committee needs, and it is why sponsors negotiate hardest on the definition of equity base, whether a return is payable on it, and the treatment of subordinated debt.

Whether a compensation formula is enforceable as drafted, how it interacts with statutory compensation or public-procurement rules, whether the grantor's payment obligation ranks ahead of other public liabilities, and what happens on the grantor's own insolvency are **matters of local law and public policy that vary fundamentally between jurisdictions**. They must be confirmed by qualified local counsel before the model treats a termination sum as a recovery.

12.2.4 Supply, interface and the hole between packages

Definition. Supply agreements secure inputs — fuel, feedstock, chemicals, grid connection, raw water — on terms whose volume, price, indexation and interruption provisions mirror the offtake in reverse. **Interface agreements** allocate responsibility where two or more contractors, utilities or authorities must physically or programmatically connect. The financier's interest in both is the same: **an input failure or an interface failure produces a revenue loss that the offtake's deduction regime charges to the project company, and the question is who reimburses it.**

Where the works are delivered under a single EPC wrap, interface risk sits inside the wrap; where they are split, it sits nowhere. That is not rhetoric but arithmetic. Had Kestrel's 48,000,000 been split into an 8,000,000 marine intake package and a 40,000,000 process plant package, a 60-day delay caused by the intake being handed over late to a specification the process contractor disputed would cost the project company `24,733.33 × 60 =` **1,484,000 — 3.092 %** of the combined contract price — with **no recovery against either contractor**, because each would demonstrate that it performed its own scope and each would claim an extension of time and prolongation cost for the other's delay. The wrap premium Domain 11 (KA 11.1.3) priced is, in part, the price of not owning that hole.

Splitting packages is nevertheless often right — it can be cheaper, it can be the only route to a specialist scope, and Domain 11 established that buying a transfer is not automatically value creating. What is never right is splitting them **without an interface regime**. The minimum package: a single **interface matrix** identifying every physical, programme, data and documentation handover with a named owner and a date; **back-to-back completion obligations** so that one contractor's milestone is defined by the other's readiness; **cross-liability or knock-for-knock provisions** stating who pays whom when a handover slips; an **interface manager** with authority inside the project company; and **a single float owner**, because unallocated float is the commonest cause of concurrent claims (Domain 8, KA 8.4). Case study B below prices exactly this omission on a digital-infrastructure project at **5,302,717**.

AI in this KA

Where it earns its place. Reading a long tariff and deduction schedule and reconstructing the payment formula as an auditable calculation is high-value machine work, and so is the reverse check: does the model's revenue line reproduce the contract's formula on a sample of historical or test-case months? Extracting every termination and compensation provision from a concession and tabulating it by cause, with the payment basis and the clause reference, turns a week of associate time into an afternoon of review. And the volume-floor computation of 12.2.2 should be automated across every revenue contract in a portfolio, because it is mechanical and nobody does it.

Where it must not go. It must not decide whether a deduction is contractually due in a disputed month, whether a force-majeure definition captures a particular event, or whether a termination notice is valid. Those are contract-interpretation questions with legal consequences and named counsel. Nor should it be trusted to normalise a bespoke tariff formula into a standard shape: the variations between formulae are where the money is, and a tidy summary is precisely the wrong output.

Verification, concretely. Re-derive the payment for two contract months by hand from the clause and compare with the extraction. Confirm that the extracted deduction regime's worst annual case has been computed and compared with covenant headroom. Sample every termination row back to its clause, and record counsel's confirmation that the compensation basis is enforceable as modelled before the recovery appears in a lender case.

KEY TERMS — KA 12.2

Term	Meaning
Take-or-pay / take-and-pay	Offtaker pays for contracted volume whether taken, or only for volume delivered.
Contracted volume floor	The minimum committed volume; a financing parameter set by the covenant, not a commercial preference.
Deduction / abatement regime	Contractual reductions in payment for unavailability or quality failure; an uncapped, annually recurring exposure.
Compensation on termination	What the grantor or offtaker pays on early termination, graded by cause.
Interface agreement	Allocation of responsibility at handovers between separately contracted packages.
Interface hole	Loss caused by a handover failure that neither contractor is liable for.

SAMPLE MCQS — KA 12.2

MCQ 12.2-A `[12.2.2 · Application]` `CFADS(x) = 9,060,000x − 2,676,000`; debt service is 5,009,635.23; the covenant is 1.20×. The minimum contracted volume that holds the covenant is: - A. 90.0000 % - B. 84.8304 % - C. 95.8892 % - D. 100.0000 %

Rationale: `(1.20 × 5,009,635.23 + 2,676,000)/9,060,000 = 0.958892`. A is the commercially negotiated floor, which delivers only 1.0935; B is the volume at which `DSCR` = 1.00, i.e. cash merely equals debt service; D is the sized case, which the covenant does not require.

MCQ 12.2-B `[12.2.2 · Analysis]` Why does a 10 % reduction in contracted volume cut `DSCR` by far more than 10 %? - A. because the covenant is tested on revenue, not `CFADS` - B. because 85 % of cash operating cost is fixed, so `CFADS` falls 14.19 %, and debt service does not fall at all - C. because interest rates rise with lower volume - D. because the tariff falls with volume

Rationale: Operating leverage compounded by a fixed denominator: 906,000 of `CFADS` lost on 6,384,000 is 14.19 %, and all of it lands on the ratio (12.2.2, and the 1.510× leverage of Domain 5, KA 5.4.3). A misstates the test basis; C and D invent mechanisms.

MCQ 12.2-C `[12.2.3 · Analysis]` A concession pays 85 % of senior debt outstanding on project-company default. Kestrel's debt outstanding is 39,510,365 at the end of year 1 and 27,965,695 at the end of year 5. The most important observation is: - A. the formula is adequate, since 85 % is a high recovery - B. the shortfall is largest early — 5,926,555 at the end of year 1 — precisely when default risk is highest - C. the shortfall is largest late, as the debt amortises - D. the formula protects equity but not lenders

Rationale: 15 % of a declining balance is largest at the start (12.2.3). C inverts the profile; D reverses the ranking — a debt-based formula protects lenders first and leaves equity at zero.

MCQ 12.2-D `[12.2.4 · Application]` Two packages are let without an interface regime; a 60-day handover dispute delays completion. The daily economic cost of delay is 24,733.33. The project company's most likely position is: - A. it recovers 1,484,000 from the contractor whose scope was late - B. it bears 1,484,000, recovers nothing, and faces prolongation claims from both contractors - C. it recovers from its insurers under a delay-in-start-up policy in all cases - D. it suffers no loss, because the offtake date moves with the works

Rationale: Each contractor shows performance of its own scope and claims for the other's delay (12.2.4). A assumes an allocation the documents do not make; C assumes cover that depends entirely on policy wording and a triggering insured peril; D assumes an offtake flexibility that date-certain revenue contracts do not grant.

SELF-CHECK — KA 12.2

1. What sets the minimum take-or-pay level? — The covenant: `x = (k × debt service + 2,676,000) ÷ 9,060,000`; for Kestrel's 1.20× that is 95.8892 %, against a negotiated 90 %.
2. What does one point of contracted volume cost in coverage? — 0.0181 of `DSCR` (90,600 ÷ 5,009,635.23), so 372,438 of headroom buys 4.11 points.
3. What does a debt-outstanding termination formula give equity? — Nothing: at the end of year 5 the lenders take 27,965,695 and the sponsors' unreturned 11,128,176 is lost.

KNOWLEDGE AREA 12.3

Guarantees, direct agreements and the security package

Topics: 12.3.1 the instrument family · 12.3.2 guarantee sufficiency: cap times credit quality · 12.3.3 direct agreements and step-in · 12.3.4 what the security package is actually worth.

12.3.1 The instrument family

Definition. The **security package** is the set of instruments and rights that convert contractual promises into recoveries. It has two halves that are routinely confused. The **credit-support instruments** stand behind counterparties' obligations to the project company; the **lenders' security** stands behind the project company's obligations to the lenders.

Instrument	Who provides it	What it does	Commercial character
On-demand bond / standby letter of credit	A bank, on the counterparty's account	Pays on a compliant demand, without proof of breach	Bank credit; near-certain payment; costs a fee
Conditional (surety) bond	A surety or insurer	Pays on proof of default and loss	Slower, defensible, cheaper
Parent company guarantee	The counterparty's group parent	Extends the obligation to a bigger balance sheet	Worth the parent's credit, no more
Retention / retention bond	Withheld from payments, or bonded	Funds defect rectification	Self-funding; releases on milestones
Sponsor support: equity commitment, cost overrun undertaking	The sponsors	Funds the SPV's own shortfalls	Domain 5, KA 5.2.3
Lenders' security: share pledge, asset security, assignment of contracts and insurances, accounts control	The SPV and its shareholders	Enables enforcement and step-in	Domain 13 perfects it

The instruments differ along three axes a financier must price separately: **how much** (face amount), **how certain** (the obligor's credit and the conditions on payment) and **how long** (expiry against the exposure it covers). The third is the most frequently missed: a performance bond that expires at provisional completion does not cover the defects-liability period, and a guarantee that expires on a fixed date rather than on discharge of the underlying obligation is a gap in the stack with a date on it.

The legal effectiveness of each instrument — the conditions for a compliant demand, whether a guarantee is primary or secondary, the effect of amendments to the underlying contract on the guarantor's liability, the perfection and priority of security interests, and their treatment in the obligor's insolvency — is **wholly jurisdiction-specific and must be confirmed by qualified counsel**. Nothing in the arithmetic below survives an instrument that cannot be called.

12.3.2 Guarantee sufficiency: cap times credit quality

The principle. A cap is a promise; a **recovery** is a cap multiplied by the probability that the obligor pays it in full when called. The financier's question is therefore never "what is the cap?" but "**what is the risk-adjusted cover, and how does it compare with the exposure?**"

WORKED EXAMPLE

Worked example 12.3.2 — is Kestrel's security package sufficient?

1. **Setup.** The exposure of 12.1.2: **12,255,673.53** in the combined 300-day, 5 %-shortfall stress. The nominal cover is the **9,600,000** aggregate cap. It is supported by an **on-demand bond of 4,800,000** (10 % of the EPC price) from an investment-grade bank, treated as payable with effective certainty, and a **parent company guarantee** up to the aggregate cap from a group parent whose probability of paying in full when called is assessed at **0.70** — an illustrative assumption standing for a mid-quality unrated corporate obligor, to be replaced in practice by the credit team's own assessment. Compute the risk-adjusted cover, the residue, and the parent-guarantee face amount that would be equivalent to the bond.
2. **Formula.** Risk-adjusted cover = Σ(instrument face × probability of payment), applying instruments in order of certainty and counting the guarantee only for the increment above the bond. Equivalent face = certain amount ÷ probability of payment.
3. **Substitution.** `4,800,000 × 1.00 + (9,600,000 − 4,800,000) × 0.70`; `12,255,673.53 − 8,160,000`; `4,800,000 ÷ 0.70`.
4. **Result.**

Layer	Face	Quality	Risk-adjusted
On-demand bond	4,800,000.00	1.00	**4,800,000.00**
Parent guarantee, incremental	4,800,000.00	0.70	**3,360,000.00**
Total cover	**9,600,000.00**		**8,160,000.00**

The credit haircut is **1,440,000**, or **15.00 %** of the nominal cap. Against the exposure, the uncovered residue rises from **2,655,674** nominal to **4,095,674** risk-adjusted — **22.75 %** of the 18,000,000 equity contribution. The parent guarantee face amount equivalent to the 4,800,000 bond is **6,857,143**, a multiple of **1.4286×**. 5. **Interpretation.** The **1.4286× equivalence** is the sentence to carry out of this KA: at a 0.70 assessment, **one dollar of unconditional bank cover is worth 1.4286 dollars of parent guarantee**, so a negotiator offered "a parent guarantee instead of the bond" is being offered a discount unless the face amount rises by at least 42.86 %. That converts an argument about instrument preference into a priced trade, which is the only form in which such arguments get settled honestly. The bond's cost makes the trade concrete: at 1.2 % per annum over a three-year construction period the 4,800,000 bond costs **172,800**, or 3.60 % of its face — which the contractor prices into the contract sum, and which is cheap against 1,440,000 of credit haircut.

Three further disciplines. **Assess the guarantor, not the counterparty.** The contracting entity in a large EPC is frequently a special-purpose or thinly capitalised subsidiary, so the whole credit question is the parent's, and the assessment must be documented, dated and owned by the credit function — not asserted in a bankability memorandum. **Test expiry against exposure**, instrument by instrument: Kestrel's delay exposure runs to the long-stop date, its performance exposure to the completion tests, and its defects exposure to the end of the defects-liability period, and any instrument expiring earlier leaves a dated hole. **And never net cover against exposure without stating both.** The residue of 4,095,674 is the number that belongs in the board paper and in the equity case; the temptation to report "fully covered up to the cap" is the reporting failure this KA exists to prevent.

The caution: probabilities of payment are judgments, they are correlated with the very events that trigger a call (a contractor group in distress is both more likely to fail and less likely to pay), and a single-point 0.70 conceals that correlation. The professional treatment is a range, a named owner for the assessment, and a structural response — an unconditional instrument — rather than a more confident number.

12.3.3 Direct agreements and step-in

Definition. A **direct agreement** is a tripartite contract among the project company, a key counterparty (the offtaker, grantor, EPC contractor or operator) and the lenders' security agent. Its function is to make the project's contracts survive the project company's own default. Its standard content: the counterparty acknowledges the assignment of the contract as security; it agrees to give the lenders **notice** of any project-company default and a **cure period** longer than the project company's own; and it agrees that the lenders (or a transferee they nominate) may **step in** and assume the contract, curing arrears, rather than have it terminated.

The commercial logic is Domain 10's, in contractual form. Enforcement that destroys the revenue contracts destroys the asset's value, leaving lenders with a plant and no offtake. Step-in converts a liquidation into a substitution: the same asset, the same contracts, a new equity owner. For the sponsors, the same instrument is the mechanism by which they can lose the project without the project failing — which is precisely the discipline it is intended to impose.

Two practical points a leader should test. **Cure periods must be operationally achievable**: a 30-day cure that requires a lenders' committee decision, an intercreditor vote and a funding call is not 30 days of cure, and Domain 10's forward-looking test discipline applies here too — the question is whether the machinery can move at the speed the clause assumes. **And step-in without a willing transferee is a right, not a remedy**: its value depends on there being an operator or sponsor able to take the asset on the terms available, which is a market question to be answered before close, not a legal question to be answered after default.

12.3.4 What the security package is actually worth

The valuation discipline for the whole package follows from 12.3.2 and takes one page:

- **Face amounts**, listed by instrument, with the exposure each is intended to cover.
- **Credit quality** of each obligor, assessed and dated, with the risk-adjusted amount.
- **Conditionality**: on-demand, conditional, or dependent on establishing breach and loss — the practical difference between money in 10 days and money in 2 years.
- **Expiry**, against the exposure period, with every gap dated.
- **Reach**: which entity in the counterparty group is actually bound, and whether the guarantee survives amendment of the underlying contract.
- **Residue**: exposure less risk-adjusted cover, stated in currency and as a share of equity.

The last line is the one that changes decisions. A package described as "10 % bond plus parent guarantee to the cap" sounds complete; the same package described as "8,160,000 of risk-adjusted cover against 12,255,674 of stress exposure, leaving 22.75 % of the equity cheque exposed" invites the questions that improve it — a larger bond, a bank guarantee in place of the parent, an uncapped indemnity for defined heads, or a smaller stress accepted with eyes open.

AI in this KA

Where it earns its place. Instrument registers are structured data trapped in unstructured documents: obligor, face amount, currency, trigger, conditions, expiry, governing law, notice address. Extracting them across a portfolio and reconciling them against the exposures they are supposed to cover — flagging every instrument that expires before its exposure ends — is machine work with an immediate and measurable payoff. Monitoring is better still: an assistant that watches expiry dates and rating actions against a live register does a job humans do badly.

Where it must not go. It must not assess credit quality as if it were a computation; the 0.70 of 12.3.2 is a judgment with an owner. It must not opine on whether a demand would be compliant, whether a guarantee has been discharged by an amendment, or whether security is perfected — all jurisdiction-specific legal questions for counsel. And it must not be allowed to net cover against exposure in a summary, because that is exactly the reporting failure 12.3.4 warns against.

Verification, concretely. Sample every face amount and expiry date back to the instrument. Require the credit function to sign the quality assessment used in any risk-adjusted number, with a date and a range. Recompute one equivalence figure by hand. And confirm counsel's sign-off that each instrument is callable as modelled before any recovery appears in a lender or equity case.

KEY TERMS — KA 12.3

Term	Meaning
On-demand bond	Bank instrument payable on a compliant demand without proof of breach; near-certain, fee-bearing.
Parent company guarantee	Group parent's assumption of a subsidiary's obligation; worth the parent's credit.
Risk-adjusted cover	Σ(face × probability of payment); the number to compare with exposure.
Equivalent face amount	Certain cover ÷ probability of payment; 1.4286× at a 0.70 assessment.
Direct agreement	Tripartite notice, cure and step-in rights making contracts survive SPV default.
Step-in	Lenders or their nominee assuming a contract instead of allowing termination.
Instrument expiry gap	The dated hole where an instrument expires before the exposure it covers.

SAMPLE MCQS — KA 12.3

MCQ 12.3-A `[12.3.2 · Application]` Cover comprises a 4,800,000 on-demand bank bond (payment effectively certain) and a parent guarantee taking total nominal cover to 9,600,000, the parent assessed at a 0.70 probability of paying in full. Risk-adjusted cover is: - A. USD 9,600,000 - B. USD 8,160,000 - C. USD 6,720,000 - D. USD 3,360,000

Rationale: `4,800,000 + 4,800,000 × 0.70 = 8,160,000`. A ignores credit quality; C applies 0.70 to the whole 9,600,000, haircutting the bank bond as well; D counts only the guarantee increment.

MCQ 12.3-B `[12.3.2 · Analysis]` A contractor offers to replace a 4,800,000 on-demand bond with a parent guarantee, the parent assessed at 0.70. The face amount that leaves the project company no worse off is: - A. USD 4,800,000 - B. USD 3,360,000 - C. USD 6,857,143 - D. any amount, since a guarantee from a large group is stronger than a bond

Rationale: `4,800,000 ÷ 0.70 = 6,857,142.86`, a 1.4286× multiple (12.3.2). A treats unequal certainty as equal; B applies the haircut in the wrong direction; D confuses balance-sheet size with payment certainty and ignores conditionality and timing.

MCQ 12.3-C `[12.3.3 · Recall]` The primary commercial purpose of a direct agreement is to: - A. increase the liability cap of the counterparty - B. give lenders notice, an extended cure period and the right to step in, so the project's contracts survive the project company's default - C. transfer the offtake obligation to the lenders - D. provide additional security over the asset

Rationale: Direct agreements preserve the contract, not the cap (12.3.3). C misstates step-in, which is a right to assume the contract, usually through a nominated transferee; D describes asset security, a separate instrument.

SELF-CHECK — KA 12.3

1. State the sufficiency test in one line. — Risk-adjusted cover, Σ(face × probability of payment), against the stress exposure; Kestrel's is 8,160,000 against 12,255,674.
2. What is a 0.70 guarantee worth against an on-demand bond? — 0.70 of its face, so the equivalent face is 1.4286× the bond — 6,857,143 for a 4,800,000 bond.
3. Which instrument attribute is most often missed? — Expiry: an instrument that expires before its exposure ends is a dated hole in the stack.

KNOWLEDGE AREA 12.4

Risk allocation, claims and change

Topics: 12.4.1 the allocation matrix as a document map · 12.4.2 change and variation mechanics · 12.4.3 the commercial arithmetic of a claim · 12.4.4 dispute pathways and the cost of the process.

12.4.1 The allocation matrix as a document map

Definition. A **risk allocation matrix** in this domain is not the register Domain 11 priced; it is the map from each priced allocation to **the clause that effects it, the instrument that backs it and the amount recoverable under it**. One row per risk, four columns beyond the risk name: the party bearing it, the document and clause, the financial limit, and the instrument standing behind that limit. The matrix's value is that it makes two failures visible that no register shows.

The orphan risk — priced as transferred in the register, but no clause transfers it. This is the commonest defect in a transaction that has moved fast, and it is created by exactly the sequence this book describes: a risk workshop allocates, a model prices the allocation, and the drafting does not follow. Domain 11's Kestrel case is instructive in reverse — five items were offered to the contractor and declined, so the register correctly shows them retained; a matrix that showed them transferred, with no clause reference, would be a 2,690,000 error hiding in a spreadsheet.

The doubly covered risk — two documents allocate the same risk to different parties, or the same cap is committed twice. Cross-liability between an EPC contract and an interface agreement is the classic case, and the aggregate-cap arithmetic of 12.1.1 is the other: a matrix that shows delay, performance and defects each backed by "the aggregate cap" has shown the reader that the cap is committed three times over.

The matrix is also the instrument of the **reviewer's discipline** of 12.A.3, and it is the artefact Domain 13's legal diligence stream produces and Domain 14's change control maintains. Adding a row is how a change order enters the financing record.

12.4.2 Change and variation mechanics

Definition. A **variation** (or change order) is an instructed alteration to scope, quality, sequence or timing, with a defined mechanism for valuing it and for adjusting time. A **compensation event** (or relief event, or excusable delay) is an occurrence for which the contract grants the contractor time, money, or both, without breach by either party.

The finance leader's interest is narrow and important: **change mechanics decide who funds the change and whether the funding exists.** Four terms carry that question.

The valuation rule. Whether variations are valued at contract rates, at agreed lump sums, on a cost-plus basis or by a defined dispute route. A cost-plus default is an open cheque against a fixed funding envelope; contract rates are only protective if the schedule of rates is complete.

The time consequence. Whether an extension of time carries prolongation cost, and at what rate. A pre-agreed daily prolongation rate removes the largest single area of claim quantification argument, and it should be negotiated at the same time as the delay damages rate, on the same evidence. Kestrel's regime is asymmetric in a way sponsors should notice: **the contractor's liability for delay is 20,000 per day, and the project company's exposure to prolongation on an owner-caused delay is a separately negotiated number** — on Kestrel's claim (12.4.3) it is 12,500 per day of site overhead before disruption and additional plant are added, which is why it should be pre-agreed on evidence rather than left to be proved after the event.

The funding link. Whether the variation is funded from contingency, from the sponsors' cost overrun undertaking, or from a further drawing — and whether the facility agreement permits it. Domain 14 (KA 14.3) operates this link at drawdown; the contractual point here is that a change mechanism that produces obligations the funding documents cannot fund is a structural defect, not an administrative one.

The threshold and approval architecture. Which changes the project company may agree alone, which require lender consent (usually any change to a material contract, per Domain 10's negative covenants), and which require the offtaker's or grantor's consent. Serial small changes below a consent threshold are how a fixed-price contract stops being fixed, which is why cumulative as well as individual thresholds belong in the drafting.

12.4.3 The commercial arithmetic of a claim

Definition. A **claim** is an assertion of entitlement to time, money or both under the contract. Its commercial value to either party is not the sum claimed. It is the **expected recovery, net of the cost of obtaining it, discounted for the time it takes** — and because disputes are slow and expensive, that arithmetic frequently inverts the negotiating intuition.

WORKED EXAMPLE

Worked example 12.4.3 — Kestrel's contractor claims 90 days and 1,870,000.

1. **Setup.** The contractor claims a **90-day** extension of time and prolongation cost, arising from a late owner-supplied approval that the contract treats as a compensation event. Quantum: site overhead **12,500 per day** for 90 days (**1,125,000**), disruption and lost productivity **480,000**, additional preliminaries and plant **265,000** — **1,870,000** in total. The project company's own assessment, on its expert's analysis, is **55 days** and **1,050,000**. Delay damages are 20,000 per day. If the matter goes to arbitration: each side's legal and expert costs **620,000**, internal management time **180,000**, an award expected in **26 months**, and assessed outcome probabilities of **0.35** that the contractor's full case succeeds, **0.25** of a split at the midpoint and **0.40** that the project company's assessment is upheld. Discount at Domain 4's **8 %**; costs are incurred evenly, so treat them as falling at the **13-month** midpoint. Each party bears its own costs (cost-shifting rules vary by jurisdiction and by arbitral rules — a matter for counsel, and one that changes this arithmetic materially).
2. **Formula.** Time impact = extension days × delay damages rate (the damages the project company forgoes by granting the extension). Disputed sum = quantum gap + time-impact gap. Expected award = Σ(probability × outcome). Present value of fighting = expected award ÷ (1.08)^2.1667 + costs ÷ (1.08)^1.0833. Settlement ceiling = present value of fighting − negotiation cost. Breakeven disputed sum: solve `k·D + PV(costs) = 0.5D + negotiation cost`, where `k = (0.35 + 0.25/2) ÷ (1.08)^2.1667`.
3. **Substitution.** Time impact `90 × 20,000 = 1,800,000`; owner's economic cost of the 90 days `24,733.33 × 90 = 2,226,000`. Gaps: `1,870,000 − 1,050,000 = 820,000` of money and `35 × 20,000 = 700,000` of forgone damages, so `D = 1,520,000`. Expected award `0.35 × 1,520,000 + 0.25 × 760,000 = 722,000`. Discount factors `(1.08)^2.1667 = 1.181458` and `(1.08)^1.0833 = 1.086949`.
4. **Result.**

Item	Amount
Contractor's quantum claimed	1,870,000.00
Value of the extension of time to the contractor (forgone delay damages)	1,800,000.00
Project company's own economic cost of the 90 days	2,226,000.00
Total exposure of the event to the project company	**4,096,000.00**
Disputed sum after the owner's assessment	**1,520,000.00**
Expected award	722,000.00
Present value of the expected award (26 months)	**611,109.54**
Present value of own costs (13 months)	**736,005.26**
Present value of fighting	**1,347,114.80**
Settling at the midpoint (760,000 + 60,000 of negotiation cost)	820,000.00
Saving from settling at the midpoint	**527,114.80**
Settlement ceiling (indifference point)	**1,287,114.80** = **84.68 %** of the disputed sum
Disputed sum at which fighting becomes rational	**6,901,234.43**
5. **Interpretation.** **The time impact is the bigger prize, and it is usually argued about	
least.** The contractor's money claim is 1,870,000; the extension of time is worth 1,800,000 to	
it in delay damages it will not pay, and the two are close to equal. Meanwhile the project	
company's own economic cost of the 90 days — 2,226,000 — exceeds both. The negotiation is	
therefore not about 1,870,000; the event is worth 4,096,000 to the project company, and any	
settlement discussion that treats the extension of time as a procedural concession has given	
away the largest number on the page.	

Fighting is expensive in a way the win probability conceals. A 0.40 chance of complete victory sounds like a strong position. It is: the expected award is only 722,000 against a disputed 1,520,000. And yet the present value of fighting is **1,347,115**, because own costs of 800,000 are certain and immediate while the award is contingent and 26 months away. The settlement ceiling is therefore **84.68 % of the disputed sum** — meaning the project company should rationally settle at anything up to 1,287,115 of a 1,520,000 dispute, and settling at the midpoint saves **527,115**. Practitioners who "never pay more than their assessment" are choosing to spend 527,115 to defend a principle, which is a legitimate choice only if it is made explicitly and for a stated reason — precedent across a portfolio of similar claims is the usual one, and it should be quantified as such.

The threshold generalises, and it is the number to carry. On these cost and probability assumptions, fighting only beats settling at the midpoint once the disputed sum exceeds **6,901,234** — roughly 4.5 times the disputed sum in front of them, and 14.4 % of the EPC price. That single figure explains the observed behaviour of the market better than any principle: small and medium claims settle because the process costs more than the gap, and only large ones are litigated. An organisation that computes its own threshold once, from its own cost base and its own historical outcome distribution, converts claims handling from temperament into policy.

The cautions are three and they are not decoration. The probabilities are judgments and should be elicited from the party's own legal advisers as a documented range, not asserted; the arithmetic is only as good as they are. Cost recovery — whether the loser pays some or all of the winner's costs — varies fundamentally by jurisdiction and by the applicable arbitral rules, and it can move this calculation by more than the disputed sum; it is a question for qualified counsel and must be answered before the threshold is relied upon. And the model omits every non-financial consequence: management distraction, the effect on a continuing working relationship, the disclosure obligation to lenders under the information covenants, and the precedent set for the remaining claims on the same project. Those belong beside the number, not inside it.

12.4.4 Dispute pathways and the cost of the process

Definition. Contracts specify an escalating sequence for resolving disagreement — typically engineer's or employer's determination, then senior executive negotiation, then mediation or a dispute board, then arbitration or litigation, with interim binding decisions in some standard forms. Each step has a cost, a duration and a probability of resolving the matter, and 12.4.3's arithmetic applies at every one.

The finance leader's contribution is to **price the pathway before it is used**. Three questions do most of the work. How long is the whole ladder? A pathway whose steps sum to 30 months has built a 30-month funding and covenant problem into every serious disagreement, and Domain 10's information covenants will require disclosure at month one. Is there an interim binding mechanism? A dispute board or interim determination that keeps cash moving while the merits are argued is worth a great deal to a project company whose covenants are tested quarterly; a pathway whose first binding outcome is a final award is not. And who funds the process? Legal and expert cost is an operating outflow the base case does not contain, and on Kestrel's arithmetic 800,000 of it is **2.148 times** the project's entire annual covenant headroom of 372,438 — a dispute budget large enough to breach the covenant on its own.

Two standing cautions. **A claim disclosed late is a covenant problem and a relationship problem at the same time** — Domain 10, KA 10.4.4's honesty asymmetry applies with full force, because a lender who learns of a 4,096,000 exposure from a compliance certificate will price the discovery into every future waiver. **And settlement is a financing event.** A settlement that changes the contract price, the completion date, the performance guarantees or the security package touches the negative covenants and the conditions precedent, so lender consent is usually required and should be sought early, not presented as a fait accompli.

The selection of forum, seat, governing law, arbitral rules, enforceability of awards and the availability of interim relief are **jurisdiction-specific matters with material commercial consequences**, and they are decided at drafting. They must be settled by qualified counsel; the financial contribution is to state what each option costs in time and money so that the legal choice is made with the commercial arithmetic in view.

AI in this KA

Where it earns its place. Claims work is document-intensive and that is where machines are strongest: assembling a chronology from correspondence, programmes and site records; identifying which notices were served within contractual time bars and which were not; mapping each claimed head to the clause relied on; and computing the 12.4.3 grid across a range of probability and cost assumptions so that the sensitivity of the settlement ceiling is visible rather than assumed. Maintaining the allocation matrix of 12.4.1 as changes are executed is another natural fit.

Where it must not go. It must not assess the merits of a claim, opine on whether a time bar has been missed with the consequence the contract states, characterise concurrent delay, or draft a settlement position. Those are legal and expert judgments; a claim consultant's and counsel's work is not a summarisation task. Nor should the probability distribution in 12.4.3 come from a model: it must come from named advisers, on the record, as a range.

Verification, concretely. Test the chronology against the primary documents on a sample of at least ten events, including every alleged notice. Recompute the expected award and one discount factor by hand. Require the probability assumptions to be attributed to a named adviser with a date. And require counsel's confirmation of the cost-shifting and limitation position before the settlement ceiling is used in a negotiation mandate.

KEY TERMS — KA 12.4

Term	Meaning
Allocation matrix	Risk → bearing party → clause → financial limit → instrument; reveals orphan and doubly covered risks.
Variation / compensation event	Instructed change; or an occurrence granting time and/or money without breach.
Prolongation cost	Time-related cost of delay, ideally at a pre-agreed daily rate.
Time impact of a claim	Extension days × delay damages rate — the damages forgone by granting time.
Settlement ceiling	Present value of fighting less negotiation cost; the rational maximum to settle at.
Breakeven disputed sum	The dispute size above which litigating beats settling at the midpoint.

SAMPLE MCQS — KA 12.4

MCQ 12.4-A `[12.4.3 · Application]` A contractor claims 90 days and 1,870,000; delay damages are 20,000 per day; the project company's daily economic cost of delay is 24,733.33. The total exposure of the event to the project company is closest to: - A. USD 1,870,000 - B. USD 3,670,000 - C. USD 4,096,000 - D. USD 2,226,000

Rationale: Quantum 1,870,000 + own economic cost `24,733.33 × 90 = 2,226,000` gives 4,096,000 (12.4.3). A counts only the money claim; B adds the forgone damages of 1,800,000 to the quantum but omits the project company's own cost; D counts only the economic cost.

MCQ 12.4-B `[12.4.3 · Analysis]` The disputed sum is 1,520,000; the present value of the expected award is 611,110 and of own costs 736,005. The rational settlement ceiling, allowing 60,000 of negotiation cost, is: - A. USD 760,000 — the midpoint - B. USD 1,287,115 - C. USD 611,110 - D. USD 1,520,000 — the full claim

Rationale: Fighting costs a present value of 1,347,115; deducting the 60,000 that settling itself costs gives an indifference point of 1,287,115, or 84.68 % of the disputed sum (12.4.3). A is one possible settlement, not the ceiling; C omits own costs, which are the larger component; D assumes no defence has value.

MCQ 12.4-C `[12.4.3 · Analysis]` Why does a 0.40 probability of complete victory still leave a settlement ceiling at 84.68 % of the disputed sum? - A. because the expected award is larger than the disputed sum - B. because own costs are certain and immediate while the award is contingent and 26 months away - C. because the discount rate is too high - D. because liquidated damages are excluded from the calculation

Rationale: PV of own costs (736,005) exceeds the PV of the expected award (611,110); certainty and timing dominate probability (12.4.3). A is arithmetically false; C confuses a parameter with the mechanism; D is untrue — the forgone damages are the largest part of the disputed sum.

MCQ 12.4-D `[12.4.1 · Analysis]` A risk register shows five construction risks as transferred; the allocation matrix shows no clause reference for any of them. The correct conclusion is: - A. the matrix is incomplete but the allocation stands - B. these are orphan risks: priced as transferred, retained in fact, and the register overstates protection - C. the risks are doubly covered - D. the contractor's aggregate cap covers them

Rationale: An allocation without a clause is not an allocation (12.4.1); the register and the model are both wrong until the drafting follows. C is the opposite defect; D assumes a cap can cover a liability the contract never created.

SELF-CHECK — KA 12.4

1. What is the largest number in a typical extension-of-time negotiation? — Usually the time impact: 90 days at 20,000 is 1,800,000 of forgone damages, against a 1,870,000 money claim and the project company's own 2,226,000 of economic cost.
2. State the settlement ceiling rule. — Settle at up to the present value of fighting less the cost of settling: 1,287,115 on a disputed 1,520,000, saving 527,115 against a midpoint settlement.
3. Why do most claims settle? — Because process cost and delay dominate: on these assumptions a dispute must exceed 6,901,234 before litigating beats settling at the midpoint.

Advanced topics — Domain 12

12.A.1 What sits outside the cap

An aggregate liability cap is routinely read as the maximum a counterparty can owe. It is not. Contracts customarily carve out defined heads from the cap — abandonment, fraud or wilful misconduct, breach of confidentiality or intellectual property, third-party death and personal injury, environmental indemnities, and sometimes the proceeds of insurance — and the carve-outs are where unbounded exposure lives, in both directions. Two consequences for a finance leader.

On the recovery side, an uncapped indemnity is worth only the obligor's balance sheet, so a carve-out that looks like unlimited protection is limited in practice by exactly the credit arithmetic of 12.3.2 — and unlike the capped heads, it has no instrument sized against it. **On the liability side**, the project company's own uncapped indemnities are the exposures most likely to exceed its equity, and they are the ones sponsors most often fail to model because they sit outside every number in the contract summary. The discipline is to list the carve-outs explicitly in the allocation matrix with the word "uncapped" in the limit column, so that they cannot be read as absent, and to test the largest of them against the insurance programme rather than the cap.

The scope, validity and effect of any carve-out — and whether a cap is displaced entirely by particular conduct — are questions of the governing law and the drafting, and they vary fundamentally between jurisdictions. Qualified counsel must confirm them; the commercial task is to ensure that each one appears in the matrix with an owner and a sizing.

12.A.2 Termination compensation against the amortisation profile

12.2.3 compared a compensation formula with the debt-outstanding profile at points in time. The advanced form is to plot both across the whole term, because the **shape** of the gap is what determines whether the structure is robust. Three shapes recur. A **debt-outstanding formula** tracks the amortisation exactly and produces no gap by construction — the lenders' preferred outcome, and the one that leaves equity at zero (11,128,176 unreturned at the end of year 5 for Kestrel). A **fixed-schedule formula** — a depreciated capital value, or a table agreed at signature — diverges from an annuity amortisation because principal repayment accelerates while straight-line depreciation does not, so the gap opens or closes depending on the tenor relationship. And a **percentage-of-debt formula** produces a gap proportional to the outstanding balance, therefore **largest in the earliest years** — 5,926,555 at the end of Kestrel's year 1 against 4,194,854 at the end of year 5 on an 85 % formula — which is the worst possible profile, because early-life default risk is the highest and the tail of the loan is the part a refinancing would have addressed anyway.

Two refinements matter in a real negotiation. **Hedge breakage** is a real cost that debt-outstanding formulae frequently exclude: a fixed-to-floating swap terminated at an unfavourable point produces a payment that is neither principal nor interest, and if the compensation clause does not name it, the lenders are not whole even under a "full debt" formula. **And the reference date matters**: compensation computed at the termination date, at the date of a valuation, or at the date of payment can differ by a full period's interest and by any accrued default interest, all of which should be specified rather than left to construction. Both are drafting points with a computable price, and both are for counsel to render enforceable and for the finance leader to size.

12.A.3 The reviewer's contract eye

The invariants to test on any contract stack before a financing relies on it:

- **Every risk shown as transferred has a clause reference**, and every clause reference resolves to language that actually transfers it (no orphan risks — 12.4.1).
- **Delay damages are calibrated on carrying cost plus forgone** `CFADS`, not revenue alone, and the calibration basis is documented (Kestrel: 24,733.33 per day, not 17,733.33 — 12.1.2).
- **The cap-binding day is computed and compared with the schedule risk analysis' credible worst-case delay** (Kestrel: day 240).
- **Sub-caps are summed against the aggregate cap**, and the aggregate exceeds the sum wherever it is intended to add protection (Kestrel's do not — 12.1.1).
- **Performance damages state which loss they restore** — bare covenant, sized coverage or value — and the model uses the same basis the contract does (562,851 / 2,980,263 / 4,835,674 — 12.1.3).
- **The application of damages proceeds is tested for tenor mismatch**: a multi-decade loss paid by a shorter-dated debt prepayment under-compensates (521,045 — 12.1.3).
- **The contracted volume floor clears the covenant**, computed as `x = (k × debt service + intercept) ÷ slope`, in the binding period, not on an annual average (95.8892 % against a negotiated 90 % — 12.2.2).
- **The maximum annual deduction under the revenue contract is computed and compared with covenant headroom** (372,438 for Kestrel — 12.2.1).
- **Termination compensation is plotted against debt outstanding across the term**, and hedge breakage is named in the formula (12.A.2).
- **Every instrument's face amount is multiplied by an owned credit assessment**, and every expiry date is tested against the exposure period it covers (8,160,000 against 9,600,000 nominal — 12.3.2).

- **Uncovered residue is reported in currency and as a share of equity**, never netted silently (2,655,674 nominal, 4,095,674 risk-adjusted, 14.75 % and 22.75 % of equity — 12.1.2, 12.3.2).
- **Carve-outs from the cap are listed as uncapped**, sized, and tested against insurance (12.A.1).
- **Every recovery the model relies on carries counsel's confirmation** that the instrument is callable and the provision enforceable in the governing jurisdiction.

Industry variations — Domain 12

- **Contracted power and availability PPPs.** The document set is the most standardised, and the binding constraint is usually the **availability and deduction regime** rather than volume: the test of 12.2.1 — maximum annual deduction against covenant headroom — replaces the volume floor of 12.2.2, and performance damages attach to heat rate or capacity rather than output.
- **Transport concessions.** Patronage risk is rarely contracted away, so there is no volume floor to compute; the load-bearing clauses instead are **compensation on termination** (12.2.3, because the grantor's covenant is the credit) and the **change-in-law and competing-infrastructure protections**, whose absence is a revenue risk no cap covers.
- **Water and desalination.** The Kestrel case: take-or-pay water purchase, output and consumption guarantees, and a heavy dependence on a single input contract (power) whose interruption provisions must mirror the offtake's force-majeure relief exactly, or a gap opens between the cost the project incurs and the revenue it may abate.
- **Digital infrastructure.** Multi-package delivery is normal — shell, mechanical and electrical fit-out, network — so **interface architecture is the dominant contractual question** (12.2.4 and Case study B), tenant contracts are shorter than the debt so termination and renewal provisions carry the refinancing case, and service-level agreements rather than output guarantees define performance.
- **Oil, gas and process industries.** Performance guarantees are multi-dimensional (throughput, yield, specification, consumption), so the buy-down arithmetic of 12.1.3 must be run per guarantee and aggregated, and licensor and process-technology agreements add a party whose liability is characteristically capped very low relative to the loss it can cause.
- **Social infrastructure and accommodation PPPs.** Payment is availability-based with detailed performance deductions and a defined **hand-back regime**: condition surveys, a hand-back reserve and rectification obligations at the end of the concession, which is a contractual exposure that falls after the loan has been repaid and is therefore invisible in every coverage ratio.

Case study — Domain 12: the cap that covered 4.96 per cent (water / desalination)

Situation. Ten weeks before financial close, Kestrel's lenders' legal adviser circulated a contract-limits table of the kind set out in 12.1.1. It showed the six numbers: a 48,000,000 fixed price, delay damages of 20,000 per day capped at 10 %, performance damages capped at 10 %, an aggregate cap of 20 %, and security comprising a 10 % on-demand bond and a parent guarantee to the aggregate cap. The sponsors' commercial team regarded the stack as market standard and the technical adviser had confirmed the plant design. Nobody had multiplied anything.

What happened. The financial adviser ran three computations in an afternoon. **The cap-binding day**: 4,800,000 ÷ 20,000 = **day 240**, against a schedule risk analysis whose P80 delay was 210 days and whose P95 was 320 — so the cap covered the P80 case and failed the P95 one, leaving **2,620,000** uncovered at 300 days and **4,104,000** at 360 (Domain 5, KA 5.4.2 computed the 360-day figure). **The performance cap's reach**: on the value basis of 12.1.3, one percentage point of permanent output shortfall is worth `90,600 × AF(0.08, 25) =` **967,134.71**, so the 4,800,000 cap covers a shortfall of **4.9631 %** — while the 1.20 covenant fails at a shortfall of **4.1108 %**. The regime was adequate by a margin of 0.85 percentage points, which nobody had designed and nobody had checked. **And the aggregate cap**: 10 % + 10 % = 20 %, so the two sub-caps consumed the aggregate exactly and a defects claim after both would recover nothing.

The combined stress — 300 days late at 95 % output — produced exposure of **12,255,673.53** against nominal cover of **9,600,000**, a residue of **2,655,673.53**. Applying the credit arithmetic of 12.3.2, with the bond at full value and the parent guarantee assessed by the lenders' credit function at 0.70, risk-adjusted cover was **8,160,000** and the residue **4,095,673.53** — **22.75 %** of the 18,000,000 equity contribution, and **16.41 %** of Domain 4's project `NPV` of 16,179,360 on the nominal measure.

How it resolved. Four changes were negotiated, each cheap because the arithmetic was explicit. The **aggregate cap moved from 20 % to 30 %** (14,400,000), so the sub-caps no longer exhausted it and a defects head had 4,800,000 of room. The **on-demand bond rose from 10 % to 15 %** (7,200,000), converting 2,400,000 of parent-guarantee cover at 0.70 into bank cover at full value and lifting risk-adjusted cover by 720,000; the contractor priced the additional bonding at about 1.2 % per annum over three years — some 86,400 on the incremental 2,400,000 — and added it to the contract sum, which the sponsors accepted as the cheapest credit enhancement available. **A termination-for-delay right at a 300-day long-stop** was added, so that the contractor's incentive did not fall to zero at the cap. And **the performance damages were re-based on the value measure** at 967,135 per percentage point, with the facility agreement amended to direct 60 % of any receipt to prepayment and 40 % to a blocked distribution account — the drafting response to the 521,044.53 tenor mismatch of 12.1.3, which cost nothing and removed a permanent equity leak.

What the domain teaches here. Every one of those four defects was visible from numbers already in the documents; none required a model, a negotiation or a new adviser. The failure was not analytical but procedural: no one had been given the job of multiplying the contract's limits against the project's stresses. That job — the six-number table, the cap-binding day, the coverage reach of each cap, and cover multiplied by credit quality — is the whole of this domain's professional content, and it takes an afternoon.

Case study B — Domain 12: the interface nobody owned (digital infrastructure)

Situation. A 24 MW hyperscale data centre in a competitive market was let as two packages to save cost and time: **shell and core at 62,000,000** and **mechanical, electrical and network fit-out at 48,000,000**, a combined **110,000,000**. The single-wrap alternative had been offered by the shell contractor at a **3.5 %** premium — **3,850,000** — and declined, correctly on Domain 11's reasoning: the shell contractor could not price the fit-out interfaces it did not control, and its loading reflected that ignorance. What the project company did not do was replace the wrap with an interface regime. There was no interface matrix, no back-to-back completion definition, no cross-liability provision, no interface manager and no single owner of programme float. Financing was **77,000,000** of senior debt at **6.5 %** against forecast operating `CFADS` of **14,600,000**.

What happened. The fit-out contractor's containment and cable-tray installation required structural penetrations at coordinates the shell contractor built to an earlier revision of the model. Each party demonstrated compliance with its own contract documents. The dispute over which revision governed the handover took eleven weeks to resolve commercially, and the practical completion date moved **74 days**.

The arithmetic was unforgiving. Daily carrying cost on the drawn debt was `77,000,000 × 0.065 ÷ 360 =` **13,902.78**; forgone `CFADS` was `14,600,000 ÷ 360 =` **40,555.56**; the daily economic cost of delay was therefore **54,458.33** and the 74 days cost **4,029,916.67**. On top of that, **both** contractors claimed extensions of time and prolongation — at 9,800 and 7,400 per day respectively, **1,272,800** in total — and both were entitled to something, because neither was in breach of its own scope. Total cost to the project company: **5,302,716.67**, or **4.821 %** of the combined contract price, with **no delay damages recoverable from anyone**.

How it resolved. The claims were settled at a discount on the 12.4.3 arithmetic — both were below the project company's own litigation threshold, and settling promptly was worth more than either defence. The interface responsibility was retrofitted mid-project: an interface manager with authority over both contracts, a matrix of 340 handovers with named owners and dates, a single model revision under change control, and back-to-back milestone definitions for the remaining handovers. The retrofit cost about **640,000** and no comparable event recurred.

What the domain teaches here. The declined wrap was not the error, and the arithmetic proves it: the wrap premium of 3,850,000 was certain, the interface loss of 5,302,717 was not, so buying the wrap would only have paid if the probability of an event of this size exceeded `3,850,000 ÷ 5,302,717 =` **72.60 %** — which no one could have argued. The error was to treat "do not buy the wrap" as a complete decision. The interface regime that would have prevented the event cost **640,000**, a breakeven probability of only **12.07 %**, and it was the obviously correct purchase on any plausible view of the risk. **Declining a transfer creates a management obligation**, and this domain's contribution is to name the instrument that discharges it — an interface matrix with owners, dates, back-to-back completion definitions, cross-liability and one owner of float — and to price it against the exposure it removes.

Executive perspective — Domain 12

What a project finance director cannot delegate in this domain:

- **The six numbers.** Contract price, damages rate, delay cap, performance cap, aggregate cap and security. Read them off the documents personally, multiply them, and know the cap-binding day (Kestrel: day 240) and the shortfall each cap reaches (4.9631 %) before signature.
- **The calibration basis of every damages provision.** Whether damages restore the covenant, the sized coverage or the value of the loss — 562,851, 2,980,263 or 4,835,674 for the same 5 % shortfall. Whoever chooses that basis chooses who bears the loss, and it is a sponsor decision, not a drafting detail.
- **The contracted volume floor.** Issue it to the commercial team as a financing constraint with its derivation attached, before negotiation: 95.8892 % for Kestrel's 1.20× covenant, not the 90 % a commercial team will otherwise agree.
- **Cover multiplied by credit quality, and the residue.** 8,160,000 against 12,255,674 of stress exposure leaves 22.75 % of the equity cheque exposed. That number belongs in the investment committee paper in those words; "covered up to the cap" does not.

- **The claims threshold, computed once.** Know the settlement ceiling rule and the disputed sum above which the organisation litigates (6,901,234 on Kestrel's assumptions). Claims policy set in advance beats claims temperament in the moment, and it removes a systematic value leak.
- **The boundary with counsel, in both directions.** Never let a financial model rely on a recovery counsel has not confirmed is available in the governing jurisdiction; and never let a legal negotiation trade a limit whose financial consequence has not been computed. The director owns the interface between the two, and it is the only place in the transaction where both errors are visible at once.

Calculation exercises — Domain 12

> **Exercise 12.1** A project has 96,000,000 of debt drawn at 7.2 %, forecast operating `CFADS` of 15,300,000 per year, and an EPC price of 120,000,000. Delay damages are 45,000 per day, capped at 8 % of the price; 30/360 applies. Compute the daily economic cost of delay, the share of it the damages recover, the cap-binding day, and the amount uncovered on a 300-day delay.

Solution. Daily interest `96,000,000 × 0.072/360 =` **19,200**; daily forgone `CFADS 15,300,000/360 =` **42,500**; daily economic cost **61,700**. Damages recover `45,000/61,700 =` **72.93 %**. Cap `120,000,000 × 0.08 =` **9,600,000**, binding at `9,600,000/45,000 =` **day 213.33**. At 300 days the economic cost is `61,700 × 300 =` **18,510,000** against a capped recovery of 9,600,000, leaving **8,910,000** uncovered — 7.425 % of the contract price. Common error: calibrating on forgone `CFADS` alone, which would set the rate at 42,500 and recover only **68.88 %** of the daily cost, omitting the carrying cost of drawn debt — the most certain component, because it accrues whether or not the plant would have performed.

> **Exercise 12.2** The same project's performance damages are capped at 8 % of the 120,000,000 price. Each percentage point of permanent output shortfall reduces annual `CFADS` by 210,000. The remaining operating life is 20 years and the sponsors' appraisal rate is 8 %. On a value basis, what output shortfall does the cap actually cover?

Solution. `AF(0.08, 20) = 9.818147`; the value of one percentage point is `210,000 × 9.818147 =` **2,061,810.87**. The cap of **9,600,000** therefore covers `9,600,000/2,061,810.87 =` **4.6561 %** of output and no more. Common error: dividing the cap by the annual `CFADS` shortfall (9,600,000/210,000 = 45.7 "points"), which values a permanent loss as though it lasted one year and overstates the cap's reach roughly tenfold.

> **Exercise 12.3** The same project is funded with 96,000,000 of senior debt at 7.2 % over 15 years. Its `CFADS` as a function of the share `x` of contracted output taken is `CFADS(x) = 16,800,000x − 3,072,000`. The covenant is **1.25×**. Compute the instalment, the `DSCR` at full contracted volume, and the minimum contracted volume floor the covenant requires.

Solution. `AF(0.072, 15) = 8.993967`; instalment `96,000,000/8.993967 =` **10,673,821.69**. At `x` = 1, `CFADS` = **13,728,000** and `DSCR` = **1.2861**. The covenant requires `CFADS ≥ 1.25 × 10,673,821.69 =` **13,342,277.11**, so `x = (13,342,277.11 + 3,072,000)/16,800,000 =` **97.7040 %**. Common error: concluding that because the sized `DSCR` of 1.2861 comfortably exceeds the 1.25 covenant, a 90 % volume floor is safe — at 90 % the `DSCR` is **1.1287**, a breach. Thin headroom converts almost any volume concession into a covenant problem, and the floor must be computed, not inferred from the base-case ratio.

> **Exercise 12.4** Stress exposure on a contract is 14,200,000. Cover comprises a 6,000,000 on-demand bank bond (payment treated as certain) and a parent company guarantee taking nominal cover to 12,000,000, the parent assessed at a 0.65 probability of paying in full when called. Compute risk-adjusted cover, the uncovered residue on both measures, and the parent-guarantee face amount equivalent to the bond.

Solution. Risk-adjusted cover `6,000,000 + (12,000,000 − 6,000,000) × 0.65 =` **9,900,000**. Uncovered: **2,200,000** on the nominal measure, **4,300,000** risk-adjusted — nearly double. Equivalent face for the bond `6,000,000/0.65 =` **9,230,769.23**, a 1.5385× multiple. Common error: applying the credit haircut to the whole 12,000,000 (giving 7,800,000), which wrongly discounts the bank bond; the haircut applies only to the layer the weaker obligor supports.

> **Exercise 12.5** A disputed sum is 2,400,000. Assessed outcomes: 0.30 that the claimant's full case succeeds, 0.30 of a split at the midpoint, 0.40 that it fails. Own legal, expert and management costs are 700,000, incurred at a 14-month midpoint; an award is expected at 30 months; the discount rate is 8 %; settling costs 50,000 of negotiation effort. Compute the present value of fighting, the saving from settling at the midpoint, the settlement ceiling, and the disputed sum above which fighting becomes rational.

Solution. Expected award `0.30 × 2,400,000 + 0.30 × 1,200,000 =` **1,080,000**. Discount factors `(1.08)^2.5 = 1.212158` and `(1.08)^(14/12) = 1.093942`. Present value of the award `1,080,000/1.212158 =` **890,972.64**; of costs `700,000/1.093942 =` **639,887.55**; present value of fighting **1,530,860.19**. Settling at the midpoint costs `1,200,000 + 50,000 =` **1,250,000**, so settling saves **280,860.19**. The settlement ceiling is `1,530,860.19 − 50,000 =` **1,480,860.19**, or **61.70 %** of the disputed sum. With `k = (0.30 + 0.15)/1.212158 = 0.371239`, fighting beats settling at the midpoint only above `(639,887.55 − 50,000)/(0.5 − 0.371239) =` **4,581,245.18**. Common error: comparing the expected award of 1,080,000 with the midpoint settlement of 1,200,000 and concluding that fighting is cheaper — which ignores both own costs, the larger term, and the 30-month delay before any recovery.

Practitioner's toolkit — Domain 12

Adoption-ready artefacts; adapt headings to your organisation, then keep them stable.

Toolkit 12.T.1 — The contract limits and calibration sheet (one per contract)

One page, filled before the contract is signed and re-filled at every amendment. Rows: contract price and change mechanics · damages rate, with the daily economic cost it was calibrated against and the share it recovers · delay cap, and **the cap-binding day** · the credible worst-case delay from the schedule risk analysis, beside it · performance damages rate and cap, with **the shortfall percentage the cap reaches** on the stated basis · the basis itself (bare covenant / sized coverage / value) named and owned · aggregate cap, with the **sum of the sub-caps** beside it · the carve-outs from the cap, each marked "uncapped" and sized · the exclusion of consequential loss, quoted · the combined stress exposure and the residue in currency and as a share of equity. Rule: no contract goes to signature until the residue line is filled and initialled by the accountable sponsor executive.

Toolkit 12.T.2 — Security package sufficiency register (one per transaction)

One row per instrument: instrument type · obligor, and which entity in the group is actually bound · face amount and currency · conditionality (on demand / conditional / requires proof of loss) · credit assessment, with the assessor's name and date · **risk-adjusted amount** · the exposure it covers and that exposure's end date · **the instrument's expiry, and the gap in days if it is earlier** · counsel's confirmation that it is callable as modelled, with date. Footer lines: total face, total risk-adjusted, stress exposure, **residue in currency and as a percentage of equity**. Rule: no lender or equity case reports cover without the risk-adjusted total beside the face total.

Toolkit 12.T.3 — Claim assessment and settlement-zone worksheet (one per claim)

Header: claim reference, event, notice served and whether within the contractual time bar (fact, with the date), clause relied on. Section 1 — **quantum**: each head claimed, the party's own assessment, and the gap. Section 2 — **time**: days claimed, days assessed, and the gap valued at the delay damages rate; plus the party's own economic cost of the days at issue. Section 3 — **process**: own legal, expert and management cost, its timing midpoint, expected date of a binding outcome, outcome probabilities attributed to a named adviser as a range and dated, and the applicable cost-shifting position confirmed by counsel. Section 4 — **arithmetic**: present value of the expected award, present value of own costs, present value of fighting, the settlement ceiling, the saving against a midpoint settlement, and the organisation's standing breakeven disputed sum. Section 5 — **non-financial**: relationship, precedent across open claims, lender disclosure obligation and date, and any consent required for a settlement that touches a material contract. Rule: the mandate given to a negotiator states the ceiling, not the aspiration.

Exam preparation — Domain 12

What is assessed. Reading a contract stack as a set of computable limits; calibrating and testing liquidated damages against the loss they cover; the interaction of sub-caps and aggregate caps; the three make-good bases for underperformance and the consequences of choosing among them; deriving a contracted volume floor from a covenant; testing termination compensation against the debt-outstanding profile; converting instrument face amounts into risk-adjusted cover and computing equivalence between instruments of different certainty; the commercial valuation of a claim including time impact and process cost; and the governance boundary between commercial arithmetic and legal conclusion.

The calculations to do under time pressure. Daily economic cost of delay = drawn debt × rate ÷ day-count basis + annual `CFADS` ÷ basis. Cap-binding day = cap ÷ daily rate. Uncovered residue = daily cost × days − min(rate × days, cap). Value of a permanent shortfall = annual `CFADS` shortfall × `AF(r, n)`, and the shortfall a cap reaches = cap ÷ value per point. Coverage-restoring buy-down = debt × (`CFADS` shortfall ÷ base `CFADS`). Contracted volume floor = (target ratio × debt service + intercept) ÷ slope. Risk-adjusted cover = Σ(face × probability of payment), and equivalent face = certain cover ÷ probability. Expected award = Σ(probability × outcome), discounted to today; settlement ceiling = present value of fighting − cost of settling.

The traps. - Negotiating the damages **rate** and ignoring the **cap-binding day** — the cap is where the structure breaks (12.1.2, Exercise 12.1). - Calibrating delay damages on forgone revenue alone, omitting carrying cost — 71.70 % coverage for Kestrel, 68.88 % in Exercise 12.1 (12.1.2). - Reading an aggregate cap as protection above sub-caps that already exhaust it (12.1.1, MCQ 12.1-B). - Valuing a permanent output shortfall as one year's loss rather than `× AF(r, n)` (Exercise 12.2). - Confusing the three make-good bases and quoting a lender-facing number as the sponsors' loss — 562,851 / 2,980,263 / 4,835,674 (12.1.3, MCQ 12.1-C). - Assuming that damages of the right **amount** compensate, regardless of how their proceeds are **applied** — the 521,045 tenor mismatch (12.1.3, MCQ 12.1-D). - Treating the contracted volume floor as a commercial preference rather than deriving it from the covenant — 95.8892 % against 90 % (12.2.2, MCQ 12.2-A, Exercise 12.3). - Inferring that a comfortable base-case `DSCR` makes a volume concession safe (Exercise 12.3). - Reading a percentage-of-debt termination formula as adequate without plotting it against the amortisation profile, where the gap is largest early (12.2.3, 12.A.2, MCQ 12.2-C). - Splitting packages without an interface regime, and assuming a loss will be recoverable from one of the contractors (12.2.4, MCQ 12.2-D, Case study B). - Reporting cap face amounts as cover, without multiplying by credit quality — 9,600,000 against 8,160,000 (12.3.2, MCQ 12.3-A, Exercise 12.4). - Accepting a parent guarantee for a bond at equal face value — the equivalence is 1.4286× at 0.70 (12.3.2, MCQ 12.3-B). - Ignoring instrument expiry against the exposure period (12.3.1, 12.3.4). - Treating an extension of time as a procedural concession rather than the largest number in the negotiation — 1,800,000 of forgone damages (12.4.3, MCQ 12.4-A). - Comparing an expected award with a settlement figure while ignoring own costs and the delay to recovery (12.4.3, MCQ 12.4-C, Exercise 12.5). - Recording a risk as transferred in the register with no clause reference — the orphan risk (12.4.1, MCQ 12.4-D). - Allowing an AI-produced contract summary, or any summary, to substitute for counsel on enforceability, and letting a model rely on a recovery counsel has not confirmed (every KA).

How the domain connects. Domain 5 established the bankability role of the EPC wrap and computed the cost of a COD slip; Domain 8 sized the contingency that a fixed-price wrap made defensible; Domain 10 supplied the coverage machinery every limit in this domain is tested against, and the covenant and lock-up triggers those tests read; Domain 11 priced the allocations this domain documents and quantified the register residue that no clause covers. Forward: Domain 13's legal, insurance and tax diligence streams verify this stack and its conditions precedent, and its model audit checks that the model's recoveries match the caps; Domain 14 operates the change and certification machinery of KA 12.4 at drawdown; Domain 15 lives with the deduction regimes, guarantees and termination provisions in operation, and reaches for them in restructuring; and Domain 16 governs the AI-assisted contract review that every KA here has bounded.

Domain 12 summary

Contracts are where Domain 11's priced allocations become enforceable limits, and those limits are numbers that can be multiplied before signature. Kestrel's stack — a 48,000,000 fixed price, delay damages of 20,000 per day against a daily economic cost of 24,733.33, a 10 % delay cap binding at **day 240**, a 10 % performance cap, a 20 % aggregate cap that the two sub-caps **exhaust exactly**, a 10 % on-demand bond and a parent guarantee — produces, against a 300-day delay and a 5 % output shortfall, exposure of **12,255,674** against nominal cover of **9,600,000** and risk-adjusted cover of **8,160,000**: a residue of **2,655,674** nominal and **4,095,674** credit-adjusted, being 14.75 % and **22.75 %** of the 18,000,000 equity cheque and 16.41 % of Domain 4's NPV of 16,179,360. Performance damages must state which loss they restore, because the same 5 % shortfall is worth **562,851** on a bare-covenant basis, **2,980,263** on the sized-coverage basis and **4,835,673.53** on the value basis — a spread of **8.591×** — and the 10 % cap reaches a shortfall of only **4.9631 %** against a covenant that fails at **4.1108 %**; even a correctly sized receipt under-compensates by **521,045** when a 25-year loss is paid by a 12-year debt prepayment, so the application clause is worth as much as the amount. On the revenue side the contracted volume floor is a **financing parameter**, not a commercial preference: Kestrel's 1.20× covenant requires **95.8892 %** of guaranteed output where the commercial team agreed 90 % (a DSCR of **1.0935**), each point of volume being worth 0.0181 of coverage; and a termination formula paying a percentage of debt outstanding fails worst early — **5,926,555** short at the end of year 1 on an 85 % formula — while a formula that covers debt in full leaves equity's **11,128,176** of unreturned capital at zero. Guarantees are worth cap × credit quality, making one dollar of on-demand bank cover equal to **1.4286** dollars of parent guarantee at a 0.70 assessment, and instrument expiry against exposure is the gap most often missed. Claims are valued on expected recovery net of process cost and time: on a disputed **1,520,000**, fighting costs a present value of **1,347,115**, the settlement ceiling is **1,287,115** — **84.68 %** of the claim — settling at the midpoint saves **527,115**, and the disputed sum must exceed **6,901,234** before litigation is rational, which is why an organisation should compute that threshold once and make it policy. Case study B prices the other half of Domain 11's lesson: declining a **3,850,000** wrap premium was defensible at a **72.60 %** breakeven probability, but failing to buy the **640,000** interface regime that would have prevented a **5,302,717** loss at a **12.07 %** breakeven was not **— declining a transfer creates a management obligation**. Throughout, the commercial arithmetic is the finance leader's and the legal conclusion is counsel's: no model may rely on a recovery that qualified counsel has not confirmed is available in the governing jurisdiction. Domain 13 verifies this stack in diligence and at financial close; Domain 14 runs its change machinery through construction; Domain 15 lives inside its deduction, guarantee and termination regimes in operation.

13

DOMAIN 13

Due Diligence and Financial Close

Group: Executing the transaction (Domain 13 of 4 in Part Three). **Target:** ~75 pages. **Binds to:** the PCI Book Pattern Specification and the shared registries (`docs/books/registries/`). This domain closes Part Three. It consumes Domain 5's condition register, Domain 6's model and model-audit economics, Domain 9's tranche structures, Domain 10's coverage machinery and Domain 12's contract set, and converts them into a dated event: money in an account. British English; USD (+SAR where useful, indicative `USD 1 ≈ SAR 3.75`).

Why this domain exists

Domains 10 to 12 sized the debt, allocated the risk and drafted the contracts — all on the assumption that the facts are as represented. Nothing so far has tested that assumption, and nothing so far has turned an agreed set of terms into a funded facility. Both jobs belong here. **Due diligence** is the organised purchase of information about a project before irrevocable commitment; **financial close** is the moment at which every condition has been satisfied, every document signed and the first drawdown is available. Between them sits the most compressed and most expensive period in a project financing, and the period in which the largest number of avoidable errors are made — not because practitioners do not know what to check, but because they treat checking as a procedural obligation rather than as an investment with a computable return.

The domain's central claim is that **diligence and close are economic decisions under a deadline, and both are governed by the price of elapsed time rather than the price of advice.** Three consequences follow, one per Knowledge Area and one across them. A diligence stream is worth running when the loss it can avoid exceeds what it costs, and because fees are small beside the cost of delay, the decisive variable is whether the stream sits inside a parallel envelope or on the critical path (KA 13.1). A model audit produces findings whose value lies entirely in their **class** — whether a finding changes a conclusion — so a finding count is not a measure of anything (KA 13.2). Conditions precedent form a **conjunction with a critical path**, which means the close date is the expected maximum of several chains and not the expected duration of the longest one, a distinction that is invisible in every close timetable drawn as a single bar (KA 13.3). And syndication converts the closed transaction into held positions, with a fee architecture and a market-flex mechanism whose value routinely exceeds the fee that sponsors spend their negotiating capital on (KA 13.4).

Learning objectives. After this domain a candidate can: name the seven standard diligence streams and state what each is uniquely able to find; price a diligence stream as an information purchase and compute its breakeven detection rate; demonstrate why running the same streams in parallel rather than in series can be the difference between creating and destroying value; classify model-audit findings by whether they change a conclusion, a reported output or nothing, and compute the debt-capacity consequence of each class; explain why a finding count and a "findings closed" percentage are misleading control metrics; build a conditions-precedent register as a dependency network with float, compute the expected close date as the expected maximum of its chains, and identify the chain whose de-risking is worth most; build an itemised close-cost budget, express it as a share of debt raised, reconcile it to the sources-and-uses statement and compute the effective cost of debt it implies; explain the underwriting, club and best-efforts syndication routes and compute the fee split, the arranger's yield differential and the cost of a failed sell-down; quantify market flex and test it against the coverage covenant; and govern AI use across data-room review, model scanning, condition tracking and investor allocation without letting a machine conclude anything that a named professional must own.

The master transaction. Kestrel Water SPC reaches the diligence phase of its financing. The mandate is for **USD 42,000,000** of senior debt at **6.0 % over 12 years** against a **USD 60,000,000** envelope funded 70/30 (Domain 6, KA 6.2.1), with an annual instalment of **USD 5,009,635.23**, first-year interest of **2,520,000** and principal of **2,489,635** (Domain 3). Documented first-year `CFADS` is **USD 6,384,000** (Domain 2), giving `DSCR` **1.2743** and, on level cash, an identical `LLCR`; `PLCR` is **1.9431** (Domain 10). The concession's expiry is fixed, so every day between mandate and close destroys one day of operating life: Domain 5 (KA 5.4.2) priced that at **USD 17,733.33 per day** of forgone `CFADS` on a 30/360 basis, which this domain uses as **USD 124,133.33 per calendar week** — the single number that governs almost every decision in it. Seven diligence streams will be commissioned for **USD 1,500,000** of fees; the full close-cost budget will come to **USD 2,709,000**; the model audit will return **34 findings**; and the conditions-precedent register will resolve into five chains whose base close date is 20 weeks away.

KNOWLEDGE AREA 13.1

The diligence streams

Topics: 13.1.1 what diligence is for · 13.1.2 the seven streams and what each uniquely sees · 13.1.3 diligence as an information purchase · 13.1.4 the parallel envelope · 13.1.5 the data room and the diligence trail.

13.1.1 What diligence is for

Definition. Due diligence is the systematic, evidenced examination of a project's technical, financial, legal, tax, insurance, environmental, social and market characteristics, undertaken before commitment, by parties who will state their findings in writing and be identifiable for them.

Three properties distinguish it from ordinary analysis, and each has a practical consequence. **It is adversarial in structure and cooperative in conduct** — the lenders' advisers are paid to find what the sponsor's team has missed or has an interest in not seeing, while everyone depends on the sponsor's cooperation to find anything at all; a diligence process that becomes genuinely adversarial in conduct simply stops producing information. **It is evidenced rather than concluded** — a diligence report's value is its trail from an assertion to a document, which is why "the technical adviser is comfortable" is not a diligence output and a numbered finding against a specific drawing revision is. And **it is time-boxed by a commercial deadline**, which is the property that makes it an economic problem rather than a research problem: unlimited diligence is available at a price nobody will pay, so the real question is always which checks, in what order, inside how long.

What diligence cannot do. It cannot transfer risk. A report identifies a ground condition; it does not pay for it (Domain 11's allocation does, and Domain 12's contract writes it down). It cannot make an unbankable project bankable (Domain 5). And it does not discharge the accountability of the person relying on it — the liability caps of 13.A.1 make that arithmetically obvious. Diligence buys **information**, and information is valuable only if a decision changes as a result. The disciplined test to apply to any proposed piece of work is therefore: what decision would a different answer change, and what is that decision worth? A stream that cannot answer that question is being commissioned for comfort, and comfort is the most expensive thing in a close timetable.

13.1.2 The seven streams and what each uniquely sees

The market has converged on seven streams. The useful way to hold them is not by discipline but by **the class of error each is uniquely able to detect** — because that is what makes them non-substitutable.

Stream	Core question	The error only this stream finds
Technical	Will the asset do what the model assumes, for as long as it assumes?	Design, capacity, availability, degradation and lifecycle-cost assumptions that no financial reviewer can challenge
Financial	Does the model implement the documents, and do the historical accounts support the forecast?	Definitional and structural defects in the model (KA 13.2)
Legal	Is the security package enforceable, are the contracts as summarised, is title good?	Enforceability, perfection and consent gaps behind a correct-looking contract summary
Tax	What cash tax will this structure actually pay, in this jurisdiction, on this timetable?	Withholding, thin capitalisation, loss-relief limits, indirect tax on construction
Insurance	Is the programme in place, and does it match the loss the documents assume is insured?	Exclusions, deductibles, sub-limits and the gap between a broker's slip and a lender's endorsement requirement
Environmental and social	What obligations, liabilities and stakeholder exposures attach to the site and the works?	Legacy contamination, resettlement, biodiversity and consent-condition obligations that survive close
Market	Will the revenue exist at the price and volume the model assumes?	Demand, price and competitive assumptions — the largest single source of realised project failure

Two structural points about the set. **The streams overlap at exactly the places where projects fail**, and the overlaps must be managed rather than eliminated: the technical adviser's availability assumption is the financial model's revenue driver and the offtake contract's liquidated-damages trigger, so three streams touch one number and the risk is that each assumes another owns it. The countermeasure is a **cross-stream interface list** — a short register naming every assumption that more than one adviser touches, with a single named owner (Toolkit 13.T.1). And **the streams are not equally reliable**. Technical, legal and tax diligence examine facts that exist; market diligence examines a forecast that does not, which is why a market consultant's report is evidence about a method rather than evidence about the future, and why Domain 7's stress-testing discipline is the proper use of it.

13.1.3 Diligence as an information purchase

The shape of the arithmetic is PML-AI's gate economics (PML-AI KA 3.3.1), reapplied to a diligence stream. A stream costs a fee and, if it sits on the critical path, elapsed time; in return it converts some probability of a costly post-close discovery into a cheap pre-close correction.

```
Expected cost without the stream = p × C
Expected cost with the stream    = fee + (weeks on the critical path × cost of delay)
                                   + p × [ d × F + (1 − d) × C ]
Net value                        = the difference
```

where `p` is the probability that a material issue of this stream's class exists, `d` the probability the stream detects it, `C` the cost if it reaches close undetected and `F` the cost of correcting it pre-close. Rearranging for the detection rate at which the stream just breaks even gives the single most useful expression in this Knowledge Area:

```
Breakeven detection rate  d* = (fee + priced elapsed time) ÷ [ p × (C − F) ]
```

Read it in words: **the breakeven detection rate is the diligence spend divided by the expected avoidable loss.** Everything about the economics of diligence follows from the fact that the numerator contains elapsed time.

WORKED EXAMPLE

Worked example 13.1.3 — Kestrel's seven streams, priced.

1. **Setup.** Illustrative parameters an organisation must estimate from its own transaction record; the arithmetic, not the parameters, is transferable. Kestrel commissions seven streams with the fees and durations below. `p`, `C` and `F` are the sponsor's own estimates from comparable closings; `C` for the financial stream is Domain 6's independently derived **2,691,071** for a material model error reaching close (KA 6.4.3), which anchors the scale of the others. Detection is assumed at **d = 0.80** for every stream. Cost of delay **124,133.33 per week**.

Stream	Fee	Weeks	p	C	F	p × C	p × (C − F)
Technical	260,000	8	0.30	6,200,000	400,000	1,860,000.00	1,740,000.00
Financial	180,000	5	0.35	2,691,071	60,000	941,874.85	920,874.85
Legal	420,000	10	0.25	9,000,000	250,000	2,250,000.00	2,187,500.00
Tax	150,000	6	0.20	4,800,000	300,000	960,000.00	900,000.00
Insurance	60,000	4	0.15	1,400,000	120,000	210,000.00	192,000.00
Environmental and social	240,000	12	0.22	7,500,000	900,000	1,650,000.00	1,452,000.00
Market	190,000	9	0.28	5,600,000	500,000	1,568,000.00	1,428,000.00
Total	**1,500,000**	**54**				**9,439,874.85**	

1. **Formula.** As above, per stream and in aggregate. The aggregate residual expected cost is `Σ p × [d × F + (1 − d) × C]`. Elapsed time is priced **once for the envelope** when streams run in parallel and **once per stream** when they run in series.

2. **Substitution.** Aggregate `Σ p × C = 9,439,874.85`. Residual `Σ p × [0.80 × F + 0.20 × C] = 2,383,574.97`. Parallel envelope `12 × 124,133.33 = 1,489,600`. Serial sequencing `54 × 124,133.33 = 6,703,200`.
3. **Result. Parallel:** expected cost `1,500,000 + 1,489,600 + 2,383,574.97 = 5,373,174.97`; net value **+USD 4,066,699.88**. **Serial:** expected cost `1,500,000 + 6,703,200 + 2,383,574.97 = 10,586,774.97`; net value **−USD 1,146,900.12**. The difference is **5,213,600**, exactly `42 weeks × 124,133.33`.
4. **Interpretation.** The same seven reviews, the same seven fees, the same detection rates — worth **+4,066,700** run inside a twelve-week envelope and **−1,146,900** run one after another. Not one hour of diligence differs between the two readings; **the entire 5,213,600 is scheduling.** That is the domain's headline result and it reframes the standard argument: a sponsor negotiating adviser fees is negotiating over 1,500,000 while the sequencing decision is worth three and a half times as much, and a lender insisting on serial sequencing "so each stream can rely on the last" is destroying value it cannot see because its own cost of delay is zero. The per-stream breakeven detection rates make the point sharper still. Inside the envelope, where only the fee is charged, the highest breakeven of the seven is **insurance at 31.25 %** and the lowest is **market at 13.31 %** — every stream pays comfortably, because a competent adviser detects far more than a third of the issues in their own discipline. Sequenced serially, the same streams need **72.02 %** (technical), **75.95 %** (legal), **86.95 %** (financial), **91.54 %** (market) and **99.42 %** (tax) — that last figure being a review that must be effectively infallible to be worth holding — while **environmental and social (119.12 %)** and **insurance (289.86 %)** cannot pay at any detection rate, because their elapsed time alone exceeds the loss they could avoid. Three cautions belong with this result. First, `d` is the least defensible parameter in the set and nobody has measured it on a sample of one project; the sensitivity of the answer to it is the reason the breakevens, not the net values, are the output a leader should carry. Second, `C` has a long right tail that expected value averages away — a legal finding can be existential rather than expensive — so a stream may be worth running below its breakeven precisely because the distribution is not symmetric. Third, several streams are **conditions precedent** rather than choices (KA 13.3): the model audit and the lenders' legal and technical reviews will be run whatever the arithmetic says, and the arithmetic's job is then to argue about **timing and scope**, which is exactly the conclusion Domain 6 reached about the audit alone.

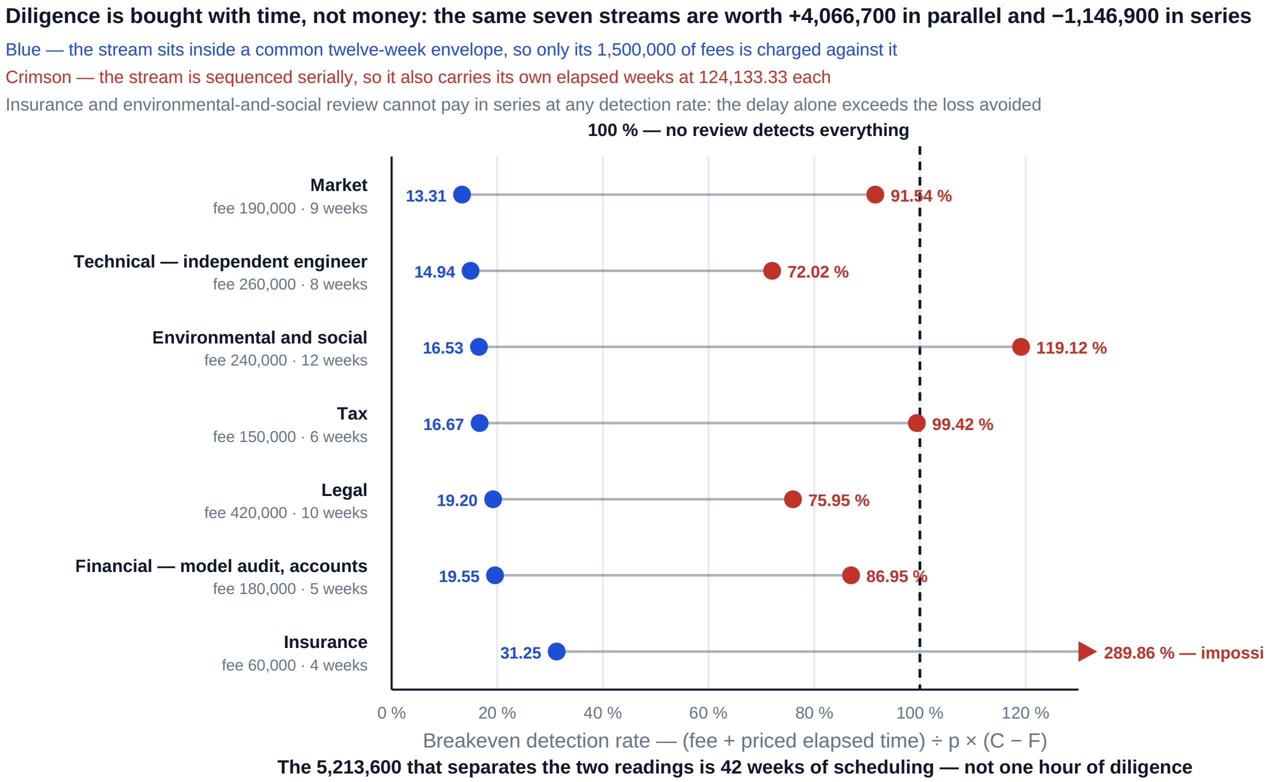

Fig 13.1.1 — The breakeven detection rate of each diligence stream, in parallel and in series

13.1.4 The parallel envelope

If elapsed time dominates the economics, then **designing the envelope is the diligence manager's principal task**, and it is a scheduling problem with four levers.

Genuine parallelism. Most streams do not depend on each other's outputs and can be commissioned on the same day. The dependencies that do exist are few and specific: the model audit needs a frozen model, the insurance adviser needs the technical report's loss scenarios, the tax adviser needs the final structure chart. Everything else is convention.

Staged deliverables instead of a single report. A stream that reports once, at week twelve, is a stream whose findings cannot be acted on inside the envelope. A stream that issues a red-flag memorandum at week three, a draft at week eight and a final report at week twelve produces the same information and gives the transaction nine weeks to respond. This is the single highest-return change available in most close processes and it costs nothing.

Scope discipline. The envelope's length is set by the longest stream — here environmental and social at twelve weeks. Shortening that stream shortens the envelope; shortening any other shortens nothing. Compression effort spent off the binding stream is spent for no return, which is the same lesson KA 13.3 draws about conditions.

Honest reliance boundaries. Parallelism creates the risk that two advisers assume different values for the same input, so each stream's report must state the inputs it took from another stream and their version. Without that, parallel diligence produces a set of internally consistent reports that are mutually inconsistent — the failure mode Domain 6's input-provenance rule exists to prevent, appearing at transaction level.

The residual honest point: parallelism has a cost of its own. Running seven streams at once requires management bandwidth the sponsor may not have, and a red-flag finding at week three against six streams already in flight produces expensive rework. The right conclusion is not "always parallel" but **"parallel by default, sequenced only where a dependency is real and priced when it is"** — and the price is 124,133.33 a week.

13.1.5 The data room and the diligence trail

A **data room** is the controlled repository through which the sponsor discloses documents to advisers and lenders, and its quality is a leading indicator of everything else: a project whose data room is complete, indexed and version-controlled is a project whose sponsor has the discipline the finance documents will require for the next twelve years. Three disciplines earn their place. **A disclosure index that is itself a document** — numbered, dated, listing what was disclosed and when — because it later determines what a lender was told, which matters if a representation is challenged. **Version control with supersession marked, not deleted** — a superseded traffic forecast left in the room without a supersession note is the exact failure Domain 6 (Case study B) priced, and a superseded forecast removed is worse, because the trail disappears. And **a question-and-answer log**, because the answers are representations in substance whether or not they are representations in form.

The **diligence trail** is what remains after close: reports, the disclosure index, the Q-and-A log, the finding registers and their close-out evidence. It is not archive material — it is the evidence base for the first covenant dispute, the first warranty claim, the first refinancing and the first sale, and Case study B shows what its absence costs.

AI in this KA

Where it earns its place. Data-room triage is now genuinely strong machine work and it attacks exactly the constraint that matters: elapsed time. Three uses are robust. **Completeness checking** — reconciling the data-room index against a required-document list and reporting what is absent, which is a set operation humans perform badly at scale. **Extraction into a comparable register** — pulling parties, dates, tenors, defined terms, termination triggers, caps, indexation formulae and governing law out of several hundred documents into the structured table Domain 5 (KA 5.3) and Domain 12 (KA 12.4.1) both depend on. And **inconsistency flagging** across documents: the same defined term with two definitions, a date that differs between the offtake and the concession, a capacity figure that differs between the technical report and the model.

Where it must not go. It must not produce a conclusion in any stream. Whether security is perfected, whether withholding applies, whether a consent condition survives close, whether an exclusion defeats an insured loss — each is a professional opinion in a named jurisdiction with a named accountable author, and a plausible machine answer to any of them is more dangerous than no answer, because it will be relied on without a liability position behind it. It must not be presented as diligence to a lender: a machine extraction is an input to a stream, never the stream. And it must not be used to shorten the envelope by removing a stream — the 13.1.3 arithmetic shows machine pre-checks are additive to review, the same conclusion Domain 6 reached about model scanning (KA 6.4.4).

Verification, concretely. Sample the extraction: for a stated sample size — twenty documents or ten per cent, whichever is larger — reconcile every extracted field to the source document and record the error rate, by field type, in the diligence file. Never accept an extraction whose error rate has not been measured on this transaction's own documents, because extraction accuracy varies by document quality and this project's documents are the only relevant population. Every flagged inconsistency is confirmed by a human before it enters a finding register. And the estimated parameters in the stream economics — `p`, `d`, `C` — are recorded with their basis, because a machine-assisted process that changes `d` without measuring it has changed the answer without evidence. **AI proposes; the professional verifies, decides and remains accountable.**

KEY TERMS — KA 13.1

Term	Meaning
Due diligence	Evidenced pre-commitment examination of a project, reported in writing by identifiable authors.
Diligence stream	One discipline's review — technical, financial, legal, tax, insurance, E&S, market.
Breakeven detection rate `d*`	`(fee + priced elapsed time) ÷ [p × (C − F)]`; the diligence spend divided by the expected avoidable loss.
Parallel envelope	The period inside which streams run concurrently, so elapsed time is priced once, not per stream.
Red-flag memorandum	An early partial deliverable that lets findings be acted on inside the envelope.
Cross-stream interface list	Register of assumptions touched by more than one adviser, each with a single named owner.
Data room / disclosure index	The controlled disclosure repository and the numbered, dated record of what was disclosed.

SAMPLE MCQS — KA 13.1

MCQ 13.1-A `[13.1.3 · Application]` A diligence stream costs a 260,000 fee, `p` = 0.30, `C` = 6,200,000 and `F` = 400,000. Run inside a parallel envelope so that it adds no elapsed time, its breakeven detection rate is: - A. 13.98 % - B. 14.94 % - C. 4.19 % - D. 72.02 %

Rationale: `260,000 / [0.30 × (6,200,000 − 400,000)] = 260,000/1,740,000 = 14.94 %`. A divides by `p × C` and omits the pre-close correction cost `F`, overstating the avoidable loss and so understating the breakeven; C divides the fee by `C` alone and ignores `p`; D is the same stream's breakeven when sequenced serially with eight weeks of delay priced in.

MCQ 13.1-B `[13.1.4 · Analysis]` Seven diligence streams with 1,500,000 of fees and durations totalling 54 weeks are worth **+4,066,700** in a twelve-week parallel envelope and **−1,146,900** sequenced serially, at a cost of delay of 124,133.33 a week. The correct reading is: - A. the streams are marginal and some should be dropped - B. the entire 5,213,600 difference is the 42 weeks of sequencing, so the sequencing decision matters roughly three and a half times more than the fee negotiation - C. the fees are too high and should be reduced by 5,213,600 - D. serial sequencing is preferable because each stream can rely on the last

Rationale: `42 × 124,133.33 = 5,213,600` and the fees, probabilities and detection rates are identical in both readings (13.1.3). A misreads a scheduling result as a scope result; C is arithmetically impossible against 1,500,000 of fees; D describes a real but rare dependency that 13.1.4 requires to be priced rather than assumed.

MCQ 13.1-C `[13.1.3 · Evaluation]` On the parameters above, insurance diligence sequenced serially has a breakeven detection rate of 289.86 %. The professional conclusion is: - A. insurance diligence should be abandoned - B. the insurance adviser is overpriced - C. insurance diligence must sit inside the envelope, where its breakeven is 31.25 % — the finding is about sequencing, not about the stream - D. the calculation is invalid because a probability cannot exceed one

Rationale: A breakeven above 100 % says the configuration cannot pay, not that the review is worthless; inside the envelope the same stream breaks even at 31.25 % (13.1.3). B is wrong because the 60,000 fee is the smallest of the seven — 496,533.33 of delay is what makes it impossible. D mistakes an impossible-configuration signal for an arithmetic error.

MCQ 13.1-D `[13.1.2 · Analysis]` Three parties rely on the same plant availability figure: the technical adviser who derives it, the financial model that earns revenue on it and the offtake agreement that pays damages against it. The specific diligence control for this is: - A. asking each adviser to confirm they are comfortable - B. a cross-stream interface list naming every multi-adviser assumption with one named owner and one version - C. running the streams serially so each can rely on the previous report - D. relying on the model audit to catch any inconsistency

Rationale: The overlap is where projects fail, and the countermeasure is single ownership of the shared assumption (13.1.2). A produces no evidence; C costs 124,133.33 a week for a control a register achieves free; D is out of scope — the audit tests the model against documents, not whether two advisers assumed the same thing.

SELF-CHECK — KA 13.1

1. State the breakeven detection rate in words. — The diligence spend divided by the expected avoidable loss, `p × (C − F)`; elapsed time is part of the spend.
2. Why does shortening the tax stream not shorten the envelope? — The envelope is set by the longest stream (environmental and social, twelve weeks); compression off the binding stream returns nothing.
3. What does diligence buy, and what does it not? — Information, and only where a decision would change; it never transfers risk and never discharges the reliant party's accountability.

KNOWLEDGE AREA 13.2

Model audit

Topics: 13.2.1 scope and what the audit is not · 13.2.2 findings by class · 13.2.3 the class-one findings priced · 13.2.4 closing findings out.

13.2.1 Scope and what the audit is not

Domain 6 (KA 6.4.3) defined the model audit, established the governance apparatus that makes it possible and **priced it** — finding that its elapsed time rather than its fee decides whether it pays, that its breakeven error rate falls from 20.10 % to 8.05 % when it moves early, and that a weak audit is worse than none below a 48.81 % detection rate. That arithmetic is not repeated here. What Domain 13 adds is the transaction-side treatment: what the audit's scope must cover, how its output must be read, and what happens to a financing when a finding lands.

The four scope layers, in the order a competent auditor works through them and in the reverse order of how often they are the problem. **Arithmetic** — do the formulae compute what they claim? This is the layer everybody expects and the layer that almost never contains the expensive error. **Structure** — do the mechanics work under stress: does the waterfall behave when a tier fails, does the tax computation behave when losses exhaust, does the schedule close at zero? **Documents** — does the model implement the finance documents: the `CFADS` definition clause by clause, the covenant test dates, the reserve mechanics, the interest convention, the distribution tests? And **assumptions and provenance** — is each input traced to a source document of the correct version, and are the conventions internally consistent (Domain 3's timing and day-count discipline, Domain 6's convention list)?

What the audit is not. It is not a valuation opinion, and it is not an endorsement of the assumptions: an auditor confirms that a tariff escalation of 2.5 % is implemented correctly and sourced to clause 14.3, not that 2.5 % is right. It is not a substitute for the technical, market or tax streams — it tests whether the model implements what those streams concluded. It is not a sponsor deliverable; it is a lenders' control, and a sponsor who treats it as a compliance exercise has misunderstood who is buying. And, critically, **it is not a warranty**: its liability position is capped and its reliance is addressed (13.A.1), so a signed audit report reduces the probability of an error and does not transfer the cost of one.

13.2.2 Findings by class, not totalled

A model audit reports findings. The universal reporting failure — in transaction status reports, in credit papers and in sponsor board packs alike — is to **total them**. "Thirty-four findings, of which thirty-one closed" is a sentence with no information content, because the findings are not commensurable. The remedy is a **class scheme applied before any count is quoted**, and the classes must be defined by consequence, not by the auditor's sense of importance:

Class	Definition	What it changes	Who must see it
Class 1 — fundamental	Changes a financing **conclusion**: debt capacity, a covenant outcome, the funding plan's sufficiency, or a distribution test	The transaction	Credit committee and sponsor board, individually and by name
Class 2 — material	Changes a reported output beyond the agreed materiality threshold, without changing a conclusion	The reported numbers	Deal team, listed individually with quantified effect
Class 3 — presentational or procedural	Changes no number: labelling, formatting, unused rows, documentation, print ranges, non-output hardcoding	Nothing	Reported in aggregate only

The class scheme requires a stated materiality threshold, and it should be expressed in the metric the transaction is judged on rather than in currency. Kestrel's was set at **0.01× of** `DSCR` — roughly 50,000 of annual `CFADS` at its debt-service level — which is the kind of threshold a credit committee can reason about, unlike "USD 100,000" whose significance depends on which line it lands in.

Kestrel's 34 findings resolved as 3 Class 1, 7 Class 2 and 24 Class 3. Three consequences that generalise:

- **Class 3 is 70.6 % of the count and 0 % of the impact.** Any metric driven by count is therefore driven by the class that does not matter, which is why "31 of 34 closed" — **91.2 %** — is a progress figure that can be true while every finding that changes the transaction remains open. The honest metric is **class-weighted**: three of three Class 1 findings open is 0 % of the impact closed, whatever the count says.
- **The seven Class 2 findings netted to +18,000 of year-one** `CFADS` — they very nearly cancelled. That is a coincidence and must never be reported as a control: each was wrong, each was corrected, and the netting tells the reader nothing except that this particular set of errors happened to point in opposing directions. A model whose Class 2 findings net to zero is not a model with no Class 2 findings.

- **Only Class 1 findings are re-audited.** Correcting a fundamental finding changes dependent lines, so the corrected model must be re-run against the audit's own test set and the golden answers re-verified (Domain 6's regression discipline). Class 2 corrections are checked; Class 3 corrections are accepted on the modeller's confirmation. Treating all three the same is how a close timetable loses two weeks to formatting.

13.2.3 The class-one findings priced

WORKED EXAMPLE

Worked example 13.2.3 — what three findings did to Kestrel's financing.

1. **Setup.** The model audit of Kestrel's financing model returns three Class 1 findings. **F-01, definitional:** the facility defines `CFADS` after the annual maintenance-reserve top-up of **260,000**; the model struck it after debt service instead, so reported `CFADS` of 6,384,000 overstates the defined figure. **F-02, convention:** the debt-service schedule was built as an **annuity due** (payments in advance) rather than in arrears as the facility provides — Domain 3 (KA 3.2.1) named this the commonest annuity mistake; here it is inside a financing model. **F-03, funding plan:** the six-month DSRA of **2,504,818** (Domain 10, KA 10.3.2) appears as an operating-period cash outflow but not in the uses of funds, so the funding plan does not fund it. The lender's requirement is a base-case `DSCR` of **1.30×**; `AF(0.06, 12) = 8.383844`.
2. **Formula.** Corrected `CFADS` = reported − the misplaced line. `DSCR` = `CFADS` ÷ debt service. `AF` for an annuity due = `AF(r, n) × (1 + r)`. Debt capacity = `CFADS ÷ target DSCR × AF(r, n)` (Domain 10, KA 10.1.2).
3. **Substitution.** F-01: `6,384,000 − 260,000`, then `÷ 5,009,635.23`. F-02: `AF_due = 8.383844 × 1.06 = 8.886875`; instalment `42,000,000 ÷ 8.886875`; then `6,384,000 ÷ that instalment`. Capacity: `6,124,000 ÷ 1.30 = 4,710,769.23`, `× 8.383844`.
4. **Result.**

Finding	The error's effect on what was reported	The corrected figure
F-02 convention	Instalment **4,726,071** — understated by **283,564**; `DSCR` reported as **1.3508**, apparently clearing the 1.30× requirement	Instalment **5,009,635.23**; `DSCR` **1.2743**
F-01 definition	`CFADS` overstated by 260,000	`CFADS` **6,124,000**; `DSCR` **1.2224**
F-01 + F-02 together	Reported `DSCR` **1.2958** on the wrong instalment	Debt capacity at 1.30× **USD 39,494,354** against 42,000,000 mandated — a resize of **USD 2,505,646**
F-03 funding plan	Uses of funds understated by **2,504,818**	The DSRA must be funded, from debt, equity or first operating cash

1. **Interpretation.** Take F-02 first, because it is the finding that should frighten a reader. A single timing convention — payments in advance instead of in arrears — understated the instalment by **283,564**, which flattered the `DSCR` from 1.2743 to **1.3508** and made the model report a project that comfortably cleared a 1.30× requirement it in fact fails. Nothing about the model looked wrong: the schedule closed at zero, total principal equalled the loan and every check block was green, because an annuity due is a perfectly valid amortisation of 42,000,000 — just not the one the facility documents. **This is the characteristic shape of an expensive model error: internally consistent, arithmetically correct and wrong against the document**, which is

precisely why the audit's document layer matters more than its arithmetic layer and why a machine scanner would not have found it (Domain 6, KA 6.4.4). Second, note the **direction of the combination**. F-01 and F-02 both flatter the ratio, so they compound rather than offset: together they take the reported 1.3508 down to a true 1.2224, and the debt the project can actually carry at the lender's requirement falls from Domain 10's **41,171,123** on uncorrected `CFADS` to **39,494,354** — a resize of **2,505,646** that must be funded as equity or not funded at all. Third, **F-03 is the finding that would have caused an event, not an argument**. A `DSCR` shortfall is negotiated; a funding plan that does not fund a mandatory reserve produces a failed condition at first drawdown or a covenant breach in the first operating year, and it is the class of error that reaches close most often, because sources-and- uses statements are reviewed for arithmetic (they always balance) rather than for completeness. Finally, the **professional caution about ordering**: had the auditor reported "34 findings, 3 fundamental" without quantifying them, the credit committee would have received a status update rather than a decision paper. The quantification — 2,505,646 of resize and 2,504,818 of unfunded reserve, on a mandate of 42,000,000 — is what converts an audit into a transaction event, and producing it is the sponsor's job as much as the auditor's.

13.2.4 Closing findings out

A finding is closed when the model has been corrected, the correction has been verified by the auditor, and the **dependent consequences have been followed through**. The third limb is where close processes fail, because a Class 1 correction propagates: correcting F-01 changes `CFADS`, which changes every coverage ratio, the sculpting profile if there is one, the distribution forecast, the lock-up test dates and the equity return. A finding closed in the model and not in the credit paper, the term sheet and the base case is not closed.

The disciplines that make close-out real: **one register, owned by one named person**, with each finding's class, quantified effect, correction, verifier and date; **a re-run of the golden-answer set after every Class 1 correction**, with any changed figure explained before it is accepted; **a final auditor's letter that lists the findings and their disposition** rather than issuing a clean opinion that conceals what was found; and **a stated position on anything not closed** — because findings are sometimes accepted rather than corrected, and an accepted finding with a written rationale is professional whereas an accepted finding recorded as "closed" is not.

AI in this KA

Where it earns its place. Domain 6 (KA 6.4.4) covered workbook scanning, version diffing, adversarial test generation and documentation drafting, and priced a pre-check at 8,000 with a 55 % detection rate. The transaction-side addition here is narrower and genuinely useful: **mapping the finance documents' defined terms onto model lines**, producing a candidate reconciliation of every `CFADS` definition limb to the cell that implements it. That is the input to Toolkit 13.T.2 and it attacks the layer where the expensive errors live.

Where it must not go. It must not assign the class. Classification is a judgment about whether a financing conclusion changes — it requires knowing the lender's requirement, the covenant level and what the credit committee approved — and a machine that grades findings by severity will grade by textual signal, systematically under-rating the quiet definitional finding that resizes the debt. It must not draft the auditor's letter or the certificate of satisfaction. And it must not be credited with detection it did not achieve: F-02 above is a document-conformance error in a structurally valid schedule, exactly the class scanners miss.

Verification, concretely. Every machine-proposed definition-to-line mapping is confirmed against the clause by a named person, with the clause reference recorded. The golden-answer set is re-run after every accepted correction, machine-assisted or not. And the finding register records, per finding, whether it was found by machine, by the auditor or by the sponsor's own review — which is the only way an organisation ever learns what its d actually is.

KEY TERMS — KA 13.2

Term	Meaning
Model audit	Independent review of a model against the transaction documents, its own logic and its arithmetic (Domain 6, KA 6.4.3).
Scope layers	Arithmetic · structure · documents · assumptions and provenance; the expensive errors sit in the last two.
Class 1 / 2 / 3 finding	Changes a financing conclusion · changes a reported output beyond materiality · changes no number.
Materiality threshold	The stated size below which a difference is not a finding; best expressed in the transaction's own metric (0.01× of DSCR).
Class-weighted closure	Progress measured by impact closed, not findings closed.
Finding close-out	Corrected, verified, and every dependent consequence followed through into papers and the base case.

SAMPLE MCQS — KA 13.2

MCQ 13.2-A [13.2.3 · Application] A model builds the debt-service schedule as an annuity due rather than in arrears. On 42,000,000 at 6 % over 12 years (AF = 8.383844) with CFADS of 6,384,000, the reported DSCR and the truth are: - A. reported 1.2743, true 1.3508 - B. reported 1.3508, true 1.2743 - C. reported and true both 1.2743 — the convention does not affect the ratio - D. reported 1.2224, true 1.2743

Rationale: `AF_due = 8.383844 × 1.06 = 8.886875`, instalment `42,000,000/8.886875 = 4,726,071` and `6,384,000/4,726,071 = 1.3508`; the correct in-arrears instalment of 5,009,635.23 gives 1.2743. A reverses the direction — an annuity due has a smaller instalment, so it flatters the ratio; C denies that the denominator moved by 283,564; D is the DSCR after the separate CFADS-definition finding, on the correct instalment.

MCQ 13.2-B [13.2.2 · Evaluation] A model audit returns 34 findings — 3 Class 1, 7 Class 2, 24 Class 3 — and the status report says "31 of 34 closed, 91.2 %". The correct challenge is: - A. the count is wrong - B. the 91.2 % is driven by the 24 Class 3 findings, which are 70.6 % of the count and none of the impact; if the three open findings are the Class 1 findings, 0 % of the impact is closed - C. Class 3 findings should not be reported at all - D. the audit should be re-run

Rationale: Findings are not commensurable, so a count-based metric measures the class that changes nothing (13.2.2). A is not the defect; C loses the aggregate record that shows model hygiene; D confuses a reporting failure with an audit failure.

MCQ 13.2-C [13.2.3 · Application] After correcting CFADS to 6,124,000, debt capacity at a 1.30× requirement over 12 years at 6 % (AF = 8.383844) is closest to: - A. USD 41,171,123 - B. USD 39,494,354 - C. USD 42,000,000 - D. USD 51,342,661

Rationale: `6,124,000/1.30 = 4,710,769.23`, `× 8.383844 = 39,494,354`. A is Domain 10's capacity on the uncorrected 6,384,000; C is the mandated amount, which the calculation rejects by 2,505,646; D omits the coverage divisor entirely (`6,124,000 × 8.383844`).

MCQ 13.2-D [13.2.1 · Analysis] Which audit scope layer would have caught a model that implements a valid amortisation the finance documents do not provide for? - A. arithmetic — the formulae are checked - B. structure — the schedule closes at zero - C. documents — the model is tested against what the facility actually says - D. assumptions and provenance — the inputs are traced

Rationale: An annuity due computes correctly and closes at zero, so layers A and B both pass (13.2.3); only conformance to the document detects it. D would trace the rate and tenor, both of which are right.

SELF-CHECK — KA 13.2

1. Why is a finding count uninformative? — Findings are not commensurable; Class 3 dominates the count and contributes nothing to impact.
2. Which audit layer holds the expensive errors, and why? — Documents and assumptions: a model can be arithmetically perfect and structurally sound while implementing something the facility does not provide for.
3. When is a Class 1 finding closed? — When corrected, verified by the auditor, and followed through into every dependent output, paper and base case.

KNOWLEDGE AREA 13.3

Conditions precedent and documentation

Topics: 13.3.1 the condition set and its categories · 13.3.2 the conjunction with a critical path · 13.3.3 float, criticality risk and where to spend · 13.3.4 the close-cost budget and sources and uses · 13.3.5 waivers, long-stop dates and post-close undertakings.

13.3.1 The condition set and its categories

Definition. A **condition precedent** is a requirement that must be satisfied — and evidenced to the lenders' satisfaction — before an obligation arises, most importantly before first drawdown. Domain 5 (KA 5.A.2) established that the bankability conditions become the CP schedule, that CPs are ordered by dependency rather than importance, that sponsor CPs behave differently from third-party CPs, and that a waived CP is not a satisfied CP. This Knowledge Area computes what that schedule costs.

The condition set divides into five practical groups, and the division matters because each group fails differently. **Corporate and authority** conditions — board and shareholder resolutions, constitutional documents, powers of attorney, authorised-signatory evidence — are within the sponsor group's control, fail through inattention rather than uncertainty, and are therefore forgivable in timetable terms. **Documentary** conditions — executed finance, security, project and direct agreements, and the legal opinions that support them — fail through negotiation, and their duration is a function of how many parties must agree. **Third-party** conditions — permits, licences, land instruments, offtaker and grantor consents, letters of credit from banks that are not the lenders — are outside the group's control and are the standard source of close slippage. **Adviser and process** conditions — the diligence reports of KA 13.1, the model audit of KA 13.2, final credit approvals — have durations the transaction can influence but not compress at will. And **financial** conditions — equity funded or committed on acceptable terms, accounts opened, insurance placed and endorsed, fees paid, the base-case model agreed and locked — are the ones most often discovered late, because they look administrative.

The professional discipline that this Knowledge Area exists to install: **a CP schedule is a dependency network, and it must be built and read as one.** A close timetable drawn as a single bar labelled "conditions precedent — 20 weeks" contains no information about what will actually make the project late.

13.3.2 The conjunction with a critical path

Close occurs when **every** condition is satisfied. That is a conjunction, and conjunctions of uncertain durations behave in a way that timetables systematically misrepresent: the close date is the **maximum** of the chains' realised durations, and the expected maximum is greater than the maximum of the expectations. The gap is not a rounding error; it is often the whole of the contingency a transaction should have carried and did not.

WORKED EXAMPLE

Worked example 13.3.2 — when will Kestrel actually close?

1. **Setup.** Kestrel's 40-odd conditions resolve into five dependency chains. Each chain's base duration is the sum of its links; each carries an independent risk of a single slip, estimated from the sponsor's own record of comparable closings. Cost of delay **124,133.33 per week**.

Chain	Contents	Base duration	Slip probability	Slip if it occurs
A	Board and shareholder resolutions → shareholders' agreement amendment → equity commitments and credit support	9 weeks	0.20	3 weeks
B	Land instrument registration → construction permit → abstraction licence	18 weeks	0.35	6 weeks
C	Diligence reports (12-week envelope) → model audit → final credit approvals	20 weeks	0.30	4 weeks
D	Finance documents agreed → security documents and perfection → legal opinions	17 weeks	0.25	5 weeks
E	Insurance adviser's report → placement and lender endorsements	7 weeks	0.15	2 weeks

1. **Formula.** Base close = `max(base durations)`. Each chain's expected duration = `base + probability × slip`. The close date = `max` over chains of the realised duration; `E[close] = Σ over all 32 outcomes of (probability × maximum duration)`. Cost of slippage = `(E[close] − base close) × cost of delay`.
2. **Substitution.** Base close = **20 weeks** (chain C). Expected durations: A 9.60, B 20.10, C 21.20, D 18.25, E 7.30. Chains A and E can never bind, since even their slipped durations (12 and 9 weeks) fall short of 20; the maximum is therefore decided by B, C and D over eight outcomes.
3. **Result.** `E[close] = 22.4075 weeks` against a **max of expectations of 21.2000**. The close-date distribution is **20 weeks with probability 34.125 %, 22 weeks with 11.375 %** and **24 weeks with 54.500 %**. Expected slip **2.4075 weeks**, costing `2.4075 × 124,133.33 =` **USD 298,851**. Chain C's own expected slip of 1.2 weeks costs **USD 148,960**, so the **conjunction premium is**

USD 149,891 — the expected cost of delay is **2.0062×** the cost of the critical chain's own expected slip.

4. **Interpretation.** Three results, in ascending order of usefulness. The first is the one that changes behaviour: **the probability that this transaction closes on its own stated date is 34.125 %** — and every individual chain owner can truthfully report that their chain is more likely than not to hold its date. That is the arithmetic behind the observation that project financings are almost never early and usually late, and it is not pessimism, incompetence or optimism bias; it is what a conjunction does. The second is the pricing consequence: **the expected cost of the CP schedule is just over double the cost of the critical chain's expected slip**, because any of three near-critical chains can become the binding one. A transaction that provisions for delay by looking at the critical path provisions for roughly half its exposure, and 149,891 is what that omission is worth here. The third is the reporting consequence: the distribution is **bimodal** — 34.1 % at 20 weeks, 54.5 % at 24 weeks, and almost nothing in between — because a slip, if it happens, is a discrete four- to six-week event rather than a smooth drift. The right output for a board is therefore not "expected close 22.4 weeks" but **"one chance in three of 20 weeks, and better than one chance in two of 24"**, which is a sentence people can plan against. The honest caveats are two: the slips are modelled as independent, and they are not — a slow grantor delays the permit chain and the consent inside the documentary chain together, which makes the true `E[close]` later than 22.4075 and the probability of the base date lower than 34.125 %; and single-slip modelling understates the right tail, since a permit that slips six weeks can slip twelve. Both errors point the same way, which is the useful thing to know about them.

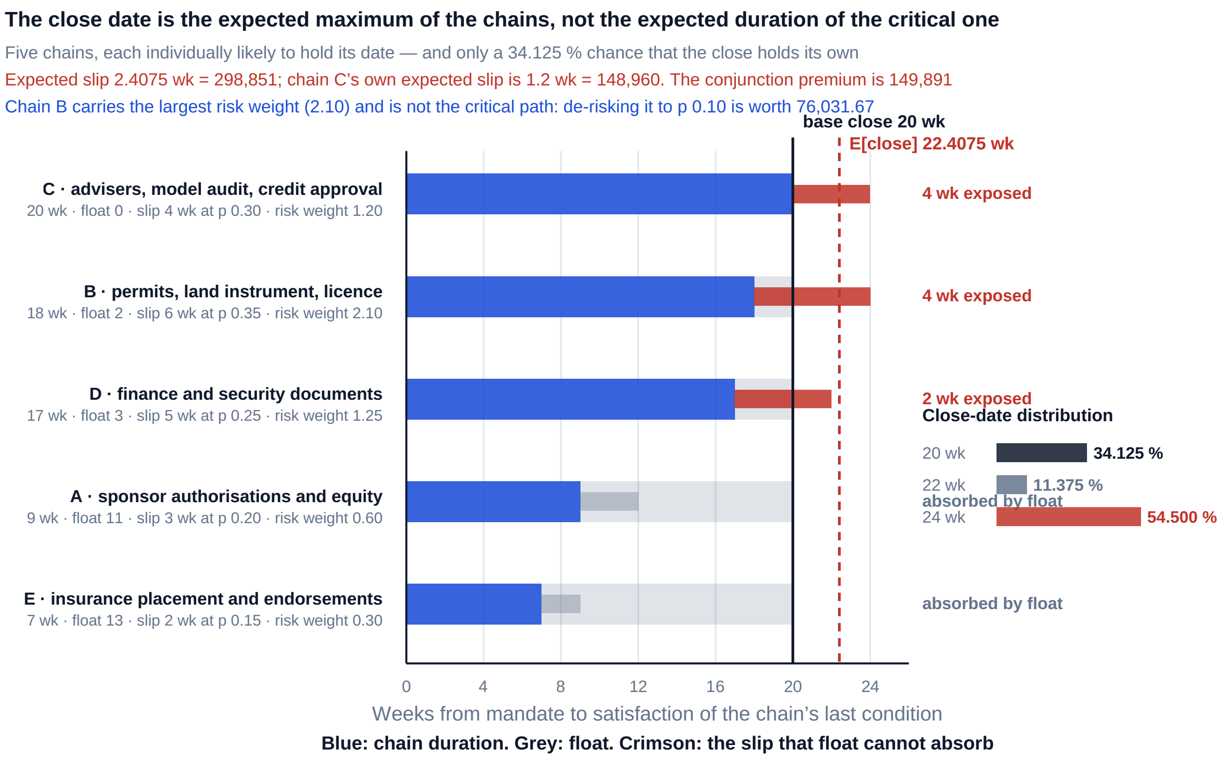

Fig 13.3.1 — Kestrel's conditions-precedent conjunction

13.3.3 Float, criticality risk and where to spend

The chart above contains the management answer, and it is counter-intuitive. **Float is real and computable**: chain A has 11 weeks of it and a 3-week slip, so its slip is absorbed entirely and costs nothing; chain E has 13 weeks of float against a 2-week slip and likewise cannot delay close. Effort spent accelerating either returns zero, however loudly the sponsor's corporate secretariat is being chased.

Criticality risk, not criticality, ranks the work. Weighting each chain by `slip probability × slip size` gives **B 2.10**, **D 1.25**, **C 1.20**, A 0.60 and E 0.30 — and adjusting for float removes A and E entirely. The nominal critical path is chain C. The chain most likely to become critical is **B**, the permit and land chain, which sits two weeks inside the base date and carries the largest slip risk in the register.

WORKED EXAMPLE

Worked example 13.3.3 — what is it worth to de-risk a chain that is not critical?

1. **Setup.** The sponsor can put a dedicated permitting team, local counsel and a pre-agreed submission protocol behind chain B, and estimates this reduces its slip probability from **0.35 to 0.10**. Nothing else changes. Cost of delay 124,133.33 per week.
2. **Formula.** Recompute `E[close]` over the same outcome set with chain B's probability changed; the value of the intervention is the reduction in expected slip cost.
3. **Substitution.** `E[close]` with `p_B = 0.10`, then `(E[close] − 20) × 124,133.33`, compared with the base 298,851.
4. **Result.** `E[close]` falls from **22.4075** to **21.7950 weeks**; expected slip cost falls from **298,851** to **USD 222,819.33**; the intervention is worth **USD 76,031.67**. The probability of closing on the base 20-week date rises from **34.125 %** to **47.250 %**, and the probability of the 24-week outcome falls from **54.500 %** to **37.000 %**.
5. **Interpretation.** The highest-value acceleration available on this transaction is on a chain that is **not** the critical path, and no critical-path analysis would have found it — which is the practical case for modelling the conjunction rather than the longest bar. Two further readings. The 76,031.67 is a **budget**, and it is the number the sponsor should compare a permitting adviser's fee against; below it the intervention pays, above it, it does not, and that is a far more useful procurement position than "permits are important". And the shift in the distribution is worth more to a board than the shift in the mean: moving the probability of the base date from roughly one in three to roughly one in two changes what can honestly be told to an offtaker, a contractor holding a price and an equity committee with a quarter-end. The caution is that reducing `p_B` to 0.10 is itself an estimate, and the honest way to present the result is as a sensitivity — if the intervention halves the risk it is worth about 38,000, if it removes two-thirds of it, about 76,000 — rather than as a single figure implying a precision the parameters do not support.

13.3.4 The close-cost budget and sources and uses

Close costs are the transaction's own price, and they are routinely provided for by a rule of thumb in the financial model — a percentage of the facility labelled "fees" — which is where the error enters. The discipline is a **line-itemised close-cost budget, reconciled to the sources-and-uses statement**, and it separates naturally into what is payable to lenders and what is not.

WORKED EXAMPLE

Worked example 13.3.4 — Kestrel's close-cost budget, and the hole it found.

1. **Setup.** The diligence fees of KA 13.1 plus the financing, sponsor-side and perfection costs. Domain 6's funding envelope (KA 6.2.1) provided **840,000** for "arrangement and financing fees" at 2.0 % of the facility and **1,800,000** for capitalised development costs, both funded at close.
2. **Formula.** Total close costs = Σ line items. Share of debt raised = total ÷ debt. Reconciliation = the envelope's provision − the itemised total.
3. **Substitution and result.**

Close-cost item	USD
Lenders' technical adviser	260,000
Lenders' legal counsel	420,000
Model auditor	180,000
Insurance adviser	60,000
Environmental and social consultant	240,000
Market adviser	190,000
Tax adviser	150,000
Diligence subtotal (KA 13.1)	1,500,000
Arrangement fee, 1.20 % of 42,000,000	504,000
Agency and account-bank set-up	40,000
Sponsor's legal counsel	380,000
Sponsor's financial adviser, success fee 0.45 %	189,000
Security perfection, registration and notarial costs	96,000
Total close costs	**2,709,000**

Total close costs are **6.45 % of debt raised** and **4.515 %** of the 60,000,000 envelope. Only **544,000** — the arrangement and agency fees — is payable to the lenders; **2,165,000** goes to third parties and the sponsor's own advisers. Against the envelope's provision of `840,000 + 1,800,000 = 2,640,000`, the budget is **69,000 short**: the fee line has **296,000 spare** (2.0 % of the facility against 544,000 of actual lender fees) while the development-cost line is **365,000 short** of the 2,165,000 it must carry. 4. **Interpretation.** Start with the number nobody expects: **6.45 % of debt raised**. That is not an error and it is not extravagance — it is what it costs to project-finance a 60,000,000 asset, and it is the strongest quantitative argument in this domain for why small projects are aggregated, standardised or financed on balance sheet instead. Split the total into a **fixed component** — the seven adviser fees, the sponsor's counsel, agency set-up and perfection, **2,016,000**, which barely varies with deal size — and a **proportional component** of **1.65 %** (the 1.20 % arrangement fee plus the 0.45 % success fee). Close cost as a share of debt is then `2,016,000 ÷ D + 1.65 %`, which is **6.45 %** at 42,000,000, **2.994 %** at 150,000,000 and **2.130 %** at 420,000,000. A facility must reach about **USD 149,333,333** before total close costs fall below **3.0 %** of debt raised — an upper bound on the scale economy, since the fixed component does grow with complexity, but the shape is right and the implication is real. Second, the **reconciliation is the finding**. A 2.0 % "financing fees" line implicitly assumed transaction costs were the arranger's fee; they are 20 % of it. The 69,000 shortfall is small in itself and must be absorbed by the balancing contingency line, taking it from Domain 6's **3,645,403** to **3,576,403**, or from 7.59 % to **7.45 %** of the EPC price — still inside the band a lender would expect, so no financing consequence follows here. But the mechanism is the one that hurts on larger transactions, and there is a second-order effect the model must pick up: a cost moved from a spend-profile line to a close-funded line carries interest for the whole construction period rather than part of it, so the reallocation also adds capitalised interest, which the same balancing line must absorb. That is why a close-cost budget belongs **inside** the model rather than beside it. Third, the **effective cost of debt**. Netting the 544,000 of lender fees from proceeds against the unchanged instalment of 5,009,635.23 gives an all-in rate of **6.2386 %** on the facility; netting the whole 2,709,000 gives **7.2376 %** — **123.8 basis points** above the 6.00 % headline. Both figures are correct and they answer different questions, which is exactly the discipline Domain 9 (KA 9.3.1) established for comparing tranches: **6.2386 % is the cost of the lender's money and 7.2376 % is the cost of raising it**, and only costs that differ between the routes being compared belong in a route comparison. Quoting the second as though the lender charged it is as misleading as quoting the headline 6.00 % as though closing were free.

13.3.5 Waivers, long-stop dates and post-close undertakings

Three instruments handle a condition that will not be satisfied in time, and all three are priced decisions rather than administrative fixes.

A **waiver** is the lenders' agreement to close without a condition, usually for a fee and often with a compensating term. Domain 5's rule stands — a waived CP is not a satisfied CP — and the professional requirement is that every waiver carries **a deadline, a named owner, evidence requirements and a stated consequence of non-satisfaction**. Case study B shows the same instrument used well and badly on one transaction.

A **long-stop date** (or availability-period expiry) is the date after which commitments lapse if close has not occurred. Its economic content is that it converts slippage into a **cliff**: up to the long-stop, delay costs the cost of delay; past it, the transaction needs re-approval, and re-approval prices at today's market rather than the market of the mandate. A leader should know the distance between `E[close]` and the long-stop in the same terms as the coverage headroom of Domain 10 — as a number of weeks, on the dashboard. Kestrel's `E[close]` of 22.4075 weeks against a 32-week long-stop leaves **9.5925 weeks** of margin, and the 24-week outcome that carries 54.5 % probability leaves 8.

A **post-close undertaking** converts a condition into a covenant: the obligation survives close with a deadline and, on breach, the consequences of Domain 10's covenant regime. This is the right answer for conditions that are genuinely mechanical and merely slow — a registration awaiting a registry, a final certificate awaiting an inspector — and the wrong answer for anything whose outcome is uncertain, because it converts a diligence question into a default risk.

AI in this KA

Where it earns its place. CP registers are long, repetitive and derived from documents, which makes three uses strong. **Extraction** of the condition list from the facility agreement's schedule into a register with clause references — the version-controlled starting point most transactions build by hand. **Dependency and duration tracking** — maintaining the register as evidence arrives, flagging conditions whose evidence is stale or whose owner has not reported, and recomputing the conjunction arithmetic of 13.3.2 whenever a duration estimate moves, which is the only way that arithmetic stays current inside a live close. And **evidence completeness checks** against each condition's stated documentary requirement.

Where it must not go. It must not determine that a condition is satisfied. Satisfaction is a determination the lenders make, on advice, against a documentary standard, and a machine's view that a condition "appears satisfied" has no standing and considerable capacity to mislead a transaction into a certificate it cannot support. It must not draft or sign the certificate of satisfaction. It must not decide whether a condition is waivable, or what a waiver should cost — that is a negotiation informed by counsel. And it must not generate the slip probabilities: `p` and the slip sizes come from an organisation's own closing record, and a plausible machine estimate would put a fabricated number at the centre of the close timetable.

Verification, concretely. The extracted register is reconciled clause by clause against the schedule by a named person before it is used, because a missing condition is invisible in a register that looks complete. Every satisfaction determination is recorded with the human who made it and the evidence relied on. The conjunction arithmetic is recomputed and reviewed at each milestone, and the estimated slip parameters are recorded with their basis and their source transaction count — so that the register carries, in the open, how much evidence sits behind its own forecast.

KEY TERMS — KA 13.3

Term	Meaning
Condition precedent (CP)	Requirement satisfied and evidenced before an obligation arises, principally first drawdown.
CP chain	A dependency sequence of conditions; its duration is the sum of its links.
Conjunction premium	The excess of the expected close slip cost over the cost of the critical chain's own expected slip.
Float	Weeks by which a chain may over-run before it affects close; absorbs slip at no cost.
Criticality risk	`slip probability × slip size`, net of float; ranks where acceleration is worth buying.
Close-cost budget	Line-itemised transaction cost, reconciled to sources and uses; fixed component plus a proportional fee rate.
Long-stop date	Date after which commitments lapse; converts slippage into re-approval at today's market.
Post-close undertaking	A condition converted into a dated covenant surviving close.

SAMPLE MCQS — KA 13.3

MCQ 13.3-A `[13.3.2 · Analysis]` Five CP chains have expected durations of 9.60, 20.10, 21.20, 18.25 and 7.30 weeks against a 20-week base close. The expected close date is: - A. 21.2000 weeks — the longest chain's expected duration - B. 22.4075 weeks — the expected maximum of the chains - C. 20.0000 weeks — the base close, since each chain is more likely than not to hold its date - D. 76.4500 weeks — the sum of the chains

Rationale: Close is a conjunction, so the date is the expected maximum, which exceeds the maximum of the expectations (13.3.2). A takes the maximum of the expectations, the standard error; C mistakes each chain's individual likelihood for the joint one — the probability of the base date is 34.125 %; D adds chains that run in parallel.

MCQ 13.3-B `[13.3.3 · Evaluation]` Chain B (18 weeks, slip 6 weeks at probability 0.35) is not the critical path; chain C (20 weeks, slip 4 at 0.30) is. Reducing chain B's slip probability to 0.10 moves `E[close]` from 22.4075 to 21.7950 weeks at a cost of delay of 124,133.33 a week. The intervention is worth: - A. nothing — chain B is not on the critical path - B. USD 76,031.67, and it is the highest-value acceleration available - C. USD 744,800 — six weeks of delay avoided - D. USD 298,851 — the whole expected slip cost

Rationale: `(22.4075 − 21.7950) × 124,133.33 = 76,031.67` (13.3.3). A is the critical-path fallacy the conjunction exists to correct; C prices the full slip as if it were certain and unavoidable; D is the total expected slip cost, which the intervention reduces but does not eliminate.

MCQ 13.3-C `[13.3.4 · Application]` Close costs total 2,709,000 on a 42,000,000 facility funding a 60,000,000 envelope, of which 544,000 is payable to lenders. Close costs as a share of debt raised are: - A. 4.515 % - B. 6.45 % - C. 1.295 % - D. 2.00 %

Rationale: `2,709,000/42,000,000 = 6.45 %`. A divides by the 60,000,000 envelope rather than the debt; C counts only the fees payable to lenders; D is the model's original rule-of-thumb provision, which the itemised budget shows to be 20 % of the true figure.

MCQ 13.3-D `[13.3.4 · Analysis]` With a fixed close-cost component of 2,016,000 and a proportional component of 1.65 % of debt, the facility size at which total close costs fall to 3.0 % of debt raised is closest to: - A. USD 67,200,000 - B. USD 149,333,333 - C. USD 122,181,818 - D. USD 90,300,000

Rationale: `2,016,000/(0.030 − 0.0165) = 2,016,000/0.0135 = 149,333,333`. A divides the fixed component by 3.0 % and ignores the proportional component; C divides by 0.0165; D applies 3.0 % to the wrong base.

MCQ 13.3-E `[13.3.5 · Recall]` A condition converted into a post-close undertaking is appropriately handled that way when: - A. its outcome is uncertain and diligence could not resolve it - B. it is mechanical and merely slow, so the only open question is timing - C. the sponsor does not wish to disclose it - D. the lenders have waived it without a deadline

Rationale: Post-close undertakings convert timing risk into a dated covenant; they convert outcome risk into default risk, which is why A is the case for not closing rather than for an undertaking (13.3.5). D describes the defect Case study B prices.

SELF-CHECK — KA 13.3

1. Why is the expected close date later than the longest chain's expected duration? — Close is a conjunction: the expected maximum exceeds the maximum of the expectations, here by 1.2075 weeks and 149,891.
2. How is acceleration effort ranked? — By criticality risk (slip probability × slip size) net of float, not by position on the critical path.
3. What two questions does the effective cost of debt answer differently? — What the lender's money costs (6.2386 % on lender fees alone) and what raising it costs (7.2376 % on the full close-cost budget).

KNOWLEDGE AREA 13.4

Syndication and financial close

Topics: 13.4.1 the three routes to a syndicate · 13.4.2 the fee architecture and the arranger's yield · 13.4.3 the failed sell-down · 13.4.4 market flex · 13.4.5 signing, close and funds flow.

13.4.1 The three routes to a syndicate

Definition. Syndication is the process by which the arranging bank or banks distribute a facility to other lenders, so that no single institution holds a position larger than its appetite or its limits allow. Three routes, distinguished entirely by **who carries the risk that the market does not turn up**:

- **Underwritten.** One or more banks commit the full amount and then sell down. The borrower has certainty of funds from the mandate; the arranger carries the distribution risk and charges for it (13.4.3 prices what that risk costs when it goes wrong).
- **Best-efforts.** The arranger undertakes to use reasonable efforts to raise the amount but does not commit it. Cheaper, and the borrower carries the risk that the book does not fill — which in a project financing means the risk that close does not happen at all, after the whole close-cost budget has been spent.
- **Club.** A small group of banks, assembled before documentation, each committing its final hold directly. No distribution risk, no underwriting fee, and — the point most often missed — no arranger with an economic interest in flexing terms, because nobody has a position to sell.

The choice is a trade between **certainty of funds**, **cost** and **flexibility**, and the sponsor's decision variable is usually the first. A transaction with a long-stop date, a contractor holding a price and a grantor's timetable cannot tolerate a book that does not fill, and will pay underwriting economics for certainty. A repeat sponsor with relationship banks and no hard deadline will club the deal and keep the fee. Kestrel's 42,000,000 sits at the size where either works, and this Knowledge Area follows it as an underwritten facility because that is the structure whose arithmetic must be understood.

13.4.2 The fee architecture and the arranger's yield

Fees in a syndicated facility are not one number. The standard architecture, and the reason for each part:

Fee	Paid to	Purpose
Praecipium (arrangement fee proper)	The arranger, exclusively	Structuring, documentation and underwriting risk; not shared with participants
Participation (pool) fee	Every lender pro rata on final allocations	Compensation for committing capital
Underwriting fee	The underwriter(s)	Explicit price of the commitment where separated from the praecipium
Commitment fee	Every lender, periodically	The undrawn balance's carrying cost (Domain 9, KA 9.2.4)
Agency fee	The facility and security agents	Ongoing administration; annual, not a close cost

WORKED EXAMPLE

Worked example 13.4.2 — who earns what on Kestrel's syndication.

1. **Setup.** A total arrangement fee of **1.20 %** on 42,000,000, of which **0.25 %** is praecipium retained by the arranger and the balance is a participation pool paid pro rata on final allocations. The arranger's target final hold is **15,000,000**; two participants take **13,500,000** each.
2. **Formula.** Total fee = facility × 1.20 %. Praecipium = facility × 0.25 %. Pool = total − praecipium, paid at `pool ÷ facility` on each lender's allocation. Fee yield = fee earned ÷ final hold.
3. **Substitution.** `42,000,000 × 1.20 % = 504,000`; `× 0.25 % = 105,000`; pool `399,000`, a rate of **0.95 %**. Arranger `105,000 + 0.95 % × 15,000,000`; each participant `0.95 % × 13,500,000`.
4. **Result.** Arranger **USD 247,500**, a fee yield of **1.650 %** on its 15,000,000 hold. Each participant **USD 128,250**, a yield of **0.950 %**. Total `247,500 + 2 × 128,250 = 504,000` ✓. The arranger earns **70.0 basis points** more per dollar held than the banks it sold to.
5. **Interpretation.** The 70 basis points is the price of two things participants do not do: structure and document the transaction, and stand behind the amount before anyone else has agreed to it. Reading it as a margin is the mistake — it is a **risk premium on distribution**, and 13.4.3 shows how quickly it is consumed. Three practical consequences for a sponsor. The fee split is **not the sponsor's business but it is the sponsor's information**: a praecipium that is a large share of the total tells you the arranger is being paid mostly for underwriting, which in turn tells you how it will behave if the book is slow — it will reach for flex. The **allocation policy is negotiable**, and a sponsor with a view on which lenders it wants in its syndicate for the next twelve years (Domain 10's waiver arithmetic: a tight syndicate consents in weeks) should express that view before the book opens, not when allocations are announced. And **fee yield, not fee, is how a lender sees the transaction**: a participant asked to take 13,500,000 at 0.950 % is comparing that yield against every other asset on its desk, which is why fees rise in a busy market even when margins do not move.

13.4.3 The failed sell-down

WORKED EXAMPLE

Worked example 13.4.3 — what an underwriter loses when the market does not turn up.

1. **Setup.** The arranger underwrites 42,000,000 with a target final hold of **15,000,000**, intending to place 27,000,000. The market takes only **20,000,000**. The arranger's internal charge for holding a project-loan position above its target hold is **90 basis points a year** over the facility's 12-year life. `AF(0.06, 12) = 8.383844`.
2. **Formula.** Residual hold = facility − amount placed. Excess = residual − target hold. Present value of the carry = excess × charge × `AF(r, n)`. Breakeven excess = the praecipium ÷ (charge × `AF`).
3. **Substitution.** Residual `42,000,000 − 20,000,000 = 22,000,000`; excess `22,000,000 − 15,000,000 = 7,000,000`; carry `7,000,000 × 0.0090 × 8.383844`.
4. **Result.** Excess hold **7,000,000**; present value of the carry **USD 528,182.17** against total fees earned of 247,500 — a **net loss of USD 280,682.17** on a transaction the arranger won, documented and closed successfully. The excess hold that exactly consumes the **praecipium** is **USD 1,391,565**, or **3.31 %** of the facility; the excess that consumes the arranger's **entire fee** is **USD 3,280,118**.
5. **Interpretation.** An underwriter can lose money on a deal that closes, and the loss threshold is startlingly low: a **3.31 % over-hold wipes out the whole underwriting premium**. That single figure explains most of the behaviour a sponsor encounters in a syndication and should be read as intelligence rather than as bad faith. It explains why arrangers insist on **market flex** (13.4.4) — flex is the instrument that lets them re-price to fill the book rather than carry the position. It explains why they **soft-sound the market before committing**, and why a sponsor who restricts pre-mandate sounding pays for that restriction in the underwriting fee. It explains why arrangers prefer **clubs in difficult markets**, where the economics of underwriting simply do not work. And it explains why an arranger's own hold level is worth watching: a bank that has just over-held on three transactions is a bank whose appetite for the fourth is about limits, not about the project. The professional caution for a sponsor is not to over-read the arithmetic — the 90 basis points is an internal capital charge, not a cash cost, and a bank may be entirely content to hold a good project-finance asset at scale. But when it is not, this is the arithmetic behind its position, and a sponsor negotiating flex without knowing it is negotiating blind against someone who does.

13.4.4 Market flex

Definition. Market flex is the arranger's contractual right, granted in the mandate or commitment letter, to change specified terms — most commonly to increase the margin or fees, and sometimes to shorten tenor, alter amortisation or reallocate between tranches — to the extent necessary to achieve a successful syndication. It is normally capped, sometimes ordered (fees before margin), and occasionally subject to a reverse flex that returns the benefit to the borrower if the book is oversubscribed.

WORKED EXAMPLE

Worked example 13.4.4 — what flex is worth, and what it costs.

1. **Setup.** Kestrel's mandate grants flex of up to **50 basis points** on the margin. Facility 42,000,000, 12 years, base rate 6.00 %, `AF(0.06, 12) = 8.383844`, `CFADS` 6,384,000, `DSCR` covenant **1.20×**.
2. **Formula.** Present value of the flex, from the lenders' side, = facility × flex × `AF(r, n)`. Its effect on the borrower = the new instalment `facility ÷ AF(r_flexed, n)` and the resulting `DSCR`. The flex the covenant survives solves `CFADS ÷ 1.20 = facility ÷ AF(r*, n)` for `r*`.
3. **Substitution.** PV of flex `42,000,000 × 0.0050 × 8.383844`. Flexed: `AF(0.065, 12) = 8.158725`; instalment `42,000,000 ÷ 8.158725`; `DSCR = 6,384,000 ÷` that. Ceiling: `6,384,000 ÷ 1.20 = 5,320,000`; `AF` required `42,000,000 ÷ 5,320,000 = 7.894737`; solve for `r*`.
4. **Result.** The 50 basis points of flex is worth **USD 1,760,607** in present value — **3.4933×** the entire 504,000 arrangement fee. Exercised in full, the instalment rises by **138,228** to **5,147,862.98** and the base-case `DSCR` falls from **1.2743** to **1.2401**. The 1.20× covenant survives up to a rate of **7.1138 %**, a flex ceiling of **111.4 basis points**, so the contractual 50 points consumes **44.9 %** of the available headroom.
5. **Interpretation. Flex is the most valuable term in the mandate letter, and it is almost never the term sponsors negotiate hardest.** At 3.49 times the arrangement fee, a sponsor who trades 10 basis points of flex for a 25 basis-point fee reduction has given away roughly 352,000 of present value to save 105,000 — and will describe the outcome as a fee win. The second reading is the covenant one, and it is the reason flex belongs in the financial model before the mandate is signed: **flex is a coverage event in disguise.** The fully-flexed `DSCR` of 1.2401 still clears the 1.20 covenant, so this transaction survives its own flex — but it does so with 44.9 % of its rate headroom consumed and nothing left for the interest-rate exposure Domain 11 (KA 11.3) priced separately, which is why the two must be stressed together rather than in sequence. The disciplined position a sponsor should hold: agree flex, **cap it at a level the model demonstrates the covenant survives with margin**, insist on ordering so fees flex before margin (a fee is a one-off, a margin is 8.38 years of annuity factor), require a **reverse flex** so that a strong book benefits the borrower, and require the arranger to evidence market conditions before exercising. What a sponsor must not do is treat flex as boilerplate, or model the base case only. And the honest note on the other side: an arranger without flex, facing the 13.4.3 arithmetic, will either not underwrite or will price the underwriting as though flex had already been exercised — so a sponsor who wins the flex negotiation outright may simply have moved the cost into the fee.

13.4.5 Signing, close and funds flow

The final mechanics, in the order they occur, and the specific way each is got wrong.

Signing is not close. Documents are executed; conditions may still be outstanding. The gap between the two is where the long-stop date lives, and a transaction that announces "financial close" on signing has misdescribed its own position — to its board, and sometimes to a market.

Satisfaction and certification. Each condition is evidenced to the agent, who confirms satisfaction against the documentary standard. The document to insist on is a **numbered certificate of satisfaction cross-referring to each condition and the evidence delivered**, because it is the record that determines, later, what was and was not satisfied. Waived and deferred conditions appear on it as such, never as satisfied (13.3.5).

The funds flow. A single statement showing every payment at close: sources by lender and by sponsor, uses by payee, net movements per account, and the day's timing. Kestrel's day-one requirement was **2,640,000** — fees and capitalised development costs, funded 70/30 as **1,848,000** of debt and **792,000** of equity (Domain 6, KA 6.2.1) — which is precisely the figure KA 13.3.4 showed must actually carry **2,709,000**. **The funds flow is where a close-cost budget error becomes visible, and it becomes visible on the day it is too late to fix.** The disciplines: reconcile the funds flow to the close-cost budget line by line before close, not on the day; confirm account signatories and payment instructions through a verified channel; and have the agent, not the sponsor's finance team, own the statement.

Equity first, or pro rata. Whether equity funds ahead of debt, alongside it or behind an equity-support instrument is a structuring decision with real consequences for the sponsor's exposure and the lenders' comfort (Domain 9, KA 9.1) — and whatever the answer, it must be the same in the documents, the model and the funds flow. **Conditions subsequent and the first drawdown.** Availability usually begins at close, but the first drawdown has its own conditions: the drawdown notice, the certification of spend, the confirmation that no default has occurred. Domain 14 takes the transaction from here.

AI in this KA

Where it earns its place. Three uses. **Term-sheet and commitment-letter normalisation** across several banks into one comparable table, which Domain 9 (KA 9.2) established and which extends naturally to flex provisions, fee structures and conditions — a genuinely tedious comparison that machines do well and humans do inconsistently. **Investor and allocation analysis**: which institutions have taken comparable paper in this sector, tenor and jurisdiction, assembled from disclosed transaction records, as a starting list for the book. And **funds-flow reconciliation**: matching a close-cost budget line by line to a funds-flow statement and reporting the differences, which is exactly the check that 13.4.5 says is done too late.

Where it must not go. It must not state market appetite or price as though it were a quote — an indicative margin generated from historical transactions is a research output, and presenting one to a sponsor as market feedback misrepresents to whoever relies on it. It must not draft or approve the certificate of satisfaction, the funds flow or any payment instruction: close-day payment instructions are the single highest-value fraud target in a project financing, and no automated channel should originate or alter one. And it must not be used to decide allocations, which are a relationship judgment with consequences over the facility's whole life.

Verification, concretely. Every normalised term-sheet field is checked against the source letter by a named person before the comparison is used, because a mis-normalised flex cap is worth 1,760,607 here. Payment instructions are verified out of band, by voice, against a previously-confirmed record — an operational control that no model output substitutes for. And the funds-flow reconciliation is signed by the agent and the sponsor's finance director, whatever produced the first draft.

KEY TERMS — KA 13.4

Term	Meaning
Underwritten / best-efforts / club	Arranger commits and sells down · uses reasonable efforts only · pre-assembled group each taking its final hold.
Praecipium	The arranger's exclusive share of the arrangement fee; the price of structuring and underwriting.
Participation (pool) fee	Fee paid pro rata on final allocations to every lender.
Final hold	The position a lender intends to retain after syndication.
Market flex	Contractual right to change specified terms to achieve syndication; capped, often ordered, sometimes reversible.
Signing vs financial close	Execution of documents vs satisfaction of all conditions and availability of funds.
Certificate of satisfaction	Numbered record of each condition, its evidence and its disposition, including waivers.
Funds flow	The statement of every payment at close, reconciled to the close-cost budget.

SAMPLE MCQS — KA 13.4

MCQ 13.4-A `[13.4.2 · Application]` A 42,000,000 facility carries a 1.20 % arrangement fee, of which 0.25 % is praecipium; the balance is a pool paid pro rata on allocations. The arranger's final hold is 15,000,000. It earns: - A. USD 180,000 - B. USD 247,500 - C. USD 105,000 - D. USD 504,000

Rationale: `105,000 + 0.95 % × 15,000,000 = 247,500`, a yield of 1.650 % against participants' 0.950 %. A applies the full 1.20 % to the hold and omits the praecipium's exclusivity; C is the praecipium alone; D is the whole fee, most of which is paid away to participants.

MCQ 13.4-B `[13.4.3 · Evaluation]` An arranger underwrites 42,000,000 with a 15,000,000 target hold, places 20,000,000, and charges itself 90 basis points a year on the excess over 12 years (`AF` = 8.383844). Against fees of 247,500 the outcome is: - A. a profit of 184,500 - B. a loss of USD 280,682.17 — the 528,182.17 carry on a 7,000,000 excess exceeds the whole fee - C. a loss of 508,500, since the entire residual hold is charged - D. break-even

Rationale: `7,000,000 × 0.0090 × 8.383844 = 528,182.17`; `247,500 − 528,182.17 = −280,682.17` (13.4.3). A charges the carry for one year only (63,000); C charges the full 22,000,000 residual rather than the excess over target hold.

MCQ 13.4-C `[13.4.4 · Application]` Fifty basis points of market flex on a 42,000,000 facility over 12 years at 6 % (`AF` = 8.383844) is worth, in present value: - A. USD 210,000 - B. USD 1,760,607 - C. USD 2,520,000 - D. USD 504,000

Rationale: `42,000,000 × 0.0050 × 8.383844 = 1,760,607`, some 3.49 times the arrangement fee. A is one year of the flex, undiscounted; C sums twelve years without discounting; D is the arrangement fee the sponsor negotiates instead.

MCQ 13.4-D `[13.4.4 · Analysis]` With `CFADS` of 6,384,000, a 42,000,000 facility over 12 years and a 1.20× `DSCR` covenant, the margin increase the covenant just survives is: - A. 50.0 basis points - B. 111.4 basis points - C. 25.0 basis points - D. unlimited, since the covenant is tested on `CFADS`

Rationale: Maximum instalment `6,384,000/1.20 = 5,320,000`, requiring `AF = 7.894737`, which solves at 7.1138 % — 111.4 basis points above 6.00 %. A is the contractual flex cap, which consumes 44.9 % of that headroom; D forgets that debt service is the denominator and rises with the rate.

MCQ 13.4-E `[13.4.5 · Recall]` A transaction has executed all its documents but two third-party consents remain outstanding. Its correct description is: - A. financially closed - B. signed, not closed — close occurs when every condition is satisfied and funds are available - C. closed subject to conditions subsequent - D. in default

Rationale: Signing and close are distinct, and the gap is where the long-stop date operates (13.4.5). C misuses a term for post-close obligations; D confuses an unsatisfied condition with a breach of an existing obligation.

SELF-CHECK — KA 13.4

1. What does the arranger's 70 basis-point yield differential pay for? — Structuring, documentation and distribution risk — the risk 13.4.3 shows a 3.31 % over-hold consumes entirely.
2. Why is flex worth more than the arrangement fee? — A margin is an annuity over the facility's life: 50 basis points is 1,760,607 of present value against a 504,000 one-off fee.
3. What is the one thing a funds flow must be reconciled to, and when? — The line-itemised close-cost budget, before close — not on the day, when the 2,640,000 provision meets a 2,709,000 requirement.

Advanced topics — Domain 13

13.A.1 Reliance, duty of care and the arithmetic of a liability cap

A diligence report is only useful to a party that can rely on it, and reliance is a legal arrangement rather than a natural consequence of reading. Three mechanisms carry it. **Addressing** — a report addressed to the sponsor gives lenders nothing; the standard remedy is a report addressed to the lenders and the agent, or a **reliance letter** extending a duty of care to named additional parties, usually for a fee and usually on the original engagement's terms. **Assignment and transferability**, which matters for syndication: a report on which only the original syndicate may rely constrains sell-down, and a secondary purchaser two years later needs a re-addressed report or its own. And **the liability cap**, which is where the arithmetic bites.

Advisers' engagement terms almost universally cap liability, commonly at a multiple of fees. Take Kestrel's technical stream: a fee of **260,000** and a cap at **three times fees** gives **780,000** against a `C` of **6,200,000** — the cap covers **12.58 %** of the exposure the stream exists to address. Across all seven streams, caps of three times fees total **4,500,000** against an aggregate expected exposure of **9,439,874.85**, or **47.67 %** — and that comparison flatters the position, because it sets the sum of maximum recoveries against the sum of expected losses, and because recovery requires proving negligence, not merely showing the finding was missed.

The conclusion is the one 13.1.1 stated and this arithmetic proves: **diligence buys information, not indemnity.** Two professional consequences follow. Negotiating a cap upwards is worth doing and is not a risk-transfer strategy — moving Kestrel's technical cap from three to five times fees adds 520,000 of theoretical recovery against a 6,200,000 exposure. And **the reliance package is a diligence deliverable in its own right**, to be listed on the CP schedule with the reports themselves; a transaction that discovers at syndication that its reports cannot be re-addressed has found a condition it cannot satisfy.

13.A.2 Vendor diligence, reliance and the price of time

A **vendor due diligence** report is commissioned by a seller and made available to bidders, typically with reliance extended for a fee. It is the standard mechanism in secondary infrastructure sales and increasingly in primary financings where a sponsor pre-packages diligence to compress a lender's timetable. The question it poses is exact: is it better to rely on someone else's diligence quickly, or to run your own slowly?

WORKED EXAMPLE

Worked example 13.A.2 — own diligence or reliance?

1. **Setup.** A buyer faces an aggregate probability **p = 0.65** that at least one material issue exists, with `C` = **9,000,000** if it reaches completion undetected and `F` = **600,000** to resolve it beforehand. **Own diligence:** 1,500,000 of fees, **12 weeks**, detection `d` = **0.90**. **Reliance on vendor diligence:** a 240,000 reliance fee, **3 weeks** of confirmatory work, detection `d` = **0.50** — lower because the scope was the seller's, the questions were the seller's and the report's cap is shared across every relying party.
2. **Formula.** Expected cost = fee + weeks × cost of delay + `p × [d × F + (1 − d) × C]`. The breakeven cost of delay is the weekly figure at which the two routes cost the same.
3. **Substitution.** Own residual `0.65 × (0.90 × 600,000 + 0.10 × 9,000,000) = 936,000`; reliance residual `0.65 × (0.50 × 600,000 + 0.50 × 9,000,000) = 3,120,000`. Breakeven cost of delay `[(1,500,000 + 936,000) − (240,000 + 3,120,000)] ÷ (3 − 12)`.
4. **Result.** At **zero** cost of delay, own diligence is better by **USD 924,000**: the 1,260,000 of fees saved does not cover the 2,184,000 of detection lost. The **breakeven cost of delay is USD 102,666.67 per week**. At Kestrel's actual **124,133.33**, reliance wins: total expected cost **3,732,400** against **3,925,600**, an advantage of **USD 193,200**.
5. **Interpretation.** The decision has no general answer and one general rule: **it is decided by the price of time, and the breakeven is computable.** An organisation that always runs its own diligence is implicitly asserting a low cost of delay; one that always relies is asserting a high one; neither has done the arithmetic. Three cautions, all about the parameters rather than the sums. The **0.50 detection rate is the weakest number here** and it is doing most of the work — it should be estimated from the vendor report's actual scope against the buyer's own question list, not assumed. The reliance route's **liability position is systematically worse** (13.A.1): a cap shared among relying parties may be worth a fraction of its face amount, which the expected-value arithmetic does not see at all. And in a **competitive process** the cost of delay is not 124,133.33 a week but the entire value of the opportunity, in which case reliance wins by a margin no amount of detection can overturn — which is why vendor diligence exists, and why a seller's decision to commission it is a decision about the sale price.

13.A.3 The reviewer's closing eye

Invariants to test on any diligence and close package. **Every diligence stream has a stated `p`, `d`, `C` and `F` with a recorded basis**, and its breakeven detection rate is on the page (13.1.3). **The envelope is drawn and the binding stream identified**, with the elapsed cost priced once, not per stream — and any serial dependency justified in writing at 124,133.33 a week (13.1.4). **Every assumption touched by more than one adviser has one named owner and one version**

(13.1.2). **The model audit's findings are classified before they are counted**, with a stated materiality threshold in the transaction's own metric, and no progress metric is quoted that is not class-weighted (13.2.2). **Every Class 1 finding is quantified in DSCR and debt-capacity terms**, and its correction is followed through into the credit paper, the term sheet and the base case (13.2.3, 13.2.4). **The CP register is a dependency network, not a list**, with a duration, an owner, an evidence requirement and a verifier per condition; float is computed per chain; and the close date is stated as the expected maximum with its distribution, not as a single date (13.3.2). **Acceleration effort is ranked by criticality risk net of float**, and the value of de-risking the highest-weighted chain is computed before any fee is paid for it (13.3.3). **The close-cost budget is line-itemised, split into fixed and proportional components, expressed as a share of debt raised, and reconciled to sources and uses and to the funds flow** — before close (13.3.4, 13.4.5). **The effective cost of debt is quoted twice**, on lender fees and on total close costs, with the question each answers stated (13.3.4). **Every waived or deferred condition carries a deadline, an owner, an evidence requirement and a consequence**, and appears as waived on the certificate of satisfaction (13.3.5). **Market flex is capped at a level the model demonstrates the coverage covenant survives with margin**, stressed jointly with interest-rate exposure, and its present value is stated beside the arrangement fee (13.4.4). **Reliance and transferability are on the CP schedule**, and every liability cap is expressed as a percentage of the exposure it addresses (13.A.1). And **every AI-assisted extraction has a measured error rate on this transaction's own documents**, with the sample size stated and a named human verifier (13.1, 13.2, 13.3, 13.4).

Industry variations — Domain 13

- **Contracted power and renewables.** Diligence is standardised and comparatively fast: technology is proven, the offtake is a known form, and the binding stream is usually grid connection and land consents rather than technical review. Model audits are routine, findings are typically Class 2 and below, and the close timetable is dominated by permit chains — so the conjunction arithmetic of 13.3.2 matters more than the stream economics of 13.1.3.
- **Merchant power, commodities and industrial.** Market diligence dominates and cannot be compressed: the report is about a method for forecasting prices, so the diligence question is the consultant's methodology and the stress cases it supports rather than its central case. Syndication is harder, flex is wider, and lenders size on stressed cases, which makes the fully-flexed coverage test of 13.4.4 the binding constraint at mandate stage.
- **Transport concessions.** Traffic and revenue diligence is the single largest stream and often the longest, so it sets the envelope; grantor-side conditions add a chain nobody in the sponsor group controls; and the documentary chain lengthens because concession, direct agreement and finance documents must be negotiated three-cornered. Expect a lower probability of the base close date than 34 % and provision accordingly.
- **Water and regulated utilities.** Environmental and social diligence is frequently the binding stream — abstraction, discharge, biodiversity and community consents — as it is for Kestrel, and regulatory reset cycles put a tariff-determination date into the CP schedule that no amount of effort moves. Legacy contamination on brownfield sites is the characteristic Class 1 E&S finding.
- **Digital infrastructure.** Diligence is tenant-credit and power-availability work as much as technical work; asset lives are short so lifecycle and technology-obsolescence assumptions carry unusual weight; and close timetables are compressed by tenant commitment dates, which raises the cost of delay and therefore favours reliance and vendor diligence (13.A.2). Case study B is drawn from this sector.

- **Social infrastructure and availability PPPs.** Documentation is closest to standardised, which compresses the documentary chain and shifts the binding constraint onto public-sector approval chains with fixed committee calendars — the governance-latency arithmetic of PML-AI (KA 3.3) applied to a close timetable. Diligence fees as a share of a small facility are the highest of any sector, making the scale arithmetic of 13.3.4 decisive: these are the transactions that are aggregated or standardised or do not get financed.

Case study — Domain 13: the audit that resized the debt (water / desalination)

Situation. Kestrel Water SPC entered diligence with a mandate for 42,000,000 of senior debt at 6.0 % over 12 years against a 60,000,000 envelope, a reported base-case `DSCR` of **1.3508** and a credit committee requiring **1.30×**. Seven diligence streams were commissioned for 1,500,000 of fees. The sponsor's first close plan sequenced them, on the arranger's preference that each stream should be able to rely on the last.

What happened. The sequencing decision was challenged with the 13.1.3 arithmetic: the same seven streams, the same fees and the same detection rates were worth **+4,066,699.88** inside a twelve-week envelope and **−1,146,900.12** run serially, the 5,213,600 difference being 42 weeks at 124,133.33. Two dependencies were real — the model audit needed a frozen model and the insurance adviser needed the technical loss scenarios — and both were accommodated inside the envelope with staged deliverables. The envelope was set at twelve weeks by the environmental and social stream and the close date at 20 weeks, and the conjunction arithmetic put the probability of holding it at **34.125 %**, with an expected close of **22.4075 weeks** and an expected slip cost of **298,851** against chain C's own **148,960**. The sponsor funded a dedicated permitting effort on chain B — which was not the critical path — reducing its slip probability from 0.35 to 0.10, worth **76,031.67** and lifting the probability of the base date to **47.250 %**.

The model audit returned **34 findings: 3 Class 1, 7 Class 2, 24 Class 3.** Two of the three fundamental findings compounded. The debt-service schedule had been built as an **annuity due**, understating the instalment by **283,564** and flattering the `DSCR` from 1.2743 to the reported 1.3508 — a model that was arithmetically correct, closed at zero, and did not implement the facility. The `CFADS` definition placed the **260,000** maintenance-reserve top-up above `CFADS`; the model placed it below, so true `CFADS` was **6,124,000** and the true `DSCR` **1.2224**. Debt capacity at the committee's 1.30× was therefore `6,124,000 ÷ 1.30 × 8.383844 =` **39,494,354**, against Domain 10's 41,171,123 on the uncorrected figure and the 42,000,000 mandated — a resize of **2,505,646**. The third finding was worse in kind if not in size: the six-month DSRA of **2,504,818** appeared as an operating outflow but not in the uses of funds, so the funding plan did not fund a mandatory reserve.

Separately, the itemised close-cost budget came to **2,709,000 — 6.45 %** of debt raised, of which only **544,000** was payable to lenders — against the model's provision of 2,640,000, a **69,000** shortfall absorbed by the contingency line (3,645,403 to **3,576,403**, 7.59 % to **7.45 %** of the EPC price).

How it resolved. The facility closed at **USD 39,494,354** of senior debt — an instalment of **4,710,769.23** and a base-case `DSCR` of exactly **1.3000** on corrected `CFADS` — with equity rising from 18,000,000 to **20,505,646**, gearing moving from 70/30 to **65.82/34.18** and debt-to-equity from 2.33:1 to **1.926:1**. The DSRA was funded from equity at commercial operations rather than from the construction envelope. Close costs fell by **41,343** on the smaller proportional base, to **2,667,657** — which, on the smaller facility, is **6.7545 %** of debt raised against the original 6.4500 %, because the fixed 2,016,000 does not shrink when the debt does. The base case was re-run and the golden-answer set re-verified before the credit paper was finalised.

What the domain teaches here. Three things, in order of how often they are got wrong. **The expensive model error is a document error, not an arithmetic error** — an annuity due is a valid amortisation and would have passed every check block, every scanner and every reasonable review that did not read the facility. **Diligence value is destroyed by sequencing, not by fees** — the 5,213,600 that separated the two close plans was more than three times the entire diligence budget and would never have appeared in a fee negotiation. And **the numbers that matter are the ones that change a conclusion**: three of thirty-four findings resized the debt by 2,505,646 and exposed an unfunded 2,504,818, while the other thirty-one, closed and reported as 91.2 % progress, changed nothing at all.

Case study B — Domain 13: the same waiver, used well and badly (digital infrastructure)

Situation. A 180,000,000 facility for a colocation data centre reached its long-stop date with two conditions unsatisfied. **CP-14** was the registration of a land instrument, complete in substance and awaiting a registry whose published processing time was 210 days. **CP-22** was a third-party interface consent from the operator of an adjacent utility corridor, which had been requested, acknowledged and not progressed. The anchor tenant's commitment expired in eleven weeks and the sponsor's cost of a day's delay to close, in forgone contracted revenue against a fixed lease term, was **240,000**.

What happened. Both conditions were waived into post-close undertakings at the same meeting, on the same paper, for the same fee. **CP-14 was waived well.** It carried a 90-day deadline, a named owner in the sponsor's legal team, a documentary standard (the registered instrument, certified), and a consequence: a **25 basis-point margin step-up** until satisfaction. Registration in fact took the registry's full 210 days, so the undertaking ran **120 days late**. The cost was the waiver fee of **0.25 % of 180,000,000 = 450,000** plus the step-up of `180,000,000 × 0.0025 × 120/360 =` **150,000**, a total of **600,000**. The counterfactual — waiting to close until registration completed — was **210 days × 240,000 = 50,400,000**. **The waiver was worth 49,800,000**, and it was the correct decision by a margin of eighty-three to one.

CP-22 was waived badly. Because it was mechanical in the sponsor's view and small in the lenders', it was recorded as waived with no deadline, no consequence and no owner — a single line in the certificate of satisfaction reading "waived". It fell out of every register within a month. Twenty-six months later, on a refinancing, the incoming lenders' legal diligence found the corridor consent outstanding and treated it as a title and access defect. Curing it — a negotiated easement, a survey, and a payment to the corridor operator now negotiating with a counterparty that had no alternative — cost **3,400,000**, **5.667 times** the entire cost of the CP-14 waiver, and delayed the refinancing by a quarter.

How it resolved. The sponsor's post-mortem changed one artefact rather than one policy: the certificate of satisfaction was rebuilt so that no condition could be marked waived without a deadline, an owner, an evidence standard and a consequence, and the waived-condition register became a standing item on the same monthly report as the covenant dashboard (Domain 10, Toolkit 10.T.2) — so that a waived condition was reported in the same place as a covenant until it was satisfied.

What the domain teaches here. The instrument was not the problem. **A waiver is a priced decision and here it was overwhelmingly right** — 600,000 against 50,400,000 — and a project finance leader who refuses to waive on principle will destroy far more value than one who waives carelessly. What distinguished the two conditions was entirely the **discipline attached to the waiver**: deadline, owner, evidence standard, consequence. Domain 5's rule that a waived CP is not a satisfied CP is not an accounting nicety; it is the difference between 600,000 and 4,000,000, and the interval over which the difference emerges is two years, which is why nobody is present to learn the lesson at the time.

Executive perspective — Domain 13

What a project finance director cannot delegate in this domain:

- **The sequencing decision.** Whether diligence runs in parallel or in series is worth **5,213,600** on Kestrel — three and a half times the entire diligence budget — and it will be decided by advisers' scheduling preferences unless the director decides it. Own the envelope, name the binding stream, and require every serial dependency to be justified at 124,133.33 a week.
- **The `CFADS` definition and the model's conformance to it.** Domain 10 made this a standing obligation; here it acquires a price. Two document-conformance findings moved Kestrel's reported `DSCR` from 1.3508 to a true 1.2224 and its debt capacity by 2,505,646, and neither was an arithmetic error. Read the definition, then read the model line that implements it.
- **The finding classification, before any count is reported.** The director's question on receiving a model audit is never "how many findings?" but "which findings change a conclusion, and by how much?" A 91.2 % closure metric with three Class 1 findings open is a report that must be sent back.
- **The close date as a distribution.** One chance in three of the stated date, and better than one in two of four weeks later, is what a board, an offtaker and a contractor holding a price need to be told. A single-date close timetable is a forecast the director knows to be 34 % likely and has chosen to present as a plan.
- **The close-cost budget and the funds flow, reconciled before close.** 6.45 % of debt raised is the real number, only 20 % of it is the lender's, and a 69,000 hole between a rule-of-thumb provision and an itemised budget becomes visible on the day it cannot be fixed.
- **The flex cap, tested against the covenant.** Fifty basis points of flex is worth 1,760,607 — 3.49 times the arrangement fee the director's team is negotiating — and consumes 44.9 % of the rate headroom the coverage covenant allows. Flex is signed at mandate stage, months before anyone models it; the director is the only person who can insist that the order be reversed.

Calculation exercises — Domain 13

> **Exercise 13.1** A diligence stream costs a **320,000** fee and takes **7 weeks**. The probability of a material issue in its class is **0.25**, the cost if it reaches close is **7,400,000** and pre-close correction costs **380,000**. The cost of delay is **96,000 per week** and detection is estimated at **0.75**. Compute the breakeven detection rate inside a parallel envelope and on the critical path, and the net value in each case. Solution. Expected avoidable loss `0.25 × (7,400,000 − 380,000) =` **1,755,000**. Inside the envelope `d* = 320,000/1,755,000 =` **18.23 %**; on the critical path the priced delay is `7 × 96,000 =` **672,000**, so `d* = 992,000/1,755,000 =` **56.52 %**. Net value: without the stream `0.25 × 7,400,000 =` **1,850,000**; with it inside the envelope `320,000 + 0.25 × (0.75 × 380,000 + 0.25 × 7,400,000) = 320,000 + 533,750 =` **853,750**, so net value **+996,250**; on the critical path add the 672,000, giving

net value **+324,250**. Common error: computing `d*` as `fee ÷ (p × C)` = 17.30 %, which ignores that a detected issue still costs `F` to fix and so overstates the loss the stream can avoid.

Exercise 13.2 Three CP chains: **A** 16 weeks, slip 4 weeks at probability 0.30; **B** 14 weeks, slip 5 at 0.40; **C** 11 weeks, slip 6 at 0.25. Cost of delay **80,000 per week**. Compute the base close, the expected close, the probability of the base date, and the conjunction premium. Solution. Base close **16 weeks** (chain A). Realised durations: A 16 or 20, B 14 or 19, C 11 or 17. Over the eight outcomes: `E[close] =` **18.1450 weeks**, with probabilities **31.50 %** at 16, **10.50 %** at 17, **28.00 %** at 19 and **30.00 %** at 20. Expected slip **2.1450 weeks**, costing **171,600**. Chain A's own expected slip is `0.30 × 4 =` 1.2 weeks, costing **96,000**, so the conjunction premium is **75,600**. Note that the max of expectations is **17.200** weeks, well inside the expected maximum. Common error: taking the longest chain's expected duration (17.200 weeks) as the expected close, which understates the slip by 0.945 weeks and the cost by 75,600 — precisely the premium the conjunction creates.

Exercise 13.3 A transaction's close costs comprise a fixed component of **1,650,000** and a proportional component of **1.85 %** of debt raised. Debt is **55,000,000**. Compute total close costs and their share of debt, and the facility size at which they would fall to 2.5 % of debt. Solution. Total `1,650,000 + 0.0185 × 55,000,000 = 1,650,000 + 1,017,500 =` **2,667,500**, which is **4.850 %** of debt raised. Setting `1,650,000/D + 0.0185 = 0.025` gives `D = 1,650,000/0.0065 =` **USD 253,846,153.85**. Common error: solving `1,650,000/D = 0.025` and reporting 66,000,000, which forgets that the proportional component consumes 1.85 of the 2.5 percentage points before the fixed component is spread at all.

Exercise 13.4 A model audit on a **36,000,000** facility at **6.5 %** over **14 years** finds that a **320,000** annual line included in `CFADS` is excluded by the facility's definition. Reported `CFADS` is **5,600,000**; the covenant is **1.25×**. Compute the reported and corrected `DSCR`, the debt capacity at 1.25× on corrected `CFADS`, and classify the finding. Solution. `AF(0.065, 14) =` **9.013842**; instalment `36,000,000/9.013842 =` **3,993,857.30**. Reported `DSCR 5,600,000/3,993,857.30 =` **1.4022**; corrected `CFADS` **5,280,000** gives **1.3220**. Debt capacity `5,280,000/1.25 × 9.013842 =` **38,074,470.00**, which **exceeds** the 36,000,000 drawn by **2,074,470**. The finding therefore moves a reported output well beyond any sensible materiality threshold but changes **no conclusion**: the covenant still holds and the debt is still supportable. It is a **Class 2** finding, not Class 1. Common error: classifying by the size of the movement (0.08 of `DSCR`, 320,000 of `CFADS`) rather than by whether a conclusion changes — the distinction on which KA 13.2.2 rests.

Exercise 13.5 A **96,000,000** facility carries a **1.35 %** arrangement fee of which **0.30 %** is praecipium; the balance is a pool paid pro rata on final allocations. The arranger's target hold is **30,000,000** and three participants take **22,000,000** each. Compute the fee split and yields. Then: the market takes only **54,000,000** and the arranger charges itself **85 basis points** a year on the excess over target hold for the facility's **14 years** at 6.5 % (`AF` = 9.013842). Compute the outcome. Solution. Total fee `96,000,000 × 1.35 % =` **1,296,000**; praecipium **288,000**; pool **1,008,000**, a rate of **1.0500 %**. Arranger `288,000 + 1.05 % × 30,000,000 =` **603,000**, a yield of **2.010 %**; each participant **231,000**, a yield of **1.050 %**; total `603,000 + 3 × 231,000 =` 1,296,000 ✓. On the failed sell-down the arranger holds `96,000,000 − 54,000,000 =` 42,000,000, an excess of **12,000,000**; the carry is `12,000,000 × 0.0085 × 9.013842 =` **919,411.92**, against fees of 603,000 — a **net loss of 316,411.92**. The excess hold that exactly consumes the praecipium is `288,000/(0.0085 × 9.013842) =` **3,758,924.52**, **3.92 %** of the facility. Common error: charging the carry on the whole 42,000,000

> residual rather than on the excess over the target hold — the arranger always intended to hold 30,000,000 and earns a fee for doing so.

Practitioner's toolkit — Domain 13

Adoption-ready artefacts; adapt headings to your organisation, then keep them stable.

Toolkit 13.T.1 — Diligence stream economics and interface sheet (one per transaction)

Part A, one row per stream: stream · adviser and named partner · fee · duration in weeks · inside the envelope or on the critical path (and if the latter, the justification and its weekly price) · `p`, `d`, `C`, `F` with the basis of each estimate and the number of comparable transactions behind it · **breakeven detection rate** · net value · deliverable schedule (red-flag date, draft date, final date) · addressee and reliance position · liability cap, and the cap as a percentage of `C`. **Part B, the cross-stream interface list:** every assumption touched by more than one adviser — availability, capacity, degradation, price, volume, tariff indexation, insured loss, tax rate, construction programme — with its single named owner, its value, its version and the streams that consume it. **Front line:** the envelope length, the binding stream, and the aggregate net value in parallel against in series.

Toolkit 13.T.2 — Model-audit finding register (one per model version)

Per finding: number · date raised · raised by (auditor / sponsor review / machine scan) · scope layer (arithmetic / structure / documents / assumptions) · description · **class 1, 2 or 3 against the stated materiality threshold** · quantified effect on `CFADS`, `DSCR`, minimum `DSCR`, debt capacity and the funding plan · correction made · dependent outputs updated (credit paper, term sheet, base case, sources and uses) · re-run of the golden-answer set (date and result) · verifier and date · disposition if accepted rather than corrected, with the written rationale. **Header block, mandatory before any count is quoted:** materiality threshold; counts by class; **class-weighted closure**; and the list of open Class 1 findings with their quantified effect. A finding count without this header block is not a reportable metric.

Toolkit 13.T.3 — Close register: conditions, costs and funds flow (one per transaction)

Part A, conditions: condition · clause reference · category (corporate / documentary / third-party / adviser / financial) · chain and predecessor · duration estimate · owner · evidence requirement · evidence received and verified by whom · status (open / satisfied / waived / deferred) · for waivers and deferrals: deadline, consequence and owner. Computed per chain: base duration, float, slip probability, slip size, **criticality risk net of float**. Computed for the transaction: base close, `E[close]` **and its distribution**, expected slip cost, the conjunction premium, weeks of margin to the long-stop, and the value of de-risking the highest-weighted chain. **Part B, costs:** every close-cost line by payee, split into fixed and proportional components, with the total as a share of debt raised and of the envelope; the reconciliation to the model's provision and to sources and uses; and the effective cost of debt on lender fees and on total close costs. **Part C, funds flow:** every payment at close by payee and account, reconciled line by line to Part B, with the reconciliation signed by the agent and the finance director **before** close day, and payment instructions verified out of band.

Exam preparation — Domain 13

What is assessed. The economics of an information purchase and the breakeven detection rate; the difference between parallel and serial diligence and how to price it; the seven streams and the error class each uniquely detects; the model audit's four scope layers and its three finding classes, with the debt-capacity consequence of a Class 1 finding computed; conditions precedent as a dependency network — float, criticality risk, the expected maximum and the conjunction premium; the close-cost budget, its fixed and proportional decomposition, its share of debt raised and its reconciliation to sources and uses and to the funds flow; the three syndication routes, the fee architecture, the arranger's yield differential and the cost of a failed sell-down; market flex and its coverage-covenant ceiling; waivers, long-stop dates and post-close undertakings; and the governance of AI across all four Knowledge Areas.

The calculations to do under time pressure. A breakeven detection rate from fee, delay, `p`, `C` and `F` (13.1.3). A stream's net value inside and outside the envelope (Exercise 13.1). A `DSCR` and debt capacity from a corrected `CFADS` at a target coverage (13.2.3, Exercise 13.4). An annuity-due instalment and the ratio it flatters (13.2.3). The expected maximum of three or four chains, with the probability of the base date (13.3.2, Exercise 13.2). Close costs as a share of debt, and the deal size at which a target share is reached (13.3.4, Exercise 13.3). A syndication fee split and the arranger's yield (13.4.2, Exercise 13.5). A carry cost on an excess hold, and the excess that consumes the praecipium (13.4.3). The present value of flex, and the margin at which a coverage covenant fails (13.4.4).

The traps. Computing `d*` as `fee ÷ (p × C)` and omitting the pre-close correction cost (Exercise 13.1) · pricing elapsed time once per stream when the streams run in parallel, or once for the envelope when they do not (13.1.3) · treating a stream's breakeven above 100 % as an arithmetic error rather than as evidence that the configuration cannot pay (MCQ 13.1-C) · quoting a finding count or a "findings closed" percentage as progress (13.2.2, MCQ 13.2-B) · classifying a finding by the size of the movement rather than by whether a conclusion changes (Exercise 13.4) · assuming an annuity due and an annuity in arrears give the same debt service — they differ by 283,564 here, and in the flattering direction (13.2.3, MCQ 13.2-A) · taking the longest chain's expected duration as the expected close (13.3.2, Exercise 13.2, MCQ 13.3-A) · spending acceleration effort on a chain with float (13.3.3, MCQ 13.3-B) · dividing close costs by the capital envelope rather than by debt raised (13.3.4, MCQ 13.3-C) · omitting the proportional component when solving for the deal size at a target close-cost share (Exercise 13.3, MCQ 13.3-D) · quoting the all-in cost of raising debt as though the lender charged it (13.3.4) · applying the full arrangement fee to the arranger's hold and ignoring the praecipium's exclusivity (MCQ 13.4-A) · charging an underwriter's carry on its residual hold rather than on the excess over target (Exercise 13.5, MCQ 13.4-B) · pricing flex as one year or as an undiscounted sum instead of an annuity (MCQ 13.4-C) · describing a signed transaction as closed (MCQ 13.4-E) · recording a waiver without a deadline, owner, evidence standard and consequence (Case study B).

How the domain connects. Domain 5's condition register becomes this domain's CP schedule and its close-cost provision; Domain 6's model and model-audit economics become KA 13.2's finding classes; Domain 9's tranche structures and effective-rate discipline become KA 13.4's syndication and close-cost arithmetic; Domain 10's `CFADS`, coverage ratios and covenant thresholds are what every Class 1 finding is measured in and what market flex is tested against; Domain 11's risk allocation and Domain 12's contracts are what the legal, technical and insurance streams verify. Forward, Domain 14 begins at the first drawdown this domain makes available and monitors the conditions this domain waived or deferred; Domain 15 operates the covenant regime the diligence file will be argued over; and Domain 16 systematises the AI controls each Knowledge Area here specified.

Domain 13 summary

Diligence and close are economic decisions under a deadline, and elapsed time — **124,133.33 per week** for Kestrel — governs both. A diligence stream is worth running when the loss it avoids exceeds what it costs, and the breakeven detection rate `d* = (fee + priced elapsed time) ÷ [p × (C − F)]` states the condition compactly: the diligence spend divided by the expected avoidable loss. Kestrel's seven streams, 1,500,000 of fees and 54 weeks of duration, are worth **+4,066,699.88** inside a twelve-week parallel envelope and **−1,146,900.12** run serially, the **5,213,600** difference being 42 weeks of scheduling and nothing else; inside the envelope no stream needs better than a **31.25 %** detection rate to pay, while in series five need **72.02 %** to **99.42 %** and two — environmental and social at **119.12 %**, insurance at **289.86 %** — cannot pay at any detection rate. Model audits are read by **class, never by count**: Kestrel's 34 findings resolved as 3 Class 1, 7 Class 2 and 24 Class 3, so the 24 that changed nothing were **70.6 %** of the count and the reported **91.2 %** closure could be true with every consequential finding open. Two document-conformance findings — an annuity-due schedule understating the instalment by **283,564** and a misplaced 260,000 reserve line — moved the reported `DSCR` from **1.3508** to a true **1.2224** and debt capacity at 1.30× to **39,494,354**, a resize of **2,505,646**, while a third left the **2,504,818** DSRA unfunded; none of the three was an arithmetic error, which is why the audit's document layer matters more than its formulae. Conditions precedent are a conjunction with a critical path, so the close date is the expected **maximum** of the chains: Kestrel's five chains give `E[close]` **22.4075 weeks** against a base of 20 and a critical chain whose own expected duration is 21.2000, a **34.125 %** probability of holding the stated date, an expected slip cost of **298,851** against the critical chain's **148,960**, and a **conjunction premium of 149,891** — and the highest-value intervention, worth **76,031.67**, is on chain B, which is not the critical path. Close costs of **2,709,000** are **6.45 %** of debt raised, only 544,000 of it payable to lenders, decomposing into a **2,016,000** fixed component and **1.65 %** proportional — so a facility must reach about **149,333,333** before close costs fall below 3.0 % of debt — and the budget reconciles to a model provision 69,000 short, implying an effective cost of debt of **6.2386 %** on lender fees and **7.2376 %** on the whole cost of raising it. Syndication pays the arranger **247,500** on a 15,000,000 hold, a **1.650 %** yield against participants' **0.950 %**, and a **3.31 %** over-hold consumes that entire premium — which is why market flex, worth **1,760,607** or **3.4933×** the arrangement fee and consuming **44.9 %** of the coverage covenant's **111.4 basis points** of rate headroom, is the term that should be negotiated hardest and almost never is. Throughout, AI shortens the envelope by extracting, reconciling and tracking, and is permitted to conclude nothing: not a legal, tax or environmental opinion, not a finding's class, not a condition's satisfaction, not a market price, and never a payment instruction. Domain 14 takes the first drawdown this domain made available.

04

PART FOUR

Operating and the future

Domains 14–16 — operations and asset management, refinancing and portfolio value, and the future of project finance: the life after close.

14

DOMAIN 14

Construction Monitoring and Drawdown (quantitative)

Group: Operating and the future (Domain 14 of 16, opening Part Four). **Target:** ~75 pages. **Binds to:** the PCI Book Pattern Specification and the shared registries (`docs/books/registries/`). This domain owns no new symbol. It composes registered ones — `CFADS`, `DSCR`, `AF(r, n)`, `NPV`, `PI`, `EAC`, `ETC`, `PoC`, `EMV`, `DSRA`, `D/E` — into the four tests a construction lender actually runs: the **draw request**, the **funds sufficiency (in-balance) test**, the **contingency coverage test on the remainder**, and the **coverage test at the first repayment date**. Where a figure was derived in an earlier domain it is **cited, not re-derived**: the construction funding profile and the pro-rata capitalised interest of **USD 2,114,597** come from Domain 6 (KA 6.2.1); the capitalised-interest area rule and the economic cost of a month of slip come from Domain 8 (KA 8.2, 8.4); the daily cost of a slip at the commercial operations date comes from Domain 5 (KA 5.4.2); the contract limits come from Domain 12 (KA 12.1); the coverage machinery comes from Domain 10. British English; USD (+SAR where useful, indicative `USD 1 ≈ SAR 3.75`).

Why this domain exists

Domain 13 closed the transaction. Every domain before it answered a question about a project that did not yet exist: is it worth building, can it carry debt, who bears which risk, what do the documents say. This domain answers the first question about a project that does exist and is being built with other people's money, and it is a narrower and harder question than any of them: **on the evidence available this month, may the next tranche of funds be released?**

That question is not "is the project going well?" and it is not "what will it finally cost?" It is a test with a binary answer, run against a defined evidence pack, on a defined date, by a person who signs. The whole apparatus of construction monitoring exists because limited-recourse debt is advanced **before** the asset that secures it exists, so the lender's only protection between financial close and commercial operations is the discipline with which each release of funds is conditioned on certified progress and on a demonstration that the remaining money still finishes the job. Everything in this domain follows from that single structural fact.

The domain's central claim is that **the construction-phase tests are not a finance version of project controls; they are a different discipline reading the same data spine, and the differences are systematic, nameable and computable.** A project controls forecast produces a range and a narrative (PML-AI Domain 7's `EAC` family). A lender's cost-to-complete produces one number, a date and, when it fails, a cash call — and it is built on a **commitment** basis, not a performance-extrapolation basis, so on a fixed-price contract the two can diverge by millions while both are correct on their own terms. Reconciling them, line by line, is the single most valuable professional skill this domain teaches, and Toolkit 14.T.2 exists to make it routine.

Learning objectives. After this domain a candidate can: restate a sources-and-uses statement at a data date into drawn, committed and available columns and explain why the identity is a construction, not a check; build a draw request from certified value through retention and advance recovery to a net funding requirement and its pro-rata debt and equity shares; compute the capitalised interest consequence of a funding order (pro rata, equity-first, debt-first) and state whose interest each order serves; compute a lender's cost-to-complete on a commitment basis and reconcile it, line by line, to a `BAC/CPI` or bottom-up `EAC` from the same data spine; explain why a blended `CPI` misattributes an overrun across fixed-price and owner-retained scope; run a funds sufficiency test and identify the double-count that makes most of them pass wrongly; compute a contingency coverage ratio on the remainder and defend the choice of denominator; value the same month's work on milestone, measured and cost-incurred certification bases and quantify the debt advanced against work not in place; compute the marginal `DSCR` of a variation and the debt share at which it becomes coverage-dilutive; distinguish the **funded** cost of a month of delay from its **economic** cost and show that one liquidated-damages rate can over-recover the first while under-recovering the second; compute the `DSCR` at a calendar-fixed first repayment date under a late commercial operations date, with its covenant, payment and reserve breakevens; compute a buy-down that restores a stated coverage level; and govern AI-assisted certification, cost-report and covenant analysis under the family's verification rule.

The master construction. Kestrel Water SPC is being built. Domain 6 (KA 6.2.1) fixed the funding: a **USD 60,000,000** envelope, **USD 42,000,000** senior debt at **6.0 %** and **USD 18,000,000** equity, funded **70/30** against certified spend over **eight construction quarters** on a profile of **6, 9, 13, 16, 17, 15, 13 and 11 per cent**; uses of EPC price **48,000,000**, owner's costs and land **3,600,000**, capitalised development **1,800,000**, arrangement and financing fees **840,000**, a balancing contingency of **3,645,403** and capitalised interest of **2,114,597**. The certified-spend base is therefore **55,245,403**, and cumulative debt drawn reaches **31,990,655** by the end of quarter six. Domain 12 (KA 12.1.1) fixed the contract limits: delay damages **20,000 per day** capped at **4,800,000** (day 240), performance damages capped at 4,800,000, aggregate liability 9,600,000. Domain 10 fixed the operating regime: annual debt service **USD 5,009,635.23**, base `CFADS` **6,384,000**, `DSCR` **1.2743**, a **1.20×** covenant biting at `CFADS` of **6,011,562** (annual headroom **372,438**), a **1.15×** lock-up and a six-month debt service reserve of **2,504,818**. This domain takes that structure to the site and runs it, quarter by quarter, to the completion tests.

KNOWLEDGE AREA 14.1

Sources and uses, draw requests and conditions

Topics: 14.1.1 the sources-and-uses statement as a control document · 14.1.2 the draw request · 14.1.3 funding order, and who pays for it.

14.1.1 The sources-and-uses statement as a control document

Definition. A **restated sources-and-uses statement** is the financial-close statement re-expressed at a data date into three columns — **drawn to date**, **remaining to be funded**, and **available commitment** — so that the identity that held at close (sources = uses) can be tested prospectively rather than merely admired retrospectively.

Domain 6 established that sources equal uses is an identity, not a check: a model can always be made to satisfy it, and what makes it informative is knowing which line balances. In construction monitoring the same point returns with teeth. **The close statement is a plan; the restated**

statement is a test. At close, the balancing line was contingency. At a data date the balancing line is whatever the sponsors will inject when the columns do not reconcile, and its size is the number this whole knowledge area exists to compute.

Three columns, and one rule for each. **Drawn to date** is a fact, reconciled to the facility agent's records and the SPV's bank statements, not to the model. **Remaining to be funded** is a forecast, and the whole of KA 14.2 is about how it is built. **Available commitment** is a contractual quantity — undrawn debt plus uncalled equity — and it carries the trap that catches most first drafts: **the undrawn commitment already includes the undrawn contingency.** Contingency is not a separate source of money; it is a use funded by the same 70/30 commitment as every other use. Adding "remaining contingency" to "undrawn debt plus uncalled equity" double-counts it, and the error flatters the in-balance test by exactly the contingency balance — which is precisely the period when the test most needs to be honest.

WORKED EXAMPLE

Worked example 14.1.1 — Kestrel's restated statement at the quarter-five data date.

1. **Setup.** Five construction quarters are complete. On Domain 6's profile the planned cumulative certified spend is `55,245,403 × 0.61 = ` **33,699,696**, funded by cumulative debt of **25,917,751** and equity of **11,107,608**, with capitalised interest of **685,663** accrued to date (quarterly amounts 27,720, 62,816, 115,682, 192,307 and 287,138) and **1,428,934** planned for quarters six to eight. Actual certified spend is higher: the owner-retained scope has cost **4,200,000** against a budget value of work done of **2,520,000**, and **465,403** of variations have been certified — so actual cumulative certified spend is **33,945,403**, or **245,707** above plan, drawn in quarter five. Restate the statement and test the balance.
2. **Formula.** Additional draw = actual − planned certified spend, split 70/30. Available commitment = (42,000,000 − cumulative debt) + (18,000,000 − cumulative equity). Additional interest = additional debt draw × 0.015 × remaining quarters. Balance = available − remaining uses.
3. **Substitution.** `245,707 × 0.70 = 171,995`; `245,707 × 0.30 = 73,712`; `(42,000,000 − 26,089,746) + (18,000,000 − 11,181,320)`; `171,995 × 0.015 × 3 = 7,740`; remaining base-scope spend `55,245,403 × 0.39 − 245,707 = 21,300,000`.
4. **Result.**

Restated statement at the quarter-five data date	Drawn (USD)	Remaining (USD)	Total (USD)
Fees and capitalised development	2,640,000	—	2,640,000
EPC contract price	29,280,000	18,720,000	48,000,000
Owner-retained scope, at budget	2,520,000	1,080,000	3,600,000
Owner-scope overrun and certified variations, from contingency	2,145,403	1,500,000	3,645,403
Interest during construction	685,663	1,436,674	2,122,337
Total uses	**37,271,066**	**22,736,674**	**60,007,740**
Senior debt	26,089,746	15,910,254	42,000,000
Equity	11,181,320	6,818,680	18,000,000
Total sources / available commitment	**37,271,066**	**22,728,934**	**60,000,000**

Remaining uses **22,736,674** against available commitment **22,728,934** — the project is **out of balance by USD 7,740** on the plan's own assumptions. 5. **Interpretation.** Read the 7,740 first, because it is the whole discipline in miniature. Nothing has gone wrong that the plan did not contain: 245,707 of contingency was drawn three quarters earlier than the model assumed it would be, and `171,995 × 0.015 × 3` is the interest that early draw costs. **Drawing contingency early is not free, and the price is computable to the dollar.** A facility whose availability is measured to the last dollar of commitment is therefore out of balance the moment any use is accelerated, which is why prudent structures either fund a small unallocated headroom above the modelled envelope or provide expressly that the sponsor tops up timing differences — and why the sponsor should know which of those two the documents say before the first draw, not during the fifth. The second reading is the composition of the remaining column. **Only 18,720,000 of the 22,736,674 is a fixed contractual commitment**; 1,080,000 is a budget for owner-retained work whose demonstrated cost efficiency is 0.60 (KA 14.2), 1,500,000 is contingency against risks not yet run, and 1,436,674 is interest that depends on the draw profile of everything above it. Three of the four lines are forecasts, and the in-balance test is only as strong as the weakest of them. Third, notice what the identity does not tell you. The columns reconcile within 7,740 while — as KA 14.2 will show — the lender's own cost-to-complete on the same data date is **1,927,740** short. **A sources-and-uses statement built from the model balances by construction; a sources-and-uses statement built from the commitments does not, and the gap between them is the report.**

14.1.2 The draw request

Definition. A **draw request** (or drawdown notice, or advance request) is the borrower's certified application for the release of funds, and it is a composite document: an arithmetic claim (this much money, split this way), an evidential claim (this work is in place, certified by this person), and a representational claim (the conditions to drawing are satisfied and the project remains in balance). Lenders decline draws far more often on the third limb than on the first.

The arithmetic runs in one direction and each step has a reason:

```
Gross certified value of work executed in the period
  − retention withheld
  − recovery of the advance payment
  + owner-retained costs incurred and evidenced
  + fees, taxes and premiums due in the period
  + interest and commitment fees accrued
  = period funding requirement
      × gearing           → senior debt draw
      × (1 − gearing)     → equity draw (or equity contribution certificate)
```

The **conditions to each drawing** are a shorter and more consequential list than the conditions precedent to first drawing (Domain 13, KA 13.3): no default or potential default continuing; the representations repeated and still true; the certificate of the independent engineer or lender's technical adviser attached; insurances in force and premiums paid; the project **in balance**; and the drawing within the availability period and within the commitment. Two of these are judgments rather than facts — "no potential default" and "in balance" — and they are where a monitoring relationship is actually conducted.

WORKED EXAMPLE

Worked example 14.1.2 — the quarter-five draw, built line by line.

1. **Setup.** Quarter five on Domain 6's profile: planned certified spend `55,245,403 × 0.17 =` **9,391,718**; interest accrued on the opening debt balance of 19,142,551 at 1.5 % is **287,138**. The additional certified contingency spend of **245,707** identified in 14.1.1 falls in this quarter. Gearing 70/30. Compute the request and the resulting balances.
2. **Formula.** Requirement = certified spend + accrued interest; debt draw = 0.70 × requirement; equity draw = 0.30 × requirement.
3. **Substitution.** `9,391,718 + 245,707 + 287,138 = 9,924,564`; `× 0.70`; `× 0.30`.
4. **Result.** Period funding requirement **USD 9,924,564**; senior debt draw **6,947,195**; equity draw **2,977,369**. Closing cumulative debt **26,089,746** (62.12 % of commitment) and equity **11,181,320** (62.12 % of commitment); undrawn commitment **22,728,934**.
5. **Interpretation.** Three things in that line are worth a leader's attention. **The interest line is a draw.** 287,138 of the request is money the project is borrowing in order to pay interest on money it has already borrowed, which is exactly what capitalised interest means and exactly why Domain 8's area rule matters: the interest row grows with the drawn balance, so it is largest in the quarters when the certified-spend row is already falling. In quarter eight Kestrel's interest row is 560,308 against a spend row of 6,076,994 — **8.44 %** of the period requirement, against **0.83 %** in quarter one. A sponsor budgeting equity calls from a construction spend curve alone will under-call late in the programme, every time. **The equity draw is a call, and calls have lead times.** 2,977,369 has to arrive from shareholders whose own approval processes (PML-AI Domain 3's governance latency) may run to weeks; a draw request that reveals the call is a draw request submitted too late. The professional practice is a rolling twelve-week funding forecast issued to shareholders, reconciled to the draw schedule, so that no call is news. **And the pro-rata split is a contractual mechanic, not an accounting one.** Each draw carries its own 70/30 split, which is why the cumulative percentages move in lockstep — 62.12 % of both commitments drawn — and why any departure from that lockstep is either a documented funding order (14.1.3) or an error worth finding immediately.

14.1.3 Funding order, and who pays for it

Definition. The **funding order** is the contractual rule determining, for each period's requirement, how much comes from debt and how much from equity. Three orders are standard: **pro rata**, in which every draw carries the gearing ratio; **equity-first**, in which the sponsors' whole commitment is exhausted before any debt is drawn; and **debt-first** (sometimes "debt-first with an equity backstop"), in which the facility is drawn to its limit before equity is called. The choice is usually settled in a single sub-clause of the facility agreement and is routinely treated as boilerplate. It is not: it moves capitalised interest, credit exposure and the sponsor's return, in different directions, by amounts that dwarf most of what is negotiated around it.

WORKED EXAMPLE

Worked example 14.1.3 — the same project under three funding orders.

1. **Setup.** Kestrel's certified-spend base **55,245,403** on the eight-quarter profile, fees and development of **2,640,000** funded at close, interest at **1.5 %** per quarter on the opening debt balance, draws treated as made at period end. Scope and contingency are held identical across the three orders so that only the sequencing changes. Compute total capitalised interest, the resulting funding requirement, and the sponsor's share of funds in the ground at each draw date.
2. **Formula.** For each period: `interest = opening debt balance × 0.015`; `requirement = certified spend + interest`; the split follows the order; total uses = 2,640,000 + 55,245,403 + Σ interest.
3. **Substitution.** Pro rata: `debt draw = 0.70 × requirement` throughout. Equity-first: `equity draw = min(requirement, 18,000,000 − equity drawn)`, the balance from debt. Debt-first: `debt draw = min(requirement, 42,000,000 − debt drawn)`, the balance from equity.
4. **Result.**

Funding order	Capitalised interest (USD)	Total uses (USD)	Debt drawn (USD)	Equity drawn (USD)	D/E
Equity first	**1,338,006**	59,223,409	41,223,409	18,000,000	2.2902
Pro rata 70/30	**2,114,597**	60,000,000	42,000,000	18,000,000	2.3333
Debt first	**2,804,070**	60,689,473	42,000,000	18,689,473	2.2473

Equity-first saves **776,591** of capitalised interest against pro rata (**36.73 %**); debt-first costs **689,473** more (**32.61 %**). The spread from best to worst is **USD 1,466,064**, or **2.4434 %** of the envelope. Under debt-first no sponsor cash is in the ground until quarter six, and at the quarter-five data date the lender's exposure is **37,323,907** against **25,917,751** pro rata — a difference of **11,406,157** — with **zero** equity contributed against **11,107,608**. 5. **Interpretation.** Start with the size of the number, because it settles a question of priorities. Kestrel's total drawn-balance exposure over the build is `2,114,597 / 0.015 =` **140,973,150** quarter-dollars, so a ten-basis-point movement in the margin is worth `140,973,150 × 0.001/4 =` **35,243** across the whole construction period. The funding-order spread of 1,466,064 is **41.6 times** that. Financings are routinely negotiated for weeks over the margin and settle the funding order by accepting the lender's draft, which is an allocation of professional attention almost exactly inverted. **Second, the three orders serve three different interests and the arithmetic says so.** Equity-first is cheapest for the project and safest for the lender, because it puts sponsor money in first — which is why lenders draft it and why sponsors resist it. Debt-first is dearest for the project and worst for the lender, and the sponsor still prefers it: discounting each order's equity draws at Domain 4's 8 % (a quarterly 1.9427 %), pro rata has a present value of **16,476,605** and debt-first **16,293,592**, so deferring the cheque is worth **183,013** of present value to the sponsor even though it costs the project 689,473 of capitalised interest, which the sponsor then funds as additional equity. **A sponsor arguing for debt-first is buying 183,013 of present value for 689,473 of nominal cost, and both statements are true.** Whether that is rational depends on the sponsor's own cost of capital, which is exactly the conversation the clause deserves and rarely gets. **Third, at a fixed envelope the saving reappears as protection.** Hold the 60,000,000 envelope and re-solve for the balancing contingency under equity-first, and contingency rises from 3,645,403 to **4,388,050 — 9.14 %** of the EPC price against 7.59 %, an extra **742,647** of funded protection bought with a sequencing clause and no additional money. That is the version of this calculation to take to a credit committee, because it converts an interest saving into the currency the committee actually cares about. **The professional caution:** none of this survives contact with a project that stops. Equity-first maximises the sponsor's irrecoverable exposure if the works are abandoned early, and a sponsor with a weak balance sheet may simply be unable to fund it — which is why the honest framing is not "equity-first is best" but "the funding order allocates construction-phase risk and cost between three parties, and the allocation must be priced before it is signed."

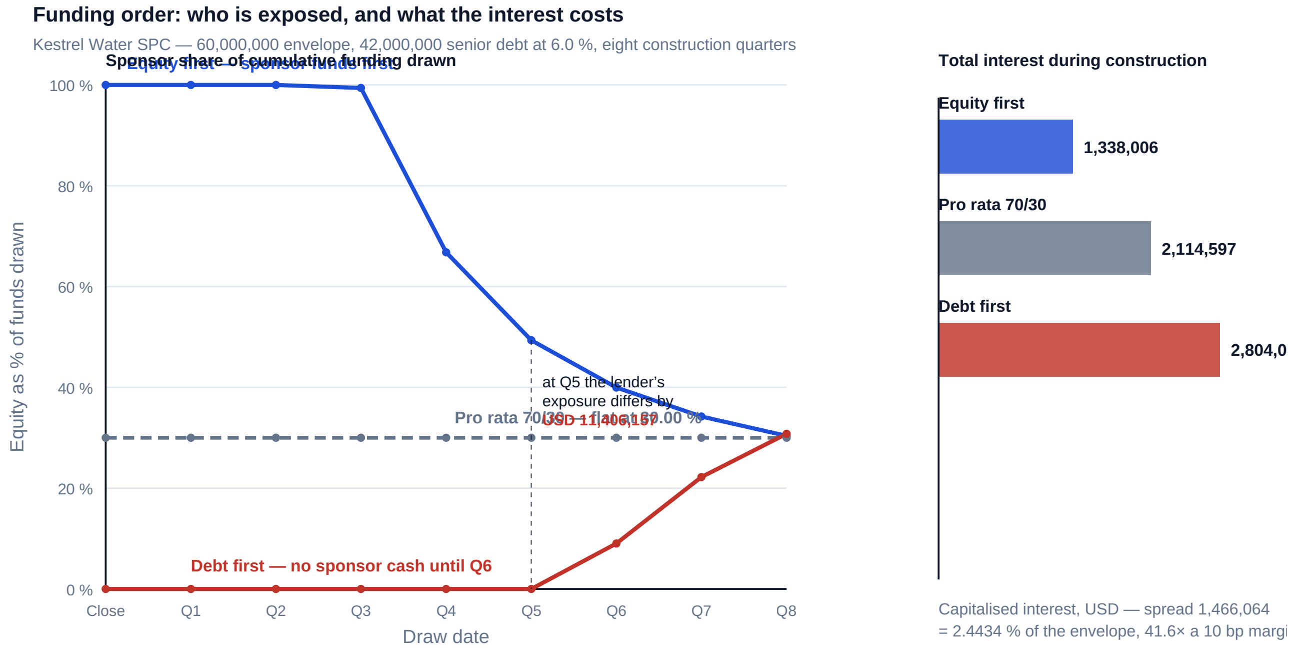

Fig 14.1.1 — Funding order: who is exposed, and what the interest costs

AI in this KA

Where it earns its place. Draw-request assembly is repetitive, document-heavy and deadline-driven, which is exactly where machine assistance pays: extracting certified quantities and values from a certifier's schedules into the draw template; reconciling the requested amount against the facility agent's drawn-to-date records and flagging every discrepancy; checking that each condition to drawing has a corresponding, in-date piece of evidence in the pack (insurance certificates, the technical adviser's certificate, the compliance certificate) and listing what is missing; and maintaining the rolling funding forecast so that no equity call is news. Recomputing the whole funding table under an alternative funding order — the arithmetic of 14.1.3 — is likewise machine work, and it is work almost nobody does by hand, which is precisely why the clause goes unnegotiated.

Where it must not go. Nothing in this knowledge area may generate a **representation**. The "no default continuing" and "in balance" statements in a draw request are certifications by named officers with consequences attached, and a model that concludes them has produced a draft for a human to verify and sign, never a signature. Nor should an assistant be permitted to reconcile a discrepancy it finds between the requested amount and the agent's records; its job is to surface the difference, because the reconciliation is where an error becomes either a correction or a misstatement.

Verification, concretely. Recompute one period's requirement by hand from the certifier's gross value — retention — advance recovery + owner costs + accrued interest, and require agreement to the dollar. Independently recompute accrued interest from the opening balance and the day-count basis (Domain 8's area rule reproduces the whole IDC line from the profile, the rate and the gearing, and disagreement is a defect not a rounding difference). Confirm that cumulative debt and equity percentages are equal under pro-rata funding, and that any divergence traces to a documented funding order. Tie drawn-to-date to the agent's statement, not to the model. **AI proposes; the professional verifies, decides and remains accountable.**

KEY TERMS — KA 14.1

Term	Meaning
Restated sources and uses	The close statement re-expressed at a data date into drawn, remaining and available columns; a test rather than an identity.
Available commitment	Undrawn debt + uncalled equity. Already includes undrawn contingency — adding contingency again double-counts.
Draw request	The certified application for funds: an arithmetic, an evidential and a representational claim.
Conditions to each drawing	The per-draw conditions: no default, representations repeated, certificates attached, insurances in force, in balance, within availability and commitment.
Funding order	Pro rata, equity-first or debt-first; allocates capitalised interest, credit exposure and sponsor return.
Equity contribution certificate	Evidence that the equity share of a draw has been provided, conditioning the debt share.

SAMPLE MCQS — KA 14.1

MCQ 14.1-A `[14.1.1 · Application]` At a data date, undrawn senior debt is 15,910,254 and uncalled equity is 6,818,680; remaining unallocated contingency is 1,500,000. Available commitment for the in-balance test is:

- A. USD 24,228,934
- B. USD 22,728,934
- C. USD 15,910,254
- D. USD 21,228,934

Rationale: Available commitment is `15,910,254 + 6,818,680 = 22,728,934`. A adds the 1,500,000 of contingency, which is a **use** funded by that same commitment — the double-count of 14.1.1, and it flatters the test by exactly the contingency balance. C counts only debt and ignores the equity commitment. D deducts the contingency instead of ignoring it, an error in the opposite direction.

MCQ 14.1-B `[14.1.3 · Analysis]` On identical scope and contingency, Kestrel's capitalised interest is 1,338,006 under equity-first funding and 2,804,070 under debt-first. The soundest reading is:

- A. debt-first is a modelling error, since total funding is fixed at 60,000,000
- B. the 1,466,064 spread is a real cost of the sequencing clause; equity-first is cheapest for the project and safest for the lender, while debt-first defers the sponsor's cheque and is worth 183,013 of present value to the sponsor
- C. the orders are economically equivalent because the sponsor funds the difference either way
- D. the difference is capitalised, so it has no effect on the project's economics

Rationale: The sequencing changes the drawn balance on which interest accrues, so it changes total uses (59,223,409 / 60,000,000 / 60,689,473) — the arithmetic of 14.1.3. A misreads a fixed envelope as a fixed requirement. C ignores that the sponsor's present value improves while the project's nominal cost worsens. D is the commonest form of the error: capitalised interest enters the depreciable base and the coverage arithmetic and is therefore paid, with interest, over the loan life.

MCQ 14.1-C `[14.1.2 · Application]` Kestrel's quarter-eight draw carries certified spend of 6,076,994 and accrued interest of 560,308. As a share of the period requirement, the interest line is closest to:

- A. 0.83 %
- B. 8.44 %
- C. 9.22 %
- D. 3.52 %

Rationale: `560,308 / (6,076,994 + 560,308) = 8.44 %`. A is the quarter-one share, the point being that the interest row grows as the spend row falls. C divides interest by certified **spend** rather than by the period requirement, the commonest slip. D is total capitalised interest as a share of the 60,000,000 envelope (Domain 6), a different denominator entirely.

MCQ 14.1-D `[14.1.2 · Recall]` Which condition to a drawing is a judgment rather than a verifiable fact?

- A. the drawing is within the availability period
- B. the requested amount is within the undrawn commitment
- C. no potential event of default is continuing, and the project remains in balance
- D. the technical adviser's certificate is attached

Rationale: A, B and D are checkable against a calendar, a ledger and a document schedule. C requires a forecast of remaining cost and a view on what constitutes a potential default — which is why those two limbs are where the monitoring relationship is conducted (14.1.2).

SELF-CHECK — KA 14.1

1. Why is a restated sources-and-uses statement a test where the close statement is an identity? — At close one line balances by construction; at a data date the columns are built from independent sources (the agent's records, certified progress, a commitment schedule), so a gap is information rather than a plug.
2. State the double-count that flatters most in-balance tests. — Adding remaining contingency to undrawn debt plus uncalled equity; contingency is a use funded by that same commitment.
3. Who benefits from each funding order? — Equity-first: the project (lowest capitalised interest, 1,338,006) and the lender (sponsor money in first). Debt-first: the sponsor's present value (+183,013) at the project's cost (+689,473). Pro rata: neither, which is why it is the common settlement.

KNOWLEDGE AREA 14.2

Cost-to-complete and contingency draw

Topics: 14.2.1 the lender's cost-to-complete, and the reconciliation to *EAC* · 14.2.2 contingency draw and coverage on the remainder · 14.2.3 the in-balance condition and the cash call.

14.2.1 The lender's cost-to-complete, and the reconciliation to EAC

Definition. A **lender's cost-to-complete** is the money still required to reach the commercial operations date, built on a **commitment** basis: remaining contractual entitlement under signed contracts, plus approved variations not yet certified, plus the assessed exposure on notified but unagreed claims, plus a bottom-up re-estimate of owner-retained scope, plus the financing costs — interest, fees, premiums — that will accrue between the data date and completion. It is not `EAC − AC`, and the difference is not an error in either number.

Domain 8 (KA 8.4.1) established the bridge in principle: a project controller's report ends at an `EAC` column and an honest range; a lender reads the same figures and produces a date and an amount. PML-AI Domain 7 (KA 7.3.3) supplies the `EAC` family — remaining work at the budgeted rate, `BAC/CPI`, the `CPI × SPI` variant, and bottom-up re-estimation. This topic supplies the missing piece: **why the two disciplines diverge systematically on a fixed-price contract, and how to reconcile them line by line so that the divergence becomes evidence instead of an argument.**

The structural reason is the wrap. Under a fixed lump-sum contract certified against milestones, the amount certified is the amount budgeted for the milestone, so on the EPC scope `EV` and `AC` are identically equal and `CPI` **on that scope is 1.000 by construction.** Cost variance on the largest single line of the project is therefore structurally invisible to earned value; the risk shows up instead as schedule, as claims, and as contractor solvency. Meanwhile the owner-retained scope — land, permits, the owner's engineering team, utility diversions, the items no contractor would price — is where the SPV bears cost risk directly, and it is small enough to be overlooked and volatile enough to be dangerous.

WORKED EXAMPLE

Worked example 14.2.1 — Kestrel's two cost-to-completes at the quarter-five data date, and the bridge between them.

1. **Setup.** The construction control accounts carry `BAC` **51,600,000** (EPC 48,000,000 plus owner-retained 3,600,000; contingency and financing sit outside the control accounts). At the data date the EPC scope is **61 %** certified, so `EV` = `AC` = **29,280,000** on that scope. The owner-retained scope is **70 %** physically complete — `EV` **2,520,000** — at an actual cost of **4,200,000**, a scope `CPI` of **0.6000**. Off the control accounts: **465,403** of variations already certified, **840,000** of variations approved but not yet certified, and notified but unagreed contractor claims whose assessed exposure the technical adviser and the SPV's quantity surveyor put at **1,260,000**. A bottom-up re-estimate of the remaining owner-retained scope comes back at **2,400,000**. Remaining capitalised interest to completion is **1,436,674** (14.1.1). Compute both cost-to-completes and reconcile them.
2. **Formula.** EVM side: `CPI = EV/AC`; `EAC(a) = AC + (BAC − EV)`; `EAC(b) = BAC/CPI`; `EAC(d) = AC + bottom-up ETC`; `CTC = EAC − AC`. Lender side: `CTC = remaining committed contract value + approved variations + assessed claim exposure + bottom-up owner scope + remaining financing costs`.
3. **Substitution.** `CPI = 31,800,000/33,480,000`; `EAC(b) = 51,600,000 × 33,480,000/31,800,000`; `EAC(d) = 48,000,000 + 4,200,000 + 2,400,000`; lender side `18,720,000 + 840,000 + 1,260,000 + 2,400,000 + 1,436,674`.
4. **Result.** `CPI` **0.949821** blended, **1.0000** on EPC scope, **0.6000** on owner-retained scope.

Forecast	EAC (USD)	CTC = EAC − AC (USD)
(a) remaining work at the budgeted rate	53,280,000	19,800,000
(b) `BAC/CPI`	54,326,038	20,846,038
(d) bottom-up re-estimate	54,600,000	21,120,000

The reconciliation to the lender's number:

Bridge from the bottom-up CTC	USD
`CTC(d)` = remaining EPC 18,720,000 + remaining owner scope 2,400,000	21,120,000
+ approved variations, outside `BAC`	840,000
+ assessed exposure on notified claims, outside `BAC`	1,260,000
+ remaining interest during construction, outside `BAC`	1,436,674
= lender's cost-to-complete	**24,656,674**

Against available commitment of **22,728,934**, the project is **out of balance by USD 1,927,740**. 5. **Interpretation.** Four readings, in order of professional value. **The bridge is the whole lesson: the two numbers differ by items each discipline is structurally blind to.** Earned value does not see capitalised interest (it is not work), it does not see approved variations until they are baselined, and it does not see claim exposure at all — 3,536,674 of the lender's number is invisible to the cost report, which is 14.34 % of it. Conversely the lender's number does not see cost-performance trend on the fixed-price scope, because there is none to see. Neither number is wrong; a report that presents one without the other is. **Second, the blended `CPI` gets close to the right total for**

entirely the wrong reasons, and that is worse than being wrong. `EAC(b)`'s uplift over `BAC` is 2,726,038, which sits plausibly between the 2,400,000 overrun implied by the owner-scope `CPI` and the 3,000,000 implied by the bottom-up. But of that uplift `48,000,000 × (1/0.949821 − 1) =` **2,535,849** is attributed to a fixed-price scope that cannot overrun, and only **190,189** to the scope that is actually overrunning by 2,400,000 to 3,000,000. A reviewer who checks the total passes it; a reviewer who checks the composition finds that the forecast is right by coincidence and will stop being right the moment the mix changes. **Disaggregate before extrapolating: a `CPI` blended across scopes with different cost-risk ownership is an average of a constant and a variable, and it forecasts neither. Third, the `CPI × SPI` method has no honest home here.** It requires an `SPI`, and a milestone-certified fixed-price scope produces a lumpy, contractually-defined value curve that does not support one; the schedule question must be answered from the programme and the independent engineer's assessment (KA 14.4), not from an index. **Fourth, the 1,927,740 is a cash call, and its date is the next draw.** The forecast has stopped being a forecast. The commercial consequence is that **the choice of basis is a negotiation, not a technique** — the sponsor will argue that claims assessed at 1,260,000 are worth far less and that the bottom-up re-estimate is pessimistic, and both arguments may be right; what cannot survive is a report that omits the lines rather than arguing about them.

14.2.2 Contingency draw and coverage on the remainder

Definition. Contingency coverage on the remainder is the ratio of unallocated contingency still available to the assessed cost of everything it must still cover:

```
Contingency coverage = remaining unallocated contingency
                       ÷ (known committed claims on contingency + P80 of the open risk register)
```

Domain 8 (KA 8.3) sized contingency at sanction and warned that the confidence level chosen is usually unnamed. This topic runs the same test during construction, where the denominator has two parts that behave completely differently, and where the standard health check is worthless.

The standard health check is the **draw rate**: contingency drawn as a percentage of contingency funded, compared with percentage progress. Kestrel at the quarter-five data date has drawn `(4,200,000 − 2,520,000) + 465,403 =` **2,145,403** of its 3,645,403, or **58.85 %**, against **61.00 %** certified progress. That looks not merely acceptable but slightly better than proportionate, and it is the number most monthly reports lead with. It is meaningless. **Contingency is not consumed pro rata to progress, because risk is not distributed pro rata to progress** — some risks retire early (ground conditions), some cluster at the end (commissioning, performance testing, permit conditions), and a draw rate that tracks progress is a coincidence, not evidence. The only test that means anything is coverage on what remains.

WORKED EXAMPLE

Worked example 14.2.2 — Kestrel's contingency, tested twice.

1. **Setup.** Contingency funded at close **3,645,403**. The sanction risk register carried six items: ground conditions at the intake structure (p 0.30, 2,600,000), a brine-outfall permit condition variation (0.25, 1,200,000), membrane supply escalation or substitution (0.35, 900,000), commissioning re-test and extended supervision (0.35, 700,000), an owner-supplied power-connection delay (0.30, 600,000) and interface and utility diversions in owner-retained scope (0.40, 800,000). By the quarter-five data date the ground-conditions and interface risks have **materialised** (they are the owner-scope overrun and the certified variations), the membrane risk has been **retired**, and four items remain open — with the claims risk restated as "notified claims settle above the assessed exposure" (p 0.40, 900,000). Known committed claims on contingency are the 840,000 of approved variations, the 1,260,000 of assessed claim exposure and the **1,320,000** by which the bottom-up owner-scope re-estimate of 2,400,000 exceeds the 1,080,000 of remaining budget. Compute coverage at close and at the data date.
2. **Formula.** Register mean = Σ `EMV` = Σ `p × impact`; variance = Σ `p(1 − p) × impact²`; P80 ≈ mean + 0.8416σ (the normal approximation of Domain 8, KA 8.3). Coverage = remaining contingency ÷ denominator.
3. **Substitution.** At close: mean `780,000 + 300,000 + 315,000 + 245,000 + 180,000 + 320,000`; variance `0.21×2,600,000² + 0.1875×1,200,000² + 0.2275×900,000² + 0.2275×700,000² + 0.21×600,000² + 0.24×800,000²`. At the data date: mean `360,000 + 300,000 + 245,000 + 180,000`; denominator `3,420,000 + P80`.
4. **Result.**

Test	Contingency available (USD)	Denominator (USD)	Coverage
At financial close, against the sanction register P80	3,645,403	3,392,416	**1.0746**
At the data date, open risk only, on the mean	1,500,000	1,085,000	**1.3825**
At the data date, open risk only, at P80	1,500,000	1,764,289	**0.8502**
At the data date, known claims + open risk at P80	1,500,000	5,184,289	**0.2893**

Sanction register: mean **2,140,000**, σ **1,488,136**, P80 **3,392,416** — so the funded contingency of 3,645,403 sat at `z = 1.0116`, about the **P84** of that register. Open register at the data date: mean **1,085,000**, σ **807,140**, P80 **1,764,289**. The shortfall on the honest test is **3,684,289**. 5. **Interpretation. The ratio a certifier reports depends entirely on what is admitted to the denominator, and the four answers span 1.38 to 0.29 on identical facts.** That is not a technical quibble; it is the reason contingency certification must be a defined procedure rather than a judgment. The 1.3825 version — remaining contingency against the mean of open risk — is the one that appears most often in practice, because it is the easiest to compute and the most comfortable to sign; it fails because contingency exists precisely to cover the tail, and covering the mean is by construction a coin-flip. The 0.8502 version is defensible arithmetic and still wrong here, because it excludes 3,420,000 of claims on contingency that are not risks at all — they are approved, assessed or re-estimated amounts. **The honest denominator adds what is already known to what might still happen.** At 0.2893 the correct professional conclusion is not "coverage is thin" but "contingency has been fully committed and the project requires additional funding" — which is the in-balance conclusion of 14.2.1 arrived at by a completely independent route, and the agreement between the two is a check worth running deliberately. **Second, the collapse from 1.0746 to 0.2893 happened without a single surprise.** Every item that materialised was on the sanction register at sanction, with a probability and an impact; ground conditions at an intake structure ran at p 0.30 and it happened. Nothing unlikely occurred. What went wrong was the confidence level: 3,645,403 was a P84 provision against the whole register, and once the two largest items materialised the remainder was funded at a level nobody had named — Domain 8's point, now with a date attached. **Third, and most practically: the draw-rate check must be struck from the report.** 58.85 % drawn at 61 % progress is the sentence that let two quarters pass without a funding conversation. Replace it with the coverage ratio, its denominator itemised, and the shortfall in currency.

14.2.3 The in-balance condition and the cash call

Definition. The **in-balance condition** is a term of the facility requiring that, at each drawdown and at each reporting date, the funds available be at least the cost to complete on the basis the documents specify. Failure is a condition failure, not a default: the consequence is that **the facility does not fund** until the position is cured. Curing is a funding event, and the documents will name its permitted forms.

Four cures, in descending order of how much they cost a sponsor. **Additional equity** — the base case, and the one every other cure is measured against. **Cost-overrun support** (Domain 5, KA 5.2.3): a pre-agreed, capped sponsor undertaking to fund overruns, which converts a negotiation into a drawing on a facility the sponsor has already granted. A **standby or contingent facility**: committed debt available only on the in-balance test, which is cheaper than equity for the sponsor and priced accordingly by lenders, and which raises gearing exactly when the project can least support it. And **reallocation** — moving a funded but unspent line (usually a residual owner's-cost or fee allocation) to the deficient line, which requires lender consent and is the cure that most often turns out on inspection to be a re-labelling of contingency that has already been committed.

Two mechanical points a leader must own. **The order of the tests matters.** A project can be in balance on the model and out of balance on the commitments (14.1.1 against 14.2.1); it can pass the in-balance test and fail the contingency-adequacy test, or the reverse. Facilities that condition drawing on one test and reporting on another produce exactly the gap Case study A exploits. And **the cure is dated by the next draw, not by the reporting date.** A cost report issued in the month after a data date has already consumed part of the time available to raise money; the practical consequence is that the sponsor's treasury must be told about a prospective shortfall from the forecast, not from the report — which is what the rolling funding forecast of 14.1.2 is for.

AI in this KA

Where it earns its place. Three tasks, all high-volume and all currently done badly. **Claims and variation registers**: extracting notified claims from correspondence, matching each to a contractual head, a quantum claimed and an assessed exposure, and keeping the register reconciled to the cost report — the register is usually maintained in three incompatible places, and a machine that reconciles them earns its keep in a month. **Disaggregated `CPI` reporting**: computing performance indices separately for fixed-price and owner-retained scope and flagging any report that blends them, which is the defect of 14.2.1 and is trivially detectable. **Register roll-forward**: tracking which sanction-register items have materialised, retired or been restated, and recomputing the coverage ratio on the remainder at every data date with the denominator itemised.

Where it must not go. An assessed claim exposure is a **legal and commercial judgment** about entitlement under a specific contract in a specific jurisdiction, and it must not be produced by a model. The same applies to the certification that contingency remaining is adequate for risk remaining: that is a signed opinion, and the arithmetic supporting it is an input to the opinion, not a substitute for it. Nor may an assistant select the denominator — the choice between 1,085,000, 1,764,289 and 5,184,289 is the whole content of the test, and a tool that picks the flattering one because it was the last one prompted for has industrialised the failure this topic exists to prevent.

Verification, concretely. Recompute the register mean and P80 by hand for two items and confirm the aggregation; require that the denominator be printed with its components, never as a single figure. Reconcile the in-balance shortfall against the contingency shortfall — the two routes should tell a consistent story, and Kestrel's do (1,927,740 of in-balance shortfall against contingency committed 1,920,000 beyond its balance, plus the 7,740 of early-draw interest). Require the bridge of 14.2.1 to be presented explicitly whenever an `EAC` and a lender's cost-to-complete appear in the same pack, with every bridge line attributed to the discipline that is blind to it.

KEY TERMS — KA 14.2

Term	Meaning
Lender's cost-to-complete	Money to completion on a commitment basis: remaining contract value + approved variations + assessed claim exposure + bottom-up owner scope + remaining financing costs.
In-balance condition	Facility term requiring available funds ≥ cost to complete; failure stops funding rather than causing default.
Assessed claim exposure	The SPV's and adviser's judgment of the probable settled cost of notified but unagreed claims.
Blended `CPI` artefact	A performance index averaged across fixed-price and owner-retained scope; misattributes an overrun and forecasts neither scope.
Contingency coverage on the remainder	Unallocated contingency ÷ (known committed claims + P80 of the open register).
Draw-rate check	Contingency drawn as a share of contingency funded, against progress; coincidental, and to be struck from reports.
Cost-overrun support	A capped sponsor undertaking to fund overruns, converting a negotiation into a drawing.

SAMPLE MCQS — KA 14.2

MCQ 14.2-A `[14.2.1 · Application]` At a data date the remaining committed EPC value is 18,720,000, approved but uncertified variations are 840,000, assessed claim exposure is 1,260,000, the bottom-up remaining owner-retained scope is 2,400,000 and remaining capitalised interest is 1,436,674. The lender's cost-to-complete is:

- A. USD 21,120,000
- B. USD 24,656,674
- C. USD 23,220,000
- D. USD 20,846,038

Rationale: All five lines sum to 24,656,674. A is the bottom-up `CTC(d)`, which omits the three lines earned value cannot see. C omits remaining capitalised interest — the single most commonly dropped line, because it is not work. D is `CTC` on the `BAC/CPI` method, a different basis entirely.

MCQ 14.2-B `[14.2.1 · Analysis]` A project's blended `CPI` is 0.949821 across a fixed-price EPC scope of 48,000,000 (certified against milestones) and owner-retained scope of 3,600,000 running at a scope `CPI` of 0.60. Applying `BAC/CPI` to the blended index:

- A. is correct, since `BAC/CPI` is the standard persistence forecast
- B. attributes 2,535,849 of uplift to a scope that cannot overrun and only 190,189 to the scope that is overrunning, so the total is right only by coincidence
- C. understates the forecast, because fixed-price scope carries the greater risk
- D. is invalid because `CPI` cannot be computed on a milestone-certified contract

Rationale: The composition of the 2,726,038 uplift is the defect (14.2.1). A treats a method as valid irrespective of the scope mix it is applied to. C inverts the risk ownership under a wrap. D overstates: `CPI` on that scope is computable and equals 1.000 by construction — which is exactly why blending destroys the signal.

MCQ 14.2-C `[14.2.2 · Analysis]` Remaining unallocated contingency is 1,500,000. Known committed claims on contingency total 3,420,000 and the open risk register has a mean of 1,085,000 and a P80 of 1,764,289. The defensible coverage ratio on the remainder is:

- A. 1.3825
- B. 0.8502
- C. 0.2893
- D. 0.4386

Rationale: `1,500,000 / (3,420,000 + 1,764,289) = 0.2893`. A tests contingency against the mean of open risk only — covering the mean is a coin-flip and it also ignores the 3,420,000 already committed. B is the P80 test on open risk only, which is defensible arithmetic on the wrong denominator. D divides by the 3,420,000 of known committed claims alone (`1,500,000/3,420,000`), omitting the open register entirely.

MCQ 14.2-D `[14.2.2 · Analysis]` A monthly report states that 58.85 % of contingency has been drawn at 61.00 % certified progress and concludes that contingency consumption is "in line". The correct professional response is:

- A. accept it; the draw rate is below progress, so consumption is favourable
- B. reject the inference — contingency is not consumed pro rata to progress, and the only meaningful test is coverage on the remainder
- C. accept it if the draw rate has tracked progress for three consecutive periods
- D. recompute the draw rate on physical rather than certified progress

Rationale: The draw rate is coincidental because risk is not uniformly distributed through a programme (14.2.2); here the honest coverage on the remainder is 0.2893. C makes a trend out of a coincidence. D refines a measure that should not be relied on at all.

■ SELF-CHECK — KA 14.2

1. Name the three lines a lender's cost-to-complete contains that an *`EAC`* cannot see. — Approved variations not yet baselined, assessed claim exposure, and remaining capitalised interest and fees: 3,536,674 of Kestrel's 24,656,674.
2. Why is *`CPI`* on a milestone-certified fixed-price scope identically 1.000, and what follows? — Certified value equals budgeted milestone value, so `EV` = `AC`; cost risk on that scope shows up as schedule, claims and contractor solvency, and blending the index with owner-retained scope destroys the only real signal.

3. State Kestrel's contingency coverage on the remainder and its denominator. — 0.2893: 1,500,000 against 3,420,000 of known committed claims plus a 1,764,289 P80 on the open register.

KNOWLEDGE AREA 14.3

Progress certification and change control

Topics: 14.3.1 certification bases, and what each one funds · 14.3.2 advance payment, retention and the cash chain · 14.3.3 change control and the marginal coverage of a variation.

14.3.1 Certification bases, and what each one funds

Definition. Progress certification is the independent determination of the value of work executed in a period, and it is the hinge of the whole drawdown mechanic: it converts physical progress into a payment entitlement and a funding requirement. Three bases are in general use and they are not interchangeable. **Milestone certification** values only milestones actually achieved, each with a defined pre-agreed value — binary, lumpy, and the only basis that ties directly to a contractual entitlement. **Measured or percentage-of-completion certification** (`PoC`) values work in place on measured quantities or an assessed completion percentage of each activity — smoother, and requiring judgment. **Cost-incurred certification** values what the contractor has spent, including materials procured but not yet delivered to site — smoothest of all, and the basis on which the SPV's money leaves the country.

Who certifies matters as much as how. In a limited-recourse financing the certifier is usually an **independent engineer** or the **lender's technical adviser**, appointed under a deed the SPV cannot vary alone, and the certificate is a condition to each drawing. The certifier's independence is the structural protection; the certification basis is the structural exposure.

WORKED EXAMPLE

Worked example 14.3.1 — one quarter's work, valued three ways.

1. **Setup.** In construction quarter six, Kestrel's EPC contractor executes work whose scheduled milestone value for the quarter is **7,200,000**. Milestones actually achieved total **5,760,000**; one further milestone worth **1,440,000** is assessed by the independent engineer as **92 %** physically complete but not achieved. The contractor has additionally procured **690,000** of membrane racks, paid for and stored at the vendor's works, not delivered. Retention is **5 %** of certified value. Gearing 70/30, quarterly interest 1.5 %, two quarters remain. Value the quarter on each basis and quantify the exposure the choice creates.
2. **Formula.** Milestone basis = Σ achieved milestone values. Measured basis = Σ achieved + `PoC` × value of the incomplete milestone. Cost-incurred basis = measured + off-site materials. Net for payment = certified × (1 − retention). Debt advanced = net × gearing. Over-certification = (cost-incurred − milestone) × (1 − retention).
3. **Substitution.** `5,760,000`; `5,760,000 + 0.92 × 1,440,000 = 7,084,800`; `7,084,800 + 690,000 = 7,774,800`; each `× 0.95`; each `× 0.70`.
4. **Result.**

Basis	Certified value (USD)	Net of 5 % retention (USD)	Senior debt advanced (USD)
Milestone	5,760,000	5,472,000	3,830,400
Measured (`PoC`)	7,084,800	6,730,560	4,711,392
Cost-incurred	7,774,800	7,386,060	5,170,242

The spread on the same quarter's work is **2,014,800** of certified value — **34.98 %** of the milestone figure. Relative to the milestone basis, the cost-incurred basis releases **1,914,060** net, of which **1,339,842** is senior debt, and accelerates that requirement by up to two quarters at a cost of `1,339,842 × 0.015 × 2 =` **40,195** of additional capitalised interest. 5. **Interpretation. The certification basis is a credit decision disguised as an accounting convention, and its magnitude is a third of a quarter's draw.** Start with what each basis buys. Milestone certification is the most conservative and the only one whose numbers are contractual: 92 % of a milestone is worth exactly nothing, which is harsh on a contractor's cash flow and perfectly aligned with the lender's security, because an unachieved milestone is an unfinished thing. Measured certification is fairer to the contractor and introduces judgment — the 92 % is an opinion, and opinions drift upward under schedule pressure. Cost-incurred certification funds the contractor's balance sheet, and it is the basis on which lenders lose money. **The 1,339,842 of senior debt advanced against work not in place is 27.91 % of the 4,800,000 performance bond**, and it is unsecured in substance unless three things are true: the SPV holds a **vesting certificate** or equivalent transferring title in the off-site materials; the materials are **identified and segregated** at the vendor's works; and they are **insured in the SPV's name** against loss and against the vendor's insolvency. Absent any one of those, the money has bought a claim in someone else's liquidation — which is precisely what Case study B costs. **Second, the basis must be stated once and held.** Drift is the real risk: a project that begins on milestones and slides toward cost-incurred as the contractor tightens is not making a series of small accommodations, it is progressively converting secured lending into unsecured lending, one certificate at a time, and nobody signs the decision. The controls are simple and worth insisting on: the basis is defined in the facility and the certifier's deed; any off-site materials certification is separately identified, capped as a percentage of the contract price, and conditioned on vesting, segregation and insurance; and the cumulative off-site balance is a reported line. **Third, note the small number.** The 40,195 of extra capitalised interest is real but trivial beside the 1,339,842 of exposure — which is the right lesson about certification: it is not a cost question, it is a security question, and pricing it as a cost question is how it gets conceded.

14.3.2 Advance payment, retention and the cash chain

Definitions. An **advance payment** (or mobilisation payment) is a sum paid to the contractor at or shortly after contract signature, secured by an **advance payment bond** and recovered by deduction from subsequent certifications. **Retention** is a percentage withheld from each certification, usually capped as a percentage of the contract price, released in tranches at completion and at the end of the defects liability period. Both are timing mechanisms, and both have a price a project finance leader must be able to state.

WORKED EXAMPLE

Worked example 14.3.2 — what Kestrel's payment terms cost in capitalised interest.

1. **Setup.** The base case is Domain 6's funding profile with certified spend paid as it arises. The alternative applies the actual contract terms: an advance payment of **10 %** of the 48,000,000 EPC price — **4,800,000** — paid at financial close and recovered pro rata against each certification, and retention of **5 %** of each EPC certification capped at **2.5 %** of the contract price, being **1,200,000**, released half at the commercial operations date and half twelve months later. Interest at 1.5 % per quarter on opening drawn balances, gearing 70/30. Compute the effect of each term and of both together on capitalised interest.
2. **Formula.** Re-run the quarterly funding model with the period requirement adjusted: `requirement = certified spend − advance recovery − retention withheld + accrued interest`, with the advance added to the close-date requirement. Compare total capitalised interest.
3. **Substitution.** Close requirement `2,640,000 + 4,800,000`; quarterly recovery `4,800,000 × profile percentage`; retention `min(0.05 × EPC certified, remaining cap)`.
4. **Result.**

Payment terms	Capitalised interest (USD)	Change vs base (USD)
Base: certified spend paid as it arises	2,114,597	—
Advance payment only (4,800,000 at close)	2,369,194	**+254,597**
Retention only (1,200,000 withheld to completion)	2,052,008	**−62,589**
Both terms, as contracted	2,306,605	**+192,008**

The advance costs **254,597** of capitalised interest — **5.30 %** of the advance itself, and **0.53 %** of the EPC contract price. Retention saves **62,589**, one quarter of what the advance costs. 5. **Interpretation. An advance payment is a loan from the project to the contractor, and 254,597 is its price.** That reframing is the whole content of the topic, and it changes the negotiation. The SPV borrows at 6.0 % to fund the advance; the contractor, whose alternative is to finance its own mobilisation, borrows at whatever its own credit commands — typically more. If the contractor's marginal cost of funds exceeds the project's, the advance is genuinely value-creating for the transaction as a whole, and the correct professional position is therefore not "refuse the advance" but **"price it"**: a contractor unwilling to reduce the contract price by at least 254,597 — 0.53 % — in exchange for a 4,800,000 advance is being lent money for nothing. That is a computation the contractor can check and argue about honestly, which is what makes it a negotiating position rather than a posture. **Second, retention is worth far less than sponsors assume, in both directions.** 62,589 of interest saving over a two-year build is immaterial; retention's value is entirely as security for defect rectification, and it should be argued on that basis, against the alternative of a retention bond, which releases the cash to the contractor and substitutes a bank's credit for the withholding. **Third, and much the most dangerous: the second retention tranche falls after the commercial operations date.** Kestrel's 600,000 becomes payable twelve months after completion, which is inside the first operating year and outside any plausible availability period. If it is paid from operating cash, `CFADS` falls to `6,384,000 − 600,000 =` **5,784,000** and the `DSCR` at the first test falls to **1.1546 — below the 1.20 covenant**. Domain 10 measured Kestrel's annual headroom at **372,438**; a 600,000 retention release exceeds it by **227,562**. So a payment term agreed by a commercial team, in a contract, two years earlier, breaches a financial covenant. The remedies are all structural and all cheap if taken early: extend availability to cover retention releases, fund the releases into a dedicated retention account at completion, take a retention bond so no cash is withheld or released, or define `CFADS` to exclude construction-contract retention releases. What is not cheap is discovering the interaction in the first operating year, which is where it is normally discovered.

14.3.3 Change control and the marginal coverage of a variation

Definition. A **variation** (change order) alters the scope, price or time of a construction contract. In a corporate setting a variation has two numbers, cost and time. In a limited-recourse financing it has **four**: the price, the time, the **funding source** (which pocket pays — the contractor's, contingency, additional debt, additional equity), and the **coverage consequence** (what the variation does to the ratios the project is judged on). The third and fourth are the ones change-control procedures usually omit, and they are the ones that cannot be fixed later.

The test that connects them is the **marginal coverage of a variation**: whether the incremental `CFADS` a variation generates supports the incremental debt service its funding creates.

```
Marginal DSCR = Δ CFADS ÷ Δ debt service,   where Δ debt service = debt-funded cost ÷ AF(r, n)

Maximum coverage-neutral debt funding = (Δ CFADS ÷ target DSCR) × AF(r, n)
```

WORKED EXAMPLE

Worked example 14.3.3 — the variation that creates value and destroys coverage.

1. **Setup.** Kestrel's operator proposes an additional membrane train: capital cost **1,850,000**, adding **240,000** per year of `CFADS` for the project's remaining 25-year operating life. The loan runs 12 years at 6.0 % (`AF(0.06, 12) = 8.383844`); the board's appraisal rate is 8.0 % (`AF(0.08, 25) = 10.674776`); base `DSCR` is 1.2743 and the covenant is 1.20×. Assess the variation on value and on coverage, and determine how it should be funded.
2. **Formula.** `NPV = ΔCFADS × AF(0.08, 25) − cost`; `PI = PV/cost`; `Δ debt service = debt-funded amount ÷ AF(0.06, 12)`; `marginal DSCR = ΔCFADS ÷ Δ debt service`; maximum coverage-neutral debt funding as above.
3. **Substitution.** `240,000 × 10.674776 = 2,561,946`; `− 1,850,000`; `1,850,000 / 8.383844 = 220,663`; `240,000 / 220,663`; `(240,000 / 1.274303) × 8.383844`.
4. **Result.** `NPV` **+USD 711,946** and `PI` **1.3848** — comfortably value-accretive on Domain 4's tests. Funded **entirely by debt**, incremental debt service is **220,663** and the **marginal** `DSCR` **is 1.0876** — below the 1.20 covenant and far below the 1.2743 base. Funded **70/30** like every other use, the debt share is 1,295,000, incremental debt service **154,464** and the marginal `DSCR` **1.5538** — accretive. The maximum debt funding that holds base coverage constant is `(240,000/1.274303) × 8.383844 =` **USD 1,578,947**, being **85.35 %** of the cost; the residual **271,053** must come from equity or contingency. To be coverage-neutral at full debt funding the variation would need **281,200** per year of `CFADS`, **41,200** more than it delivers.
5. **Interpretation. A variation can be value-accretive and coverage-dilutive at the same time, and the funding decision follows the coverage test, not the** `NPV`**.** That is the sentence to carry out of this topic. The reason the two tests disagree is structural: `NPV` discounts 25 years of incremental cash at 8 %, while `DSCR` compares one year of incremental cash against debt service compressed into 12 years at 6 %. A long-lived, modestly-returning addition therefore looks good on value and thin on coverage, every time — and the ratio that binds is the one in the facility agreement. **The general rule is worth memorising in the form that makes it usable:** a variation is coverage-neutral if its debt-funded share does not exceed `ΔCFADS/target DSCR × AF(r, n)` — here 85.35 % of cost — so at Kestrel's 70 % gearing the variation is comfortably accretive and at 100 % debt funding it is not. **Most variations are silently funded at 100 % from whatever is undrawn, which is the funding order nobody chose.** Third, the aggregate effect is small and therefore easy to wave through: fully debt-funded, this variation moves the blended `DSCR` from 1.2743 to **1.2665**, the covenant cash trigger from 6,011,562 to **6,276,357** and annual headroom from 372,438 to **347,643** — a loss of **24,795** of headroom, 6.66 % of it. One variation is nothing. Eight are a covenant. **The control that works is a threshold in the change-control procedure**: every variation above a stated value carries a computed marginal `DSCR` and a named funding source before approval, and the cumulative headroom consumed by approved variations is a reported line — the financing analogue of PML-AI Domain 4's baseline-drift test. **The professional caution:** the 240,000 of incremental `CFADS` is a forecast produced by the party proposing the variation, and a marginal `DSCR` of 1.0876 is inside the error bar of most operating forecasts. Where the coverage test is close, the honest conclusion is that the variation should not be debt-funded at all.

AI in this KA

Where it earns its place. Certification packs are large, repetitive and structured, which makes them good machine territory: cross-checking claimed quantities against the measurement schedule and the previous certificate; detecting **certification drift** by tracking, period by period, the share of certified value attributable to achieved milestones, assessed percentages and off-site materials — a trend no monthly report currently shows and which a machine can produce for nothing; verifying that every off-site materials certification has a matching vesting certificate, segregation confirmation and insurance endorsement, and listing the exceptions; and computing the four numbers of a variation (price, time, funding source, marginal `DSCR`) at the point the variation is logged, so that the coverage consequence exists before the approval rather than after it.

Where it must not go. A **certification is a professional opinion with liability attached** and must not be generated, and an assessed percentage of completion must not be inferred from photographs or progress narrative and presented as a measurement. Entitlement questions — whether a claimed variation is a variation at all, whether the contractor bears the cost, whether a delay is excusable — are contractual and jurisdictional and belong to the certifier and to counsel.

Verification, concretely. Recompute one certificate end to end from measured quantities to net payment, including advance recovery and retention, and require agreement to the dollar; confirm the retention cap has not been exceeded cumulatively. Reconcile cumulative certified value to cumulative milestone value and separately identify the off-site balance. For every variation above the threshold, recompute `Δ debt service` and the marginal `DSCR` by hand and check the funding source recorded matches the funding actually drawn. Require that any tool computing a marginal `DSCR` print the `AF(r, n)` it used and the target ratio it compared against — those two inputs carry the answer.

KEY TERMS — KA 14.3

Term	Meaning
Milestone certification	Value of milestones actually achieved; binary, lumpy, tied to contractual entitlement.
Measured / `PoC` certification	Value of work in place on measured quantities or assessed completion; requires judgment.
Cost-incurred certification	Value of contractor spend including off-site materials; funds the contractor's balance sheet.
Vesting certificate	Instrument transferring title in off-site materials to the SPV; with segregation and insurance, the condition of any off-site certification.
Certification drift	Progressive migration from milestone toward cost-incurred certification; converts secured lending into unsecured lending without a decision.
Advance payment / bond	Pre-mobilisation payment recovered from certifications, secured by a bond; a loan from the project to the contractor.
Retention / retention bond	Percentage withheld from certifications, capped and released in tranches; a bond substitutes bank credit for the withholding.
Marginal `DSCR` of a variation	`ΔCFADS ÷ Δ debt service`; coverage-neutral debt funding = `(ΔCFADS/target DSCR) × AF(r, n)`.

SAMPLE MCQS — KA 14.3

MCQ 14.3-A `[14.3.1 · Application]` A quarter's work comprises achieved milestones of 5,760,000 and one milestone worth 1,440,000 assessed at 92 % complete; the contractor has also procured 690,000 of off-site materials. Retention is 5 %. The senior debt advanced at 70 % gearing under a cost-incurred basis exceeds that under a milestone basis by:

- A. USD 2,014,800

- B. USD 1,914,060
- C. USD 1,339,842
- D. USD 690,000

Rationale: `(7,774,800 − 5,760,000) × 0.95 × 0.70 = 1,339,842`. A is the gross certified spread, before retention and before the gearing split. B applies retention but not gearing. D counts only the off-site materials and omits the 92 % milestone.

MCQ 14.3-B `[14.3.2 · Analysis]` Kestrel's second retention tranche of 600,000 falls due twelve months after the commercial operations date and is paid from operating cash. The consequence is:

- A. none; retention is part of the EPC price already funded
- B. `CFADS` falls to 5,784,000 and the `DSCR` to 1.1546, breaching the 1.20 covenant, because the release exceeds the 372,438 of annual headroom by 227,562
- C. the `DSCR` is unaffected, since retention is a balance-sheet movement
- D. the release is deducted from debt service, so coverage improves

Rationale: The release is a cash outflow in an operating period, and coverage is computed on cash (Domain 10). A confuses being funded with being available — the availability period has ended. C misstates a cash payment as an accrual. D reverses the direction of the effect.

MCQ 14.3-C `[14.3.3 · Application]` A variation costs 1,850,000 and adds 240,000 a year of `CFADS`. The loan factor is `AF(0.06, 12) = 8.383844` and base `DSCR` is 1.2743. The maximum debt-funded amount that leaves base coverage unchanged is closest to:

- A. USD 1,850,000
- B. USD 1,578,947
- C. USD 1,295,000
- D. USD 2,012,123

Rationale: `(240,000/1.274303) × 8.383844 = 1,578,947`, being 85.35 % of the cost. A is full debt funding, which yields a marginal `DSCR` of 1.0876. C is the 70 % pro-rata debt share — inside the limit but not the limit. D applies `AF` to the full `ΔCFADS` without the coverage divisor, the sizing error of Domain 10, Exercise 10.1.

MCQ 14.3-D `[14.3.3 · Analysis]` A variation has an `NPV` of +711,946 at the board's 8 % rate and a marginal `DSCR` of 1.0876 against a 1.20× covenant if fully debt-funded. The correct conclusion is:

- A. approve and fund from the facility; a positive `NPV` is decisive
- B. reject; a marginal `DSCR` below the covenant disqualifies the variation
- C. approve, but cap debt funding at the coverage-neutral 1,578,947 and fund the residual 271,053 from equity or contingency
- D. approve and renegotiate the covenant

Rationale: Value and coverage answer different questions and both bind; the funding mix is the lever that satisfies both (14.3.3). A ignores the covenant. B discards 711,946 of value that a funding decision can capture. D proposes to renegotiate a covenant over 271,053, which no lender would price kindly.

SELF-CHECK — KA 14.3

1. State the three conditions that make off-site materials certification defensible. — Vesting of title in the SPV, identification and segregation at the vendor's works, and insurance in the SPV's name including against vendor insolvency.

2. What does Kestrel's 10 % advance payment cost, and what is the negotiating consequence? — 254,597 of capitalised interest, 5.30 % of the advance and 0.53 % of the EPC price; the advance should be priced, not simply refused or granted.
3. Give the coverage-neutral funding rule for a variation. — Debt funding must not exceed `ΔCFADS/ target DSCR × AF(r, n)` — for Kestrel's train, 1,578,947, or 85.35 % of cost.

KNOWLEDGE AREA 14.4

Schedule delay, lender reporting and completion tests

Topics: 14.4.1 the funded cost of a month of slip · 14.4.2 coverage at a calendar-fixed first repayment date · 14.4.3 lender reporting and the completion tests.

14.4.1 The funded cost of a month of slip

Definition. The **funded cost of delay** is the cash the project must raise, per unit of time of slip, to keep paying while it is not yet earning. It is a strict subset of the **economic cost of delay**, which adds the cash the project would have earned. The distinction sounds pedantic and is worth millions: **the drawdown and in-balance tests see only the funded cost, and the liquidated damages that under-recover the economic cost can substantially over-recover the funded one.**

Domain 5 (KA 5.4.2) priced a slip at the commercial operations date at **24,733.33 per day** — 7,000 of interest on a fully drawn 42,000,000 plus 17,733.33 of forgone `CFADS`, on a 30/360 basis — and Domain 8 (KA 8.4.2) priced a slip declared during construction, where escalation on unbought scope appears and forgone revenue is deferred rather than lost. This topic completes the family by splitting the same event along a different axis: **not economic against contractual, but funded against forgone.**

WORKED EXAMPLE

Worked example 14.4.1 — a month of extension at Kestrel's commercial operations date.

1. **Setup.** At the scheduled commercial operations date the facility is fully drawn at **42,000,000** at **6.0 %**. Prolongation costs the SPV bears directly: extended owner's team and site supervision **138,000** per month, extended construction all-risks and delay-in-start-up premium **46,000**, and independent engineer, facility agent and technical adviser monitoring fees **21,000**. Operating `CFADS` would have been **6,384,000** per year. Delay damages are **20,000 per day**, capped at **4,800,000** (Domain 12). The availability period runs to **six months** beyond the scheduled commercial operations date. Compute the funded and economic costs per month, the recovery each enjoys, and the exposure the mismatch creates.
2. **Formula.** Funded cost = interest on drawn debt + prolongation costs = `42,000,000 × 0.06/12 + 138,000 + 46,000 + 21,000`. Forgone `CFADS` = `6,384,000/12`. Full economic cost = funded cost + forgone `CFADS`. Recovery = damages ÷ cost. Months covered = damages cap ÷ monthly cost.
3. **Substitution.** `210,000 + 138,000 + 46,000 + 21,000`; `6,384,000/12 = 532,000`; `415,000 + 532,000`; `600,000/415,000`; `600,000/947,000`; `4,800,000/415,000`; `4,800,000/947,000`.
4. **Result.**

Per month of slip at the commercial operations date	USD	Damages recovery
Interest on drawn debt (42,000,000 at 6.0 %)	210,000	
Extended owner's team and site supervision	138,000	
Extended all-risks and delay-in-start-up premium	46,000	
Independent engineer, agent and adviser fees	21,000	
Funded cost of delay	**415,000**	**144.58 %**
Forgone CFADS (6,384,000 ÷ 12)	532,000	
Full economic cost of delay	**947,000**	**63.36 %**
Domain 5's basis (interest + forgone *CFADS* only)	742,000	80.86 %
Delay damages at 20,000 per day (30 days)	600,000	

Damages of 4,800,000 cover **11.5663 months** of funded cost but only **5.0686 months** of full economic cost. Over an eight-month slip — the day the cap binds — the funded cost is **3,320,000** against damages of 4,800,000, a **surplus of 1,480,000**, while the full economic cost is **7,576,000**, leaving **2,776,000** uncovered. The availability period funds only `6 × 415,000 =` **2,490,000** of extension cost. 5. **Interpretation. One damages rate, three coverage percentages — 144.58 %, 80.86 % and 63.36 % — and which one you quote depends entirely on which cost basis you are defending.** That is the professional content, and it explains a pattern practitioners see and misdiagnose: a delayed project whose drawdown tests keep passing while equity value is being destroyed. The in-balance test measures funds against cost-to-complete, and forgone revenue is in neither column; a delay that over-recovers the funded cost therefore leaves the facility comfortable and the sponsor poorer, month after month, with no covenant to notice it. **The report must carry both numbers or it is not a report.** Second, note the two constraints that bind before the damages cap does, because they are the ones sponsors miss. **The availability period binds at six months, not eleven and a half**: beyond month six the facility no longer funds the extension and the sponsor pays 415,000 a month in cash even though damages will eventually reimburse it — a **timing** exposure of up to `(11.5663 − 6) × 415,000 =` about **2,310,000** that is invisible in a coverage table. And **damages are recovered later than the cost is incurred**, often much later and sometimes only after adjudication, so the sponsor is financing the contractor's breach in the meantime. Third, the composition is instructive: **prolongation costs the SPV bears directly — 205,000 a month — are 49.4 % of the funded cost and are entirely absent from Domain 5's calibration basis.** A damages rate calibrated on interest plus forgone `CFADS` alone therefore understates the loss by those 205,000, and the correct calibration statement is that Kestrel's full monthly cost is 947,000, or **31,566.67 per day** — against which 20,000 recovers 63.36 %, not the 80.86 % Domain 5's narrower basis suggests. Both figures are honest; the negotiation should use the wider one, and say so.

14.4.2 Coverage at a calendar-fixed first repayment date

The mechanic. Facility agreements set repayment dates. Some set them by reference to the **actual** commercial operations date; many set them by **calendar date**, fixed at financial close with a cushion for expected slip. A calendar-fixed first repayment date has a consequence that almost no construction report models: **a slip does not postpone the first instalment, it shortens the operating period that funds it.** With `m` months of slip, the first annual test sees `(12 − m)/12` of a year's `CFADS` against a full year's debt service.

WORKED EXAMPLE

Worked example 14.4.2 — Kestrel's commercial operations date slips four months.

1. **Setup.** Debt service **USD 5,009,635.23** falls due twelve months after the scheduled commercial operations date, by calendar date. Annual operating CFADS **6,384,000**, accruing evenly from the actual commercial operations date. The covenant is **1.20×**; the six-month debt service reserve holds **2,504,818**. Delay damages of **20,000 per day** are recoverable, capped at 4,800,000. Compute the DSCR at the first test for a four-month slip, the reserve consumption, and the slip at which each threshold is crossed.
2. **Formula.** `CFADS at first test = 6,384,000 × (12 − m)/12`; `DSCR = CFADS ÷ 5,009,635.23`. Breakevens solve `6,384,000 × (12 − m)/12 = k` for `k` equal to the covenant cash trigger (1.20 × debt service), to debt service, and to debt service less the reserve.
3. **Substitution.** `6,384,000 × 8/12 = 4,256,000`; `÷ 5,009,635.23`. Covenant: `12 × (1 − 6,011,562.28/6,384,000)`. Payment: `12 × (1 − 5,009,635.23/6,384,000)`. With reserve: `12 × (1 − (5,009,635.23 − 2,504,817.62)/6,384,000)`.
4. **Result.**

Slip (months)	CFADS at first test (USD)	DSCR	Shortfall vs debt service (USD)	Share of DSRA consumed
0	6,384,000	**1.2743**	—	—
1	5,852,000	**1.1681**	—	—
2	5,320,000	**1.0620**	—	—
3	4,788,000	**0.9558**	221,635	8.85 %
4	4,256,000	**0.8496**	753,635	30.09 %
6	3,192,000	**0.6372**	1,817,635	72.57 %
8	2,128,000	**0.4248**	2,881,635	115.04 %

Breakevens: the **1.20× covenant is breached beyond 0.7001 months — 21.00 days**; the project can no longer pay debt service from operating cash beyond **2.5834 months (77.50 days)**; and it can no longer pay from cash plus a fully funded reserve beyond **7.2917 months (218.75 days)**. If delay damages are creditable to CFADS, a four-month slip yields `20,000 × 120 =` **2,400,000** of damages, lifting CFADS to **6,656,000** and the DSCR to **1.3286 — 0.4791 higher**, and above the undelayed base case. 5. **Interpretation. Three weeks of slip breaches the covenant.** That is the result, and it is the single most under-modelled number in construction-phase finance. Nothing has gone wrong with the project: the plant works, the tariff is contracted, CFADS is running exactly at forecast on an annualised basis, and the covenant fails at the first test because the test period contains less operating time than the obligation period. Sponsors discover this at the first compliance certificate, which is the worst possible moment, and the discovery is unforced. **Second, whether delay damages count in CFADS is worth 0.4791 of coverage — more than the entire distance from the base case to the covenant.** Domain 10 established that CFADS is a defined term and that its documented construction governs every ratio built on it; this is that principle at its most valuable. Damages at 20,000 a day are 600,000 a month against forgone CFADS of 532,000, so they **over-compensate** the coverage effect of delay, and the sign of the delay's impact on the first test flips on a definitional clause. A leader who negotiates one thing in the CFADS definition should negotiate this. Note the tail carefully, though: once the damages cap binds at eight months the damages line stops growing while the CFADS line keeps shrinking, so the damages-credited coverage peaks at **1.3829** and then falls, crossing 1.20 at **9.7226 months** and 1.00 at **11.6059** — the cap converts a comfortable position into a cliff. **Third, the reserve is smaller protection than it looks.** A six-month debt service reserve does not survive six months of slip; it survives **7.2917 months** only because operating cash covers most of the obligation, and it is consumed 30.09 % by a four-month slip that also breaches the covenant and must then be replenished ahead of any distribution (Domain 10, KA 10.3.3). **Fourth, the fixes are all structural and all cheap at signing.** Set the first repayment date by reference to the actual commercial operations date with a long-stop; or annualise the first test on a look-forward basis; or size the reserve on the delay scenario rather than on a convention of six months; or provide expressly that delay damages and delay-in-start-up insurance proceeds are `CFADS`. Each is a drafting point worth more than the margin. **The professional caution:** none of these fixes should be pursued quietly as a technical amendment. A lender asked to move a repayment date late in a delayed project reads the request as news about the project, which is why the structure is negotiated before the delay and the disclosure discipline of Domain 10, KA 10.4.4 applies from the first month of slip.

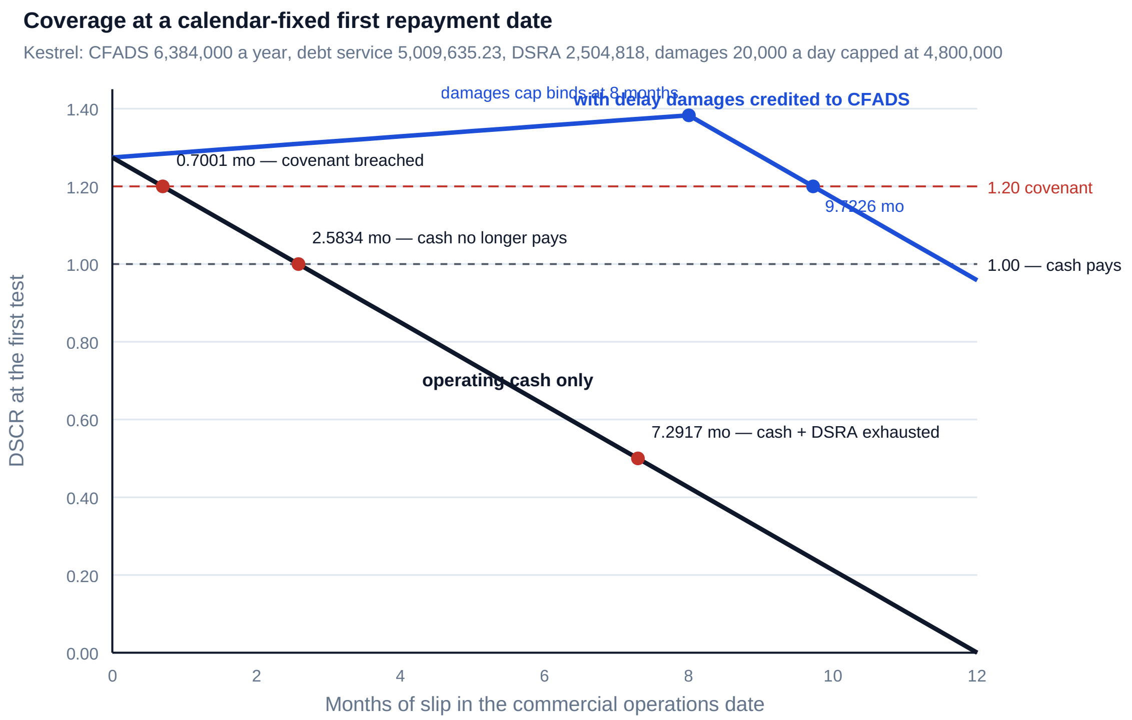

Fig 14.4.1 — Coverage at a calendar-fixed first repayment date

14.4.3 Lender reporting and the completion tests

The reporting obligation. Construction-phase information covenants are heavier than operating ones because the security does not yet exist. The standard pack, monthly or quarterly: certified progress against the programme, with the certifier's own commentary; cost incurred against the sources and uses, restated per 14.1.1; the cost-to-complete and the in-balance statement (14.2.1); the contingency position with coverage on the remainder (14.2.2); the variation and claims registers with movements; the schedule position, critical path and forecast commercial operations date; health, safety and environmental performance and any reportable incident; insurance status and claims; and the compliance certificate signed by named officers. Two disciplines make the pack useful rather than ceremonial: **the same numbers reconcile across sections** — the contingency balance in the contingency section equals the contingency line in the restated sources and uses, and a pack in which they differ has told the lender something the sponsor did not intend — and **the forecast commercial operations date is stated as a single date with a stated confidence, not as "on programme"**, because "on programme" is the phrase that precedes every delay notification.

The completion sequence. Completion is not one event, and conflating its stages is the commonest drafting and reporting error of the phase. **Mechanical completion** confirms the plant is built and can be safely energised. **Commissioning and reliability testing** demonstrates that it runs. **Performance testing** demonstrates that it runs to guarantee — output, quality, efficiency, consumption. **Provisional acceptance or the commercial operations date** is the contractual moment from which revenue accrues and, in the financing, the moment the construction facility converts to term debt, the operating covenant regime begins, the completion guarantees and cost-overrun support fall away, the first retention tranche is released and the debt service reserve must be funded. **Final acceptance** follows the defects liability period and releases the second retention tranche. The financing attaches consequences to each, and the sponsor who has read only the EPC contract's definitions will be surprised by the facility's.

Conditional and partial completion. Plants routinely complete late, or on time with a punch list, or on time at less than guaranteed output. The first is KA 14.4.1 and 14.4.2. The second is managed by valuing the punch list and withholding, which is arithmetic. The third is the interesting case, because a permanent output shortfall is a permanent `CFADS` shortfall and therefore a permanent coverage problem, remedied by **performance liquidated damages** or a **buy-down** — a lump sum, usually applied to prepay debt, calibrated so that the financing survives a smaller plant (Domain 12, KA 12.1.3).

WORKED EXAMPLE

Worked example 14.4.3 — the buy-down that restores which coverage?

1. **Setup.** Kestrel completes on time but the performance test settles at an operating `CFADS` of **5,900,000** rather than 6,384,000. Debt outstanding at completion **42,000,000**, `AF(0.06, 12) = 8.383844`, covenant **1.20×**, base `DSCR` **1.2743**, performance damages cap **4,800,000**. Compute the coverage as completed and the buy-down that restores (i) the covenant and (ii) the coverage the lenders originally priced.
2. **Formula.** `DSCR = CFADS ÷ (debt ÷ AF)`. Supportable debt at a target ratio = `(CFADS ÷ target) × AF`. Buy-down = 42,000,000 − supportable debt.
3. **Substitution.** `5,900,000 / (42,000,000/8.383844)`; `(5,900,000/1.20) × 8.383844`; `(5,900,000/1.274303) × 8.383844`.
4. **Result.** As completed the `DSCR` is **1.1777** — the **covenant is breached from the first test**. Restoring the 1.20× covenant requires debt of **41,220,566**, a buy-down of **779,434**. Restoring the 1.2743 base coverage requires debt of **38,815,789**, a buy-down of **3,184,211**. Both sit inside the 4,800,000 performance cap; the full cap would take debt to 37,200,000 and coverage to **1.3297**.
5. **Interpretation. The buy-down that restores the covenant is a quarter of the buy-down that restores the coverage the lenders priced, and the difference — 2,404,777 — is settled by drafting, not by negotiation after the test.** That is the transferable lesson. A buy-down formula expressed as "the amount required to restore the `DSCR` to the covenant level" is worth 779,434 here; one expressed as "to restore the base-case `DSCR` in the financial model at close" is worth 3,184,211; and the two clauses look almost identical to a reader who is not computing them. The sponsor should also see what the difference costs it: at 41,220,566 of debt the project sits exactly on its covenant with zero headroom, so the first bad operating year breaches — a position no sponsor should accept in exchange for keeping 2,404,777 of contractor money that would otherwise have deleveraged the project. **Second, note that the buy-down is a prepayment, not compensation.** It repairs coverage by shrinking debt, so it repairs the lender's position directly and the sponsor's only through the coverage headroom it restores; the equity return still falls, because the plant is permanently smaller. **Third, the cap is the binding question again.** Here both buy-downs fit inside 4,800,000, but a 10 % shortfall rather than a 7.6 % one would not, and the residue would sit on equity exactly as Domain 12 computed for the delay head. The professional sequence at a failed performance test is therefore: compute the permanent `CFADS` effect; compute both buy-downs; read the clause to see which one you are entitled to; check it against the cap and against the aggregate cap after any delay claim; and only then open the negotiation.

AI in this KA

Where it earns its place. The construction report pack is a reconciliation problem, and machines reconcile. Cross-checking that every figure appearing in two sections of the pack agrees, and listing the exceptions, is the single highest-value automation available in this knowledge area, and it catches the defect that most embarrasses sponsors. Beyond that: recomputing the coverage-at-first-repayment table (14.4.2) at every data date from the current forecast commercial operations date, so that the covenant breakeven in days of slip is a live number rather than an annual discovery; maintaining the delay-cost ledger with funded and economic columns and the cumulative damages position against the cap and its binding day; and drafting the narrative sections of the pack from the underlying registers, which reduces the drift between what the numbers say and what the commentary says.

Where it must not go. A **forecast commercial operations date** is a schedule opinion belonging to the certifier and the delivery organisation, and a model that produces one from trend data has produced an input to that opinion, not a substitute for it. Whether a delay is excusable, whether an extension of time is due, whether a completion test has been passed, and what remedies a failed test entitles anyone to are contractual determinations for the certifier and counsel. And no machine output may constitute the compliance certificate: that is signed.

Verification, concretely. Recompute the `DSCR` at the first repayment date by hand for the current forecast slip and confirm the covenant breakeven in days; require that any tool reporting it state whether delay damages have been treated as `CFADS` and cite the definition clause relied on. Recompute the monthly funded and economic costs of delay independently and confirm the damages recovery percentages against both. For a buy-down, recompute the supportable debt at both target ratios and check which the clause specifies. Tie the cumulative damages position to the cap and its binding day, and confirm nothing in the pack assumes recovery beyond it.

KEY TERMS — KA 14.4

Term	Meaning
Funded cost of delay	Cash the project must raise per period of slip: interest on drawn debt plus prolongation costs. Excludes forgone `CFADS`.
Economic cost of delay	Funded cost plus forgone `CFADS`; the correct calibration basis for delay damages.
Availability period	The window in which the facility may be drawn; often binds on a delay long before the damages cap does.
Calendar-fixed repayment date	A first instalment set by date rather than by actual completion; a slip shortens the operating period that funds it.
Mechanical completion / commissioning / performance test / commercial operations date / final acceptance	The completion sequence; the financing attaches distinct consequences to each.
Buy-down	A lump sum, usually applied to prepay debt, calibrated so the financing survives a permanently smaller plant.

SAMPLE MCQS — KA 14.4

MCQ 14.4-A `[14.4.2 · Application]` Kestrel's first instalment of 5,009,635.23 falls on a fixed calendar date twelve months after the scheduled commercial operations date. `CFADS` accrues evenly from actual completion at 6,384,000 a year. If completion slips four months, the `DSCR` at the first test is:

- A. 1.2743
- B. 0.8496
- C. 1.1681
- D. 0.4248

Rationale: `6,384,000 × 8/12 = 4,256,000`; `÷ 5,009,635.23 = 0.8496`. A assumes the test date moves with completion. C is a one-month slip. D is an eight-month slip.

MCQ 14.4-B [14.4.2 · Analysis] With a 1.20× covenant, debt service of 5,009,635.23 and CFADS of 6,384,000 a year accruing from actual completion, the slip at which the first covenant test fails is closest to:

- A. six months
- B. three weeks
- C. two months
- D. it cannot fail, since annualised CFADS is unchanged

Rationale: `12 × (1 − 6,011,562.28/6,384,000) = 0.7001` months, 21.00 days. A and C overstate the tolerance by an order of magnitude. D is the error the calculation exists to kill: the covenant is tested on the period, not on an annualised run rate.

MCQ 14.4-C [14.4.1 · Analysis] Kestrel's funded cost of a month of slip at completion is 415,000 and its full economic cost 947,000; delay damages are 600,000 a month. The most useful observation for a board is:

- A. damages recover 80.86 % of the cost of delay
- B. damages over-recover the funded cost by 44.58 % while recovering only 63.36 % of the economic cost, so the drawdown tests keep passing while equity value is destroyed
- C. damages fully cover the delay, so no action is required
- D. the delay is cost-neutral because damages exceed the funded cost

Rationale: The two bases give 144.58 % and 63.36 % recovery, and only the second is a statement about value (14.4.1). A quotes Domain 5's narrower basis, which omits the 205,000 of monthly prolongation cost. C and D mistake a passing funding test for an absence of loss.

MCQ 14.4-D [14.4.3 · Application] A performance test settles CFADS at 5,900,000 against debt of 42,000,000, `AF(0.06, 12) = 8.383844`. The buy-down required to restore the base-case DSCR of 1.2743 is closest to:

- A. USD 779,434
- B. USD 3,184,211
- C. USD 4,800,000
- D. nil, since 1.1777 exceeds 1.00

Rationale: `42,000,000 − (5,900,000/1.274303) × 8.383844 = 3,184,211`. A restores only the 1.20× covenant, leaving zero headroom. C is the performance damages cap, not the calibrated amount. D confuses paying debt service with satisfying a covenant (Domain 10, KA 10.2.1).

SELF-CHECK — KA 14.4

1. Distinguish the funded and economic costs of a month of Kestrel's delay, with figures. — Funded 415,000 (interest 210,000 plus 205,000 of prolongation); economic 947,000, adding 532,000 of forgone CFADS. Damages of 600,000 recover 144.58 % and 63.36 % respectively.
2. Why does a three-week slip breach a 1.20× covenant on an otherwise perfect project? — A calendar-fixed first repayment date shortens the operating period funding a full year's debt service; `12 × (1 − 6,011,562/6,384,000) = 0.7001` months.
3. What is the drafting difference between the two buy-downs, in money? — 2,404,777: restoring the 1.20× covenant costs 779,434, restoring the 1.2743 base coverage costs 3,184,211.

Advanced topics — Domain 14

14.A.1 The availability period, long-stop dates and the commitment that lapses

An **availability period** is the window in which a facility may be drawn; a **long-stop date** is the date by which completion must occur before defined consequences follow. Both are dates set at close against a programme that has not yet started, and both are routinely set with a cushion calibrated to optimism. The construction-phase consequence is asymmetric and often misunderstood: **an expired availability period does not accelerate the loan, it simply stops funding** — the undrawn commitment lapses and every remaining use becomes an equity obligation. For Kestrel, a six-month cushion funds `6 × 415,000 =` **2,490,000** of extension cost against delay damages that would cover **11.5663 months**, so the binding constraint on a delay is the availability period and not the damages cap, and the exposure is a **timing** exposure of roughly 2,310,000 rather than a loss. Three practices follow. Model the availability period against the P80 completion date, not the programme date. Check that the availability period covers every use that falls after completion — Kestrel's second retention tranche of 600,000 falls twelve months after the commercial operations date and is outside any plausible window (KA 14.3.2). And treat an availability-period extension as a pre-agreed mechanic where possible, because a request made during a delay is priced as news.

14.A.2 Certifying value while liability is disputed

The hardest recurring problem in construction monitoring is that certification and entitlement move on different clocks. Work is in place, the SPV disputes that the contractor is entitled to be paid for it, and the draw request must nonetheless state a number. Three conventions are in use and each has a cost. **Certify and reserve**: the certifier values the work and the certificate records the dispute, so cash moves and the argument continues — which protects the programme and funds a position the SPV may later have to recover. **Withhold and escalate**: nothing is certified, which protects the money and stops the site, converting a commercial dispute into a schedule loss at 415,000 a month of funded cost and 947,000 of economic cost. **Certify into escrow**: the funds are drawn and held, which costs the interest and preserves both positions — the compromise, and the one that requires the facility to permit a drawing into an account the SPV does not control. The finance leader's contribution here is not legal but arithmetical: **price the three options before the meeting**, because the ordinary failure mode is to withhold on principle for two months and discover that principle cost 1,894,000 of economic value against a claim worth 1,260,000.

14.A.3 The reviewer's drawdown eye

Invariants to test on any construction monitoring pack. Drawn-to-date ties to the facility agent's records, not to the model. Cumulative debt and equity percentages of commitment are equal under pro-rata funding, and any divergence traces to a documented funding order. Available commitment is undrawn debt plus uncalled equity and **excludes** contingency, which is a use. The restated sources-and-uses columns reconcile, and any imbalance is explained in cash terms rather than absorbed. Capitalised interest to date reproduces Domain 8's area rule from the profile, the rate and the gearing, to the dollar; interest is computed on opening, not closing, balances. `CPI` is reported **separately** for fixed-price and owner-retained scope, and equals 1.000 on milestone-certified fixed-price scope; no blended index feeds a `BAC/CPI` forecast. The lender's cost-to-complete contains remaining committed value, approved variations, assessed claim exposure, a bottom-up owner-scope estimate **and** remaining financing costs, and reconciles to the `EAC`-based `CTC` through an explicit bridge. Contingency coverage on the remainder is reported with its denominator itemised, and the in-balance shortfall and the contingency shortfall tell a consistent story. Certified value reconciles to milestone value with the off-site materials balance separately identified, and every off-site amount has vesting, segregation and insurance. Retention held has not exceeded its cap, and each release tranche has a funding source that is not operating cash. Every variation above the threshold carries a marginal DSCR and a funding source, and cumulative headroom consumed by variations is reported. The delay ledger shows funded **and** economic costs and the cumulative damages position against the cap and its binding day. And the DSCR at the first repayment date is computed from the current forecast completion date, with the covenant breakeven stated in days and the treatment of delay damages in CFADS cited to a clause.

Industry variations — Domain 14

- **Water and desalination.** Long commissioning and reliability-test periods mean the gap between mechanical completion and the commercial operations date is months rather than weeks, so the calendar-fixed repayment-date problem of KA 14.4.2 is at its sharpest; permit conditions (discharge, brine outfall) generate owner-retained variations late in the programme, exactly where contingency coverage on the remainder is thinnest.
- **Solar and wind, contracted power.** Modular construction gives smooth, measurable progress and short commissioning, so certification is comparatively easy — but equipment is procured early and in bulk, which makes **off-site materials certification** the dominant exposure and vesting the dominant control. Grid-connection dates are owner-retained risk and are the usual cause of a commercial operations date slip the EPC contractor does not pay for.
- **Transport concessions.** Long linear works, extensive land and utility interfaces, and measured rather than milestone certification, so PoC judgment dominates and the certifier's independence is the whole control. Variations are numerous and small, which is precisely the regime in which cumulative headroom consumption (KA 14.3.3) escapes notice.
- **Digital infrastructure — data centres.** Very long lead times on electrical and cooling equipment, so advance payments and off-site certification are structural rather than exceptional, and the vesting question of Case study B is a first-order credit issue. Phased handover means the completion sequence is repeated per hall, and the facility must define whether revenue from an early phase is CFADS before overall completion.
- **Oil, gas and process plant.** Performance testing is elaborate and multi-parameter (throughput, yield, energy consumption, emissions), so partial completion and buy-down arithmetic (KA 14.4.3) is a normal rather than exceptional outcome, and the interaction of delay and performance sub-caps under an aggregate cap (Domain 12, KA 12.1.2) is tested in practice.
- **Social infrastructure PPPs — availability-based.** Payment begins on availability rather than output, so a delay is pure revenue deferral with no volume dimension; the sharp edge is that unitary-charge deductions begin immediately at service commencement, so a project that completes late and commissions imperfectly is penalised twice, and the contingency held for the commissioning period must cover both.

Case study — Domain 14: the quarter-five draw that did not fund (water / desalination)

Situation. Kestrel Water SPC reached the end of construction quarter five with certified progress at **61 %**, cumulative certified spend of **33,945,403**, and a monthly report whose headline was that contingency was **58.85 % drawn against 61.00 % progress** — "consumption proportionate, no funding issue arising". The same report showed a blended `CPI` of **0.949821** and a `BAC/CPI` forecast of **54,326,038** against a `BAC` of 51,600,000, described as "a 2.7 million forecast overrun within the funded contingency of 3,645,403". Two quarters had passed on that reading. The quarter-five draw request was submitted for **9,924,564**.

What happened. The lenders' technical adviser declined to certify. Three findings, each independently sufficient. **The `CPI` was blended across scopes with different cost-risk ownership.** Of the 2,726,038 of uplift in the `BAC/CPI` forecast, **2,535,849** was attributed to a fixed lump-sum EPC scope certified against milestones — where `CPI` is 1.000 by construction and cannot overrun — and only **190,189** to the owner-retained scope, which was running at a scope `CPI` of **0.6000** and had already consumed **1,680,000** of contingency. Disaggregated, the bottom-up remaining owner-retained estimate was **2,400,000** against a remaining budget of 1,080,000. **The cost-to-complete omitted three lines earned value cannot see.** On a commitment basis it was `18,720,000 + 840,000 + 1,260,000 + 2,400,000 + 1,436,674 =` **24,656,674** against available commitment of `15,910,254 + 6,818,680 =` **22,728,934** — out of balance by **1,927,740**. **And the contingency test failed independently.** Known committed claims on contingency were **3,420,000** (840,000 of approved variations, 1,260,000 of assessed claim exposure, 1,320,000 of bottom-up owner-scope uplift) against remaining contingency of **1,500,000**; adding the open register's P80 of **1,764,289** gave coverage of **0.2893**. The report's 58.85 % draw-rate check had been a coincidence throughout: risk had never been distributed pro rata to progress, and the two largest sanction-register items — ground conditions at the intake and owner-retained interface diversions — had both materialised, leaving the remainder funded at a confidence level nobody had computed. The original 3,645,403 had been a **P84** provision against the whole register; what remained was a P-nothing.

How it resolved. The sponsors injected **1,927,740** of additional equity as a condition to the quarter-five drawing, taking equity to **19,927,740** and total funding to **61,927,740**. Because senior debt was unchanged at 42,000,000, **every coverage ratio was untouched** — `DSCR` stayed at 1.2743 — and the whole overrun landed on equity: gearing moved from 70.0/30.0 to **67.82/32.18** (`D/E` **2.1076**), the equity cheque rose **10.71 %**, and Domain 4's project `NPV` of +16,179,360 fell to **+14,251,620** on the same operating forecasts. The monitoring regime was rewritten with three changes and no new money: disaggregated `CPI` reporting by scope with the blended figure banned; a contingency coverage ratio on the remainder replacing the draw-rate check, with the denominator itemised at every data date; and the in-balance test computed on the commitment basis at every draw and every reporting date, reconciled against the contingency shortfall as a cross-check.

What the domain teaches here. Two independent tests reached the same conclusion — 1,927,740 of in-balance shortfall against 1,920,000 of contingency committed beyond its balance, plus the 7,740 of interest on an early draw — and the monthly report contained neither. What it contained instead were two numbers that looked reassuring and were arithmetically empty: a draw rate compared with progress, and a blended `CPI` that happened to produce a plausible total by attributing 93 % of an overrun to the one scope incapable of overrunning. The lesson is not that the sponsors were negligent; it is that **the reassuring figures are the default outputs of an ordinary cost report, and the tests that matter have to be specified, itemised and demanded.** That is the entire purpose of Toolkit 14.T.2.

Case study B — Domain 14: the switchgear that was never ours (digital infrastructure)

Situation. A data-centre SPV financed a first phase at an envelope of **420,000,000 — 294,000,000** senior debt and **126,000,000** equity — with a funded contingency of 12,600,000. Lead times on medium-voltage switchgear and generators ran to fourteen months, and the electrical subcontractor's cash flow could not carry them. With the general contractor's support, the certification basis was allowed to drift: over five months the independent engineer certified **38,400,000** of off-site equipment at the vendors' works. Vesting certificates, segregation confirmations and insurance endorsements in the SPV's name were obtained for 22,000,000 of it. For **16,400,000 — 3.90 %** of the envelope — they were not; the paperwork was described in the monthly report as "in progress with the vendors".

What happened. The electrical subcontractor entered insolvency proceedings sixteen months into a twenty-two-month programme. The vested equipment was traced and released. Of the 16,400,000 that had been certified without vesting, segregation or insurance, the administrator treated the equipment as the subcontractor's asset; **4,100,000** was ultimately recovered through negotiated release of identifiable items, a **25 %** recovery. The loss was **12,300,000**. Of the certified amount, **11,480,000** had been advanced as senior debt at 70 % gearing, and it had been outstanding for fourteen months at 5.5 %, adding **736,633** of capitalised interest carried on an asset that did not exist. Total cost **13,036,633 — 10.35 %** of the equity cheque. Against remaining contingency of **5,600,000** the in-balance test failed by **6,700,000**, which the sponsors funded as equity; gearing moved to **68.90/31.10**. The plant completed five months late, by which point the delay damages cap had bound and the availability period had been extended once, for a fee.

How it resolved. The immediate remedy was procedural and should have been in place from the first draw: certification of off-site materials was capped at a stated percentage of the contract price; each certification was conditioned on a vesting certificate, a segregation confirmation from an independent inspector and an insurance endorsement naming the SPV, delivered before the certificate rather than promised after it; and the cumulative off-site balance became a reported line with an ageing analysis. The certification-basis mix — achieved milestones, assessed percentages, off-site materials — was reported every month thereafter, which made drift visible within one period instead of five.

What the domain teaches here. Certification drift is never decided; it accumulates. No single certificate was unreasonable, every one had a commercial justification, and the cumulative effect was to convert 11,480,000 of secured lending into an unsecured claim in a third party's insolvency — a decision no committee took and no minute records. The arithmetic that would have stopped it is trivial and was never run: **16,400,000 certified without vesting was 3.90 % of the envelope and 130.16 % of the whole 12,600,000 of contingency funded at close**, and its exposure was fully computable on the day of each certificate. Compare Kestrel: the exposure there was measured on the same basis at 1,339,842 in a single quarter, or 27.91 % of the performance bond. The control is not sophistication; it is a reported line and a condition that must exist before, not after, the money moves.

Executive perspective — Domain 14

What a project finance director cannot delegate in this domain:

- **The certification basis, and the off-site balance.** Which basis the facility specifies, whether the mix has drifted, and how much has been certified against work not in place — with vesting, segregation and insurance confirmed before the money moves, never promised after it (Case study B, where the answer was 16,400,000 and cost 13,036,633).
- **The in-balance test on the commitment basis, at every draw.** Not the model's balance, which reconciles by construction, but the commitment number: Kestrel's 24,656,674 against

22,728,934, and the double-count that would have hidden it by adding contingency to available funds.

- **The contingency coverage on the remainder, with its denominator itemised.** 0.2893, not the 1.3825 that appears when only the mean of open risk is counted, and never the draw-rate check that let two quarters pass.
- **The funding order.** A single sub-clause worth 1,466,064 of capitalised interest and 11,406,157 of mid-build credit exposure — 41.6 times the ten-basis-point margin the same team spent three weeks negotiating.
- **The date the first instalment falls due, and what counts as `CFADS` when it does.** A calendar-fixed date breaches a 1.20× covenant on **three weeks** of slip, and whether delay damages are `CFADS` is worth **0.4791** of coverage at that test. Both are drafting decisions taken before there is a delay.
- **The two costs of delay, stated together.** 415,000 funded and 947,000 economic a month, with the damages recovering 144.58 % of the first and 63.36 % of the second — because a project whose drawdown tests keep passing while equity value drains is the failure mode this domain exists to make visible.

Calculation exercises — Domain 14

Exercise 14.1 A project has a total funding envelope of 180,000,000, of which 96,000,000 has been drawn. At the data date the remaining committed contract value is 74,000,000; approved variations not yet certified are 3,200,000; assessed exposure on notified claims is 2,600,000; remaining owner's costs are 4,100,000; and remaining capitalised interest and fees are 2,900,000. Remaining unallocated contingency is 5,000,000. Run the in-balance test. Solution. Available commitment `= 180,000,000 − 96,000,000 =` **84,000,000**. Cost-to-complete `= 74,000,000 + 3,200,000 + 2,600,000 + 4,100,000 + 2,900,000 =` **86,800,000**. The project is **out of balance by 2,800,000**, which must be cured before the next drawing. Common error: adding the 5,000,000 of remaining contingency to available funds, giving 89,000,000 and a comfortable surplus of 2,200,000 — contingency is a **use** funded by the same undrawn commitment, so the addition double-counts it and reverses the answer.

Exercise 14.2 A four-quarter construction programme spends 20,000,000 per quarter, funded 75/25 at 8.0 % per annum, interest accruing at 2.0 % per quarter on the opening drawn debt balance with draws at period end. Compute total capitalised interest under pro-rata funding and under equity-first funding. Solution. Pro rata: interest accrues from quarter two on a growing 75 %-geared balance, giving **1,818,068** and total uses of 81,818,068 (debt 61,363,551, equity 20,454,517). Equity-first: the 20,000,000 equity commitment funds the whole of quarter one, so debt is first drawn in quarter two and first accrues interest in quarter three, giving **1,208,000** and total uses of 81,208,000. The saving is **610,068**, or **33.56 %**. Common error: computing interest on closing rather than opening balances, which charges a full period of interest to every draw and overstates capitalised interest — Domain 8's Exercise 8.2 measured that error at 42 % on a comparable profile.

Exercise 14.3 An EPC contract has a price of 90,000,000, an advance payment of 15 % recovered pro rata against certified value, and retention of 10 % of each certification capped at 5 % of the contract price. Cumulative certified value before this month is 36,000,000; this month's certified value is 7,200,000. Compute this month's net payment and the cumulative retention held. Solution. Advance `= 13,500,000`; recovery this month `= 0.15 × 7,200,000 =` **1,080,000**. Retention cap `= 0.05`

> `× 90,000,000 =` 4,500,000; held before `= 0.10 × 36,000,000 =` 3,600,000, so the cap permits a further 900,000 and this month's 10 % of 7,200,000 = **720,000** fits. Net payment `= 7,200,000 − 1,080,000 − 720,000 =` **5,400,000**; cumulative retention held **4,320,000**. Common error: applying retention to the amount net of advance recovery — `0.10 × (7,200,000 − 1,080,000) = 612,000`, giving a net payment of 5,508,000 and **over-paying by 108,000**. Retention is withheld from gross certified value; advance recovery is a separate deduction.

> **Exercise 14.4** A variation costs 6,200,000 and will add 600,000 a year of `CFADS`. The loan runs 15 years at 6.5 % and the target `DSCR` is 1.25×. Compute the maximum coverage-neutral debt funding, the equity or contingency required, and the marginal `DSCR` if the variation were fully debt-funded. Solution. `AF(0.065, 15) = 9.402669`. Maximum debt funding `= (600,000/1.25) × 9.402669 =` **4,513,281**, being **72.79 %** of the cost; the residual **1,686,719** must come from equity or contingency. Fully debt-funded, incremental debt service `= 6,200,000/9.402669 =` **659,387** and the marginal `DSCR = 600,000/659,387 =` **0.9099** — the variation would not even service the debt raised to build it. Common error: sizing on the full `ΔCFADS` without the coverage divisor (`600,000 × 9.402669 = 5,641,601`), which is the Domain 10 sizing error transplanted into change control and would leave the project 1,128,320 over-levered on this variation alone.

> **Exercise 14.5** A facility has annual debt service of 11,200,000 falling due on a calendar-fixed date twelve months after scheduled completion, a 1.15× covenant, a six-month debt service reserve of 5,600,000, and operating `CFADS` of 14,400,000 a year accruing evenly from actual completion. Completion slips three months. Compute the `DSCR` at the first test, the reserve consumption, and the three breakeven slips. Solution. `CFADS` at the first test `= 14,400,000 × 9/12 =` **10,800,000**; `DSCR = 10,800,000/11,200,000 =` **0.9643**. Shortfall **400,000**, being **7.14 %** of the reserve. Breakevens: covenant cash trigger `= 1.15 × 11,200,000 = 12,880,000`, so the covenant fails beyond `12 × (1 − 12,880,000/14,400,000) =` **1.2667 months**; operating cash ceases to cover debt service beyond `12 × (1 − 11,200,000/14,400,000) =` **2.6667 months**; cash plus a fully funded reserve ceases to cover it beyond `12 × (1 − 5,600,000/14,400,000) =` **7.3333 months**. Common error: computing the `DSCR` on annualised `CFADS` of 14,400,000 (giving 1.2857 and a comfortable pass) — the covenant is tested on the period's cash, not on a run rate, which is the whole content of KA 14.4.2.

Practitioner's toolkit — Domain 14

Adoption-ready artefacts; adapt headings to your organisation, then keep them stable.

Toolkit 14.T.1 — Draw request pack and certification control sheet

Part A, the arithmetic, one row per line with a source reference: gross certified value this period, split into value of milestones achieved, value of assessed percentage completion, and value of off-site materials; less retention this period, with cumulative retention against its cap; less advance recovery this period, with cumulative recovery against the advance; plus owner-retained costs incurred and evidenced; plus fees, taxes and premiums; plus interest and commitment fees accrued, computed on **opening** balances with the day-count basis stated; equals the period funding requirement, split by the documented funding order into the debt draw and the equity draw. **Part B, the conditions**, one row per condition to drawing with the evidence reference, its expiry date, and the named person who confirmed it — no default or potential default, representations repeated, certifier's certificate, insurances in force and premiums paid, in balance, within the availability period, within the commitment. **Part C, the certification-mix trend**: the three components of Part A's gross certified value as percentages, this period and cumulatively, with the off-site balance aged. **Rules:** the off-site line is nil unless vesting, segregation and insurance are all in place for the amount certified; the certification mix is reported whether or not anyone asks; and cumulative drawn is tied to the facility agent's statement, never to the model.

Toolkit 14.T.2 — Restated sources and uses, the in-balance test, and the `EAC` bridge

Sheet 1, restated statement: every use in three columns — drawn to date, remaining to be funded, total — against sources in the same three columns, with available commitment stated as undrawn debt plus uncalled equity and a printed warning that contingency is **not** added. Any imbalance is explained in cash terms with its cause named. **Sheet 2, lender's cost-to-complete:** remaining committed contract value; approved variations not yet certified; assessed exposure on notified claims, with the assessor named; bottom-up re-estimate of owner-retained scope; remaining capitalised interest, fees, premiums and taxes to the forecast commercial operations date; total; available commitment; surplus or shortfall. **Sheet 3, the bridge:** the `EAC`-based `CTC` from the delivery organisation's cost report on each published method, then one line per reconciling item — each labelled with the discipline that is blind to it — arriving at the lender's number. **Sheet 4, disaggregated performance:** `EV`, `AC` and `CPI` separately for fixed-price and owner-retained scope, with the blended index shown only to be marked "not to be used for forecasting". **Rule:** no cost report is issued, and no draw request signed, without Sheets 2 and 3 complete.

Toolkit 14.T.3 — Contingency coverage and change-control register

Contingency section: funded amount; drawn to date, itemised by cause with the register item each draw retires; remaining unallocated. **Denominator, itemised:** approved variations not yet certified; assessed claim exposure; bottom-up re-estimate above budget on owner-retained scope; then the open risk register with `p`, impact, `EMV`, and the computed mean, σ and P80. **Coverage on the remainder**, computed as remaining ÷ (known committed + P80), with the two partial ratios (against the mean, and against open risk only) shown beside it and labelled as insufficient tests. **Shortfall in currency. Change-control section**, one row per variation above the threshold: price; time effect; funding source, named; `ΔCFADS` claimed and by whom; `Δ debt service` at the debt-funded share; **marginal** `DSCR`; coverage-neutral maximum debt funding; headroom consumed; cumulative headroom consumed by all approved variations against the base-case headroom. **Rule:** the draw-rate check — contingency drawn against progress — does not appear on this form; and no variation is approved without a funding source and a marginal `DSCR`.

Exam preparation — Domain 14

What is assessed. The restated sources-and-uses statement and the in-balance test on a commitment basis; the composition of a draw request from gross certified value to the debt and equity draws; the funding-order comparison and its capitalised-interest, exposure and present-value consequences; the lender's cost-to-complete and its line-by-line reconciliation to an `EAC`-based `CTC`; why a blended `CPI` misattributes an overrun where one scope is fixed-price and milestone-certified; the contingency coverage ratio on the remainder and the defensible choice of denominator; the three certification bases and the exposure created by off-site certification without vesting; advance payment and retention arithmetic, including the retention release that falls in the first operating year; the marginal `DSCR` of a variation and the coverage-neutral debt share; the funded versus economic cost of a month of slip and the three damages-recovery percentages; the `DSCR` at a calendar-fixed first repayment date with its covenant, payment and reserve breakevens; and the buy-down that restores a stated coverage level.

The calculations to do under time pressure. Available commitment and the in-balance result from a five- or six-line cost-to-complete. A period funding requirement and its 70/30 split, with accrued interest on the opening balance. Certified value on three bases, net of retention, times gearing. Net payment after advance recovery and retention, with the cap tested. `Δ debt service = cost ÷ AF(r, n)` and the marginal `DSCR`; and the coverage-neutral maximum, `(ΔCFADS/target) × AF(r, n)`. `CFADS × (12 − m)/12 ÷ debt service`, and the three breakeven slips by inverting it. Supportable debt at a target ratio, and the buy-down as the difference from outstanding debt.

The traps. Adding remaining contingency to available commitment (Exercise 14.1, MCQ 14.1-A) · omitting remaining capitalised interest from a cost-to-complete because it is not work (MCQ 14.2-A) · applying `BAC/CPI` to a blended index across fixed-price and owner-retained scope (14.2.1, MCQ 14.2-B, Case study A) · reading the contingency draw rate against progress as evidence of health (14.2.2, MCQ 14.2-D) · testing contingency against the mean of open risk rather than the P80 plus known commitments (MCQ 14.2-C) · computing capitalised interest on closing rather than opening balances (Exercise 14.2, imported from Domain 8) · certifying off-site materials without vesting, segregation and insurance (14.3.1, Case study B) · applying retention to the amount net of advance recovery (Exercise 14.3) · assuming a retention release falling after completion is already funded because it is inside the contract price (14.3.2, MCQ 14.3-B) · sizing a debt-funded variation on full `ΔCFADS` without the coverage divisor (Exercise 14.4, MCQ 14.3-C) · treating an `NPV`-positive variation as automatically fundable by debt (14.3.3, MCQ 14.3-D) · quoting one damages-recovery percentage without naming the cost basis (14.4.1, MCQ 14.4-C) · computing the first-test `DSCR` on annualised rather than period `CFADS` (Exercise 14.5, MCQ 14.4-B) · assuming a slip postpones a calendar-fixed repayment date (MCQ 14.4-A) · and confusing the buy-down that restores a covenant with the one that restores base coverage (14.4.3, MCQ 14.4-D).

How the domain connects. Domain 6 built the construction funding model this domain operates; Domain 8 supplied the capitalised-interest area rule, the escalation treatment and the economic cost of a slip during construction; Domain 5 priced a slip at the commercial operations date and established cost-overrun support; Domain 12 fixed the contract limits — damages rates, caps, bonds and guarantees — that this domain draws against; Domain 10 supplied the coverage machinery, the reserve and the covenant that KA 14.4 tests; Domain 13 delivered the conditions precedent that KA 14.1's conditions to drawing continue; and PML-AI Domain 7 supplied the `EAC` family that KA 14.2 reconciles to. Forward, Domain 15 takes the project past completion into operation — the waterfall, the covenant regime, refinancing and restructuring — and Domain 16 governs the automation that this domain's reporting cycle invites.

Domain 14 summary

Construction monitoring is the discipline of releasing other people's money against evidence, and it reduces to four tests. The **draw request** converts certified value into a funding requirement: Kestrel's quarter five was `9,391,718 + 245,707 + 287,138 =` **9,924,564**, split 70/30, of which the interest line — **0.83 %** of the requirement in quarter one and **8.44 %** in quarter eight — is money borrowed to pay interest on money already borrowed. The **funding order** that governs that split is a single sub-clause worth **1,466,064** of capitalised interest across three orders (equity-first **1,338,006**, pro rata **2,114,597**, debt-first **2,804,070**), **11,406,157** of mid-build lender exposure, **183,013** of present value to the sponsor, and — at a fixed envelope — **742,647** of additional funded contingency; it is **41.6 times** the ten-basis-point margin move the same negotiation spends weeks on. The **in-balance test** on a commitment basis is not `EAC − AC`: Kestrel's lender cost-to-complete of **24,656,674** exceeds an `EAC`-based `CTC` of 21,120,000 by three lines earned value cannot see — approved variations, assessed claim exposure and remaining capitalised interest — and exceeds available commitment of **22,728,934** by **1,927,740**, while a blended `CPI` of 0.949821 was attributing

2,535,849 of overrun to a fixed-price scope where CPI is 1.000 by construction and only **190,189** to the owner-retained scope actually running at 0.6000. The **contingency test** fails independently and earlier: 1,500,000 remaining against 3,420,000 of known committed claims plus a 1,764,289 P80 gives coverage of **0.2893**, where the comfortable versions of the same ratio read 1.3825 and 0.8502, and where a draw rate of 58.85 % against 61.00 % progress had read as health for two quarters. Certification is where security is quietly converted: the same quarter's work is worth **5,760,000** on milestones, **7,084,800** measured and **7,774,800** cost-incurred, and the cost-incurred basis advances **1,339,842** of senior debt against work not in place — 27.91 % of the performance bond, and the mechanism that cost Case study B **13,036,633**, or 10.35 % of its equity. Payment terms carry prices: a 10 % advance costs **254,597** of capitalised interest, 0.53 % of the EPC price, and should be priced rather than granted; a 600,000 retention release falling twelve months after completion takes CFADS to 5,784,000, the DSCR to **1.1546**, and breaches a 1.20× covenant by exceeding Domain 10's 372,438 of headroom. Variations carry four numbers, not two: a 1,850,000 train adding 240,000 of CFADS has an NPV of **+711,946** and a marginal DSCR of **1.0876** if fully debt-funded, so the coverage-neutral limit is **1,578,947**, or 85.35 % of cost. And delay must be costed twice — **415,000** funded and **947,000** economic a month, recovered by the same 20,000-a-day damages at **144.58 %** and **63.36 %** respectively — because a project whose drawdown tests keep passing while equity drains is invisible on one basis and obvious on the other. At the first, calendar-fixed repayment date the arithmetic is unforgiving: **three weeks** of slip (0.7001 months) breaches the 1.20× covenant, 2.5834 months exhausts operating cash, 7.2917 months exhausts cash plus the 2,504,818 reserve, a four-month slip gives a DSCR of **0.8496** and consumes 30.09 % of that reserve — and whether delay damages count as CFADS moves the same test by **0.4791**, from 0.8496 to 1.3286. Domain 15 takes the plant into operation, where the waterfall, the covenant regime and the reserve replenishment this domain has already committed begin to run.

15

DOMAIN 15

Operations, Performance and Restructuring

Group: Operating and the future (Part Four). **Target:** ~78 pages. **Binds to:** the PCI Book Pattern Specification and the shared registries (`docs/books/registries/`). This domain operates the machinery the rest of the book built: Domain 2's accrual-to-cash bridge, Domain 3's amortisation schedule, Domain 10's `CFADS`, `DSCR`, `LLCR`, `PLCR`, reserves and covenants, Domain 13's model audit and Domain 14's completion tests all arrive here as monthly obligations rather than closing conditions. British English; USD (+SAR where useful, indicative `USD 1 ≈ SAR 3.75`).

Why this domain exists

Every preceding domain answered a question that can be closed. Is the project worth building (Domain 4)? Can it carry the debt (Domain 10)? Are the risks allocated (Domain 11), the contracts sound (Domain 12), the diligence complete (Domain 13), the construction funded and certified (Domain 14)? Financial close answers all of them once. **Operations answers them again every quarter for twenty-five years**, against outturn rather than forecast, and with the project's contractual machinery now live: tests that trap cash, reserves that must be topped up before shareholders are paid, and remedies that engage automatically when a threshold is crossed. The question this domain closes is the one the others left open — what actually happens to the cash, period by period, and who decides.

The central claim is that **the operating phase is governed by the distribution test, not by the covenant.** Domain 10 established the covenant, the lock-up and the reserve as the lender's control architecture, and computed Kestrel Water's headroom to breach as USD 372,437.72 of annual cash. That number is real but it is not the one that binds. The condition a sponsor must satisfy to be paid is stricter than the condition it must satisfy to avoid default, it is tested forwards as well as backwards, and it sits below a reserve top-up that has first call on the same cash. A project can be fully compliant, paying every obligation on time, and still pay its shareholders nothing for four consecutive years — which is exactly what Kestrel does below. From there the domain follows the money outwards: monitoring and covenant testing (KA 15.1), the waterfall and the distributions that fall out of it (KA 15.2), refinancing and the negotiated amendments that change the terms mid-life (KA 15.3), and the arithmetic of distress, restructuring, exit and handback (KA 15.4).

Learning objectives. After this domain a candidate can: build an operating bridge from physical performance drivers to `CFADS` and reconcile it to the reported financial statements; convert a covenant, lock-up and distribution threshold into a cash trigger and then into each driver's own units; distinguish backward-looking, rolling and forward-looking tests and explain why a rolling test lags a deterioration by its own window length; operate a full cash waterfall period by period, including reserve top-ups, block accounts and the distributable amount that falls out; explain and compute why deferred distributions cost equity more than the cash deferred; price a refinancing as a net present value against break costs and fees and solve for its breakeven margin; price a waiver, an equity cure and a covenant amendment against one another and explain why cure rights are an option with a scarcity value; compare a maturity extension, a principal haircut and an equity injection on both lender recovery and equity value, and locate the enforcement floor that bounds the negotiation; size a handback reserve as a sinking fund and explain why early funding can cost equity more in present value than late funding; value an exit and separate genuine value creation from the arithmetic flattery of early crystallisation; and govern AI-assisted monitoring, covenant certification and restructuring analysis.

The master financing, now operating. Kestrel Water SPC reached commercial operations with the structure Domains 3, 9 and 10 built: capital cost **USD 60,000,000**, funded **70/30** as **USD 42,000,000** of senior debt at **6.0 %** over **12 years** and **USD 18,000,000** of equity; annual instalment **USD 5,009,635.23**, of which year one is interest **2,520,000** and principal **2,489,635**; a **25-year** operating life; documented first-year CFADS of **USD 6,384,000** (**6,984,000** before working-capital movements) on EBITDA of **7,500,000** and EBIT of **5,100,000**; a base-case DSCR of **1.2743**, identically equal to LLCR, with PLCR at **1.9431** (Domain 10, KA 10.2.2). The facility carries a **1.20×** DSCR covenant, a **1.15×** lock-up trigger, a six-month debt service reserve of **2,504,817.62**, and — the term this domain adds — a **distribution condition requiring 1.25× on both a backward and a forward-looking test**. This domain builds the operating bridge that produces the 6,384,000, runs the waterfall for six years of real outturn, refinances at the end of year five, prices the amendment that resolves a breach, restructures the project on a permanent-deterioration branch, and funds an **8,000,000** handback obligation.

KNOWLEDGE AREA 15.1

Operational monitoring, financial reporting and covenant testing

Topics: 15.1.1 what changes at commercial operations · 15.1.2 the operating bridge from drivers to *CFADS* · 15.1.3 triggers in driver units · 15.1.4 backward, rolling and forward-looking tests.

15.1.1 What changes at commercial operations

Definition. The **operating régime** is the standing set of reporting, testing and certification obligations that replaces the construction régime once the completion tests of Domain 14 are satisfied. Its four components are worth naming because each has a different failure mode. **Information covenants** require delivery of management accounts, audited statements, an annual operating budget and a periodic update of the financial model, each by a contractual date; failure is a breach in its own right, independent of performance. **Compliance certificates** are signed statements that the tested ratios have been computed on the documented definitions and that no default subsists — the point at which a number stops being management information and becomes a representation. **Financial covenant testing** measures the ratios on defined dates. **Distribution conditions** are the positive tests a borrower must pass before it may pay anything to shareholders.

Three consequences follow that surprise first-time operators. The **calendar becomes contractual**: test and delivery dates are fixed and do not move because a system implementation slipped. The **model becomes a deliverable**, not an internal tool — which is why Domain 6's model governance and Domain 13's model audit are contractual rather than merely prudent. And the **definitions**

become load-bearing in the other direction: at close, a favourable `CFADS` definition raised debt capacity; in operations, the same definition determines whether a distribution is lawful.

15.1.2 The operating bridge from drivers to `CFADS`

Definition. The **operating bridge** is the reconciliation that carries physical performance — availability, output, unit costs — through the accrual statements to the cash figure the covenant is computed on. It matters because the two ends are owned by different people. Plant availability belongs to the operator; `CFADS` belongs to the finance director; and the covenant is breached by the first and reported by the second. Without an explicit bridge, nobody can say what a one-point loss of availability does to a distribution.

Worked example 15.1.2 — Kestrel's first operating year, from membranes to `CFADS`.

1. **Setup.** Contracted capacity **30,000 m³/day**; guaranteed availability **95.0 %**; capacity payment **USD 7,300,000** per year at guaranteed availability, abated pro rata below it; volume payment **USD 0.55/m³** on water delivered. Year one delivered **10,000,000 m³** at **95.0 %** availability. Operating costs: fixed operations and maintenance **2,100,000**, energy **0.26/m³**, chemicals and consumables **0.04/m³**, insurance **200,000**. Depreciation is straight-line over the 25-year life. Cash tax is **20 %** of profit before tax. Working capital is receivables at 45 days of revenue (360-day convention), payables at 90 days of cash operating cost, and an inventory of chemicals and membrane spares of **325,000**, built from nil during the first year.
2. **Formula.** Revenue = capacity payment × min(1, availability ÷ guaranteed) + volume price × volume. `EBITDA` = revenue − cash operating cost. `EBIT` = `EBITDA` − depreciation. Cash tax = 20 % × (`EBIT` − interest). `CFADS` = `EBITDA` − cash tax ± movement in working capital, per the facility's documented definition (Domain 10, KA 10.1.1).
3. **Substitution.** Capacity payment `7,300,000 × 1.00`; volume payment `0.55 × 10,000,000`; operating cost `2,100,000 + 0.26 × 10,000,000 + 0.04 × 10,000,000 + 200,000`; depreciation `60,000,000 ÷ 25`; tax `0.20 × (5,100,000 − 2,520,000)`; working capital `12,800,000 × 45/360 − 5,300,000 × 90/360 + 325,000`.
4. **Result.**

Line	USD	Note
Capacity payment	7,300,000	availability at guarantee, no abatement
Volume payment	5,500,000	0.55 × 10,000,000 m³
Revenue	**12,800,000**	
Cash operating cost	(5,300,000)	2,100,000 + 2,600,000 + 400,000 + 200,000
`EBITDA`	**7,500,000**	
Depreciation	(2,400,000)	60,000,000 ÷ 25
`EBIT`	**5,100,000**	
Interest	(2,520,000)	Domain 3's year-one interest
Profit before tax	**2,580,000**	
Cash tax at 20 %	(516,000)	
`CFADS` **before working capital**	**6,984,000**	`EBITDA` − cash tax
Movement in working capital	(600,000)	1,600,000 − 1,325,000 + 325,000, built from nil
`CFADS` **as defined**	**6,384,000**	`DSCR` **1.2743**

1. **Interpretation.** The table is the domain's single most useful artefact because it is the only place where an engineer's number and a lender's number are visibly the same number. Read downwards, it prices performance: the **cash-to-revenue gearing is 0.80** — a dollar of lost revenue costs only eighty cents of `CFADS`, because cash tax falls by twenty cents with it, which means every headroom figure expressed in `CFADS` understates the revenue loss the project can absorb by a quarter. Read upwards, it prices the definition: the 600,000 working-capital line is the same choice Domain 2 (KA 2.3.1) used to move the reported `DSCR` from 1.39 to 1.27, and it is now a live monitoring exposure rather than a modelling convention — because working capital moves with revenue. When revenue falls, receivables fall and working capital releases cash, flattering `CFADS` in the year of decline; when revenue recovers, working capital rebuilds and penalises `CFADS` in the year of recovery. A deteriorating project therefore reports a covenant ratio that is better than its trading and a recovering project one that is worse, and a finance director who does not decompose the movement will misread both. The professional caution is to report `CFADS` in two lines — before and after working capital — in every internal pack, while certifying only the defined figure to the lenders.

15.1.3 Triggers in driver units

Definition. A **trigger in driver units** is a covenant, lock-up or distribution threshold restated in the physical or commercial quantity that management actually controls. Domain 10 established the first translation, from ratio to cash. This is the second, and it is the one that puts a covenant on an operations dashboard rather than a finance dashboard.

Kestrel's three thresholds, against annual debt service of 5,009,635.23:

Test	Ratio	CFADS trigger	Headroom from 6,384,000	% of CFADS
Distribution condition	1.25×	**6,262,044.04**	**121,955.96**	1.9103 %
Financial covenant	1.20×	**6,011,562.28**	**372,437.72**	5.8339 %
Distribution lock-up	1.15×	**5,761,080.51**	**622,919.49**	9.7575 %

The first line is the domain's opening claim in arithmetic. **The distribution condition binds at 1.9103 % of CFADS; the covenant at 5.8339 %.** Kestrel is three times closer to losing its dividend than to breaching its loan, and only one of those two numbers appeared in the financing paper.

Translating the covenant headroom of 372,437.72 into drivers divides by the 0.80 cash-to-revenue gearing to reach **465,547.16** of pre-tax revenue or cost, and then by each driver's own unit economics:

Driver	Unit economics	Move that reaches the 1.20× covenant	Move that reaches the 1.25× distribution test
Availability	76,842.11 per percentage point (7,300,000 ÷ 95)	**6.0585 points** — availability floor **88.9415 %**	**1.9839 points** — floor **93.0161 %**
Delivered volume	0.25/m³ cash contribution (0.55 − 0.26 − 0.04)	**1,862,188.62 m³**, 18.6219 % of output	**609,779.81 m³**, 6.0978 %
Energy unit cost	10,000,000 m³ exposed	**+0.046555/m³** (+17.9 % on 0.26)	**+0.015244/m³** (+5.9 %)

Two readings of that table are worth carrying. First, **the drivers are not equally dangerous**: a 6-point availability loss and a 19 % volume loss both cost the same cash, but one is a plausible outcome of membrane fouling and the other is not, so the availability line is the one that belongs in a monthly review. Second, **the energy column is where an unhedged input becomes a covenant risk** — a 5.9 % rise in the delivered cost of power, which no operator would call a crisis, removes the entire dividend.

15.1.4 Backward, rolling and forward-looking tests

Definitions. A **backward-looking test** measures the period that has ended. A **rolling test** measures the trailing window — commonly four quarters — and is the standard form because it removes seasonality. A **forward-looking test** measures the projected next window on an agreed basis. Most facilities test the covenant on a rolling backward basis and the distribution condition on both bases, and the reason is the arithmetic below.

WORKED EXAMPLE

Worked example 15.1.4 — the four quarters a rolling test cannot see.

1. **Setup.** Kestrel's reported quarterly CFADS in year one is **1,700,000 · 1,640,000 · 1,560,000 · 1,484,000** (summing to the documented 6,384,000) and in year two **1,610,000 · 1,520,000 · 1,440,000 · 1,393,894.11** (summing to 5,963,894.11). Because the loan is a level annuity paid annually, **every rolling twelve-month window contains exactly one instalment of 5,009,635.23**, so the denominator is constant. At the year-one test date the board's re-forecast of year two is **5,500,000**.
2. **Formula.** Rolling DSCR = Σ (last four quarters' CFADS) ÷ 5,009,635.23. Forward DSCR = forecast next-twelve-month CFADS ÷ 5,009,635.23.
3. **Substitution.** Successive four-quarter sums: `6,384,000`; `6,294,000`; `6,174,000`; `6,054,000`; `5,963,894.11`. Forward: `5,500,000 ÷ 5,009,635.23`.
4. **Result.**

Test date	Rolling four-quarter CFADS	Backward DSCR	1.25× distribution	1.20× covenant	1.15× lock-up
End Y1 Q4	6,384,000.00	**1.2743**	pass	pass	clear
End Y2 Q1	6,294,000.00	**1.2564**	pass	pass	clear
End Y2 Q2	6,174,000.00	**1.2324**	**fail**	pass	clear
End Y2 Q3	6,054,000.00	**1.2085**	**fail**	pass	clear
End Y2 Q4	5,963,894.11	**1.1905**	**fail**	**breach**	clear

The forward test at end Y1 Q4, on the 5,500,000 re-forecast, gives **1.0979** — failing both the distribution condition and the covenant threshold **four quarters before the backward test registers anything at all**. 5. **Interpretation.** This is the most important table in the knowledge area, and its lesson is structural rather than numerical: **a rolling backward test lags a deterioration by the length of its own window.** Kestrel's cash begins falling in the second quarter of year one and its covenant does not break until the fourth quarter of year two — by which time the problem is eighteen months old, the remediation window has closed, and the negotiation happens from a position of fact rather than forecast. The forward test exists precisely to close that gap, and it does: it fails at the first available date. Three professional consequences follow. **Whose forecast counts becomes a covenant question** — Domain 10 (KA 10.A.1) named this, and here it has a price, because at the end of year one the sponsors' own re-forecast is the instrument that stops their dividend. **A forward test punishes candour**, which is the uncomfortable truth of the mechanism: a management team that revises its forecast honestly triggers a consequence a team that does not revise avoids for a year, and the only defence against that perverse incentive is a documented re-forecast basis with a named owner and an independent review. And **the outturn will differ from the forecast anyway**: year two actually delivered **5,963,894.11**, some **463,894.11 above** the re-forecast that stopped the dividend. The distribution was correctly withheld on the information available, and the information available was wrong by nearly half a million dollars. Reporting that reconciliation — forecast, outturn, variance, cause — is what converts a forward test from a trap into a control.

AI in this KA

Where it earns its place. Operating monitoring is the strongest genuine application of machine work in this book: high-frequency, high-volume, structurally repetitive. Three uses are straightforwardly valuable. **Driver-level anomaly detection** — flagging that specific energy consumption per cubic metre has drifted 3 % above its trailing distribution weeks before the quarterly pack would show it, because that drift is the leading indicator of the covenant. **Automated bridge reconciliation** — checking that the operating bridge of 15.1.2 still ties from the meter readings to the ledger to the reported `CFADS`, every month, and raising the specific line that broke. **Reporting-calendar assurance** — tracking every information covenant against its contractual date across a portfolio, which is pure administration and a real source of avoidable breach.

Where it must not go. An assistant must not compute the certified ratio, and must not draft the compliance certificate. The certificate is a representation by a named officer that the number was computed on the documented definition; a model that has read the definition well is still a model that has read it, and the distinction between a plausible reading and the correct one is the entire content of Domain 10's KA 10.1.1. Nor should a model produce the forward forecast that a distribution test will be measured on — that forecast has a contractual consequence and therefore needs an owner who can be questioned.

Verification, concretely. Recompute one period's certified `DSCR` by hand from the statements, every test date, and tie it to the bridge of 15.1.2. Reconcile the machine's `CFADS` line to the facility definition clause by clause once per model version (Toolkit 15.T.1). Back-test the anomaly detector against the last eight quarters and record its false-negative rate, because a monitoring tool that has never been measured is an assumption, not a control. **AI proposes; the professional verifies, decides and remains accountable.**

KEY TERMS — KA 15.1

Term	Meaning
Operating régime	The standing reporting, testing and certification obligations that follow completion.
Compliance certificate	A signed representation that tested ratios were computed on the documented definitions and no default subsists.
Operating bridge	The reconciliation from physical drivers through the accrual statements to defined `CFADS`.
Cash-to-revenue gearing	The fraction of a revenue movement that reaches `CFADS` after cash tax — 0.80 for Kestrel.
Trigger in driver units	A ratio threshold restated in availability points, volume or unit cost.
Rolling test	A covenant measured on a trailing window; it lags a deterioration by the window length.
Forward-looking test	A covenant or distribution condition measured on a projected window on an agreed basis.

SAMPLE MCQS — KA 15.1

MCQ 15.1-A `[15.1.3 · Application]` Kestrel's debt service is 5,009,635.23 and `CFADS` is 6,384,000. Which figure states the headroom to the **distribution condition** of 1.25×? - A. USD 372,437.72 - B. USD 121,955.96 - C. USD 622,919.49 - D. 0.0243 of a ratio point

Rationale: `6,384,000 − 5,009,635.23 × 1.25 = 121,955.96`, or 1.9103 % of `CFADS`. A is the headroom to the 1.20× covenant and is the standard confusion this KA exists to remove; C is the headroom to the

1.15× lock-up; D expresses the gap in ratio points, which conveys no magnitude (Domain 10, KA 10.2.1).

MCQ 15.1-B [15.1.2 · Analysis] Kestrel's revenue falls by 500,000 in a year. Holding working capital constant, CFADS falls by: - A. USD 500,000 - B. USD 400,000 - C. USD 625,000 - D. USD 100,000

Rationale: Revenue falls to EBITDA one-for-one, profit before tax falls by the same amount and cash tax falls by 20 % of it, so CFADS falls by `0.80 × 500,000 = 400,000` — the cash-to-revenue gearing of 15.1.2. A ignores the tax shield entirely; C divides by 0.80 instead of multiplying, which is the correct arithmetic run backwards (it converts a CFADS gap into a revenue gap); D is the tax saving mistaken for the cash effect.

MCQ 15.1-C [15.1.4 · Analysis] A project's CFADS begins declining in the second quarter of year one. Its covenant is a rolling four-quarter DSCR. The earliest date on which that covenant can fail, and the reason, is: - A. the same quarter, because the test is continuous - B. up to four quarters later, because a trailing window dilutes each weak quarter with three earlier stronger ones - C. never, provided debt service is paid - D. immediately, because rolling tests are more sensitive than annual tests

Rationale: Kestrel's window carries the decline for four quarters before the sum falls through the threshold (15.1.4). A misdescribes a trailing measure; C confuses breach with payment default (Domain 10, KA 10.2.1); D reverses the effect — smoothing reduces sensitivity, which is why the forward test is needed alongside.

MCQ 15.1-D [15.1.3 · Application] Kestrel's capacity payment is 7,300,000 at a guaranteed availability of 95.0 %, abated pro rata. Availability alone must fall to roughly what level before the 1.20× covenant is breached? - A. 93.02 % - B. 88.94 % - C. 90.15 % - D. 87.43 %

Rationale: Covenant headroom 372,437.72 ÷ 0.80 = 465,547.16 of revenue, ÷ 76,842.11 per point = 6.0585 points below 95.0 %. A is the floor for the 1.25× distribution condition (1.9839 points); C omits the 0.80 cash-to-revenue gearing altogether (372,437.72 ÷ 76,842.11 = 4.8468 points); D applies the gearing twice (465,547.16 ÷ 0.80 = 581,933.95, giving 7.5731 points).

SELF-CHECK — KA 15.1

1. Why does a deteriorating project report a flattered covenant ratio? — Falling revenue releases working capital, which is added to CFADS in the year of decline and reversed in the year of recovery (15.1.2).
2. State the lag structure of a rolling four-quarter covenant. — It cannot register a deterioration for up to four quarters; the forward test exists to close that gap and failed for Kestrel four quarters earlier, at 1.0979.
3. Which of Kestrel's three thresholds binds first, and by how much cash? — The 1.25× distribution condition, at 121,955.96 or 1.9103 % of CFADS.

KNOWLEDGE AREA 15.2

The cash waterfall in operation, reserves and distributions

Topics: 15.2.1 the operating waterfall in priority order · 15.2.2 reserve top-ups and the block account · 15.2.3 the distributable amount that falls out · 15.2.4 what a distribution drought costs equity.

15.2.1 The operating waterfall in priority order

Definition. The **operating cash waterfall** is the contractual priority order in which each period's project cash is applied. Domain 10 (KA 10.3.3) built its top — operating costs, taxes, senior debt service — and stated the principle that reserve top-ups rank above distributions. This is the rest of it, in the order the accounts actually work:

```
Revenue collected into the proceeds account
  1  operating and maintenance costs, insurance, SPV administration
  2  cash taxes
  3  senior fees, agency and account-bank charges
  4  senior interest
  5  senior scheduled principal
  6  debt service reserve top-up, to the required balance
  7  maintenance reserve top-up, to the required balance
  8  handback / decommissioning reserve top-up, from the defined year
  9  mandatory prepayment and cash sweep, if triggered
 10  subordinated and shareholder debt: interest, then principal
 11  distributions to equity — only if every distribution condition is satisfied
 12  otherwise: retained in the distribution-block account
```

Three features of the order do the work. **Steps 1 to 5 are obligations**; steps 6 to 9 are **restorations**, which is a different thing — they do not fall due to a counterparty but they have first call on cash and no negotiating partner. **Steps 10 and 11 are permissions**: the economic meaning of subordination is not a lower ranking in an insolvency but the absence of an entitlement in the ordinary course. And **step 12 is where the sponsor's money goes when a test fails —** not away, but into an account it cannot reach, which is the mechanism the rest of this knowledge area is about.

Kestrel's reserve requirements: a **debt service reserve** of six months' forward debt service, **2,504,817.62** (constant, because a level annuity's next six months never change), and a **maintenance reserve** funded at **600,000 per year** as the levelised charge for a **3,000,000** membrane replacement every five years. There is no subordinated debt: the 18,000,000 of equity is subscribed as share capital, so step 10 is empty and step 11 is where all sponsor cash arises.

15.2.2 Reserve top-ups and the block account

Definition. A **top-up** is the application of period cash to restore a reserve to its required balance; the **distribution-block account** is where cash that fails a distribution test is held. The pair is what makes the waterfall a liquidity system rather than a payment order, because cash trapped in one period funds a restoration in another.

WORKED EXAMPLE

Worked example 15.2.2 — six years of Kestrel's waterfall, and what the block account paid for.

1. **Setup.** Kestrel's actual outturn over six operating years, built from the drivers of 15.1.2. Year two suffers an unhedged power-price shock (energy $0.26 \rightarrow 0.40/m^3$) with availability at 92.0 % on 9,500,000 m^3; year three worsens as membranes foul (88.0 %, 9,000,000 m^3, energy 0.42); year four is the remediation year (94.5 %, 9,900,000 m^3, energy hedged at 0.32) in which the membrane replacement is brought forward from year five; years five and six run at 95.0 % and 10,000,000 m^3 with energy at 0.30. Debt service is the level 5,009,635.23; the maintenance-reserve charge is 600,000 a year; the distribution condition requires 1.25× on both a backward and a forward test.
2. **Formula.** For each year: residual = `CFADS` − debt service − maintenance-reserve top-up. If the distribution conditions are satisfied, distribution = residual + block-account balance, and the block account resets to nil; otherwise the residual is added to the block account. A top-up that the year's residual cannot fund is drawn from the block account.
3. **Substitution.** Year three: `5,202,936.48 − 5,009,635.23 = 193,301.25`, against a 600,000 top-up, leaving `600,000 − 193,301.25 = 406,698.75` to be drawn from the block account. Year five: `6,480,627.99 − 5,009,635.23 − 600,000 = 870,992.76`, plus the block-account balance of 1,206,931.59.
4. **Result.**

Year	CFADS	Backward DSCR	Residual after service and reserve	Distribution	Block account, closing
1	6,384,000.00	1.2743	774,364.77	**nil**	774,364.77
2	5,963,894.11	**1.1905**	354,258.88	**nil**	1,128,623.65
3	5,202,936.48	**1.0386**	(406,698.75)	**nil**	721,924.90
4	6,094,641.91	1.2166	485,006.68	**nil**	1,206,931.59
5	6,480,627.99	1.2936	870,992.76	**2,077,924.35**	nil
6	6,495,588.34	1.2966	885,953.11	**885,953.11**	nil

Year four additionally required a sponsor injection of **1,500,000**: the membrane replacement brought forward cost 3,000,000 against a maintenance-reserve balance of 1,800,000, a gap of **1,200,000**, with the remaining **300,000** restoring the reserve. 5. **Interpretation.** Four observations, in ascending order of professional value. First, **the debt service reserve was never drawn.** The worst year's `CFADS` of 5,202,936.48 exceeded debt service by 193,301.25; the 2,504,817.62 reserve sat untouched throughout. Everything that went wrong went wrong above the reserve's pay-line, which is the general case: reserves are sized for the failure that does not happen, and the failures that do happen bind on the distribution test. Second, **year one is the paradox this domain exists to teach.** The project met its covenant at 1.2743 and passed the backward distribution test at 1.25×, yet paid nothing, because the forward test failed on the sponsors' own re-forecast (15.1.4). There was 774,364.77 of cash, contractually available to nobody. Third, **the block account is not a penalty box; it is pre-funding.** The 406,698.75 that year three could not contribute to the maintenance reserve came out of the cash years one and two had been forced to retain — the trapped dividend paid for the reserve shortfall that the deterioration caused. A sponsor that had successfully argued for a weaker distribution test in negotiation would have paid that

774,364.77 out in year one and been asked for it back, as new equity, in year three. Fourth, **the reserve top-up is the ranking that hurts.** In year three the maintenance charge consumed the entire residual and more; had the replacement not been brought forward, the same cash would still have been unavailable. A distribution forecast that models debt service but not reserve restoration overstates equity cash by the full reserve charge in every year — 600,000 a year for Kestrel, which is 77.5 % of the base-case dividend.

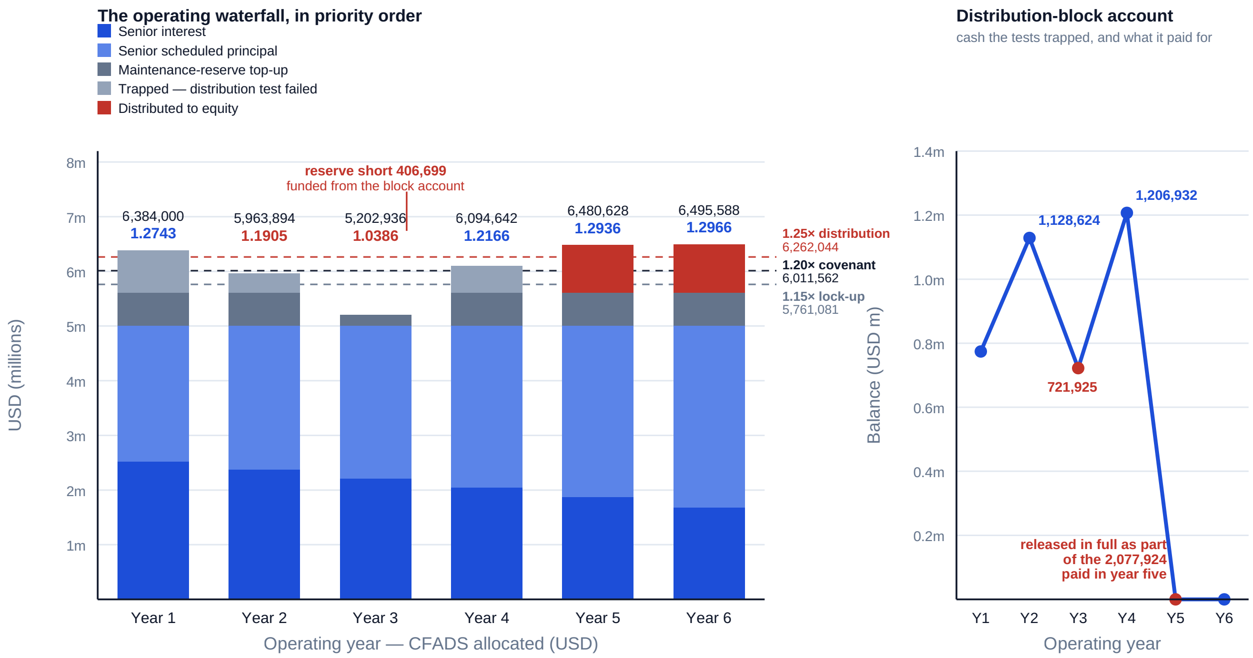

Fig 15.2.1 — Kestrel's operating waterfall and the block account it fills

15.2.3 The distributable amount that falls out

Definition. The **distributable amount** is what remains after every prior step, and in a level base case it is a residual with a closed form:

```
Distributable = CFADS − debt service − reserve top-ups   (subject to every distribution condition)
```

For Kestrel's base case: `6,384,000 − 5,009,635.23 − 600,000 =` **774,364.77** per year, a cash yield of **4.3020 %** on the 18,000,000 subscribed. That number deserves to be sat with, because it is the honest shape of project-finance equity and it surprises people. A project appraised at a Domain 4 `NPV` of +16,179,360 with an `IRR` of 12.19 % pays its owners a 4.30 % cash yield for twelve years. The reason is arithmetic, not disappointment: while the loan is outstanding, debt service consumes 78.5 % of `CFADS`, and equity is the residual claimant on a leveraged asset. Once the loan retires the same `CFADS` produces **5,784,000** a year — a **32.13 %** yield on the original subscription — and from year sixteen the handback charge of **697,844.05** (KA 15.4) reduces it to **5,086,155.95**, before rising to **5,686,155.95** once the maintenance-reserve charge ends after year twenty.

The whole 25-year base-case profile, discounted at the 8.0 % the board owns, gives an equity `NPV` of **+5,027,733.03** and an **equity `IRR` of 9.8591 %**, on total distributions of **80,505,936.71** — a money multiple of **4.4726×**. Two cautions belong beside those figures. They are **not** Domain 4's appraisal restated: Domain 4 valued the unlevered project over a 15-year horizon on a different net-inflow basis, and comparing the two numbers as though one should reproduce the other is a category error that appears in real board packs. And the 9.8591 % is **back-end loaded to an extreme degree — 88.5 %** of total distributions (71,213,559.47 of 80,505,936.71) fall in the last thirteen years — which is why the exit and refinancing questions of KA 15.3 and 15.4 dominate sponsor behaviour in a way that a level-yield instrument would not.

15.2.4 What a distribution drought costs equity

Definition. A **distribution drought** is a run of periods in which cash is generated but not distributable. Its cost to equity has two components that must be separated, because only one of them is a performance failure.

WORKED EXAMPLE

Worked example 15.2.4 — decomposing Kestrel's six-year cost to equity.

1. **Setup.** Base case: 774,364.77 distributed in each of years 1–6. Actual: nil in years 1–4, 2,077,924.35 in year 5, 885,953.11 in year 6, less a 1,500,000 injection at the end of year 4. Discount rate 8.0 %.
2. **Formula.** Present value of each profile at 8 %; the difference is the total cost. The cash cost is the undiscounted difference; the balance is the timing cost.
3. **Substitution.** Base `774,364.77 × AF(0.08, 6)`. Actual `2,077,924.35 × DF(0.08, 5) + 885,953.11 × DF(0.08, 6) − 1,500,000 × DF(0.08, 4)`.
4. **Result.** Base-case present value **3,579,795.15**. Actual present value **1,972,501.14**, less the injection's **1,102,544.78**, giving **869,956.36**. **Loss of present value 2,709,838.79.** The undiscounted cash difference is **3,182,311.16**, which reconciles exactly: the six-year `CFADS` shortfall of **1,682,311.16** (36,621,688.84 actual against 38,304,000 base) plus the **1,500,000** injection.
5. **Interpretation.** The reconciliation is the point. Of the 3,182,311.16 of cash equity lost, only **52.9 %** is a trading shortfall; **47.1 %** is a capital call caused by a reserve that was not big enough for an event brought forward by one year. That decomposition changes who is accountable for what: the operator owns the 1,682,311.16, and the finance director owns the 1,500,000, because a maintenance reserve sized on the assumption that a five-year replacement cycle will run to five years is a financing decision, not an engineering one. Note also that the present-value loss of 2,709,838.79 is **less** than the cash loss of 3,182,311.16, because **2,963,877.46** of distributions were ultimately paid — four and five years late rather than never. Deferral is expensive and it is not destruction, and conflating the two produces the commoner of the two errors in distressed-project board papers: treating a lock-up as a write-off. The other error is the mirror image, treating it as a timing matter of no consequence, and the arithmetic settles that one decisively: **2,709,838.79 is 53.90 % of the entire 25-year equity `NPV` of 5,027,733.03.** Six poor years out of twenty-five removed more than half the base-case equity value, because they fell at the front of the profile where discounting bites least.

AI in this KA

Where it earns its place. Waterfall computation is deterministic, rule-heavy and tedious — ideal machine work, and the natural home of a well-tested calculation engine. Two applications add real value beyond speed. **Forward waterfall projection under scenarios**: running the distribution conditions, block-account mechanics and reserve top-ups across hundreds of driver scenarios to produce a distribution distribution rather than a point forecast, which is the only honest answer to "when will we be paid?". And **block-account and reserve reconciliation**: tying account-bank statements to the modelled balances every period, which catches the operational errors — a top-up made from the wrong account, a release taken before the test date — that no ratio would reveal.

Where it must not go. A model must not determine whether a distribution is permitted. That determination rests on the satisfaction of every condition, including qualitative ones ("no default subsisting", "no material adverse change"), and it is a decision with legal consequence for the officers who authorise the payment. Nor should generated code implement the waterfall without independent reconstruction: the ordering of steps 6 to 11 is where an assistant will confidently produce a defensible-looking but wrong sequence, because the conventional order and the documented order differ in most facilities by at least one step.

Verification, concretely. Reconstruct one period's waterfall by hand from the accounts and tie every step to its clause. Prove the closing identity in every period — opening balances plus receipts less applications equals closing balances, account by account. Test the engine on the three adversarial cases: a period where the residual cannot fund a top-up (Kestrel's year three); a period where a test fails on the forward leg only (year one); and a period where a release and a top-up fall due simultaneously. An engine that has not been run against those three has not been tested.

KEY TERMS — KA 15.2

Term	Meaning
Operating cash waterfall	The contractual priority order for applying each period's project cash.
Top-up / restoration	Application of cash to restore a reserve to its required balance; ranks above distributions.
Distribution-block account	Where cash that fails a distribution test is held; a pre-funding mechanism, not a penalty.
Distributable amount	`CFADS` less debt service less reserve top-ups, subject to every distribution condition.
Cash yield	Distribution ÷ equity subscribed — 4.3020 % for Kestrel while the loan runs, 32.13 % after.
Distribution drought	A run of periods generating cash that is not distributable; costs equity in cash and in timing.

SAMPLE MCQS — KA 15.2

MCQ 15.2-A `[15.2.3 · Application]` Kestrel's `CFADS` is 6,384,000, debt service 5,009,635.23 and the maintenance-reserve charge 600,000. The base-case distributable amount is: - A. USD 1,374,364.77 - B. USD 774,364.77 - C. USD 6,384,000 - D. USD 774,364.77 less the six-month debt service reserve of 2,504,817.62

Rationale: `6,384,000 − 5,009,635.23 − 600,000 = 774,364.77`. A omits the reserve top-up, which is the single most common distribution-forecasting error (15.2.2); C is `CFADS` itself; D double-counts a reserve that was funded at close and requires no top-up while debt service is level.

MCQ 15.2-B `[15.2.2 · Analysis]` In Kestrel's year one the backward `DSCR` is 1.2743 against a 1.20× covenant and a 1.25× distribution condition, and 774,364.77 of cash remains after debt service and the reserve charge. Nothing is distributed. The correct explanation is: - A. the covenant was breached - B. the distribution condition also requires a forward-looking test, which failed at 1.0979 on the sponsors' re-forecast - C. the debt service reserve had to be topped up first - D. distributions are prohibited in the first operating year

Rationale: Both backward tests passed; the forward leg of the distribution condition failed (15.1.4, 15.2.2). A is contradicted by 1.2743 > 1.20; C is false — a level annuity's six-month reserve never needs topping up; D invents a term.

MCQ 15.2-C [15.2.4 · Analysis] Kestrel's equity lost 3,182,311.16 of cash over six years but only 2,709,838.79 of present value at 8 %. The reason is: - A. an arithmetic inconsistency between the two measures - B. 1,463,877.46 of the withheld cash was eventually distributed, so part of the loss is deferral rather than destruction - C. the discount rate should have been the loan rate of 6 % - D. the 1,500,000 injection should not be counted as a cost to equity

Rationale: Deferral costs the time value, not the principal (15.2.4). C would change both figures without changing the relationship; D is wrong because an injection is cash equity provides and does not get back other than through later distributions already counted.

MCQ 15.2-D [15.2.1 · Recall] In the operating waterfall, a handback or decommissioning reserve top-up ranks: - A. above senior debt service, being a statutory obligation - B. below distributions to equity, being a long-dated liability - C. above distributions to equity and below senior debt service, alongside the other reserve restorations - D. at the same level as cash taxes

Rationale: Restorations sit between obligations and permissions (15.2.1). A inverts the security architecture; B is the error that leaves a handback obligation unfunded (Case study B); D confuses a reserve with a period cost.

SELF-CHECK — KA 15.2

1. Why was Kestrel's debt service reserve never drawn across six deteriorating years? — Worst-year CFADS of 5,202,936.48 still exceeded debt service by 193,301.25; the failures bound on the distribution test, above the reserve's pay-line.
2. What did the block account actually pay for? — The 406,698.75 maintenance-reserve shortfall in year three, out of dividends trapped in years one and two.
3. State Kestrel's base-case cash yield before and after the loan retires. — 4.3020 % on 774,364.77 while the loan runs; 32.13 % on 5,784,000 afterwards.

KNOWLEDGE AREA 15.3

Refinancing, waivers and amendments

Topics: 15.3.1 why operating-phase debt is mispriced · 15.3.2 pricing a refinancing · 15.3.3 waiver, cure and amendment, priced against one another · 15.3.4 the cash sweep as the price of consent.

15.3.1 Why operating-phase debt is mispriced

Definition. Refinancing is the replacement of an existing facility with a new one, and in project finance its economic driver is specific: the original facility was priced for a risk that has since been retired. A construction-phase margin compensates a lender for completion risk, technology risk and the possibility of a project that never operates. Once the completion tests of Domain 14 are passed and two or three years of operating history exist, none of those risks remain, but the margin does — because a term loan's price is fixed at signing and the market's view is not.

Three structural features make the opportunity larger than a margin comparison suggests. **The tail**: Domain 10 (KA 10.2.2) measured Kestrel's PLCR at 1.9431 against an LLCR of 1.2743, quantifying thirteen years of project life beyond the loan's maturity; a refinancing can lend against part of that tail, which no margin reduction can replicate. **The amortisation profile**: a schedule sculpted for a construction-phase risk view is usually faster than an operating asset requires. And **the covenant package**: the ratios, reserves and distribution conditions negotiated when the project was a drawing can be reset against observed performance. The professional discipline is therefore to price the whole new term sheet, not the margin — and the worked example exists to show how badly the margin alone misleads.

15.3.2 Pricing a refinancing

WORKED EXAMPLE

Worked example 15.3.2 — Kestrel refinances at the end of year five.

1. **Setup.** Senior debt outstanding **27,965,694.77** with **7 years** remaining at an all-in **6.00 %** (a 3.00 % base rate swapped fixed for the full tenor, plus a 300 basis point margin). Annual service 5,009,635.23. The market will lend against the operating asset at a **145 basis point** margin, an all-in **4.45 %**. Costs: the interest-rate swap must be broken — the seven-year market swap rate has fallen to **2.20 %**, so the borrower is paying 80 basis points above market on the amortising notional; an arrangement fee of **1.00 %**; a prepayment fee under the existing facility of **0.50 %**; and **320,000** of legal, model-audit, technical-adviser and agency cost. Two structures are on offer: the same 7-year tenor, or a **10-year** tenor lending three years into the tail. Discount incremental cash to equity at 8.0 %.
2. **Formula.** New instalment = outstanding ÷ `AF(new rate, tenor)`. Incremental cash to equity in each year = old service ceasing less new service arising. Swap break cost = Σ (old fixed − market) × notional in each remaining year, discounted at the market rate. Net present value = present value of incremental cash − total transaction cost.
3. **Substitution.** Break cost: `0.80 % ×` the seven remaining opening balances (27,965,694.77 · 24,634,001.23 · 21,102,406.07 · 17,358,915.21 · 13,390,814.89 · 9,184,628.55 · 4,726,071.04), each discounted at 2.20 % → 218,909.55 + 188,678.82 + 158,150.03 + 127,294.31 + 96,082.11 + 64,483.14 + 32,466.39. Fees: `0.0100 × 27,965,694.77` and `0.0050 × 27,965,694.77`. Instalments: `27,965,694.77 ÷ AF(0.0445, 7)` and `÷ AF(0.0445, 10)`.
4. **Result.** Swap break cost **886,064.34**; arrangement fee **279,656.95**; prepayment fee **139,828.47**; advisers **320,000 — total transaction cost 1,625,549.77**.

Option	New instalment	Base-case DSCR	PV of incremental cash at 8 %	Net present value
4.45 % over 7 years	**4,737,139.41**	1.3476	1,418,714.07	**(206,835.69)**
4.45 % over 10 years	**3,525,588.23**	1.8108	2,425,030.89	**+799,481.12**

Breakeven all-in rate on the 7-year option: **4.2206 %**, a margin of **122.06 basis points**. Breakeven all-in rate on the 10-year option: **5.1308 %**. 5. **Interpretation.** The headline is the trap. **A refinancing that cuts the margin by 155 basis points destroys 206,835.69 of value.** The reason is that the margin saving is a thin annual flow on a rapidly amortising balance — 272,495.82 in year one, falling every year — while the costs are a thick lump today, and the swap break alone consumes 54.5 % of them. The professional discipline that follows is to solve for the breakeven before opening a negotiation: Kestrel's margin must fall by **177.94 basis points**, not 155, to pay for the transaction, so the market's best offer is **22.94 basis points short**. That single figure is the whole mandate brief. What rescues the transaction is not price but **structure**: extending three years into the tail turns the same margin into +799,481.12, and the decomposition shows why — of the 2,425,030.89 of present value, **1,418,714.07 is the margin, 586,108.78 is the tenor measured alone, and 420,208.04 is the interaction** between the two, which exists only because the extension is priced at the lower rate. None of the three components pays for the transaction by itself. Two cautions complete the analysis. The discount rate matters to magnitude but not to sign here, and a refinancing paper should state it and test it, because debt-service savings discounted at the cost of debt look larger than the same savings discounted at the cost of equity and neither is wrong in the abstract. And the extension is not free of consequence: the LLCR rises to **1.8108** and the PLCR to **2.9824**, coverage a lender will not simply hand over — which is what KA 15.3.4 is about.

15.3.3 Waiver, cure and amendment, priced against one another

Definitions. A **waiver** is the lenders' consent to disregard a specific breach, usually for a fee. An **equity cure** is the sponsors' contractual right to inject cash treated as CFADS (or applied to prepayment) so that a ratio is restored; facilities cap the number available over the loan's life. An **amendment** changes the terms — most commonly resetting a covenant level for a defined period — in exchange for a fee, a margin uplift and usually tighter controls.

Kestrel breaches at two successive test dates, and the two situations look identical and are not.

	End of year 2	End of year 3
CFADS	5,963,894.11	5,202,936.48
Backward DSCR	**1.1905**	**1.0386**
Covenant cash requirement at 1.20×	6,011,562.28	6,011,562.28
Equity cure required	**47,668.16**	**808,625.80**
Waiver fee at 15 basis points on debt outstanding	**55,307.03**	—
Amendment: 25 bp fee on 36,871,351.43	**92,178.38**	—
Amendment: present value at 8 % of a 40 bp margin uplift on years 3–12	**651,258.24**	—
Total amendment cost	**743,436.62**	—

WORKED EXAMPLE

Worked example 15.3.3 — the cheapest option that was the wrong one.

1. **Setup.** At the year-two test date the breach is marginal: DSCR 1.1905 against 1.20. The sponsors hold **two** equity cures over the loan's life and have used none. Three responses are available on the numbers above. At the year-three date, nine months later, the breach is severe.
2. **Formula.** Cure = covenant cash requirement − actual CFADS. Waiver fee = basis points × debt outstanding. Amendment cost = fee + Σ (margin uplift × opening balance in each remaining year), discounted at 8 %.
3. **Substitution.** Year two: `6,011,562.28 − 5,963,894.11`. Year three: `6,011,562.28 − 5,202,936.48`. Uplift: `0.0040 ×` each opening balance from year three to year twelve (147,485.41 · 136,295.99 · 124,435.21 · 111,862.78 · 98,536.00 · 84,409.62 · 69,435.66 · 53,563.26 · 36,738.51 · 18,904.28), discounted at 8 %.
4. **Result.** The year-two cure of **47,668.16** is the cheapest cash response — **7,638.87 less than the waiver** and 695,768.46 less than the amendment. The year-three cure would cost **808,625.80**, which is **808,625.80 ÷ 47,668.16 = 17.0 times** the year-two requirement; put the other way, the year-two breach was **5.8950 %** of the year-three breach. And the amendment at 743,436.62 is within **5,290.97** — seven-tenths of one per cent — of the present value of that year-three cure (**748,727.59** discounted one year at 8 %).
5. **Interpretation.** Two conclusions, and the second is the one that separates a practitioner from an analyst. First, on price, **the amendment and the second cure are indistinguishable**: a difference of 5,290.97 on obligations of three-quarters of a million is inside the error of any forecast underlying either, so the decision cannot be taken on cost and anyone who presents it as a cost comparison has not understood it. The real differences are structural — the amendment resets four test dates and costs 40 basis points for the remaining life; the cure fixes one date, leaves the covenant profile intact, and consumes an irreplaceable option. And note an asymmetry the raw comparison hides: **a cure is an investment and a fee is not.** Cure cash enters the project and is recovered through later distributions; a waiver fee and a margin uplift leave permanently. On a like-for-like basis the cure is better than 5,290.97 suggests. Second, on the actual decision, **spending a cure on the year-two breach was the error.** The sponsors paid 47,668.16 and preserved 7,638.87 of fee, and in doing so consumed half of a facility that nine months later would have been worth 808,625.80 of cover. The correct posture is a rule, not a calculation: **cure rights are scarce options and must be spent in proportion to what they buy** — a marginal breach is waived, and the incremental cost of doing so (7,638.87, or **0.9447 %** of the value of the option preserved) is the cheapest insurance premium in this book. The reflex to reach for the cheapest cash response at each test date, in isolation, is exactly how sponsors arrive at the third breach with no options and no negotiating position.

A third caution belongs here. **An amendment must be sized against the stressed case, not the current one.** A covenant reset to 1.10× negotiated at the year-two date, on the then-current 1.1905, would have been breached again at year three's 1.0386 — a second breach, a second negotiation and a second fee, from a materially worse position. The amendment Kestrel actually took reset the covenant to 1.00× for two test dates before stepping to 1.10× and 1.20×, which is what "sized against the stress" means in practice.

15.3.4 The cash sweep as the price of consent

Definition. A **cash sweep** applies a defined share of cash that would otherwise be distributed to mandatory prepayment. Domain 10 (KA 10.1.3) named it; here it is priced, because a sweep is how lenders take a share of a refinancing or amendment gain, and it is routinely under-priced by sponsors who read it as a liquidity provision rather than a transfer of value.

Take the successful 10-year refinancing of 15.3.2. Post-refinancing distributable cash rises from 774,364.77 to **2,258,411.77** a year — a 12.5467 % cash yield where there had been 4.3020 %. A lender asked to lend three years further into the tail will want a share of that, and the standard ask is **50 % of cash available for distribution** applied to prepayment.

Structure	Loan retires	Equity IRR	Equity NPV at 8 %
No refinancing	year 13	9.8591 %	5,027,733.03
Refinanced, no sweep, net of the 1,625,549.77 costs	year 15	—	**5,571,846.45**
Refinanced, 50 % sweep, net of costs	year 13	—	**4,925,518.59**

The sweep costs equity **646,327.86** of present value and 39.76 basis points of IRR on the pre-cost profile, and it removes two years of debt life for the lender. Against a refinancing gain of **544,113.42** at time zero (the 799,481.12 net present value of 15.3.2 discounted five years at 8 %), that is decisive: **with a 50 % sweep the refinancing is worth 102,214.44 less to equity than doing nothing.** The **breakeven sweep share is 40.3334 %** — above it, the transaction destroys equity value.

The professional content is a habit rather than a formula. **Never evaluate a refinancing on the margin and the tenor and then negotiate the sweep separately.** The sweep is part of the price, its effect is second-order in appearance and first-order in magnitude, and the breakeven share is computable before the first meeting. A sponsor who walks into a refinancing knowing that 40.33 % is its indifference point negotiates from a position no amount of relationship can substitute for.

AI in this KA

Where it earns its place. Three uses are strong. **Solving for breakevens** across the transaction's whole parameter space — margin, tenor, fee, sweep share, break cost — is exactly the sort of repeated root-finding a machine should do, and it produces the single most useful output of a refinancing analysis, which is the boundary of the acceptable rather than a point answer. **Term-sheet comparison** across competing offers, normalising to a common present value, catches the option that looks cheapest on margin and is not. **Amendment-history extraction** across a portfolio — every covenant reset, fee and margin uplift ever granted, with dates — builds the institutional memory that makes the next negotiation cheaper.

Where it must not go. A model must not determine whether a cure right remains available, or how many, or on what conditions. That is a reading of the facility's cure clause and its history of use, and it is the specific fact on which the year-two decision above turned. Nor should AI draft the request that goes to a lender: the tone, sequencing and candour of a waiver request is a relationship instrument, and Domain 10 (KA 10.4.4) established that the relationship is the asset that determines the terms of every future consent.

Verification, concretely. Recompute the break cost by hand on the notional schedule and tie each period's notional to the amortisation table (Domain 3's checks). Confirm that the incremental cash-flow set is complete — old service ceasing, new service arising, fees, and the sweep — because the characteristic machine error is an omitted leg, not a wrong multiplication. Re-derive the breakeven independently and check that the base case sits on the correct side of it. Where a model has read an amendment or waiver clause, check its reading against the document before anyone quotes it.

KEY TERMS — KA 15.3

Term	Meaning
Refinancing	Replacement of a facility priced for a risk since retired; value comes from margin, tenor and covenant reset together.
Swap break cost	Present value of the difference between the contracted fixed rate and the market rate over the remaining notional profile.
Breakeven margin	The new all-in rate at which a refinancing's net present value is nil — 4.2206 % for Kestrel's 7-year option.
Waiver	Consent to disregard a specific breach, usually for a fee.
Equity cure	Contractual injection restoring a ratio; a scarce option, capped in number.
Amendment	A change of terms — usually a covenant reset — for a fee, a margin uplift and tighter controls.
Cash sweep	A defined share of distributable cash applied to mandatory prepayment; part of the price of consent.

SAMPLE MCQS — KA 15.3

MCQ 15.3-A `[15.3.2 · Application]` Kestrel's 7-year refinancing at 4.45 % saves 272,495.82 a year with a present value of 1,418,714.07 against total costs of 1,625,549.77. The correct conclusion is: - A. proceed — a 155 basis point margin saving is material - B. reject on these terms: net present value is (206,835.69), and the margin must fall 177.94 basis points to break even - C. proceed — the `DSCR` improves from 1.2743 to 1.3476 - D. reject — refinancing an operating asset is never economic

Rationale: `1,418,714.07 − 1,625,549.77 = (206,835.69)`; the breakeven all-in rate is 4.2206 %, a 122.06 bp margin (15.3.2). A prices the headline and not the transaction; C cites a coverage improvement that has no cash value to equity; D overgeneralises — the 10-year variant is worth +799,481.12.

MCQ 15.3-B `[15.3.2 · Analysis]` Of the 2,425,030.89 present value in Kestrel's 10-year refinancing, how much is attributable to the margin reduction measured alone? - A. USD 2,425,030.89 - B. USD 1,418,714.07 - C. USD 586,108.78 - D. USD 420,208.04

Rationale: The margin component is the 7-year option's present value (15.3.2). A is the total, which treats the whole gain as margin; C is the tenor component measured alone; D is the interaction term that exists only because the extension is priced at the lower rate.

MCQ 15.3-C `[15.3.3 · Analysis]` The sponsors hold two equity cures. At a test date the `DSCR` is 1.1905 against a 1.20× covenant, requiring a cure of 47,668.16; a waiver is available for 55,307.03. The better decision, and why, is: - A. cure — it is 7,638.87 cheaper in cash - B. waive — paying 7,638.87 more preserves an option worth 808,625.80 of cover nine months later, or 0.9447 % of the value preserved - C. neither — a marginal breach may be disregarded - D. cure — cure cash counts as `CFADS` and so also improves the reported ratio

Rationale: Cures are scarce options and must be spent in proportion to what they buy (15.3.3). A optimises one test date in isolation, which is precisely the error; C invents a materiality threshold facilities do not contain; D is true of the mechanics and irrelevant to the choice.

MCQ 15.3-D [15.3.4 · Application] The lenders consent to a 10-year refinancing worth 544,113.42 to equity at time zero, in exchange for a 50 % sweep of distributable cash that costs equity 646,327.86 of present value. Equity should: - A. accept — the tenor extension improves coverage to 1.8108 - B. decline or renegotiate: net of the sweep the transaction destroys 102,214.44, and the breakeven sweep share is 40.3334 % - C. accept — a sweep only accelerates repayment and does not reduce total distributions - D. accept — the IRR cost of 39.76 basis points is immaterial

Rationale: The sweep exceeds the gain (15.3.4). A cites coverage, which is the lenders' benefit; C is false — accelerated prepayment reduces the present value of distributions even where the undiscounted total is similar; D judges materiality on the wrong measure, since 646,327.86 exceeds the whole gain.

SELF-CHECK — KA 15.3

1. Why does a 155 basis point margin saving fail to pay for Kestrel's refinancing? — The saving is a thin flow on a rapidly amortising balance; the breakeven reduction is 177.94 basis points.
2. What actually makes the transaction work? — Tenor: three years into the tail, plus the interaction of tenor with the lower rate — 586,108.78 and 420,208.04 of the 2,425,030.89.
3. State the rule for spending an equity cure. — In proportion to what it buys; a marginal breach is waived, because the cure is a capped option with a scarcity value.

KNOWLEDGE AREA 15.4

Distress, restructuring, exit and handback

Topics: 15.4.1 what distress is, financially · 15.4.2 the three restructurings, compared · 15.4.3 the enforcement floor and the bargaining range · 15.4.4 exit and the arithmetic of crystallisation · 15.4.5 handback and the residual obligation.

15.4.1 What distress is, financially

Definition. A project is in **financial distress** when its forecast cash cannot sustain its contracted debt service on the agreed profile — which is a different condition from a covenant breach (a test failure at a date) and from insolvency (an inability to pay). The distinction is operational: a breach is cured, waived or amended; distress is **restructured**, because no fee or one-off injection fixes a profile mismatch that persists for the life of the loan.

The diagnostic is simple and worth stating as a rule. Compute the **sustainable debt service** — revised CFADS divided by the target coverage — and compare it with the scheduled service. If scheduled service exceeds sustainable service for a single period, the project has a liquidity problem and reserves exist for it. If it exceeds sustainable service for the remaining life, the project has a **capital structure problem**, and only three levers can close it: **time** (extend), **principal** (reduce), or **new money** (inject).

15.4.2 The three restructurings, compared

WORKED EXAMPLE

Worked example 15.4.2 — Kestrel on the permanent-deterioration branch.

1. **Setup.** Suppose the year-three deterioration of KA 15.2 had proved **permanent** rather than remediable: revised `CFADS` of **5,202,936.48** flat for years 4 to 25. Senior debt outstanding at the end of year three is **34,073,997.28**, with **9 years** remaining at 6.00 % and scheduled service of 5,009,635.23 — a `DSCR` of **1.0386**. The lenders require **1.20×** restored. Lender recovery is measured as the present value of what they receive, both at the contract rate of 6.00 % and at a **7.00 %** risk-adjusted required return; equity value is the present value at 8.0 % of distributions after the 600,000 maintenance charge to year twenty and the 697,844.05 handback charge from year sixteen.
2. **Formula.** Sustainable service = revised `CFADS` ÷ 1.20. Extension: solve for the tenor `n` with `outstanding ÷ AF(0.06, n) ≤` sustainable service. Haircut: supported debt = sustainable service × `AF(0.06, 9)`. Injection: prepay the difference and re-amortise over the original 9 years.
3. **Substitution.** Sustainable service `5,202,936.48 ÷ 1.20 = 4,335,780.40`. Required annuity factor `34,073,997.28 ÷ 4,335,780.40 = 7.858792`, against `AF(0.06, 10) = 7.360087` and `AF(0.06, 11) = 7.886875` → 11 years. Haircut: `4,335,780.40 × 6.801692`.
4. **Result.**

	A — extend to 11 years	B — principal haircut	C — sponsor injection
Amount written off / injected	—	**4,583,353.23** (13.4512 %)	**4,583,353.23**
New instalment	**4,320,342.23**	**4,335,780.40**	**4,335,780.40**
Restored `DSCR`	**1.2043**	**1.2000**	**1.2000**
Lender recovery, PV at 6.00 %	**100.0000 %**	**86.5488 %**	**100.0000 %**
Lender recovery, PV at 7.00 %	**95.0779 %**	**82.9037 %**	**96.3549 %**
Equity value, PV at 8 %	**14,898,548.64**	**18,656,183.22**	**14,072,829.99**
Annual equity cash while debt runs	282,594.25	267,156.08	267,156.08

1. **Interpretation.** The table separates three questions that boards routinely merge. **What does each side prefer?** Lenders rank C, then A, then B; equity ranks B (18,656,183.22), then A (14,898,548.64), then C (14,072,829.99). The rankings are exactly opposed on the haircut, which is why haircuts are demanded and rarely granted. **Why does equity prefer the extension over the injection**, when the injection produces a lower debt service? Because the injection costs 4,583,353.23 today to buy a debt-service saving whose present value at 8 % is less — a gap of **825,718.65**, which is the arithmetic reason sponsors resist new money even when they have it, and the reason a lender should read a refusal to inject as information about the sponsor's discount rate rather than about its confidence. **And what does the extension cost the lenders?** At the contract rate, nothing: extending a loan at its own coupon returns exactly par, because the present value of a longer annuity at the same rate is the same principal. That is the arithmetic identity behind every "amend and extend", and it is why extensions are the default restructuring.

But at the 7.00 % the lenders now require, the extension recovers **95.0779 %** — a real loss of 4.92 points of present value, which the accounting may or may not recognise. The professional caution is precisely there: **an extension is described as par and is not**, and a credit committee that measures recovery only at the contract rate cannot see what it has given away. The sensitivity is worth carrying as a rule of thumb — for Kestrel, every 100 basis points of required-return increase costs roughly five points of extension recovery.

15.4.3 The enforcement floor and the bargaining range

Definition. The **enforcement floor** is the recovery a lender would achieve by enforcing its security and realising the asset, net of costs and delay. It is the boundary of every restructuring negotiation, because no lender rationally accepts a proposal recovering less than the floor, and no sponsor rationally accepts one leaving less than the value of its equity elsewhere.

WORKED EXAMPLE

Worked example 15.4.3 — computing the floor, and what it excludes.

1. **Setup.** On enforcement, Kestrel would be sold to a buyer who assumes lower performance and a higher cost base than the incumbent — `CFADS` of **4,900,000** over the remaining 22 years — and who requires **13.0 %** on a project acquired out of enforcement with a broken operations and maintenance arrangement. Enforcement and sale costs are **6 %** of proceeds, and the process takes **twelve months**, over which the lenders discount at their 6.00 % contract rate.
2. **Formula.** Floor = [buyer's `CFADS` × `AF`(buyer's rate, remaining life) × (1 − costs)] ÷ (1 + lender rate).
3. **Substitution.** `4,900,000 × AF(0.13, 22) = 4,900,000 × 7.169513`; `× 0.94`; `÷ 1.06`.
4. **Result.** Gross enterprise value **35,130,615.34**; costs **2,107,836.92**; net **33,022,778.42**; discounted twelve months **31,153,564.55** — a recovery of **91.4291 %**.
5. **Interpretation.** The floor decides the negotiation before it starts. Option B's haircut recovers **82.9037 %** on the lenders' own required return — **8.5254 points below** what enforcement pays — so it is not a hard bargain, it is **outside the feasible set**, and a sponsor proposing it has spent credibility on an option that could never be accepted. Working the constraint in reverse gives the more useful number: the **maximum haircut a lender can accept is 2,920,432.73**, or **8.5709 %** of the outstanding, at which recovery equals the floor. But that haircut supports an instalment of **4,580,266.69**, a `DSCR` of only **1.1359** — still below the 1.20 required. **The largest haircut the lenders can grant does not solve the problem**, which is the decisive finding of the whole analysis: on these numbers the restructuring must include an extension, and the only remaining questions are how long and at what price. Two professional qualifications. The floor is a **model of a distressed sale**, and every input is contestable — the buyer's `CFADS`, the required return, the cost percentage, the delay — so the discipline is to present it as a range and to state which input the conclusion depends on (here, the buyer's required return: at 10.0 % the floor would exceed par and no restructuring would be rational at all). And the floor systematically **overstates** what most lenders will really do, because enforcement carries consequences no discounted cash flow captures: the offtaker's step-in rights (Domain 12's direct agreements), regulatory and political exposure, reputational cost, and the simple institutional fact that enforcing leaves a bank owning a desalination plant. That is why Domain 10 (KA 10.4.3) observed that lenders rarely accelerate a fundamentally sound project — the floor bounds the negotiation, but the negotiation is conducted well inside it.

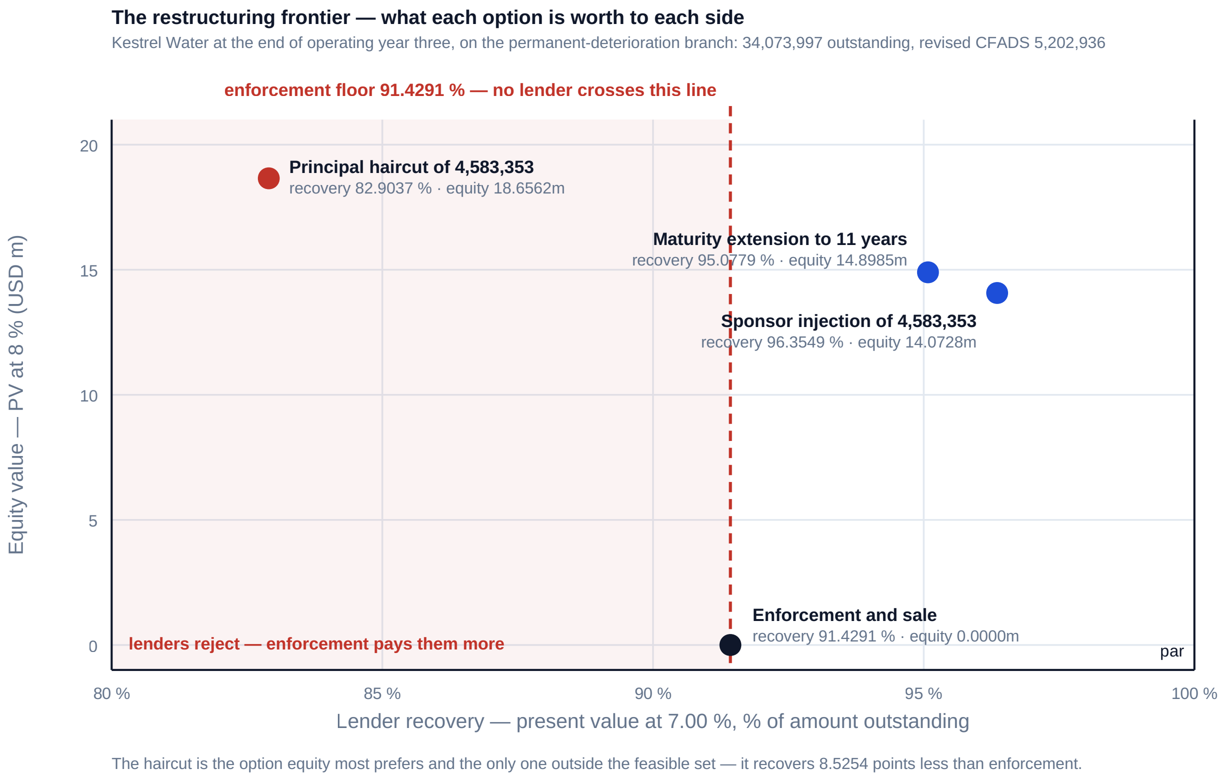

Fig 15.4.1 — The restructuring frontier

15.4.4 Exit and the arithmetic of crystallisation

Definition. An **exit** is the sale of the sponsor's equity interest, priced as the present value of the distributions the buyer expects at the buyer's required return. It is the ordinary end of a sponsor's involvement — development-stage sponsors sell to long-term infrastructure holders as a matter of business model — and it is where IRR is most often used to claim credit for arithmetic.

WORKED EXAMPLE

Worked example 15.4.4 — Kestrel sold at the end of year eight.

1. **Setup.** The base-case profile of 15.2.3. By the end of year eight the sponsors have received **6,194,918.16** of distributions on 18,000,000 subscribed. A long-term holder will buy the remaining seventeen years of distributions. Price it at required returns of 6.50 %, 7.00 %, 7.50 % and at the sponsors' own 8.00 %.
2. **Formula.** Price = Σ (distributions in years 9–25) discounted at the buyer's required return to the end of year eight. Seller's IRR = the rate zeroing the flow (18,000,000) at time zero, distributions in years 1–8, and the price at year eight.
3. **Substitution.** Discount 774,364.77 (years 9–12), 5,784,000 (13–15), 5,086,155.95 (16–20) and 5,686,155.95 (21–25) at each rate.
4. **Result.**

Buyer's required return	Price at end of year 8	Seller's IRR	Seller's NPV at 8 %
6.50 %	**39,260,418.77**	13.4196 %	7,661,177.40
7.00 %	**37,541,787.43**	12.8568 %	6,732,654.36
7.50 %	**35,919,098.27**	12.3059 %	5,855,965.90
8.00 %	**34,386,097.04**	11.7666 %	**5,027,733.03**
Hold to maturity	—	**9.8591 %**	**5,027,733.03**

1. **Interpretation.** The bottom two rows contain the lesson, and it is one of the most useful invariants in this book. **Selling at the sponsors' own 8.00 % discount rate raises the reported IRR from 9.8591 % to 11.7666 % and creates exactly nothing** — the NPV at 8 % is **identical at 5,027,733.03**, to the cent, because a sale at the discount rate is by construction a swap of the same value at a different date. The entire IRR uplift of 1.9075 points is the arithmetic of early crystallisation: a positive NPV realised sooner produces a higher internal rate, which is Domain 4's reinvestment fiction (KA 4.1.2) appearing in a transaction rather than a forecast. What is real is the difference between the buyer's required return and the sponsors': selling at 7.50 % rather than 8.00 % is worth **1,533,001.23** at the end of year eight, or **828,232.87** at time zero — which reconciles exactly to the NPV difference of 828,232.87 between the two rows. That 828,232.87 is **yield compression**, and it is a view about the capital market, not an achievement of the project. The professional discipline is therefore to report an exit in three separated numbers: the value created by operating the asset (which is the base-case NPV), the value created by yield compression (which is a market call, and should be attributed as one), and the IRR, which explains neither and should never carry the credit for both. A sponsor whose realised IRR of 12.3059 % is presented as operating performance has misdescribed a 9.8591 % project sold into a favourable market — and the same arithmetic will run against them when the market moves the other way, at which point the honest baseline they did not establish is what they will need.

15.4.5 Handback and the residual obligation

Definition. Handback is the transfer of the asset to the grantor or offtaker at the end of the concession, in a condition the agreement specifies. The **residual obligation** is the cost of achieving that condition, and it is the last item in the waterfall and the first one forgotten, because it falls due after every person who negotiated it has left.

Kestrel's obligation is an independently estimated **8,000,000** in year-25 money — plant condition works, a final membrane and pump replacement to a defined residual standard, and restoration of the intake structure — funded by a reserve earning **3.00 %**.

WORKED EXAMPLE

Worked example 15.4.5 — when to start funding, and the invariant that decides it.

1. **Setup.** Two funding profiles: level annual contributions over **years 16 to 25** (ten years), or over **years 6 to 25** (twenty years). Equity discounts at 8.0 %.
2. **Formula.** Contribution = obligation ÷ `FVAF`(reserve rate, years). Present cost to equity = contribution × `AF(0.08, years)` × `DF(0.08, years before the first contribution)`.
3. **Substitution.** `8,000,000 ÷ FVAF(0.03, 10) = 8,000,000 ÷ 11.463879`; `8,000,000 ÷ FVAF(0.03, 20) = 8,000,000 ÷ 26.870374`. Then `× AF(0.08, 10) × DF(0.08, 15)` and `× AF(0.08, 20) × DF(0.08, 5)`.
4. **Result.**

Profile	Annual contribution	Total contributed	Present cost at 8 %
Years 16–25 (10 years)	**697,844.05**	6,978,440.53	**1,476,147.78**
Years 6–25 (20 years)	**297,725.66**	5,954,513.22	**1,989,422.56**

Starting ten years earlier more than halves the annual charge and reduces the undiscounted total by 1,023,927.31 — and costs equity **513,274.78 more** in present value, **34.7712 %** more. 5. **Interpretation.** The result is counterintuitive until the mechanism is named, and then it is obvious and permanently useful. **A pre-funded reserve is equity lending money to itself at the reserve's earning rate.** Kestrel's reserve earns 3.00 %; its equity requires 8.00 %; every dollar contributed early is a dollar invested at a 500 basis point negative spread. The invariant that proves it: **when the reserve rate equals the discount rate, the funding profile is irrelevant** — the present cost of any profile that reaches 8,000,000 at year 25 is exactly `8,000,000 × DF(0.08, 25) =` **1,168,143.24**, and the indifference rate is therefore precisely the discount rate, 8.0000 %. That single fact organises the whole negotiation: below the discount rate, sponsors rationally defer and grantors rationally insist; above it, the interests align and the argument disappears. It also tells a leader where to spend effort — not on the funding profile, which is worth 513,274.78, but on **the credit quality of the reserve's investments and the rate they earn**, which is worth more. And it names the reason grantors insist anyway, which is not arithmetic but counterparty risk: a reserve funded in the last ten years of a concession depends on the concessionaire's solvency in years 16 to 25, and the grantor has no claim on a sponsor that has already sold. The resolution used in practice — and the recommendation here — is a **reserve plus a handback bond or parent guarantee**, which separates the funding question from the credit question instead of using the first as a substitute for the second.

Three further mechanics complete the topic. The obligation is fixed by a **condition survey**, typically two to five years before expiry, and the survey is where the estimate meets reality — Case study B prices what happens when it does. **Residual obligations survive handback**: a defects period, a retention against the handback works, and warranties on replaced equipment. And **the asset's residual value belongs to whoever the concession says it belongs to**, which is usually the grantor and never automatically the sponsor: Domain 4's appraisal treated the tail as project cash, and the concession may treat the terminal asset as worth nothing at all to equity.

AI in this KA

Where it earns its place. Restructuring analysis is combinatorial — tenor, haircut, injection, sweep, covenant reset and fee interact — and enumerating the option space against two objective functions (lender recovery, equity value) is exactly the work a machine should do. Generating the **frontier** of

Fig 15.4.1, and locating the enforcement floor across a range of buyer assumptions, converts a negotiation about positions into an inspection of a feasible set. Two further uses: **early-warning classification** across a portfolio, ranking assets by distance to the sustainable-service condition of 15.4.1 rather than by current ratio; and **handback obligation tracking**, holding every asset's survey date, estimate vintage and reserve balance in one place, because the failure mode in Case study B is an information failure a register would have prevented.

Where it must not go. Nothing in the legal characterisation of distress may be delegated: whether facts constitute an event of default, what remedies follow, what a security package actually secures, and how an insolvency régime would treat any of it are questions for qualified counsel in the relevant jurisdiction, and they differ fundamentally between jurisdictions in ways no general model should be trusted to reflect. Nor should a model be used to select the restructuring proposal: the choice among feasible options is a judgment about counterparties, relationships and reputational consequence over decades, and it belongs to named people.

Verification, concretely. Re-derive the sustainable-service test by hand; confirm that each option's restored ratio meets the requirement stated. Check the extension identity — present value at the contract rate must return exactly par — as the cheapest single audit of any restructuring model. Test the enforcement floor's sensitivity to the buyer's required return and report the range, not the point. For handback, verify that the sinking fund reaches the obligation in the final year and that the indifference-rate invariant holds in the model. Record every assumption about a distressed buyer as an assumption, with an owner.

KEY TERMS — KA 15.4

Term	Meaning
Financial distress	Forecast cash cannot sustain contracted debt service for the remaining life — a capital structure problem, not a liquidity one.
Sustainable debt service	Revised `CFADS` ÷ target coverage; the diagnostic that separates distress from a bad period.
Maturity extension	Lengthening the tenor; returns par at the contract rate and less at any higher required return.
Principal haircut	Writing down principal; preferred by equity, feasible only above the enforcement floor.
Enforcement floor	Net recovery from enforcing security and realising the asset; bounds every negotiation.
Yield compression	Value created by selling at a lower required return than the seller's own — a market view, not performance.
Handback	Transfer of the asset at concession end in a specified condition; the residual obligation and its reserve.
Condition survey	The pre-expiry inspection that fixes the handback obligation against the funded estimate.

SAMPLE MCQS — KA 15.4

MCQ 15.4-A `[15.4.2 · Analysis]` A lender extends a distressed loan at its existing contract rate so that coverage is restored. Measured at the contract rate, its recovery is: - A. below par, because payment is deferred - B. exactly par, because the present value of a longer annuity at the same rate is the same principal - C. above par, because more interest is received in total - D. indeterminate without the sponsor's discount rate

Rationale: The extension identity of 15.4.2 — which is why "amend and extend" is the default and why a committee measuring recovery only at the contract rate sees no loss. The real cost appears at a higher required return: 95.0779 % at 7.00 % for Kestrel. C confuses total nominal interest with present value; D is irrelevant to the lender's measure.

MCQ 15.4-B `[15.4.3 · Analysis]` Kestrel's enforcement floor is a 91.4291 % recovery. A proposed haircut recovers 82.9037 % on the lenders' 7.00 % required return. The correct characterisation is: - A. an aggressive but negotiable proposal - B. outside the feasible set — enforcement pays the lenders 8.5254 points more - C. acceptable, because enforcement destroys value for everyone - D. acceptable, because equity value of 18,656,183.22 is the highest of the options

Rationale: No lender accepts less than the floor (15.4.3). A misreads infeasibility as negotiating distance; C is true of the parties jointly and irrelevant to the lender's individual choice; D states equity's preference, which is not a constraint on lenders.

MCQ 15.4-C `[15.4.4 · Analysis]` A sponsor holding a 9.8591 % `IRR` project sells at the end of year eight at exactly its own 8.0 % discount rate and reports an `IRR` of 11.7666 %. The value created is: - A. the 1.9075 percentage point `IRR` uplift - B. nil — the `NPV` at 8 % is identical at 5,027,733.03; the uplift is the arithmetic of early crystallisation - C. USD 828,232.87 - D. USD 34,386,097.04

Rationale: A sale at the discount rate is a swap of equal value at a different date (15.4.4). A mistakes a rate for value; C is the value of yield compression at 7.50 % rather than 8.00 %, which is a different transaction; D is the price, not the gain.

MCQ 15.4-D `[15.4.5 · Application]` An 8,000,000 handback obligation at year 25 can be funded by a reserve earning 3.00 % over years 16–25 (697,844.05 a year) or years 6–25 (297,725.66 a year). At an 8 % equity discount rate, the earlier profile: - A. is cheaper, because the total contributed is 1,023,927.31 lower - B. costs equity 513,274.78 more in present value, because the reserve earns 500 basis points less than equity requires - C. costs the same, because both reach 8,000,000 at year 25 - D. is cheaper, because contributions are smaller

Rationale: Pre-funding is equity lending to itself at the reserve rate (15.4.5). A and D compare undiscounted amounts across different decades; C would be true only if the reserve earned the 8 % discount rate — the indifference-rate invariant, at which every profile costs `8,000,000 × DF(0.08, 25) = 1,168,143.24`.

SELF-CHECK — KA 15.4

1. What single test distinguishes distress from a bad period? — Whether scheduled service exceeds sustainable service (`CFADS` ÷ target coverage) for one period or for the remaining life.
2. Why could Kestrel's restructuring not be solved by a haircut alone? — The largest haircut above the enforcement floor is 2,920,432.73, which supports a `DSCR` of only 1.1359 against the 1.20 required.
3. Why does a grantor insist on early handback funding when it costs the sponsor more? — Not arithmetic but counterparty risk: a late-funded reserve depends on the concessionaire's solvency in the final decade. The clean answer is a reserve plus a bond or guarantee.

Advanced topics — Domain 15

15.A.1 The restatement problem: when a test date moves under you

Operating covenants are tested on reported figures, and reported figures change — through an audit adjustment, a prior-period error, a reclassification of maintenance cost between capital and revenue, or a superseded forecast basis. Facilities handle this unevenly, so three questions must be settled before they matter: **is a certified test re-opened if the underlying figures are restated**, and within what period; **does a later-discovered breach rank from the original test date or from discovery**, which determines whether the cure and waiver windows have already expired; and **whose basis governs a forward test that has been superseded** — the forecast as made, or as it should have been made. The protection is procedural rather than contractual: certify only on figures that have passed the same controls as the statutory accounts, keep every certified ratio's computation and inputs frozen and retrievable, and disclose a material re-forecast when it is made rather than when it is tested. Domain 10 (KA 10.4.4) established that early disclosure is the asset; this is where it earns its return, because a lender told in advance treats a restated breach as administration and a lender told afterwards treats it as a discovery.

15.A.2 Portfolio operations and the shared-covenant problem

The disciplines of this domain scale badly when each asset is run in isolation, and three portfolio effects deserve naming. **Test-date clustering**: facilities testing on the same quarter-end turn one adverse market event into simultaneous breaches and simultaneous negotiations, exactly when negotiating capacity is scarcest — staggering test dates at the point of documentation costs nothing. **Cross-default and cross-acceleration**: a breach in one ring-fenced project reaches other assets through guarantees, holding-company facilities or change-of-control provisions, so the true exposure of a project-level covenant is a portfolio question no project-level model can see. **Cure-capacity concentration**: a sponsor with four assets each holding two cures does not have eight cures, it has as many as its liquidity supports, which in a correlated downturn is fewer. The response is one consolidated covenant register across the portfolio — every test, date, threshold, cash trigger, cure remaining and cross-default link — owned by one person with authority over all of it. Its absence is the commonest reason a group of individually sound projects experiences a group-level liquidity event.

15.A.3 The reviewer's operating eye

Invariants to test on any operating, waterfall or restructuring model — each one a defect indicator if it fails:

- The operating bridge ties: revenue less cash opex equals `EBITDA`; `EBITDA` less cash tax equals `CFADS` before working capital; the working-capital movement equals the change in the balance sheet position, not a plug.
- The same `CFADS` line feeds the covenant, the lock-up, every distribution test and the reported ratio (Domain 10, KA 10.A.3), and the certified figure is the defined one.
- Every rolling test window contains exactly the debt service the documents say it does; for a level annuity paid annually, exactly one instalment.
- Forward-looking tests are computed on a stated, dated forecast with a named owner, and the outturn is reconciled to it afterwards.
- The waterfall closes in every period: opening balances plus receipts less applications equals closing balances, account by account, with no residual.

- Reserve top-ups rank above distributions, and a period whose residual cannot fund a top-up draws from the block account rather than reducing the top-up.
- Distributions are nil in any period where any condition fails, including the forward leg and the qualitative conditions — a model that distributes on the backward test alone is wrong.
- `LLCR` equals `DSCR` wherever cash is level and service is an annuity at the loan rate (Domain 10, KA 10.2.2) — still true after a refinancing, at the new rate.
- Refinancing arithmetic is complete: old service ceasing, new service arising, break cost on the actual notional profile, every fee, and the sweep.
- A maturity extension at the contract rate returns exactly par in present value; if the model says otherwise, the model is wrong.
- Restructuring options are tested against the enforcement floor before they are presented, and the floor's sensitivity to the distressed buyer's required return is shown as a range.
- The handback sinking fund reaches the obligation in its final year; the funding profile is irrelevant when the reserve rate equals the discount rate.
- Minimum, not average, coverage over the remaining life is reported, and the distribution profile is reported after tests rather than before.

Industry variations — Domain 15

- **Contracted water and availability-based PPPs.** Deductions are formulaic and performance-linked, so the operating bridge is unusually tight and the covenant is predictable from operating data — Kestrel's case. The distinctive operating risk is an **unhedged input**: a pass-through that was assumed and is not contractual, which is how a 5.9 % power-price rise removes an entire dividend.
- **Contracted power and renewables.** Resource variability makes a single year's coverage uninformative, so structures use rolling multi-year tests and larger reserves; the characteristic operating-phase event is a **curtailment or grid-constraint régime change** that no contract allocated, and the characteristic financing event is a refinancing into a tighter market once operating history exists.
- **Transport concessions.** Patronage risk plus high operating leverage means `CFADS` falls faster than revenue, and ramp-up years dominate the covenant profile. Handback obligations are large, physical and surveyed — pavement, structures, tolling systems — which makes KA 15.4.5 the sector's defining operating-phase discipline rather than a footnote (Case study B).
- **Regulated utilities.** Regulatory reset cycles produce step-changes in allowed revenue, so covenant testing and reserve sizing must straddle resets; the operating-phase negotiation is frequently a **covenant reset timed to the determination**, and a rolling test that spans a reset measures two different businesses.
- **Digital infrastructure.** Short asset lives compress the tail, so refinancing capacity is limited and the tenor lever of KA 15.3.2 is weaker; contract renewal rather than performance is the covenant risk, and technology obsolescence makes the handback and residual-value questions genuinely open rather than formulaic.
- **Mining and resources.** Price exposure dominates and distress is cyclical rather than idiosyncratic, so restructurings are negotiated against a commodity view; the enforcement floor is unusually volatile, which widens the bargaining range and makes the sensitivity discipline of 15.4.3 essential rather than good practice.

Case study — Domain 15: the dividend the forward test cancelled (water / desalination)

Situation. Kestrel Water reached the end of its first operating year having done everything right. Availability had held at the 95.0 % guarantee, 10,000,000 m³ had been delivered, CFADS of **6,384,000** matched the base case to the dollar, and the backward DSCR of **1.2743** cleared both the 1.20× covenant and the 1.25× distribution condition. After debt service of 5,009,635.23 and the 600,000 maintenance-reserve charge, **774,364.77** sat in the proceeds account, and the board's papers showed it as the first dividend. Between the year-end and the test date, the operator revised its power-cost forecast: a supply contract had rolled onto merchant pricing, and the delivered cost of energy would rise from 0.26 to roughly 0.40 per cubic metre. Management re-forecast year two at **5,500,000** of CFADS.

What happened. The distribution condition required 1.25× on a forward-looking as well as a backward-looking basis. The re-forecast gave a forward DSCR of **1.0979**, and the dividend was not payable. The 774,364.77 went into the distribution-block account. The board's first reaction was that the test had been misapplied — the project had met every covenant — and the second was that the re-forecast should be revisited. Both reactions were understandable and the second was dangerous: the finance director's position, that a forecast prepared to stop a dividend and a forecast prepared to support one cannot be different documents, held, and the re-forecast stood. Year two delivered CFADS of **5,963,894.11 — 463,894.11 above** the forecast that cancelled the dividend, and still a DSCR of **1.1905**, a covenant breach. The sponsors cured it for **47,668.16**, the cheapest available response, consuming one of two lifetime cures. Year three was worse: fouling took availability to 88.0 %, CFADS fell to **5,202,936.48**, DSCR to **1.0386**, and the year's residual after debt service was **193,301.25** against a 600,000 reserve charge — so **406,698.75** of the maintenance top-up was drawn from the block account.

How it resolved. With one cure left and a breach requiring **808,625.80** to cure, the sponsors negotiated rather than paid: a covenant reset to 1.00× for two test dates stepping to 1.10× and 1.20×, a **40 basis point** margin uplift with a present value of **651,258.24**, an amendment fee of **92,178.38**, monthly reporting, and a committed **1,500,000** of remediation equity. The membrane replacement was brought forward from year five and cost 3,000,000 against a reserve of 1,800,000 — the **1,200,000** gap and **300,000** of reserve restoration being exactly what the injection funded. Power was hedged at 0.32 for year four and 0.30 thereafter. Year four recovered to CFADS **6,094,641.91** (DSCR 1.2166 — still below the 1.25 distribution test), and year five cleared both legs at **1.2936** backward and **1.2966** forward, releasing the whole **1,206,931.59** block-account balance alongside the year's **870,992.76** as a **2,077,924.35** distribution. Over six years equity lost **3,182,311.16** of cash — **1,682,311.16** of trading shortfall and **1,500,000** of capital call — and **2,709,838.79** of present value at 8 %.

What the domain teaches here. Three things, in order of how often they are got wrong. The project was never in danger: the debt service reserve of 2,504,817.62 was never touched, because even the worst year's cash exceeded debt service by 193,301.25. What bound was the distribution test, at 1.9103 % of CFADS against the covenant's 5.8339 % — and the sponsors had modelled the covenant. The block account was not a penalty but the pre-funding that paid for year three's reserve shortfall out of year one's cancelled dividend, which is the strongest available argument against negotiating a weaker distribution test. And the 47,668.16 cure was the expensive decision of the whole sequence: paying **7,638.87** more for a waiver would have preserved an option worth 808,625.80 nine months later, and the amendment that replaced it cost **743,436.62**.

Case study B — Domain 15: the handback nobody had surveyed (transport concession)

Situation. A thirty-year urban toll-road concession approached expiry. The concession agreement required the concessionaire to hand the asset back in a defined condition and to fund a handback reserve over the **final five years**, against an independent estimate to be prepared in **year 25**. That estimate came in at **42,000,000**, producing a contribution of **7,910,892.00** a year into a reserve earning 3.00 % — comfortably affordable against distributable cash of **21,000,000**, leaving **13,089,108.00** a year for shareholders. The estimate was prepared from design records and a visual inspection. No intrusive condition survey was commissioned, because the agreement did not require one until year 28.

What happened. The year-28 survey found the obligation was **58,000,000**, not 42,000,000 — an understatement of **16,000,000**, or **38.0952 %**. Two causes, both foreseeable: pavement rehabilitation was required over a substantially longer length than the design records implied, because a mid-life resurfacing had been thinner than specified; and the tolling system needed wholesale replacement, which the original estimate had treated as the incoming operator's cost on a reading of the agreement that the grantor did not share. By the end of year 28 the reserve held **24,451,776.08**, which would grow to **25,940,889.24** by expiry — leaving a gap of **32,059,110.76** to be funded from two remaining years.

How it resolved. The required additional contribution was **15,792,665.40** a year, taking the total reserve charge to **23,703,557.40** — **112.8741 %** of distributable cash. The final two years therefore produced no distributions at all and a shortfall of **2,703,557.40** a year, met by a sponsor injection of **5,407,114.79** in aggregate — the additional contribution of **31,585,330.80** over two years consuming the **26,178,216.00** of dividends those years would have paid and 5,407,114.79 more besides. The grantor additionally held a retention of **2,900,000** — 5 % of the works — for twelve months against defects. The concession completed and handed back on time, and the sponsor's realised return on a thirty-year asset was set, in the end, by an estimate prepared from records in year 25.

What the domain teaches here. The obligation was affordable and the timing was not. Had the 58,000,000 been known at year 20 and funded over ten years, the charge would have been **5,059,369.38** a year — **24.0922 %** of distributable cash, absorbed without comment. The **26,178,216.00** of forgone distributions and **5,407,114.79** of injected equity was therefore the price of an **information** failure, not a cash failure: a survey commissioned eight years earlier than the agreement required would have cost a fraction of one year's contribution. Two further lessons. An estimate prepared from design records is an estimate of the asset as designed, and a thirty-year asset is not the asset as designed. And a reserve is not a substitute for a credit instrument: the grantor's protection came from the retention and the sponsor's willingness to inject, not from a reserve that was funded on a schedule the agreement itself had made too late.

Executive perspective — Domain 15

What a project finance director cannot delegate in this domain:

- **The distribution conditions, in full and in cash.** Not the covenant — the test that stops the dividend, including its forward leg, its qualitative limbs and the reserve charges that rank above it. For Kestrel that is 6,262,044.04 of `CFADS`, 1.9103 % below base case, and it is the number that belongs at the top of the operating dashboard.

- **The re-forecast basis, and its owner.** A forward-looking test makes the company's own forecast the instrument that stops its own dividend. The basis, the timing, the reviewer and the reason a forecast prepared to stop a payment is the same document as one prepared to support it are the director's to establish before the first test date, not during it.
- **The reserve sizing assumption.** Kestrel's 1,500,000 capital call — 47.1 % of the six-year cost to equity — was caused by a maintenance reserve levelised on the assumption that a five-year replacement cycle runs to five years. That is a financing judgment wearing engineering clothes, and it is the director's.
- **The cure inventory.** How many cures remain, what each is worth, and the standing rule that a marginal breach is waived rather than cured. Kestrel's 47,668.16 decision cost an option worth 808,625.80, and it was taken by someone optimising a single test date.
- **The whole term sheet on any refinancing or amendment.** Margin, tenor, fees, break cost and sweep priced together as one net present value, with the breakeven computed before the first meeting. A 155 basis point saving that destroys 206,835.69 of value, and a consent whose sweep costs more than the gain it consents to, are both signed by directors who priced one line.
- **The handback obligation, from the first operating year.** Its estimate vintage, the date of the next intrusive survey, the reserve's earning rate, and the credit instrument standing behind it. It falls due after everyone who negotiated it has gone, which is precisely why it needs an owner now.

Calculation exercises — Domain 15

> **Exercise 15.1** Annual debt service is 7,200,000 and `CFADS` is 9,100,000. The facility has a 1.20× covenant, a 1.10× lock-up and a 1.30× distribution condition. The cash tax rate is 25 %, and the capacity payment abates by 120,000 per percentage point of availability. State each trigger in cash, the headroom, and the availability movement that reaches the covenant and the distribution test.

Solution. `DSCR = 9,100,000/7,200,000 =` **1.2639**. Covenant `7,200,000 × 1.20 =` **8,640,000**, headroom **460,000** (**5.0549 %** of `CFADS`); lock-up `× 1.10 =` **7,920,000**, headroom **1,180,000** (**12.9670 %**); distribution `× 1.30 =` **9,360,000**, a **shortfall of 260,000** (**−2.8571 %**) — the project already fails its distribution test while comfortably inside its covenant. Cash-to-revenue gearing is `1 − 0.25 = 0.75`, so revenue headroom to the covenant is `460,000/0.75 =` **613,333.33**, which is `÷ 120,000 =` **5.1111** availability points; the distribution test is already breached by `346,666.67` of revenue, i.e. **2.8889 points** of availability must be recovered to restore it. Common error: computing availability movement from the `CFADS` headroom directly (460,000 ÷ 120,000 = 3.83 points), which omits the tax shield and understates the tolerance by a quarter.

> **Exercise 15.2** In a period, `CFADS` is 8,450,000, debt service 7,200,000 and the required maintenance-reserve top-up 450,000. The distribution condition is 1.30× backward and forward; the forward twelve-month forecast is 7,650,000. State the distributable amount and what happens to it.

Solution. After debt service **1,250,000**; after the reserve top-up **800,000**. Backward `DSCR 8,450,000/7,200,000 =` **1.1736**; forward `7,650,000/7,200,000 =` **1.0625**. The test requires `7,200,000 × 1.30 =` **9,360,000** on both legs, so **both fail** and the **800,000 is trapped in the distribution-block account**, available for a future reserve top-up or release when the tests are next satisfied. Common error: reporting 1,250,000 as the distributable amount by omitting the reserve top-up that ranks above equity — the single most common error in operating distribution forecasts (15.2.2).

> **Exercise 15.3** A facility has 31,500,000 outstanding with 8 years remaining at an all-in 5.75 %. The market offers 4.55 % over the same 8 years. Costs: arrangement fee 1.00 %, prepayment fee 0.25 %, advisers 260,000, swap break 640,000. Discount incremental cash at 8 %. Compute the net present value and the breakeven all-in rate.

Solution. `AF(0.0575, 8) = 6.271705`, so the existing instalment is `31,500,000/6.271705 =` **5,022,557.82**. `AF(0.0455, 8) = 6.582433`, so the new instalment is **4,785,464.53** and the annual saving **237,093.28**. `AF(0.08, 8) = 5.746639`, giving a present value of **1,362,489.49**. Costs `315,000 + 78,750 + 260,000 + 640,000 =` **1,293,750**. **Net present value +68,739.49.** The breakeven all-in rate is **4.6112 %** — the margin must fall by **113.88 basis points** against the **120** on offer, leaving **6.12 basis points** of cushion. Common error: discounting the saving over the loan's full original tenor rather than its remaining 8 years, which inflates the present value and can turn a marginal transaction into an apparently comfortable one.

> **Exercise 15.4** A distressed project has 24,000,000 outstanding with 7 years remaining at 5.5 %, and revised `CFADS` of 3,600,000 flat. Lenders require 1.25× restored and will not accept a recovery below an enforcement floor of 90.0 %. Determine the extension tenor required, the haircut the coverage requirement implies, and whether a haircut is feasible.

Solution. Sustainable service `3,600,000/1.25 =` **2,880,000** against the current instalment of `24,000,000/AF(0.055, 7) = 24,000,000/5.682967 =` **4,223,146.03** (`DSCR` **0.8524** — deeply distressed). Required annuity factor `24,000,000/2,880,000 =` **8.333333**; `AF(0.055, 11) = 8.092536` (instalment 2,965,695.68, `DSCR` **1.2139** — insufficient) and `AF(0.055, 12) = 8.618518` (instalment **2,784,701.55**, `DSCR` **1.2928**), so the tenor must extend to **12 years**, five years beyond the current maturity. A haircut retaining the 7-year tenor supports only `2,880,000 × 5.682967 =` **16,366,945.30**, a haircut of **7,633,054.70** (**31.8044 %**) and a recovery of **68.1956 %** — far below the 90.0 % floor, so **infeasible**. The largest feasible haircut is `24,000,000 × 10.0 % =` **2,400,000**, which supports an instalment of **3,800,831.42** and a `DSCR` of only **0.9472**: as with Kestrel, **the haircut cannot solve the problem and an extension is unavoidable**. Common error: computing the required annuity factor and rounding the tenor down (11 years), which leaves coverage at 1.2139 and the covenant still breached — restructuring tenors round up.

> **Exercise 15.5** A 12,000,000 handback obligation falls due at the end of year 25. A reserve earns 2.5 %; equity discounts at 8 %. Compare funding over years 16–25, years 11–25 and years 6–25 on present cost, and state the rate at which the choice ceases to matter.

Solution. `FVAF(0.025, 10) = 11.203382` → **1,071,105.16** a year, present cost `1,071,105.16 × AF(0.08, 10) × DF(0.08, 15) =` **2,265,706.06**. `FVAF(0.025, 15) = 17.931927` → **669,197.47** a year, present cost **2,653,163.73**. `FVAF(0.025, 20) = 25.544658` → **469,765.54** a year, present cost **3,139,004.45**. Latest funding is cheapest; funding twenty years ahead costs **873,298.39 more** than funding ten. The choice ceases to matter when the reserve rate equals the discount rate, at which point every profile costs `12,000,000 × DF(0.08, 25) =` **1,752,214.86**. Common error: ranking the profiles by total contributions (**9,395,310.90** over twenty years against **10,711,051.58** over ten) and concluding that early funding is cheaper — a comparison of undiscounted sums across different decades, which reverses the correct answer.

Practitioner's toolkit — Domain 15

Adoption-ready artefacts; adapt headings to your organisation, then keep them stable.

Toolkit 15.T.1 — The operating bridge (one per reporting period)

A single sheet running from physical performance to certified CFADS, in fixed rows: availability and its abatement effect · delivered volume and its unit contribution · each unit cost with its exposure quantity · revenue · cash operating cost · EBITDA · depreciation · EBIT · interest · profit before tax · cash tax · CFADS before working capital · working-capital movement, decomposed into receivables, payables and inventory · CFADS **as defined**, with the clause reference. Two columns beside each row: budget and prior period. Two rules: the working-capital movement is a derived figure from balance-sheet positions and never a plug; and the sheet carries the cash-to-revenue gearing at the foot, so every variance can be read straight into covenant headroom.

Toolkit 15.T.2 — Operating covenant and distribution register (one per facility, per test date)

Per test: name and clause · test date and frequency · backward, rolling or forward-looking · window length · threshold · computed value · **the CFADS level at which it triggers** · headroom in currency and as a percentage of base case · **the same headroom in each driver's own units** · reserve balances against required levels · block-account balance and what it is earmarked for · cures used and remaining, with the value of the largest breach each remaining cure would cover · information covenants due and delivered · cross-default links to other facilities (15.A.2). Front line, in this order: **which test binds first, by how much cash, in which driver, and how many cures remain.**

Toolkit 15.T.3 — Transaction decision sheet (refinancing, amendment or restructuring)

- [ ] Incremental cash-flow set complete: existing service ceasing, new service arising, every fee, break costs on the actual notional profile, and the sweep.
- [ ] Discount rate stated, owned and sensitivity-tested; the sign of the answer confirmed at both the cost of debt and the cost of equity.
- [ ] **Breakeven computed before the first meeting** — margin, tenor, fee or sweep share, whichever is being negotiated.
- [ ] Value decomposed: how much is margin, how much tenor, how much interaction, how much covenant reset.
- [ ] For an amendment: sized against the stressed case, not the current one; the reset tested against the downside forecast at every date it must survive.
- [ ] For a restructuring: sustainable service computed; each option's restored coverage verified; the extension identity checked (par at the contract rate); recovery reported at the contract rate **and** at a risk-adjusted rate.
- [ ] Enforcement floor computed, with its sensitivity to the distressed buyer's required return shown as a range; any option below the floor removed before presentation.
- [ ] Equity value and lender recovery reported for every option, on one page (Fig 15.4.1).
- [ ] Cure inventory stated; the option value of each remaining cure quantified.
- [ ] AI-produced analysis: the breakeven re-derived independently, one period recomputed by hand, every clause reading checked against the document, verifier named.

Exam preparation — Domain 15

What is assessed. Whether a candidate can operate a financing rather than arrange one: build and reconcile an operating bridge; convert every threshold into cash and into driver units; distinguish backward, rolling and forward tests and reason about the lag; run a waterfall period by period including reserve top-ups and a block account; separate the cash cost of a drought from its timing cost; price a refinancing, a waiver, a cure, an amendment and a sweep against one another; compare restructuring options on both sides of the table and locate the enforcement floor; size a handback sinking fund; and value an exit without confusing crystallisation with performance.

The calculations to do under time pressure. Trigger in cash (debt service × threshold) and headroom, then headroom ÷ cash-to-revenue gearing to reach the driver. Distributable = CFADS − debt service − reserve top-ups. Rolling four-quarter sums from a quarterly series. Refinancing net present value = saving × AF(discount rate, remaining years) − total costs, and the breakeven rate by inspection of AF. Sustainable service = revised CFADS ÷ target coverage, then the required annuity factor and the tenor from the factor table. Haircut = outstanding − sustainable service × AF. Sinking-fund contribution = obligation ÷ FVAF.

The traps, each cross-referenced. Omitting the reserve top-up from a distribution forecast (15.2.2, Exercise 15.2) · quoting covenant headroom when the distribution test binds (15.1.3, MCQ 15.1-A) · converting cash headroom into driver units without the tax gearing (15.1.2, Exercise 15.1) · expecting a rolling backward test to register a current deterioration (15.1.4, MCQ 15.1-C) · reading a working-capital release as improved trading (15.1.2) · treating a lock-up as a write-off, or as a matter of no consequence (15.2.4) · pricing a refinancing on the margin alone (15.3.2, MCQ 15.3-A) · discounting a refinancing saving over the original rather than the remaining tenor (Exercise 15.3) · negotiating a sweep separately from the transaction it pays for (15.3.4) · spending a scarce cure on a marginal breach (15.3.3, MCQ 15.3-C) · sizing an amendment against the current case rather than the stress (15.3.3) · reporting an extension as par without a risk-adjusted recovery (15.4.2, MCQ 15.4-A) · proposing a haircut below the enforcement floor (15.4.3, MCQ 15.4-B) · rounding a restructuring tenor down (Exercise 15.4) · claiming an IRR uplift from early crystallisation as value created (15.4.4, MCQ 15.4-C) · ranking handback funding profiles on undiscounted totals (15.4.5, Exercise 15.5).

How the domain connects. Backwards: Domain 2 supplies the accrual-to-cash bridge and the CFADS definition this domain reports on; Domain 3 the amortisation schedule whose opening balances drive every break-cost and margin-uplift calculation; Domain 4 the appraisal outturn is measured against, and the IRR pathologies that reappear in the exit arithmetic of 15.4.4; Domain 6 the model that is now a contractual deliverable; Domain 7 the revenue mechanics behind the operating bridge; Domain 10 every ratio, reserve and covenant this domain operates; Domain 12 the security package that makes the enforcement floor real; Domain 13 the model audit certification depends on; Domain 14 the completion tests that start the operating clock. Forwards: Domain 16 governs the automation of KA 15.1 and 15.2 — where most organisations will actually put AI to work in finance.

Domain 15 summary

The operating phase is governed by the distribution test, not the covenant. Kestrel Water's covenant binds at CFADS of **6,011,562.28 — 372,437.72**, or 5.8339 %, below base case — but its distribution condition binds at **6,262,044.04**, only **121,955.96** or **1.9103 %** below, and that is the number that decides whether shareholders are paid. The operating bridge from availability and volume through EBITDA of 7,500,000 and cash tax of 516,000 to the defined CFADS of **6,384,000** makes the translation possible in both directions: a **0.80** cash-to-revenue gearing means every headroom figure understates the revenue tolerance by a quarter, so the covenant is reached by **6.0585** points of availability and the distribution test by **1.9839**. Rolling backward tests lag by their own window — Kestrel's covenant does not fail until 1.1905 four quarters after the decline begins, while the forward test fails at **1.0979** immediately — which is why the forward leg exists and why the re-forecast basis is a governance matter. Running the waterfall for six real years produces the domain's central demonstration: the **2,504,817.62** debt service reserve was never drawn, because even the worst year's **5,202,936.48** of CFADS exceeded debt service by 193,301.25, while the distribution test withheld four consecutive dividends and the block account rose to **1,206,931.59** before releasing into a **2,077,924.35** payment in year five. That trapped cash was not lost but deployed — the **406,698.75** maintenance-reserve shortfall of year three was funded from the dividend cancelled in year one — and the six-year cost to equity of **3,182,311.16** in cash and **2,709,838.79** in present value decomposes cleanly into **1,682,311.16** of trading shortfall and **1,500,000** of capital call, which are two different people's accountabilities. Base-case equity earns **774,364.77** a year, a **4.3020 %** cash yield rising to 32.13 % once the loan retires, for a 25-year IRR of **9.8591 %** and an NPV at 8 % of **5,027,733.03**. Mid-life, arithmetic discipline replaces enthusiasm: a **155 basis point** margin reduction destroys **206,835.69** because the breakeven reduction is **177.94** basis points, and only the tenor extension — 1,418,714.07 of margin, 586,108.78 of tenor and 420,208.04 of interaction — makes the transaction worth **+799,481.12**, which a 50 % cash sweep costing **646,327.86** then more than takes back (breakeven sweep share **40.3334 %**). A waiver at **55,307.03** and a cure at **47,668.16** differ by 7,638.87 and by an option worth **808,625.80**; an amendment at **743,436.62** and that second cure differ by **5,290.97**, which means the decision was never about price. In distress the three levers are time, principal and new money, and the enforcement floor of **91.4291 %** decides among them: the extension recovers **95.0779 %** at a risk-adjusted 7 % and leaves equity **14,898,548.64**; the injection recovers **96.3549 %** and leaves **14,072,829.99**; the haircut equity most wants recovers **82.9037 %** and is infeasible, and even the largest feasible haircut of **2,920,432.73** restores only a **1.1359** DSCR — so an extension is unavoidable. An exit at the sponsor's own discount rate raises the IRR from 9.8591 % to **11.7666 %** while the NPV stays identical at 5,027,733.03, and only the **828,232.87** of yield compression is real. And the last obligation is the first forgotten: an **8,000,000** handback funded over the final decade costs **1,476,147.78** in present value against **1,989,422.56** funded over two, because a reserve earning 3 % is equity lending to itself at a 500 basis point negative spread — an argument that disappears entirely when the reserve rate reaches the discount rate, at which every profile costs **1,168,143.24**. Domain 16 governs the automation of all of it.

16

DOMAIN 16

Data, Automation and Responsible AI in Finance (systematic AI treatment)

Group: Operating and the future (Domain 16 of 16, closing Part Four). **Target:** ~78 pages. **Binds to:** the PCI Book Pattern Specification and the shared registries (`docs/books/registries/`). Every earlier domain carried an **AI in this KA** treatment; this domain is where the treatment becomes **arithmetic**. It composes registered symbols — `CFADS`, `DSCR`, `NPV`, `AF(r, n)`, `EMV`, the hundred-per-cent rule, the mesh-versus-layer interface count `n(n−1)/2`, and PML-AI's governance latency `E[wait] = M/2 + L` — and it proposes for the registry the six automation and assurance measures it derives: **cost per reviewed item**, **automation breakeven volume**, **total misclassification cost**, **break-even posterior**, **zero-failure validation sample size** and **revalidation interval**. Where a figure was derived earlier it is **cited, not re-derived**: `CFADS` **6,384,000**, debt service **5,009,635.23**, `DSCR` **1.2743**, the 1.20× covenant biting at `CFADS` **6,011,562** and the annual headroom of **372,438** come from Domain 10 (KA 10.1–10.2); the 3,250,352 debt-capacity consequence of a one-cell tax error and the year-twelve minimum `DSCR` of 1.1851 come from Domain 6 (KA 6.4.1); the working-capital treatment worth **600,000** of `CFADS` comes from Domain 2 (KA 2.3.1); model risk as a priced register line comes from Domain 11 (KA 11.4.3). British English; USD (+SAR where useful, indicative `USD 1 ≈ SAR 3.75`). Rates used throughout, stated once: a loaded analyst hour **USD 96.00** (**USD 1.60** a minute), a senior modeller or reviewer hour **USD 150.00**, a legal-and-finance specialist hour **USD 240.00**.

Why this domain exists

Fifteen domains have each ended with a paragraph on where machine assistance earns its place and where it must not go. Those paragraphs were judgments — well-founded, but judgments. They left one question open, and it is the question a finance director actually has to answer: **at what point, measured in money, does automating a piece of the finance function stop being a slide and start being a decision that survives audit?** Nothing so far has priced an automation, chosen an alert threshold, sized a validation suite, or stated what a human approval costs and what it buys. This domain does all four, and it does them with the same decimal discipline the book applies to a debt schedule, because an automation programme that cannot be costed cannot be governed.

The domain's central claim is that **responsible AI in finance is a quantitative discipline, and the numbers usually point somewhere other than intuition.** Four results carry the chapter, and each contradicts something a practitioner is likely to believe. Automation economics turn on the **cost of the errors the automation makes**, not on the labour it displaces — so the more consequential the work, the more volume automation needs before it pays, which is the opposite of the usual argument for automating the important things first. An anomaly detector tuned to maximise accuracy is tuned wrongly whenever a false positive and a missed error cost different amounts, and the gap between the accuracy-optimal and the cost-optimal threshold is a real annual number somebody is paying. Validation by testing proves far less than it appears to: the suite an organisation can afford bounds the defect rate at a level that still admits hundreds of errors a year at production volume, which is why continuous monitoring and human approval — not pre-deployment testing — carry the assurance load. And the value of a control, like the value of a loosened threshold, is a **marginal** quantity: a blanket four-eyes rule is close to value-neutral, while the same rule above a derivable threshold is strongly positive.

Underneath all four sits the principle the suite has carried since its first page, now with an arithmetic meaning rather than a rhetorical one: **AI proposes; the professional verifies, decides and remains accountable.** Verification is not a moral posture in this domain. It is a line item whose expected return, in every worked example below, runs between **eighteen and four hundred and thirty times** its cost — and that is the finding a leader should take away, because it survives disagreement about the technology.

Learning objectives. After this domain a candidate can: describe a financial-data spine and price the architecture choice against a pairwise-reconciliation estate using the registered interface count; compute the all-in cost per reviewed item for a manual and an automated process including the cost of the errors each makes, and derive the breakeven volume, the volume at which a specific asset does not justify automation, the automated error rate at which the case reverses, and the consequence cost at which it reverses; explain why the breakeven rises with the consequence of an error; read an anomaly detector as a classifier, compute precision, recall, accuracy and total misclassification cost at several thresholds, identify the cost-minimising threshold, demonstrate that it differs from the accuracy-maximising threshold, quantify the annual cost of choosing the latter, and derive the break-even posterior probability and the marginal precision test that locates the optimum; derive the number of scenarios required to surface a failure mode of stated probability at stated confidence and explain why a reported worst case from a generated set is a percentile rather than a stress; compute the probability that a multi-item document extraction is entirely correct, derive the per-item accuracy a stated sweep confidence requires, and defend a verification scope on the load-bearing subset; compute the real and apparent saving from machine-assisted model building and price the review that is habitually omitted; test an explanation for attributional completeness using the hundred-per-cent rule; derive the zero-failure validation sample size for a stated confidence and defect-rate bound, adjust it for one observed failure, and translate a validated defect rate into expected annual production errors; identify and quantify label bias in a detector trained on a prior process's output, and size the blind audit sample that measures it; build a model inventory tiered by expected annual loss and derive each model's revalidation interval; price a managed against a private deployment and state the breach consequence at which the answer flips; derive a dual-approval threshold from the approver's cost and the loss it prevents, and compute the value destroyed by applying the control blanket-fashion; price approval latency using the registered governance-latency formula; and set out an AI governance frame — inventory, tiering, human approval, incident and rollback, disclosure — that a lender's information covenants can be satisfied against.

The master estate. Kestrel Water SPC is in operation, with the structure Domains 10 and 15 established. Its sponsor group owns **six operating assets** and runs one finance function across them, and that is the unit this domain analyses, because automation economics are a portfolio question rather than a project question. The estate produces **56,400 consumption-and-billing records a year** (Kestrel alone contributes **9,400**) that feed the revenue forecast, and **48,000 payments and journal entries a year** of which **2.5 per cent** are genuinely erroneous. Nine finance-relevant systems hold the data. A proposed financial-data spine costs **USD 900,000** to build and **USD 140,000** a year to run; a separate review-automation platform commits **USD 148,000** a year against a manual review pipeline costing **USD 13.60** a record in labour. Against that estate the domain prices every judgment the earlier fifteen made in words: Kestrel's covenant headroom of **USD 372,438** a year is the yardstick every data-quality figure is measured against, because it is the amount of cash the project can lose before a covenant fails, and a forecast error, a missed anomaly or a misread definition that exceeds it is not an operational nuisance but a credit event.

KNOWLEDGE AREA 16.1

Financial-data architecture, forecast automation and anomaly detection

Topics: 16.1.1 the financial-data spine · 16.1.2 forecast automation and the economics of a review · 16.1.3 anomaly detection as a classifier.

16.1.1 The financial-data spine

Definition. A **financial-data spine** is a single governed layer through which every system publishes and consumes financial facts, so that each fact has one authoritative representation. Four properties make it a spine rather than a warehouse. **Grain** — every table states the level at which one row is one fact (one meter reading, one invoice line, one period's `CFADS`), because most reconciliation failures are grain mismatches rather than arithmetic errors. **Golden source** — exactly one system is authoritative for each fact, named, and the others are derived. **Lineage** — every reported figure can be traced to the transactions beneath it without human reconstruction. And the property that matters most in a financing and is most often absent, the **definitional layer**: the facility's defined terms are implemented **once**, as code, with a clause reference, so that `CFADS` means in the data what it means in the document.

That last property is the finance-specific one, and it is why this topic is not an information technology digression. Domain 2 (KA 2.3.1) showed Kestrel's `CFADS` moving by **600,000** on one working-capital treatment, and Domain 10 (KA 10.1.1) showed that the same trading year therefore supports a `DSCR` of either 1.39 or 1.27. In a nine-system estate with no definitional layer, `CFADS` is implemented as many times as it is needed, each implementation drifts, and the organisation discovers the drift when two papers disagree in front of a credit committee.

WORKED EXAMPLE

Worked example 16.1.1 — pricing the architecture, then pricing the thing that actually matters.

1. **Setup.** Nine finance-relevant systems: the billing platform, the meter-data historian, the general ledger, the treasury system, the fixed-asset register, the payroll system, the procurement and payables system, the model repository and the lender-reporting pack. Today each pair that must agree is reconciled directly. Each **pairwise** reconciliation consumes **12 hours a month** (a monthly tie-out plus investigation of differences); a **spine feed** consumes **4 hours a month** (a load and an exception report). The loaded analyst hour is **USD 96.00**. The spine costs **USD 900,000** to build and **USD 140,000** a year to run. Appraise it at the board's **8 per cent** over **10 years**.
2. **Formula.** The registered interface count (PML-AI Domain 4, "mesh versus layered interfaces"): a mesh needs `n(n−1)/2` pairwise interfaces where a layer needs `n`. Annual cost = interfaces × hours a month × 12 × rate. Then `NPV = net annual saving × AF(0.08, 10) − build`.
3. **Substitution.** Mesh interfaces `9 × 8 / 2 = 36`; spine feeds `9`. Mesh `36 × 12 × 12 × 96.00`; spine labour `9 × 4 × 12 × 96.00`, plus the 140,000 run cost. `AF(0.08, 10) = 6.710081`.
4. **Result.** Mesh **5,184 hours a year**, costing **USD 497,664**. Spine **432 hours a year**, costing **USD 41,472** of labour and **USD 181,472** all-in. Net annual saving **USD 316,192**; simple payback **2.8464 years**; `NPV` **= 316,192 × 6.710081 − 900,000 = +USD 1,221,674**.
5. **Interpretation.** The case passes, but read why it passes, because the reason is the convexity rather than the level. The mesh grows as the square of the estate: adding a tenth system adds **9** pairwise reconciliations — **1,296 hours** and **USD 124,416** a year — against **48 hours** and **USD 4,608** for one more spine feed, a ratio of **27 to 1**. At fifteen systems the mesh needs **105** reconciliations and **USD 1,451,520** a year against the spine's **USD 69,120**. So the spine is not a labour saving on today's estate; it is the only architecture whose cost is linear in an estate that will grow, and the appraisal should be run at the estate size the organisation expects rather than the one it has. Two cautions, and the second is the professional one. **The saving is only realised if the pairwise reconciliations are actually retired** — estates routinely acquire a spine and keep the reconciliations, which converts a 316,192 saving into a 181,472 cost — the spine's own 41,472 of feed labour plus its 140,000 run cost, paid on top of an unchanged mesh — and that failure is a governance failure rather than a technical one. And **the labour case is not the case that matters.** A single definitional divergence in `CFADS` is worth **600,000** of reported cash (Domain 2, KA 2.3.1) — **1.6110 times** Kestrel's entire annual covenant headroom of 372,438 — and a definition implemented wrongly in code does not misreport once: it misreports every period until somebody finds it, so over the same ten-year appraisal it is worth `600,000 × AF(0.08, 10) =` **USD 4,026,049** in present value, **3.2955 times** the 1,221,674 the hours produced. The business case that gets built is the labour one; the business case that decides is the definitional one. State both, and be honest that the second is the reason.

16.1.2 Forecast automation and the economics of a review

Definitions. Forecast automation replaces a manual cycle — pull the period's consumption and billing records, screen them for plausibility, correct what is wrong, re-run the revenue forecast — with a pipeline that ingests, screens, refers exceptions to a person, and publishes. **Cost per reviewed item** is the all-in cost of processing one record, and it has two parts that must both be counted: the **processing cost** (labour, platform, inference) and the **error cost**, being the residual undetected-error rate multiplied by the consequence of an undetected error. A comparison that omits the second part is not a comparison.

WORKED EXAMPLE

Worked example 16.1.2 — the breakeven volume, and why it moves the wrong way.

1. **Setup.** The estate's consumption-and-billing pipeline. **Manual: 8.5 minutes** a record at **USD 1.60** a minute; the manual process leaves **1.80 per cent** of records with an undetected error; an undetected error costs **USD 320** (re-billing, credit notes, the downstream forecast correction and the reconciliation to the lender pack). **Automated:** committed fixed cost **USD 148,000** a year (licence, platform, model maintenance, monitoring and the governance overhead this domain later specifies); **USD 0.50** a record of platform and inference; **5.00 per cent** of records referred to a person for adjudication at **6.75 minutes** each (adjudication is faster than full review because the pipeline presents the exception with its context); residual undetected-error rate **2.60 per cent** — higher than the manual process, which is the assumption most automation cases quietly omit.
2. **Formula.** Cost per item = processing cost + (undetected-error rate × consequence). Total annual cost = fixed + volume × cost per item. Breakeven volume `V*` where the two totals are equal: `V* = fixed ÷ (manual cost per item − automated cost per item)`.
3. **Substitution.** Manual `8.5 × 1.60 = 13.60`, plus `0.018 × 320 = 5.76`. Automated `0.50 + 0.05 × (6.75 × 1.60) = 0.50 + 0.54 = 1.04`, plus `0.026 × 320 = 8.32`. Advantage `19.36 − 9.36`. Breakeven `148,000 ÷ 10.00`.
4. **Result.** Manual **USD 19.36** a record; automated **USD 9.36** a record of variable cost; advantage exactly **USD 10.00**; **breakeven 14,800 records a year**. At Kestrel's own **9,400** records the automation costs **USD 54,000 a year more** than the manual process. Across the six-asset portfolio's **56,400** records it saves **USD 416,000 a year** (manual 1,091,904 against 675,904 all-in, or **USD 11.9841** a record).
5. **Interpretation.** Three readings, in ascending order of usefulness. **First, the answer is an estate answer, not an asset answer.** The same platform is a 54,000 annual loss at one project and a 416,000 annual gain at six, and no amount of enthusiasm at the project changes that; the project's honest options are to wait for the estate, to buy a service priced per record rather than per year, or to decline. Automation cases fail at the asset level far more often than they fail technically. **Second, the error-cost term is where the case is decided, and omitting it flatters automation.** Ignore both error rates and the advantage rises to `13.60 − 1.04 = 12.56`, so the breakeven appears at **11,783 records — 3,017 records**, or **20.3822 per cent**, lower than the truth. The sensitivities are worth carrying in the head: at the portfolio's 56,400 records the case survives an automated undetected-error rate up to **4.9050 per cent** (against the 2.60 modelled, so it tolerates nearly double) and a consequence cost up to **USD 1,241.99** an error. **Third — and this is the result that changes behaviour — the breakeven rises with the consequence of an error.** Reprice an undetected error at **USD 1,200** rather than 320 and both processes get dearer, but the automated one gets dearer faster, because its error rate is the higher of the two: manual **35.20**, automated **32.24**, advantage only **2.96**, and the **breakeven moves to 50,000 records a year**. At the portfolio's 56,400 the whole programme is then worth **USD 18,944** a year — arithmetically positive, professionally indistinguishable from nil. The intuition to abandon is "automate the important work first". The defensible rule is the reverse: **automate the high-volume, low-consequence work, and spend the freed capacity on the low-volume, high-consequence work where a human error rate of 1.80 per cent beats a machine's 2.60.** The caution that belongs beside every such case: both error rates are measured quantities, and an organisation that has not measured its manual error rate has no automation case at all, only an aspiration with a spreadsheet.

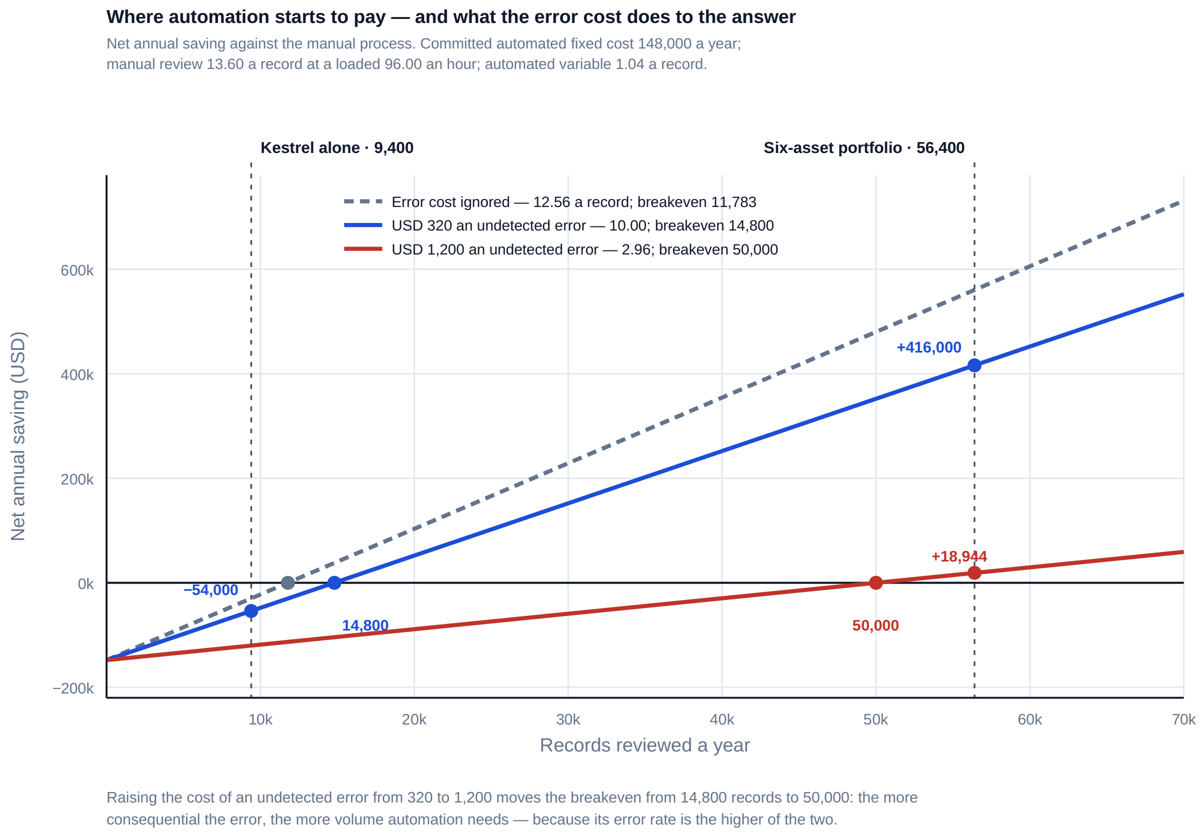

Fig 16.1.2 — Where automation starts to pay, and what the error cost does to the answer

The forecast-quality translation. The pipeline's purpose is a better forecast, and forecast quality has to be expressed in covenant terms to mean anything to a board. Measured over the estate's last three years, the manual cycle's revenue forecast carried a mean absolute percentage error of **4.8 per cent** and the automated pipeline's **3.1 per cent**. On Kestrel's `CFADS` of **6,384,000** those are error bands of **USD 306,432** and **USD 197,904 — 82.2773 per cent** and **53.1374 per cent** of the annual covenant headroom of 372,438. That is the sentence that belongs in the paper: the improvement is not "1.7 points of accuracy", it is **the difference between a forecast band that nearly consumes the covenant headroom and one that consumes half of it.** Note also what the comparison does not license — a lower average error does not mean a smaller tail, and the covenant is tested in periods (Domain 10, KA 10.2.1), so the figure to monitor is the worst period's error, not the mean.

16.1.3 Anomaly detection as a classifier

Definitions. An anomaly detector assigns each item a score and raises an **alert** above a threshold. Four counts follow: **true positives** (`TP`, real errors caught), **false positives** (`FP`, clean items investigated), **false negatives** (`FN`, real errors missed) and **true negatives** (`TN`). From them: **recall** = `TP/(TP + FN)`, the share of real errors caught; **precision** = `TP/(TP + FP)`, the share of alerts that are real; **accuracy** = `(TP + TN)/total`, the share of all items classified correctly. And the measure that should govern the choice of threshold, which none of those three is: **total misclassification cost** = `FP × cost of a false positive + FN × cost of a false negative`.

WORKED EXAMPLE

Worked example 16.1.3 — the threshold, chosen properly.

1. **Setup.** The estate's payment-and-journal detector: **48,000** items a year, of which **2.5 per cent — 1,200** are genuinely erroneous and **46,800** are clean. Investigating a false positive costs **USD 40** (25 minutes of an analyst). A missed error costs **USD 320**. Five candidate thresholds have been measured on a held-out year.
2. **Formula.** Recall, precision and accuracy as defined; total cost `= FP × 40 + FN × 320`. The decision rule for one marginal item: act when the probability the item is a real error exceeds the **break-even posterior** `= cost of a false positive ÷ cost of a false negative`.
3. **Result.**

Threshold	TP	FP	FN	Recall	Precision	Accuracy	Alerts	Total cost
T1 (score ≥ 0.90)	600	110	600	50.0 %	84.5070 %	98.5208 %	710	196,400
T2 (≥ 0.80)	720	205	480	60.0 %	77.8378 %	**98.5729 %**	925	161,800
T3 (≥ 0.70)	840	400	360	70.0 %	67.7419 %	98.4167 %	1,240	131,200
T4 (≥ 0.60)	960	950	240	80.0 %	50.2618 %	97.5208 %	1,910	**114,800**
T5 (≥ 0.45)	1,080	2,900	120	90.0 %	27.1357 %	93.7083 %	3,980	154,400

1. **Result, stated.** Total cost is minimised at **T4 — USD 114,800 a year**. Accuracy is maximised at **T2 — 98.5729 per cent**, where the total cost is **USD 161,800**. Choosing the accuracy-maximising threshold therefore costs **USD 47,000 a year**.
2. **Interpretation.** The headline is the one the spine of this domain demands: **maximising accuracy is the wrong objective whenever a false positive and a false negative cost different amounts**, and here they differ eight-fold. Accuracy weights every item equally, so it is dominated by the 46,800 clean items and rewards a detector for leaving them alone; cost weights each item by its consequence, and a missed error costs eight investigations. The 47,000 is not an abstraction — it is an annual sum the finance function is paying for a metric choice nobody minuted. Three further readings do the professional work. **The optimum looks bad on precision.** At T4 only **50.2618 per cent** of alerts are real, so the team working the queue experiences a coin flip and will lobby to tighten the threshold; the leader's job is to explain that a queue which is half false is the correct queue at these costs, and to resource the **1,910 alerts** — at 25 minutes each, about **795.83 hours** a year, or **USD 76,400** of analyst time — rather than tune the model to make the queue feel better. **The decision rule is marginal, not average.** The break-even posterior is `40 ÷ 320 = 12.5 per cent`: act on an item whenever there is at least a **one in eight** chance it is real. Average precision at every threshold in the table exceeds 12.5 per cent, including T5's 27.1357 per cent — yet T5 is worse than T4, because what matters is the quality of the **additional** alerts a looser threshold buys. From T3 to T4 the step adds 120 real errors and 550 false ones: marginal precision **17.9104 per cent**, above 12.5, benefit `120 × 320 = 38,400` against cost `550 × 40 = 22,000`, net **+16,400** — take it. From T4 to T5 the step adds 120 real and **1,950** false: marginal precision **5.7971 per cent**, benefit 38,400 against cost **78,000**, net **−39,600** — refuse it. The optimum is where marginal precision crosses the break-even posterior, and every step's net exactly reconciles to the change in the cost column, which is the arithmetic check to run on any such table. **And the two cost inputs are the real argument.** They are estimates, and the threshold is a function of their ratio alone, so the productive discussion in the room is "what does a missed error actually cost us?" rather than "which model is best". Change the ratio to 1:4 and the optimum tightens to T3; change it to 1:20 and it loosens to T5, the last

threshold measured. State the ratio, own it, and revisit it when the consequence changes — for instance when the same detector begins to screen items that feed a covenant certificate rather than a management report.

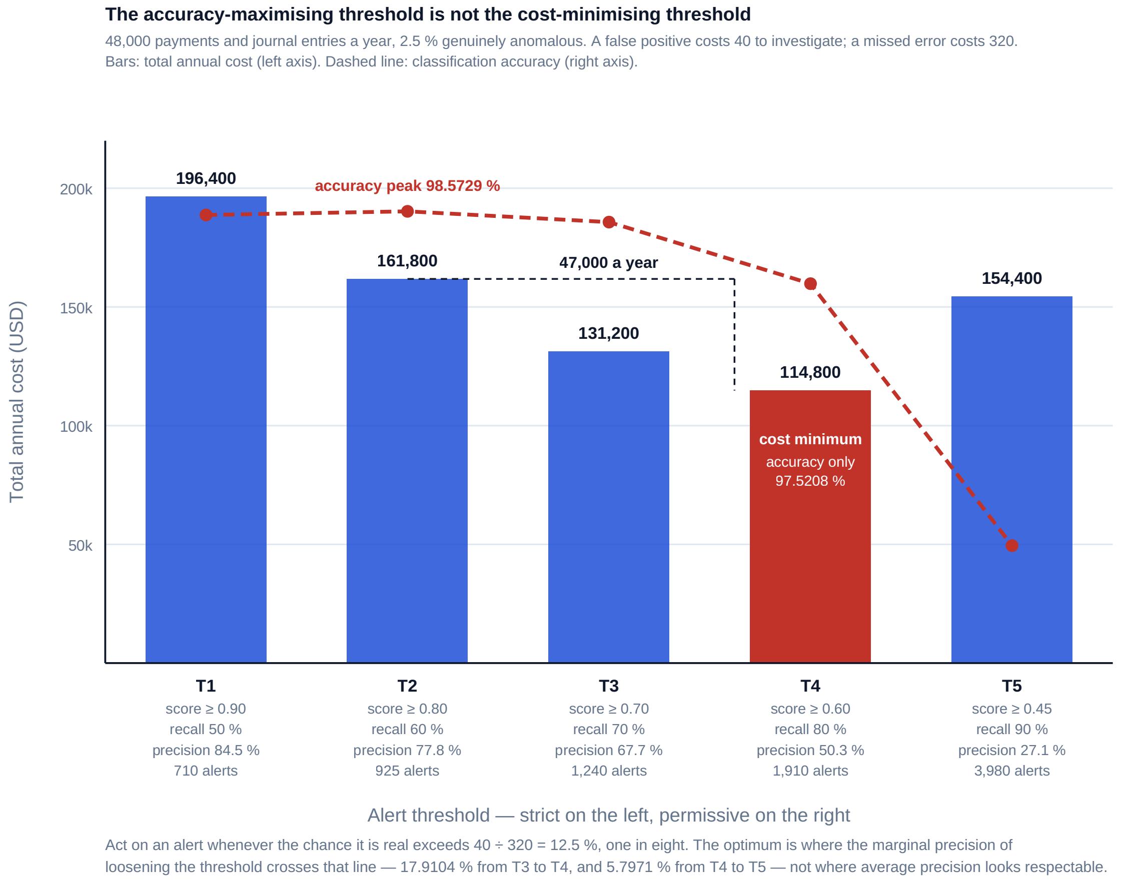

Fig 16.1.3 — The accuracy-maximising threshold is not the cost-minimising threshold

AI in this KA

Where it earns its place. Everything in this knowledge area is machine work by construction, and the highest-value application is the one organisations skip: using the pipeline's own output to **measure** the quantities the business case assumed. The manual error rate, the automated error rate, the referral rate, the adjudication time and the realised consequence of an undetected error are all observable once the pipeline is instrumented, and a business case that is re-run on measured inputs each year is a governed automation rather than a purchased one. Machine assistance also belongs in the threshold work — scoring a held-out year at fifty thresholds and tabulating cost is trivial for a machine and tedious for a person.

Where it must not go. No model may set its own operating threshold, because the threshold encodes the organisation's view of what a missed error costs, and that is a judgment with an owner. Nor may a detector's alert be closed by the same system that raised it: adjudication is the human step that converts a score into a finding, and a pipeline that both flags and clears has no control in it at all (16.4.2 prices exactly this). And no automation business case may be approved on processing costs alone — the omission of the error-cost term is the specific, nameable defect that makes automation look 20.3822 per cent better than it is.

Verification, concretely. Recompute one period's cost per item by hand from the loaded rate, the minutes and the measured error rate, and require agreement to the cent. Confirm that every threshold row's total cost reconciles to the previous row's plus the marginal benefit less the marginal cost — an arithmetic identity that catches transcription errors instantly. Confirm the prevalence used in the table against a blind audit rather than against the detector's own labels (16.3.3 shows what happens when this is skipped). Reconcile the pipeline's published forecast to the ledger for one period, line by line, and express the residual as a percentage of covenant headroom rather than of revenue. **AI proposes; the professional verifies, decides and remains accountable.**

KEY TERMS — KA 16.1

Term	Meaning
Financial-data spine	A single governed publishing layer; `n` feeds in place of `n(n−1)/2` reconciliations.
Grain	The level at which one row is one fact; most reconciliation failures are grain failures.
Golden source	The one system authoritative for a fact; all others are derived.
Definitional layer	The facility's defined terms implemented once, as code, with clause references.
Cost per reviewed item	Processing cost + (undetected-error rate × consequence of an error).
Automation breakeven volume	Committed fixed cost ÷ the per-item cost advantage.
Recall / precision / accuracy	Real errors caught ÷ real errors · real alerts ÷ alerts · all correct ÷ all items.
Total misclassification cost	`FP` × false-positive cost + `FN` × false-negative cost; the objective to minimise.
Break-even posterior	False-positive cost ÷ false-negative cost; the probability above which acting pays.
Marginal precision	Additional true positives ÷ additional alerts from loosening a threshold.

SAMPLE MCQS — KA 16.1

MCQ 16.1-A `[16.1.2 · Application]` Manual review costs 13.60 a record and leaves 1.80 % of records with an undetected error; automated processing costs 1.04 a record and leaves 2.60 %; an undetected error costs 320; the automated platform's committed fixed cost is 148,000 a year. The breakeven volume is: - A. 11,783 records a year - B. 14,800 records a year - C. 15,812 records a year - D. 142,308 records a year

Rationale: All-in costs are `13.60 + 0.018 × 320 = 19.36` and `1.04 + 0.026 × 320 = 9.36`; the advantage is 10.00 and `148,000 ÷ 10.00 = 14,800`. A omits the error-cost term from both sides (advantage 12.56, `148,000 ÷ 12.56 = 11,783`) — the specific defect of 16.1.2. C divides the fixed cost by the automated all-in cost per record (`148,000 ÷ 9.36`) instead of by the advantage, treating the comparison as an absorption problem rather than a differential one. D makes the same error against the automated processing cost alone (`148,000 ÷ 1.04`), ignoring the manual alternative entirely.

MCQ 16.1-B `[16.1.2 · Analysis]` An organisation reprices an undetected error from USD 320 to USD 1,200 on the same two processes — manual review 13.60 a record leaving 1.80 % of records with an undetected error, automated processing 1.04 a record leaving 2.60 %, on a committed fixed cost of 148,000 a year. The breakeven volume: - A. falls, because errors are now more expensive and automation catches more of them - B. is unchanged, because the error rates are unchanged - C. rises from 14,800 to 50,000, because the automated process has the higher error rate so a larger consequence erodes its advantage - D. becomes irrelevant, because at a high enough consequence neither process is acceptable

Rationale: Manual becomes `13.60 + 0.018 × 1,200 = 35.20` and automated `1.04 + 0.026 × 1,200 = 32.24`; the advantage falls from 10.00 to 2.96 and `148,000 ÷ 2.96 = 50,000`. A assumes the automation is the more accurate process, which the given rates contradict. B confuses the rates with their monetised effect. D is a governance observation, not the arithmetic asked for — and at 56,400 records the automation is still positive, by 18,944.

MCQ 16.1-C `[16.1.3 · Analysis]` A detector's five thresholds give accuracies of 98.5208, 98.5729, 98.4167, 97.5208 and 93.7083 per cent and total misclassification costs of 196,400, 161,800, 131,200, 114,800 and 154,400. The threshold that should be operated, and why: - A. the second — it maximises accuracy, which is the standard classification metric - B. the fifth — it maximises recall, and missing an error is the expensive outcome - C. the fourth — it minimises total cost, which is the only objective that reflects the different consequences of the two error types - D. the first — its precision of 84.5070 % gives the investigation team the most reliable queue

Rationale: Accuracy weights all items equally and is dominated by the 46,800 clean ones; cost weights each by consequence, and the minimum is 114,800 at T4 — choosing T2 on accuracy costs 47,000 a year. B over-corrects: T5's extra 120 catches cost 78,000 of investigation to save 38,400. D optimises the queue's comfort rather than the firm's loss.

MCQ 16.1-D `[16.1.3 · Analysis]` A false positive costs 40 and a missed error 320. Loosening the threshold from T4 to T5 would add 120 true positives and 1,950 false positives. The correct conclusion is: - A. accept — the marginal alerts still have 27.1357 % precision overall, well above the 12.5 % break-even - B. refuse — marginal precision is 5.7971 %, below the 12.5 % break-even, and the step costs 78,000 to save 38,400 - C. accept — recall rises from 80 % to 90 %, and recall is the measure lenders ask about - D. indeterminate without knowing the anomaly base rate

Rationale: The test is on the marginal alerts: `120 ÷ (120 + 1,950) = 5.7971 %`, and the net is −39,600, which reconciles exactly to the cost column rising from 114,800 to 154,400. A quotes T5's average precision, which is the standing error in threshold decisions. C treats a ratio as an objective. D is false — the counts given already embed the base rate.

SELF-CHECK — KA 16.1

1. Why does the automation breakeven rise as the consequence of an undetected error rises? — Because the automated process has the higher residual error rate, so a larger consequence penalises it more than the manual process, eroding the per-item advantage: 14,800 records at USD 320 an error, 50,000 at USD 1,200.
2. State the one-line decision rule for acting on an alert, and the test that locates the optimal threshold. — Act when the probability the item is real exceeds the false-positive cost divided by the false-negative cost (here 12.5 per cent, one in eight); the optimum is where the **marginal** precision of loosening crosses that figure, not where average precision or accuracy looks best.
3. What does a data spine actually buy in a financing, beyond hours? — One implementation of each defined term, so `CFADS` means in the data what it means in the document; the 600,000 definitional divergence of Domain 2 is 1.6110 times Kestrel's annual covenant headroom, and because a mis-implemented definition recurs every period it is worth 4,026,049 in present value over the ten-year appraisal — 3.2955 times the whole labour case.

KNOWLEDGE AREA 16.2

Scenario generation, document review and model assistance

Topics: 16.2.1 scenario generation and the coverage question · 16.2.2 document review and the load-bearing subset · 16.2.3 model assistance and the verification that must replace the build.

16.2.1 Scenario generation and the coverage question

Definition. Scenario generation is the machine production of many coherent states of the world — price paths, demand paths, availability profiles, macro sets — for evaluation through the financial model. It is genuinely valuable and genuinely easy to misreport, because a generated set answers a different question from a **defined case**. Domain 6 (KA 6.4.2) distinguished sensitivity from scenario and Domain 7 built the revenue stresses; neither is re-derived here. What is new is the question a generated set forces: **how many scenarios, and what does the worst one mean?**

WORKED EXAMPLE

Worked example 16.2.1 — coverage, and the worst case that is not a stress.

1. **Setup.** An analyst reports: "we ran 40 scenarios; three breached the covenant, and our worst case shows a `DSCR` of 1.04." A second analyst reports a 500-scenario run and quotes its minimum `DSCR` as the downside case. Assess both statements. Assume each scenario independently has a **5 per cent** probability of containing the failure mode of interest.
2. **Formula.** Probability that a set of k independent scenarios contains **at least one** instance of a mode of probability p: $1 - (1 - p)^k$. The k needed for a stated detection confidence $1 - \alpha$: $k \geq \ln(\alpha) \div \ln(1 - p)$. The expected percentile of the minimum of k draws: approximately $1 \div (k + 1)$.
3. **Substitution.** $1 - 0.95^{40}$; then $\ln(0.05) \div \ln(0.95)$; then $\ln(0.10) \div \ln(0.99)$; then $1 \div 501$.
4. **Result.** A 40-scenario set contains at least one instance of a 5 per cent mode with probability **0.871488** — so "three of our forty breached" is close to the expected outcome and carries almost no information. To be **95 per cent** confident of surfacing a 5 per cent mode requires **59** scenarios; to be **90 per cent** confident of surfacing a **1 per cent** mode requires **230**. The minimum of 500 draws sits at about the **0.1996th percentile**.
5. **Interpretation.** Two errors are being made, and they are opposite. The first analyst has **under-covered and over-interpreted**: forty scenarios cannot demonstrate the absence of a 1-in-100 mode (which needs 230 for even 90 per cent confidence), and the presence of three breaches is unsurprising rather than alarming. The second has **mislabelled a percentile as a stress**: the minimum of 500 draws is a 1-in-500 event whose value depends almost entirely on the tail of the input distributions, which is the part of a generated model nobody has validated. A defined bank case is a different object — it is a chosen set of assumptions with an owner, which is exactly why lenders insist on it (Domain 6, KA 6.4.1b: the flat case is demanded not because anyone believes it but because it is the case that finds the defect). The professional discipline follows in three parts. **Report k with every scenario statement**, and report the share of scenarios that breach rather than the count, because a count without k is meaningless. **Choose k from the smallest probability you need to see**, not from a run time — $\ln(\alpha) \div \ln(1 - p)$ takes ten seconds and settles the argument. And **never let a generated minimum replace a defined case in a credit paper**; present both, labelled, with the generated set describing shape and the defined case carrying the covenant test. The standing caution: the independence this arithmetic assumes is usually false. Scenarios drawn from one correlated generator explore less of the space than their count suggests, so k computed this way is a **floor**, and Domain 11 (KA 11.4.1) already priced what a correlation assumption is worth when it is wrong.

16.2.2 Document review and the load-bearing subset

Definition. Machine document review extracts structured facts — defined terms, thresholds, test dates, notice periods, caps — from long agreements. On a project financing this is among the most valuable applications available, because the documents are long, the terms are numerous and the consequences of missing one are quantified elsewhere in this book. It is also the application with the most misleading performance statistics, and the reason is combinatorial.

WORKED EXAMPLE

Worked example 16.2.2 — why 92 per cent per item is worthless, and what to do instead.

1. **Setup.** A facility agreement contains **340** defined terms. **26** of them are load-bearing: they feed the model, a covenant test or a reported ratio. An extraction tool is measured at **92 per cent** per-item accuracy. The question a reviewer must answer is not "how accurate is the tool?" but "what is the probability the deliverable is right?"
2. **Formula.** For `m` independent items each correct with probability `a`, the probability that **all** are correct is `aᵐ`. The per-item accuracy required for a stated sweep confidence `c` is `a ≥ c^(1/m)`.
3. **Substitution.** `0.92²⁶`; then `0.95^(1/26)`.
4. **Result.** The probability that all 26 load-bearing definitions are extracted correctly is **0.114415 —** so there is an **88.5585 per cent** chance that at least one is wrong. Across all 340 terms the expected number of errors is **27.2**. To be **95 per cent** confident of a clean sweep of 26 items, per-item accuracy must reach **99.8029 per cent**.
5. **Interpretation.** This is the single most important arithmetic in the domain for anyone tempted to rely on an extraction, and it generalises: **per-item accuracy is the wrong statistic whenever the deliverable requires every item to be right.** A tool reported as "92 per cent accurate" delivers a correct 26-item summary about one time in nine. And the required 99.8029 per cent is not a procurement target — no honest vendor will offer it, and no buyer should believe it — so the resolution is not a better model. It is a **change of scope**: do not require the machine to be right, require the load-bearing subset to be **verified**. Price that. Twenty-six definitions at half an hour each of a legal-and-finance specialist at USD 240.00 an hour is **USD 3,120**. Set it against one known consequence: Domain 2's working-capital treatment of `CFADS` is worth **600,000** of reported cash, which is **1.6110 times** Kestrel's annual covenant headroom of 372,438, and the verification is **192.3077 times** cheaper than that single error. The professional posture that follows is precise rather than pious. Let the tool read all 340 terms and produce the register — that is real and large value, because a human reading 340 terms will also miss some and will take days. Then **verify the 26 against the document, by a named person, with the date recorded**, and treat the other 314 as indicative until one of them becomes load-bearing. Two cautions. **The subset must be chosen before the tool runs**, or the tool's own confidence scores will choose it, and a model is least confident where it is wrong in ways it can detect — not where it is wrong in ways it cannot. And **superseded drafts are the commonest failure mode in practice**, not misreading: an extraction from the wrong version is 100 per cent accurate against the wrong document, which no accuracy statistic can see. Case study B is that failure, priced.

16.2.3 Model assistance and the verification that must replace the build

Definition. Model assistance is machine help in constructing a financial model — formula generation, schedule construction, restructuring a workbook, writing the check block, translating a term sheet into a debt module. Domain 6 built the model and its six invariants; this topic asks only what changes when the model is built with assistance, and the answer is that **the verification burden moves and grows.**

The mechanism is worth stating precisely, because it is usually described as a productivity gain and it is really a redistribution. A modeller who builds a schedule by hand acquires, as a by-product, a mental model of how it works — where the circularity sits, which cell drives which, what a wrong answer would look like. A reviewer of a machine-built schedule has none of that, so review must reconstruct it, and reconstruction is slower than construction was for the person who did it. The saving is real; it is just much smaller than the build-time comparison suggests.

WORKED EXAMPLE

Worked example 16.2.3 — the real saving, the apparent saving, and the price of the difference.

1. **Setup.** A sculpted debt module. **Hand-built:** 40 hours to build, 8 hours to review. **Machine-assisted:** 6 hours to produce, and — because the reviewer must reconstruct the construction logic — **26** hours to review properly. A senior modeller or reviewer costs **USD 150.00** an hour. The organisation's observed record is that an unreviewed machine-assisted module carries a material defect with probability **0.35**; Domain 6 (KA 6.4.1) priced one such defect — a one-cell tax error — at **USD 3,250,352** of lost debt capacity.
2. **Formula.** Cost = hours × rate. `EMV` = probability × consequence (the registered expected-monetary-value form, PML-AI KA 8.2).
3. **Substitution.** Hand `(40 + 8) × 150`; machine with full review `(6 + 26) × 150`; machine with the habitual 8-hour review `(6 + 8) × 150`. `EMV = 0.35 × 3,250,352`.
4. **Result.** Hand-built **USD 7,200**; machine-assisted with proper review **USD 4,800 — a real saving of USD 2,400, or 33.3333 per cent**. Reviewed at the habitual 8 hours the cost is **USD 2,100**, an **apparent saving of 70.8333 per cent**. The omitted 18 hours of review are worth **USD 2,700**. The expected cost of the defect they would have caught is **USD 1,137,623.20 — 421.3419 times** the review.
5. **Interpretation.** The number to carry into a management conversation is **a third, not two thirds**. An organisation that books the 70.8333 per cent has not made a saving; it has converted USD 2,700 of certain cost into USD 1,137,623.20 of expected cost, and it has done so invisibly, because the omission leaves no artefact. Three consequences follow. **Fund the review explicitly, as a line, at the ratio the organisation has measured** — here roughly four hours of review per hour of assisted build, against one per five hand-built; the ratio is the governable quantity and it should be in the model-governance standard, not left to the reviewer's diary. **Require the assistant to produce the audit trail, not just the model**: a written statement of what each block does, the invariants it should satisfy, and the check block itself, which is work a machine does well and which cuts reconstruction time — this is the single most effective way to make the 26 hours smaller honestly. **And keep the accountability where Domain 6 put it**: the modeller who ships the module owns its defects regardless of who typed the formulae, and the organisation that treats "the tool built it" as mitigation has no model governance. The caution worth stating plainly: the 0.35 defect probability is this organisation's measured figure on its own work, not a general property of assisted modelling, and a firm that has not measured its own should assume a figure high enough to make the review unarguable and then measure.

AI in this KA

Where it earns its place. All three topics are the application, so the useful question is what machine assistance adds to their **governance**, and the answer is measurement and coverage. Machines are good at computing what a scenario set covers, at diffing an extraction against a previous version of the same agreement (which catches the superseded-draft failure directly), and at generating the invariant tests a reviewer would otherwise write by hand. A second, genuinely strong use: having one model **critique** another's output — an adversarial pass over an extraction register or a generated scenario set, asked to find the internally inconsistent entries — because inconsistency detection is cheap for a machine and does not require the machine to be right about the substance.

Where it must not go. No machine may choose the load-bearing subset (16.2.2), decide `k` (16.2.1) or sign a model off (16.2.3), because each of those is a scoping decision that determines what will not be checked. Nor may a generated scenario minimum be presented as a bank case, or a critique pass be treated as verification: an adversarial model that finds nothing has not established that there is nothing, and treating a silent critique as assurance is the most seductive failure in this knowledge area.

Verification, concretely. Recompute a^m for the actual subset size before relying on any extraction summary — one line of arithmetic that reframes the whole procurement. Diff the extraction against the executed document's defined-terms clause, not against a data room copy, and record the version identifier. For scenario work, publish `k`, the share of scenarios breaching, and the defined case separately. For assisted models, run Domain 6's six invariants plus the effective-tax-rate check before any output is quoted, and record the review hours actually spent against the standard's ratio — an unfunded review is the defect this knowledge area exists to prevent. **AI proposes; the professional verifies, decides and remains accountable.**

KEY TERMS — KA 16.2

Term	Meaning
Scenario coverage	$1 - (1 - p)^k$: the chance `k` scenarios contain a mode of probability `p`.
Required `k`	`ln(α) ÷ ln(1 − p)`: scenarios needed to surface a mode at `1 − α` confidence.
Generated minimum	The worst of `k` draws; a percentile (≈ `1/(k+1)`), not a defined stress case.
Load-bearing subset	The extracted items that feed a model, covenant or reported ratio; the verification scope.
Sweep probability	a^m: the chance an `m`-item extraction is entirely correct at per-item accuracy `a`.
Reconstruction cost	The review time a machine-built artefact needs because no one holds its construction logic.
Superseded-draft error	Extraction from the wrong version; 100 per cent accurate against the wrong document.

SAMPLE MCQS — KA 16.2

MCQ 16.2-A `[16.2.2 · Application]` A tool extracts defined terms at 92 % per-item accuracy. Twenty-six of them are load-bearing. The probability that all 26 are correct is closest to: - A. 92.00 % - B. 11.44 % - C. 88.56 % - D. 99.80 %

Rationale: $0.92^{26} = 0.114415$. A mistakes per-item accuracy for deliverable accuracy — the error the topic exists to correct. C is the complement, the probability that at least one is wrong. D is the per-item accuracy that would be required for a 95 % clean sweep, `0.95^(1/26)`.

MCQ 16.2-B `[16.2.2 · Analysis]` Given that result, the defensible professional response is: - A. reject machine extraction and read all 340 terms manually - B. procure a tool with 99.8029 % per-item accuracy - C. use the tool for all 340 terms, then verify the 26 load-bearing terms against the executed document by a named person on a recorded date — about USD 3,120 against a single known 600,000 exposure - D. accept the register and rely on the tool's confidence scores to flag which entries to check

Rationale: The remedy is a change of scope, not of tool: verification of the load-bearing subset is 192.3077 times cheaper than one 600,000 definitional error. A discards the tool's genuine value on the other 314 terms. B specifies an unachievable target. D lets the model choose its own verification scope, and a model is least confident where it can detect its own error — not where it cannot.

MCQ 16.2-C `[16.2.1 · Analysis]` An analyst reports "40 scenarios run, 3 breached, worst-case `DSCR` 1.04". The soundest critique is: - A. 40 is too few to demonstrate the absence of a low-probability mode — 230 are needed for 90 % confidence on a 1 % mode — and the worst of 40 draws is a percentile, not a defined stress case - B. 3 breaches out of 40 is a 7.5 % breach probability, which should be reported as the covenant risk - C. the run is adequate; scenario counts above 30 are conventionally sufficient - D. the analyst should have reported the mean `DSCR` across the 40 scenarios

Rationale: Coverage and labelling are the two defects: `ln(0.10) ÷ ln(0.99) = 230`, and a generated minimum is roughly the `1/(k+1)` percentile of an unvalidated tail. B treats a sample frequency from an unvalidated generator as a probability. C invents a convention. D reports the statistic covenants never test (Domain 10: the minimum, not the average).

MCQ 16.2-D `[16.2.3 · Application]` A module takes 40 hours to build and 8 to review by hand, or 6 hours assisted with 26 hours of review. At USD 150 an hour, the saving from assistance is: - A. 70.8333 % - B. 33.3333 % - C. 85.0000 % - D. nil, since total hours are similar

Rationale: `(48 − 32) × 150 = 2,400` on 7,200, or 33.3333 %. A is the apparent saving when the review is left at the hand-built 8 hours — the omission worth 2,700 of review against 1,137,623.20 of expected defect cost. C compares build hours only (6 against 40). D ignores the 16-hour difference.

SELF-CHECK — KA 16.2

1. Why is per-item extraction accuracy the wrong statistic? — Because the deliverable needs every load-bearing item right: 92 per cent per item gives 0.114415 for 26 items, and a 95 per cent sweep would need 99.8029 per cent per item.
2. How is k chosen for a scenario run? — From the smallest probability the run must be able to surface: `k ≥ ln(α) ÷ ln(1 − p)` — 59 for a 5 per cent mode at 95 per cent confidence, 230 for a 1 per cent mode at 90 per cent — and the result is a floor, because correlated generators explore less than their count implies.
3. What does machine assistance do to a model's total cost? — It moves cost from build to review and reduces the total by about a third, not two thirds; booking the larger figure converts USD 2,700 of review into USD 1,137,623.20 of expected defect cost.

KNOWLEDGE AREA 16.3

Explainability, validation, bias and model risk

Topics: 16.3.1 explainability as an accounting identity · 16.3.2 validation — how many test cases, honestly derived · 16.3.3 bias in a finance model · 16.3.4 model-risk governance: inventory, tiering, revalidation.

16.3.1 Explainability as an accounting identity

Definition. Explainability in a finance context is not a philosophical property of a model; it is the ability to state, for a specific output, **how much of it came from what** — in the units of the output, adding up. A global statement about which inputs matter in general is interpretability and is useful for design. What a decision, an audit or a lender's question requires is a **local, additive attribution** of this period's number.

The test is therefore one the book already owns. PML-AI's **hundred-per-cent rule** — Σ children − parent = 0 at every level — applies unchanged: an explanation whose components do not reconcile to the output is not an explanation, it is a commentary. Suppose an automated forecast publishes `CFADS` of **6,384,000** and attributes it across drivers — volume, tariff, escalation, cash operating cost, cash tax, working-capital movement — summing to **6,190,000**. The residual is **USD 194,000**: **3.0388 per cent** of the forecast, and **52.0892 per cent** of Kestrel's annual covenant headroom of 372,438. The professional reading is exact: **the model has an unexplained component larger than half the project's covenant headroom**, so any conclusion drawn from the attribution about which driver to manage is unsafe, and the correct response is to find the residual, not to describe it as model complexity. Three further disciplines belong here. **Attribution must be in currency, not in importance scores** — a ranking cannot be reconciled and therefore cannot be audited. **The counterfactual must be stated**: "volume contributed 1.2m" means nothing without "relative to what", and the baseline is a choice with an owner. And **explanation is not justification**: a faithful account of how a model reached a number says nothing about whether the number is right, which is 16.3.2's job. Where a model cannot be given an additive local attribution at all, the honest options are to use it only for triage and never for a reported figure, or to replace it with one that can — and for a covenant-relevant number the second is usually the right answer.

16.3.2 Validation — how many test cases, honestly derived

Definition. Validation is the evidence that a model performs as claimed on cases it has not seen. The question a governance committee always asks and rarely gets answered is how many test cases are enough, and it has a derivable answer under stated assumptions.

WORKED EXAMPLE

Worked example 16.3.2 — the sample size, and the assurance it does not give.

1. **Setup.** A covenant-certificate assembler must be validated before deployment. The committee wants to be **95 per cent confident** that its defect rate is **below 1 per cent**. Test cases are independent and representative of production; all of them pass. Then: what changes if one case fails; and what does the validated bound imply at a production volume of **56,400** uses a year?
2. **Formula.** If the true defect rate were p, the probability that n independent tests all pass is $(1 - p)^n$. Requiring that probability to be at most α gives $n \geq \ln(\alpha) \div \ln(1 - p)$. With one failure permitted, the condition becomes $(1 - p)^n + n \cdot p \cdot (1 - p)^{n-1} \leq \alpha$, solved numerically. Expected annual production errors = $p \times \text{uses}$.
3. **Substitution.** $\ln(0.05) \div \ln(0.99)$; then the two-term condition at $p = 0.01$; then $0.01 \times 56{,}400$; then $\ln(0.05) \div \ln(1 - 10/56{,}400)$.
4. **Result.** $n \geq 298.0729$**, so 299 test cases**, all passing. The sample-size ladder, at 95 per cent confidence: **59** cases bound the rate below 5 per cent, **149** below 2 per cent, **299** below 1 per cent, **598** below 0.5 per cent, **2,995** below 0.1 per cent. Raising confidence to 99 per cent at the 1 per cent bound requires **459**. Permitting **one** failure at the 1 per cent bound and 95 per cent confidence requires **473**. And a model validated to "below 1 per cent" and used 56,400 times a year is consistent with **564 errors a year**; to bound expected production errors at **10** a year the defect rate must be below **0.017730 per cent**, which requires **16,895** passing test cases.
5. **Interpretation.** The ladder is the useful artefact — it converts an unbounded argument into a choice — but the honest content of this example is the last line, and it is uncomfortable. **The test suite an organisation can afford proves far less than the assurance it wants.** Nobody is going to construct 16,895 representative test cases for a covenant-certificate assembler; 299 is already a serious programme. So pre-deployment testing cannot be the control that makes a high-volume model safe, and an organisation that behaves as though it were has mis-sited its assurance. The load must be carried by three things instead: **continuous monitoring** of the production error rate, which accumulates evidence at production volume rather than at test volume and reaches 16,895 observations in **about four months** (16,895 ÷ 56,400 of a year, 109 days), against the years a test programme would need; **human approval** at the points where consequence is concentrated (KA 16.4.3, and the tiering of 16.3.4, which is how one decides where); and **rollback**, because the realistic response to a discovered defect is to stop, not to have prevented it. Four cautions on the arithmetic itself, all of which a reviewer should press. The formula assumes **independence** — 299 near-duplicate cases are not 299 tests, and the commonest inflation of a validation claim is a suite generated from one template. It assumes **representativeness** — a suite drawn from historical cases cannot bound the error rate on the cases the model will actually meet, which is 16.3.3's problem. It gives a **one-sided bound on a binary outcome**, so it says nothing about the size of a defect: 299 passes are consistent with a rare but catastrophic failure, which is why tiering is by consequence and not by rate. And a single failure moves the requirement from 299 to **473** — a **58.19 per cent** increase — which is the arithmetic reason "we fixed it and re-ran the failing case" is not a validation: the suite must grow, not be repaired.

16.3.3 Bias in a finance model

Definition. Bias, in this domain, is a systematic difference between what a model was trained or calibrated on and what it is used on. It is a measurable property, not a moral one, and in project finance it takes four concrete forms worth naming. **Training-period bias** — a demand model calibrated on a period whose conditions no longer hold, which Domain 7's demand work must guard against. **Survivorship bias** — a comparables set of completed, financed, surviving projects, which systematically overstates achievable performance because the failures are absent. **Label bias** — a detector trained on the errors a previous process happened to find, so it inherits that process's blind spots. And **attribute bias with legal exposure** — a counterparty- or location-derived score that operates as a proxy for a protected characteristic; what is lawful here differs materially between jurisdictions and this book does not state any jurisdiction's position, but the professional obligation is constant: know which attributes the model uses, know what they proxy for, and take legal advice before a score of this kind affects access to credit or employment.

Label bias is the one that bites hardest in an automated finance function, because it is invisible in every metric the model reports.

WORKED EXAMPLE

Worked example 16.3.3 — the recall that was never 80 per cent.

1. **Setup.** The detector of 16.1.3, operating at T4: **48,000** items, **1,200** assumed anomalies, **960** caught, **950** false positives, reported recall **80 per cent**, reported total cost **USD 114,800**. The labels it was trained on came from the previous manual process. A **blind audit** is run over the **46,090** items the detector did not flag: **1,000** sampled, **17** genuine errors found.
2. **Formula.** Extrapolated missed errors = unflagged population × sample error rate. True population = caught + extrapolated. True recall = caught ÷ true population. Restated cost = `FP × 40 + true FN × 320`. Audit sample size for a `±e` bound at `1 − α` confidence on a proportion `p`: `n ≥ z²·p·(1 − p) ÷ e²`.
3. **Substitution.** `17 ÷ 1,000 = 1.70 per cent`; `46,090 × 0.0170`; then `960 ÷ 1,744`; then `950 × 40 + 784 × 320`; and for the sample size `1.96² × 0.025 × 0.975 ÷ 0.01²`.
4. **Result.** Extrapolated missed errors **784**, against the 240 the model's own arithmetic assumed. True anomaly population **1,744**, **45.3333 per cent** above the assumed 1,200. True recall **55.0459 per cent**, not 80. Restated total cost **USD 288,880** against the reported 114,800 — a factor of **2.5164** — with the false-negative cost understated by **USD 174,080**. The audit that revealed this cost **USD 13,600** — 1,000 items re-reviewed at 8.5 minutes each, **USD 13.60** at the loaded analyst rate, a desk re-review rather than the 25-minute investigation an alert requires — and identified missed errors worth **USD 250,880** a year, a return of **18.4471 times**. A sample of **937** would have sufficed for a ±1 percentage-point bound at 95 per cent confidence on a 2.5 per cent rate.
5. **Interpretation.** The mechanism deserves stating slowly, because it is the most under-appreciated failure in automated controls. **A detector trained on a prior process's findings learns to reproduce that process, including what it could not see.** Its measured recall is recall against the labels, and if the labels are 60 per cent complete then a bounding estimate of true recall is `0.80 × 0.60 = 48 per cent` — the audit's 55.0459 per cent sits between that bound and the claim, which is what one should expect. Every metric in 16.1.3's table was computed correctly and every one was measuring the wrong population. Three consequences. **No threshold change fixes this.** If the 784 missed errors are precisely the kinds the old process could not detect, loosening the threshold does not reach them: scaling the T4-to-T5 marginal benefit by the 1.4533 population factor gives **USD 55,808** against an unchanged USD 78,000 of investigation cost, so T4 remains optimal and the answer is a **model change** — new features, and the audit's 17 confirmed errors as a new label source — not a tuning change. **The blind audit is the only instrument that measures this, and it must be blind**: a sample drawn from items the model scored highly, or reviewed by the people who set the labels, reproduces the bias. Budget it as a permanent line, sized by `z²p(1 − p)/e²` for the precision the decision needs, and re-run it annually. **And restate the business case afterwards.** The automation's claimed saving in 16.1.2 was computed on assumed error rates; the audit is how those become measured, and a case that is never restated is a case that was never governed. The standing caution: extrapolating 17 findings to 784 carries real sampling error — the 95 per cent interval around a 1.70 per cent rate on 1,000 draws is wide enough that the point estimate should be reported with its bound, and the decision should be robust to the low end of it.

16.3.4 Model-risk governance: inventory, tiering, revalidation

Definitions. Model risk is, on the shared registry's definition, the risk of loss from decisions based on flawed, misused or misunderstood models — financial or AI. Domain 11 (KA 11.4.3) established it as a priced line in the operating risk register and made the point that its probability is a function of governance rather than of nature; that analysis is not repeated. What this topic adds is the **governance instrument**: a **model inventory** (every model in use, with its purpose, owner, inputs, outputs, dependencies and the decisions it touches), a **tier** for each, and a **revalidation interval** derived from the tier rather than set by convention.

Tiering by "materiality" is the usual practice and it is unfalsifiable. Tiering by **expected annual loss** is derivable: `EMV` per year = uses a year × defect probability × consequence per defect, on the registered expected-monetary-value form. Set a tolerance for the loss an organisation is willing to accumulate between validations, and the interval follows: `revalidation interval = tolerance ÷ annual EMV`.

WORKED EXAMPLE

Worked example 16.3.4 — the inventory that sets its own calendar.

1. **Setup.** Five models in the estate's finance function, with the uses, defect probabilities and consequences measured or estimated as shown. The tolerance for accumulated expected loss between validations is **USD 50,000** per model. Consequences are drawn from figures already established: Kestrel's annual covenant headroom **372,438** (a covenant-forecast error consumes it), the forecast band **306,432** at the manual error level (16.1.2), the **600,000** definitional exposure (Domain 2), a **325,000** waiver fee (Case study B) and **320** per undetected payment error (16.1.3).
2. **Formula.** `annual EMV = uses × p × consequence`; `interval = 50,000 ÷ annual EMV`.
3. **Result.**

Model	Uses a year	p	Consequence (USD)	Annual EMV	Interval	Policy outcome
M1 Covenant-forecast model	12	0.030	372,438	134,077.68	0.372918 yr = **136.12 d**	Quarterly
M2 Payment anomaly detector	48,000	0.001	320	15,360.00	3.255208 yr = 1,188.15 d	Annual (policy floor)
M3 Consumption forecaster	12	0.050	306,432	183,859.20	0.271947 yr = **99.26 d**	Quarterly
M4 Defined-term extractor	4	0.885585	600,000	2,125,404.00	0.023525 yr = **8.59 d**	**Verify every use**
M5 Covenant-certificate assembler	4	0.020	325,000	26,000.00	1.923077 yr = 701.92 d	Annual (policy floor)

1. **Interpretation.** The instrument's value is that it produces a **calendar an auditor can challenge on its inputs** rather than a tier label nobody can argue with. Read three results. **M4's interval of 8.59 days is shorter than the interval between its uses** — four facilities a year, roughly 91 days apart — which means the rule's own arithmetic says the model must be verified at **every** use. That is exactly the conclusion 16.2.2 reached from the combinatorics of a 26-item sweep, arrived at independently from a governance rule, and the agreement of two derivations is

the strongest evidence in this domain that the conclusion is right. **The high-volume model is the low-tier one.** M2 touches 48,000 items but each defect costs 320, so its expected annual loss is the smallest in the inventory and annual revalidation is generous — while M1 and M3, used twelve times a year each, need quarterly attention because one bad output consumes a covenant headroom. The instinct to govern by volume is precisely inverted. **And the policy floor matters as much as the formula.** M2 and M5 compute to intervals above a year, and the standard should nonetheless cap the interval at twelve months, because the formula prices known failure modes and a year of undetected drift is the unpriced one. Two cautions. The tolerance of 50,000 is a policy parameter with an owner, and halving it halves every interval, so it belongs in the standard and not in a spreadsheet. And each `p` is an estimate that the monitoring of 16.3.2 should be replacing with a measurement — the inventory is a living instrument, and a model whose `p` has never been measured should carry the pessimistic figure until it has.

AI in this KA

Where it earns its place. Assurance work has its own automatable core, and it is worth taking: generating a validation suite's boundary and adversarial cases from a specification; monitoring production error rates and raising a drift alert; computing the inventory's `EMV` and interval columns from the monitoring feed so the calendar updates itself; producing the additive attribution of 16.3.1 for every published figure; and diffing a model's behaviour between versions so a change that alters an output nobody expected to change is surfaced before release rather than after.

Where it must not go. A model may not validate itself, in any of the several forms this takes: generating its own test cases from its own training distribution, scoring its own explanations for plausibility, or setting the `p` in its own tier. It may not decide the tolerance, the tier or the interval, which are risk-appetite decisions. And an explanation must never be **generated as narrative** — a fluent paragraph describing why a model produced a figure, unconstrained by the additive reconciliation of 16.3.1, is the most dangerous artefact in this knowledge area, because it is persuasive precisely where it is unverifiable.

Verification, concretely. Reconcile every attribution to its output and require the residual to be nil or explained, in currency (16.3.1). Recompute the validation sample size from the stated bound and confidence, and check the suite for independence by asking how many distinct templates generated it. Confirm the audit sample of 16.3.3 was drawn from unflagged items and reviewed by people outside the labelling process. Recompute one row of the inventory's `EMV` and interval by hand. And require that every `p` in the inventory names its source — measured, estimated, or inherited — because an inventory of estimates presented as measurements is worse than no inventory. **AI proposes; the professional verifies, decides and remains accountable.**

KEY TERMS — KA 16.3

Term	Meaning
Local additive attribution	An explanation of one output in the output's own units, reconciling to it.
Attribution residual	Output − Σ attributions; must be nil or explained (the hundred-per-cent rule).
Zero-failure sample size	`n ≥ ln(α) ÷ ln(1 − p)`: tests needed to bound a defect rate at `1 − α` confidence.
Label bias	A detector trained on a prior process's findings inherits its blind spots.
Blind audit	A sample of unflagged items reviewed independently, to measure what the model cannot see.
Model inventory	Every model in use, with purpose, owner, dependencies, decisions touched and tier.
Revalidation interval	Tolerance for accumulated expected loss ÷ annual `EMV`, capped by policy.

SAMPLE MCQS — KA 16.3

MCQ 16.3-A `[16.3.2 · Application]` How many independent, representative test cases, all passing, are needed to be 95 % confident that a model's defect rate is below 1 %? - A. 100 - B. 299 - C. 95 - D. 459

Rationale: `n ≥ ln(0.05) ÷ ln(0.99) = 298.0729`, so 299 — the "rule of three" at `3/p`. A is the reciprocal of the bound, which gives only a 63 % confidence. C confuses the confidence level with a count. D is the requirement at 99 % confidence, not 95 %.

MCQ 16.3-B `[16.3.2 · Analysis]` That model passes its 299 cases and is then used 56,400 times a year. The correct statement is: - A. the validation shows the model will produce fewer than 10 errors a year - B. the validation is consistent with up to about 564 errors a year, so monitoring, human approval at concentrated-consequence points and rollback must carry the assurance - C. the validation is invalid because the test volume is smaller than the production volume - D. the validation guarantees a 99 % success rate in production

Rationale: A 1 % bound at 56,400 uses admits `0.01 × 56,400 = 564` errors; bounding expected errors at 10 would need 16,895 passing cases. A inverts the bound. C misstates the requirement — test volume need not match production volume, it must be independent and representative. D reads a one-sided confidence bound as a point guarantee.

MCQ 16.3-C `[16.3.3 · Analysis]` A detector reports 80 % recall at a total cost of 114,800. A blind audit of 1,000 of the 46,090 unflagged items finds 17 genuine errors. The most important consequence is: - A. the threshold should be loosened until recall reaches 90 % - B. the true anomaly population is about 1,744 rather than 1,200, true recall is 55.0459 %, and the restated cost is 288,880 — the missed errors are the kinds the labels never contained, so the answer is a model change, not a threshold change - C. the audit sample is too small to act on - D. the detector should be withdrawn

Rationale: Extrapolating 1.70 % over 46,090 gives 784 missed errors; recall and cost restate to 55.0459 % and 288,880. Loosening (A) does not reach errors the model cannot see — scaling the T4→T5 benefit by 1.4533 gives 55,808 against 78,000 of cost, so T4 remains optimal. C is wrong: 937 would suffice for a ±1-point bound at 95 % confidence. D discards a control that still avoids substantial loss.

MCQ 16.3-D `[16.3.4 · Application]` A model is used 4 times a year, fails with probability 0.885585 per use, and a failure costs 600,000. With a tolerance of 50,000 of accumulated expected loss between validations, the revalidation interval is: - A. annual, since four uses a year is low volume - B. about 8.59 days — shorter than the interval between uses, so the model must be verified at every use - C. quarterly, matching the use frequency - D. about 3.26 years, since the annual `EMV` is small

Rationale: `EMV = 4 × 0.885585 × 600,000 = 2,125,404`; `50,000 ÷ 2,125,404 = 0.023525` years. A and C tier by volume or convenience rather than by expected loss — the inversion the topic exists to correct. D is the interval for the payment detector (M2), whose `EMV` is 15,360.

SELF-CHECK — KA 16.3

1. What makes an explanation auditable? — Local additive attribution in the output's own units, reconciling to the output; a 194,000 residual on a 6,384,000 forecast is 52.0892 per cent of covenant headroom and disqualifies any conclusion drawn from the attribution.
2. Why can pre-deployment testing not make a high-volume model safe? — 299 passing cases bound the defect rate below 1 per cent, which at 56,400 uses still admits 564 errors a year; bounding at 10 would need 16,895 cases, so monitoring, human approval and rollback carry the load.
3. What does a blind audit measure that no model metric can? — The errors absent from the labels: 1,000 unflagged items yielding 17 findings restated the population from 1,200 to 1,744, recall from 80 to 55.0459 per cent, and cost from 114,800 to 288,880.

KNOWLEDGE AREA 16.4

Privacy, cybersecurity, human approval and AI governance

Topics: 16.4.1 confidentiality, privacy and the deployment choice · 16.4.2 cybersecurity of an automated finance function · 16.4.3 human approval — what it is, and what it costs · 16.4.4 the AI governance frame.

16.4.1 Confidentiality, privacy and the deployment choice

Definitions. A project financing generates three distinguishable classes of sensitive data, and conflating them produces bad controls. **Commercially confidential information** — the financial model, the price deck, the bank case, draft documents, the offtaker's consumption profile — is protected by contract, and its disclosure damages a live negotiation immediately and irreversibly. **Personal data** — employee, contractor and sometimes customer information — is protected by law that differs materially between jurisdictions; this book does not state any jurisdiction's requirements, and the professional obligation is to establish which regimes apply to each data set before it is processed, and to take advice. **Statutorily or contractually restricted information** — price-sensitive information about a listed sponsor, or data whose location is restricted by a licence or a concession — carries its own consequences independent of the other two.

Two practical rules cut across all three. **Minimisation beats protection**: the surest control on a diligence data set is that the personal data was never loaded, and redaction at ingestion is cheaper than every control downstream. And **retention by a processor is the risk that gets missed** — whether a service retains inputs, for how long, whether they train on them, and in which jurisdiction they rest are contractual questions to settle before use, not afterwards.

WORKED EXAMPLE

Worked example 16.4.1 — the deployment choice, and the number at which it flips.

1. **Setup.** The automation platform of 16.1.2 can run as a **managed service** at **USD 148,000** a year or as a **private deployment** inside the sponsor's own environment at **USD 410,000** a year. The security function assesses the annual probability of a confidentiality breach involving model or deal data at **0.80 per cent** on the managed service and **0.15 per cent** privately. The consequence of such a breach during a live financing — re-pricing, remedial legal and communications cost, and the relationship damage Domain 10 (KA 10.4.4) priced qualitatively — is assessed at **USD 6,500,000**.
2. **Formula.** `EMV` avoided = probability differential × consequence. Compare with the cost differential. Breakeven consequence = cost differential ÷ probability differential.
3. **Substitution.** `(0.008 − 0.0015) × 6,500,000`; `410,000 − 148,000`; `262,000 ÷ 0.0065`.
4. **Result.** `EMV` avoided **USD 42,250** a year against a cost differential of **USD 262,000** — so on these figures the private deployment is **not** justified, by USD 219,750 a year. The breakeven consequence is **USD 40,307,692**.
5. **Interpretation.** The value of this calculation is not its answer but the sentence it forces: **state the consequence at which your choice flips, then argue about whether the consequence is really that large.** USD 40.3 million is a large number for a routine operating pipeline, which is why the managed service is defensible for one — and it is not an implausible number for the information set of a live acquisition, a refinancing at scale or a competitive bid, where the loss is the transaction rather than the data. So the professionally correct architecture is usually neither of the two options as posed: **segment the data**, run the high-volume operating pipeline on the managed service, and keep the live-transaction information set — model, price deck, bank case, draft documents — inside the private environment, where the volume is low and the 262,000 differential applies to a fraction of the work. Three cautions. The probabilities are **assessments, not measurements**, and no organisation has enough breach history to measure them, so the arithmetic's honest role is to structure the judgment rather than to settle it — present it as a breakeven, never as an `NPV`. The consequence is **not fully monetisable**: a breach that costs a licence, or a regulatory consequence in one jurisdiction, is not on this scale at all, and where that is possible the calculation should not be run — the answer is the private deployment regardless of arithmetic. And a private deployment is **not automatically safer**: it moves the risk from a specialist provider's controls to the organisation's own, and an under-resourced private environment can carry the higher probability, which is a real inversion and should be tested rather than assumed.

16.4.2 Cybersecurity of an automated finance function

Definition. Domain 11 (KA 11.4.1) established that cyber risk on an industrial project is primarily an **availability** risk on the operational-technology network and therefore a `CFADS` risk with the same shape as any outage. This topic addresses the different exposure created by automating the **finance** function, which is not availability but **integrity and authority**: the risk that money moves, or a figure is reported, on an instruction nobody competent gave.

Three attack surfaces are specific to an automated finance function and none of them existed in a manual one. **Instruction injection through ingested content**: an assistant that reads supplier invoices, emails or documents will read whatever is in them, including text crafted to look like instructions, so any pipeline that both reads external content and can act must treat ingested text as data and never as direction. **Label and data poisoning**: an adversary who can influence what a detector learns — by shaping which items get labelled, or by operating just below a known threshold — degrades the control quietly, and 16.3.3's blind audit is the only routine instrument that would notice. **Authority accumulation**: automation tends to concentrate permissions, because it is easier to give a pipeline broad access than to scope it, and the result is a single identity that can read the ledger, initiate a payment and update the record of what it did.

The control that answers the third is separation of duties, and it should be sized rather than declared.

WORKED EXAMPLE

Worked example 16.4.2 — the dual-approval threshold, derived.

1. **Setup.** The estate makes **4,800** payments a year totalling **USD 68,000,000**. A second approver costs **15 minutes** at **USD 1.60** a minute — **USD 24.00** a payment. **0.20 per cent** of payments are erroneous or fraudulent, and a second approval catches **85 per cent** of those, so **0.17 per cent** of payment value is protected per payment reviewed. **1,150** payments, worth **USD 63,400,000**, exceed the threshold derived below; the remaining **3,650**, worth **USD 4,600,000**, fall under it.
2. **Formula.** Expected loss avoided on one payment = caught rate × payment value. The control pays where that exceeds the approver's cost, so the threshold is `V* = approver cost ÷ caught rate`. Net value of a policy = `caught rate × value reviewed − approver cost × payments reviewed`.
3. **Substitution.** `24.00 ÷ 0.0017`; then the two policies on the value and count above and below.
4. **Result. Threshold `V*` = USD 14,117.65**, so dual approval pays on any payment above about **USD 14,118**. **Blanket dual approval on all 4,800 payments** costs **USD 115,200** and avoids **USD 115,600** — a net value of **USD 400 a year**. **Thresholded dual approval on the 1,150 payments above 14,118** costs **USD 27,600** and avoids **USD 107,780** — a net value of **USD 80,180**. The 3,650 small payments cost **USD 87,600** to approve and avoid **USD 7,820**, destroying **USD 79,780**, which reconciles exactly to the difference between the two policies.
5. **Interpretation.** The blanket rule — the rule almost every finance function actually has — is **value-neutral to three significant figures**, and it achieves that by combining a strongly positive control on large payments with a strongly negative one on small ones. That is the general lesson of this domain restated in a third setting: **the value of a control, like the value of a looser alert threshold and the value of a verification step, is a marginal quantity, and averaging destroys the information.** Four practical consequences. **Derive the threshold and put it in the policy**, with its two inputs visible — the approver's cost and the caught rate — because both are arguable and neither is usually written down. **Spend the released capacity on the exposure that has no control**: 87,600 a year of approver time bought 7,820 of protection, and the same money spent on the blind audit of 16.3.3 bought a 18.4471-times return. **Do not read this as an argument against controls on small payments** — a sampling control on the sub-threshold population costs a fraction of blanket approval and preserves deterrence, which the arithmetic above does not model and which matters, because a published threshold is an instruction to an adversary to stay below it. That last point is the professional caution, and it is why the threshold should be reviewed against the observed distribution of attempts, not set once. **And the caught rate is the input to defend**: at 85 per cent the control is strong; a second approver who signs without looking has a caught rate near nil, at which point the whole 115,200 is waste and the arithmetic says so.

16.4.3 Human approval — what it is, and what it costs

Definition. Human approval is a named person taking responsibility for an output before it has an effect. It has three levels that must not be confused. **Review** — a person examines the output and can stop it. **Approval** — a person is accountable for it having proceeded. **Certification** — a person makes a representation to a third party, with the consequences that attach; Domain 14's treatment of draw-request certifications stands, and nothing in an automated pipeline may generate a representation.

Approval architecture is usually discussed as though it were free, and it is not. PML-AI's registered governance-latency formula gives the price directly: the expected wait for a committee decision is `E[wait] = M/2 + L`, for a meeting interval `M` and a paper lead time `L`.

WORKED EXAMPLE

Worked example 16.4.3 — what an approval gate costs while it waits.

1. **Setup.** A change to the automation pipeline of 16.1.2 would deliver the **USD 416,000** a year of saving established there. Two approval designs: a **quarterly AI governance committee** (`M` = 60 days of effective interval between usable decision slots, `L` = 10 days of paper lead time) and a **weekly delegated panel** with a defined mandate (`M` = 7 days, `L` = 2 days).
2. **Formula.** `E[wait] = M/2 + L`. Value forgone = `E[wait] × annual benefit ÷ 365`.
3. **Substitution.** `60/2 + 10 = 40` days; `7/2 + 2 = 5.5` days; `416,000 ÷ 365 = 1,139.7260` a day.
4. **Result.** The committee design forgoes **USD 45,589.04** of value per change; the delegated panel forgoes **USD 6,268.49**. The difference — **USD 39,320.55 per change** — is the price of the governance design, paid whether or not the committee changes the decision.
5. **Interpretation.** The point is emphatically **not** that approval should be removed. It is that **approval must be tiered by consequence, because the tier is where the cost sits**, and an organisation that routes every model change to one quarterly committee is paying roughly 39,000 a change for assurance it does not need on most of them while giving inadequate attention to the few that matter. The design this domain endorses has three tiers derived from the instruments already built. Changes to a model in the **top tier** of 16.3.4's inventory — M1 and M3, where one bad output consumes a covenant headroom — go to the committee, and the 45,589 is well spent because the exposure is 372,438 a year of headroom. Changes to **threshold parameters** go to a delegated panel with a written mandate, because 16.1.3 showed the decision is a cost-ratio judgment that a small standing group can take competently and repeatedly. Changes that alter **what is reported to a lender** go nowhere near either: they are an information-covenant matter (Domain 10, KA 10.4.1) and require the finance director's own decision. Two cautions worth holding. **Latency is not the only cost of a gate** — PML-AI's gate-net-value form prices the defect the gate catches against the delay it imposes, and a gate whose catch rate is nil is pure cost — so measure what the committee actually changes. And **the delegated mandate must be written**, naming which parameters may move, within what bounds, and what triggers escalation, because an undocumented delegation is not a tier, it is an absence of control that will be described as one after an incident.

16.4.4 The AI governance frame

Definition. AI governance in a finance function is the set of arrangements that makes the use of models decidable, reversible and attributable. Stripped of framework language, it answers three questions for every model: **what is it for**, **who verifies its output before it has an effect**, and **who is accountable for the decision it informs**. A programme that cannot answer those three for every entry in its inventory does not have governance, whatever documentation it holds.

Six components carry the weight, and each has been derived rather than asserted in this domain. The **inventory and tiering** of 16.3.4, with intervals that follow from expected loss and a policy floor. The **validation and monitoring** regime of 16.3.2, with monitoring sized to carry the assurance that testing cannot. The **human-approval architecture** of 16.4.3, tiered so that its latency cost is spent where consequence is. The **audit trail**: for every published figure, the model version, the input data version, the attribution (16.3.1), and the named verifier with a date — which is what makes an output defensible a year later, when the person has moved on. **Incident response and rollback**, including the number nobody computes: 16.A.2 prices a quarter's suspension of the estate's pipeline at **USD 141,000**, and a rollback plan without that figure has not been tested. And **disclosure**, which is the component most often forgotten in a financing: where a model materially affects a reported figure or a covenant certificate, whether and how that is disclosed to lenders is an information-covenant question, and a sponsor who is asked about it for the first time during a waiver negotiation has lost the initiative Domain 10 (KA 10.4.4) identified as the most valuable asset in a covenant relationship.

On external frameworks, accurately. Several published frameworks are useful reference points and are named here in the book's own words, with no text reproduced. **ISO/IEC 42001** specifies requirements for an artificial-intelligence management system, in the same management-system idiom as ISO/IEC 27001 for information security, and is therefore the natural home for the inventory, tiering and approval arrangements above. **ISO/IEC 23894** offers guidance on managing risk in artificial-intelligence systems. The **NIST AI Risk Management Framework**, published by the United States National Institute of Standards and Technology, is a voluntary, function-based framework widely used as a structuring device. The **European Union's Artificial Intelligence Act** is a risk-tiered regulatory regime whose obligations depend on how a system is classified and on the role a party plays in relation to it, applying in phases; whether and how it reaches a particular project company is a legal question that depends on facts and jurisdiction, and it is not answered in this book. In addition, banking supervisors in several jurisdictions publish model-risk management expectations for regulated lenders. A project company is not usually within their scope — but **its lenders are**, and the expectation therefore propagates to the borrower through diligence questionnaires and information covenants, which is why a sponsor with a defensible model inventory has a commercial advantage and not merely a tidy file. None of these frameworks is endorsed by the Institute, none is reproduced, and none substitutes for advice on the law applicable to a specific project.

AI in this KA

Where it earns its place. Governance work has real automatable substance: maintaining the inventory from the deployment pipeline so that no model is in production without an entry; monitoring for models in use that are not in the inventory, which is the commonest governance failure and is detectable from access logs; assembling the audit trail automatically at publication time, because an audit trail assembled later is an assertion; tracking approvals against mandates and flagging the ones that exceeded them; and drafting the disclosure language a lender pack needs, for a human to settle.

Where it must not go. No model may approve, certify or disclose. No model may set its own tier, tolerance or mandate. And no model may write the governance framework it will be governed by — a recursion that sounds absurd and is nonetheless the most likely way an organisation ends up with a policy that reads well, cites the right frameworks, and imposes no constraint anyone has costed. The test to apply to any such document is the one this domain has applied throughout: **which number in it would change a decision?** A governance document with no numbers governs nothing.

Verification, concretely. Reconcile the inventory against production access logs and require the difference to be nil. Sample five published figures and attempt to reproduce each from the recorded model version, data version and attribution alone; a figure that cannot be reproduced from its own trail is not auditable. Test the rollback by running it, and record the cost. Confirm that every approval in the last quarter was within a written mandate, and that the exceptions were escalated. And confirm that the disclosure position on model-derived figures has been stated to lenders in a document, not in a conversation. **AI proposes; the professional verifies, decides and remains accountable.**

KEY TERMS — KA 16.4

Term	Meaning
Minimisation	Not loading the sensitive data; the cheapest and strongest confidentiality control.
Breakeven consequence	Cost differential ÷ probability differential; the loss at which a deployment choice flips.
Instruction injection	Ingested content crafted to be read as direction; treat all ingested text as data.
Label poisoning	Influencing what a detector learns; detectable only by blind audit.
Authority accumulation	One automated identity able to read, act and record; the separation-of-duties failure.
Dual-approval threshold	Approver cost ÷ caught rate; the value above which a second approval pays.
Review / approval / certification	Can stop it · accountable for it · represents it to a third party.
Governance latency	`E[wait] = M/2 + L`; the value forgone while a gate waits.

SAMPLE MCQS — KA 16.4

MCQ 16.4-A `[16.4.2 · Application]` A second approver costs USD 24 a payment; 0.20 % of payments are bad and a second approval catches 85 % of those. The payment value above which dual approval pays is closest to: - A. USD 12,000 - B. USD 14,118 - C. USD 120,000 - D. USD 80,000

Rationale: Caught rate `0.0020 × 0.85 = 0.0017`; `24.00 ÷ 0.0017 = 14,117.65`. A omits the catch rate and divides by 0.0020. C misplaces a decimal in the bad-payment rate, using 0.02 % rather than 0.20 % (`24.00 ÷ 0.0002`). D uses the **15 %** a second approval misses instead of the 85 % it catches (`24.00 ÷ 0.0003`) — the sign-of-the-control error.

MCQ 16.4-B `[16.4.2 · Analysis]` On the same figures, 4,800 payments a year total 68,000,000, of which 1,150 payments worth 63,400,000 exceed the threshold. Comparing blanket dual approval with thresholded dual approval: - A. blanket is better, because it catches more bad payments in total - B. they are equivalent, because the same catch rate applies - C. thresholded is worth 80,180 a year against blanket's 400, because the 3,650 small payments cost 87,600 to approve and protect only 7,820 - D. neither is worthwhile, since the blanket policy nets only 400

Rationale: Blanket costs 115,200 and avoids 115,600; thresholded costs 27,600 and avoids 107,780. A is true and irrelevant — the extra catches cost more than they save. B ignores that the value protected varies with payment size while the approver's cost does not. D draws the wrong conclusion from the blanket result: the control is strongly positive where it is targeted.

MCQ 16.4-C `[16.4.1 · Analysis]` A managed service costs 148,000 a year with an assessed 0.80 % annual breach probability; a private deployment costs 410,000 at 0.15 %. The most defensible use of the arithmetic is: - A. compute the `EMV` avoided of 42,250, conclude the managed service wins, and close the question - B. state the breakeven consequence of 40,307,692 and then debate whether the information at risk is worth that — which segments the answer: operating pipeline managed, live-transaction data private - C. choose the private deployment, because confidentiality cannot be quantified - D. choose the managed service, because the probability differential is only 0.65 percentage points

Rationale: The arithmetic's role is to locate the argument, not to end it: 262,000 ÷ 0.0065 = 40,307,692, a figure implausible for a routine pipeline and entirely plausible for a live transaction. A over-reads an assessment as a measurement. C abandons a usable structure. D confuses a small differential with a small consequence.

MCQ 16.4-D `[16.4.3 · Application]` A change worth 416,000 a year waits for a committee with a 60-day effective interval and a 10-day paper lead time. The value forgone while it waits is closest to: - A. USD 6,268 - B. USD 45,589 - C. USD 68,384 - D. nil — the benefit accrues once approved

Rationale: `E[wait] = 60/2 + 10 = 40` days at `416,000 ÷ 365 = 1,139.7260` a day. A is the delegated panel's 5.5-day wait. C prices the full 60-day meeting interval in place of `M/2 + L`, dropping both the expected-half rule and the paper lead time. D ignores that a delayed benefit is a forgone benefit.

SELF-CHECK — KA 16.4

1. How is a dual-approval threshold derived, and what does a blanket rule do? — Approver cost ÷ caught rate: USD 24.00 ÷ 0.0017 = USD 14,118. A blanket rule nets USD 400 a year by combining a strongly positive control above the threshold with an USD 79,780 loss below it.
2. What does the deployment breakeven of USD 40,307,692 tell a director? — That the managed service is defensible for a routine operating pipeline and indefensible for a live-transaction information set, so the answer is to segment the data rather than to choose once.
3. What are the three levels of human involvement, and which may never be automated? — Review (can stop it), approval (accountable for it), certification (represents it to a third party). Certification may never be generated by a model; a pipeline produces a draft for a person to verify and sign.

Advanced topics — Domain 16

16.A.1 Agentic automation and the authorisation boundary

An **agent** differs from an assistant in one respect that changes every calculation in this domain: it can **act**, not merely propose. The temptation is obvious — the 16.4.2 pipeline already identifies which payments are sound, so allowing it to release them removes the approver's cost entirely. Price that. Removing the thresholded human approval saves **USD 27,600** of approver time and adds **USD 107,780** of expected loss, a net destruction of **USD 80,180** a year, which is precisely the value of the control. Removing approval altogether takes expected loss from **USD 20,400** (0.20 per cent of value, 15 per cent of it uncaught) to **USD 136,000**. The inversion is exact and general: **an agent that replaces a control does not inherit the control's effect, it inherits its absence**, and the saving it books is the control's cost while the loss it creates is the control's benefit.

Three design rules follow, each with an arithmetic basis in this domain. **Scope authority by consequence, using the threshold that already exists**: an agent may release below the 14,118 threshold, where human approval was destroying value anyway, and may only prepare above it. **Keep the recording separate from the acting** — an identity that can both move money and amend the record of having moved it defeats reconstruction, and 16.4.4's audit trail is the control being defeated. And **treat every ingested document as data**: an agent with release authority that reads supplier correspondence has an instruction-injection surface directly onto the payment run, which is the single most consequential new exposure in an automated finance function. The honest summary is that agentic automation is defensible exactly where the human control was uneconomic, and nowhere else — which is a narrower claim than the technology's advocates make and a broader one than its sceptics allow.

16.A.2 The automation you cannot switch off

Every automation case in this domain is a comparison with a manual process, and the comparison assumes the manual process still exists. It does not, after about a year. The **fallback capacity** question is therefore part of the case, and it is a staffing lead-time question rather than a cost question. Restoring manual review over the estate's **56,400** records at **8.5 minutes** each requires **7,990 hours** a year — **4.99375 full-time equivalents** at 1,600 productive hours — and that capability cannot be assembled inside a reporting cycle, whatever the budget.

The number that belongs in the rollback plan is the cost of a **suspension**, because the fixed cost is committed while the manual cost returns. Suspending the pipeline for one quarter means paying the quarter's committed fixed cost of **USD 37,000** and reviewing 14,100 records manually at 19.36 all-in — **USD 272,976** — for a total of **USD 309,976**, against the automated quarter's **USD 168,976**: an incremental **USD 141,000** for one quarter's suspension. Two consequences. **Rollback is a funded option, not a right**, and a plan that does not name its cost has not been tested — the 141,000 should sit in the incident procedure beside the technical steps. And **capability decay should be resisted deliberately**: retaining each pipeline's manual procedure as documented, exercised work — a blind sample reviewed manually every period, which the audit discipline of 16.3.3 already requires and prices — keeps the fallback real at almost no marginal cost. That is the rare control that satisfies two purposes at once, and it is the strongest argument for the audit line surviving its first budget round.

16.A.3 The reviewer's automation eye

Invariants to test on any automation, model or governance claim in a finance function:

- Every automation business case carries an **error-cost line for both processes**, and both error rates are measured rather than assumed; omitting them flatters automation by 20.3822 per cent on 16.1.2's figures.
- The **breakeven volume** is stated, and the case is assessed at the volume the entity actually has — 9,400 records is a 54,000 loss where 56,400 is a 416,000 gain.
- The breakeven is **recomputed at the consequence cost the work actually carries**; a case built at 320 an error does not transfer to work worth 1,200 an error.
- Any threshold is justified by **total misclassification cost**, never by accuracy; and the optimum is confirmed by **marginal** precision against the break-even posterior (false-positive cost ÷ false-negative cost).
- Each threshold row's total cost **reconciles** to the adjacent row plus marginal benefit less marginal cost.

- The **prevalence** underlying any precision or recall figure comes from a **blind audit**, not from the model's own labels; recall against incomplete labels overstates itself (80 per cent claimed, 55.0459 per cent measured).
- Scenario statements carry `k` and the **share** breaching, and no generated minimum is presented as a defined case.
- Extraction reliance is assessed on `aᵐ` **for the load-bearing subset**, and that subset is chosen **before** the tool runs, verified against the **executed** document, with the version identifier recorded.
- Every published figure has a **local additive attribution** that reconciles to it, with a nil or explained residual, in currency.
- Every validation claim states its **confidence and bound**, and the sample size reconciles to `ln(α) ÷ ln(1 − p)`; the suite's **independence** is evidenced; and the claim is translated into **expected annual production errors**.
- Every model in production has an **inventory entry**, a tier derived from `uses × p × consequence`, a revalidation interval capped by policy, and a named owner — and the inventory reconciles to production access logs.
- Every approval sits within a **written mandate**; certifications are never machine-generated; and the **latency cost** of the gate is known.
- The **rollback cost** is computed and the rollback has been exercised.

Industry variations — Domain 16

- **Water and regulated utilities.** Telemetry is dense and consumption data is regulated, so the data spine's value is high and the deployment question of 16.4.1 is constrained by licence conditions on data location rather than by the arithmetic. Regulatory reset cycles break the training period, so the training-period bias of 16.3.3 is structural: a consumption model calibrated across a reset is calibrated on two different worlds.
- **Contracted power and availability payments.** Revenue is formulaic, so forecast automation is the easiest win in the estate and the anomaly detector's prevalence is low — which raises the false-positive burden at any recall, because a low base rate means most alerts are false however good the model. Threshold economics dominate, and the 16.1.3 arithmetic is the whole discussion.
- **Transport concessions.** Patronage data is high-volume, granular and genuinely non-stationary, so 56,400-record-scale automation pays easily while forecast automation is the hardest case in the book: the model that fitted last year's ramp is the model that will mislead through the next disruption, and validation must be on out-of-period data or it proves nothing.
- **Digital infrastructure.** Contract terms are numerous, bespoke and frequently amended, which makes document review the highest-value application and the superseded-draft failure of 16.2.2 the highest-probability one — Case study B is drawn from this sector for that reason. Tenant data is commercially sensitive third-party data, so minimisation is a contractual obligation, not a preference.
- **Mining and resources.** Price-deck scenario generation is central and the sign-change appraisal pathologies of Domain 4 make generated-scenario reporting especially treacherous; a generated minimum across a commodity path distribution is a statement about the tail of an assumed price process, and should be labelled as such.

- **Public-sector and development-finance-supported projects.** Explainability is not optional: a figure that affects a public payment must be reconstructable by an auditor from its own trail, so 16.3.1's additive attribution and 16.4.4's audit trail move from good practice to condition of funding, and models that cannot support them are simply unusable however accurate.

Case study — Domain 16: the automation that lost at one asset and won at six (water)

Situation. Kestrel Water's sponsor group approved an automation platform for its consumption-and-billing review on a business case built at the Kestrel project, where the finance team had the appetite and the data. The case compared **USD 13.60** a record of manual review with **USD 1.04** a record of automated processing and a committed **USD 148,000** a year, and showed a saving. It carried no error-cost line for either process.

What happened. Three things, in sequence, each worth a number.

The volume error. At Kestrel's **9,400** records a year the platform cost **USD 235,984** against the manual process's **USD 181,984** — **USD 54,000 a year more**, not less. Restated properly, with the manual undetected-error rate measured at **1.80 per cent** and the automated at **2.60 per cent** against a consequence of **USD 320**, the all-in costs are **19.36** and **9.36** a record and the breakeven is **14,800 records a year**. The original case, omitting error costs, had implied a breakeven of **11,783** — **20.3822 per cent** flattering — and even that was above Kestrel's volume. The programme was rescoped to the six-asset estate's **56,400** records, where it saves **USD 416,000 a year**. The rescoping was the right answer arrived at for the wrong reason: nobody had computed a breakeven at all.

The threshold error. The payment-and-journal detector deployed alongside it was tuned, on the vendor's recommendation, to maximise accuracy: **98.5729 per cent** at threshold T2, with **925** alerts a year and a total misclassification cost of **USD 161,800**. Recomputed on the estate's own costs — **USD 40** to investigate a false positive, **USD 320** for a missed error — the cost-minimising threshold is **T4**, at **USD 114,800**, with **1,910** alerts and precision of only **50.2618 per cent**. Moving the threshold saved **USD 47,000 a year** and was resisted for two quarters by the team working the queue, whose objection — that half the alerts were now false — was factually correct and economically irrelevant, since the break-even posterior is `40 ÷ 320 = 12.5 per cent`.

The label-bias discovery. In the second year the finance director commissioned a **blind audit** of the items the detector had not flagged: **1,000** of **46,090**, reviewed independently of the labelling team at 8.5 minutes an item, a cost of **USD 13,600**. It found **17** genuine errors — a **1.70 per cent** rate, extrapolating to **784** missed errors a year against the **240** the detector's own arithmetic assumed. The true anomaly population was therefore about **1,744**, not 1,200 — **45.3333 per cent** higher — the detector's true recall was **55.0459 per cent**, not 80, and the restated total cost at T4 was **USD 288,880**, a factor of **2.5164** on the reported figure. The false-negative cost had been understated by **USD 174,080** a year.

How it resolved. The estate adopted four changes, each traceable to one of the numbers above. The business case is now restated annually on measured error rates rather than assumed ones. The threshold is a delegated-panel decision within a written mandate, reviewed when either cost input moves, rather than a vendor default. The blind audit became a permanent annual line, sized at **1,000** records (above the **937** a ±1-percentage-point bound at 95 per cent confidence requires), which returned **18.4471 times** its cost in its first year; the same discipline was extended as a periodic manual sample across the billing pipeline, which is how the fallback capacity of 16.A.2 is kept exercised. And the detector was **retrained** rather than retuned, using the audit's confirmed findings as a new label source, because scaling the T4-to-T5 marginal benefit by the 1.4533 population factor gives **USD 55,808** against an unchanged **USD 78,000** of investigation cost — no threshold reaches errors the labels never contained.

What the domain teaches here. Three failures, one cause. The volume error, the threshold error and the label bias were all failures to compute something computable, and each was defended with a qualitative argument — the platform is strategic, accuracy is the standard metric, the model reports 80 per cent recall. **Every one of those arguments dissolved on contact with arithmetic that took under an hour.** The programme's total annual improvement from arithmetic alone, before any technology changed, was **USD 470,000** of rescoping and **USD 47,000** of retuning, against a restated loss estimate that was **USD 174,080** worse than believed — and the honesty of the third number is what made the first two credible to the board.

Case study B — Domain 16: the definition read from the wrong draft (digital infrastructure)

Situation. A hyperscale data-centre financing carried **USD 130,000,000** of senior debt with annual debt service of **USD 17,000,000**, a **1.25×** DSCR covenant and a **1.20×** distribution lock-up. The borrower's finance team used a document-extraction tool to build its covenant register from the facility agreement's **340** defined terms, and reported the first annual compliance certificate on that register.

What happened. The register's CFADS definition had been extracted from a **late negotiation draft** rather than the executed agreement. In the executed version, **capitalised maintenance** was struck **above** CFADS; in the draft it sat below. The reported CFADS was **USD 22,780,000** and the reported DSCR **1.3400** — comfortably inside covenant. On the executed definition, CFADS was **USD 20,230,000**, a difference of **USD 2,550,000**, and the DSCR was **1.1900**.

The consequences followed mechanically. The **1.25× covenant** required CFADS of **USD 21,250,000**, so the project was short by **USD 1,020,000** and in breach from the first test date. The **1.20× lock-up** trigger of **USD 20,400,000** was also crossed, by **USD 170,000**, so distributions were trapped — and the sponsor had already declared a dividend against the 1.3400 figure. The tool's per-item accuracy was never the issue: the extraction was faithful to the document it was given, which is the failure mode no accuracy statistic can detect, and the combinatorial arithmetic of 16.2.2 had already said what to expect — at 92 per cent per-item accuracy the probability that all **26** load-bearing definitions were right was **0.114415**.

How it resolved. The sponsors injected an equity cure of **USD 1,020,000** to restore the covenant, and paid a waiver fee of **25 basis points** on the outstanding **USD 130,000,000 — USD 325,000 —** for the historic breach and the mis-stated certificate. Total direct cost **USD 1,345,000**, before the declared dividend was reversed and before the lenders moved the facility to monthly reporting for a year. The verification that would have prevented it — **26** load-bearing definitions at half an hour each of a legal-and-finance specialist at **USD 240.00** an hour — costs **USD 3,120**. The direct cost was **431.0897 times** the verification.

What the domain teaches here. The extraction was accurate, the arithmetic was correct, and the certificate was false. **The only control that would have caught it operates on the executed document and records the version identifier**, and it is the control 16.3.4's tiering rule independently demands: with four uses a year, an 88.5585 per cent chance of at least one wrong load-bearing term and a 600,000 consequence, the model's annual EMV is **USD 2,125,404** and its revalidation interval is **8.59 days** — shorter than the interval between its uses, which is the formula's way of saying verify every time. Two derivations, from combinatorics and from expected loss, converge on the same instruction, and the borrower had implemented neither.

Executive perspective — Domain 16

What a project finance director cannot delegate in this domain:

- **The error-cost line in every automation case.** Not the labour saving — the two error rates and the consequence per error. Their omission flattered Case study A's breakeven by 20.3822 per cent and hid a 54,000 annual loss at the asset that sponsored the programme.
- **The threshold, because it encodes what a missed error costs us.** The number is a cost-ratio judgment with an owner, not a model setting: 12.5 per cent break-even posterior, T4 not T2, 47,000 a year.
- **The blind audit, and its permanence.** It is the only instrument that measures what the model cannot see, and it is the first line cut in a budget round. It restated a claimed 114,800 loss to 288,880 and returned 18.4471 times its cost.
- **The verification scope on load-bearing items.** Twenty-six definitions, named verifier, recorded date, executed document, version identifier — USD 3,120 against the 431.0897 multiple Case study B paid.
- **The approval architecture and its price.** 45,589 a change through a quarterly committee against 6,268 through a mandated panel: tier it by consequence, write the mandate, and keep certification and lender disclosure to the director's own hand.
- **The rollback, funded and exercised.** USD 141,000 for a quarter's suspension and 4.99375 full-time equivalents to restore manual capacity — figures that must exist before an incident, not after.

Calculation exercises — Domain 16

> **Exercise 16.1** Manual review takes 7 minutes a record at a loaded USD 96.00 an hour and leaves 2.20 per cent of records with an undetected error. An automated pipeline costs USD 0.95 a record in processing and leaves 3.50 per cent undetected, on a committed fixed cost of USD 96,000 a year. An undetected error costs USD 500. Find the breakeven volume. Solution. Manual `7 × 1.60 = 11.20` plus `0.022 × 500 = 11.00`, total **USD 22.20**. Automated `0.95` plus `0.035 × 500 = 17.50`, total **USD 18.45**. Advantage **USD 3.75**; breakeven `96,000 ÷ 3.75 =` **25,600 records a year**. Common error: omitting the error-cost terms, which gives an advantage of `11.20 − 0.95 = 10.25` and an apparent breakeven of 9,366 records — 63.4 per cent too low, and the specific defect of 16.1.2.

> **Exercise 16.2** A detector screens 30,000 items a year of which 900 are genuinely erroneous. A false positive costs USD 60 to investigate; a missed error costs USD 900. Three thresholds give: A — 450 true positives, 90 false; B — 675 true positives, 300 false; C — 855 true positives, 1,500 false. Compute accuracy and total cost at each, choose the threshold, and confirm the choice marginally. Solution. Accuracy: A `(450 + 29,010)/30,000 =` **98.2000 %**; B `(675 + 28,800)/30,000 =` **98.2500 %**; C `(855 + 27,600)/30,000 =` **94.8500 %**. Total cost: A `90 × 60 + 450 × 900 =` **410,400**; B `300 × 60 + 225 × 900 =` **220,500**; C `1,500 × 60 + 45 × 900 =` **130,500**. **Choose C** — accuracy is maximised at B, and choosing B would cost **USD 90,000 a year**. Marginal confirmation: B to C adds 180 true and 1,200 false positives, marginal precision `180/1,380 =` **13.0435 %**, above the

> break-even posterior `60 ÷ 900 =` **6.6667 %** (one in 15); benefit `180 × 900 = 162,000` against cost `1,200 × 60 = 72,000`, net **+90,000**, which reconciles exactly to the fall in total cost. Common error: selecting B on accuracy — the 90,000 mistake this domain exists to prevent.

> **Exercise 16.3** How many independent, representative test cases, all passing, are needed to be 95 per cent confident that a model's defect rate is below 2 per cent? And at 99 per cent confidence? Solution. $n \ge \ln(\alpha) \div \ln(1 - p)$. At 95 per cent: $\ln(0.05) \div \ln(0.98) = 148.2837$, so **149** cases. At 99 per cent: $\ln(0.01) \div \ln(0.98) = 227.9482$, so **228**. Common error: using $1/p = 50$ or $100/p$-style reasoning; a defect rate bound of 2 per cent needs roughly $3/p$ cases at 95 per cent confidence, not $1/p$, and the difference between 50 and 149 is the difference between a claim and evidence.

> **Exercise 16.4** An extraction tool is 95 per cent accurate per item. Eighteen extracted terms are load-bearing. What is the probability the deliverable is entirely correct, and what per-item accuracy would a 99 per cent clean sweep require? Solution. 0.95^{18} = **0.397214**, so a **39.7214 per cent** chance all eighteen are right and a **60.2786 per cent** chance at least one is wrong. For a 99 per cent sweep, `a ≥ 0.99^(1/18) =` **99.9442 per cent** per item. Common error: reporting the tool's 95 per cent as the reliability of the register — the mistake that produced Case study B's false certificate.

> **Exercise 16.5** An estate makes 3,600 payments a year totalling USD 52,000,000. A second approver costs 12.5 minutes at USD 96.00 an hour. 0.25 per cent of payments are bad and a second approval catches 80 per cent of them. Derive the dual-approval threshold, then value blanket against thresholded approval given that 820 payments worth USD 48,900,000 exceed the threshold. Solution. Approver cost `12.5 × 1.60 =` **USD 20.00**; caught rate `0.0025 × 0.80 =` **0.0020**; threshold `20.00 ÷ 0.0020 =` **USD 10,000**. Blanket: cost `3,600 × 20 =` **72,000**, avoided `0.0020 × 52,000,000 =` **104,000**, net **+32,000**. Thresholded: cost `820 × 20 =` **16,400**, avoided `0.0020 × 48,900,000 =` **97,800**, net **+81,400**. The 2,780 sub-threshold payments cost **55,600** and avoid **6,200**, destroying **49,400** — exactly the difference between the two policies. Common error: concluding from the blanket policy's positive 32,000 that the policy is sound; the control is strongly positive above the threshold and strongly negative below it, and only the marginal view shows it.

Practitioner's toolkit — Domain 16

Adoption-ready artefacts; adapt headings to your organisation, then keep them stable.

Toolkit 16.T.1 — The automation business case, on one page

Mandatory lines, in this order, with no line permitted to be blank: the task and the **volume** the deciding entity actually has (asset, estate, group); **manual cost per item** = minutes × loaded rate, with the rate stated; **measured manual undetected-error rate** and its source; **consequence per undetected error** and its derivation; **manual all-in cost per item**; **automated committed fixed cost** a year; **automated processing cost per item**, including the referral rate and adjudication time; **measured or assumed automated undetected-error rate**, flagged as which; **automated all-in cost per item**; **advantage per item**; **breakeven volume**; **net annual value at the actual volume**; the **automated error rate** and the **consequence cost** at which the case reverses; the **rollback cost** for one period's suspension; and the **fallback capacity** in full-time equivalents and lead time. Rule: a case with no error-cost line is returned unread.

Toolkit 16.T.2 — Threshold and alert-queue sheet (per detector, per review)

Per candidate threshold: score cut · `TP` · `FP` · `FN` · `TN` · recall · precision · accuracy · alert count · adjudication hours · **total misclassification cost**. Above the table: the false-positive cost and the false-negative cost, each with its owner and derivation, and the derived **break-even posterior**. Below it: the **marginal** true positives, false positives, precision, benefit, cost and net for each step, with the arithmetic check that each row's net reconciles to the change in total cost. Footer lines: the operating threshold and who approved it; the date of the last **blind audit** of unflagged items, its sample size, findings, extrapolated missed errors and restated cost; and the prevalence in use, marked labelled or audited.

Toolkit 16.T.3 — Model inventory, tier and approval record

One row per model in production, reconciled quarterly to production access logs — an unreconciled inventory is the governance failure this artefact exists to prevent. Columns: model and version · purpose · owner · inputs and their golden sources · outputs and the decisions they touch · whether any output reaches a lender or a certificate · **uses a year** · **defect probability** `p`, marked measured / estimated / inherited · **consequence per defect** and its derivation · **annual** `EMV` · **revalidation interval** = tolerance ÷ `EMV`, with the policy cap applied · date last validated, sample size, confidence and bound · monitoring in place and the drift trigger · **approval tier and written mandate reference** · attribution method and the last reconciliation residual · rollback procedure, its cost, and the date it was last exercised · named verifier for the current period. Header: the tolerance parameter and its owner.

Exam preparation — Domain 16

What is assessed. All-in cost per reviewed item for a manual and an automated process, and the breakeven volume; the effect on the breakeven of the consequence per error and of each process's error rate; the volume, error-rate and consequence breakevens as sensitivities; a classifier's recall, precision, accuracy and total misclassification cost across thresholds, the cost-minimising threshold, the annual cost of choosing the accuracy-maximising one, the break-even posterior, and the marginal-precision test; scenario coverage `1 − (1 − p)ᵏ` and the required `k` for a stated confidence; the sweep probability `aᵐ` and the per-item accuracy a stated sweep confidence demands; the real against the apparent saving from machine-assisted model building, and the expected cost of an omitted review; attributional completeness and the residual as a share of covenant headroom; zero-failure validation sample size, the one-failure adjustment, and the translation of a validated bound into expected annual production errors; label bias, the extrapolation from a blind audit, the restated recall and cost, and the audit sample size for a stated bound; model tiering by annual `EMV` and the derived revalidation interval; the deployment breakeven consequence; the dual-approval threshold and the value of blanket against thresholded application; and governance latency `E[wait] = M/2 + L` priced against a daily benefit.

The calculations to do under time pressure. `minutes × rate + rate of error × consequence` for each process, then `fixed ÷ advantage`. The consequence and error-rate breakevens by rearranging the same equation. `FP × cost + FN × cost` down a threshold column, then the marginal step: `Δ TP × FN cost − Δ FP × FP cost`, and `FP cost ÷ FN cost` for the posterior. `ln(α) ÷ ln(1 − p)` for both scenario coverage and validation sample size — the same formula twice, which is worth noticing. `aᵐ` and `c^(1/m)`. `uses × p × consequence`, then `tolerance ÷ EMV`. `approver cost ÷ caught rate`, then cost and avoided loss on each population. `M/2 + L`, times `annual benefit ÷ 365`.

The traps. Omitting the error-cost term from an automation case (Exercise 16.1, MCQ 16.1-A, Case study A) · assuming the automated error rate equals or beats the manual one when the case depends on it (16.1.2) · appraising an automation at the estate's volume when the deciding entity is one asset, or the reverse (16.1.2, Case study A) · assuming a higher consequence per error strengthens the automation case when it weakens it (MCQ 16.1-B) · selecting a threshold on accuracy (Exercise 16.2, MCQ 16.1-C, Case study A) · testing a threshold step on average rather than marginal precision (MCQ 16.1-D) · quoting recall or precision computed against incomplete labels (16.3.3, MCQ 16.3-C, Case study A) · attempting to fix label bias by moving the threshold (16.3.3) · drawing an audit sample from flagged items or having it reviewed by the labelling team (16.3.3) · reporting a scenario count without `k`, or a generated minimum as a bank case (16.2.1, MCQ 16.2-C) · reading per-item extraction accuracy as deliverable accuracy (Exercise 16.4, MCQ 16.2-A, Case study B) · extracting from a data-room draft rather than the executed document (16.2.2, Case study B) · letting the tool's confidence scores choose the verification scope (MCQ 16.2-B) · booking the apparent saving from assisted model building while leaving the review at its hand-built hours (16.2.3, MCQ 16.2-D) · treating a validation bound as a production guarantee (MCQ 16.3-B) · repairing a failing test case instead of enlarging the suite from 299 to 473 (16.3.2) · tiering models by transaction volume rather than by expected loss (MCQ 16.3-D, 16.3.4) · omitting the policy cap on a computed revalidation interval (16.3.4) · presenting a deployment `EMV` as a decision rather than its breakeven consequence as a question (MCQ 16.4-C) · deriving a dual-approval threshold without the catch rate (MCQ 16.4-A) · concluding that a blanket control is sound because its net is positive (Exercise 16.5, MCQ 16.4-B) · pricing an approval gate at zero (MCQ 16.4-D) · and assuming an agent inherits the effect of the control it replaces rather than its absence (16.A.1).

How the domain connects. Domain 2 supplied the definitional exposure — 600,000 of `CFADS` on one working-capital treatment — that makes the definitional layer of 16.1.1 and the verification scope of 16.2.2 worth their cost. Domain 6 built the model, its six invariants and the effective-rate check that 16.2.3's review must run, and supplied the 3,250,352 defect consequence that prices it. Domain 10 supplied `CFADS`, the debt service, the 1.20× covenant and the 372,438 of annual headroom that every data-quality figure in this domain is measured against, and the information covenants that make model disclosure a contractual matter. Domain 11 established model risk as a priced register line whose probability is a function of governance, which is the claim 16.3.4 turns into a calendar. Domain 13's model audit is the external counterpart of 16.3.2's validation, and Domain 14 established that no pipeline may generate a representation. PML-AI supplied three registered forms used unchanged: the mesh-versus-layer interface count, `EMV`, and governance latency. Backwards, this domain gives every earlier **AI in this KA** section its arithmetic; forwards, it is the last domain, and what it hands on is not another technique but a standard of proof — that a claim about automation, like a claim about coverage, is worth exactly what its arithmetic can carry.

Domain 16 summary

Responsible AI in finance is a quantitative discipline, and its four central results all contradict a common belief. **Automation is decided by error costs, not labour.** Manual review of the estate's consumption-and-billing records costs **USD 13.60** a record plus **1.80 per cent** of undetected errors at **USD 320 — USD 19.36** all-in — against the pipeline's **USD 1.04** plus **2.60 per cent**, or **USD 9.36**, so on a committed **USD 148,000** the breakeven is **14,800 records a year**: a **USD 54,000** annual loss at Kestrel's 9,400 and a **USD 416,000** annual gain across the estate's 56,400. Omitting the error terms moves the apparent breakeven to **11,783** and flatters automation by **20.3822 per cent**; repricing an undetected error at **USD 1,200** moves the real breakeven to **50,000** and leaves the whole programme worth **USD 18,944 —** so the more consequential the work, the more volume automation needs, and the rule is to automate high-volume, low-consequence work rather than important work. **Accuracy is the wrong objective for a detector.** Over 48,000 payments and journals at a 2.5 per cent anomaly rate, with a false positive costing **USD 40** and a missed error **USD 320**, cost is minimised at **USD 114,800** (threshold T4, 80 per cent recall, precision only **50.2618 per cent**) while accuracy peaks at **98.5729 per cent** two thresholds away, where cost is **USD 161,800 —** a **USD 47,000** annual price for a metric choice. The rule is a **12.5 per cent** break-even posterior, one in eight, tested on marginal precision: **17.9104 per cent** from T3 to T4 earns 16,400 and **5.7971 per cent** from T4 to T5 loses 39,600. **Validation proves less than it appears. 299** passing independent cases bound a defect rate below 1 per cent at 95 per cent confidence — and at 56,400 uses a year that still admits **564** errors; bounding expected errors at ten would need **16,895** cases, and one observed failure raises the 299 to **473 —** which is why monitoring, tiered human approval and a funded rollback carry the assurance, not testing. **And controls, thresholds and verification are marginal quantities.** Blanket dual approval nets **USD 400** a year; the same control above a derived **USD 14,118** threshold nets **USD 80,180**, because 3,650 small payments cost 87,600 to approve and protect 7,820. Around those four results sit the domain's other derivations: a data spine costing **USD 900,000** returns an `NPV` of **+USD 1,221,674** on hours while its real value is the single implementation of `CFADS` whose divergence is worth **1.6110** times Kestrel's annual covenant headroom of **372,438**; a 92 per cent-accurate extractor gets all **26** load-bearing definitions right **11.4415 per cent** of the time, so a 95 per cent sweep would demand **99.8029 per cent** per item and the answer is **USD 3,120** of human verification instead — the control Case study B omitted at a cost of **USD 1,345,000**, or **431.0897** times; machine-assisted model building saves a real **33.3333 per cent**, not the apparent 70.8333, and the omitted 18 hours of review are **USD 2,700** against **USD 1,137,623.20** of expected defect cost; a blind audit of **1,000** unflagged items restated the detector's recall from 80 to **55.0459 per cent** and its cost from 114,800 to **USD 288,880**, returning **18.4471** times its cost; a model inventory tiered on `uses × p × consequence` gives the defined-term extractor an annual `EMV` of **USD 2,125,404** and an **8.59-day** interval that independently reproduces "verify every use"; a deployment choice turns on a breakeven consequence of **USD 40,307,692** rather than on an `EMV`; and an approval gate costs **USD 45,589.04** a change through a quarterly committee against **USD 6,268.49** through a mandated panel. Every one of those verification steps returns between eighteen and four hundred and thirty times its cost, which is the domain's real finding and the one that survives every disagreement about the technology: **AI proposes; the professional verifies, decides and remains accountable — and the verification is the cheapest line in the budget.**

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